Contents

Preface 2

1 Chapter 1 4

The Natural Numbers 4
1.0.1 Roadmap . . . 5
1.0.2 Named moves . . . 5
1.0.3 Threads . . . 5
1.0.4 Summary . . . 6
1.0.5 Terms . . . 6

1.1 Peano's Axioms and Induction 6
1.1.1 What counting needs . . . 7
1.1.2 The one assumption . . . 8
1.1.3 Induction in the form we use . . . 10
1.1.4 Every natural number except 0 has one predecessor . . . 11
1.1.5 Exercises . . . 13
1.1.5.1 A. Check your understanding . . . 13
1.1.5.2 B. Practice . . . 13
1.1.5.3 C. Going deeper . . . 15

1.2 The Recursion Theorem; Addition and Multiplication 17
1.2.1 Recursion on $\mathbb{N}$ . . . 17
1.2.2 Addition and multiplication . . . 20
1.2.3 The laws of arithmetic . . . 21
1.2.4 Exercises . . . 25
1.2.4.1 A. Check your understanding . . . 25
1.2.4.2 B. Practice . . . 26
1.2.4.3 C. Going deeper . . . 27

1.3 Order and Well-Ordering of the Natural Numbers 29
1.3.1 Counting up from m to n . . . 30
1.3.2 Order and arithmetic . . . 30
1.3.3 A total order . . . 32
1.3.4 Well-ordering . . . 34
1.3.5 Powers and parity . . . 35
1.3.6 Exercises . . . 37
1.3.6.1 A. Check your understanding . . . 37
1.3.6.2 B. Practice . . . 37
1.3.6.3 C. Going deeper . . . 38

1.4 All Peano Systems Are Alike 40

1.4.1 Two Peano systems, and what "alike" should mean . . . 40
1.4.2 The categoricity theorem . . . 41
1.4.3 Arithmetic and order travel along the isomorphism . . . 44
1.4.4 Identifying the two sets of natural numbers . . . 45
1.4.5 How later chapters cite *Logic and Proof* and *Sets and Functions* . . . 47
1.4.6 Exercises . . . 48
1.4.6.1 A. Check your understanding . . . 48
1.4.6.2 B. Practice . . . 49
1.4.6.3 C. Going deeper . . . 51

1.5 Where ℕ Comes From **52**
1.5.1 Sets as the raw material . . . 52
1.5.2 Zero and the successor as sets . . . 53
1.5.3 The least inductive set . . . 53
1.5.4 Pairs as sets . . . 54
1.5.5 Exercises . . . 55
1.5.5.1 A. Check your understanding . . . 55
1.5.5.2 B. Practice . . . 55
1.5.5.3 C. Going deeper . . . 56
1.5.5.4 R. Review . . . 57

2 Chapter 2 **59**

The Integers **59**
2.0.1 Roadmap . . . 60
2.0.2 Named moves . . . 60
2.0.3 Threads . . . 60
2.0.4 Summary . . . 61
2.0.5 Terms . . . 61

2.1 The Integers as Classes of Pairs **61**
2.1.1 What we use about the natural numbers . . . 62
2.1.2 A naive attempt . . . 62
2.1.3 Pairs that record a difference . . . 62
2.1.4 Pictures of classes . . . 64
2.1.5 Why this definition . . . 66
2.1.6 Exercises . . . 68
2.1.6.1 A. Check your understanding . . . 68
2.1.6.2 B. Practice . . . 68
2.1.6.3 C. Going deeper . . . 69

2.2 Well-Definedness, and the Arithmetic of the Integers **71**
2.2.1 Definitions through representatives . . . 71
2.2.2 The lemma . . . 71
2.2.3 Sum, product and negative of integers . . . 73
2.2.4 Rules that fail . . . 76
2.2.5 The natural numbers inside the integers . . . 77
2.2.6 Exercises . . . 79
2.2.6.1 A. Check your understanding . . . 79
2.2.6.2 B. Practice . . . 79
2.2.6.3 C. Going deeper . . . 81

2.3 Commutative Rings; the Integers Are One **82**
2.3.1 The laws of a ring . . . 83
2.3.2 The integers form a commutative ring . . . 84
2.3.3 Exercises . . . 88
2.3.3.1 A. Check your understanding . . . 88
2.3.3.2 B. Practice . . . 89
2.3.3.3 C. Going deeper . . . 90

2.4 Ordered Rings and the Rules for Inequalities **91**
2.4.1 Ordered rings . . . 91
2.4.2 The order of the integers . . . 93
2.4.3 Rules for inequalities . . . 95
2.4.4 Least elements, and a ring that cannot be ordered . . . 98
2.4.5 Exercises . . . 99
2.4.5.1 A. Check your understanding . . . 99
2.4.5.2 B. Practice . . . 100
2.4.5.3 C. Going deeper . . . 101

2.5 Division with Remainder **102**
2.5.1 Divisibility on the constructed integers . . . 103
2.5.2 Division with remainder . . . 105
2.5.3 Even and odd . . . 107
2.5.4 Exercises . . . 108
2.5.4.1 A. Check your understanding . . . 108
2.5.4.2 B. Practice . . . 108
2.5.4.3 C. Going deeper . . . 109
2.5.4.4 R. Review . . . 110

3 Chapter 3 **112**

The Rationals **112**
3.0.1 Roadmap . . . 113
3.0.2 Named moves . . . 113
3.0.3 Threads . . . 113
3.0.4 Summary . . . 114
3.0.5 Terms . . . 114

3.1 The Rationals as Classes of Fractions **114**
3.1.1 Fractions as pairs . . . 114
3.1.2 Adding and multiplying classes . . . 117
3.1.3 The rationals form a field . . . 119
3.1.4 The integers inside the rationals . . . 120
3.1.5 Exercises . . . 122
3.1.5.1 A. Check your understanding . . . 122
3.1.5.2 B. Practice . . . 122
3.1.5.3 C. Going deeper . . . 124

3.2 The Rationals as an Ordered Field; the Archimedean Property **127**
3.2.1 Which fractions are positive . . . 127
3.2.2 The rationals form an ordered field . . . 129
3.2.3 Between two rationals . . . 131
3.2.4 The Archimedean property . . . 131

3.2.5 The integer part . . . 133
3.2.6 Exercises . . . 134
3.2.6.1 A. Check your understanding . . . 134
3.2.6.2 B. Practice . . . 135
3.2.6.3 C. Going deeper . . . 136

3.3 The Gaps in the Rationals 139
3.3.1 No rational squares to two . . . 139
3.3.2 The set below the gap . . . 141
3.3.3 A rational sequence aimed at the gap . . . 144
3.3.4 Exercises . . . 145
3.3.4.1 A. Check your understanding . . . 145
3.3.4.2 B. Practice . . . 146
3.3.4.3 C. Going deeper . . . 148
3.3.4.4 R. Review . . . 150

4 Chapter 4 153

Ordered Fields and Sequences in an Ordered Field 153
4.0.1 Roadmap . . . 154
4.0.2 Named moves . . . 155

4.1 Working in an Ordered Field 155
4.1.1 The standing setup . . . 155
4.1.2 Fractions . . . 156
4.1.3 Halving and comparing . . . 157
4.1.4 Absolute value . . . 158
4.1.5 Exercises . . . 161
4.1.5.1 A. Check your understanding . . . 161
4.1.5.2 B. Practice . . . 161
4.1.5.3 C. Going deeper . . . 162

4.2 The Rationals inside an Ordered Field; the Archimedean Property 163
4.2.1 The natural numbers in an ordered field . . . 164
4.2.2 A copy of the rationals in every ordered field . . . 166
4.2.3 The Archimedean property . . . 169
4.2.4 Integer part and density . . . 170
4.2.5 A field that is not Archimedean . . . 171
4.2.6 Exercises . . . 172
4.2.6.1 A. Check your understanding . . . 172
4.2.6.2 B. Practice . . . 173
4.2.6.3 C. Going deeper . . . 174

4.3 Sequences and Limits in an Ordered Field 176
4.3.1 Sequences . . . 176
4.3.2 The limit of a sequence . . . 177
4.3.3 Proving a limit from the definition . . . 180
4.3.4 Archimedes and the sequence $1/(n+1)$ . . . 182
4.3.5 Changing finitely many terms . . . 183
4.3.6 A sequence has at most one limit . . . 184
4.3.7 Convergent sequences are bounded . . . 185
4.3.8 Exercises . . . 186

4.3.8.1 A. Check your understanding . . . 186
4.3.8.2 B. Practice . . . 186
4.3.8.3 C. Going deeper . . . 188

4.4 The Algebra and Order of Limits 190
4.4.1 Two tools . . . 190
4.4.2 The limit laws . . . 191
4.4.3 Limits and order . . . 194
4.4.4 Exercises . . . 195
4.4.4.1 A. Check your understanding . . . 195
4.4.4.2 B. Practice . . . 196
4.4.4.3 C. Going deeper . . . 198

4.5 Cauchy Sequences 199
4.5.1 Terms that agree with one another . . . 200
4.5.2 Convergent sequences are Cauchy, and Cauchy sequences are bounded . . . 201
4.5.3 Sums and products of Cauchy sequences . . . 203
4.5.4 The sign of a Cauchy sequence that is not null . . . 204
4.5.5 Sign stability and reciprocals . . . 206
4.5.6 Small steps do not make a Cauchy sequence . . . 207
4.5.7 Exercises . . . 208
4.5.7.1 A. Check your understanding . . . 208
4.5.7.2 B. Practice . . . 209
4.5.7.3 C. Going deeper . . . 210

4.6 A Cauchy Sequence with No Rational Limit 212
4.6.1 The sequence again . . . 212
4.6.2 The theorem . . . 213
4.6.3 What this chapter did not prove . . . 215
4.6.4 Exercises . . . 216
4.6.4.1 A. Check your understanding . . . 216
4.6.4.2 B. Practice . . . 217
4.6.4.3 C. Going deeper . . . 218
4.6.4.4 R. Review . . . 220

5 Chapter 5 223

Constructing the Real Numbers from Cauchy Sequences 223
5.0.1 Roadmap . . . 224
5.0.2 Named moves . . . 224
5.0.3 Threads . . . 225
5.0.4 Summary . . . 225
5.0.5 Terms . . . 226

5.1 Sequences That Should Converge 226
5.1.1 Two sequences aimed at one hole . . . 226
5.1.2 Cauchy sequences of rationals . . . 228
5.1.3 When two sequences name the same number . . . 229
5.1.4 The truncations and the Newton sequence . . . 231
5.1.5 Finite changes, and shifts . . . 233
5.1.6 Exercises . . . 234
5.1.6.1 A. Check your understanding . . . 234

5.1.6.2 B. Practice . . . 235
5.1.6.3 C. Going deeper . . . 236

5.2 The Real Numbers and Their Arithmetic 237
5.2.1 The real numbers as classes . . . 237
5.2.2 Arithmetic on representatives . . . 238
5.2.3 The laws of arithmetic . . . 240
5.2.4 Inverses . . . 241
5.2.5 Exercises . . . 243
5.2.5.1 A. Check your understanding . . . 243
5.2.5.2 B. Practice . . . 244
5.2.5.3 C. Going deeper . . . 245

5.3 Positive Classes and the Order 247
5.3.1 Comparing terms does not compare classes . . . 247
5.3.2 Positive sequences . . . 248
5.3.3 The positive classes . . . 251
5.3.4 The order of the reals . . . 252
5.3.5 Comparing classes through their terms . . . 253
5.3.6 Exercises . . . 254
5.3.6.1 A. Check your understanding . . . 254
5.3.6.2 B. Practice . . . 255
5.3.6.3 C. Going deeper . . . 257

5.4 The Rationals Inside the Reals 259
5.4.1 A copy of the rationals . . . 260
5.4.2 Archimedes in the reals . . . 261
5.4.3 Between two reals there is a rational . . . 262
5.4.4 The square root of two . . . 263
5.4.5 Exercises . . . 264
5.4.5.1 A. Check your understanding . . . 264
5.4.5.2 B. Practice . . . 265
5.4.5.3 C. Going deeper . . . 267

5.5 Least Upper Bounds 269
5.5.1 Complete ordered fields . . . 269
5.5.2 Powers of two in an ordered field . . . 270
5.5.3 Bisection with rational ends . . . 271
5.5.4 What we built, and what comes next . . . 275
5.5.5 Exercises . . . 276
5.5.5.1 A. Check your understanding . . . 276
5.5.5.2 B. Practice . . . 277
5.5.5.3 C. Going deeper . . . 278
5.5.5.4 R. Review . . . 280

6 Chapter 6 284

The Complete Ordered Field Is Unique 284
6.0.1 Roadmap . . . 285
6.0.2 Named moves . . . 286
6.0.3 Threads . . . 286
6.0.4 Summary . . . 286

6.0.5 Terms 287

6.1 What Completeness Gives at Once **287**
6.1.1 Maps that respect everything 287
6.1.2 A complete ordered field is Archimedean 289
6.1.3 The rationals below an element 291
6.1.4 Constructing a point as a supremum 293
6.1.5 Exercises 296
6.1.5.1 A. Check your understanding 296
6.1.5.2 B. Practice 297
6.1.5.3 C. Going deeper 298

6.2 Building the Map from the Rationals Below **299**
6.2.1 What the map must be 299
6.2.2 Sums and products, measured by rationals 300
6.2.3 The map into a complete field 301
6.2.4 Exercises 304
6.2.4.1 A. Check your understanding 304
6.2.4.2 B. Practice 305
6.2.4.3 C. Going deeper 306

6.3 Two Complete Ordered Fields, One Isomorphism **308**
6.3.1 The uniqueness theorem 309
6.3.2 One real line 310
6.3.3 Automorphisms 312
6.3.4 What the isomorphism carries 314
6.3.5 Assume the axioms or build a model 316
6.3.6 Exercises 318
6.3.6.1 A. Check your understanding 318
6.3.6.2 B. Practice 320
6.3.6.3 C. Going deeper 321
6.3.6.4 R. Review 323

Preface

The two earlier volumes used numbers as familiar objects. Their arithmetic and order were school facts, named where they were used and never proved. This volume proves them. It starts from Peano's axioms for the natural numbers and builds the integers, the rationals and the real numbers, each out of the one before. At the end the real numbers form a complete ordered field, and any two complete ordered fields are the same up to exactly one isomorphism.

Who it is for. The reader we have in mind has worked through *Logic and Proof* and *Sets and Functions*, or has their habits from elsewhere. That means negating a quantified statement, running an induction, and checking that a rule on equivalence classes does not depend on the representative chosen. The last of these is used in almost every chapter. The chapters point back to both volumes where they use a definition or a rule, so a reader who is unsure of a step can look it up.

What it covers.

Chapter	Title	What it does
1	The Natural Numbers	$\mathbb{N}$ as a Peano system: induction, the recursion theorem, addition and multiplication, order and well-ordering, any two Peano systems alike; a short closing section on where $\mathbb{N}$ comes from in set theory
2	The Integers	$\mathbb{Z}$ as classes of pairs of natural numbers, well-defined operations, commutative rings, ordered rings and the rules for inequalities, division with remainder
3	The Rationals	$\mathbb{Q}$ as classes of fractions, an Archimedean ordered field, and the gap in $\mathbb{Q}$ where $\sqrt{2}$ should be
4	Ordered Fields and Sequences	Limits and Cauchy sequences in an arbitrary ordered field, the Archimedean property, and a Cauchy sequence of rationals with no rational limit

Chapter	Title	What it does
5	The Real Numbers from Cauchy Sequences	$\mathbb{R}$ as classes of rational Cauchy sequences, an ordered field in which every bounded set has a supremum
6	The Complete Ordered Field Is Unique	Exactly one order-preserving isomorphism between any two complete ordered fields, and what *Analysis on the Real Line* takes from this volume

Chapters 1–3 build the numbers that algebra needs, one chapter for each. Chapters 4 and 5 fill the gap that Chapter 3 found in $\mathbb{Q}$: first the rules for sequences, proved without the number they aim at, then the numbers made from the sequences themselves. Chapter 6 shows that the construction does not matter.

How to read it. The main line is Chapters 1–6; read them in order, since each uses the ones before it. Every section has Quick checks with folded answers; answer each one before you unfold it. The exercises of a section come in three groups, A to check understanding, B for practice and C to go deeper, and the last section of each chapter adds R items that reach back to earlier chapters. A chapter's opening page lists what it needs, with links, and the time it takes. Second-reading folds, in most chapters, hold edge cases and longer justifications. They can stay closed on a first reading, but they are not optional, since later proofs may use what they contain. Results marked optional can be skipped, and nothing outside them depends on them.

Time. About 120 hours on the core track (all of A, one B and one C per section, the exit checks) and about 180 hours for everything. Chapter by chapter, core and full: Chapter 1, 20 and 30 hours; Chapter 2, 20 and 25; Chapter 3, 15 and 25; Chapter 4, 25 and 40; Chapter 5, 25 and 35; Chapter 6, 15 and 25. The figures come from word counts and exercise counts, not from reader logs, and will be replaced by the median of logged hours once there are some.

What comes next. *Analysis on the Real Line* does not need this volume. It starts from the axioms of a complete ordered field, the axioms this volume proves for its $\mathbb{R}$, and Chapter 6 explains why it does not matter which model of them one has in mind. A reader who wants to know that those axioms describe something real, and only one thing, will find both answers here. *Algebraic Preliminaries* builds on Chapters 2, 5 and 6: the integers with division with remainder (its Chapter 1 takes up divisibility, prime factorization and congruences), the real numbers as a complete ordered field, and square roots in it, together with the ring and field rules of Chapters 2, 3 and 4 that come with them.

Chapter 1

The Natural Numbers

The two earlier volumes counted with the natural numbers and never asked what they are. *Logic and Proof* took induction as an axiom about numbers it did not build, and *Sets and Functions* described $\mathbb{N}$ as a subset of the school $\mathbb{R}$ and proved the recursion theorem there. A volume that builds the real numbers cannot start from a subset of $\mathbb{R}$. This chapter takes one assumption instead: there is a **Peano system**, a set with a start 0, a step S, and three demands on them (the start is not a step, the step never merges two numbers, and the only subset that contains the start and is closed under the step is everything). From that one assumption the chapter derives induction, the recursion theorem, addition and multiplication with their laws, the order, well-ordering, and powers.

A reader may fear that the choice of Peano system matters. Section 1.4 shows it does not: any two Peano systems are alike in exactly one way, so every statement about one is a statement about the other, and the $\mathbb{N}$ of this chapter, the $\mathbb{N}$ of *Logic and Proof* and the $\mathbb{N}$ of *Sets and Functions* are identified by a citing rule that says which earlier results carry over. Section 1.5, short and without proofs, says where a Peano system comes from: sets are the raw material, the natural numbers form the least inductive set, and pairs are sets. The proofs belong to *Axiomatic Set Theory*.

What you need. *Logic and Proof*, Chapter 4 (induction and strong induction: Logic and Proof, Axiom 4.2.1, Logic and Proof, Theorem 4.2.2), *Logic and Proof*, Chapter 2 (reductio: Logic and Proof, Definition 2.5.1), *Logic and Proof*, Section 7.6 (non-standard models, for the comparison after the independence example: Logic and Proof, Example (A Model of Arithmetic with an Infinite Number)) and *Logic and Proof*, Chapter 8 (the catalogue of what a proof may not do, for the find-the-error exercises: Logic and Proof, Example (What is not allowed)). *Sets and Functions*, Chapter 1 (sets cut out by a property, ordered pairs and products: Sets and Functions, Definition 1.2.3, Sets and Functions, Definition 1.4.1, Sets and Functions, Definition 1.4.2, Sets and Functions, Proposition 1.3.2); *Sets and Functions*, Chapter 2 (functions, injective, surjective and bijective maps, composition, inverses: Sets and Functions, Definition 2.1.1, Sets and Functions, Proposition 2.1.1, Sets and Functions, Definition 2.2.1, Sets and Functions, Definition 2.2.2, Sets and Functions, Definition 2.2.3, Sets and Functions, Definition 2.1.2, Sets and Functions, Theorem 2.4.4); *Sets and Functions*, Chapter 3 (induction, the recursion theorem, well-ordering and the three proof boxes: Sets and Functions, Theorem 3.2.1, Sets and Functions, Theorem 3.3.1, Sets and Functions, Theorem 3.1.3, Sets and Functions, Lemma 3.1.1, Sets and Functions, Example (Method: finding $P(n)$ inside $P(n+1)$), Sets and Functions, Example (Method: the least element), Sets and Functions, Example (Method: the smallest counterexample)); *Sets and Functions*, Chapter 4 (finite and infinite sets, for the citing rule of Section 1.4: Sets and Functions, Definition 4.1.2, Sets and Functions, Definition 4.1.3, Sets and Functions, Theorem 4.1.9); and *Sets and Functions*, Chapter 5 (partial orders: Sets and Functions,

Definition 5.3.1).

Time. About 20 hours on the core track (all of A, one B and one C per section, the exit checks); about 30 hours for everything.

1.0.1 Roadmap

- **Peano's Axioms and Induction.** What counting needs, and why the smallest closed subset of the school $\mathbb{R}$ does not do; a Peano system (M, e, s) and the assumption that one exists, which names its set $\mathbb{N}$; the three clauses (P3), (P4), (P5) shown independent by three small structures, one of them two copies of $\mathbb{N}$; induction in the set form and the statement form; and the predecessor, the proof that every natural number except 0 is a successor of exactly one, and that no number is its own successor.
- **The Recursion Theorem; Addition and Multiplication.** A function on $\mathbb{N}$ defined by a start and a step, existing and unique, by the smallest closed set of pairs; the sum and product defined by recursion, the numerals, successor cancellation, and the laws of arithmetic (associativity, commutativity, distributivity) proved by induction on the recursion variable.
- **Order and Well-Ordering of the Natural Numbers.** $m \leq n$ when $n = m + j$; the order and its compatibility with sum and product, cancellation in sums and products, and $n - m$ for $m \leq n$; a strict total order; every non-empty set has a least element; powers and the parity of $n \cdot n$.
- **All Peano Systems Are Alike.** The categoricity theorem: any two Peano systems are isomorphic by exactly one isomorphism, which carries sum, product and order along; the identification of this $\mathbb{N}$ with the $\mathbb{N}$ of *Logic and Proof* and *Sets and Functions*, and the citing rule that lets a later chapter cite their results about $\mathbb{N}$.
- **Where $\mathbb{N}$ Comes From.** Seven axioms of set theory said in one line each, zero and the successor as sets, ω as the least inductive set and why it is a Peano system, and pairs as sets. No proofs: they are in *Axiomatic Set Theory*.

1.0.2 Named moves

- **Induction through a closed subset** (Example (Method: induction through a closed subset)). To prove a property of every natural number, let A be the set of numbers that have it; show $0 \in A$ and that $n \in A$ gives $\mathrm{S}(n) \in A$; then $A = \mathbb{N}$ by (P5). It is used for induction in Section 1.1, and in Section 1.4 to show that the map out of $\mathbb{N}$ is onto (apply (P5) to its image). (Section 1.1)
- **Fix the others, induct on the recursion variable** (Example (Method: fix the others, induct on the recursion variable)). A law about a sum or product is proved by induction on the variable the definition is a recursion on, with the other variables fixed. It proves the laws of arithmetic in Section 1.2 and is reused in Section 1.3. (Section 1.2)
- **Translate $\leq$ into a sum.** Replace $m \leq n$ by $n = m + j$ for some j, and every order claim becomes a claim about sums, which Section 1.2 already controls. (Section 1.3)
- **Build by recursion, compare by induction** (Example (Method: build by recursion, compare by induction)). To compare two structures, build the map between them by the recursion theorem and prove it respects everything by induction. It proves categoricity in Section 1.4.

1.0.3 Threads

- **Induction.** *Logic and Proof* took it as an axiom, *Sets and Functions* kept that axiom for its $\mathbb{N} \subseteq \mathbb{R}$, and here it is clause (P5); Section 1.4 shows the three agree.

- **Recursion.** The recursion theorem of Section 1.2 builds the sum, the product and the power, and Section 1.4 uses it to build the isomorphism between two Peano systems. Chapter 2 uses it again for multiples in a ring.
- **Well-ordering and least elements.** Section 1.3 proves it for $\mathbb{N}$; Section 2.4 extends it to sets of integers bounded below, and Section 2.5 uses it to find remainders.
- **The citing rule.** Section 1.4 says which results of *Logic and Proof* and *Sets and Functions* may be cited about this $\mathbb{N}$ and why; every later chapter cites through it with Corollary 1.4.3.

1.0.4 Summary

A Peano system is a set M with an element e and a function $s\colon M \to M$ such that e is not a value of s, s is injective, and the only subset of M that contains e and is closed under s is M (Definition 1.1.1). The chapter assumes one and calls its set $\mathbb{N}$ (Axiom 1.1.1); the last clause is induction (Theorem 1.1.2), and every element but 0 has exactly one predecessor (Proposition 1.1.3). A start and a step define exactly one function on $\mathbb{N}$ (Theorem 1.2.1), which defines the sum and the product (Definition 1.2.1); they satisfy cancellation (Corollary 1.2.2) and the usual laws (Theorem 1.2.3). The order $m \leq n$ when $n = m+j$ (Definition 1.3.1) is compatible with the operations (Proposition 1.3.1), is a strict total order (Theorem 1.3.2), and every non-empty set has a least element (Theorem 1.3.3); powers and parity follow (Proposition 1.3.4). Any two Peano systems are isomorphic by exactly one isomorphism (Theorem 1.4.1), which preserves sums, products and order (Proposition 1.4.2), so the $\mathbb{N}$ of the earlier volumes may be identified with this one (Corollary 1.4.3). Section 1.5 describes, without proofs, how set theory supplies a Peano system: ω, the least inductive set, whose elements are the von Neumann numerals, with pairs built as sets.

What is open: nothing here lets us subtract 5 from 3. Chapter 2 builds the integers from pairs of natural numbers to fix that.

1.0.5 Terms

- **Peano system**: a set with a start and a step satisfying (P3), (P4), (P5) (Definition 1.1.1).
- **natural number**, $\mathbb{N}$, 0, S, **successor**: the elements, start and step of the assumed Peano system (Axiom 1.1.1).
- **predecessor**: the one m with $\mathrm{S}(m) = n$, for $n \neq 0$ (Proposition 1.1.3).
- **defined by recursion**: determined by a start and a step (Theorem 1.2.1).
- **sum, product, numeral**: the operations defined by recursion, and $1 \coloneqq \mathrm{S}(0)$, $2 \coloneqq \mathrm{S}(1)$, and so on (Definition 1.2.1).
- $\leq$, $<$: the order on $\mathbb{N}$ (Definition 1.3.1).
- **well-ordering**: every non-empty subset has a least element (Theorem 1.3.3).
- **power**: b^n defined by recursion on n (Proposition 1.3.4).
- **isomorphism of Peano systems**: a bijection that carries start to start and step to step (Theorem 1.4.1).

1.1 Peano's Axioms and Induction

Sets and Functions, Chapter 3 took the natural numbers as a subset $\mathbb{N}$ of the familiar real numbers, and borrowed the arithmetic of $\mathbb{R}$ whenever it was convenient. This volume builds the integers, the rationals and the real numbers. If we started from the $\mathbb{N}$ inside $\mathbb{R}$, the real numbers would rest on the real numbers. So we need $\mathbb{N}$ on its own. We make one assumption: there is a set with a start and a step that obey three rules. This section shows that each rule is needed,

proves induction in the form we use, and proves that every natural number except 0 has exactly one predecessor.

1.1.1 What counting needs

Two descriptions of $\mathbb{N}$ come to mind at once, and neither will do.

The first is $\mathbb{N} = \{0, 1, 2, 3, \dots\}$. It names four elements and then writes three dots, and the dots carry all the content. They say "keep going one at a time, and nothing else", and to say what that means is exactly the job of a definition of $\mathbb{N}$.

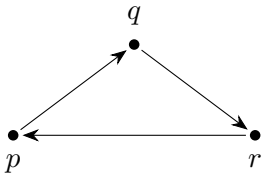
The second is more careful: $\mathbb{N}$ is the smallest subset of $\mathbb{R}$ that contains 0 and is closed under $x \mapsto x + 1$. A smallest such subset exists, by Sets and Functions, Proposition 1.3.2 with start 0 and step $x \mapsto x + 1$, and the set form of Logic and Proof, Axiom 4.2.1 shows that it is the $\mathbb{N}$ of *Sets and Functions*. Either way the description needs $\mathbb{R}$. Since $\mathbb{R}$ is where this volume is going, it cannot be where $\mathbb{N}$ comes from.

So we describe $\mathbb{N}$ by what counting has to do, not by where its elements live. Counting needs

- (P1) a **start**: an element where counting begins;
- (P2) a **step**: a rule sending each element to a next element of the same set.

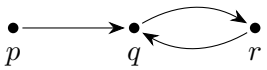
That is a set M, an element $e \in M$, and a function $s\colon M \to M$. A start and a step are not enough, and here are three ways they fail to count. In each picture an arrow goes from x to $s(x)$.

A loop. Let $M \coloneqq \{p, q, r\}$, start at p, and step $p \mapsto q \mapsto r \mapsto p$.



Counting comes back to its start: three steps after p we are at p again, as on a clock.

A merge. Let $M \coloneqq \{p, q, r\}$ again, start at p, and step $p \mapsto q \mapsto r \mapsto q$.



Counting never returns to p, but two different elements, p and r, step to the same q. After r the count repeats q, and runs in a circle from there.

An extra row. Take the counting numbers, and beside them a second copy with its own step, which the start never reaches. Every element still has one next element, no two elements step to the same one, and nothing steps back to the start. Yet starting at the start and stepping, we never meet the second copy. Counting cannot see those elements, and a description of the natural numbers must exclude them.

Each failure is ruled out by one clause: the step never returns to the start; the step never sends two elements to the same one; and nothing is there that steps from the start cannot reach. The last clause needs care, since "reached by steps" is the three dots again. The way around it is the one Sets and Functions, Proposition 1.3.2 takes: speak of subsets. A subset that contains the start and is closed under the step contains everything the steps reach. If the only such subset is

the whole set, nothing is left over.

A Peano system is a set with a start and a step, where the step is one-to-one, never returns to the start, and reaches everything.

Definition 1.1.1. Peano system

A **Peano system** (M, e, s) consists of a set M, an element $e \in M$, and a function $s\colon M \to M$, such that:

- (P3) $s(x) \neq e$ for **every** $x \in M$;
- (P4) s is injective;
- (P5) **every** subset $A \subseteq M$ with $e \in A$ and $s(x) \in A$ for every $x \in A$ is equal to M.

In words: (P1) and (P2) are built into the data, since $e \in M$ and s maps M into M. (P3) says the step never leads back to the start. (P4) says two different elements never step to the same element. (P5) is induction for M: a subset that contains the start and is closed under the step is everything, so every element of M is reached from e by steps. We write (M, e, s) and not (N, e, s), so that the letter never suggests $\mathbb{N}$. The notation (M, e, s) lists the three pieces of data; it is not a set built from them (if one object is wanted, take the pair $((M, e), s)$ of Sets and Functions, Definition 1.4.1).

The merge breaks (P4), since $s(p) = q = s(r)$ with $p \neq r$. The extra row breaks (P5): the first copy contains the start, is closed under the step, and is not everything. The loop is the subject of the first Quick check.

Quick check. For the loop ($M = \{p, q, r\}$, $e = p$, $s(p) = q$, $s(q) = r$, $s(r) = p$), which of (P3), (P4), (P5) fail?

Solution. Only (P3): $s(r) = p = e$. (P4) holds, since the three values $s(p) = q$, $s(q) = r$, $s(r) = p$ are different. (P5) holds: a subset containing p and closed under s contains $s(p) = q$, then $s(q) = r$, so it is M.

1.1.2 The one assumption

The natural numbers are to be a Peano system. That is the whole of what we assume about them.

Axiom 1.1.1. The Natural Numbers

There is a Peano system.

We fix one Peano system for the rest of the volume and write it $(\mathbb{N}, 0, \mathrm{S})$. Its elements are the **natural numbers**, 0 is **zero**, and $\mathrm{S}(n)$ is the **successor** of n. Spelled out, $\mathbb{N}$ is a set, $0 \in \mathbb{N}$, $\mathrm{S}\colon \mathbb{N} \to \mathbb{N}$ is a function, and:

- (P3) $\mathrm{S}(n) \neq 0$ for every $n \in \mathbb{N}$;
- (P4) $\mathrm{S}(m) = \mathrm{S}(n)$ implies $m = n$;
- (P5) every $A \subseteq \mathbb{N}$ that contains 0 and is closed under S is $\mathbb{N}$.

Why believe the assumption? If the school real numbers are trusted, the $\mathbb{N}$ of *Sets and Functions*, with start 0 and step $x \mapsto x+1$, is a Peano system; Section 1.4 checks this. And *Axiomatic Set Theory*, Chapter 1 builds a Peano system from sets alone, assuming nothing about numbers; Section 1.5 describes how. Which Peano system we fix does not matter: Section 1.4 proves that any two are copies of each other, matched by exactly one bijection that sends start to start and step to step.

From here on, every fact about $\mathbb{N}$ is proved from Axiom 1.1.1. We do not borrow the order, the subtraction or the arithmetic of $\mathbb{R}$. Addition and multiplication are not defined yet; Section 1.2 defines them. Even the numeral 1 is not available until then, and the element after 0 is written $\mathrm{S}(0)$.

Why these three clauses and no others? Each one does work that the other two cannot do: for each clause there is a structure satisfying the other two and failing it.

Example. Each axiom is needed

Show that each of the following structures (M, e, s), with $s\colon M \to M$, satisfies two of (P3), (P4), (P5) and fails the third.

(i) $M := \{0\}$, $e := 0$, $s(0) := 0$.

(ii) $M := \{0, \mathrm{S}(0)\}$, $e := 0$, $s(0) := \mathrm{S}(0)$, $s(\mathrm{S}(0)) := \mathrm{S}(0)$.

(iii) (**Two copies of $\mathbb{N}$.**) Let $a \neq b$ be two objects, $M := \mathbb{N} \times \{a, b\}$, $e := (0, a)$, and $s(n, t) := (\mathrm{S}(n), t)$ for $n \in \mathbb{N}$ and $t \in \{a, b\}$.

Solution.

(i) **Fails (P3)**: $s(0) = 0 = e$. (P4) holds: M has one element, so two inputs with equal outputs are equal. (P5) holds: a subset of $\{0\}$ containing $e = 0$ is $\{0\}$.

(ii) Here $0 \neq \mathrm{S}(0)$ by (P3) for $\mathbb{N}$, so M has two elements. **Fails (P4)**: $s(0) = s(\mathrm{S}(0))$ with $0 \neq \mathrm{S}(0)$. (P3) holds: both values are $\mathrm{S}(0) \neq 0 = e$. (P5) holds: if $A \subseteq M$ has 0 as an element and is closed under s, then $s(0) = \mathrm{S}(0) \in A$, so $A = M$.

(iii) First, s is a function $M \to M$. Each element of M is (n, t) for some $n \in \mathbb{N}$ and $t \in \{a, b\}$ (Sets and Functions, Definition 1.4.2), and for exactly one such pair (Sets and Functions, Definition 1.4.1), so the formula names one value; and $(\mathrm{S}(n), t) \in M$, since S maps $\mathbb{N}$ into $\mathbb{N}$.

(P3) holds. If $s(n, t) = (0, a)$, then $(\mathrm{S}(n), t) = (0, a)$ gives $\mathrm{S}(n) = 0$ (Sets and Functions, Definition 1.4.1), against (P3) for $\mathbb{N}$.

(P4) holds. If $s(n, t) = s(n', t')$, then $(\mathrm{S}(n), t) = (\mathrm{S}(n'), t')$ gives $t = t'$ and $\mathrm{S}(n) = \mathrm{S}(n')$, so $n = n'$ by (P4) for $\mathbb{N}$. Hence $(n, t) = (n', t')$.

(P5) fails. Let $A := \mathbb{N} \times \{a\}$. It contains $e = (0, a)$, and it is closed under s, since $s(n, a) = (\mathrm{S}(n), a) \in A$. But $(0, b) \in M$ is not in A, because its second coordinate is $b \neq a$. So $A \neq M$.

The picture shows (iii), with an arrow from each element to its image under s.

$$(0,a) \xrightarrow{e} (\mathrm{S}(0),a) \longrightarrow (\mathrm{S}(\mathrm{S}(0)),a) \longrightarrow \cdots$$

$$(0,b) \longrightarrow (\mathrm{S}(0),b) \longrightarrow (\mathrm{S}(\mathrm{S}(0)),b) \longrightarrow \cdots$$

This is the extra row of the start of the section. The element $(0,b)$ has no arrow coming in, and it is not e. The top row is a copy of $\mathbb{N}$, and so is the bottom row; but the bottom row cannot be reached from e. (P3) and (P4) cannot prevent this, and (P5) is the clause that says "nothing else".

(P5) speaks of **every** subset of M, and that is what lets it exclude every kind of extra row. A first-order induction schema speaks only of subsets defined by formulas, and *Logic and Proof*, Section 7.6 showed (Logic and Proof, Example (A Model of Arithmetic with an Infinite Number)) that the sentences true in $\mathbb{N}$ also hold in a structure with an element beyond every numeral; its closing remark promised the comparison made here. Exercise C2 builds an extra row in which, unlike $(0,b)$ above, every element has an arrow coming in.

1.1.3 Induction in the form we use

(P5) is induction, stated about subsets of a Peano system. For $\mathbb{N}$ we record it in the two forms the rest of the volume cites: a set form, and a form about properties.

Theorem 1.1.2. Induction on $\mathbb{N}$

(a) (Set form.) Let $A \subseteq \mathbb{N}$. If $0 \in A$, and $\mathrm{S}(n) \in A$ for every $n \in A$, then $A = \mathbb{N}$.

(b) (Property form.) Let $P(n)$ be a property of natural numbers, possibly with other variables as parameters. If $P(0)$ holds and, **for every** $n \in \mathbb{N}$, $P(n)$ implies $P(\mathrm{S}(n))$, then $P(n)$ holds **for every** $n \in \mathbb{N}$.

Proof. *(a)* This is (P5) of Definition 1.1.1 for the Peano system $(\mathbb{N}, 0, \mathrm{S})$ of Axiom 1.1.1.

(b) Let $A := \{n \in \mathbb{N} : P(n)\}$ (Sets and Functions, Definition 1.2.3). Then $0 \in A$, because $P(0)$ holds. Let $n \in A$. Then $P(n)$, so $P(\mathrm{S}(n))$ by hypothesis, and $\mathrm{S}(n) \in \mathbb{N}$ because S maps $\mathbb{N}$ into $\mathbb{N}$; hence $\mathrm{S}(n) \in A$. By (a), $A = \mathbb{N}$, which says that $P(n)$ holds for every $n \in \mathbb{N}$. ■

Compare part (b) with the principle of induction, Logic and Proof, Axiom 4.2.1. Read with $\mathrm{S}(n)$ in place of $n+1$, the two statements are the same, and Section 1.2 shows that $n+1 = \mathrm{S}(n)$ once addition is defined. The difference lies in their status. In *Logic and Proof*, Chapter 4, induction was taken on trust, for an $\mathbb{N}$ that came with the school arithmetic attached. Here (P5) is part of the one assumption, and arithmetic is not assumed at all: Section 1.2 builds it, and proves its laws by this theorem.

Warning. **Both halves of the hypothesis are needed.** The set $\mathbb{N} \setminus \{0\}$ is closed under S, by (P3), and it is not $\mathbb{N}$. The set $\{0\}$ contains 0 and is not closed under S, since $\mathrm{S}(0) \neq 0$.

Quick check. Let $P(n)$ be "$n = 0$". Does the base case hold? Does the step hold? What does Theorem 1.1.2 (b) conclude?

Solution. The base case holds: $P(0)$ says $0 = 0$. The step fails at $n = 0$: $P(0)$ holds, while $P(\mathrm{S}(0))$ says $\mathrm{S}(0) = 0$, which is false by (P3). So the theorem concludes nothing, and indeed P fails at $\mathrm{S}(0)$.

1.1.4 Every natural number except 0 has one predecessor

Sets and Functions used that $n - 1 \in \mathbb{N}$ for every natural number $n \neq 0$ (Sets and Functions, Lemma 3.1.1 (c)), with the subtraction done in $\mathbb{R}$. Our $\mathbb{N}$ has no subtraction. Yet Sections 1.2–1.4 split, again and again, into the cases "$n = 0$" and "$n = \mathrm{S}(k)$ for some k". For that we need every $n \neq 0$ to be a successor, and of exactly one k. The fact holds in every Peano system, and the proof is no harder there; Exercises B3 and C2 use it in that generality.

Proposition 1.1.3. Predecessors in a Peano System

Let (M, e, s) be a Peano system.

(a) For every $x \in M$ with $x \neq e$ there is **exactly one** $y \in M$ with $s(y) = x$.

(b) $s(x) \neq x$ for every $x \in M$.

Idea (Scratch work). For (a), "at most one" is (P4): if $s(y) = x = s(y')$, then $y = y'$. "At least one" is the real question. Of the three clauses, only (P5) can prove that every element of M has a given property, and (P5) speaks of subsets. So we must choose a subset A of M and show $A = M$. That choice is the new point. Inside an abstract M nothing can be computed: there is no subtraction that would produce y from x.

The first candidate is the set of values of s, $\{z \in M : z = s(y) \text{ for some } y \in M\}$. It fails at the start, since e is not a value, by (P3). The cure is the shape of Sets and Functions, Lemma 3.1.1 (c): carry "$z = e$ or ...".

Givens	Goal
(M, e, s) a Peano system; $x \in M$, $x \neq e$	some $y \in M$ with $s(y) = x$
$A := \{z \in M : z = e \text{ or } z = s(y) \text{ for some } y \in M\}$	$A = M$; then $x \in A$ and $x \neq e$ give y
(P5) for A	$e \in A$; and $z \in A \Rightarrow s(z) \in A$, since $s(z)$ is a value of s

The step never uses $z \in A$, as in Sets and Functions, Example (An induction that ignores its hypothesis).

Idea. Both parts apply (P5) to a subset chosen for the purpose: (a) to e together with the values of s, and (b) to the set where $s(x) \neq x$. In (b), unlike (a), the step uses its hypothesis, through (P4).

Proof. (a) Let $x \in M$ with $x \neq e$. *At most one:* if $s(y) = x$ and $s(y') = x$, then $y = y'$ by (P4). *At least one:* let

$$A := \{z \in M : z = e \text{ or } z = s(y) \text{ for some } y \in M\}$$

(Sets and Functions, Definition 1.2.3). Then $e \in A$, by the first alternative. Let $z \in A$. Then $s(z) \in M$ is the value of s at z, so $s(z) \in A$, by the second alternative with $y := z$. By (P5), $A = M$. Hence $x \in A$; since $x \neq e$, the second alternative holds, and $x = s(y)$ for some $y \in M$.

(b) Let $A' := \{x \in M : s(x) \neq x\}$. Then $e \in A'$, since $s(e) \neq e$ by (P3). Let $x \in A'$, so that $s(x) \neq x$. If $s(s(x)) = s(x)$, then (P4), applied to the inputs $s(x)$ and x, gives $s(x) = x$, a contradiction. So $s(s(x)) \neq s(x)$, that is, $s(x) \in A'$. By (P5), $A' = M$. This proves the proposition.

■

For $\mathbb{N}$, part (a) says that every natural number $n \neq 0$ is $\mathrm{S}(k)$ for exactly one $k \in \mathbb{N}$; this k is the **predecessor** of n. Part (b) says that no natural number is its own successor.

The two proofs make the same move with one difference. In (a) the closure step proves $s(z) \in A$ outright, whatever z is; in (b) it needs $x \in A'$, and (P4) turns the hypothesis $s(x) \neq x$ into the conclusion $s(s(x)) \neq s(x)$. The move has a name.

Example. Method: induction through a closed subset

Template. (1) Name $A := \{x \in M : P(x)\}$. (2) Show $e \in A$. (3) Take $x \in A$ and show $s(x) \in A$, using $P(x)$ if needed. (4) Conclude $A = M$ by (P5). If P fails at e, carry "$x = e$ or $P(x)$" (the shape of Sets and Functions, Lemma 3.1.1 (c)).

When to use it. A statement about every element of a Peano system whose elements are not known one by one: an abstract (M, e, s), the image of a map out of $\mathbb{N}$, a set of pairs.

The usual failure. A property false at e. Closure checked only for elements of a special form ($x = s(y)$), forgetting e. (P5) applied to a set that is not a subset of M.

On $\mathbb{N}$ the method is Theorem 1.1.2 (b), and Sets and Functions, Example (Method: finding $P(n)$ inside $P(n+1)$) still finds $P(n)$ inside $P(\mathrm{S}(n))$ in the step; the box adds the choice of the set.

Quick check. In structure (ii) of Example (Each axiom is needed), which element has two predecessors, and which conclusions of Proposition 1.1.3 fail?

Solution. The element S(0): $s(0) = \mathrm{S}(0) = s(\mathrm{S}(0))$. So "exactly one" in (a) fails at $x = \mathrm{S}(0)$, because (P4) fails; and (b) fails at $x = \mathrm{S}(0)$, since $s(\mathrm{S}(0)) = \mathrm{S}(0)$.

Quick check. Does the loop ($e = p$, $s(p) = q$, $s(q) = r$, $s(r) = p$) satisfy the conclusion of Proposition 1.1.3 (a)? What does the answer say about a converse?

Solution. Yes. Every element, e included, has exactly one element stepping to it: r to p, p to q, q to r. Yet the loop is not a Peano system, since it fails (P3). So the conclusion of (a) does not give back the axioms; Exercise C2 shows that it does not even together with (P3) and (P4).

The next section uses each of (P3), (P4) and (P5) to prove that a function on $\mathbb{N}$ can be defined by recursion, and then defines addition and multiplication.

1.1.5 Exercises

1.1.5.1 A. Check your understanding

Exercise. A1

(a) For each of the three structures in Example (Each axiom is needed), which of (P3), (P4), (P5) fails?

(b) In the proof of Proposition 1.1.3 (a), which axiom gives at least one y, and which gives at most one?

(c) Is $\mathbb{N} \setminus \{0\}$ closed under S? Is it $\mathbb{N}$? Which hypothesis of Theorem 1.1.2 (a) fails for it?

(d) True or false: in a Peano system (M, e, s), the function s is a bijection $M \to M$.

Solution.

(a) (i) fails (P3); (ii) fails (P4); (iii) fails (P5). Each satisfies the other two (Example (Each axiom is needed)).

(b) At least one comes from (P5), applied to the set A of elements that are e or a value of s. At most one is (P4).

(c) Yes: for $n \in \mathbb{N} \setminus \{0\}$, $\mathrm{S}(n) \in \mathbb{N}$ and $\mathrm{S}(n) \neq 0$ by (P3). No: $0 \in \mathbb{N}$ is not in it. The hypothesis that fails is $0 \in A$.

(d) False. The function s is injective, by (P4), but not surjective: e is not a value of s, by (P3) (Sets and Functions, Definition 2.2.2).

1.1.5.2 B. Practice

Exercise. B1: Put the proof in order

Let (M, e, s) be a Peano system. The following lines contain a proof that $s(s(x)) \neq x$ **for**

every $x \in M$, shuffled, together with two lines that do not belong. Write the six correct lines in an order that makes a proof, say which other orders are also admissible, and say why each of the other two lines fails.

(i) Let $x \in A$; we show $s(x) \in A$.
(ii) Let $A := \{x \in M : s(s(x)) \neq x\}$.
(iii) Then $s(s(x)) = x$ by (P4), contradicting $x \in A$.
(iv) $e \in A$, since $s(s(e))$ is a value of s, so it is not e, by (P3).
(v) Suppose $s(s(s(x))) = s(x)$.
(vi) So $s(s(s(x))) \neq s(x)$, that is, $s(x) \in A$; by (P5), $A = M$.
(vii) Since s is a bijection, write $x = s(y)$ and apply the hypothesis to y.
(viii) Since $s(s(e)) \neq e$, and every element is reached from e by steps, $s(s(x)) \neq x$ for every x.

Solution. An admissible order is (ii), (iv), (i), (v), (iii), (vi):

> Let $A := \{x \in M : s(s(x)) \neq x\}$. Then $e \in A$, since $s(s(e))$ is a value of s, so it is not e, by (P3). Let $x \in A$; we show $s(x) \in A$. Suppose $s(s(s(x))) = s(x)$. Then $s(s(x)) = x$ by (P4), contradicting $x \in A$. So $s(s(s(x))) \neq s(x)$, that is, $s(x) \in A$; by (P5), $A = M$.

Line (ii) comes first, since every other line mentions A. Lines (i), (v), (iii), (vi) come in that order: (i) opens the closure step, (v) opens a supposition, (iii) applies (P4) to the inputs $s(s(x))$ and x and reaches a contradiction, and (vi) closes the supposition and applies (P5), which needs both $e \in A$ and the closure step. So (iv) may stand right after (ii) or right after (i), before the supposition (v) opens; it must come before (vi), which uses it. Stating the base case first, as above, is the convention.

Line (vii) cites a false fact: s is not surjective, since e is not a value, by (P3). Citing an unavailable "fact" is (N9) of Logic and Proof, Example (What is not allowed). Line (viii) is an "and so on" with nothing behind it: "every element is reached from e by steps" is (P5) only if the step from x to $s(x)$ is proved, and line (viii) never proves it. That is (N13).

Exercise. B2: Two steps at a time

Let $d := \mathrm{S} \circ \mathrm{S} \colon \mathbb{N} \to \mathbb{N}$ (Sets and Functions, Definition 2.1.2), so that $d(n) = \mathrm{S}(\mathrm{S}(n))$. Show that $(\mathbb{N}, 0, d)$ satisfies (P3) and (P4) of Definition 1.1.1, and that it does not satisfy (P5).

Hint: look at the set $\mathbb{N} \setminus \{\mathrm{S}(0)\}$.

Solution. (P3): for $n \in \mathbb{N}$, $d(n) = \mathrm{S}(\mathrm{S}(n)) \neq 0$ by (P3) for $\mathbb{N}$, applied to $\mathrm{S}(n) \in \mathbb{N}$.

(P4): if $d(m) = d(n)$, then $\mathrm{S}(\mathrm{S}(m)) = \mathrm{S}(\mathrm{S}(n))$, so $\mathrm{S}(m) = \mathrm{S}(n)$ by (P4) for $\mathbb{N}$, and then $m = n$ by (P4) again. (Equivalently: d is a composite of two injections, so it is injective by Sets and Functions, Theorem 2.2.1 (a).)

(P5) fails. Let $A := \mathbb{N} \setminus \{\mathrm{S}(0)\}$. It contains 0, since $0 \neq \mathrm{S}(0)$ by (P3). It is closed under d: let $n \in A$; then $d(n) \in \mathbb{N}$, and $d(n) \neq \mathrm{S}(0)$, because $\mathrm{S}(\mathrm{S}(n)) = \mathrm{S}(0)$ would give $\mathrm{S}(n) = 0$ by (P4), against (P3). But $A \neq \mathbb{N}$, since $\mathrm{S}(0) \in \mathbb{N}$ and $\mathrm{S}(0) \notin A$. So a subset of $\mathbb{N}$ with 0 as an

element and closed under d need not be all of $\mathbb{N}$. The step d skips $\mathrm{S}(0)$, and starting from 0 it meets only 0, $\mathrm{S}(\mathrm{S}(0))$, $\mathrm{S}(\mathrm{S}(\mathrm{S}(\mathrm{S}(0))))$, ...: every second number.

Exercise. B3: Prove or disprove

Let (M, e, s) be a Peano system. Prove or disprove each statement.

(a) If $A \subseteq M$ contains $s(e)$ and is closed under s, then $A \cup \{e\} = M$.
(b) If $A \subseteq M$ is non-empty and closed under s, then $A = M$.
(c) Every $x \neq e$ in M is $s(y)$ for some $y \in M$ with $y \neq x$.
(d) If $f\colon M \to M$ satisfies $f(e) = e$ and $f(s(x)) = s(f(x))$ for every $x \in M$, then $f = \mathrm{id}_M$.
(e) Every $x \in M$ is e or $s(s(y))$ for some $y \in M$.

Solution.

(a) **True.** The set $A \cup \{e\}$ is a subset of M containing e. It is closed under s: for $x = e$, $s(e) \in A$ by hypothesis; for $x \in A$, $s(x) \in A$ by closure. By (P5), $A \cup \{e\} = M$.
(b) **False.** Let $A := M \setminus \{e\}$. It contains $s(e)$, since $s(e) \neq e$ by (P3), so it is non-empty. It is closed under s, since no value of s is e, by (P3). It misses e, so $A \neq M$.
(c) **True.** Let $x \neq e$. By Proposition 1.1.3 (a) there is $y \in M$ with $s(y) = x$. If $y = x$, then $s(x) = x$, against Proposition 1.1.3 (b). So $y \neq x$.
(d) **True**, by Example (Method: induction through a closed subset). Let $A := \{x \in M : f(x) = x\}$. Then $e \in A$, since $f(e) = e$. If $x \in A$, then $f(s(x)) = s(f(x)) = s(x)$, so $s(x) \in A$. By (P5), $A = M$. So f and id_M have the same domain, the same codomain and the same value at every $x \in M$, and $f = \mathrm{id}_M$ by Sets and Functions, Proposition 2.1.1.
(e) **False.** Take $x := s(e)$. It is not e, by (P3). If $s(e) = s(s(y))$ for some $y \in M$, then $e = s(y)$ by (P4), against (P3). So $s(e)$ is neither e nor of the form $s(s(y))$.

1.1.5.3 C. Going deeper

Exercise. C1: A Peano system with a different start

Give an example of a Peano system (M, e, s) with $M \subseteq \mathbb{N}$ and $e \neq 0$. Check every clause of Definition 1.1.1.

Solution. Any correct answer has: a set $M \subseteq \mathbb{N}$; an element $e \in M$ with $e \neq 0$; a function $s\colon M \to M$, so that every value lies in M; and the three clauses (P3), (P4), (P5) for these data.

Take $M := \mathbb{N} \setminus \{0\}$, $e := \mathrm{S}(0)$, and $s(n) := \mathrm{S}(n)$ for $n \in M$.

- **The data.** $\mathrm{S}(0) \in M$, since $\mathrm{S}(0) \in \mathbb{N}$ and $\mathrm{S}(0) \neq 0$ by (P3) for $\mathbb{N}$. For $n \in M$, $\mathrm{S}(n) \in \mathbb{N}$ since S maps $\mathbb{N}$ into $\mathbb{N}$, and $\mathrm{S}(n) \neq 0$ by (P3) for $\mathbb{N}$, so $\mathrm{S}(n) \in M$. So $s\colon M \to M$ is a function, since S is one and its values on M lie in M (Sets and Functions, Definition 2.1.1).
- **(P3).** If $s(n) = \mathrm{S}(0)$ for some $n \in M$, then $\mathrm{S}(n) = \mathrm{S}(0)$, so $n = 0$ by (P4) for $\mathbb{N}$; but

$0 \notin M$.

- **(P4).** If $\mathrm{S}(m) = \mathrm{S}(n)$ with $m, n \in M$, then $m = n$ by (P4) for $\mathbb{N}$.
- **(P5).** Let $A \subseteq M$ with $\mathrm{S}(0) \in A$ and $\mathrm{S}(n) \in A$ for every $n \in A$. The set $A' \coloneqq A \cup \{0\} \subseteq \mathbb{N}$ contains 0 and is closed under S: $\mathrm{S}(0) \in A \subseteq A'$; and for $n \in A$, $\mathrm{S}(n) \in A \subseteq A'$. By Theorem 1.1.2 (a), $A' = \mathbb{N}$. Every $n \in M$ lies in A' and is not 0, so $n \in A$. Hence $A = M$.

Other answers can be checked against the four properties at the start. For instance, $M \coloneqq \mathbb{N} \setminus \{0, \mathrm{S}(0)\}$, $e \coloneqq \mathrm{S}(\mathrm{S}(0))$, $s(n) \coloneqq \mathrm{S}(n)$ also works, by the same argument with $\{0, \mathrm{S}(0)\}$ added in (P5).

Exercise. C2: Every element but one is a successor, used later

Let a, b, c be three different objects, $M \coloneqq \mathbb{N} \times \{a, b, c\}$, $e \coloneqq (0, a)$, and, for $n, k \in \mathbb{N}$,

$$s(n, a) \coloneqq (\mathrm{S}(n), a), \qquad s(n, b) \coloneqq (\mathrm{S}(n), b),$$
$$s(0, c) \coloneqq (0, b), \qquad s(\mathrm{S}(k), c) \coloneqq (k, c).$$

(a) Explain why s is a function $M \to M$.

(b) Show that (M, e, s) satisfies (P3) and (P4).

(c) Show that every $x \in M$ with $x \neq e$ is a value of s.

(d) Show that (M, e, s) fails (P5).

(e) Which conclusions of Proposition 1.1.3 hold in (M, e, s)?

Solution (Hint 1). For (a), apply Proposition 1.1.3 (a) to $\mathbb{N}$.

Solution (Hint 2). For (d), try the set $\mathbb{N} \times \{a\}$.

Solution.

(a) Each element of M is (n, t) for some $n \in \mathbb{N}$ and $t \in \{a, b, c\}$ (Sets and Functions, Definition 1.4.2), and for exactly one such pair (Sets and Functions, Definition 1.4.1). For $t = a$ and $t = b$, one formula applies and names one value. For $t = c$, the first coordinate is 0 or $\mathrm{S}(k)$ for some $k \in \mathbb{N}$, by Proposition 1.1.3 (a) for $\mathbb{N}$; never both, by (P3) for $\mathbb{N}$; and the k is unique. So exactly one line of the definition applies to each element, and it names one value. The values lie in M, since $\mathrm{S}(n), 0, k \in \mathbb{N}$.

(b) Sort the values by their second coordinates (Sets and Functions, Definition 1.4.1):

$$\begin{aligned} &\text{second coordinate } a: && s(n, a) = (\mathrm{S}(n), a), \\ &\text{second coordinate } b: && s(n, b) = (\mathrm{S}(n), b) \quad \text{and} \quad s(0, c) = (0, b), \\ &\text{second coordinate } c: && s(\mathrm{S}(k), c) = (k, c). \end{aligned}$$

(P3) holds. A value equal to $e = (0, a)$ would have second coordinate a, so it would be $(\mathrm{S}(n), a)$ with $\mathrm{S}(n) = 0$, against (P3) for $\mathbb{N}$.
(P4) holds. Suppose $s(x) = s(x')$. The two values have the same second coordinate,

so they come from the same row of the display. In the first row, $(\mathrm{S}(n), a) = (\mathrm{S}(n'), a)$ gives $n = n'$ by (P4) for $\mathbb{N}$. In the third row, $(k, c) = (k', c)$ gives $k = k'$, so $\mathrm{S}(k) = \mathrm{S}(k')$. In the second row, $(\mathrm{S}(n), b) \neq (0, b)$ because $\mathrm{S}(n) \neq 0$; so either both values are of the form $(\mathrm{S}(n), b)$, and $n = n'$ as before, or both are $(0, b)$, and both inputs are $(0, c)$. In every case $x = x'$.

(c) Let $x = (m, t) \neq e$. If $t = a$, then $m \neq 0$, so $m = \mathrm{S}(n)$ for some n (Proposition 1.1.3 (a) for $\mathbb{N}$), and $x = s(n, a)$. If $t = b$, then $x = (0, b) = s(0, c)$, or $m = \mathrm{S}(n)$ and $x = s(n, b)$. If $t = c$, then $x = (m, c) = s(\mathrm{S}(m), c)$.

(d) The set $\mathbb{N} \times \{a\}$ contains e, is closed under s, since $s(n, a) = (\mathrm{S}(n), a)$, and misses $(0, b)$, since $b \neq a$. So it is a subset of M containing e, closed under s, and not equal to M.

(e) Both. Part (a) of Proposition 1.1.3 holds: a predecessor exists by (c), and it is unique by (P4), proved in (b). Part (b) holds: $(\mathrm{S}(n), a) \neq (n, a)$ and $(\mathrm{S}(n), b) \neq (n, b)$, since $\mathrm{S}(n) \neq n$ in $\mathbb{N}$ (Proposition 1.1.3 (b)); $s(0, c) = (0, b) \neq (0, c)$, since $b \neq c$; and $s(\mathrm{S}(k), c) = (k, c) \neq (\mathrm{S}(k), c)$, since $k \neq \mathrm{S}(k)$. So neither conclusion of the proposition gives back (P5), even together with (P3) and (P4).

The picture to keep in mind: the elements with second coordinate a form a copy of $\mathbb{N}$, starting at e. The elements with second coordinates b and c form one chain running forever in both directions. Read (n, b) as "n" and (k, c) as "$-(k+1)$": the step moves one place to the right everywhere, with $(0, c)$, playing -1, stepping to $(0, b)$, playing 0. Every element of the chain has an arrow coming in, and none of it is reached from e.

1.2 The Recursion Theorem; Addition and Multiplication

The natural numbers now have a zero, a successor map and induction (Theorem 1.1.2), and nothing else: no addition, no multiplication. *Sets and Functions*, Section 3.3 had the same need and met it with Sets and Functions, Theorem 3.3.1, a theorem that turns two equations into a function. Its proof built partial solutions on the sets $\{0, \dots, n\}$ and compared them using the order of the real numbers. The natural numbers of Axiom 1.1.1 have no order yet; the order is defined in the next section, from the addition of this one. So this section proves the recursion theorem by a construction that needs no order at all: it takes the smallest set of pairs that the two equations force, and shows that this set is the graph of a function. Then it defines addition and multiplication as school does, and proves the laws of arithmetic by one move and its consequences.

1.2.1 Recursion on $\mathbb{N}$

A starting value and a rule for the next value determine exactly one function on $\mathbb{N}$.

Theorem 1.2.1. Recursion Theorem on $\mathbb{N}$

Let X be a set, let $a \in X$, and let $g\colon \mathbb{N} \times X \to X$ be a function. Then there is **exactly one** function $f\colon \mathbb{N} \to X$ such that

$$f(0) = a \quad \text{and} \quad f(\mathrm{S}(n)) = g(n, f(n)) \quad \text{for every } n \in \mathbb{N}.$$

This is the statement of Sets and Functions, Theorem 3.3.1, with $\mathbb{N}$ the Peano system of Ax-

iom 1.1.1 and $\mathrm{S}(n)$ in place of $n+1$. The set $\mathbb{N} \times X$ is the Cartesian product of Sets and Functions, Definition 1.4.2. As before, a is the starting value and g turns the stage n and the current value into the next value; the conclusion has an existence half and a uniqueness half.

Idea (Scratch work). Uniqueness will be an induction: two solutions agree at 0, and the second equation passes agreement from n to $\mathrm{S}(n)$. Existence is the hard half. "Define f by the two equations" names f on both sides of the second one, so it is not yet a definition. A definition must produce an object, and the object a function is made of is its graph (Sets and Functions, Definition 2.1.1). So ask what the graph must contain. It must contain the pair $(0, a)$. Whenever it contains a pair (n, x), meaning $f(n) = x$, it must also contain $(\mathrm{S}(n), g(n, x))$.

That is a description of a closed set: in $\mathbb{N} \times X$, take the starting element $(0, a)$ and the step $(n, x) \mapsto (\mathrm{S}(n), g(n, x))$, which maps $\mathbb{N} \times X$ into itself because g takes its values in X. By Sets and Functions, Proposition 1.3.2 there is a smallest closed subset; call it R. It contains exactly what the equations force and nothing more. The question is whether R is a graph: does each n have exactly one partner x with $(n, x) \in R$?

Givens	Goal
R is the smallest closed subset of $\mathbb{N} \times X$	each n has exactly one x with $(n, x) \in R$
induction on n	base: the only pair at 0 is $(0, a)$; step: the only pair at $\mathrm{S}(n)$ is $(\mathrm{S}(n), g(n, x_n))$, where x_n is the partner of n
an extra pair $(n, y) \in R$	$R \setminus \{(n, y)\}$ is still closed, against "smallest"

The last row is the gap the proof closes. A smallest closed set has no spare elements: if one pair could be removed without breaking closure, the smaller set would be closed, and R would lie inside it. Removing a pair is safe only if no step produces it from a pair that stays. That is where (P3) and (P4) enter.

Before reading on: where is each of them used?

Proof. **Step 1 (uniqueness).** Let $f, f' \colon \mathbb{N} \to X$ both satisfy the two equations, and let $P(n)$ be "$f(n) = f'(n)$". $P(0)$ holds, since $f(0) = a = f'(0)$. If $P(n)$ holds, then

$$f(\mathrm{S}(n)) = g(n, f(n)) = g(n, f'(n)) = f'(\mathrm{S}(n)),$$

so $P(\mathrm{S}(n))$ holds. By Theorem 1.1.2 (b), $P(n)$ holds for every $n \in \mathbb{N}$. The two functions have the same domain and codomain, so $f = f'$ by Sets and Functions, Proposition 2.1.1.

Step 2 (the set of pairs). Call a subset $C \subseteq \mathbb{N} \times X$ *closed* if $(0, a) \in C$ and $(\mathrm{S}(n), g(n, x)) \in C$ for every $(n, x) \in C$. The step sends $\mathbb{N} \times X$ into itself, since $\mathrm{S}(n) \in \mathbb{N}$ and $g(n, x) \in X$. By Sets and Functions, Proposition 1.3.2 there is a closed set $R \subseteq \mathbb{N} \times X$ that lies inside every closed set.

Step 3.

Claim. For every $n \in \mathbb{N}$ there is exactly one $x \in X$ with $(n, x) \in R$.

Proof of Claim. We use Theorem 1.1.2 (b) on n.

Base case. $(0, a) \in R$, since R is closed. Suppose $(0, b) \in R$ with $b \neq a$, and let $R' \coloneqq R \setminus \{(0, b)\}$. Then $(0, a) \in R'$, because $(0, a) \neq (0, b)$ (Sets and Functions, Definition 1.4.1). Let $(m, x) \in R'$. Then $(m, x) \in R$, so $(\mathrm{S}(m), g(m, x)) \in R$; and this pair is not $(0, b)$, because $\mathrm{S}(m) \neq 0$ by (P3). So it lies in R', and R' is closed. Hence $R \subseteq R'$, which is false, since $(0, b) \in R$ and $(0, b) \notin R'$. So a is the only partner of 0.

Inductive step. Let $n \in \mathbb{N}$, and suppose n has exactly one partner x_n in R. Since $(n, x_n) \in R$ and R is closed, $(\mathrm{S}(n), g(n, x_n)) \in R$. Suppose $(\mathrm{S}(n), y) \in R$ with $y \neq g(n, x_n)$, and let $R' \coloneqq R \setminus \{(\mathrm{S}(n), y)\}$. Then $(0, a) \in R'$, because $\mathrm{S}(n) \neq 0$ by (P3). Let $(m, x) \in R'$; then $(\mathrm{S}(m), g(m, x)) \in R$. If this pair were $(\mathrm{S}(n), y)$, then $\mathrm{S}(m) = \mathrm{S}(n)$, so $m = n$ by (P4); then $(n, x) \in R$, so $x = x_n$ by the induction hypothesis; and then $y = g(m, x) = g(n, x_n)$, against the choice of y. So $(\mathrm{S}(m), g(m, x)) \in R'$, and R' is closed. As in the base case, $R \subseteq R'$ is false. So $g(n, x_n)$ is the only partner of $\mathrm{S}(n)$, which proves the claim.

■

Step 4 (the function). R is a subset of $\mathbb{N} \times X$ in which each $n \in \mathbb{N}$ has exactly one partner, so $f \coloneqq (R, X)$ is a function $\mathbb{N} \to X$ by Sets and Functions, Definition 2.1.1, and $f(n)$ is the partner of n. Since $(0, a) \in R$, $f(0) = a$. For $n \in \mathbb{N}$, $(n, f(n)) \in R$, so closure gives $(\mathrm{S}(n), g(n, f(n))) \in R$, that is, $f(\mathrm{S}(n)) = g(n, f(n))$. This proves existence, and with Step 1, the theorem.

■

Each Peano axiom has a visible job in the proof. (P3) appears in the base case of Step 3, where no step can produce a pair with first coordinate 0, and again in its inductive step, where it keeps $(0, a)$ in R'. (P4) appears in the inductive step: a pair that steps to first coordinate $\mathrm{S}(n)$ has first coordinate n, where the induction hypothesis applies. (P5) enters through Theorem 1.1.2, in Steps 1 and 3. No choice is made: R is the one smallest closed set, and each partner is the only one. From now on a definition "$f(0) \coloneqq a$, $f(\mathrm{S}(n)) \coloneqq g(n, f(n))$" on $\mathbb{N}$ is legitimate whenever $a \in X$ and $g\colon \mathbb{N} \times X \to X$ is a function, and f is **defined by recursion**.

Quick check. For $X = \mathbb{N}$, $a = 0$ and $g(n, x) = \mathrm{S}(\mathrm{S}(x))$, which pairs of R have first coordinate 0, 1 or 2? (Here 1, 2, 3, 4 stand for $\mathrm{S}(0)$, $\mathrm{S}(1)$, $\mathrm{S}(2)$, $\mathrm{S}(3)$.)

Solution. $(0, 0)$, since R contains the starting pair; then $(1, 2)$, since $g(0, 0) = \mathrm{S}(\mathrm{S}(0)) = 2$; then $(2, 4)$, since $g(1, 2) = \mathrm{S}(\mathrm{S}(2)) = 4$. There are no others with these first coordinates, by the claim of Step 3.

Quick check. In the inductive step of the claim, why does $R' \coloneqq R \setminus \{(\mathrm{S}(n), y)\}$ still contain $(0, a)$? Which axiom is used?

Solution. (P3): $\mathrm{S}(n) \neq 0$, so the removed pair has a different first coordinate from $(0, a)$ and is not $(0, a)$ (Sets and Functions, Definition 1.4.1).

Warning. The hypothesis that g takes its values in X cannot be dropped. Let $X = \{0\}$, $a = 0$, and $g(n, x) := \mathrm{S}(0)$ for $n \in \mathbb{N}$, $x \in X$, so that g maps $\mathbb{N} \times X$ into $\mathbb{N}$ but not into X. Any $f\colon \mathbb{N} \to X$ with the two equations would need $f(\mathrm{S}(0)) = g(0, f(0)) = \mathrm{S}(0)$, and $\mathrm{S}(0) \notin X$ since $\mathrm{S}(0) \neq 0$ by (P3); so no such f exists. The proof breaks at Step 2: this step does not map $\mathbb{N} \times X$ into itself, so Sets and Functions, Proposition 1.3.2 does not apply.

1.2.2 Addition and multiplication

With the theorem, addition is the old school definition made official. To add n to m, start at m and take n successor steps: $m + 0$ is m, and $m + \mathrm{S}(n)$ is one step further than $m + n$. Multiplication is repeated addition in the same way.

Adding n to m is taking n successor steps from m.

Definition 1.2.1. Addition and Multiplication on $\mathbb{N}$

(a) For each $m \in \mathbb{N}$, let $A_m\colon \mathbb{N} \to \mathbb{N}$ be the function given by Theorem 1.2.1 with $X = \mathbb{N}$, $a = m$ and $g(n, x) = \mathrm{S}(x)$. The **sum** of m and n is $m + n := A_m(n)$. Thus, for all $m, n \in \mathbb{N}$,

$$m + 0 = m, \qquad m + \mathrm{S}(n) = \mathrm{S}(m + n).$$

(b) For each $m \in \mathbb{N}$, let $M_m\colon \mathbb{N} \to \mathbb{N}$ be the function given by Theorem 1.2.1 with $X = \mathbb{N}$, $a = 0$ and $g(n, x) = x + m$. The **product** of m and n is $m \cdot n := M_m(n)$, also written mn. Thus, for all $m, n \in \mathbb{N}$,

$$m \cdot 0 = 0, \qquad m \cdot \mathrm{S}(n) = m \cdot n + m.$$

(c) The **numerals** are $1 := \mathrm{S}(0)$, $2 := \mathrm{S}(1)$, $3 := \mathrm{S}(2)$, and so on. For every $n \in \mathbb{N}$, $n + 1 = \mathrm{S}(n)$; in particular $1 + 1 = 2$ and $k + 2 = (k + 1) + 1$ for every $k \in \mathbb{N}$.

In words: (a) and (b) are the two clauses we would write for a school definition, and the theorem guarantees that they define exactly one function A_m, respectively M_m, for each m. Multiplication binds more tightly than addition, so $m \cdot n + m$ is $(m \cdot n) + m$.

Well-definedness. Each g maps $\mathbb{N} \times \mathbb{N}$ into $\mathbb{N}$: $\mathrm{S}(x) \in \mathbb{N}$ because S maps $\mathbb{N}$ into $\mathbb{N}$, and $x + m = A_x(m) \in \mathbb{N}$ because A_x maps into $\mathbb{N}$. So (b) uses (a), and (a) uses nothing but the successor. The sum is then a function $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$, with graph $\{((m, n), k) \in (\mathbb{N} \times \mathbb{N}) \times \mathbb{N} : k = A_m(n)\}$ (Sets and Functions, Definition 1.2.3): each (m, n) gets exactly one k, because each A_m exists and is unique. Likewise for the product.

The successor is "plus one". Proof of the last sentence of (c): for every $n \in \mathbb{N}$,

$$n + 1 = n + \mathrm{S}(0) = \mathrm{S}(n + 0) = \mathrm{S}(n),$$

by (c) and the two equations of (a). From now on we write $n + 1$ for $\mathrm{S}(n)$ whenever $n \in \mathbb{N}$; the defining equations then read as in school:

$$m + 0 = m, \quad m + (n + 1) = (m + n) + 1,$$
$$m \cdot 0 = 0, \qquad m \cdot (n + 1) = m \cdot n + m.$$

Two small computations are used repeatedly later: $1 + 1 = \mathrm{S}(1) = 2$, and, for every $k \in \mathbb{N}$,

$$k + 2 = k + \mathrm{S}(1) = \mathrm{S}(k + 1) = (k + 1) + 1.$$

A non-example by minimal change: keep $a = m$ in (a), but replace $g(n, x) = \mathrm{S}(x)$ by $g(n, x) = \mathrm{S}(n)$, which ignores the current value. This g is still a function $\mathbb{N}\times\mathbb{N} \to \mathbb{N}$, so Theorem 1.2.1 gives exactly one $f\colon \mathbb{N} \to \mathbb{N}$ with $f(0) = m$ and $f(\mathrm{S}(n)) = \mathrm{S}(n)$. The first equation $m + 0 = m$ still holds for this "sum", but the second, $m + \mathrm{S}(n) = \mathrm{S}(m + n)$, fails: with $m = 2$, $f(1) = \mathrm{S}(0) = 1$, while $\mathrm{S}(f(0)) = \mathrm{S}(2) = 3$. So it would give $2 + 1 = 1$, and it is not addition.

The defining equations immediately give the two Peano facts in the "plus one" notation.

Corollary 1.2.2. Successor Cancellation

For all $k, l \in \mathbb{N}$: if $k + 1 = l + 1$, then $k = l$; and $k + 1 \neq 0$.

Proof. By (c) of Definition 1.2.1, $k + 1 = \mathrm{S}(k)$ and $l + 1 = \mathrm{S}(l)$. So the first claim is (P4) for $\mathbb{N}$, and the second is (P3) (Axiom 1.1.1). ∎

Quick check. Compute $2 + 2$ and $2 \cdot 2$ from the defining equations alone.

Solution. By (a) of Definition 1.2.1, three times,

$$2 + 2 = \mathrm{S}(2 + 1) = \mathrm{S}(\mathrm{S}(2 + 0)) = \mathrm{S}(\mathrm{S}(2)) = \mathrm{S}(3) = 4.$$

By (b), twice and then at 0,

$$2 \cdot 2 = 2 \cdot 1 + 2 = (2 \cdot 0 + 2) + 2 = (0 + 2) + 2,$$

and $0 + 2 = \mathrm{S}(0 + 1) = \mathrm{S}(\mathrm{S}(0 + 0)) = \mathrm{S}(\mathrm{S}(0)) = 2$. So $2 \cdot 2 = 2 + 2 = 4$.

1.2.3 The laws of arithmetic

Every law of arithmetic that *Sets and Functions*, Chapter 3 took from the real numbers must now be proved, and the only tools are the four defining equations and induction. The equations are lopsided: they say what happens when the **second** argument goes up by one, and nothing about the first. So in every induction in the proof of Theorem 1.2.3 we fix all variables but one and induct on the variable that the recursion runs on, the second argument of $+$ or $\cdot$.

Idea (Scratch work). Take the first law that school arithmetic uses without a thought: $0 + n = n$. The defining equations do not give it. They give $m + 0 = m$, with the 0 in the **second** place, and $0 + n$ has its 0 in the first. Which variable to induct on? The equations speak about 0 and successors in the second place, and in $0 + n$ that place is held by n. So induct on n: then the base case is a defining equation, and the step can open $0 + (n + 1)$ with the other one.

Givens	Goal
$m+0=m$ and $m+(n+1)=(m+n)+1$ for all m,n	$0+n=n$ for every $n\in\mathbb{N}$
$0+0=0$: the first equation with $m=0$	base closed; step: $0+(n+1)=n+1$
IH: $0+n=n$	$0+(n+1)=(0+n)+1=n+1$: closed

The second equation, with $m=0$, opens the term at $n+1$; the hypothesis applies to the inner $0+n$; nothing is left to do. Had we tried to induct with 0 and n in the other roles, there would be no equation to open the term with.

The scratch work proves the first half of (a) in the theorem below; the rest of the theorem is proved the same way.

Theorem 1.2.3. Laws of Arithmetic on $\mathbb{N}$

For all $k,m,n\in\mathbb{N}$:

(a) $0+n=n$, and $(m+1)+n=(m+n)+1$;
(b) $m+n=n+m$;
(c) $(k+m)+n=k+(m+n)$;
(d) $m(n+k)=mn+mk$ and $(n+k)m=nm+km$;
(e) $0\cdot n=0$, $(m+1)\cdot n=mn+n$, and $n\cdot 1=1\cdot n=n$;
(f) $mn=nm$;
(g) $(km)n=k(mn)$.

The move of the scratch work is worth naming, because the identities of this section are proved with it.

Example. Method: fix the others, induct on the recursion variable

Template. (1) Find the variable the definition is a recursion on: the second argument of $+$ and $\cdot$, the exponent of a power u^n (*Sets and Functions*, Section 3.3; on $\mathbb{N}$ in the next section). (2) Fix the other variables. (3) Induct on it: the base is the defining equation at 0; the step opens the term at $n+1$ by the other equation, applies the hypothesis inside, and closes up. (4) If the variable sits where no equation applies (a first argument), first prove the lemma that moves it.

When to use it. Identities between terms built by recursion: laws of $+$ and $\cdot$, of powers, of sums.

The usual failure. Inducting on a first argument (the warning after the proof); citing a law not yet proved, such as commutativity before (a).

Idea. Every clause except the second half of (d) and the last part of (e), which follow from earlier clauses directly, is proved by Example (Method: fix the others, induct on the recursion

variable), in an order where each uses only earlier ones. Commutativity (b) needs both halves of (a), which move a successor out of the **first** argument, where no defining equation reaches.

Proof. Each induction is Theorem 1.1.2 (b), applied to the property stated in the clause, with the other variables fixed. "Def" refers to the equations of Definition 1.2.1, "IH" to the induction hypothesis.

(a) The first half, $0+n=n$, is the scratch work above: the base is Def with $m=0$, and the step is $0+(n+1)=(0+n)+1=n+1$, by Def and IH. For the second half, fix $m\in\mathbb{N}$ and induct on n. *Base case:* $(m+1)+0=m+1=(m+0)+1$, by Def twice. *Inductive step:* if $(m+1)+n=(m+n)+1$, then

$$(m+1)+(n+1)=\big((m+1)+n\big)+1=\big((m+n)+1\big)+1=\big(m+(n+1)\big)+1,$$

by Def, IH, and Def read from right to left inside the outer $+1$.

(c) We prove associativity of addition in full. Fix $k,m\in\mathbb{N}$, and let $P(n)$ be "$(k+m)+n=k+(m+n)$". *Base case:* by the first equation of (a) of Definition 1.2.1, applied twice,

$$(k+m)+0=k+m=k+(m+0).$$

Inductive step: suppose $P(n)$. Then

$$\begin{aligned}(k+m)+(n+1)&=\big((k+m)+n\big)+1=\big(k+(m+n)\big)+1\\&=k+\big((m+n)+1\big)=k+\big(m+(n+1)\big),\end{aligned}$$

where the first equality is Def with first argument $k+m$, the second is $P(n)$, the third is Def with first argument k and second argument $m+n$, and the fourth is Def with first argument m. So $P(n+1)$ holds, and by Theorem 1.1.2 (b), $P(n)$ holds for every $n\in\mathbb{N}$.

(b), (d)–(g) follow by the same move, in the order of the table in the fold below.

■

↻ **Second reading** · *Proofs of (b) and (d)–(g)*

"Def" refers to the equations of Definition 1.2.1, "IH" to the induction hypothesis. The second half of (d) and the last part of (e) are derived from earlier clauses without induction; for the first half of (d) the variable is k, the recursion variable of $m(n+k)$.

The other clauses follow the same pattern; the table gives, for each, the property inducted on, the base case, the step, and the earlier clauses used. They are proved in the order of the table, so each row uses only (a), (c) and rows above it.

Clause, variable	Base case	Inductive step (IH at the marked equality)	Uses
(b) $m+n=n+m$; m fixed, on n	$m+0=m=0+m$	$m+(n+1)=(m+n)+1\overset{\text{IH}}{=}(n+m)+1=(n+1)+m$	Def; (a), both halves

Clause, variable	Base case	Inductive step (IH at the marked equality)	Uses
(d) $m(n+k) = mn+mk$; m, n fixed, on k	$m(n+0) = mn = mn+0 = mn+m\cdot 0$	$m(n+(k+1)) = m((n+k)+1) = m(n+k)+m \overset{\text{IH}}{=} (mn+mk)+m = mn+(mk+m) = mn+m(k+1)$	Def; (c)
(e) $0\cdot n = 0$; on n	$0\cdot 0 = 0$ (Def)	$0\cdot(n+1) = 0\cdot n+0 \overset{\text{IH}}{=} 0+0 = 0$	Def
(e) $(m+1)n = mn+n$; m fixed, on n	$(m+1)\cdot 0 = 0 = 0+0 = m\cdot 0+0$	$(m+1)(n+1) = (m+1)n+(m+1) \overset{\text{IH}}{=} (mn+n)+(m+1) = mn+(n+(m+1)) = mn+((n+m)+1) = mn+((m+n)+1) = mn+(m+(n+1)) = (mn+m)+(n+1) = m(n+1)+(n+1)$	Def; (b), (c)
(e) $n\cdot 1 = n$, $1\cdot n = n$; no induction		$n\cdot 1 = n\cdot 0+n = 0+n = n$; $1\cdot n = (0+1)\cdot n = 0\cdot n+n = 0+n = n$	Def; (a); first two parts of (e)
(f) $mn = nm$; m fixed, on n	$m\cdot 0 = 0 = 0\cdot m$	$m(n+1) = mn+m \overset{\text{IH}}{=} nm+m = (n+1)m$	Def; (e)
(d), second half; no induction		$(n+k)m = m(n+k) = mn+mk = nm+km$	(d), first half; (f)
(g) $(km)n = k(mn)$; k, m fixed, on n	$(km)\cdot 0 = 0 = k\cdot 0 = k(m\cdot 0)$	$(km)(n+1) = (km)n+km \overset{\text{IH}}{=} k(mn)+km = k(mn+m) = k(m(n+1))$	Def; (d)

In the step of (b), the last equality is (a), second half, read from right to left. In the long step of (e), the equalities after IH use, in order, (c), Def, (b), Def, (c), Def. In the last part of (e), $1 = 0+1$ is the first half of (a), and $(0+1)\cdot n = 0\cdot n+n$ is the second part of (e) with $m = 0$. In (g), $k(mn+m) = k(mn)+km$ is the first half of (d). This proves the theorem.

The laws (a)–(g) are the rules of school arithmetic, now proved from Axiom 1.1.1. The next section defines the order from addition and proves its properties with these laws.

Warning. **Do not assume $0+n = n$.** It is clause (a) of Theorem 1.2.3, proved by induction; the defining equation is $n+0 = n$. For the same reason, inducting on m in $m+n = n+m$ gets stuck: the step would need $(m+1)+n$, and the definition says nothing about a sum whose **first** argument is a successor until (a) is proved.

The order of the table is forced by the lopsided definition. Commutativity, which lets us move a successor from the second place to the first, comes only after the two facts in (a) that move it by hand; once it is available, (e) and (f) follow the same way for multiplication.

Quick check. Which variable would you induct on to prove $m(n+k) = mn+mk$, and which earlier law lets the step close?

Solution. On k, the second argument of $n+k$ and of mk, where the recursion runs; m and n are fixed. The step is

$$m(n+(k+1)) = m((n+k)+1) = m(n+k)+m = (mn+mk)+m = mn+(mk+m) = mn+m(k+1),$$

by Def of $+$, Def of $\cdot$, IH, associativity (c), and Def of $\cdot$ again. Law (c) is the one that lets the step close.

1.2.4 Exercises

1.2.4.1 A. Check your understanding

Exercise. A1

(a) In the proof of Theorem 1.2.1, where are (P3) and (P4) used?

(b) Why does the uniqueness half need only an induction, while existence needs the set R?

(c) Name the move, and the variable, for proving $0 \cdot n = 0$.

(d) State the idea of the existence proof in two sentences.

Solution.

(a) (P3) is used in the base case of the claim of Step 3: no step produces a pair with first coordinate 0, since $\mathrm{S}(m) \neq 0$, so a second pair $(0, b)$ could be removed without breaking closure. It is used again in the inductive step, where $\mathrm{S}(n) \neq 0$ keeps $(0, a)$ in R'. (P4) is used in the inductive step: a pair that steps to first coordinate $\mathrm{S}(n)$ has first coordinate n, where the induction hypothesis applies.

(b) Uniqueness compares two functions that are **given**, value by value, and comparing values at $n+1$ through the values at n is exactly an induction on n. Existence asks for a function that is not given; induction proves statements and builds nothing. So existence needs an object first, the set R, and induction then proves a statement about it (the claim of Step 3).

(c) Fix the others, induct on the recursion variable (Example (Method: fix the others, induct on the recursion variable)). There are no others to fix; the recursion variable of $0 \cdot n$ is n. Base: $0 \cdot 0 = 0$ by Def. Step: $0 \cdot (n+1) = 0 \cdot n + 0 = 0 + 0 = 0$, by Def, the hypothesis, and Def.

(d) The smallest set of pairs closed under the two equations exists. Each n has exactly one partner in it, because a second partner could be removed without breaking closure, so the set is the graph of the required function.

1.2.4.2 B. Practice

Exercise. B1: Small products

Compute $3 \cdot 2$ and $2 \cdot 3$ from the defining equations of Definition 1.2.1 alone, without using Theorem 1.2.3. Here $5 := \mathrm{S}(4)$ and $6 := \mathrm{S}(5)$.

Solution. For $3 \cdot 2$, by (b) twice and then at 0:

$$3 \cdot 2 = 3 \cdot 1 + 3 = (3 \cdot 0 + 3) + 3 = (0 + 3) + 3.$$

By (a), $0 + 3 = \mathrm{S}(0 + 2) = \mathrm{S}(\mathrm{S}(0 + 1)) = \mathrm{S}(\mathrm{S}(\mathrm{S}(0 + 0))) = \mathrm{S}(\mathrm{S}(\mathrm{S}(0))) = 3$. Then

$$3 + 3 = \mathrm{S}(3 + 2) = \mathrm{S}(\mathrm{S}(3 + 1)) = \mathrm{S}\Big(\mathrm{S}(\mathrm{S}(3 + 0))\Big) = \mathrm{S}(\mathrm{S}(\mathrm{S}(3))) = 6.$$

So $3 \cdot 2 = 6$.

For $2 \cdot 3$, by (b), $2 \cdot 3 = 2 \cdot 2 + 2$, and $2 \cdot 2 = 4$ by the Quick check above. Then

$$4 + 2 = \mathrm{S}(4 + 1) = \mathrm{S}(\mathrm{S}(4 + 0)) = \mathrm{S}(\mathrm{S}(4)) = 6.$$

So $2 \cdot 3 = 6 = 3 \cdot 2$, as (f) of Theorem 1.2.3 predicts; but the two computations took different routes, and neither used a law.

Exercise. B2: Distributivity, last step blank

Here is the proof of $m(n + k) = mn + mk$ for all $m, n, k \in \mathbb{N}$, with the last step left blank. *Name the move, then complete it.*

> Fix $m, n \in \mathbb{N}$ and induct on k. Base case: $m(n+0) = mn = mn+0 = mn+m \cdot 0$. Inductive step: suppose $m(n + k) = mn + mk$. Then
>
> $m(n+(k+1)) = m((n+k)+1) = m(n+k)+m = (mn+mk)+m =$ _____.

Justify each equality you add.

Solution. The move is **fix the others, induct on the recursion variable** (Example (Method: fix the others, induct on the recursion variable)): in $m(n + k)$ the recursion of $+$ runs on k, so m and n are fixed and the induction is on k.

The completion is

$$(mn + mk) + m = mn + (mk + m) = mn + m(k + 1),$$

where the first equality is associativity, (c) of Theorem 1.2.3, and the second is the defining equation $m(k + 1) = mk + m$ of Definition 1.2.1 (b), read from right to left. This is the statement for $k + 1$, so by Theorem 1.1.2 (b) it holds for every $k \in \mathbb{N}$. (This is clause (d) of Theorem 1.2.3, already proved; the exercise practices the move.)

Exercise. B3: Prove or disprove

Prove each true statement and refute each false one with a counterexample. All letters stand for natural numbers, and you may use Theorem 1.2.3.

(a) If $m + n = 0$, then $m = n = 0$.
(b) If $mn = m$, then $n = 1$.
(c) $m \cdot (n + 1) = m \cdot n + n$.
(d) If $m + n = m$, then $n = 0$.
(e) If $n \cdot n = n$, then $n = 0$.

Solution.

(a) **True.** Suppose $n \neq 0$. Then $n = \mathrm{S}(k) = k + 1$ for some $k \in \mathbb{N}$ (Proposition 1.1.3 (a), and (c) of Definition 1.2.1), and

$$m + n = m + (k + 1) = (m + k) + 1 \neq 0$$

by Def and Corollary 1.2.2. So $m + n = 0$ forces $n = 0$, and then $m = m + 0 = 0$.

(b) **False.** Take $m = n = 0$: $0 \cdot 0 = 0 = m$, but $n = 0 \neq 1$, since $1 = \mathrm{S}(0) \neq 0$ by (P3).

(c) **False.** This is a look-alike of the defining equation $m \cdot (n + 1) = m \cdot n + m$ of Definition 1.2.1 (b), whose last term is m, not n. Take $m = 0$, $n = 1$: the left side is $0 \cdot 2 = 0$ by law (e), and the right side is $0 \cdot 1 + 1 = 0 + 1 = 1$, and $1 \neq 0$.

(d) **True.** Induct on m for the property "for every n, if $m + n = m$, then $n = 0$". *Base:* $0 + n = n$ by law (a), so $0 + n = 0$ gives $n = 0$. *Step:* suppose the property for m, and let $(m+1)+n = m+1$. By law (a), $(m+1)+n = (m+n)+1$, so $(m+n)+1 = m+1$, and Corollary 1.2.2 gives $m + n = m$. By the hypothesis, $n = 0$.

(e) **False.** Take $n = 1$: $1 \cdot 1 = 1 \cdot 0 + 1 = 0 + 1 = 1$, by Def and law (a), but $1 \neq 0$.

1.2.4.3 C. Going deeper

Exercise. C1: Exit check: addition is commutative

From the two defining equations of addition, $m + 0 = m$ and $m + (n + 1) = (m + n) + 1$, and Theorem 1.1.2 alone, prove that $m + n = n + m$ for all $m, n \in \mathbb{N}$. Prove every auxiliary fact you need.

Solution (Hint 1). The definition says what happens when the second argument is 0 or a successor. To use it on $n + m$, you need to know the same for the first argument: prove two lemmas first.

Solution (Hint 2). Start with the first lemma, $0 + n = n$, by induction on n; the base is the first equation with $m = 0$. Then find the second lemma, about $(m + 1) + n$, yourself.

Solution. All inductions are by Theorem 1.1.2 (b).

Lemma 1: $0 + n = n$ *for every* n. Induct on n. Base: $0 + 0 = 0$, by the first equation. Step: if $0 + n = n$, then $0 + (n + 1) = (0 + n) + 1 = n + 1$, by the second equation and the hypothesis.

Lemma 2: $(m+1)+n = (m+n)+1$ *for all* m, n. Fix m and induct on n. Base: $(m+1)+0 = m + 1 = (m + 0) + 1$, by the first equation twice. Step: if $(m + 1) + n = (m + n) + 1$, then

$$(m + 1) + (n + 1) = \big((m + 1) + n\big) + 1 = \big((m + n) + 1\big) + 1 = \big(m + (n + 1)\big) + 1,$$

by the second equation, the hypothesis, and the second equation read from right to left inside the outer $+1$.

Main induction. Fix m and induct on n for "$m + n = n + m$". Base: $m + 0 = m = 0 + m$, by the first equation and Lemma 1. Step: if $m + n = n + m$, then

$$m + (n + 1) = (m + n) + 1 = (n + m) + 1 = (n + 1) + m,$$

by the second equation, the hypothesis, and Lemma 2 (with n in the role of m and m in the role of n), read from right to left. Hence $m + n = n + m$ for all $m, n \in \mathbb{N}$.

Marking scheme. (10 points) - 2: proves $0+n = n$ by induction on n, from the first equation (base) and the second (step). - 3: proves, by an induction with the other variable fixed, a lemma that moves a successor out of the first argument (the book's $(m+1)+n = (m+n)+1$, or $1 + n = n + 1$ together with associativity of addition), base and step both justified. - 2: in the main induction, the base case $m + 0 = 0 + m$, using the first lemma. - 2: in the main induction, the step, using the second equation, the hypothesis and that lemma. - 1: names the variable inducted on in each induction and fixes the other. Fatal errors (0 for the step): citing commutativity, or any law of Theorem 1.2.3, without proving it from the two equations; treating $0 + n = n$ as if it were a defining equation. A second route inducts on m in the main induction with n fixed; it needs the same two lemmas, and its base and step map to the same 2 + 2 points. A third route proves associativity and $1 + n = n + 1$ (each by induction on n, from the equations and $0 + n = n$) and then closes the step as $m + (n + 1) = (m + n) + 1 = (n + m) + 1 = n + (m + 1) = n + (1 + m) = (n + 1) + m$; those two lemmas together earn the 3 points.

Exercise. C2: Associativity of multiplication, last half blank

Prove that $(km)n = k(mn)$ for all $k, m, n \in \mathbb{N}$. The first half is given; *name the move, then complete it.*

> Fix $k, m \in \mathbb{N}$ and induct on n. Base case: $(km) \cdot 0 = 0 = k \cdot 0 = k(m \cdot 0)$.
> Inductive step: _____

You may use (a)–(f) of Theorem 1.2.3. If you read the fold after its proof, the last row of the table compresses this step into one line; write it out without looking, and give the reason for each equality.

Solution. The move is **fix the others, induct on the recursion variable**: in $(km)n$ and $k(mn)$ the variable n is the second argument of the outer and the inner product, where the

recursion runs, so k and m are fixed.

Inductive step. Suppose $(km)n = k(mn)$. Then

$$\begin{aligned}(km)(n+1) &= (km)n + km = k(mn) + km \\ &= k(mn+m) = k(m(n+1)),\end{aligned}$$

where the first equality is the defining equation of Definition 1.2.1 (b) with first factor km, the second is the induction hypothesis, the third is (d) of Theorem 1.2.3 read from right to left ($k(x+y) = kx + ky$ with $x = mn$, $y = m$), and the fourth is the defining equation with first factor m. By Theorem 1.1.2 (b), $(km)n = k(mn)$ for every n, and k, m were arbitrary.

The base case used the defining equation $x \cdot 0 = 0$ three times, with $x = km$, $x = k$ and $x = m$. Commutativity (f) is not needed for this route; a proof that first rewrites km as mk would need it.

Exercise. C3: Doubling

Show that there is exactly one function $D\colon \mathbb{N} \to \mathbb{N}$ with $D(0) = 0$ and $D(n+1) = (D(n)+1)+1$ for every $n \in \mathbb{N}$, and prove that $D(n) = n + n$ for every $n \in \mathbb{N}$.

Solution. Existence and uniqueness. The function $g\colon \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, $g(n, x) := \mathrm{S}(\mathrm{S}(x)) = (x+1)+1$, maps into $\mathbb{N}$ since S maps $\mathbb{N}$ into $\mathbb{N}$. By Theorem 1.2.1 with $X = \mathbb{N}$ and $a = 0$, there is exactly one $D\colon \mathbb{N} \to \mathbb{N}$ with $D(0) = 0$ and $D(n+1) = g(n, D(n)) = (D(n)+1)+1$. (Its graph is the set R of the proof for these data, which contains $(1, 2)$ and $(2, 4)$ by the first Quick check after Theorem 1.2.1; so $D(1) = 2$ and $D(2) = 4$.)

The formula. Induct on n by Theorem 1.1.2 (b) for "$D(n) = n+n$". Base: $D(0) = 0 = 0+0$, by the defining equation $m + 0 = m$. Step: suppose $D(n) = n + n$. Then

$$\begin{aligned}(n+1)+(n+1) &= ((n+1)+n)+1 = ((n+n)+1)+1 \\ &= (D(n)+1)+1 = D(n+1),\end{aligned}$$

where the first equality is the defining equation of Definition 1.2.1 (a), the second is (a) of Theorem 1.2.3, $(m+1)+n = (m+n)+1$ with $m = n$, the third is the hypothesis, and the last is the equation of D. Hence $D(n) = n + n$ for every $n \in \mathbb{N}$.

Alternatively, $n \mapsto n + n$ satisfies the two equations that define D (the computation above, read without the hypothesis, shows $(n+1)+(n+1) = ((n+n)+1)+1)$, so it is D by the uniqueness half of Theorem 1.2.1. Either way the double $n + n$ is $2n$ too: $2n = n \cdot 2 = n \cdot 1 + n = n + n$ by (f), the defining equation of $\cdot$, and (e) of Theorem 1.2.3.

1.3 Order and Well-Ordering of the Natural Numbers

Sets and Functions, Chapter 3 ordered the natural numbers by the order of the real line, and proved from that order and induction that every non-empty set of natural numbers has a least element (Sets and Functions, Theorem 3.1.3). The natural numbers of Axiom 1.1.1 lie on no

line. They do have addition, though, and addition already says when one number comes before another: m comes before n when we can count up from m to n. This section takes that as the definition of the order, ties it to the arithmetic of the previous section, proves that it is a total order and a well-ordering, and records the facts about powers and parity that later sections cite.

1.3.1 Counting up from m to n

The first idea is to copy *Sets and Functions*: $m \leq n$ when $n - m \geq 0$. On $\mathbb{N}$ this says nothing yet. There is no subtraction, and "≥ 0" is the very order we are trying to define. What we do have is the sum. In school, $2 \leq 5$ because three more steps lead from 2 to 5; that statement uses only $+$.

$m \leq n$ when n is reached from m by adding a natural number.

Definition 1.3.1. Order on $\mathbb{N}$

For $m, n \in \mathbb{N}$, we write $m \leq n$ if $n = m + j$ for some $j \in \mathbb{N}$, and $m < n$ if $m \leq n$ and $m \neq n$. As usual, $n \geq m$ means $m \leq n$, and $n > m$ means $m < n$.

Two examples. First, $2 \leq 5$, with $j = 3$: by the equations of Definition 1.2.1 (a),

$$2 + 3 = \mathrm{S}(2 + 2) = \mathrm{S}(\mathrm{S}(2 + 1)) = \mathrm{S}(\mathrm{S}(\mathrm{S}(2 + 0))) = \mathrm{S}(\mathrm{S}(3)) = 5.$$

Second, $0 \leq n$ for every $n \in \mathbb{N}$, since $n = 0 + n$ by Theorem 1.2.3 (a).

The definition is short, and so it promises little. Nothing in it says that $5 \leq 2$ fails, or that $\leq$ behaves like an order at all. Antisymmetry will need the fact that a sum is 0 only when both terms are, and totality, that any two numbers are comparable, will need an induction. We first collect what the definition says about sums and products, since those facts are the tools for everything else.

1.3.2 Order and arithmetic

Cancellation in sums, $m + k = n + k \Rightarrow m = n$, needs no order at all, and it is what makes the j of the definition unique. With it, the order clauses (c) and (e) follow one pattern: translate $m < n$ into an equation $n = m + j$ with $j \neq 0$, compute with the laws of Theorem 1.2.3, and translate back; the rest are inductions or successor arguments.

Proposition 1.3.1. Order and Arithmetic on $\mathbb{N}$

For all $k, m, n, b \in \mathbb{N}$:

(a) $m \leq n$ if and only if $n = m + j$ for some $j \in \mathbb{N}$; this j is then unique, and is written $n - m$. Moreover $m < n$ if and only if $n = m + j$ for some $j \neq 0$.

(b) If $m + k = n + k$, then $m = n$.

(c) If $m < n$, then $m + k < n + k$.

(d) If $mn = 0$, then $m = 0$ or $n = 0$.

(e) If $m < n$ and $k \neq 0$, then $mk < nk$.

(f) If $mk = nk$ and $k \neq 0$, then $m = n$.

(g) *(Induction from b.)* Let $P(n)$ be a property of natural numbers. If $P(b)$ holds and, for every $n \in \mathbb{N}$ with $n \geq b$, $P(n)$ implies $P(n + 1)$, then $P(n)$ holds for every $n \in \mathbb{N}$ with

$n \geq b$.

Idea. Cancellation (b) needs no order: induct on k and peel off one successor at a time with Corollary 1.2.2. The equivalence in (a) is the definition, and uniqueness is (b). Then (c) and (e) translate an inequality $m < n$ into an equation $n = m + j$ with $j \neq 0$, compute, and translate back; (d) writes m and n as successors and uses that a successor is never 0. Clause (f) is proved by induction on m, the first factor, using law (e) to open $(m+1)k$ (step (4) of Example (Method: fix the others, induct on the recursion variable)). Finally (g) shifts the start of an induction from 0 to b.

Proof. All inductions are by Theorem 1.1.2 (b), and "Law (x)" refers to clause (x) of Theorem 1.2.3. We prove (b) first, since (a) uses it.

(b) Induct on k for "for all m, n: $m + k = n + k \Rightarrow m = n$". For $k = 0$, $m + 0 = m$ and $n + 0 = n$. If the statement holds for k and $m + (k + 1) = n + (k + 1)$, then $(m + k) + 1 = (n + k) + 1$ by Definition 1.2.1 (a), so $m + k = n + k$ by Corollary 1.2.2, and $m = n$ by the hypothesis.

(c) The equivalence is Definition 1.3.1.

Uniqueness. If $m + j = m + j'$, then $j + m = j' + m$ by Law (b), so $j = j'$ by (b) of this proposition.

The strict case. If $n = m + j$, then $j = 0$ if and only if $n = m$: if $j = 0$, then $n = m + 0 = m$; if $n = m$, then $m + j = m = m + 0$, so $j = 0$ by uniqueness. Since $m < n$ holds exactly when $m \leq n$ and $m \neq n$ (by Definition 1.3.1), $m < n$ holds if and only if $n = m + j$ with $j \neq 0$.

(c) Let $m < n$, so $n = m + j$ with $j \neq 0$ by (a). Then

$$n + k = (m + j) + k = m + (k + j) = (m + k) + j$$

by Laws (c), (b) and (c). Since $j \neq 0$, (a) gives $m + k < n + k$.

(d) Let $mn = 0$ and $n \neq 0$; we show $m = 0$. By Proposition 1.1.3 (a), $n = \mathrm{S}(j) = j + 1$ for some $j \in \mathbb{N}$, so $0 = mn = mj + m$ by Definition 1.2.1 (b). If $m \neq 0$, then likewise $m = i + 1$ for some $i \in \mathbb{N}$, and

$$0 = mj + (i + 1) = (mj + i) + 1,$$

by Definition 1.2.1 (a), which contradicts Corollary 1.2.2. So $m = 0$.

(e) Let $m < n$ and $k \neq 0$. By (a), $n = m + j$ with $j \neq 0$, so $nk = (m + j)k = mk + jk$ by Law (d). By (d), $jk \neq 0$, since $j \neq 0$ and $k \neq 0$. By (a), $mk < nk$.

(f) Fix $k \neq 0$, and induct on m for "for every $n \in \mathbb{N}$, $mk = nk \Rightarrow m = n$". For $m = 0$: if $0 \cdot k = nk$, then $nk = 0$ by Law (e), so $n = 0$ by (d), since $k \neq 0$. Suppose the statement holds for m, and let $(m + 1)k = nk$. By Proposition 1.1.3 (a), $k = i + 1$ for some $i \in \mathbb{N}$, so

$$(m + 1)k = mk + k = mk + (i + 1) = (mk + i) + 1 \neq 0$$

by Law (e), Definition 1.2.1 (a) and Corollary 1.2.2. So $nk \neq 0$, hence $n \neq 0$ (as $0 \cdot k = 0$), and $n = n' + 1$ for some $n' \in \mathbb{N}$ (Proposition 1.1.3 (a)). Then $mk + k =$

$(m+1)k = nk = n'k+k$ by Law (e), so $mk = n'k$ by (b), and $m = n'$ by the hypothesis. Hence $m + 1 = n$.

(g) Let $Q(i)$ be "$P(b + i)$", for $i \in \mathbb{N}$. $Q(0)$ is $P(b + 0) = P(b)$. If $Q(i)$ holds, then $b + i \geq b$ by (a), so the hypothesis gives $P((b + i) + 1)$; and $(b + i) + 1 = b + (i + 1)$ by Definition 1.2.1 (a), so $Q(i + 1)$ holds. By Theorem 1.1.2 (b), $Q(i)$ holds for every $i \in \mathbb{N}$. Finally, if $n \geq b$, then $n = b + i$ for some i by (a), and $P(n)$ is $Q(i)$.

■

Clause (a) is how subtraction appears on $\mathbb{N}$: $n - m$ is defined only when $m \leq n$, as the unique j with $m + j = n$. For example $5 - 2 = 3$, by the example after Definition 1.3.1, and $n - 0 = n$. Clause (g) is the "induction from a base" that *Sets and Functions*, Chapter 3 used as Sets and Functions, Theorem 3.2.1, now proved on $\mathbb{N}$.

Quick check. Show that $3 \leq 5$, naming the j. Is $n \leq n + 1$? Is $n + 1 \leq n$?

Solution. $3 + 2 = \mathrm{S}(3 + 1) = \mathrm{S}(\mathrm{S}(3 + 0)) = \mathrm{S}(\mathrm{S}(3)) = 5$, so $3 \leq 5$ with $j = 2$, and $5 - 3 = 2$. Yes, $n \leq n+1$, with $j = 1$. No: if $n = (n+1)+j$, then $n+0 = n+(j+1)$ by Theorem 1.2.3 (c) and (b), so $j + 1 = 0$ by the uniqueness in (a), against Corollary 1.2.2.

1.3.3 A total order

The definition of $\leq$ came from arithmetic, so its properties as an order must come from arithmetic too. The next theorem proves them, and then describes the sets $\{0, \ldots, n\}$ and $\{1, \ldots, n\}$ that *Sets and Functions* used everywhere, now as subsets of $\mathbb{N}$.

Theorem 1.3.2. Order on $\mathbb{N}$

(a) $<$ is a strict total order on $\mathbb{N}$: for all $k, m, n \in \mathbb{N}$, never $n < n$; if $k < m$ and $m < n$, then $k < n$; and **exactly one** of $m < n$, $m = n$, $n < m$ holds.

(b) For $m, n \in \mathbb{N}$, $m < \mathrm{S}(n)$ if and only if $m \leq n$. Hence no $k \in \mathbb{N}$ satisfies $n < k < \mathrm{S}(n)$.

(c) For $n \in \mathbb{N}$, the set $\{0, \ldots, n\} \coloneqq \{k \in \mathbb{N} : k \leq n\}$ satisfies $\{0, \ldots, 0\} = \{0\}$ and $\{0, \ldots, \mathrm{S}(n)\} = \{0, \ldots, n\} \cup \{\mathrm{S}(n)\}$, with $\mathrm{S}(n) \notin \{0, \ldots, n\}$.

(d) $0 \leq n$ for every $n \in \mathbb{N}$; if $n \neq 0$, then $1 \leq n$; and $0 < 1$.

(e) For $n \in \mathbb{N}$, the set $\{1, \ldots, n\} \coloneqq \{k \in \mathbb{N} : 1 \leq k \leq n\}$ satisfies $\{1, \ldots, 0\} = \emptyset$ and $\{1, \ldots, \mathrm{S}(n)\} = \{1, \ldots, n\} \cup \{\mathrm{S}(n)\}$.

Idea. First $\leq$ is a partial order: reflexive ($j = 0$), transitive (associativity), antisymmetric (cancel, then use that a sum is 0 only when both terms are). Then it is total, by induction on n with "for every m" kept inside the property. The three properties of $<$ in (a) follow from these four facts, and the rest unwinds (b).

Proof. "Law (x)" refers to Theorem 1.2.3, and "Order (x)" to Proposition 1.3.1.

Zero sums. If $j + i = 0$, then $i = 0$ and $j = 0$. Indeed, if $i \neq 0$, then $i = i' + 1$ for some i' (Proposition 1.1.3 (a)), and $j+i = (j+i')+1 \neq 0$ by Definition 1.2.1 (a) and Corollary 1.2.2. So $i = 0$, and $j = j + 0 = 0$.

$\leq$ *is a partial order* (Sets and Functions, Definition 5.3.1). (PO1) $n = n + 0$, so $n \leq n$. (PO3) If $m = k + i$ and $n = m + j$, then $n = k + (i + j)$ by Law (c), so $k \leq n$. (PO2) If $n = m + j$ and $m = n + i$, then $m + 0 = m = m + (j + i)$ by Law (c), so $j + i = 0$ by the uniqueness in Order (a). By zero sums $i = 0$, and $m = n + 0 = n$.

$\leq$ *is total.* We induct on n for "for every $m \in \mathbb{N}$, $m \leq n$ or $n \leq m$". For $n = 0$: $0 \leq m$, since $m = 0 + m$ by Law (a). Suppose the property holds for n, and let $m \in \mathbb{N}$. If $m \leq n$, say $n = m + j$, then $n + 1 = m + (j + 1)$ by Definition 1.2.1 (a), so $m \leq n + 1$. If $n \leq m$, say $m = n + j$, there are two cases. If $j = 0$, then $m = n$, and $n + 1 = m + 1$ gives $m \leq n + 1$. Otherwise $j \neq 0$, so $j = j' + 1$ for some $j' \in \mathbb{N}$ (Proposition 1.1.3 (a)), and

$$m = n + (j' + 1) = (n + j') + 1 = (n + 1) + j'$$

by Definition 1.2.1 (a) and Law (a), so $n + 1 \leq m$.

(a) *Irreflexive:* $n < n$ would need $n \neq n$. *Transitive:* let $k < m$ and $m < n$. Then $k \leq n$ by (PO3). If $k = n$, then $k \leq m$ and $m \leq k$, so $k = m$ by (PO2), against $k < m$. So $k < n$. *Trichotomy:* at least one of the three holds, since by totality $m \leq n$ or $n \leq m$, and each of these is "$<$ or $=$". At most one: $m < n$ excludes $m = n$ by definition, and $m < n$ together with $n < m$ would give $m = n$ by (PO2).

(b) ($\Leftarrow$) If $n = m+j$, then $\mathrm{S}(n) = n+1 = m+(j+1)$, and $j+1 \neq 0$, so $m < \mathrm{S}(n)$ by Order (a). ($\Rightarrow$) If $m < \mathrm{S}(n)$, then by Order (a) and Proposition 1.1.3 (a), $\mathrm{S}(n) = m+(i+1) = \mathrm{S}(m + i)$ for some i, so $n = m + i$ by (P4), and $m \leq n$. Finally, if $n < k < \mathrm{S}(n)$, then $k \leq n$, so $k < n$ or $k = n$, and either contradicts $n < k$ by trichotomy.

(c) If $k \leq 0$, then $0 = k + j$ for some j, and $k = 0$ by zero sums; and $0 \leq 0$. So $\{0, \ldots, 0\} = \{0\}$. By (b), $k \leq \mathrm{S}(n)$, that is, $k < \mathrm{S}(n)$ or $k = \mathrm{S}(n)$, holds if and only if $k \leq n$ or $k = \mathrm{S}(n)$; this is the union. Since $n \leq n$, (b) gives $n < \mathrm{S}(n)$, so $\mathrm{S}(n) \leq n$ fails by trichotomy.

(d) $0 \leq n$, since $n = 0 + n$ by Law (a). If $n \neq 0$, then $n = k + 1$ for some k (Proposition 1.1.3 (a)), and $k + 1 = 1 + k$ by Law (b), so $1 \leq n$. Finally $1 = 0 + 1$ and $1 \neq 0$ (Corollary 1.2.2), so $0 < 1$ by Order (a).

(e) For $k \in \mathbb{N}$, $1 \leq k$ holds if and only if $k \neq 0$: by (d) if $k \neq 0$, and $1 \leq 0$ fails by (c), since $1 \neq 0$. So $\{1, \ldots, n\} = \{0, \ldots, n\} \setminus \{0\}$, and (c) gives $\{1, \ldots, 0\} = \emptyset$ and $\{1, \ldots, \mathrm{S}(n)\} = \{1, \ldots, n\} \cup \{\mathrm{S}(n)\}$, as $\mathrm{S}(n) \neq 0$ by (P3). This proves the theorem.

■

So the notation $\{0, \ldots, n\}$ and $\{1, \ldots, n\}$ of *Sets and Functions*, Chapter 3 keeps its meaning on $\mathbb{N}$: by (c) and (e) each set grows by exactly one new element at each step, and $\{1, \ldots, 0\} = \emptyset$. In the "plus one" notation of Definition 1.2.1, (b) says that $n + 1$ is the next number after n.

Quick check. Is there a $k \in \mathbb{N}$ with $4 < k < 5$? List $\{0, \ldots, 3\}$ using (c).

Solution. No: $5 = \mathrm{S}(4)$, and (b) says that nothing lies strictly between 4 and $\mathrm{S}(4)$. By (c), three times, $\{0, \ldots, 3\} = \{0\} \cup \{1\} \cup \{2\} \cup \{3\} = \{0, 1, 2, 3\}$.

Warning. **Cancelling a factor needs it to be non-zero.** $5 \cdot 0 = 2 \cdot 0$, yet $5 \neq 2$: clause (f) of Proposition 1.3.1 carries $k \neq 0$ for this reason. Clause (e) carries it too, since $2 < 5$ while $2 \cdot 0 < 5 \cdot 0$ fails.

1.3.4 Well-ordering

In *Sets and Functions*, Chapter 3, Sets and Functions, Theorem 3.1.3 was derived from Logic and Proof, Axiom 4.2.1, which there was an axiom, by way of strong induction. On $\mathbb{N}$, induction is a theorem (Theorem 1.1.2), so well-ordering becomes one too. Here a direct induction does it, on the sets $\{0, \ldots, n\}$ of Theorem 1.3.2 (c).

Theorem 1.3.3. Well-Ordering of $\mathbb{N}$

Every **non-empty** subset $A \subseteq \mathbb{N}$ has a least element: an $a \in A$ with $a \leq x$ for every $x \in A$. The least element is unique.

Idea. Suppose A has no least element. Walk up through $0, 1, 2, \ldots$: the first element of A met on the way would be least. So no element of $\{0, \ldots, n\}$ lies in A, and this is proved by induction on n. Since every a lies in $\{0, \ldots, a\}$, A is empty. Compare Sets and Functions, Example (Method: the least element), which *uses* a least element; here we produce one.

Proof. Let $A \subseteq \mathbb{N}$ be non-empty, and suppose, for a contradiction, that A has no least element. We use Theorem 1.1.2 (b) on n, for the property $B(n)$: "$\{0, \ldots, n\} \cap A = \emptyset$".

Base case. By Theorem 1.3.2 (c), $\{0, \ldots, 0\} \cap A = \{0\} \cap A$. If $0 \in A$, then $0 \leq x$ for every $x \in A$ by Theorem 1.3.2 (d), so 0 is a least element of A, contrary to our supposition. So $B(0)$ holds.

Inductive step. Suppose $B(n)$. By Theorem 1.3.2 (c),

$$\{0, \ldots, \mathrm{S}(n)\} \cap A = (\{0, \ldots, n\} \cap A) \cup (\{\mathrm{S}(n)\} \cap A) = \{\mathrm{S}(n)\} \cap A.$$

Suppose $\mathrm{S}(n) \in A$, and let $x \in A$. By $B(n)$, $x \leq n$ fails, so $x < \mathrm{S}(n)$ fails by Theorem 1.3.2 (b), and $\mathrm{S}(n) \leq x$ by trichotomy (Theorem 1.3.2 (a)). Hence $\mathrm{S}(n)$ is a least element of A, contrary to our supposition. So $\mathrm{S}(n) \notin A$, and $B(\mathrm{S}(n))$ holds.

By induction, $B(n)$ holds for every $n \in \mathbb{N}$. Since A is non-empty, there is some $a \in A$. Then $a \leq a$, so $a \in \{0, \ldots, a\} \cap A$, against $B(a)$. This contradiction shows, by RAA (Logic and Proof, Definition 2.5.1), that A has a least element.

Uniqueness. If a and a' are least elements, then $a \leq a'$ and $a' \leq a$. If $a \neq a'$, these give $a < a'$ and $a' < a$, so $a < a$ by transitivity, contradicting irreflexivity (Theorem 1.3.2 (a)). So $a = a'$.

■

The proof concludes an existence statement from a contradiction, so its last step is RAA, not ¬I: what is refuted is "A has no least element". The statement is that of Sets and Functions, Theorem 3.1.3, read on $\mathbb{N}$; there the proof went through strong induction on $\mathbb{N} \subseteq \mathbb{R}$, here through the property $B(n)$.

Both hypotheses are needed. The empty set is a subset of $\mathbb{N}$ with no least element, since a least element must be an element of the set. For sets of numbers that are not subsets of $\mathbb{N}$, the warning after Sets and Functions, Theorem 3.1.3 gives $\mathbb{Z}$ and the positive rationals.

Quick check. What is the least element of $\{n \in \mathbb{N} : n \cdot n > 10\}$, and why?

Solution. It is 4. First, $3 \cdot 3 = 9$, $4 \cdot 4 = 16$, and $9 < 10 < 16$, since $10 = 9 + 1$ and $16 = 10 + 6$. So 4 lies in the set. Any $x < 4$ lies in $\{0, \dots, 3\} = \{0, 1, 2, 3\}$ (Theorem 1.3.2 (b), (c)), where $x \cdot x$ is 0, 1, 4 or 9, so $x \cdot x \leq 9 < 10$ and x is not in the set. By trichotomy every element x of the set has $4 \leq x$.

1.3.5 Powers and parity

Sets and Functions, Chapter 4 counted with powers $2^m 3^n$ and with even and odd numbers, and took both from the real numbers and the integers. Here are the facts it used, proved on $\mathbb{N}$. Section 1.4 cites them in its rule for reading *Sets and Functions* on $\mathbb{N}$, and Chapter 4 of this volume uses them again.

For $b \in \mathbb{N}$, the powers b^n ($n \in \mathbb{N}$) are defined by Theorem 1.2.1 with $X = \mathbb{N}$, $a = 1$ and $g(n, x) = x \cdot b$:

$$b^0 := 1, \qquad b^{n+1} := b^n \cdot b.$$

These are the two equations of Sets and Functions, Definition 3.3.2, now on $\mathbb{N}$; so $b^1 = 1 \cdot b = b$ by Theorem 1.2.3 (e).

Quick check. Compute 2^3 from the two equations.

Solution. $2^0 = 1$, $2^1 = 1 \cdot 2 = 2$, $2^2 = 2 \cdot 2 = 4$, $2^3 = 4 \cdot 2 = 8$.

Proposition 1.3.4. Powers and Parity on $\mathbb{N}$

For all $b, m, n \in \mathbb{N}$:

(a) $b^{m+n} = b^m \cdot b^n$;

(b) if $b \neq 0$, then $b^n \neq 0$;

(c) if $b > 1$ and $m < n$, then $b^m < b^n$;

(d) there is exactly one pair (q, r) with $q \in \mathbb{N}$, $r \in \{0, 1\}$ and $n = 2q + r$. In particular, no $n \in \mathbb{N}$ is both of the form $2q$ and of the form $2q' + 1$.

Idea.

(a) and (b) are inductions on n following the definition of b^n, by the recursion-variable method of Example (Method: fix the others, induct on the recursion variable). For (c), first $b^j > 1$ for $j \geq 1$, by induction from 1; then write $n = m + j$ and use (a). For (d), existence is an induction (a number is even or odd, and adding one swaps the two), and uniqueness reduces, after canceling, to "$r = 2j + r'$ with $j \geq 1$", which makes $r \geq 2$.

↻ **Second reading** · *Proof of the proposition*

Proof. "Law (x)" refers to Theorem 1.2.3, and "Order (x)" to Proposition 1.3.1.

(a) Fix m and induct on n (Theorem 1.1.2 (b)). For $n = 0$: $b^{m+0} = b^m = b^m \cdot 1 = b^m b^0$, by Law (e). If $b^{m+n} = b^m b^n$, then

$$b^{m+(n+1)} = b^{(m+n)+1} = b^{m+n} b = (b^m b^n) b = b^m (b^n b) = b^m b^{n+1},$$

by Definition 1.2.1 (a), the definition of powers, the hypothesis, Law (g), and the definition again.

(b) Let $b \neq 0$; induct on n. $b^0 = 1 \neq 0$ by Corollary 1.2.2 ($1 = 0 + 1$). If $b^n \neq 0$, then $b^{n+1} = b^n b \neq 0$ by Order (d).

(c) Let $b > 1$. Then $b > 0$, by $0 < 1$ (Theorem 1.3.2 (d)) and transitivity, so $b \neq 0$. First, $b^j > 1$ for every $j \geq 1$, by Order (g) with base 1: $b^1 = b > 1$; and if $b^j > 1$, then

$$b^{j+1} = b^j b > 1 \cdot b = b > 1$$

by Order (e) (with $b \neq 0$), Law (e) and transitivity. Now let $m < n$. By Order (a), $n = m + j$ with $j \neq 0$, so $j \geq 1$ by Theorem 1.3.2 (d), and $b^j > 1$. By (b), $b^m \neq 0$. Then, by Order (e), Law (e), Law (f) and (a),

$$b^m = 1 \cdot b^m < b^j b^m = b^m b^j = b^{m+j} = b^n.$$

(d) *Existence,* by Theorem 1.1.2 (b). For $n = 0$: $0 = 2 \cdot 0 + 0$. Suppose $n = 2q + r$ with $r \in \{0, 1\}$. If $r = 0$, then $n = 2q$ and $n + 1 = 2q + 1$. If $r = 1$, then

$$n + 1 = (2q + 1) + 1 = 2q + 2 = 2q + 2 \cdot 1 = 2(q + 1) = 2(q + 1) + 0,$$

by Law (c) and $1 + 1 = 2$ (Definition 1.2.1 (c)), Law (e), and Law (d).

Uniqueness. Let $2q + r = 2q' + r'$ with $r, r' \in \{0, 1\}$. Suppose first that $q < q'$. By Order (a), $q' = q + j$ with $j \neq 0$, so

$$2q + r = 2(q + j) + r' = 2q + (2j + r')$$

by Laws (d) and (c), and canceling $2q$ (Order (b), after Law (b)) gives $r = 2j + r'$. Since $j \geq 1$ (Theorem 1.3.2 (d)), Order (a) gives $j = 1 + i$ for some $i \in \mathbb{N}$, and $2j + r' = (2 + 2i) + r' =$

$2 + (2i + r')$ by Laws (d), (e) and (c). So $r = 2 + (2i + r') \geq 2$ by Order (a). But $r = 0$ or $r = 1$, and both are $< 2 = \mathrm{S}(1)$ by Theorem 1.3.2 (b); this contradicts trichotomy. So $q < q'$ is impossible, and $q' < q$ is impossible by the same argument with the roles exchanged. By trichotomy $q = q'$, and then $r = r'$ by Order (b) (after Law (b)). The pairs (q, r) and (q', r') are therefore equal (Sets and Functions, Definition 1.4.1).

If $n = 2q = 2q + 0$ and $n = 2q' + 1$, the pairs $(q, 0)$ and $(q', 1)$ would both represent n, and they differ since $0 \neq 1$; so no n is of both forms.

■

Any other definition of powers on $\mathbb{N}$ by the same two equations, $c(0) = 1$ and $c(n+1) = c(n) \cdot b$, gives these powers, by the uniqueness half of Theorem 1.2.1. Section 1.4 and Chapter 4 of this volume use the proposition.

1.3.6 Exercises

1.3.6.1 A. Check your understanding

Exercise. A1

(a) True or false: for $m, n \in \mathbb{N}$, $m < n$ implies $m + 1 \leq n$. Justify your answer.

(b) Is $5 \leq 2$? Justify your answer.

(c) Which clause of Theorem 1.3.2 says that $\{0, \ldots, n\}$ grows by one element at each step? List $\{0, \ldots, 2\}$ and $\{1, \ldots, 2\}$.

(d) Which clause of Proposition 1.3.1 defines $n - m$? Compute $5 - 2$.

Solution.

(a) True. By Proposition 1.3.1 (a), $n = m + j$ with $j \neq 0$. By Theorem 1.3.2 (d), $1 \leq j$, so $j = 1 + i$ for some $i \in \mathbb{N}$, by Proposition 1.3.1 (a) again. Then $n = m + (1 + i) = (m + 1) + i$ by Theorem 1.2.3 (c), so $m + 1 \leq n$.

(b) No. $5 = 2 + 3$ with $3 \neq 0$ (the example after Definition 1.3.1), so $2 < 5$ by Proposition 1.3.1 (a). By trichotomy (Theorem 1.3.2 (a)), neither $5 < 2$ nor $5 = 2$, so $5 \leq 2$ fails.

(c) Clause (c). $\{0, \ldots, 2\} = \{0, 1, 2\}$, and $\{1, \ldots, 2\} = \{1, 2\}$ by clause (e).

(d) Clause (a): $n - m$ is the unique j with $n = m + j$. Since $5 = 2 + 3$, $5 - 2 = 3$.

1.3.6.2 B. Practice

Exercise. B1: Order as inclusion

Prove that for all $m, n \in \mathbb{N}$, $m \leq n$ if and only if $\{0, \ldots, m\} \subseteq \{0, \ldots, n\}$, and that $m < n$ if and only if this inclusion is proper.

Solution. ($\Rightarrow$) Let $m \leq n$, say $n = m + j$, and let $k \in \{0, \ldots, m\}$, say $m = k + i$. Then $n = k + (i + j)$ by Theorem 1.2.3 (c), so $k \leq n$ and $k \in \{0, \ldots, n\}$.

($\Leftarrow$) Since $m = m + 0$, $m \in \{0, \ldots, m\} \subseteq \{0, \ldots, n\}$, so $m \leq n$.

The strict case. Let $m < n$. The inclusion holds by the first part, and $n \in \{0, \ldots, n\}$ while $n \notin \{0, \ldots, m\}$, because $n \leq m$ would contradict $m < n$ by trichotomy (Theorem 1.3.2 (a)). So the inclusion is proper. Conversely, if the inclusion is proper, then $m \leq n$ by the first part, and $m \neq n$, since $m = n$ would make the two sets equal. So $m < n$.

Exercise. B2: A product is at least its factor

Prove that for all $m, n \in \mathbb{N}$ with $n \geq 1$, $m \leq mn$. Show by an example that the hypothesis $n \geq 1$ is needed.

Solution. Since $1 \leq n$, Proposition 1.3.1 (a) gives $n = 1 + j$ for some $j \in \mathbb{N}$. Then

$$mn = m(1 + j) = m \cdot 1 + mj = m + mj,$$

by Theorem 1.2.3 (d) and (e). By Proposition 1.3.1 (a) again, $m \leq m + mj = mn$.

The hypothesis is needed: for $m = 1$ and $n = 0$, $mn = 0$, and $1 \leq 0$ is false, since $0 < 1$ (Theorem 1.3.2 (d)) and trichotomy.

1.3.6.3 C. Going deeper

Exercise. C1: Strong induction on $\mathbb{N}$

Let $P(n)$ be a property of natural numbers. Suppose that, for every $n \in \mathbb{N}$, if $P(k)$ holds for every $k \in \mathbb{N}$ with $k < n$, then $P(n)$ holds. Prove that $P(n)$ holds for every $n \in \mathbb{N}$.

Solution (Hint 1). Consider the set of natural numbers at which P fails.

Solution (Hint 2). If that set is non-empty, Theorem 1.3.3 gives it a least element a. What do you know about $P(k)$ for $k < a$?

Solution. Let $A := \{n \in \mathbb{N} : P(n) \text{ fails}\}$, a set by Sets and Functions, Definition 1.2.3, and suppose, for a contradiction, that $A \neq \emptyset$. By Theorem 1.3.3, A has a least element a. Let $k \in \mathbb{N}$ with $k < a$. If $k \in A$, then $a \leq k$, and with $k < a$ this gives $k < k$ (by transitivity, or directly if $a = k$), contradicting irreflexivity (Theorem 1.3.2 (a)). So $k \notin A$, that is, $P(k)$ holds. This holds for every $k < a$, so the hypothesis gives $P(a)$, contradicting $a \in A$. Hence $A = \emptyset$, and $P(n)$ holds for every $n \in \mathbb{N}$.

This is the statement of Logic and Proof, Theorem 4.2.2, read on $\mathbb{N}$. There it was derived from Logic and Proof, Axiom 4.2.1 directly; here the route goes through well-ordering, which

was itself proved from Theorem 1.1.2. The proof is the move of Sets and Functions, Example (Method: the smallest counterexample): a least failure, below which everything holds. As in *Logic and Proof*, the hypothesis at $n = 0$ is vacuous (no $k < 0$, since $k \leq 0$ forces $k = 0$ by Theorem 1.3.2 (c)), so it hands us $P(0)$ outright.

Marking scheme. (10 points) - 2: considers the set of failures (or argues by contradiction from a failure) and makes it a set. - 3: applies Theorem 1.3.3, with the set non-empty, to obtain a least failure a. - 3: shows $P(k)$ for every $k < a$, using that a is least and the order facts. - 2: applies the hypothesis at a and reaches the contradiction. Fatal errors (0 for the step): citing Logic and Proof, Theorem 4.2.2 (a statement about the natural numbers of *Logic and Proof*, Chapter 4, not proved from Axiom 1.1.1) instead of proving the result; taking a least element of a set not shown to be non-empty. A route by Theorem 1.1.2 (b) on "$P(k)$ for every $k < n$" earns the same points: 2 for the property $Q(n)$: "$P(k)$ for every $k < n$"; 2 for the vacuous base (no $k < 0$, by Theorem 1.3.2 (c)); 4 for the step ($P(n)$ from the hypothesis, then $k < n + 1$ gives $k \leq n$ by Theorem 1.3.2 (b), so $k < n$ or $k = n$); 2 for deducing $P(n)$ from $Q(n + 1)$.

Exercise. C2: Find the error

Here is an attempted proof that any two natural numbers are comparable. The proof is meant to replace the totality step in the proof of Theorem 1.3.2, so it may cite only results stated before that theorem. Is the conclusion true? Is the proof correct? If not, give the first line that fails, the error, and a counterexample to the step.

> **Claim.** For all $m, n \in \mathbb{N}$: $m \leq n$ or $n \leq m$.
>
> (1) Fix m; we use Theorem 1.1.2 (b) on n, for the property "$m \leq n$ or $n \leq m$". (2) For $n = 0$: $0 \leq m$, since $m = 0 + m$. (3) Suppose that $m \leq n$ or $n \leq m$. (4) If $n = m + j$, then $n + 1 = m + (j + 1)$, so $m \leq n + 1$. (5) If $n \leq m$, then $n + 1 \leq m$, because $n < n + 1$ and nothing lies strictly between n and $n + 1$. (6) So $m \leq n + 1$ or $n + 1 \leq m$. (7) By induction, the claim holds.

Solution. The conclusion is **true**: it is the totality proved inside Theorem 1.3.2, and it follows from trichotomy, Theorem 1.3.2 (a). The proof is **not** correct.

The failing line is (5), and it fails for two independent reasons.

First, $n \leq m$ includes the case $n = m$, where the conclusion $n + 1 \leq n$ is false (the Quick check after Proposition 1.3.1). This is a missing case, entry (N7) of Logic and Proof, Example (What is not allowed). *Counterexample to the step:* $m = n = 0$. The hypothesis of line (3) holds, since $0 \leq 0$, and so does $n \leq m$; but $n + 1 = 1 \leq 0$ is false, since $0 < 1$ (Theorem 1.3.2 (d)).

Second, even when $n < m$, the stated reason does not give $n + 1 \leq m$ without comparing m with $n + 1$: "nothing lies strictly between n and $n + 1$" excludes $n < m < n + 1$, and one still needs that $m < n + 1$, $m = n + 1$ or $n + 1 < m$. That is trichotomy, and Theorem 1.3.2 (b), which the reason quotes, was itself proved from trichotomy, that is, from the claim being proved. So the line rests on a fact not available at that point of the proof of Theorem 1.3.2: entry (N9).

Repair. Replace (5) by two cases. If $n = m$, then $m \leq n + 1$, since $n + 1 = m + 1$. If

$n < m$, then $m = n + j$ with $j \neq 0$ (Proposition 1.3.1 (a)), so $j = j' + 1$ for some j', and $m = (n + 1) + j'$ by Definition 1.2.1 (a) and Theorem 1.2.3 (a); this uses no comparison. Lines (1)–(4), (6) and (7) are correct; fixing m is harmless, since the step never needs the property for another m.

Marking scheme. (10 points) - 2: says that the conclusion is true and that the proof is not correct. - 3: names line (5) as the failing line, with either defect: the case $n = m$, or the hidden comparison of m with $n + 1$. - 2: classifies the error as (N7), a missing case, or as (N9), a line resting on an unavailable fact (the claim itself). - 2: gives a counterexample to the step, such as $m = n = 0$ (a case with $n = m$; the (N9) defect alone has none, since the conclusion of (5) is true when $n < m$). - 1: gives a repair through $m = n + j$ with $j \neq 0$. Fatal errors (0 for the step): declaring the conclusion false; blaming line (1).

1.4 All Peano Systems Are Alike

Section 1.1 fixed one Peano system and called it $(\mathbb{N}, 0, \mathrm{S})$. It is not the only one. The natural numbers of *Sets and Functions*, Chapter 3, sitting inside the familiar real numbers, form another; the numbers from 1 on, $(\mathbb{N} \setminus \{0\}, 1, \mathrm{S})$, form a third; and *Axiomatic Set Theory* builds a fourth out of sets alone (Section 1.5 describes how). So a question is left: does the choice matter? This section answers no. Any two Peano systems are copies of each other, by exactly one matching, and that matching also carries $+$, $\cdot$ and $<$ across. Then it says how the later chapters read what *Logic and Proof*, Chapter 4, and *Sets and Functions*, Chapters 3 and 4 proved about natural numbers.

1.4.1 Two Peano systems, and what "alike" should mean

Here is a second Peano system, inside $\mathbb{N}$ itself: $(\mathbb{N} \setminus \{0\}, 1, \mathrm{S})$, the natural numbers from 1 on, with the same successor. Exercise C1 of Section 1.1 asked for the check; here it is, in three lines.

- S maps $\mathbb{N} \setminus \{0\}$ into itself, since $\mathrm{S}(n) \neq 0$ by (P3) for $\mathbb{N}$.
- (P3) holds with $e = 1$: if $\mathrm{S}(n) = 1 = \mathrm{S}(0)$, then $n = 0$ by (P4) for $\mathbb{N}$, and $0 \notin \mathbb{N} \setminus \{0\}$. (P4) holds because S is injective on all of $\mathbb{N}$.
- (P5) holds: if $A \subseteq \mathbb{N} \setminus \{0\}$ contains 1 and is closed under S, then $\{0\} \cup A$ contains 0 and is closed under S (as $\mathrm{S}(0) = 1 \in A$), so it is $\mathbb{N}$ by Theorem 1.1.2 (a), and $A = \mathbb{N} \setminus \{0\}$.

Its starting point is 1, not 0, yet it counts the same way: $n \mapsto \mathrm{S}(n)$ pairs $0, 1, 2, \ldots$ with $1, 2, 3, \ldots$, sends start to start, and turns one step into one step.

That is the right meaning of "alike". For Peano systems (M, e, s) and (M', e', s'), call a bijection $\psi\colon M \to M'$ an **isomorphism** if

$$\psi(e) = e' \quad \text{and} \quad \psi(s(x)) = s'(\psi(x)) \quad \text{for every } x \in M.$$

In words: ψ renames the elements, sends start to start, and stepping then renaming is the same as renaming then stepping.

Warning. Being equinumerous is not enough. The bijection $\tau\colon \mathbb{N} \to \mathbb{N}$ that exchanges 1 and 2 and fixes every other number satisfies $\tau(0) = 0$, but $\tau(\mathrm{S}(0)) = \tau(1) = 2$, while $\mathrm{S}(\tau(0)) = \mathrm{S}(0) = 1$. So τ is a bijection from a Peano system to itself that is **not** an isomorphism: it does not respect the successor. Categoricity, below, is a statement about

isomorphisms, and it says much more than "same size".

1.4.2 The categoricity theorem

The theorem says that every Peano system is an isomorphic copy of $\mathbb{N}$, and so any two are copies of each other, in only one way.

Theorem 1.4.1. Categoricity of the Peano Axioms

Let (M, e, s) be a Peano system.

(a) There is **exactly one** function $\varphi\colon \mathbb{N} \to M$ with $\varphi(0) = e$ and $\varphi(\mathrm{S}(n)) = s(\varphi(n))$ for every $n \in \mathbb{N}$.

(b) This φ is a bijection.

(c) For any two Peano systems (M, e, s) and (M', e', s') there is **exactly one** bijection $\psi\colon M \to M'$ with $\psi(e) = e'$ and $\psi(s(x)) = s'(\psi(x))$ for every $x \in M$.

Try it first (ten minutes): given a Peano system (M, e, s), which map $\mathbb{N} \to M$ would you build, with which theorem, and how would you show that it is onto?

Idea (Scratch work). The attempt meets three obstacles, one for each part.

① **Which map.** It is forced: 0 must go to e, then 1 to $s(e)$, then 2 to $s(s(e))$, and so on. But "and so on" is not a definition. Theorem 1.2.1 turns it into one, with $X = M$, starting value e and step $x \mapsto s(x)$; it gives existence and uniqueness at once.

② **Onto.** We know nothing about the elements of M one by one, so we cannot show "each $x \in M$ is $s(\ldots s(e))$" directly. What we have is (P5) for M, which speaks about subsets. So look at the image of φ: it contains e and is closed under s, and (P5) does the rest, by Example (Method: induction through a closed subset).

③ **One-to-one.** Prove "$\varphi(m) = \varphi(n) \Rightarrow m = n$" by induction on n. If m is fixed before the induction, the hypothesis at n speaks about that one m, while the step needs it at the predecessor of m. So "for every m" goes inside the property. The base and the step then use (P3) and (P4) for M: a successor is never e, and equal successors have equal predecessors.

For (c), compose: go back from M to $\mathbb{N}$ along φ^{-1}, then forward to M' along φ'.

Givens	Goal
(M, e, s) a Peano system	$\varphi\colon \mathbb{N} \to M$, $\varphi(0) = e$, $\varphi(\mathrm{S}(n)) = s(\varphi(n))$, bijective
Theorem 1.2.1 with $X = M$, $a = e$, $g(n, x) = s(x)$	(a): such a φ exists, and only one
the image $\varphi(\mathbb{N})$ contains e and is closed under s	onto, by (P5) for M
$Q(n)$: for every m, $\varphi(m) = \varphi(n) \Rightarrow m = n$	one-to-one: base by (P3), step by (P4) for M

Before reading on: which axiom of M gives surjectivity, and which give injectivity?

Proof. (a) Let $g\colon \mathbb{N}\times M\to M$ be $g(n,x)\coloneqq s(x)$, a function because s is. By Theorem 1.2.1 with $X=M$ and $a=e$, there is exactly one $\varphi\colon \mathbb{N}\to M$ with $\varphi(0)=e$ and $\varphi(\mathrm{S}(n))=g(n,\varphi(n))=s(\varphi(n))$ for every $n\in\mathbb{N}$.

(b), onto. Let $B\coloneqq\varphi(\mathbb{N})\subseteq M$, the image of $\mathbb{N}$ under φ (Sets and Functions, Definition 2.5.1). Then $e=\varphi(0)\in B$. If $x\in B$, say $x=\varphi(n)$ with $n\in\mathbb{N}$, then $\mathrm{S}(n)\in\mathbb{N}$, since S maps $\mathbb{N}$ into $\mathbb{N}$, and $s(x)=s(\varphi(n))=\varphi(\mathrm{S}(n))\in B$. So B contains e and is closed under s, and (P5) for M gives $B=M$. Hence φ is surjective (Sets and Functions, Definition 2.2.2).

(b), one-to-one. Let $Q(n)$ be "for every $m\in\mathbb{N}$, if $\varphi(m)=\varphi(n)$, then $m=n$". We prove $Q(n)$ for every $n\in\mathbb{N}$ by Theorem 1.1.2 (b).

Base case. Let $m\in\mathbb{N}$ with $\varphi(m)=\varphi(0)=e$. Suppose $m\neq 0$. Then $m=\mathrm{S}(k)$ for some $k\in\mathbb{N}$ (Proposition 1.1.3 (a)), and $\varphi(m)=s(\varphi(k))\neq e$ by (P3) for M, a contradiction. So $m=0$.

Inductive step. Let $n\in\mathbb{N}$ with $Q(n)$, and let $m\in\mathbb{N}$ with $\varphi(m)=\varphi(\mathrm{S}(n))=s(\varphi(n))$. If $m=0$, then $s(\varphi(n))=\varphi(0)=e$, against (P3) for M. So $m\neq 0$, and $m=\mathrm{S}(k)$ for some $k\in\mathbb{N}$ (Proposition 1.1.3 (a)). Then $s(\varphi(k))=\varphi(m)=s(\varphi(n))$, so $\varphi(k)=\varphi(n)$ by (P4) for M, and $k=n$ by $Q(n)$, applied to k in the place of m. Hence $m=\mathrm{S}(k)=\mathrm{S}(n)$, which is $Q(\mathrm{S}(n))$.

So φ is injective (Sets and Functions, Definition 2.2.1), and with the first half it is a bijection (Sets and Functions, Definition 2.2.3).

(c) Let $\varphi\colon\mathbb{N}\to M$ and $\varphi'\colon\mathbb{N}\to M'$ be the maps of (a) for the two systems; both are bijections by (b). By Sets and Functions, Theorem 2.4.4, φ has an inverse $\varphi^{-1}\colon M\to\mathbb{N}$, and we set $\psi\coloneqq\varphi'\circ\varphi^{-1}$ (Sets and Functions, Definition 2.1.2). The two equations $\varphi\circ\varphi^{-1}=\mathrm{id}_M$ and $\varphi^{-1}\circ\varphi=\mathrm{id}_{\mathbb{N}}$ of Sets and Functions, Definition 2.4.2 also say that φ is an inverse of φ^{-1}, so φ^{-1} is a bijection by Sets and Functions, Theorem 2.4.4. Hence ψ is a bijection as a composite of bijections (Sets and Functions, Theorem 2.2.1 (c)).

ψ has the two properties. Since $\varphi(0)=e$, we have $\varphi^{-1}(e)=0$, so $\psi(e)=\varphi'(0)=e'$. Let $x\in M$ and $n\coloneqq\varphi^{-1}(x)$, so $x=\varphi(n)$. Then $s(x)=\varphi(\mathrm{S}(n))$, so $\varphi^{-1}(s(x))=\mathrm{S}(n)$, and

$$\psi(s(x))=\varphi'(\mathrm{S}(n))=s'(\varphi'(n))=s'(\psi(x)).$$

Uniqueness. Let $\chi\colon M\to M'$ be **any** function with $\chi(e)=e'$ and $\chi(s(x))=s'(\chi(x))$ for every $x\in M$. Then $\chi\circ\varphi\colon\mathbb{N}\to M'$ satisfies $(\chi\circ\varphi)(0)=\chi(e)=e'$ and

$$(\chi\circ\varphi)(\mathrm{S}(n))=\chi(s(\varphi(n)))=s'((\chi\circ\varphi)(n))$$

for every $n\in\mathbb{N}$. By the uniqueness in (a), for the system M', $\chi\circ\varphi=\varphi'$. So for every $x\in M$, $\chi(x)=\chi(\varphi(\varphi^{-1}(x)))=\varphi'(\varphi^{-1}(x))=\psi(x)$, and $\chi=\psi$ by Sets and Functions, Proposition 2.1.1. This proves the theorem.

■

The proof used one pattern three times: a map out of $\mathbb{N}$ came from the recursion theorem, two maps out of $\mathbb{N}$ were shown equal by the uniqueness half of that theorem, and every property of

the map came from an induction or from (P5). It deserves a name.

Example. Method: build by recursion, compare by induction

Template. (1) To make a map out of $\mathbb{N}$, give its value at 0 and say how the value at $\mathrm{S}(n)$ comes from the value at n; Theorem 1.2.1 builds the map, and only one. (2) To show two maps out of $\mathbb{N}$ equal, check that both satisfy the same two equations, and cite the uniqueness. (3) To prove a property of the map (onto, one-to-one, preserving an operation), induct on n, or apply (P5) to its image.

When to use it. Maps between Peano systems; operations defined by recursion; recognizing a formula as the map a recursion defines.

The usual failure. "And so on" in place of the theorem, entry (N13) of Logic and Proof, Example (What is not allowed). Citing uniqueness when the equations differ (another start or another step). Fixing m outside $Q(n)$, as ③ of the scratch work warned.

The uniqueness in (c) is stronger than stated: no function at all, bijective or not, other than ψ respects e and s.

Quick check. Functions $f, g\colon \mathbb{N} \to \mathbb{N}$ satisfy $f(0) = g(0) = 3$, $f(n+1) = f(n) + 2$ and $g(n+1) = g(n) + 2$ for every n. Which step of Example (Method: build by recursion, compare by induction) gives $f = g$? What is $f(2)$?

Solution. Step (2): both satisfy the equations of Theorem 1.2.1 with $X = \mathbb{N}$, $a = 3$ and step $h(n, x) = x + 2$, so they are equal by its uniqueness. From the equations, $f(1) = 3 + 2 = 5$ and $f(2) = 5 + 2 = 7$.

Step (2) also finds the map of the theorem without computing it. For the Peano system $(\mathbb{N} \setminus \{0\}, 1, \mathrm{S})$ of the first subsection, the map $n \mapsto \mathrm{S}(n)$ sends $\mathbb{N}$ into $\mathbb{N} \setminus \{0\}$ by (P3), sends 0 to $\mathrm{S}(0) = 1$, the starting point, and its value $\mathrm{S}(\mathrm{S}(n))$ at $\mathrm{S}(n)$ is the step $s = \mathrm{S}$ applied to its value $\mathrm{S}(n)$ at n. By the uniqueness in Theorem 1.4.1 (a), $\varphi(n) = \mathrm{S}(n) = n + 1$ (Definition 1.2.1 (c)). In particular $\varphi(3) = 4$; step by step, $\varphi(0) = 1$, $\varphi(1) = \mathrm{S}(1) = 2$, $\varphi(2) = 3$, $\varphi(3) = 4$.

Second reading · *Each axiom has a job*

Each axiom of M has a job in (b), and the structures of Example (Each axiom is needed) show each job failing. These are not Peano systems, so the theorem does not apply to them. But for each of them Theorem 1.2.1, with $X = M$, $a = e$ and $g(n, x) = s(x)$ exactly as in the proof of (a), still gives exactly one $\varphi\colon \mathbb{N} \to M$ with $\varphi(0) = e$ and $\varphi(\mathrm{S}(n)) = s(\varphi(n))$.

- In structure (i), $M = \{0\}$ with $s(0) = 0$, which fails (P3) only, φ is the constant map 0, not injective: $\varphi(1) = s(\varphi(0)) = 0 = \varphi(0)$.
- In structure (ii), $M = \{0, 1\}$ with $s(0) = s(1) = 1$, which fails (P4) only, $\varphi(1) = \varphi(2) = 1$, again not injective.
- In the two copies of $\mathbb{N}$, structure (iii), which fail (P5) only, φ is $n \mapsto (n, a)$ (step (2) of the box: this map sends 0 to $e = (0, a)$ and $(\mathrm{S}(n), a) = s(n, a)$). It is injective by Sets and

Functions, Definition 1.4.1, and it misses $(0, b)$, so it is not onto.

Exercise C2 asks the reverse question for the structure of Exercise (C2: Every element but one is a successor, used later): is there a map from it to $\mathbb{N}$ that respects the two structures?

Quick check. Let ψ be an isomorphism $(\mathbb{N}, 0, \mathrm{S}) \to (\mathbb{N}, 0, \mathrm{S})$. What is $\psi(3)$?

Solution. $\psi(3) = 3$. The identity $\mathrm{id}_{\mathbb{N}}$ is such an isomorphism, and by Theorem 1.4.1 (c) it is the only one, so $\psi = \mathrm{id}_{\mathbb{N}}$.

1.4.3 Arithmetic and order travel along the isomorphism

An isomorphism matches e and s. Addition and multiplication on $\mathbb{N}$ were built from 0 and S alone, by recursion (Definition 1.2.1), and $<$ can be described by $+$ (Proposition 1.3.1 (a)). So we expect the matching to carry them across, as long as the other system has operations defined by the same equations.

Proposition 1.4.2. Isomorphisms Preserve Arithmetic and Order

Let (M, e, s) be a Peano system, and let $\oplus, \otimes \colon M \times M \to M$ be functions such that, for all $x, y \in M$,

$$x \oplus e = x, \quad x \oplus s(y) = s(x \oplus y),$$
$$x \otimes e = e, \quad x \otimes s(y) = (x \otimes y) \oplus x.$$

Let $\varphi \colon \mathbb{N} \to M$ be the bijection of Theorem 1.4.1. Then for all $m, n \in \mathbb{N}$:

(a) $\varphi(m + n) = \varphi(m) \oplus \varphi(n)$;
(b) $\varphi(mn) = \varphi(m) \otimes \varphi(n)$;
(c) $m < n$ if and only if $\varphi(n) = \varphi(m) \oplus z$ for some $z \in M$ with $z \neq e$.

Idea. For (a) and (b), this is step (3) of Example (Method: build by recursion, compare by induction): fix m and induct on n, the variable the recursions run on. Each side obeys the same two equations, the base from 0 and the step from S, and φ turns one set of equations into the other. For (c), translate Proposition 1.3.1 (a), "$m < n$ iff $n = m + j$ with $j \neq 0$", through (a), using that φ is a bijection with $\varphi(0) = e$.

Proof. *(a)* Fix $m \in \mathbb{N}$; we induct on n (Theorem 1.1.2 (b)). *Base case:* $\varphi(m + 0) = \varphi(m) = \varphi(m) \oplus e = \varphi(m) \oplus \varphi(0)$, by Definition 1.2.1 (a) and the first equation for $\oplus$. *Step:* if $\varphi(m + n) = \varphi(m) \oplus \varphi(n)$, then

$$\begin{aligned} \varphi(m + \mathrm{S}(n)) &= \varphi(\mathrm{S}(m + n)) = s(\varphi(m + n)) \\ &= s(\varphi(m) \oplus \varphi(n)) = \varphi(m) \oplus s(\varphi(n)) = \varphi(m) \oplus \varphi(\mathrm{S}(n)), \end{aligned}$$

by Definition 1.2.1 (a), the equation of φ, the hypothesis, the second equation for $\oplus$, and the equation of φ again.

(b) Fix $m \in \mathbb{N}$ and induct on n. *Base case:* $\varphi(m \cdot 0) = \varphi(0) = e = \varphi(m) \otimes e = \varphi(m) \otimes \varphi(0)$, by Definition 1.2.1 (b) and the first equation for $\otimes$. *Step:* if $\varphi(mn) = \varphi(m) \otimes \varphi(n)$, then

$$\begin{aligned}\varphi(m \cdot \mathrm{S}(n)) &= \varphi(mn + m) = \varphi(mn) \oplus \varphi(m) \\ &= (\varphi(m) \otimes \varphi(n)) \oplus \varphi(m) = \varphi(m) \otimes s(\varphi(n)) = \varphi(m) \otimes \varphi(\mathrm{S}(n)),\end{aligned}$$

by Definition 1.2.1 (b), part (a) just proved, the hypothesis, the second equation for $\otimes$, and the equation of φ.

(c) By Proposition 1.3.1 (a), $m < n$ if and only if $n = m + j$ for some $j \in \mathbb{N}$ with $j \neq 0$.

($\Rightarrow$) Let $n = m + j$ with $j \neq 0$. By (a), $\varphi(n) = \varphi(m) \oplus \varphi(j)$. Put $z := \varphi(j)$. Since φ is injective (Theorem 1.4.1 (b)), $j \neq 0$ gives $z \neq \varphi(0) = e$.

($\Leftarrow$) Let $z \in M$ with $z \neq e$ and $\varphi(n) = \varphi(m) \oplus z$. Since φ is onto (Theorem 1.4.1 (b)), $z = \varphi(j)$ for some $j \in \mathbb{N}$, and $j \neq 0$ because $\varphi(0) = e \neq z$. By (a), $\varphi(n) = \varphi(m) \oplus \varphi(j) = \varphi(m + j)$, so $n = m + j$ by injectivity. Hence $m < n$. This proves the proposition.

■

So once a Peano system carries an addition and a multiplication obeying the recursive equations, φ is an isomorphism for them too, and the order defined from $\oplus$ as in (c) is carried along as well. Nothing about $\oplus$ and $\otimes$ was assumed beyond the four equations.

Quick check. For $(\mathbb{N} \setminus \{0\}, 1, \mathrm{S})$ and its map $\varphi(n) = n + 1$: $\varphi(2 + 3) = 6$, but $\varphi(2) + \varphi(3) = 3 + 4 = 7$. Does this contradict Proposition 1.4.2 (a)?

Solution. No. The proposition uses the $\oplus$ of the target system, not the $+$ of $\mathbb{N}$. That system carries one: $x \oplus y := x + j$ where $y = \mathrm{S}(j)$ (Proposition 1.1.3 (a)) lies in $\mathbb{N} \setminus \{0\}$, since $x + 0 = x \neq 0$ and $x + \mathrm{S}(k) = \mathrm{S}(x + k) \neq 0$ by (P3); it satisfies $x \oplus 1 = x$ and $x \oplus \mathrm{S}(y) = \mathrm{S}(x \oplus y)$ by Definition 1.2.1 (a). From these equations, $3 \oplus 1 = 3$, $3 \oplus 2 = \mathrm{S}(3 \oplus 1) = 4$, $3 \oplus 3 = 5$ and $3 \oplus 4 = 6$. So $\varphi(2) \oplus \varphi(3) = 3 \oplus 4 = 6 = \varphi(5)$, as the proposition says.

1.4.4 Identifying the two sets of natural numbers

Now apply all this to the natural numbers of *Sets and Functions*, Chapter 3. To keep them apart from $\mathbb{N}$, write $\mathbb{N}_{\mathbb{R}}$ for them. Two different things are true. Clause (a) below is about $\mathbb{N}_{\mathbb{R}}$, a subset of the familiar real numbers: it is a Peano system, so it is a copy of $\mathbb{N}$, and the proof uses the school arithmetic of $\mathbb{R}$. Clauses (b) and (c) are about $\mathbb{N}$ alone: the principles that *Logic and Proof*, Chapter 4 assumed and *Sets and Functions*, Chapter 3 proved inside $\mathbb{R}$ (induction, recursion, and the arithmetic of Sets and Functions, Lemma 3.1.1) are also theorems about $\mathbb{N}$, proved from Axiom 1.1.1 alone.

Corollary 1.4.3. Our $\mathbb{N}$ and the $\mathbb{N}$ of Sets and Functions

(a) **About $\mathbb{N}_{\mathbb{R}}$.** Write $\mathbb{N}_{\mathbb{R}} \subseteq \mathbb{R}$ for the set of natural numbers as *Sets and Functions*,

Section 3.1 describes it, with (($\mathbb{N}$)0), ($\mathbb{N}$S) and Logic and Proof, Axiom 4.2.1. With $e \coloneqq 0$ and $s(n) \coloneqq n+1$, it is a Peano system. There is exactly one bijection $\varphi\colon \mathbb{N} \to \mathbb{N}_{\mathbb{R}}$ with $\varphi(0) = 0$ and $\varphi(\mathrm{S}(n)) = \varphi(n) + 1$ for every $n \in \mathbb{N}$. For all $m, n \in \mathbb{N}$ it satisfies

$$\varphi(m+n) = \varphi(m) + \varphi(n), \qquad \varphi(mn) = \varphi(m)\varphi(n),$$

and $m < n$ if and only if $\varphi(m) < \varphi(n)$; and it sends each numeral of $\mathbb{N}$ to the natural number of the same name.

(b) **About $\mathbb{N}$, proved from Axiom 1.1.1.** The statements of Logic and Proof, Axiom 4.2.1 and Sets and Functions, Theorem 3.3.1, read on $\mathbb{N}$ with $n + 1 \coloneqq \mathrm{S}(n)$, are Theorem 1.1.2 (b) and Theorem 1.2.1. Induction from a base $b \in \mathbb{N}$ (Sets and Functions, Theorem 3.2.1 with $n_0 = b$) is Proposition 1.3.1 (g).

(c) Each clause of Sets and Functions, Lemma 3.1.1 holds on $\mathbb{N}$, with $n - m$, for $m \leq n$, read as the unique $j \in \mathbb{N}$ with $n = m + j$ of Proposition 1.3.1 (a).

Idea. For (a), check (P3)–(P5) for $\mathbb{N}_{\mathbb{R}}$ with the tools of *Sets and Functions*, Section 3.1, and the equations of Proposition 1.4.2 with school arithmetic; then Theorem 1.4.1 and that proposition do the rest. Parts (b) and (c) are a matter of reading statements side by side and pointing to the label that proves each one on $\mathbb{N}$.

Before reading on: which clause do the later chapters actually use, when they build $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$?

Proof. *(a)* The map s sends $\mathbb{N}_{\mathbb{R}}$ into $\mathbb{N}_{\mathbb{R}}$ by ($\mathbb{N}$S). (P3): for $n \in \mathbb{N}_{\mathbb{R}}$, $n \geq 0$ by Sets and Functions, Lemma 3.1.1 (a), so $n + 1 \geq 1 > 0$ and $n + 1 \neq 0$. (P4): $m + 1 = n + 1$ gives $m = n$, by subtracting 1 in $\mathbb{R}$. (P5): a subset of $\mathbb{N}_{\mathbb{R}}$ that contains 0 and is closed under $n \mapsto n + 1$ is $\mathbb{N}_{\mathbb{R}}$; this is the set form of Logic and Proof, Axiom 4.2.1, the remark at the start of *Sets and Functions*, Section 3.1. So $(\mathbb{N}_{\mathbb{R}}, 0, s)$ is a Peano system (Definition 1.1.1), and Theorem 1.4.1 (a), (b) give exactly one φ as stated, and it is a bijection.

For the operations, take $\oplus$ and $\otimes$ to be the addition and multiplication of $\mathbb{N}_{\mathbb{R}}$; they are functions $\mathbb{N}_{\mathbb{R}} \times \mathbb{N}_{\mathbb{R}} \to \mathbb{N}_{\mathbb{R}}$ by Sets and Functions, Lemma 3.1.1 (b). The four equations of Proposition 1.4.2 read $x + 0 = x$, $x + (y + 1) = (x + y) + 1$, $x \cdot 0 = 0$ and $x(y + 1) = xy + x$, which hold in $\mathbb{R}$. That proposition gives $\varphi(m + n) = \varphi(m) + \varphi(n)$ and $\varphi(mn) = \varphi(m)\varphi(n)$.

For the order we first show: for $x, y \in \mathbb{N}_{\mathbb{R}}$, $x < y$ if and only if $y = x + z$ for some $z \in \mathbb{N}_{\mathbb{R}}$ with $z \neq 0$. If $x < y$, then $z \coloneqq y - x \in \mathbb{N}_{\mathbb{R}}$ by Sets and Functions, Lemma 3.1.1 (d), and $z \neq 0$ since $y \neq x$. Conversely, $z \neq 0$ gives $z \geq 1 > 0$ by Sets and Functions, Lemma 3.1.1 (a), so $y = x + z > x$. Now Proposition 1.4.2 (c), with $e = 0$, says that $m < n$ if and only if $\varphi(n) = \varphi(m) + z$ for some $z \in \mathbb{N}_{\mathbb{R}}$ with $z \neq 0$, that is, if and only if $\varphi(m) < \varphi(n)$.

Finally, $\varphi(0) = 0$, and each numeral of $\mathbb{N}$ is the successor of the one before (Definition 1.2.1 (c)), so $\varphi(1) = \varphi(\mathrm{S}(0)) = 0 + 1 = 1$, $\varphi(2) = \varphi(\mathrm{S}(1)) = 1 + 1 = 2$, and so on, one step for each numeral written.

(b) By Definition 1.2.1 (c), $n + 1 = \mathrm{S}(n)$ for every $n \in \mathbb{N}$. So, read on $\mathbb{N}$, the statement of Logic and Proof, Axiom 4.2.1 becomes Theorem 1.1.2 (b), and the statement of Sets and Functions, Theorem 3.3.1 is word for word Theorem 1.2.1. Induction from a base $b \in \mathbb{N}$ is Proposition 1.3.1 (g).

(c) We take the clauses of Sets and Functions, Lemma 3.1.1 in turn, for $m, n \in \mathbb{N}$.

- (a) "$n \geq 0$; if $n \neq 0$, then $n \geq 1$" is Theorem 1.3.2 (d).
- (b) "$m + n \in \mathbb{N}$ and $mn \in \mathbb{N}$" holds because $+$ and $\cdot$ are functions $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ (Definition 1.2.1).
- (c) If $n \neq 0$, then $1 \leq n$ by Theorem 1.3.2 (d), so $n = 1 + j$ for exactly one $j \in \mathbb{N}$ by Proposition 1.3.1 (a); this j is $n - 1$, and it lies in $\mathbb{N}$.
- (d) If $m \leq n$, then $n - m \in \mathbb{N}$ by Proposition 1.3.1 (a).
- (e) No $k \in \mathbb{N}$ has $n < k < \mathrm{S}(n)$, by Theorem 1.3.2 (b). For the second form, let $m < n$. By trichotomy (Theorem 1.3.2 (a)), $\mathrm{S}(m) < n$, $\mathrm{S}(m) = n$ or $n < \mathrm{S}(m)$. The last gives $n \leq m$ by Theorem 1.3.2 (b), and then $n \leq m < n$ gives $n < n$ by transitivity, which (a) of that theorem rules out. So $m + 1 = \mathrm{S}(m) \leq n$.

This proves the corollary.

■

Clause (a) is the honest sense in which the choice of Peano system does not matter: $\mathbb{N}$ and $\mathbb{N}_{\mathbb{R}}$ are copies of each other by a unique isomorphism, which also matches $+$, $\cdot$ and $<$. But (a) is a theorem about $\mathbb{N}_{\mathbb{R}}$, and its proof used the school arithmetic of $\mathbb{R}$ (in (P3), (P4), the four equations, the step describing $<$ by $+$, and the numerals). The answer to the prompt is therefore (b), with (c): when a later chapter applies induction or recursion to $\mathbb{N}$, it uses a theorem proved here from Axiom 1.1.1, and $\mathbb{R}$ never enters.

Warning. **The isomorphism carries 0, the successor, $+$, $\cdot$ and $<$, nothing else.** In $\mathbb{N}_{\mathbb{R}}$, $3 - 5$ and $3/5$ are real numbers; in $\mathbb{N}$ neither exists until Chapters 2 and 3 build them. In the construction from sets of Section 1.5, $2 \in 3$ is true; in $\mathbb{N}_{\mathbb{R}}$ it means nothing. No later theorem asks such questions.

1.4.5 How later chapters cite *Logic and Proof* and *Sets and Functions*

In the rest of this volume, $\mathbb{N}$ is the Peano system of Axiom 1.1.1, with the 0, $+$, $\cdot$ and $<$ of this chapter. The later chapters build $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ from it, and they cite results of *Logic and Proof*, Chapter 4, and *Sets and Functions*, Chapters 3 and 4 about natural numbers. Those proofs used the arithmetic of the real numbers freely (*Sets and Functions*, Chapter 3 takes the arithmetic and order of $\mathbb{R}$ as school facts, and Sets and Functions, Theorem 4.3.1, for example, uses $1-b$), so they cannot be moved to $\mathbb{N}$ word for word. The **citing rule** is this: each fact about natural numbers such a proof uses is read through its counterpart proved in this chapter, namely Corollary 1.4.3 (b) and (c) for induction, recursion and Sets and Functions, Lemma 3.1.1, Theorem 1.3.3 for Sets and Functions, Theorem 3.1.3, Theorem 1.2.3, Theorem 1.3.2 and Proposition 1.3.1 for the rest of school arithmetic and order (closure, cancellation, trichotomy, $n - m$ for $m \leq n$), and Proposition 1.3.4 for powers and parity. Read this way, the results that later chapters cite fall into three groups, which later chapters refer to by number: (i) results that run on $\mathbb{N}$ through the citing rule, the facts about finite and countable sets, citable anywhere; (ii) results that also need Proposition 1.3.4 and the cancellation of Proposition 1.3.1 (b), (f), citable only in review exercises; (iii) templates used later over an ordered field. The fold lists each group in full.

Second reading · *The three groups in full*

(i) **Results that run on $\mathbb{N}$ through the citing rule above.** These are Sets and Functions, Proposition 3.3.8 (for its $\mathbb{N}$ side, which is Sets and Functions, Theorem 3.2.1 with base 1 and Sets and Functions, Lemma 3.1.1 (e)), Sets and Functions, Definition 4.1.2, Sets and Functions, Definition 4.1.3, Sets and Functions, Corollary 4.1.3, Sets and Functions, Theorem 4.1.9, Sets and Functions, Lemma 4.2.1, Sets and Functions, Theorem 4.2.2, Sets and Functions, Proposition 4.2.3, Sets and Functions, Proposition 4.2.4 and Sets and Functions, Theorem 4.3.1 (where $\{0,1\}^{\mathbb{N}}$ is indexed by $\mathbb{N}$, and $1-b$ on $\{0,1\}$ is read by Proposition 1.3.1 (a)), and Sets and Functions, Exercise (C2: Counting products and functions) (a), cited later only from an exercise solution (its $\mathbb{N}$ side is an induction on m, read by Corollary 1.4.3 (b), $mn \in \mathbb{N}$, read by Corollary 1.4.3 (c), and $0 = 0 \cdot n$ and $mn+n = (m+1)n$, read by Theorem 1.2.3 (e)). Running a result on $\mathbb{N}$ includes the results its proof cites: Sets and Functions, Theorem 4.1.2, Sets and Functions, Proposition 4.1.5, Sets and Functions, Proposition 4.1.6, Sets and Functions, Corollary 4.1.4, Sets and Functions, Corollary 4.1.7 and the count "$\{0,\dots,x\}$ has $x+1$ elements" in the examples after Sets and Functions, Definition 4.1.2 run through the same citing rule, and they belong to group (i) themselves. A later chapter may cite the results of group (i) anywhere.

(ii) **Results that use school facts, and need Proposition 1.3.4 on $\mathbb{N}$.** These are Sets and Functions, Theorem 4.2.6 (powers by Sets and Functions, Definition 3.3.2 and Sets and Functions, Proposition 3.3.4 (a), (d), (e) over the school $\mathbb{R}$, parity through Sets and Functions, Theorem 3.2.3 on the school $\mathbb{Z}$, and cancellation in $\mathbb{R}$), Sets and Functions, Proposition 4.2.5 (parity through Sets and Functions, Theorem 3.2.3), and through them Sets and Functions, Corollary 4.2.7 (a). On $\mathbb{N}$, the powers are read as Proposition 1.3.4 (a)–(c), the parity claim as its (d), and cancellation as Proposition 1.3.1 (b), (f); with that, each of these proofs runs on $\mathbb{N}$. A later chapter may cite them only in a review (R) exercise; anywhere else, a construction that needs one of them proves it on $\mathbb{N}$ first. Part (b) of Sets and Functions, Corollary 4.2.7 is not in this group and is not checked on $\mathbb{N}$ here: *Sets and Functions*, Chapter 4 treats its list $A_1,\dots,A_k$ as a family in the everyday sense of Sets and Functions, Definition 1.3.3, with no set holding it, and such families are left to *Axiomatic Set Theory*. A later chapter that needs (b) proves it on $\mathbb{N}$ first. Sets and Functions, Theorem 4.2.8 is a statement about the school $\mathbb{Z}$ and $\mathbb{Q}$, and is never applied to $\mathbb{N}$.

(iii) **Templates.** Sets and Functions, Definition 3.3.1 and Sets and Functions, Proposition 3.3.4 are used later only as patterns, with the school $\mathbb{R}$ replaced by an ordered field F. Their $\mathbb{N}$ side is read through the citing rule above, and their field side is proved again from the axioms of F. The parts that need $\mathbb{Z}$ (Sets and Functions, Proposition 3.3.4 (d) for an integer base, and the general index of Sets and Functions, Definition 3.3.1, which uses Sets and Functions, Proposition 3.1.2) are not part of the pattern.

So everything that *Logic and Proof*, Chapter 4, and *Sets and Functions*, Chapter 3 assumed about natural numbers (that 0 is one, that $n+1$ is one whenever n is, induction, recursion, and the arithmetic of Sets and Functions, Lemma 3.1.1) is now proved from the one assumption Axiom 1.1.1. Section 1.5 says where a Peano system comes from, and Chapter 2 builds $\mathbb{Z}$ from $\mathbb{N}$, with pairs and classes of pairs.

1.4.6 Exercises

1.4.6.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Theorem 1.4.1.

(a) Where in the proof is (P5) for M used, and where are (P3) and (P4) for M used?

(b) Where is (P4) for $\mathbb{N}$ used? (It is not visible in the proof itself.)

(c) Run φ of part (a) for the system $(\mathbb{N} \setminus \{0\}, 1, \mathrm{S})$ on 0, 1 and 2, from the two equations alone.

(d) State the idea of the proof of (a) and (b) in two sentences.

Solution.

(a) (P5) for M is used once, for surjectivity: the image $\varphi(\mathbb{N})$ contains e and is closed under s, so it is M. (P3) for M is used for injectivity, in the base case ($\varphi(\mathrm{S}(k)) = s(\varphi(k)) \neq e$, so only $m = 0$ has $\varphi(m) = e$) and at the start of the inductive step (to exclude $m = 0$). (P4) for M is used in the inductive step, to pass from $s(\varphi(k)) = s(\varphi(n))$ to $\varphi(k) = \varphi(n)$.

(b) Inside Theorem 1.2.1: in its claim's step, a pair stepping to first coordinate $\mathrm{S}(n)$ has first coordinate n, by (P4) for $\mathbb{N}$.

(c) $\varphi(0) = 1$; $\varphi(1) = \varphi(\mathrm{S}(0)) = \mathrm{S}(\varphi(0)) = \mathrm{S}(1) = 2$; $\varphi(2) = \varphi(\mathrm{S}(1)) = \mathrm{S}(2) = 3$.

(d) The map is forced at 0 and at each successor, so the recursion theorem builds it and makes it unique. It is onto because its image contains e and is closed under s, so (P5) for M applies; it is one-to-one by an induction whose base and step are (P3) and (P4) for M.

1.4.6.2 B. Practice

Exercise. B1: The even numbers

Let $E := \{2n : n \in \mathbb{N}\}$, with $e := 0$ and $s(x) := x + 2$ for $x \in E$. Show that $(E, 0, s)$ is a Peano system, and that the map φ of Theorem 1.4.1 is $\varphi(n) = 2n$.

Solution. Here $E = \{x \in \mathbb{N} : x = 2n \text{ for some } n \in \mathbb{N}\}$. By Definition 1.2.1 (b), $2 \cdot \mathrm{S}(n) = 2n + 2$ for every $n \in \mathbb{N}$. So s maps E into E: $s(2n) = 2n + 2 = 2 \cdot \mathrm{S}(n) \in E$.

(P3): $2n + 2 = 2n + \mathrm{S}(1) = \mathrm{S}(2n + 1)$ by Definition 1.2.1 (a), and a successor is never 0 ((P3) for $\mathbb{N}$). (P4): if $x + 2 = y + 2$, then $x = y$ by Proposition 1.3.1 (b). (P5): let $A \subseteq E$ with $0 \in A$ and $x + 2 \in A$ for every $x \in A$. Let $B := \{n \in \mathbb{N} : 2n \in A\}$. Then $0 \in B$, since $2 \cdot 0 = 0 \in A$; and if $n \in B$, then $2 \cdot \mathrm{S}(n) = 2n + 2 \in A$, so $\mathrm{S}(n) \in B$. By Theorem 1.1.2 (a), $B = \mathbb{N}$, so every $2n$ lies in A, and $A = E$.

The map $n \mapsto 2n$ from $\mathbb{N}$ to E sends 0 to $2 \cdot 0 = 0 = e$, and $2 \cdot \mathrm{S}(n) = 2n + 2 = s(2n)$. By the uniqueness in Theorem 1.4.1 (a), it is φ. In particular, by (b) of that theorem, $n \mapsto 2n$ is a bijection $\mathbb{N} \to E$.

Exercise. B2: The identification at work

Let $\varphi\colon \mathbb{N} \to \mathbb{N}_{\mathbb{R}}$ be the bijection of Corollary 1.4.3 (a).

(a) Compute $\varphi(3)$ step by step from the two equations of φ.

(b) Compute $\varphi(2 \cdot 3)$ in two ways: from (a) of the corollary, and by unwinding $2 \cdot 3$ in $\mathbb{N}$ to a numeral first.

Solution.

(a) Using $\varphi(0) = 0$ and $\varphi(\mathrm{S}(n)) = \varphi(n) + 1$, with $3 = \mathrm{S}(2)$, $2 = \mathrm{S}(1)$, $1 = \mathrm{S}(0)$ (Definition 1.2.1 (c)):

$$\varphi(3) = \varphi(2) + 1 = (\varphi(1) + 1) + 1 = ((0 + 1) + 1) + 1 = 3.$$

(b) By the corollary, $\varphi(2 \cdot 3) = \varphi(2)\varphi(3) = 2 \cdot 3 = 6$ in $\mathbb{N}_{\mathbb{R}}$, where $\varphi(2) = 2$ as in (a). In $\mathbb{N}$, $2 \cdot 3 = 2 \cdot 2 + 2 = (2 \cdot 1 + 2) + 2 = ((2 \cdot 0 + 2) + 2) + 2 = 6$, by Definition 1.2.1 (b) three times and then $2 \cdot 0 = 0$, $0 + 2 = 2$, $2 + 2 = 4$, $4 + 2 = 6$ from the equations for $+$. Then $\varphi(6) = 6$ by six steps as in (a). The two answers agree, as the corollary says they must.

Exercise. B3: Prove or disprove

Prove each true statement by Example (Method: build by recursion, compare by induction), and refute each false one with a counterexample.

(a) Exactly one function $f\colon \mathbb{N} \to \mathbb{N}$ has $f(0) = 1$ and $f(n + 1) = 2f(n)$ for every n.

(b) Exactly one function $f\colon \mathbb{N} \to \mathbb{N}$ has $f(0) = 0$ and $f(n + 1) = f(n) + 2n + 1$ for every n, and it is $f(n) = n \cdot n$.

(c) Some injective function $f\colon \mathbb{N} \to \mathbb{N}$ has $f(n + 1) = f(n)$ for every n.

(d) If $f, g\colon \mathbb{N} \to \mathbb{N}$ satisfy $f(0) = g(0)$, $f(n + 1) = f(n) + n$ and $g(n + 1) = g(n) + n$ for every n, then $f = g$.

(e) Every bijection $\psi\colon \mathbb{N} \to \mathbb{N}$ with $\psi(0) = 0$ is an isomorphism $(\mathbb{N}, 0, \mathrm{S}) \to (\mathbb{N}, 0, \mathrm{S})$.

Solution.

(a) **True.** Step (1) of the box: Theorem 1.2.1 with $X = \mathbb{N}$, $a = 1$ and $g(n, x) = 2x$ gives exactly one such f.

(b) **True.** Theorem 1.2.1 with $X = \mathbb{N}$, $a = 0$ and $g(n, x) = x + 2n + 1$ gives exactly one such f. The map $n \mapsto n \cdot n$ satisfies both equations: $0 \cdot 0 = 0$, and

$$\begin{aligned}(n + 1)(n + 1) &= (n + 1)n + (n + 1) = (n \cdot n + n) + (n + 1) \\ &= (n \cdot n + (n + n)) + 1 = (n \cdot n + 2n) + 1,\end{aligned}$$

by Definition 1.2.1 (b), Theorem 1.2.3 (e), associativity (c) twice, and $2n = (1 + 1)n = 1 \cdot n + n = n + n$ by (e). So it is f, by step (2) of the box.

(c) **False.** For any such f, $f(1) = f(0)$ with $1 \neq 0$, so f is not injective.

(d) **True.** Both satisfy the equations of Theorem 1.2.1 with $X = \mathbb{N}$, $a = f(0)$ and step

$h(n, x) = x + n$; by its uniqueness (step (2) of the box), $f = g$.

(e) **False.** The bijection τ of the first warning exchanges 1 and 2 and has $\tau(0) = 0$, but $\tau(\mathrm{S}(0)) = 2 \neq 1 = \mathrm{S}(\tau(0))$.

1.4.6.3 C. Going deeper

Exercise. C1: Writing out categoricity

Without looking at the text, prove part (b) of Theorem 1.4.1: for a Peano system (M, e, s), the map $\varphi\colon \mathbb{N} \to M$ with $\varphi(0) = e$ and $\varphi(\mathrm{S}(n)) = s(\varphi(n))$ is a bijection.

Solution (Hint 1). For surjectivity, use (P5) for M on a set built from φ. For injectivity, induct on n; decide first where "for every m" goes.

Solution (Hint 2). The induction for injectivity starts: "Let $Q(n)$ be: for every $m \in \mathbb{N}$, $\varphi(m) = \varphi(n)$ implies $m = n$."

Solution. Onto. The image $B := \varphi(\mathbb{N})$ (Sets and Functions, Definition 2.5.1) contains $e = \varphi(0)$, and $x = \varphi(n) \in B$ gives $s(x) = \varphi(\mathrm{S}(n)) \in B$. By (P5) for M, $B = M$.

One-to-one. By Theorem 1.1.2 (b) we prove, for every n, $Q(n)$: "for every $m \in \mathbb{N}$, $\varphi(m) = \varphi(n)$ implies $m = n$". For $Q(0)$, let $\varphi(m) = e$; an $m \neq 0$ is $\mathrm{S}(k)$ (Proposition 1.1.3 (a)), with $\varphi(m) = s(\varphi(k)) \neq e$ by (P3) for M; so $m = 0$. For the step, assume $Q(n)$ and $\varphi(m) = s(\varphi(n))$. Then $m \neq 0$, as e is not a value of s (P3 for M); so $m = \mathrm{S}(k)$, and $s(\varphi(k)) = s(\varphi(n))$ gives $\varphi(k) = \varphi(n)$ by (P4) for M, hence $k = n$ by $Q(n)$ used at k, and $m = \mathrm{S}(n)$. So φ is injective, and a bijection.

Marking scheme. (10 points) - 3: surjectivity: shows the image of φ contains e and is closed under s, and cites (P5) for M. - 2: sets up an induction (on either variable, or by least counterexample) whose hypothesis is strong enough to use at the predecessor, e.g. $Q(n)$ with "for every m" inside, or the difference $j \neq 0$ fixed and induction on m. - 2: base case, via $m = \mathrm{S}(k)$ for $m \neq 0$ and (P3) for M. - 3: inductive step, using (P3) for M to get $m \neq 0$, (P4) for M to cancel s, and the induction hypothesis at the predecessor k. Fatal errors (0 for the part concerned): an induction whose hypothesis is then applied at a value it does not cover (typically: m fixed outside Q, then used at k); using injectivity of φ to prove injectivity; using (P3), (P4) or (P5) **for** $\mathbb{N}$ where the axiom for M is needed. A proof of injectivity by Theorem 1.3.3 (a least n with some $m \neq n$, $\varphi(m) = \varphi(n)$) earns the same 7 points, mapped to base, step and hypothesis in the same way. So does the fixed-difference route (reducing to $\varphi(m + j) \neq \varphi(m)$ for $j \neq 0$ by Theorem 1.3.2 (a) and Proposition 1.3.1 (a), and inducting on m): its base uses (P3) for M, its step (P4) for M and the hypothesis at m, and the points are mapped in the same way.

Exercise. C2: Without induction, no recursion

Let (M, e, s) be the structure of Exercise (C2: Every element but one is a successor, used

later): $M = \mathbb{N} \times \{a, b, c\}$, $e = (0, a)$, $s(n, a) = (\mathrm{S}(n), a)$, $s(n, b) = (\mathrm{S}(n), b)$, $s(0, c) = (0, b)$ and $s(\mathrm{S}(k), c) = (k, c)$. Show that there is **no** function $f\colon M \to \mathbb{N}$ with $f(e) = 0$ and $f(s(x)) = \mathrm{S}(f(x))$ for every $x \in M$. Which Peano axiom, missing in M, is responsible?

Solution. Suppose such an f exists. For every $k \in \mathbb{N}$, $s(\mathrm{S}(k), c) = (k, c)$, so

$$f((k, c)) = \mathrm{S}\Big(f((\mathrm{S}(k), c))\Big).$$

Let $T \coloneqq \{f((k, c)) : k \in \mathbb{N}\} \subseteq \mathbb{N}$; it is non-empty, since it contains $f((0, c))$. By Theorem 1.3.3, T has a least element $t = f((k_0, c))$. But $u \coloneqq f((\mathrm{S}(k_0), c)) \in T$, and $t = \mathrm{S}(u) = u + 1$, so $u < t$ (Proposition 1.3.1 (a)). This contradicts the leastness of t, since $t \leq u$ and $u < t$ would give $u < u$ (Theorem 1.3.2 (a)). So no such f exists. (The condition $f(e) = 0$ was not even used: no map at all turns s into S.)

M satisfies (P3) and (P4) and fails only (P5) (Exercise (C2: Every element but one is a successor, used later) (b), (d)). So (P5) is what makes the recursion theorem possible: the downward chain $(0, c), (1, c), (2, c), \ldots$, each element the successor of the next, would need values in $\mathbb{N}$ that decrease forever.

1.5 Where $\mathbb{N}$ Comes From

Section 1.1 assumed one thing: that a Peano system exists (Axiom 1.1.1). Why should we believe it? Inside the real numbers there is one, the set $\mathbb{N}_{\mathbb{R}}$ of *Sets and Functions* (Corollary 1.4.3 (a)); but that answer presupposes $\mathbb{R}$, an object far richer than the one it is meant to support. *Axiomatic Set Theory* gives a better answer: it builds a Peano system out of sets alone. This section describes that construction, without proofs. Nothing later in the book depends on it.

1.5.1 Sets as the raw material

Sets and Functions used sets as a working language. Its Section 1.2 stated a short list of working rules (W1)–(W6) that say which sets exist: a set may be cut out of a given set by a property (Sets and Functions, Definition 1.2.3), finitely many objects may be listed, sets have unions, the subsets of a set form a set (Sets and Functions, Definition 1.3.2), pairs from two sets form a product, and the number sets are sets. It did not derive these rules from more basic principles, and it took numbers and pairs as given objects. Two questions were left open: where these rules come from, and whether everything else, numbers and pairs included, is a set too.

Set theory answers both questions with a list of axioms. The construction below uses seven of them, each of which can be said in one line:

- **Extensionality.** Two sets with the same elements are equal (Sets and Functions, Definition 1.2.1, now an axiom).
- **Empty set.** There is a set with no elements, written $\emptyset$.
- **Pairing.** For any sets a, b there is a set $\{a, b\}$ whose elements are exactly a and b.
- **Union.** For any set $\mathcal{S}$ of sets there is a set $\bigcup \mathcal{S}$ of the elements of its members; with pairing it gives $a \cup b$.
- **Power set.** For any set X there is a set $\mathcal{P}(X)$ of all its subsets.
- **Specification.** For any set A and any property, the elements of A with that property form a set.

- **Infinity.** Some set contains $\emptyset$ and is closed under one particular step, the successor below.

Axiomatic Set Theory states the rest of the list (replacement, foundation and choice) and proves everything this section only describes.

Two features of the list matter for us. First, every axiom but extensionality says that **some set exists**. All of them except the empty set and infinity build their set out of sets already given; infinity is the one axiom that hands us a set no finite chain of building steps reaches. Second, there is nothing on the list except sets: no numbers, no points, no pairs as separate objects. So if natural numbers are to exist in this world, they must be particular sets. The question is which ones.

The answer is a matter of choice, not discovery. We need a set ω, an element of it to call zero and a function on it to call the successor, such that the three satisfy (P3), (P4) and (P5) of Definition 1.1.1. By Theorem 1.4.1, any two such choices are copies of each other, so it does not matter much which one we make. It only has to be made from the empty set by the operations on the list.

1.5.2 Zero and the successor as sets

There is only one set we are given outright, so it plays zero. For the successor we need a step that always produces something new, and the axioms offer a natural one: add the set itself as a new element. From here through Exercise B2, 0 and S stand for these sets, except in a sentence that names $\mathbb{N}$, $\mathbb{N}_{\mathbb{R}}$ or $\mathbb{R}$, where they keep their usual meanings; elsewhere in the section, and in the rest of the book, they mean what they meant before:

$$0 \coloneqq \emptyset, \qquad \mathrm{S}(x) \coloneqq x \cup \{x\}.$$

The right-hand side is a set by pairing (with $a = b = x$) and union. The first few numerals then come out as follows:

$$1 = \mathrm{S}(0) = \emptyset \cup \{\emptyset\} = \{0\},$$
$$2 = \mathrm{S}(1) = \{0\} \cup \{1\} = \{0, 1\},$$
$$3 = \mathrm{S}(2) = \{0, 1\} \cup \{2\} = \{0, 1, 2\}.$$

The pattern is the point of the choice: *each numeral is the set of the numerals before it.* So n has exactly n elements, and "m comes before n" can be read as $m \in n$. Thus $2 \in 3$, and also $2 \subseteq 3$, since every element of 2 is an element of 3.

This reading is a feature of this construction, not of the natural numbers. In $\mathbb{N}_{\mathbb{R}}$ the question "is $2 \in 3$?" has no meaning, and a warning in Section 1.4 records that no theorem of this book asks it. These sets are called the **von Neumann numerals**.

Quick check. Write 3 using only $\emptyset$ and braces. Is $1 \subseteq 3$? Is $3 \in 1$?

Solution. $3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$. Yes, $1 \subseteq 3$: the only element of $1 = \{0\}$ is 0, and $0 \in 3$. No, $3 \notin 1$: the only element of 1 is $\emptyset$, and $3 \neq \emptyset$, since 3 has elements.

1.5.3 The least inductive set

We now want a set holding all the numerals and nothing else. The other axioms build 0, then 1, then 2, one at a time, each in finitely many steps. A set holding all of them at once is what

the axiom of infinity is for, and its "particular step" is S.

A set I is **inductive** if $\emptyset \in I$ and $\mathrm{S}(x) \in I$ whenever $x \in I$. The axiom of infinity says that an inductive set exists. It says nothing more, and in particular it does not say that its inductive set holds only numerals.

The cure is the move of Sets and Functions, Proposition 1.3.2, with I as the ambient set, $\emptyset$ as the starting element and S as the step. The inductive subsets of I form a set, cut out of $\mathcal{P}(I)$ by specification, and their intersection is the smallest inductive subset of I. *Axiomatic Set Theory* checks that it lies inside every inductive set, not only inside those contained in I, and so does not depend on which I the axiom supplied. This least inductive set is called ω.

Is $(\omega, \emptyset, \mathrm{S})$ a Peano system? The three axioms fare differently.

- **(P5)** is leastness itself. A subset of ω that contains $\emptyset$ and is closed under S is inductive, so it contains the least inductive set ω; hence it is ω.
- **(P3)** is immediate. Every $\mathrm{S}(x)$ has x as an element, and $\emptyset$ has none, so $\mathrm{S}(x) \neq \emptyset$.
- **(P4)** is the real work. From $x \cup \{x\} = y \cup \{y\}$ we want $x = y$, and for arbitrary sets this needs care. For elements of ω it holds because each $n \in \omega$ has two properties that the pattern above suggests: an element of an element of n is an element of n, and $n \notin n$. *Axiomatic Set Theory* proves both by induction on ω, and (P4) follows from them.

With all three in hand, $(\omega, \emptyset, \mathrm{S})$ is a Peano system. By Theorem 1.4.1 it is a copy of $(\mathbb{N}, 0, \mathrm{S})$, by exactly one isomorphism, which carries the von Neumann numeral n to the natural number n. So the assumption Axiom 1.1.1 is not a new article of faith: it follows from the axioms for sets. A reader who trusts them may read Axiom 1.1.1 as a theorem.

Warning. **An inductive set need not be ω.** The axiom of infinity supplies *some* inductive set, which may hold much else. In the same way, $\mathbb{R}$ contains 0 and is closed under $x \mapsto x+1$, and $\mathbb{R}$ is not $\mathbb{N}_{\mathbb{R}}$. Arguments about natural numbers use leastness, that is (P5), and never closure alone.

1.5.4 Pairs as sets

One more object from *Sets and Functions* has to be found among the sets: the ordered pair. Sets and Functions, Definition 1.4.1 took (a, b) as a new kind of object and asked only for its characteristic property, that $(a, b) = (c, d)$ exactly when $a = c$ and $b = d$. Set theory has no new kinds of objects, so a pair must be a set built from a and b.

The first idea, $\{a, b\}$, fails at once: $\{a, b\} = \{b, a\}$, so it cannot record which coordinate came first. The standard repair marks the first coordinate by giving it a set of its own:

$$(a, b) \coloneqq \{\{a\}, \{a, b\}\}.$$

This is the **Kuratowski pair**. The singleton $\{a\}$ names the first coordinate; the other member adds the second. When $a = b$, both members are $\{a\}$, and $(a, a) = \{\{a\}\}$ has a single element. *Axiomatic Set Theory* proves the characteristic property for this definition, by comparing the members of two such sets case by case.

Once pairs are sets, so is everything built from them. The product $A \times B$ is a set of pairs, cut out of $\mathcal{P}(\mathcal{P}(A \cup B))$ by specification. A function is a set of pairs (Sets and Functions, Definition 2.1.1), so $\mathbb{N} \times \mathbb{N}$, the operations $+$ and $\cdot$ of Section 1.2, and the isomorphisms of Section 1.4 are all sets. Chapter 2 of this volume needs none of this: it uses pairs only through the characteristic property of Sets and Functions, Definition 1.4.1.

Quick check. Is $\{\{0\}\}$ an ordered pair? If so, of what?

Solution. Yes: $(0,0) = \{\{0\},\{0,0\}\} = \{\{0\}\}$, since $\{0,0\} = \{0\}$. As a set, it is also $\{1\}$, because $1 = \{0\}$.

1.5.5 Exercises

1.5.5.1 A. Check your understanding

Exercise. A1

(a) List the elements of S(3).

(b) True or false: $2 \in 3$ and $2 \subseteq 3$, for the von Neumann numerals.

(c) Why can the set $\{a,b\}$ not serve as the ordered pair (a,b)?

(d) Which Peano axiom is "ω is the least inductive set" in disguise?

Solution.

(a) $S(3) = 3 \cup \{3\}$ has the elements 0, 1, 2, 3; it is the numeral 4.

(b) Both are true: $3 = \{0,1,2\}$ has 2 as an element, and the elements 0, 1 of 2 are elements of 3.

(c) It forgets the order: $\{1,2\} = \{2,1\}$, while the pairs $(1,2)$ and $(2,1)$ must differ, by the characteristic property, since $1 \neq 2$.

(d) (P5): a subset of ω containing $\emptyset$ and closed under S is inductive, so by leastness it is all of ω.

1.5.5.2 B. Practice

Exercise. B1: Numerals by hand

(a) Write S(4) as a set of numerals. How many elements does the numeral n have, for $n \leq 5$?

(b) Compute $2 \cup 3$ and $2 \cap 3$.

(c) Is $\{1\} \subseteq 3$? Is $\{1\}$ one of the numerals $0, 1, \ldots, 5$?

Solution.

(a) $S(4) = \{0,1,2,3\} \cup \{4\} = \{0,1,2,3,4\}$, the numeral 5. The numeral n is $\{0,1,\ldots,n-1\}$, and it has n elements: $0 = \emptyset$ has none, $1 = \{0\}$ has one, and so on, going up one numeral at a time. The step $S(k) = k \cup \{k\}$ gives one more element than k has, because $k \notin k$: the elements of k are earlier numerals, which have fewer elements than k, so none of them is k. So $0, 1, \ldots, 5$ have $0, 1, \ldots, 5$ elements, and in particular they are all different.

(b) $2 = \{0,1\}$ and $3 = \{0,1,2\}$, so $2 \cup 3 = \{0,1,2\} = 3$ and $2 \cap 3 = \{0,1\} = 2$: the union is the larger numeral and the intersection the smaller.

(c) Yes, $\{1\} \subseteq 3$, since $1 \in 3$. But $\{1\}$ is none of the numerals $0, \ldots, 5$. It has one element, so by (a) the only candidate is $1 = \{0\}$; and $\{1\} \neq \{0\}$, since their only elements 1 and 0 differ (1 has an element, 0 has none).

Exercise. B2: Pairs as sets

Write $(0,1)$, $(1,0)$ and $(1,1)$ as Kuratowski pairs, using numerals where you can. Show directly that $(0,1) \neq (1,0)$. How many elements does (a,b) have when $a \neq b$, and when $a = b$?

Solution. $(0,1) = \{\{0\},\{0,1\}\} = \{1,2\}$, since $\{0\} = 1$ and $\{0,1\} = 2$. Next, $(1,0) = \{\{1\},\{1,0\}\} = \{\{1\},2\}$, and $(1,1) = \{\{1\},\{1,1\}\} = \{\{1\}\}$.

The set 1 is an element of $(0,1)$. It is not an element of $(1,0)$: it is not $\{1\}$, since $0 \in 1$ while the only element of $\{1\}$ is $1 \neq 0$; and it is not 2, since $1 \in 2$ while $1 \notin 1 = \{0\}$. So the two sets have different elements, and $(0,1) \neq (1,0)$.

If $a \neq b$, then $\{a\} \neq \{a,b\}$, because b lies in the second and not in the first; so (a,b) has two elements. If $a = b$, then $(a,a) = \{\{a\}\}$ has one. (Note also that $(0,1) = \{1,2\}$ is not a numeral, although both its elements are: it is not $0 = \emptyset$, since it has elements, and every other numeral $\{0,1,\ldots,n-1\}$ contains 0, which $\{1,2\}$ does not, since 1 and 2 have elements and 0 has none.)

1.5.5.3 C. Going deeper

Exercise. C1: Closed is not least

Each of the sets $\mathbb{R}$, $[0,\infty)$ and $\mathbb{N}_{\mathbb{R}} \cup \{n + \frac{1}{2} : n \in \mathbb{N}_{\mathbb{R}}\}$ contains 0 and is closed under $x \mapsto x+1$. For each, with $e = 0$ and $s(x) = x+1$, decide which of (P3), (P4), (P5) hold. Then find the least subset of $\mathbb{R}$ that contains 0 and is closed under $x \mapsto x+1$, and prove that it is least.

Solution. This exercise is about subsets of the real numbers of *Sets and Functions*, so it uses their arithmetic freely, as Corollary 1.4.3 (a) does; facts about $\mathbb{N}_{\mathbb{R}}$ itself come from Sets and Functions, Lemma 3.1.1.

(P4) holds in all three: $x+1 = y+1$ gives $x = y$.

(P3) fails only in $\mathbb{R}$, where $s(-1) = -1+1 = 0$. In the other two sets, $x+1 = 0$ would need $x = -1$. But every $n \in \mathbb{N}_{\mathbb{R}}$ has $n \geq 0$ (Sets and Functions, Lemma 3.1.1 (a)), so $-1 \notin \mathbb{N}_{\mathbb{R}}$, $-1 \notin [0,\infty)$, and $n + \frac{1}{2} \geq \frac{1}{2} > -1$.

(P5) fails in all three. The subset $\mathbb{N}_{\mathbb{R}}$ of each contains 0 and is closed under $x \mapsto x+1$ (it lies in $[0,\infty)$ by the same clause (a)), but it is not the whole set: it misses $-1 \in \mathbb{R}$ (as above), and $\frac{1}{2}$, which lies in $[0,\infty)$ and in the third set, by Sets and Functions, Lemma 3.1.1 (e) with $n = 0$ (no natural number lies strictly between 0 and 1).

The least such set is $\mathbb{N}_{\mathbb{R}}$. It contains 0 and is closed under $x \mapsto x+1$. Let $T \subseteq \mathbb{R}$ be any

set with these two properties. Then $T \cap \mathbb{N}_{\mathbb{R}}$ contains 0, and it is closed: if $n \in T \cap \mathbb{N}_{\mathbb{R}}$, then $n+1 \in T$ and $n+1 \in \mathbb{N}_{\mathbb{R}}$. By the set form of Logic and Proof, Axiom 4.2.1 (the remark at the start of *Sets and Functions*, Section 3.1), the only such subset of $\mathbb{N}_{\mathbb{R}}$ is $\mathbb{N}_{\mathbb{R}}$ itself, so $T \cap \mathbb{N}_{\mathbb{R}} = \mathbb{N}_{\mathbb{R}}$, that is, $\mathbb{N}_{\mathbb{R}} \subseteq T$.

The three sets show the point of the warning above: closure under the step is cheap, and only the least closed set is the natural numbers.

1.5.5.4 R. Review

Exercise. R1

Prove that $\mathbb{N} \setminus \{0\}$ is infinite.

Solution. We read Sets and Functions, Definition 4.1.3 and Sets and Functions, Theorem 4.1.9 on subsets of $\mathbb{N}$, through the citing rule of Section 1.4 (group (i)). Write $A \coloneqq \mathbb{N} \setminus \{0\}$ and $B \coloneqq \mathbb{N} \setminus \{0, 1\}$, where $1 = \mathrm{S}(0)$ (Definition 1.2.1 (c)). Consider S restricted to A, with codomain B. Its values lie in B: for $n \in A$, $\mathrm{S}(n) \neq 0$ by (P3) for $\mathbb{N}$ (Axiom 1.1.1), and $\mathrm{S}(n) \neq \mathrm{S}(0) = 1$ by (P4), since $n \neq 0$. It is injective by (P4). It is onto B: if $m \in B$, then $m \neq 0$, so $m = \mathrm{S}(k)$ for some $k \in \mathbb{N}$ by Proposition 1.1.3 (a), and $k \neq 0$ because $m \neq 1 = \mathrm{S}(0)$; so $k \in A$. Hence it is a bijection $A \to B$. The set B is a proper subset of A: $B \subseteq A$ is clear, and $1 \in A$, since $1 = \mathrm{S}(0) \neq 0$ by (P3), while $1 \notin B$. By Sets and Functions, Theorem 4.1.9, a finite set is not equinumerous with any proper subset; so A is not finite, that is, it is infinite (Sets and Functions, Definition 4.1.3).

Exercise. R2

Let $D \colon \mathbb{N} \to \mathbb{N}$ be $D(n) \coloneqq \mathrm{S}(\mathrm{S}(n))$. Show that D has a left inverse and no right inverse. Write a left inverse down explicitly. Is it the only one?

Solution. No right inverse. A function with a right inverse is surjective (Sets and Functions, Proposition 2.4.2), and D is not: 0 is not a value, since $D(n) = \mathrm{S}(\mathrm{S}(n)) \neq 0$ by (P3) for $\mathbb{N}$.

A left inverse. Write $1 = \mathrm{S}(0)$ (Definition 1.2.1 (c)). Let $L \colon \mathbb{N} \to \mathbb{N}$ have the graph

$$G \coloneqq \{(n, k) \in \mathbb{N} \times \mathbb{N} : (n \in \{0, 1\} \text{ and } k = 0) \text{ or } n = \mathrm{S}(\mathrm{S}(k))\}.$$

Each $n \in \mathbb{N}$ has exactly one partner. If $n = 0$, the partner $k = 0$ works, and no k has $0 = \mathrm{S}(\mathrm{S}(k))$, by (P3). If $n = 1$, the partner $k = 0$ works, and no k has $\mathrm{S}(0) = \mathrm{S}(\mathrm{S}(k))$: by (P4) that would give $0 = \mathrm{S}(k)$, against (P3). If $n \notin \{0, 1\}$, then $n \neq 0$, so $n = \mathrm{S}(m)$ for exactly one m, by Proposition 1.1.3 (a); this m is not 0, since $n \neq \mathrm{S}(0)$, so $m = \mathrm{S}(k)$ for exactly one k, by Proposition 1.1.3 (a) again. Thus $n = \mathrm{S}(\mathrm{S}(k))$ for exactly one k. So L is a function (Sets and Functions, Definition 2.1.1) with $L(0) = L(1) = 0$ and $L(\mathrm{S}(\mathrm{S}(k))) = k$. Hence $L(D(n)) = n$ for every $n \in \mathbb{N}$, that is, $L \circ D = \mathrm{id}_{\mathbb{N}}$: L is a left inverse of D (Sets and Functions, Definition 2.4.1 (a)). This matches Sets and Functions, Theorem 2.4.1: D is injective, by (P4) applied twice, and $\mathbb{N} \neq \emptyset$.

Not unique. The values of a left inverse at 0 and 1 are never used, since neither is of the form $D(n)$; setting $L'(0) := 5$, $L'(1) := 0$ and $L'(\mathrm{S}(\mathrm{S}(k))) := k$ gives another left inverse.

Exercise. R3

Let $M = \{0, 1\}$, $e = 0$ and $s(0) = s(1) = 1$. Show that there is no function $f\colon M \to \mathbb{N}$ with $f(0) = 0$ and $f(s(x)) = \mathrm{S}(f(x))$ for every $x \in M$. Which of the Peano axioms play the part, for definitions on $\mathbb{N}$, that a theorem about reading formulas plays for definitions on formulas?

Solution. Suppose f exists. Taking $x = 1$, $f(1) = f(s(1)) = \mathrm{S}(f(1))$. But no $n \in \mathbb{N}$ has $n = \mathrm{S}(n)$ (Proposition 1.1.3 (b)). So no such f exists. (Taking $x = 0$ instead gives $f(1) = \mathrm{S}(0) = 1$, so the two equations at the input 1 also contradict each other.)

The trouble is that 1 is built in two ways, as $s(0)$ and as $s(1)$, and the two ways ask for different values. In *Logic and Proof*, Chapter 4, Logic and Proof, Theorem 4.3.4 defines functions on formulas, and what makes it work is Logic and Proof, Theorem 4.3.3: each compound formula is built from its parts in exactly one way. On $\mathbb{N}$ the same job is done by (P3) and (P4): 0 is not a successor, so its value is not asked for twice, and a successor is the successor of exactly one number, so the equation at $\mathrm{S}(n)$ asks for one thing. The structure M fails (P4), since $s(0) = s(1)$, though it satisfies (P5).

Chapter 2

The Integers

In the natural numbers, "subtract 5 from 3" has no answer. The school integers answer it by a stroke of the pen: add negative numbers, and trust that everything still works. This chapter builds them instead, with no new assumption. A pair (a, b) of natural numbers records the difference $a - b$ without performing it, two pairs record the same difference when $a + d = b + c$, and an integer is a class of such pairs. The quotient set of *Sets and Functions*, Section 5.2 makes this a set, and the number of pairs per integer is infinite, so nothing about an integer may depend on which pair we use to write it.

That is the danger of the whole chapter. A formula that looks fine on pairs can give different answers for two pairs of the same class, and then it is no operation on integers. Section 2.2 proves one lemma that tells us what to check, and the check is done once, one argument at a time. With the operations in hand, Section 2.3 names the structure they form, a commutative ring, and checks every law by pushing it down to the natural numbers. Section 2.4 adds the order, from one choice, the set of positive elements, and proves the rules for inequalities that every later chapter uses. Section 2.5 closes with division with remainder and the even-odd split. At the end $\mathbb{Z}$ is an ordered ring in which every set bounded below has a least element, and it is still not a field.

What you need. Chapter 1: induction and the recursion theorem on $\mathbb{N}$ (Theorem 1.1.2, Theorem 1.2.1), with the citing rule of Section 1.4 for results of the earlier volumes (Corollary 1.4.3), the arithmetic and order of $\mathbb{N}$ (Definition 1.2.1, Corollary 1.2.2, Theorem 1.2.3, Proposition 1.3.1, Theorem 1.3.2), well-ordering (Theorem 1.3.3), and parity (Proposition 1.3.4). *Sets and Functions*: ordered pairs, products, functions and their equality (Sets and Functions, Definition 1.4.1, Sets and Functions, Definition 1.4.2, Sets and Functions, Definition 2.1.1, Sets and Functions, Proposition 2.1.1); equivalence relations, classes, quotient sets and the partition theorem, with the method for checking a rule on representatives, from its Section 5.2 (Sets and Functions, Definition 5.2.1, Sets and Functions, Definition 5.2.2, Sets and Functions, Definition 5.2.4, Sets and Functions, Theorem 5.2.1, Sets and Functions, Example (Method: checking that a definition on classes is well defined), Sets and Functions, Example (A rule on classes that is not a function)); the least-element method (Sets and Functions, Example (Method: the least element)); for Section 2.4's C1 only, finite and infinite sets from its Section 4.1 (Sets and Functions, Definition 4.1.2, Sets and Functions, Definition 4.1.3, Sets and Functions, Corollary 4.1.3, Sets and Functions, Proposition 4.1.6); and, for Section 2.5 only, the school facts about division and parity that it compares with (Sets and Functions, Theorem 3.2.3, Sets and Functions, Theorem 1.1.2, Sets and Functions, Definition 1.1.2, Sets and Functions, Definition 1.1.3). *Logic and Proof*: the square of an even integer (Logic and Proof, Theorem 2.1.3), for a comparison in Section 2.5.

Time. About 20 hours on the core track (all of A, one B and one C per section, the exit checks); about 25 hours for everything.

2.0.1 Roadmap

- **The Integers as Classes of Pairs.** Why $3 - 5$ needs a new kind of number; pairs of natural numbers with $(a, b) \sim (c, d)$ when $a + d = b + c$, checked to be an equivalence relation; $\mathbb{Z}$ as the quotient set; pictures of the classes on a staircase; and the normal form, by which every class contains exactly one pair of the form $(n, 0)$ or $(0, n)$.
- **Well-Definedness, and the Arithmetic of the Integers.** The lemma on functions defined on classes through representatives, with its one-argument-at-a-time check; the sum, product and negative of integers, shown well defined; rules that look like functions and are not; and the copy of $\mathbb{N}$ inside $\mathbb{Z}$, which turns the class $[(a, b)]$ into the difference $a - b$.
- **Commutative Rings; the Integers Are One.** The ring and field axioms; the integers a commutative ring with $1 \neq 0$, every law checked by the move that reduces it to the level below; the algebra that follows from the axioms alone.
- **Ordered Rings and the Rules for Inequalities.** Ordered rings and fields; an order from a positive set; the order of $\mathbb{Z}$; the rules for inequalities, including multiplication by a positive and by a negative element; multiples; least and greatest elements of sets of integers; and a ring that cannot be ordered.
- **Division with Remainder.** Divisibility on the constructed integers and its rules; division with remainder, existence and uniqueness, with the existence proof by a least element; and even and odd integers, exactly one of the two.

2.0.2 Named moves

- **Check on representatives, one argument at a time** (Example (Method: check on representatives, one argument at a time), with Sets and Functions, Example (Method: checking that a definition on classes is well defined) and Theorem 2.2.1). Change one representative with the other fixed; each change is one identity at the level below. A formula symmetric in its two inputs gets the second change free. (Section 2.2)
- **Reduce to the level below** (Example (Method: reduce to the level below)). Write a law about classes on representatives; the two sides become the same pair, or two related pairs, by an identity in $\mathbb{N}$. (Section 2.3)
- **Turn the comparison into one positive element** (Example (Method: turn the comparison into one positive element)). To compare a and b, look at $b - a$ and ask whether it is in the positive set. (Section 2.4)
- **Order from a positive set** (Proposition 2.4.1). Say which elements are positive, check (P1) and (P2), and read off every rule for inequalities. (Section 2.4)
- **Least element** (Sets and Functions, Example (Method: the least element)). A set of integers bounded below has a least element; it finds the remainder in Section 2.5 and the integer part in Chapter 3. (Sections 2.4 and 2.5)

2.0.3 Threads

- **Well-definedness, the level below and positive sets.** Section 2.1 makes integers from pairs of naturals, Section 3.1 makes rationals from pairs of integers, and Chapter 5 makes real numbers from classes of sequences of rationals, each time by the lemma and the check of Section 2.2, the move of Section 2.3 and the order of Section 2.4.
- **Least elements.** Chapter 1 proves it for $\mathbb{N}$, Section 2.4 for sets of integers bounded below, Section 2.5 uses it for the remainder, and *Algebraic Preliminaries*, Chapter 1 takes it up

for divisibility and the Euclidean algorithm.
- **The copy of $\mathbb{N}$.** Section 2.2 identifies n with $[(n, 0)]$; every later chapter writes $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ without further comment.

2.0.4 Summary

A pair (a, b) of natural numbers records the difference $a - b$, and the relation $a + d = b + c$ is an equivalence relation (Proposition 2.1.1); its classes form $\mathbb{Z}$ (Definition 2.1.1), and each class has a normal form $(n, 0)$ or $(0, n)$ (Proposition 2.1.2). A rule defined on classes through representatives is a function when it passes the check of Theorem 2.2.1; the sum, product and negative do (Definition 2.2.1, Proposition 2.2.2), and $n \mapsto [(n, 0)]$ copies $\mathbb{N}$ into $\mathbb{Z}$ (Proposition 2.2.3). $\mathbb{Z}$ is a commutative ring with $1 \neq 0$ (Definition 2.3.1, Theorem 2.3.1), and the algebra of a ring follows (Proposition 2.3.2). The positive classes make $\mathbb{Z}$ an ordered ring (Definition 2.4.1, Proposition 2.4.1, Definition 2.4.2, Theorem 2.4.2) with the usual rules for inequalities (Proposition 2.4.3); a set of integers bounded below has a least element (Proposition 2.4.4). Divisibility has its usual rules (Proposition 2.5.1), division with remainder exists and is unique (Theorem 2.5.2), and every integer is even or odd and not both (Corollary 2.5.3).

What is open: $\mathbb{Z}$ is not a field, since $2x = 1$ has no solution. Chapter 3 repeats the construction with pairs of integers to supply quotients.

2.0.5 Terms

- **integer**: a class $[(a, b)]$ of pairs of natural numbers under $a + d = b + c$ (Definition 2.1.1).
- **normal form**: the one pair $(n, 0)$ or $(0, n)$ in a class (Proposition 2.1.2).
- **well defined**, **induced function**: a rule on classes that does not depend on the representatives, and the function it gives (Theorem 2.2.1).
- **commutative ring**, **field**: a set with addition and multiplication satisfying the laws (F1)–(F7), (F9), and also $1 \neq 0$ and (F8) for a field (Definition 2.3.1).
- **ordered commutative ring**, **ordered field**: a ring or field with a relation $<$ satisfying (O1)–(O4) (Definition 2.4.1).
- **positive set**: a subset P closed under sums and products, with every element exactly one of positive, zero or negative (Proposition 2.4.1).
- **discreteness**: no integer lies strictly between n and $n + 1$ (Theorem 2.4.2).
- **multiple** $n \cdot a$: a added n times, defined by recursion (Section 2.4).
- **divides**, **prime**: $a \mid b$ when $b = ka$ for an integer k; p is prime when $p \geq 2$ and its only positive divisors are 1 and p (Proposition 2.5.1).
- **quotient, remainder**: the q and r with $a = qn + r$ and $0 \leq r < n$ (Theorem 2.5.2).
- **even, odd**: $a = 2q$ or $a = 2q + 1$ (Corollary 2.5.3).

2.1 The Integers as Classes of Pairs

Chapter 1 took the natural numbers to be a Peano system and built on it addition, multiplication and the order, with the laws of arithmetic and well-ordering; by Section 1.4 any two Peano systems are alike, and from here on $\mathbb{N}$ is that system (Corollary 1.4.3, with the citing rule of Section 1.4). In $\mathbb{N}$ the equation $5 + x = 3$ has no solution: $5 \leq 5 + x$ for every $x \in \mathbb{N}$ by Proposition 1.3.1 (a), while $3 < 5$ by the same result, as $5 = 3 + 2$ with $2 \neq 0$. So $5 + x = 3$ would give $5 \leq 3 < 5$, hence $5 < 5$, which Theorem 1.3.2 (a) rules out. This section builds a larger set $\mathbb{Z}$, out of $\mathbb{N}$ alone, with the one tool of *Sets and Functions*, Section 5.2: the quotient of a set by an equivalence relation.

2.1.1 What we use about the natural numbers

Every fact about $\mathbb{N}$ in this section and the next is one of the following results of Chapter 1, cited where a proof uses it. The defining equation $n+0=n$ and the evaluation of numeral sums and comparisons such as $2+3=5$ and $0 \neq 2$ rest on Definition 1.2.1 and Theorem 1.3.2 and are not cited each time.

- **Defining equations** (Definition 1.2.1 (a), (b)): $n+0=n$, $m+\mathrm{S}(n)=\mathrm{S}(m+n)$, $n\cdot 0=0$; the numerals $1 \coloneqq \mathrm{S}(0)$, $2 \coloneqq \mathrm{S}(1)$, …, with $n+1=\mathrm{S}(n)$ (Definition 1.2.1 (c)).
- **Laws of arithmetic** (Theorem 1.2.3): $+$ and $\cdot$ are commutative and associative, $\cdot$ distributes over $+$, and $0+n=n$, $1\cdot n=n$, $0\cdot n=0$.
- **Cancellation** (Proposition 1.3.1 (b), (f)): $m+k=n+k$ gives $m=n$, and so does $mk=nk$ with $k\neq 0$.
- **Order** (Theorem 1.3.2 (a)): $<$ is a strict total order, so exactly one of $m<n$, $m=n$, $n<m$ holds.
- **Order and addition** (Proposition 1.3.1 (a), (c)): $m \leq n$ if and only if $n=m+j$ for some $j\in\mathbb{N}$, and $m<n$ if and only if this holds with $j\neq 0$; adding preserves $<$.
- **Small facts and well-ordering**: $0<1$ (Theorem 1.3.2 (d)); nothing lies strictly between n and $\mathrm{S}(n)$ (Theorem 1.3.2 (b)); every non-empty subset of $\mathbb{N}$ has a least element (Theorem 1.3.3).

That is the whole list for now. Nothing about the integers or the rationals of school enters a proof: from here on they are the sets this chapter and the next build, and their familiar properties become theorems.

2.1.2 A naive attempt

The school picture of an integer is "a natural number with a sign": $+3$, -2, 0. It is tempting to take that literally and let an integer be a pair (s,n) with a sign $s\in\{0,1\}$ (read 0 as "plus" and 1 as "minus") and a size $n\in\mathbb{N}$. Two things go wrong at once.

First, zero gets two names, $(0,0)$ and $(1,0)$: a "plus zero" and a "minus zero". Either both stay, and every theorem must treat them as the same, or one is banned, and every definition must check that it never produces the banned one.

Second, addition needs cases. Adding $+m$ and $-n$ should give $+(m-n)$ when $n\leq m$ and $-(n-m)$ when $m<n$, where $m-n$ is the subtraction of Proposition 1.3.1 (a), defined on $\mathbb{N}$ only when $n\leq m$. There are four sign patterns for a sum, and two of them split again by the order of the sizes. A proof that addition is associative would split by the signs of all three terms and then by the order of their sizes, and every case would subtract in $\mathbb{N}$ in its own direction.

The trouble in both is the same: the definition **performs** a subtraction that $\mathbb{N}$ can only do one way round. The construction below never performs it. It records the subtraction as a pair, and lets an equivalence relation decide when two recorded subtractions should count as equal.

2.1.3 Pairs that record a difference

Think of a pair $(a,b)\in\mathbb{N}\times\mathbb{N}$ as a note saying "a minus b", to be computed later or never. The note $(3,5)$ is legitimate even though $3-5$ is not a natural number. Many notes say the same thing: $(3,5)$, $(0,2)$ and $(10,12)$ all describe "two less than nothing", which school calls -2.

When do two notes (a,b) and (c,d) describe the same thing? School would write $a-b=c-d$, and then add $b+d$ to both sides to get $a+d=b+c$. That last equation is the one we keep,

because it uses only addition, which $\mathbb{N}$ has for all inputs. The computation with minus signs is a **heuristic**, borrowed from school to find the definition; the definition does not depend on it.

An integer is a difference you have not computed yet.

Definition 2.1.1. The Integers

For $(a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$, let

$$(a, b) \sim (c, d) \iff a + d = b + c.$$

The set of **integers** is the quotient set

$$\mathbb{Z} := (\mathbb{N} \times \mathbb{N})/\sim$$

of Sets and Functions, Definition 5.2.4, and an **integer** is an element of $\mathbb{Z}$. The class of the pair (a, b) is written $[(a, b)]$:

$$[(a, b)] = \{(c, d) \in \mathbb{N} \times \mathbb{N} : (c, d) \sim (a, b)\}.$$

In words: two pairs are related when their **cross sums** agree, the first coordinate of each added to the second coordinate of the other. The relation is a subset of $(\mathbb{N} \times \mathbb{N}) \times (\mathbb{N} \times \mathbb{N})$, described by a property (Sets and Functions, Definition 1.4.3, Sets and Functions, Definition 1.4.2). An integer is one whole class: the set of all pairs that record the same difference. The notation $[(a, b)]$ is the class notation $[x]$ of Sets and Functions, Definition 5.2.2 with $x = (a, b)$; no new bracket is introduced.

Well-definedness. Sets and Functions, Definition 5.2.4 is stated for an **equivalence relation**. Until $\sim$ is shown to be one, the definition of $\mathbb{Z}$ is a plan, not a definition. The next proposition supplies the missing check, and it is placed directly after the definition for that reason.

An integer is a set of pairs. It is not a natural number with a sign, and it is not a pair. *Sets and Functions*, Chapter 1 adopted the convention that a pair is not a number (after Sets and Functions, Definition 1.4.1). The same care is needed here: $[(0, 2)]$ is the set $\{(0, 2), (1, 3), (2, 4), \dots\}$, and the school -2 is what those pairs have in common. The construction takes "what they have in common" to be **the set of them**.

Proposition 2.1.1. The Relation Behind the Integers

The relation $\sim$ of Definition 2.1.1 is an equivalence relation on $\mathbb{N} \times \mathbb{N}$ (Sets and Functions, Definition 5.2.1).

Idea. This is the check of Sets and Functions, Example (Method: proving or disproving a property of a relation): three clauses, each unwound to an equation in $\mathbb{N}$. Reflexivity and symmetry are rearrangements of one equation, by commutativity. Transitivity has two hypotheses, $a + d = b + c$ and $c + f = d + e$, and a goal $a + f = b + e$ in which c and d do not appear. In school one would subtract them. In $\mathbb{N}$ we add the two equations, so that $c + d$ appears on both sides, and then remove it by cancellation.

Before reading on: which line uses cancellation in $\mathbb{N}$, and why is it more direct than subtracting $c + d$?

Proof. We check (Eq1)–(Eq3) of Sets and Functions, Definition 5.2.1. Let $(a,b),(c,d),(e,f) \in \mathbb{N}\times\mathbb{N}$.

(Eq1) $a+b=b+a$ by commutativity (Theorem 1.2.3 (b)), so $(a,b)\sim(a,b)$.

(Eq2) Suppose $(a,b)\sim(c,d)$, that is, $a+d=b+c$. By commutativity, $c+b=b+c=a+d=d+a$, and $c+b=d+a$ says $(c,d)\sim(a,b)$.

(Eq3) Suppose $(a,b)\sim(c,d)$ and $(c,d)\sim(e,f)$, that is,

$$a+d=b+c \quad \text{and} \quad c+f=d+e.$$

Adding the two equations gives $(a+d)+(c+f)=(b+c)+(d+e)$. By associativity and commutativity (Theorem 1.2.3 (b), (c)), the left side is $(a+f)+(c+d)$ and the right side is $(b+e)+(c+d)$, so

$$(a+f)+(c+d)=(b+e)+(c+d).$$

By cancellation (Proposition 1.3.1 (b)) with $k=c+d$, $a+f=b+e$. That is $(a,b)\sim(e,f)$.

This proves that $\sim$ is an equivalence relation.

■

The answer to the prompt is the last line of (Eq3). Subtracting $c+d$ would need a check that the difference exists in $\mathbb{N}$ (Proposition 1.3.1 (a)), and reading the difference off uses uniqueness, itself proved by cancellation. Cancellation removes $c+d$ for every value, with no check.

So $\mathbb{Z}=(\mathbb{N}\times\mathbb{N})/\!\sim$ is now defined, and by Sets and Functions, Theorem 5.2.1 its elements partition $\mathbb{N}\times\mathbb{N}$: every pair lies in exactly one integer, and $[(a,b)]=[(c,d)]$ if and only if $a+d=b+c$.

Quick check. Which classes contain $(4,0)$, $(0,4)$ and $(4,4)$? Is any pair in two of them?

Solution. Each pair lies in its own class, and $(4,4)\sim(0,0)$ (cross sums $4+0=4+0$), so $[(4,4)]=[(0,0)]$, the class of zero. The three classes are distinct: for $(4,0)$ and $(0,4)$ the cross sums are $4+4\neq 0+0$; for $(4,0)$ and $(0,0)$, $4+0\neq 0+0$; for $(0,4)$ and $(0,0)$, $0+0\neq 4+0$. Distinct classes are disjoint (Sets and Functions, Theorem 5.2.1), so no pair lies in two of them.

2.1.4 Pictures of classes

The quarter plane $\mathbb{N}\times\mathbb{N}$ is a grid of lattice points, and the classes are easy to see in it.

Example. Three classes, listed and drawn

List the elements (x,y) with $x\le 4$ of each of the classes $[(3,5)]$, $[(0,0)]$ and $[(4,1)]$, and describe each class as a set of lattice points.

Solution. *The class* $[(3,5)]$. A pair (x,y) lies in it when $x+5=y+3$. This holds if and only if $y=x+2$. Indeed, if $y=x+2$, then $y+3=x+(2+3)=x+5$ by associativity (Theorem 1.2.3 (c)). Conversely, $x+5=(x+2)+3$, so $x+5=y+3$ gives $(x+2)+3=y+3$, and cancellation (Proposition 1.3.1 (b)) gives $y=x+2$. So

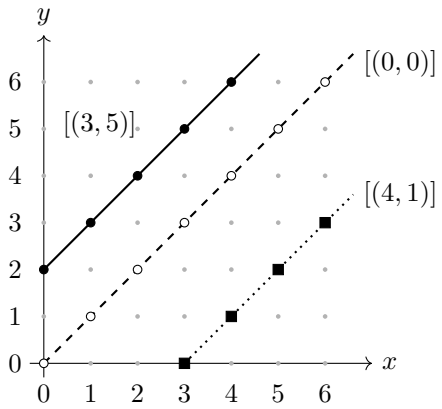
$$[(3,5)]=\{(x,x+2):x\in\mathbb{N}\},$$

and, since the natural numbers $x\leq 4$ are exactly $0,1,2,3,4$ (Theorem 1.3.2 (c)), the elements with $x\leq 4$ are $(0,2)$, $(1,3)$, $(2,4)$, $(3,5)$, $(4,6)$.

The class $[(0,0)]$. Here the condition is $x+0=y+0$, that is, $x=y$, since $n+0=n$ (Definition 1.2.1 (a)). So $[(0,0)]=\{(x,x):x\in\mathbb{N}\}$, the diagonal, and, likewise, the elements with $x\leq 4$ are $(0,0)$, $(1,1)$, $(2,2)$, $(3,3)$, $(4,4)$.

The class $[(4,1)]$. The condition $x+1=y+4$ holds if and only if $x=y+3$, by the same two steps as for $[(3,5)]$. So $[(4,1)]=\{(y+3,y):y\in\mathbb{N}\}$. Its elements with $x\leq 4$ are $(3,0)$ and $(4,1)$ only: $x=y+3=3+y$ gives $3\leq x$ (Proposition 1.3.1 (a)); with $x\leq 4=\mathrm{S}(3)$, x is 3 or 4, since no natural number lies strictly between 3 and $\mathrm{S}(3)$ (Theorem 1.3.2 (b)); and then $y+3=3=0+3$ or $y+3=4=1+3$, so y is 0 or 1 by cancellation.

Each of these three classes is the set of lattice points on one diagonal line $y=x+k$ or $x=y+k$ of the quarter plane, starting on an axis and running up and to the right.



The picture shows the three classes as three diagonals: $[(3,5)]$ solid with filled dots, $[(0,0)]$ dashed with open dots, $[(4,1)]$ dotted with squares. Every pair lies in exactly one class (Sets and Functions, Theorem 5.2.1), so no two of these diagonals share a lattice point; after the normal form below we see that every class is a diagonal of this kind, so every lattice point lies on exactly one of them. The class $[(0,0)]$ is the degenerate case: it is the integer zero, and it is the only class whose line passes through the origin.

A non-example by minimal change. Replace the sums in the definition by products: on $\mathbb{N}\times\mathbb{N}$, let $(a,b)\approx(c,d)$ mean $ad=bc$. This is still reflexive ($ab=ba$) and symmetric ($ad=bc$ gives $cb=da$), both by commutativity of $\cdot$ (Theorem 1.2.3 (f)). It is **not** transitive:

$$(1,1)\approx(0,0),\qquad (0,0)\approx(2,1),\qquad \text{but}\qquad (1,1)\not\approx(2,1),$$

since $1\cdot 0=1\cdot 0$, $0\cdot 1=0\cdot 2$ and $1\cdot 1\neq 1\cdot 2$. The clause that fails is (Eq3), and the line of the proof above that has no counterpart is cancellation. Additive cancellation holds for every k; multiplicative cancellation (Proposition 1.3.1 (f)) needs $k\neq 0$, and the pair $(0,0)$ supplies a middle factor that is 0. When Section 3.1 builds fractions with this product relation, it excludes zero second coordinates for exactly this reason.

2.1.5 Why this definition

Why cross sums. "$a - b = c - d$" cannot be the definition, because $a - b$ is defined in $\mathbb{N}$ only when $b \leq a$. The cross-sum equation $a + d = b + c$ says the same thing whenever both differences exist, and it makes sense for every two pairs.

Why whole classes. Each class has a simplest pair, one with a coordinate 0, as the next proposition shows. One could define $\mathbb{Z}$ as the set of these simplest pairs instead. But then the sum of $(3, 0)$ and $(0, 5)$ would have to be brought back to a simplest pair, $(0, 2)$, by a subtraction in $\mathbb{N}$ in the right direction: the case analysis of the naive attempt returns. With classes, any representatives may be combined, and the equivalence relation absorbs the bookkeeping. The price is the check of the next section: a definition through representatives must not depend on which ones were used. The class of $(0, 0)$ is the diagonal; it will be the zero of $\mathbb{Z}$.

The first result about the new set says that each class has exactly one simplest name.

Proposition 2.1.2. Normal Form of an Integer

(a) Every class $[(a, b)] \in \mathbb{Z}$ contains **exactly one** pair of the form $(n, 0)$ or $(0, n)$ with $n \in \mathbb{N}$ (the pair $(0, 0)$ has both forms and counts once). Precisely, for each $(a, b) \in \mathbb{N} \times \mathbb{N}$ **exactly one** of the following holds:
- $a = b + k$ for some $k \neq 0$, and then $[(a, b)] = [(k, 0)]$;
- $a = b$, and then $[(a, b)] = [(0, 0)]$;
- $b = a + k$ for some $k \neq 0$, and then $[(a, b)] = [(0, k)]$.

(b) For all $m, n \in \mathbb{N}$: if $[(m, 0)] = [(n, 0)]$, then $m = n$; and if $[(0, m)] = [(0, n)]$, then $m = n$.

Idea. The three cases are the three cases of trichotomy for a and b, and Proposition 1.3.1 (a) turns each inequality into an equation with a non-zero k. In each case one line of arithmetic shows that (a, b) is related to the simplest pair. For "exactly one pair", two simplest pairs of the same form are handled by (b), so we prove (b) first. Two of opposite forms, $(k, 0)$ and $(0, l)$ with $k, l \neq 0$, would need $k + l = 0$, which the order forbids.

Proof. (b) Suppose $[(m, 0)] = [(n, 0)]$. By Sets and Functions, Theorem 5.2.1 (b), $(m, 0) \sim (n, 0)$, that is, $m + 0 = 0 + n$. Since $m + 0 = m$ (Definition 1.2.1 (a)) and $0 + n = n$ (Theorem 1.2.3 (a)), $m = n$. Likewise $[(0, m)] = [(0, n)]$ gives $0 + n = m + 0$, so $n = m$.

(a), the three cases. Let $(a, b) \in \mathbb{N} \times \mathbb{N}$. By Theorem 1.3.2 (a), exactly one of $b < a$, $a = b$, $a < b$ holds. By Proposition 1.3.1 (a), $b < a$ holds if and only if $a = b + k$ for some $k \neq 0$, and $a < b$ if and only if $b = a + k$ for some $k \neq 0$. So exactly one of the three listed conditions holds. In each case the stated equality of classes follows from Sets and Functions, Theorem 5.2.1 (b), once the two pairs are shown to be related:

- if $a = b + k$, then $a + 0 = a = b + k$ (Definition 1.2.1 (a)), so $(a, b) \sim (k, 0)$;
- if $a = b$, then $a + 0 = b + 0$, so $(a, b) \sim (0, 0)$;
- if $b = a + k$, then $a + k = b = b + 0$ (Definition 1.2.1 (a)), so $(a, b) \sim (0, k)$.

In particular every class contains a pair of the form $(n, 0)$ or $(0, n)$.

(a), only one such pair. Let C be a class containing two such pairs p and p'. Since both lie

in C, $[p] = C = [p']$ by Sets and Functions, Theorem 5.2.1 (b). If $p = (m, 0)$ and $p' = (n, 0)$, then $m = n$ by (b), so $p = p'$; likewise if both have the form $(0, \cdot)$. This covers every case in which one of the pairs is $(0, 0)$, since $(0, 0)$ has both forms. There remains $p = (k, 0)$ and $p' = (0, l)$ with $k \neq 0$ and $l \neq 0$. Then $p \sim p'$ says $k + l = 0 + 0 = 0$. Since $l \neq 0$, Proposition 1.3.1 (a) gives $k < 0$; and since $k = 0 + k$ (Theorem 1.2.3 (a)) with $k \neq 0$, it also gives $0 < k$. By transitivity $k < k$, which Theorem 1.3.2 (a) rules out. So this case cannot occur, and the pair is unique.

■

So every integer has a **normal form**: it is $[(n, 0)]$ for a unique $n \in \mathbb{N}$, or $[(0, n)]$ for a unique $n \in \mathbb{N}$ with $n \neq 0$. In the picture, every class is a diagonal: $(x, y) \sim (n, 0)$ means $x + 0 = y + n$, that is, $x = y + n$, and $(x, y) \sim (0, n)$ means $x + n = y + 0$, that is, $y = x + n$ (Definition 1.2.1 (a)), so

$$[(n, 0)] = \{(y + n, y) : y \in \mathbb{N}\} \quad \text{and} \quad [(0, n)] = \{(x, x + n) : x \in \mathbb{N}\}.$$

The normal form is where the class's diagonal meets an axis. The next section reads $[(n, 0)]$ as the natural number n and $[(0, n)]$ as its negative; this is how the description of *Sets and Functions*, Section 3.1, "an integer is a natural number or the negative of one", becomes a theorem about the constructed $\mathbb{Z}$.

Quick check. Find the normal forms of $[(17, 9)]$ and $[(9, 17)]$, and a third pair in each class.

Solution. Since $17 = 9 + 8$, $[(17, 9)] = [(8, 0)]$ and $[(9, 17)] = [(0, 8)]$ (Proposition 2.1.2 (a)). Third pairs: $(18, 10)$ and $(10, 18)$, since $17 + 10 = 9 + 18$ gives both $(17, 9) \sim (18, 10)$ and $(9, 17) \sim (10, 18)$.

Warning. A property of a class must not be read off **one** representative. "The integer $[(a, b)]$ is small when $a \leq 3$" sounds like a property of integers and is not one: $[(3, 3)] = [(5, 5)]$, the first pair has $a = 3 \leq 3$, and the second has $a = 5$. The class is "small" or not according to the name we happened to use. The property "$[(a, b)]$ is positive when $b < a$" looks the same and happens to be safe: if $(a, b) \sim (a', b')$ and $b < a$, then $b' < a'$. That needs a proof, and Section 2.4 gives it before it calls any integer positive.

Quick check. Is $(2, 7) \sim (10, 15)$? Is $(2, 7) \sim (15, 10)$?

Solution. Yes: the cross sums are $2 + 15 = 17$ and $7 + 10 = 17$. No: the cross sums are $2 + 10 = 12$ and $7 + 15 = 22$, and $12 \neq 22$. In school terms, $(2, 7)$ and $(10, 15)$ both record "five less", while $(15, 10)$ records "five more". The normal forms agree: $7 = 2 + 5$ and $15 = 10 + 5$, so $[(2, 7)] = [(0, 5)] = [(10, 15)]$, while $[(15, 10)] = [(5, 0)]$, a different class by Proposition 2.1.2 (a).

The set $\mathbb{Z}$ exists, and each element has a simplest name. It has no arithmetic yet. The next section defines the sum and product of two integers by formulas on representatives, and the

warning above is the question it must answer for each formula.

2.1.6 Exercises

2.1.6.1 A. Check your understanding

Exercise. A1

(a) State the relation $\sim$ and the set $\mathbb{Z}$ of Definition 2.1.1 in symbols.

(b) Which pairs with a coordinate equal to 0 lie in $[(6, 9)]$?

(c) Why does the proof of Proposition 2.1.1 cancel $c + d$ rather than subtract it?

(d) True or false, with a reason: $[(1, 1)] = [(5, 5)]$.

Solution.

(a) For $(a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$: $(a, b) \sim (c, d) \iff a + d = b + c$; and $\mathbb{Z} := (\mathbb{N} \times \mathbb{N})/\sim = \{[(a, b)] : (a, b) \in \mathbb{N} \times \mathbb{N}\}$.

(b) Only $(0, 3)$. A pair $(x, 0)$ in the class would need $x+9 = 0+6 = 6$ (Theorem 1.2.3 (a)); but $9 \leq x+9$ by Proposition 1.3.1 (a) (with Theorem 1.2.3 (b) to write $x+9 = 9+x$), and $6 < 9$, so $x + 9 = 6$ would give $9 \leq 6 < 9$, and then $9 < 9$, which Theorem 1.3.2 (a) excludes. A pair $(0, y)$ needs $0 + 9 = y + 6$, that is, $3 + 6 = y + 6$, so $y = 3$ by cancellation (Proposition 1.3.1 (b)). This agrees with Proposition 2.1.2 (a): $9 = 6 + 3$, so $[(6, 9)] = [(0, 3)]$, and the class has exactly one pair with a coordinate 0.

(c) Subtraction in $\mathbb{N}$ is defined only when the number subtracted is at most the other (Proposition 1.3.1 (a)). Here $c + d$ is at most both sides, so subtracting would be possible, but only after that check, and reading off the differences uses the uniqueness in (a), which is itself proved by cancellation. Cancellation (Proposition 1.3.1 (b)) holds for every $k \in \mathbb{N}$ with no check, so the proof uses it directly.

(d) True. $(1, 1) \sim (5, 5)$ because $1 + 5 = 1 + 5$, so the classes are equal by Sets and Functions, Theorem 5.2.1 (b). Both are the zero class $[(0, 0)]$.

2.1.6.2 B. Practice

Exercise. B1: Three look-alike relations

On $\mathbb{N} \times \mathbb{N}$, determine which of the following relations are equivalence relations. For each one, decide which of (Eq1)–(Eq3) hold. Justify your answer.

(a) $(a, b) \approx_1 (c, d)$ if and only if $a + c = b + d$.

(b) $(a, b) \approx_2 (c, d)$ if and only if $a + 2d = 2b + c$.

(c) $(a, b) \approx_3 (c, d)$ if and only if $a + d \leq b + c$.

Solution.

(a) **Symmetric, but neither reflexive nor transitive.** Symmetric: $a + c = b + d$ gives $c + a = d + b$ by commutativity (Theorem 1.2.3 (b)). Not reflexive: $(0, 1) \approx_1 (0, 1)$ would need $0 + 0 = 1 + 1$, that is, $0 = 2$, which is false. Not transitive: $(0, 1) \approx_1 (1, 0)$ since $0 + 1 = 1 + 0$, and $(1, 0) \approx_1 (0, 1)$ since $1 + 0 = 0 + 1$; transitivity would

give $(0,1) \approx_1 (0,1)$, which we just saw is false. Notice that **every** pair is related to something, since $(a,b) \approx_1 (b,a)$ by commutativity. A symmetric relation in which every element is related to something still need not be reflexive: "x related to y, so y related to x, so x related to x" uses transitivity, and that is what fails here.

(b) **An equivalence relation.** (Eq1) $a + 2b = 2b + a$. (Eq2) $a + 2d = 2b + c$ gives $c + 2b = 2d + a$, by commutativity. (Eq3) Suppose $a + 2d = 2b + c$ and $c + 2f = 2d + e$. Adding, and rearranging by Theorem 1.2.3 (b), (c),

$$(a + 2f) + (2d + c) = (2b + e) + (2d + c),$$

and cancellation (Proposition 1.3.1 (b)) gives $a + 2f = 2b + e$. In school terms the pair (a,b) records $a - 2b$, and the relation says $a - 2b = c - 2d$.

(c) **Reflexive and transitive, not symmetric.** (Eq1) $a + b \leq b + a$, since the two sides are equal. (Eq3) First, for $x, y, u, v \in \mathbb{N}$, $x \leq y$ and $u \leq v$ give $x + u \leq y + v$: by Proposition 1.3.1 (a), $y = x + i$ and $v = u + j$ for some $i, j \in \mathbb{N}$, so $y + v = (x + u) + (i + j)$ by Theorem 1.2.3 (b), (c), and (a) again gives $x + u \leq y + v$. Now let $a + d \leq b + c$ and $c + f \leq d + e$. Adding and rearranging,

$$(a + f) + (c + d) \leq (b + e) + (c + d).$$

If $b + e < a + f$, then $(b + e) + (c + d) < (a + f) + (c + d)$ by Proposition 1.3.1 (c), contradicting the display by trichotomy (Theorem 1.3.2 (a)). So $a + f \leq b + e$, by trichotomy again. Not symmetric: $(0,1) \approx_3 (0,0)$ since $0 + 0 \leq 1 + 0$, that is, $0 \leq 1$; but $(0,0) \approx_3 (0,1)$ would need $0 + 1 \leq 0 + 0$, that is, $1 \leq 0$, which is false since $0 < 1$ (Theorem 1.3.2 (d)).

Exercise. B2: A class in set-builder form

Let $(a,b) \in \mathbb{N} \times \mathbb{N}$. Describe the class $[(a,b)]$ by a set-builder expression, in the two cases $b \leq a$ and $a \leq b$. Hence describe $[(7,2)]$ and $[(2,7)]$.

Solution. Case $b \leq a$. Write $a = b + k$ with $k \in \mathbb{N}$ (Proposition 1.3.1 (a)). We claim

$$[(a,b)] = \{(y + k, y) : y \in \mathbb{N}\}.$$

A pair (x,y) lies in $[(a,b)]$ when $x + b = y + a$, that is, $x + b = (y + k) + b$ by Theorem 1.2.3 (c), (b). By cancellation (Proposition 1.3.1 (b)) this holds if and only if $x = y + k$; the converse direction is substitution. So the elements of the class are exactly the pairs $(y + k, y)$, $y \in \mathbb{N}$.

Case $a \leq b$. Write $b = a + k$ (Proposition 1.3.1 (a)). In the same way, $(x,y) \sim (a,b)$ means $x + (a + k) = y + a$, which by rearranging and canceling a holds if and only if $y = x + k$. So $[(a,b)] = \{(x, x + k) : x \in \mathbb{N}\}$.

The two descriptions agree when $a = b$ (then $k = 0$), as they must. For the examples, $7 = 2 + 5$, so $[(7,2)] = \{(y + 5, y) : y \in \mathbb{N}\}$, and $[(2,7)] = \{(x, x + 5) : x \in \mathbb{N}\}$.

2.1.6.3 C. Going deeper

Exercise. C1: Exit check: the relation that builds $\mathbb{Z}$

Let $\sim$ be the relation of Definition 2.1.1 on $\mathbb{N} \times \mathbb{N}$.

(a) Prove that $\sim$ is an equivalence relation.

(b) Prove that if $(a, b) \sim (a', b')$ and $(c, d) \sim (c', d')$, then $(a + c, b + d) \sim (a' + c', b' + d')$.

(c) Say in one sentence what (b) means for a rule that adds two integers by adding representatives coordinatewise.

Name the property of $\mathbb{N}$ behind each step.

Solution.

(a) This is Proposition 2.1.1; here is the proof once more. Reflexive: $a + b = b + a$ by commutativity. Symmetric: if $a + d = b + c$, then $c + b = d + a$ by commutativity. Transitive: let $a+d = b+c$ and $c+f = d+e$. Adding the two equations and rearranging by associativity and commutativity gives $(a + f) + (c + d) = (b + e) + (c + d)$, and cancellation of $c + d$ (Proposition 1.3.1 (b)) gives $a + f = b + e$.

(b) The hypotheses are $a + b' = b + a'$ and $c + d' = d + c'$. Adding them,

$$(a + b') + (c + d') = (b + a') + (d + c').$$

By associativity and commutativity (Theorem 1.2.3 (b), (c)), the left side is $(a + c) + (b' + d')$ and the right side is $(b + d) + (a' + c')$. So

$$(a + c) + (b' + d') = (b + d) + (a' + c'),$$

which is the cross-sum equation for $(a + c, b + d) \sim (a' + c', b' + d')$.

(c) The class of $(a + c, b + d)$ does not depend on which representatives of $[(a, b)]$ and $[(c, d)]$ were used, so "add representatives coordinatewise, then take the class" is **well defined**: it assigns exactly one integer to each pair of integers.

Marking scheme. (10 points)

- 2: reflexive and symmetric, each citing commutativity of $+$ in $\mathbb{N}$.
- 3: transitivity: combines the two hypotheses into one equation with a common summand on both sides (by adding them, or by substituting one into the other) and removes that summand by **cancellation** in $\mathbb{N}$, named as such.
- 2: in (b), combines the two hypotheses (by adding them, or by substituting both into a rearranged sum).
- 2: in (b), rearranges by associativity and commutativity to the cross-sum equation for the two sums.
- 1: (c) concludes that the coordinatewise sum is well defined on classes.

A proof of (b) that changes one argument at a time (first (a, b) to (a', b') with (c, d) fixed, then the other) and joins the two by transitivity earns the four points of (b) if each step is justified. Fatal errors (0 for the step): subtracting in $\mathbb{N}$ (for instance "subtract d from both sides") without showing that the difference exists; proving only reflexivity and symmetry and calling the relation an equivalence relation.

2.2 Well-Definedness, and the Arithmetic of the Integers

Section 2.1 built $\mathbb{Z}$ as the set of classes of $\sim$ on $\mathbb{N} \times \mathbb{N}$, and gave each integer a normal form (Proposition 2.1.2). The integers cannot yet be added. Every operation on classes will be defined by a formula on representatives, and *Sets and Functions*, Section 5.2 showed what can go wrong: a formula that gives two answers for one class (Sets and Functions, Example (A rule on classes that is not a function)). This section proves, once and for all, the lemma that says when such a formula defines a function. It then uses the lemma for the sum, product and negative of integers, and finds a copy of $\mathbb{N}$ inside $\mathbb{Z}$.

2.2.1 Definitions through representatives

In *Sets and Functions*, Section 5.2 the addition of $\mathbb{Z}/n$ was defined by $[a] + [b] \coloneqq [a + b]$, and Sets and Functions, Proposition 5.2.5 checked that the right side depends only on the classes, not on the integers a and b used to name them. In Sets and Functions, Example (A rule on classes that is not a function) the same kind of formula, $[a]_6 \mapsto [a]_4$, gave one class two values, and so defined no function.

The same situation will recur throughout this volume: for $\mathbb{Z}$ in this section, for $\mathbb{Q}$ in Section 3.1, and for $\mathbb{R}$ later in this volume. Each time, a set is a quotient $X/\!\sim$, and something is defined on it by a formula on elements of X. Rather than repeat the argument of Sets and Functions, Proposition 5.2.5 for every formula, we separate it into two parts: a general lemma, proved here once, which turns a **check on representatives** into a function on classes; and the check itself, which is different for each formula and is all that remains to do each time.

A rule "$[x] \mapsto$ something computed from x" is called **well defined** when representatives of the same class always give the same result. The lemma below says that being well defined is enough, in three settings: a map from one quotient to another, a binary operation on a quotient, and a property of classes.

2.2.2 The lemma

A rule on classes is a function as soon as equivalent inputs give equivalent outputs.

Theorem 2.2.1. Functions on Classes

Let $\sim$ be an equivalence relation on a set X, and $\approx$ an equivalence relation on a set Y. Classes are taken for $\sim$ in X and for $\approx$ in Y.

(a) Let $f\colon X \to Y$ be a function such that, **for all** $x, x' \in X$, $x \sim x'$ implies $f(x) \approx f(x')$. Then there is **exactly one** function $F\colon X/\!\sim \to Y/\!\approx$ with $F([x]) = [f(x)]$ for every $x \in X$.

(b) Let $g\colon X \times X \to X$ be a function such that, **for all** $x, x', y, y' \in X$,

$$\begin{aligned} x \sim x' &\implies g(x,y) \sim g(x',y), \\ y \sim y' &\implies g(x,y) \sim g(x,y'). \end{aligned}$$

Then there is **exactly one** function $G\colon (X/\!\sim) \times (X/\!\sim) \to X/\!\sim$ with $G([x],[y]) = [g(x,y)]$ for all $x, y \in X$.

(c) Let $S \subseteq X$ be such that, for all $x, x' \in X$, $x \sim x'$ and $x \in S$ imply $x' \in S$. Let $\bar{S} \coloneqq \{[x] : x \in S\} \subseteq X/\sim$. Then, **for every** $x \in X$, $[x] \in \bar{S}$ if and only if $x \in S$.

In words: (a) says that a map on elements that respects the two relations passes to a map on classes; (b) says that a binary operation that respects $\sim$ **in each argument separately** passes to an operation on classes; (c) says that a property that respects $\sim$ can be asked of a class, and the answer is the answer for any of its representatives. Clause (a) will give the negative of an integer, (b) its sum and product, and (c) the positive integers in Section 2.4.

Idea. Clause (a) is Sets and Functions, Example (Method: checking that a definition on classes is well defined) turned into a theorem: steps (1)–(3) of that method are its hypothesis, and step (4) is the graph built below. A function is a graph with **exactly one** output for each input (Sets and Functions, Definition 2.1.1). The obvious graph is the set of all pairs $([x], [f(x)])$. It has at least one output for each class, because every class is some $[x]$. The only danger is two outputs for one class, coming from two representatives x and x' of it; Sets and Functions, Theorem 5.2.1 (b) turns "same class" into $x \sim x'$, the hypothesis turns that into $f(x) \approx f(x')$, and Sets and Functions, Theorem 5.2.1 (b) turns it back into "same output". For (b) the same plan works, once the two one-argument hypotheses are chained into one statement about both arguments by transitivity.

Proof. *(a)* Let

$$G_F \coloneqq \{(C, D) \in (X/\sim) \times (Y/\approx) : C = [x] \text{ and } D = [f(x)] \text{ for some } x \in X\},$$

a set by specification (Sets and Functions, Definition 1.2.3). We check the two clauses of Sets and Functions, Definition 2.1.1. (Fn1): every $C \in X/\sim$ is $[x]$ for some $x \in X$ (Sets and Functions, Definition 5.2.4), and then $(C, [f(x)]) \in G_F$. (Fn2): suppose $(C, D) \in G_F$ and $(C, D') \in G_F$, say $C = [x]$, $D = [f(x)]$, and $C = [x']$, $D' = [f(x')]$. Then $[x] = [x']$, so $x \sim x'$ by Sets and Functions, Theorem 5.2.1 (b). By the hypothesis, $f(x) \approx f(x')$, so $[f(x)] = [f(x')]$ by Sets and Functions, Theorem 5.2.1 (b), that is, $D = D'$. So F, the function with graph G_F and codomain $Y/\approx$, is a function $X/\sim \to Y/\approx$, and $F([x]) = [f(x)]$ because $([x], [f(x)]) \in G_F$.

For uniqueness, let $F' \colon X/\sim \to Y/\approx$ also satisfy $F'([x]) = [f(x)]$ for every x. Every $C \in X/\sim$ is some $[x]$, and then $F'(C) = [f(x)] = F(C)$. The two functions have the same domain and codomain, so $F' = F$ by Sets and Functions, Proposition 2.1.1.

(b) First, for all $x, x', y, y' \in X$ with $x \sim x'$ and $y \sim y'$,

$$g(x, y) \sim g(x', y) \quad \text{and} \quad g(x', y) \sim g(x', y'),$$

by the first hypothesis (with y fixed) and the second (with x' fixed). By transitivity, (Eq3) of Sets and Functions, Definition 5.2.1, $g(x, y) \sim g(x', y')$.

The rest is the proof of (a) with pairs of classes in place of classes: the graph of all $(([x], [y]), [g(x, y)])$ has an output for each pair of classes, since each pair of classes is $([x], [y])$ for some x, y (Sets and Functions, Definition 1.4.2, Sets and Functions, Definition 5.2.4), and only one, by the paragraph above, Sets and Functions, Definition 1.4.1 (equal pairs of classes have equal entries) and Sets and Functions, Theorem 5.2.1 (b); so it defines a function G, and G is unique as in (a) (Sets and Functions, Proposition 2.1.1).

Second reading · *The graph of G, checked as in (a)*

Now let G_G be the set of $(P, E) \in ((X/\sim) \times (X/\sim)) \times (X/\sim)$ such that $P = ([x], [y])$ and $E = [g(x, y)]$ for some $x, y \in X$, a set by specification. (Fn1): an element of $(X/\sim) \times (X/\sim)$ is a pair (C, D) of classes (Sets and Functions, Definition 1.4.2), and $C = [x]$, $D = [y]$ for some $x, y \in X$ (Sets and Functions, Definition 5.2.4); then $((C, D), [g(x, y)]) \in G_G$. (Fn2): suppose (P, E) and (P, E') lie in G_G, with $P = ([x], [y])$, $E = [g(x, y)]$, and $P = ([x'], [y'])$, $E' = [g(x', y')]$. From $([x], [y]) = ([x'], [y'])$ the characteristic property of pairs (Sets and Functions, Definition 1.4.1) gives $[x] = [x']$ and $[y] = [y']$, so $x \sim x'$ and $y \sim y'$ by Sets and Functions, Theorem 5.2.1 (b). By the first paragraph, $g(x, y) \sim g(x', y')$, so $E = E'$ by Sets and Functions, Theorem 5.2.1 (b). So G_G, with codomain $X/\sim$, defines a function G with $G([x], [y]) = [g(x, y)]$. Uniqueness is as in (a), by Sets and Functions, Proposition 2.1.1.

(c) ($\Leftarrow$) If $x \in S$, then $[x] \in \bar{S}$ by the definition of $\bar{S}$. ($\Rightarrow$) If $[x] \in \bar{S}$, then $[x] = [x']$ for some $x' \in S$. By Sets and Functions, Theorem 5.2.1 (b), $x \sim x'$, so $x' \sim x$ by (Eq2). The hypothesis, applied to $x' \sim x$ and $x' \in S$, gives $x \in S$.

This proves the theorem.

■

The functions F and G are said to be **induced** by f and g. In practice we write the conclusion of (a) as "the rule $[x] \mapsto [f(x)]$ is well defined", and of (b) as "the operation $[x] * [y] := [g(x, y)]$ is well defined", and the work is always the check of the hypotheses.

Remark. **Each hypothesis has a job.** Without the hypothesis of (c) the conclusion fails. On $\mathbb{N} \times \mathbb{N}$ with the relation $\sim$ of Definition 2.1.1, let $S := \{(a, b) : 0 < a\}$. Then $(1, 1) \in S$, so $[(1, 1)] \in \bar{S}$; and $[(0, 0)] = [(1, 1)]$, so $[(0, 0)] \in \bar{S}$, while $(0, 0) \notin S$. Membership in $\bar{S}$ does not tell us whether a particular representative lies in S. The hypotheses of (a) and (b) are needed in the same way, and the examples below show them failing.

2.2.3 Sum, product and negative of integers

To guess the formulas, borrow the school picture once more. If (a, b) records "$a - b$" and (c, d) records "$c - d$", then school algebra gives

$$\begin{aligned} (a - b) + (c - d) &= (a + c) - (b + d), \\ (a - b)(c - d) &= (ac + bd) - (ad + bc), \\ -(a - b) &= b - a. \end{aligned}$$

These identities are a **heuristic**: they come from the school integers, which this chapter does not assume, and they are used only to find the definition. Each right side is again a difference of two natural numbers, so it can be recorded as a pair.

Definition 2.2.1. Sum, Product and Negative of Integers

For all $(a,b),(c,d) \in \mathbb{N}\times\mathbb{N}$, let

$$\begin{aligned}[(a,b)]+[(c,d)] &\coloneqq [(a+c,\ b+d)],\\ [(a,b)]\cdot[(c,d)] &\coloneqq [(ac+bd,\ ad+bc)],\\ -[(a,b)] &\coloneqq [(b,a)].\end{aligned}$$

The **sum** $+\colon \mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}$ and **product** $\cdot\colon \mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}$ are the functions that Theorem 2.2.1 (b), with $X=\mathbb{N}\times\mathbb{N}$ and the equivalence relation $\sim$ of Proposition 2.1.1, induces from the maps $g_+,\ g_\cdot\colon (\mathbb{N}\times\mathbb{N})\times(\mathbb{N}\times\mathbb{N})\to\mathbb{N}\times\mathbb{N}$ given by the pairs on the right; the **negative** $-\colon\mathbb{Z}\to\mathbb{Z}$ is the function that Theorem 2.2.1 (a) induces from $(a,b)\mapsto(b,a)$. We also write xy for $x\cdot y$.

In words: to add integers, add representatives coordinatewise; to multiply, use the cross formula; to negate, swap the coordinates. All the arithmetic on the right happens in $\mathbb{N}$, where it is always possible. The definition is legitimate only if the hypotheses of Theorem 2.2.1 hold for the three maps on pairs; the next proposition checks them, and until then the three formulas are rules, not yet functions.

The proof below uses the move this chapter is built on, after Sets and Functions, Example (Method: checking that a definition on classes is well defined):

Example. Method: check on representatives, one argument at a time

Template. To show that $[x]*[y]\coloneqq[g(x,y)]$ is well defined: (1) fix y, take $x\sim x'$, and derive $g(x,y)\sim g(x',y)$ from the one equation that $x\sim x'$ unwinds to; (2) fix x', take $y\sim y'$, and derive $g(x',y)\sim g(x',y')$; if $g(x,y)=g(y,x)$ always, (2) is (1) with the roles exchanged; (3) conclude by Theorem 2.2.1 (b). A rule on single classes needs only (1), and then clause (a).

When to use it. Every operation on a quotient given by a formula on representatives: the sum and product of $\mathbb{Z}$, and every later quotient built the same way.

The usual failure. Checking one numerical instance; using a law of $\mathbb{Z}$ to check an operation of $\mathbb{Z}$ (the computation stays in $\mathbb{N}$); claiming symmetry for a formula that lacks it, such as $[(a+d,\ b+c)]$.

Quick check. Does $[(a,b)]\mapsto[(a+a,\ b)]$ define a function $\mathbb{Z}\to\mathbb{Z}$?

Solution. No. $(0,0)\sim(1,1)$, since $0+1=0+1$, but the outputs are $(0+0,0)=(0,0)$ and $(1+1,1)=(2,1)$, and $(0,0)\not\sim(2,1)$, since $0+1\neq 0+2$. In school terms the rule computes $2a-b$, which is not a function of $a-b$.

Proposition 2.2.2. The Operations on Integers Are Well Defined

The maps g_+ and $g_\cdot$ of Definition 2.2.1 satisfy the hypotheses of Theorem 2.2.1 (b), and the map $(a,b)\mapsto(b,a)$ satisfies those of Theorem 2.2.1 (a), with $\sim$ on both sides. Hence the sum, product and negative of Definition 2.2.1 are well-defined functions.

Idea (Scratch work). The sum is one line; the product is the real check. Take the first argument (step (1) of the box):

Givens	Goal
$(a,b) \sim (a',b')$; (c,d) fixed	$g.((a,b),(c,d)) \sim g.((a',b'),(c,d))$
$a+b' = b+a'$	$(ac+bd)+(a'd+b'c) = (ad+bc)+(a'c+b'd)$
every term of the goal contains c or d; the given contains neither	multiply the given by c, and, read from right to left, by d
$ac+b'c = bc+a'c$ and $bd+a'd = ad+b'd$	add the two and regroup: the goal

Both formulas are unchanged when the two input pairs are exchanged, so step (2) comes free.

Before reading on: which hypothesis of Theorem 2.2.1 (b) does this table check?

Proof. Let $(a,b),(a',b'),(c,d) \in \mathbb{N}\times\mathbb{N}$ with $(a,b)\sim(a',b')$, that is,

$$a+b' = b+a'. \tag{2.2.1}$$

All rearrangements below use Theorem 1.2.3: commutativity and associativity of $+$, commutativity of $\cdot$, and distributivity.

Sum, first argument. We need $(a+c,b+d)\sim(a'+c,b'+d)$, that is, $(a+c)+(b'+d) = (b+d)+(a'+c)$. The left side is $(a+b')+(c+d)$, which by Eq. 2.2.1 equals $(b+a')+(c+d)$, and that is the right side rearranged.

Product, first argument. We need

$$(ac+bd,\ ad+bc) \sim (a'c+b'd,\ a'd+b'c),$$

that is, the cross-sum equation

$$(ac+bd)+(a'd+b'c) = (ad+bc)+(a'c+b'd). \tag{2.2.2}$$

Multiply both sides of Eq. 2.2.1 by c and use distributivity: $ac+b'c = bc+a'c$. Multiply both sides of Eq. 2.2.1, read from right to left, by d: $bd+a'd = ad+b'd$. Adding these two equations,

$$(ac+b'c)+(bd+a'd) = (bc+a'c)+(ad+b'd).$$

The left side rearranges to the left side of Eq. 2.2.2, and the right side to its right side. This proves Eq. 2.2.2.

Second arguments. By commutativity in $\mathbb{N}$, $g_+((a,b),(c,d)) = (a+c,b+d) = (c+a,d+b) = g_+((c,d),(a,b))$, and likewise

$$g.((a,b),(c,d)) = (ac+bd,\ ad+bc) = (ca+db,\ cb+da) = g.((c,d),(a,b)).$$

So if $(c,d) \sim (c',d')$, then, by the first-argument case applied with the roles of the two pairs exchanged,

$$g((a,b),(c,d)) = g((c,d),(a,b)) \sim g((c',d'),(a,b)) = g((a,b),(c',d'))$$

for $g = g_+$ and $g = g_\cdot$. This is the second hypothesis of Theorem 2.2.1 (b).

Negative. The pair (b,a) is related to (b',a') when $b + a' = a + b'$, and that is Eq. 2.2.1 read from right to left. This is the hypothesis of Theorem 2.2.1 (a).

By Theorem 2.2.1 (b) and (a), the three rules define functions $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ and $\mathbb{Z} \to \mathbb{Z}$, as claimed.

■

The table checks the first hypothesis of Theorem 2.2.1 (b), $g(x,y) \sim g(x',y)$, for the product; symmetry gives the second, and the first paragraph of that proof joins the two by transitivity of $\sim$. Changing both at once is also possible, but the computation then mixes four equations in eight unknowns; one argument at a time keeps each computation to a single hypothesis.

2.2.4 Rules that fail

Every check happens in the hypotheses of the lemma. Here they fail, next to a rule where they hold.

Example. The first coordinate is not a function of the class

(a) Does the rule $[(a,b)] \mapsto a$ define a function $\mathbb{Z} \to \mathbb{N}$?

(b) Does the rule $[(a,b)] \mapsto [(a+1,b)]$ define a function $\mathbb{Z} \to \mathbb{Z}$?

(c) A student argues: "Each integer x has a pair with least first coordinate, by Theorem 1.3.3 applied to the non-empty set of first coordinates of the pairs in x; call that least first coordinate $h(x)$. Then h is a function $\mathbb{Z} \to \mathbb{N}$, since each x determines its own least first coordinate. So $[(a,b)] \mapsto a$ is a function, namely h, with $h([(a,b)]) = a$ for every (a,b)." Find the error.

Solution.

(a) No. $(0,2) \sim (1,3)$, since $0 + 3 = 2 + 1$, so $[(0,2)] = [(1,3)]$ by Sets and Functions, Theorem 5.2.1 (b). Read with the representative $(0,2)$, the rule gives 0; read with $(1,3)$, it gives 1. One input gets two values, and (Fn2) of Sets and Functions, Definition 2.1.1 fails. The hypothesis of Theorem 2.2.1 (a), with $f(a,b) := a$ and equality on $\mathbb{N}$ as $\approx$ (so that $f(x) \approx f(x')$ means $f(x) = f(x')$), fails at $x = (0,2)$, $x' = (1,3)$.

(b) Yes. Let $(a,b) \sim (a',b')$, so $a + b' = b + a'$. Then

$$(a+1) + b' = (a + b') + 1 = (b + a') + 1 = b + (a' + 1)$$

by Theorem 1.2.3 (b), (c), which is $(a+1,b) \sim (a'+1,b')$. By Theorem 2.2.1 (a), the rule is a function. In fact it is $x \mapsto x + [(1,0)]$, since $[(a,b)] + [(1,0)] = [(a+1,b+0)] = [(a+1,b)]$.

(c) The conclusion is false, by (a), so some step fails. The function h is fine: the least first coordinate is computed from the set x alone. The failing step is the last clause,

"$h([(a,b)]) = a$ for every (a,b)". It holds only when (a,b) happens to be the pair with least first coordinate. The argument chose a particular representative to define h, and then used an **arbitrary** one. Counterexample to the step: $x = [(1,3)] = [(0,2)]$ contains $(0,2)$, so $h(x) = 0$, while the representative $(1,3)$ has first coordinate 1.

Warning. **A formula that looks right need not be well defined.** "$[(a,b)] \cdot [(c,d)] \coloneqq [(ac,bd)]$, like the sum" is not a definition: $(1,0) \sim (2,1)$, since $1+1 = 0+2$, but with $(c,d) = (1,2)$ the outputs are $(1\cdot 1, 0\cdot 2) = (1,0)$ and $(2\cdot 1, 1\cdot 2) = (2,2)$, and $(1,0) \not\sim (2,2)$, since $1+2 \neq 0+2$. **And well defined is necessary, not sufficient.** The rule $[(a,b)] * [(c,d)] \coloneqq [(ad+bc,\ ac+bd)]$ is well defined, since it is the negative of the product; but it gives $[(0,1)] * [(0,1)] = [(0+0, 0+1)] = [(0,1)]$, so the "square" of the class of $(0,1)$ is that class itself, not $[(1,0)]$. Well-definedness says that a rule is a function; whether it is the right function is a separate question.

2.2.5 The natural numbers inside the integers

The integers were built to contain the natural numbers and more. What $\mathbb{Z}$ contains is a **copy** of $\mathbb{N}$: the classes in normal form $[(n,0)]$, each a set of pairs. Whether a natural number n is itself an element of $\mathbb{Z}$ depends on what the elements of a Peano system are, which the axioms leave open (Section 1.5); the copy is all we use, and it has the same arithmetic.

Proposition 2.2.3. The Natural Numbers Embed in the Integers

Let $\iota\colon \mathbb{N} \to \mathbb{Z}$, $\iota(n) \coloneqq [(n,0)]$.

(a) ι is injective, and for all $m, n \in \mathbb{N}$,

$$\iota(m+n) = \iota(m) + \iota(n), \qquad \iota(mn) = \iota(m)\,\iota(n).$$

(b) For every $x \in \mathbb{Z}$, $x + \iota(0) = x$ and $\iota(1)\cdot x = x$. For every $(a,b) \in \mathbb{N}\times\mathbb{N}$, $[(a,b)] = \iota(a) + (-\iota(b))$.

(c) $\mathbb{Z} = \iota(\mathbb{N}) \cup \{-\iota(n) : n \in \mathbb{N}\}$.

Idea. The identities in (a) and (b) are computations with the formulas of Definition 2.2.1 in which at least one operand is some $\iota(n) = [(n,0)]$, and its zero second coordinate makes most terms vanish. Injectivity and (c) are the two halves of Proposition 2.1.2: (b) of it says that different natural numbers give different classes, and (a) says that every class is in normal form.

Proof. *(a)* If $\iota(m) = \iota(n)$, then $[(m,0)] = [(n,0)]$, so $m = n$ by Proposition 2.1.2 (b); hence ι is injective (Sets and Functions, Definition 2.2.1). By Definition 2.2.1,

$$\iota(m) + \iota(n) = [(m+n,\ 0+0)] = [(m+n,0)] = \iota(m+n),$$

since $0+0=0$ (Definition 1.2.1 (a)), and

$$\iota(m)\,\iota(n) = [(mn+0\cdot 0,\ m\cdot 0+0\cdot n)] = [(mn,0)] = \iota(mn),$$

since $m\cdot 0=0$ (Definition 1.2.1 (b)), $0\cdot n=0$ (Theorem 1.2.3 (e)) and $k+0=k$.

(b) Let $x=[(a,b)]$. Then $x+\iota(0)=[(a+0,b+0)]=[(a,b)]=x$, and

$$\iota(1)\cdot x = [(1\cdot a+0\cdot b,\ 1\cdot b+0\cdot a)] = [(a,b)] = x,$$

by Theorem 1.2.3 (e) ($1\cdot n=n$, $0\cdot n=0$) and $n+0=n$. Finally, $-\iota(b)=-[(b,0)]=[(0,b)]$, so $\iota(a)+(-\iota(b))=[(a+0,\ 0+b)]=[(a,b)]$, using $0+b=b$ (Theorem 1.2.3 (a)).

(c) ($\supseteq$) Both sets on the right consist of integers. ($\subseteq$) Let $x\in\mathbb{Z}$. By Proposition 2.1.2 (a), $x=[(k,0)]$ or $x=[(0,k)]$ for some $k\in\mathbb{N}$. In the first case $x=\iota(k)$. In the second, $x=[(0,k)]=-[(k,0)]=-\iota(k)$ by Definition 2.2.1. So x lies in the right side.

■

Identification. From now on we write n for $\iota(n)$ and regard $\mathbb{N}$ as a subset of $\mathbb{Z}$. This is a convenient abuse of notation, and (a) makes it honest: because ι is injective, different natural numbers stay different; because ι preserves $+$ and $\cdot$, a sum or product of natural numbers is the same whether computed in $\mathbb{N}$ or in $\mathbb{Z}$. One could rebuild $\mathbb{Z}$ so that it contains $\mathbb{N}$ literally, by putting each n in place of $\iota(n)$, but nothing later needs it. The order of $\mathbb{N}$ is matched to an order on $\mathbb{Z}$ in Section 2.4.

With this convention, (b) reads $[(a,b)]=a+(-b)$: the pair (a,b) records the difference it was meant to record. The notation $a-b$ is defined in Section 2.3, where $x-y:=x+(-y)$ is set up in every ring at once. Clause (c) reads

$$\mathbb{Z} = \mathbb{N}\cup\{-n : n\in\mathbb{N}\},$$

which is word for word the definition of the integers in *Sets and Functions*, Section 3.1. There it was a definition inside the familiar real numbers; here it is a theorem about a set built from $\mathbb{N}$. The positive integers keep the definition of *Sets and Functions*, Section 3.1,

$$\mathbb{Z}_{>0} := \{n\in\mathbb{N} : n\neq 0\},$$

which mentions only $\mathbb{N}$, and so is now a subset of $\mathbb{Z}$. In particular $5+x=3$ has the solution $x=[(0,2)]=-2$ in $\mathbb{Z}$, since $[(5,0)]+[(0,2)]=[(5,2)]=[(3,0)]$ by Proposition 2.1.2 (a).

Quick check. Compute $[(2,5)]\cdot[(4,1)]$ from Definition 2.2.1, and write the two factors and the product as n or $-n$ with $n\in\mathbb{N}$.

Solution. $[(2,5)]\cdot[(4,1)]=[(2\cdot 4+5\cdot 1,\ 2\cdot 1+5\cdot 4)]=[(13,22)]$. By Proposition 2.1.2 (a): $5=2+3$, so $[(2,5)]=[(0,3)]=-[(3,0)]=-3$ (the negative of Definition 2.2.1); $4=1+3$, so $[(4,1)]=[(3,0)]=3$; and $22=13+9$, so $[(13,22)]=[(0,9)]=-[(9,0)]=-9$. So the product of -3 and 3 is -9, as school predicts; here it is a computation in $\mathbb{N}$, not a rule of signs.

The integers now have a sum, a product and a negative, each a function on classes, and they

contain a copy of $\mathbb{N}$ with its arithmetic. Section 2.3 checks that these operations obey the usual laws, by reducing each law to a law of $\mathbb{N}$; Section 2.4 orders $\mathbb{Z}$.

2.2.6 Exercises

2.2.6.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Proposition 2.2.2.

(a) Which hypothesis of Theorem 2.2.1 (b) does the computation leading to Eq. 2.2.2 verify?

(b) Run that computation with $(a,b) = (1,3)$, $(a',b') = (4,6)$ and $(c,d) = (2,0)$: check that $(a,b) \sim (a',b')$, compute both outputs, and check that they are equivalent.

(c) Where in the proof are the laws of arithmetic in $\mathbb{N}$ used?

(d) Give the idea of the proof in two sentences.

Solution.

(a) The first: for all x, x', y, $x \sim x'$ implies $g.(x,y) \sim g.(x',y)$, with $x = (a,b)$, $x' = (a',b')$, $y = (c,d)$.

(b) $1 + 6 = 7 = 3 + 4$, so $(1,3) \sim (4,6)$. The outputs are

$$(ac + bd,\ ad + bc) = (2 + 0,\ 0 + 6) = (2,6),$$
$$(a'c + b'd,\ a'd + b'c) = (8 + 0,\ 0 + 12) = (8,12),$$

and $2 + 12 = 14 = 6 + 8$, so they are equivalent. The two intermediate equations are $ac + b'c = bc + a'c$, here $2 + 12 = 6 + 8$, and $bd + a'd = ad + b'd$, here $0 = 0$, since $d = 0$. Both outputs are representatives of -4, and indeed $(-2) \cdot 2 = -4$.

(c) In every rearrangement: commutativity and associativity of $+$ (to regroup the four-term sums), distributivity (to multiply Eq. 2.2.1 by c and by d), and commutativity of $\cdot$ and $+$ (in the second-argument step, to see that both formulas are symmetric in the two input pairs).

(d) A change of representative in one argument is an equation in $\mathbb{N}$, and multiplying it by c and by d and adding gives exactly the cross-sum equation for the two products. The second argument follows by symmetry of the formula, and transitivity (inside Theorem 2.2.1 (b)) combines the two changes.

2.2.6.2 B. Practice

Exercise. B1: Two rules that pass the check

Prove that each rule defines a function $\mathbb{Z} \to \mathbb{Z}$, and identify it in terms of the operations of Definition 2.2.1.

(a) $[(a,b)] \mapsto [(b,a)]$.

(b) $[(a,b)] \mapsto [(a+b,\ a+b)]$.

Solution.

(a) Let $(a,b) \sim (a',b')$, that is, $a+b' = b+a'$. The target $(b,a) \sim (b',a')$ is $b+a' = a+b'$, the same equation read from right to left. By Theorem 2.2.1 (a), the rule is a function. It is the negative $x \mapsto -x$ of Definition 2.2.1.

(b) Every output is the zero class: $(a+b, a+b) \sim (0,0)$, since $(a+b)+0 = (a+b)+0$. So for $(a,b) \sim (a',b')$ the outputs $(a+b, a+b)$ and $(a'+b', a'+b')$ are both related to $(0,0)$, hence to each other by (Eq2) and (Eq3) of Proposition 2.1.1, and Theorem 2.2.1 (a) applies. The function is the constant map $x \mapsto 0$, where $0 = \iota(0) = [(0,0)]$. In terms of Definition 2.2.1 it is $x \mapsto x+(-x)$, since $[(a,b)]+[(b,a)] = [(a+b,\ b+a)] = [(a+b,\ a+b)]$ by Theorem 1.2.3 (b).

Exercise. B2: Determine which

Determine which of the following rules define a function. Justify your answer.

(a) $\mathbb{Z} \to \mathbb{N}$, $[(a,b)] \mapsto a+b$.

(b) $\mathbb{Z} \to \mathbb{Z}$, $[(a,b)] \mapsto [(2a, 2b)]$.

(c) $\mathbb{Z} \to \mathbb{Z}$, $[(a,b)] \mapsto [(a^2, b^2)]$, where $n^2 := n \cdot n$.

Solution.

(a) No. $[(0,0)] = [(1,1)]$, since $0+1 = 0+1$, but the rule gives $0+0 = 0$ for the first representative and $1+1 = 2$ for the second.

(b) Yes. If $a+b' = b+a'$, then by distributivity (Theorem 1.2.3 (d))

$$2a+2b' = 2(a+b') = 2(b+a') = 2b+2a',$$

which is $(2a, 2b) \sim (2a', 2b')$; Theorem 2.2.1 (a) applies. The function is $x \mapsto x+x$: $[(a,b)]+[(a,b)] = [(a+a, b+b)]$, and $2n = (1+1)n = n+n$ by $1+1 = 2$ (Definition 1.2.1 (c)) and Theorem 1.2.3 (d), (e).

(c) No. $[(0,1)] = [(1,2)]$, since $0+2 = 1+1$. The rule gives $[(0,1)]$ for the first representative and $[(1,4)]$ for the second, and $(0,1) \not\sim (1,4)$, since $0+4 = 4 \neq 2 = 1+1$.

Exercise. B3: Prove or disprove

Prove or disprove that each rule defines a function.

(a) $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $([(a,b)], [(c,d)]) \mapsto [(a+d,\ b+c)]$.

(b) $\mathbb{Z} \to \mathbb{Z}$, $[(a,b)] \mapsto [(b,\ a+b)]$.

(c) $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $([(a,b)], [(c,d)]) \mapsto [(ad,\ bc)]$.

(d) $\mathbb{Z} \to \mathbb{Z}$, $[(a,b)] \mapsto [(a \cdot a + b \cdot b,\ a \cdot b + b \cdot a)]$.

Solution.

(a) **True.** The formula is not symmetric in the two input pairs, so both steps of Exam-

ple (Method: check on representatives, one argument at a time) are needed. *First argument:* if $a+b'=b+a'$, then

$$\begin{aligned}(a+d)+(b'+c) &= (a+b')+(c+d)\\ &= (b+a')+(c+d) = (b+c)+(a'+d),\end{aligned}$$

which is $(a+d,b+c)\sim(a'+d,b'+c)$. *Second argument:* if $c+d'=d+c'$, then

$$\begin{aligned}(a+d)+(b+c') &= (a+b)+(d+c')\\ &= (a+b)+(c+d') = (b+c)+(a+d'),\end{aligned}$$

which is $(a+d,b+c)\sim(a+d',b+c')$. The rearrangements are by Theorem 1.2.3 (b), (c), and Theorem 2.2.1 (b) concludes. The rule is $(x,y)\mapsto x+(-y)$, since $[(a,b)]+[(d,c)]=[(a+d,\ b+c)]$.

(b) **False.** $(0,0)\sim(1,1)$, but the outputs are $(0,0+0)=(0,0)$ and $(1,1+1)=(1,2)$, and $(0,0)\not\sim(1,2)$, since $0+2\neq 0+1$. The heuristic predicts the failure: $b-(a+b)=-a$ depends on a, not on $a-b$ alone.

(c) **False.** $(1,0)\sim(2,1)$, since $1+1=0+2$. With $(c,d)=(0,1)$ in the second argument, the outputs are $(1\cdot 1,\ 0\cdot 0)=(1,0)$ and $(2\cdot 1,\ 1\cdot 0)=(2,0)$, and $(1,0)\not\sim(2,0)$, since $1+0\neq 0+2$.

(d) **True.** The output is $g.(x,x)$ for $x=(a,b)$. If $x\sim x'$, then $g.(x,x)\sim g.(x',x)$ and $g.(x',x)\sim g.(x',x')$, by the two hypotheses checked in Proposition 2.2.2; transitivity, (Eq3) of Proposition 2.1.1, joins them, and Theorem 2.2.1 (a) applies. The rule is $x\mapsto x\cdot x$.

2.2.6.3 C. Going deeper

Exercise. C1: Properties of classes, and an operation that half works

(a) Prove Theorem 2.2.1 (c), and give an example, different from the one in the remark after the theorem, of a subset $S\subseteq\mathbb{N}\times\mathbb{N}$ for which its conclusion fails for $\sim$ of Definition 2.1.1.

(b) Give an example of an equivalence relation $\sim$ on $\mathbb{N}$ and a function $g\colon\mathbb{N}\times\mathbb{N}\to\mathbb{N}$ such that the first hypothesis of Theorem 2.2.1 (b) holds and the second fails. Check each property, and show that no function G as in the conclusion of (b) exists for your example.

Solution.

(a) *Proof.* Let $x\in X$. If $x\in S$, then $[x]\in\bar{S}$ by the definition of $\bar{S}$. Conversely, if $[x]\in\bar{S}$, there is $x'\in S$ with $[x]=[x']$; then $x\sim x'$ (Sets and Functions, Theorem 5.2.1 (b)), so $x'\sim x$ by symmetry, and the hypothesis on S, applied to $x'\sim x$ and $x'\in S$, gives $x\in S$.
Example. Let $S:=\{(a,b)\in\mathbb{N}\times\mathbb{N}:a=0\}$. Then $(0,0)\in S$, so $[(0,0)]\in\bar{S}$; and $[(1,1)]=[(0,0)]$, so $[(1,1)]\in\bar{S}$, but $(1,1)\notin S$. The hypothesis fails at $(0,0)\sim(1,1)$. Any example must contain a pair of S and an equivalent pair outside S, since that is exactly the negation of the hypothesis.

(b) *Example.* On $\mathbb{N}$, let $x\sim y$ mean "$x=0$ if and only if $y=0$". This is an equivalence relation: it says that x and y have the same value under the function sending 0 to 0

and every other number to 1, so it is an equivalence relation by Sets and Functions, Example (The fibers of a function). Its two classes are $\{0\}$ and $\mathbb{N} \setminus \{0\}$. Let

$$g(x,y) := \begin{cases} x & \text{if } y = 1, \\ 0 & \text{if } y \neq 1. \end{cases}$$

First hypothesis holds. Let $x \sim x'$ and $y \in \mathbb{N}$. If $y = 1$, then $g(x,y) = x \sim x' = g(x',y)$. If $y \neq 1$, both values are 0, and $0 \sim 0$.

Second hypothesis fails. $1 \sim 2$, since neither is 0; but $g(5,1) = 5$ and $g(5,2) = 0$, and $5 \not\sim 0$.

No induced G. If $G([x],[y]) = [g(x,y)]$ held for all x, y, then, since $[1] = [2]$, we would get $[5] = [g(5,1)] = G([5],[1]) = G([5],[2]) = [g(5,2)] = [0]$, and $[5] \neq [0]$ because $5 \not\sim 0$.

What any example must have. (i) $\sim$ is not equality: some class has two elements $y \sim y'$ with $y \neq y'$, else the second hypothesis holds trivially. (ii) $\sim$ has at least two classes, else every two outputs are related and both hypotheses hold. (iii) For some x and some $y \sim y'$, the values $g(x,y)$ and $g(x,y')$ lie in different classes, while for every y, changing x within its class never moves $g(x,y)$ out of its class. A different answer can be checked against (i)–(iii).

Exercise. C2: One at a time or both at once

Let $\sim$ be an equivalence relation on X and $g\colon X \times X \to X$ a function. Prove that the hypotheses of Theorem 2.2.1 (b) hold if and only if, for all $x, x', y, y' \in X$,

$$x \sim x' \text{ and } y \sim y' \implies g(x,y) \sim g(x',y').$$

Name the property of $\sim$ that each direction uses.

Solution. ($\Rightarrow$) Let $x \sim x'$ and $y \sim y'$. The first hypothesis, with y fixed, gives $g(x,y) \sim g(x',y)$; the second, with x' fixed, gives $g(x',y) \sim g(x',y')$. By **transitivity** (Eq3), $g(x,y) \sim g(x',y')$. This direction is the first paragraph of the proof of Theorem 2.2.1 (b).

($\Leftarrow$) Let $x \sim x'$ and $y \in X$. By **reflexivity** (Eq1), $y \sim y$, so the displayed condition with $y' = y$ gives $g(x,y) \sim g(x',y)$: the first hypothesis. Likewise $x \sim x$ and $y \sim y'$ give $g(x,y) \sim g(x,y')$: the second.

So the two forms are interchangeable. The one-at-a-time form is the one to check, because each check involves a single hypothesis $x \sim x'$; the both-at-once form is the one the proof of the lemma uses.

2.3 Commutative Rings; the Integers Are One

Sections 2.1 and 2.2 built $\mathbb{Z}$ with well-defined operations $+$, $\cdot$ and $-$. Why was it built at all? $\mathbb{N}$ already adds and multiplies by all the usual laws but one: it has no negatives, since $1 + x = x + 1 \neq 0$ for every $x \in \mathbb{N}$ (Theorem 1.2.3 (b), Corollary 1.2.2). This section collects the laws of school algebra into one definition, a **commutative ring**, checks them for $\mathbb{Z}$ by reducing

each to a law of $\mathbb{N}$, and derives the consequences that hold in every commutative ring.

2.3.1 The laws of a ring

Expanding brackets or moving a term across an equation uses a short list of laws: sums and products may be reordered and regrouped, 0 and 1 do nothing, every element has a negative, and products distribute over sums. The list holds in $\mathbb{Z}$, it will hold in $\mathbb{Q}$, and a result proved from the list alone holds wherever the list does. It deserves a name.

A commutative ring is a set where you may add, subtract and multiply by the usual rules; a field is one where you may also divide by anything non-zero.

Definition 2.3.1. Commutative Ring, Field

A **commutative ring** is a set R with two functions, **addition** $+\colon R \times R \to R$ and **multiplication** $\cdot\colon R \times R \to R$ (written $a \cdot b = ab$), such that clauses (F1)–(F7) and (F9) of the following list hold; clause (F8) is used only for fields.

(F1) $a + b = b + a$ **for all** $a, b \in R$.
(F2) $(a + b) + c = a + (b + c)$ **for all** $a, b, c \in R$.
(F3) **There is** an element $0 \in R$ with $a + 0 = a$ **for every** $a \in R$.
(F4) **For every** $a \in R$ there is an element $-a \in R$ with $a + (-a) = 0$.
(F5) $(ab)c = a(bc)$ **for all** $a, b, c \in R$.
(F6) $ab = ba$ **for all** $a, b \in R$.
(F7) **There is** an element $1 \in R$ with $a \cdot 1 = a$ **for every** $a \in R$.
(F8) **For every non-zero** $a \in R$ there is an element $a^{-1} \in R$ with $aa^{-1} = 1$.
(F9) $a(b + c) = ab + ac$ **for all** $a, b, c \in R$.

A **field** is a commutative ring in which, **in addition**, $1 \neq 0$ and (F8) holds. We write $a - b := a + (-b)$, and in a field $a/b := ab^{-1}$ for $b \neq 0$.

In words: (F1)–(F4) say that addition is commutative and associative, with a neutral element 0 and negatives; (F5)–(F7) say the first three things about multiplication; (F9) links the two and lets us expand brackets. (F8) asks for inverses of **non-zero** elements only, and only in a field. The notations $-a$ and a^{-1} anticipate that these elements are unique, which we prove below. These are the axioms (F1)–(F9) of the linear-algebra companion, Chapter 0, and of the ordered fields of *Analysis on the Real Line*, Chapter 1, in the same numbering, except that there (F7) includes $1 \neq 0$.

Examples.

- $\mathbb{Z}$, with the operations of Definition 2.2.1: the next theorem.
- **The zero ring** $R := \{0\}$, with $0 + 0 := 0$, $0 \cdot 0 := 0$ and $1 := 0$. Every axiom other than (F8) is an equation in R, and every such equation reads $0 = 0$. So R is a commutative ring, with $1 = 0$, and not a field. This degenerate ring is why several results below assume $1 \neq 0$.

A non-example by minimal change. The set $\mathbb{N}$, with its addition and multiplication from Chapter 1, satisfies (F1)–(F3), (F5)–(F7) and (F9), by Theorem 1.2.3 and Definition 1.2.1 (a). That one missing clause, (F4), is why $\mathbb{Z}$ had to be built.

Why these clauses. No axiom says that $ab = 0$ forces $a = 0$ or $b = 0$: that is true in $\mathbb{Z}$ but

fails in other commutative rings (the warning below), so it must be proved where it holds. The condition $1 \neq 0$ is kept out of the ring axioms, so that the zero ring is a ring, and put into the field axioms, so that $\{0\}$, where (F8) holds vacuously, is not a field.

Warning. A commutative ring may have **zero divisors**: non-zero a, b with $ab = 0$. The integers modulo 6 of *Sets and Functions*, Section 5.2 (built there from the school integers, with well-defined operations by Sets and Functions, Proposition 5.2.5) form a commutative ring: each of (F1)–(F7) and (F9) passes from the school integers to classes through representatives, a check we do not need and do not carry out here. In this ring $[2] \neq [0]$ and $[3] \neq [0]$, yet $[2][3] = [6] = [0]$. So "$ab = 0$, hence $a = 0$ or $b = 0$" is not a consequence of the ring axioms (F1)–(F7) and (F9). In a field it is, since $b = a^{-1}(ab)$ when $a \neq 0$ (Section 4.1). In an **ordered** ring it is a theorem, proved in Section 2.4 from the order axioms.

2.3.2 The integers form a commutative ring

To prove a law for classes, compute both sides through representatives; the two pairs of natural numbers are then equal or related, and the classes agree.

Theorem 2.3.1. The Integers Form a Commutative Ring

$\mathbb{Z}$, with the addition, multiplication and negation of Definition 2.2.1, is a commutative ring, with zero $[(0,0)]$ and one $[(1,0)]$, and in it $1 \neq 0$.

Idea (Scratch work). Take (F9), the law with the most terms.

Givens	Goal
$x, y, z \in \mathbb{Z}$	$x(y+z) = xy + xz$
$x = [(a,b)]$, $y = [(c,d)]$, $z = [(e,f)]$: every class has a representative (Sets and Functions, Definition 5.2.4)	two classes, each computed by Definition 2.2.1 from these representatives
left: $(a(c+e)+b(d+f),\ a(d+f)+b(c+e))$; right: $((ac+bd)+(ae+bf),\ (ad+bc)+(af+be))$	equal pairs, or related pairs
distribute in $\mathbb{N}$, then regroup four terms	the same pair: the classes are equal

The operations are functions (Proposition 2.2.2), so any representatives may be used. Only (F4) will need $\sim$; only (F5) is long.

Proof. By Proposition 2.2.2, $+$ and $\cdot$ are functions $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$. Let $x, y, z \in \mathbb{Z}$; by Sets and Functions, Definition 5.2.4, $x = [(a,b)]$, $y = [(c,d)]$, $z = [(e,f)]$ with $a, \ldots, f \in \mathbb{N}$. By Definition 2.2.1, each side of each axiom is the class of the pair shown in the table. "Equal" means the same pair of natural numbers, by Theorem 1.2.3 and Definition 1.2.1 (a), (b). "Related" means related by $\sim$ (Definition 2.1.1: $(p,q) \sim (r,s)$ means $p + s = q + r$), so the classes agree by Sets and Functions, Theorem 5.2.1 (b).

Axiom	left side, as a pair	right side, as a pair	why the classes agree
(F1) $x+y=y+x$	$(a+c, b+d)$	$(c+a, d+b)$	equal: $+$ is commutative on $\mathbb{N}$
(F2) $(x+y)+z = x+(y+z)$	$((a+c)+e, (b+d)+f)$	$(a+(c+e), b+(d+f))$	equal: $+$ is associative on $\mathbb{N}$
(F3) $x+[(0,0)]=x$	$(a+0, b+0)$	(a,b)	equal: $m+0=m$
(F4) $x+(-x)=[(0,0)]$	$(a+b, b+a)$	$(0,0)$	related: $(a+b)+0=(b+a)+0$
(F5) $(xy)z=x(yz)$	see below	see below	equal, after expanding (below)
(F6) $xy=yx$	$(ac+bd, ad+bc)$	$(ca+db, cb+da)$	equal: $\cdot$ and $+$ are commutative
(F7) $x\cdot[(1,0)]=x$	$(a\cdot1+b\cdot0, a\cdot0+b\cdot1)$	(a,b)	equal: $n\cdot1=n$, $n\cdot0=0$, $n+0=0+n=n$
(F9) $x(y+z)=xy+xz$	$(a(c+e)+b(d+f), a(d+f)+b(c+e))$	$((ac+bd)+(ae+bf), (ad+bc)+(af+be))$	equal: distribute, then reorder the four terms

For (F5), write $xy=[(g,h)]$ with $g \coloneqq ac+bd$, $h \coloneqq ad+bc$. Then $(xy)z=[(ge+hf, gf+he)]$, and in $\mathbb{N}$

$$ge+hf = ace+bde+adf+bcf,$$
$$gf+he = acf+bdf+ade+bce.$$

On the other side, $yz=[(ce+df, cf+de)]$, and $x(yz)$ is the class of the pair whose coordinates are

$$a(ce+df)+b(cf+de) = ace+adf+bcf+bde,$$
$$a(cf+de)+b(ce+df) = acf+ade+bce+bdf.$$

Both first coordinates are sums of ace, adf, bcf, bde, and both second coordinates of acf, ade, bce, bdf, so the pairs are equal (Theorem 1.2.3 (b)–(g)).

Finally, $1 \neq 0$: if $[(1,0)]=[(0,0)]$, then $1=0$ by Proposition 2.1.2 (b). But $0<1$ in $\mathbb{N}$ (Theorem 1.3.2 (d)), and $<$ is irreflexive (Theorem 1.3.2 (a)), so $1 \neq 0$. This proves the theorem.

■

Seven of the eight rows end in "equal". Only (F4) needs $\sim$: the pair $(a+b, b+a)$ records "$(a+b)-(b+a)$", which is zero, without performing the subtraction, and that is the whole point of passing to classes.

The proof used one move eight times. It has a name.

Example. Method: reduce to the level below

Template. To prove a law about classes whose operations are defined on representatives: (1) write each element as a class $[p]$; (2) compute each side by the defining formulas, with any representatives, since the operations are well defined; (3) compare the two representatives at the level below, by its laws: equal, or related; (4) conclude that the classes are equal (Sets

and Functions, Theorem 5.2.1 (b)).

When to use it. Laws in $\mathbb{Z}$, with $\mathbb{N}$ below; any structure whose operations are formulas in a known ring (Exercise C1 below).

The usual failure. Claiming equal pairs where only $\sim$ holds (the row (F4)); computing with representatives before the operation is known to be a function; using a law of $\mathbb{Z}$ inside the computation in $\mathbb{N}$, which is circular.

Remark. The zero and one of this ring are $[(0,0)] = \iota(0)$ and $[(1,0)] = \iota(1)$ (Proposition 2.2.3), so after the identification of Section 2.2 they are the elements 0 and 1 of $\mathbb{N}$. With $a - b \coloneqq a + (-b)$, the formula $[(a,b)] = \iota(a) + (-\iota(b))$ of Proposition 2.2.3 reads $[(a,b)] = a - b$ for $a, b \in \mathbb{N}$: the subtraction the pair recorded is now performed in $\mathbb{Z}$.

Quick check. Which class is the negative of $[(2,7)]$? Show by reducing to $\mathbb{N}$ that no other class $[(c,d)]$ satisfies $[(2,7)] + [(c,d)] = [(0,0)]$.

Solution. $-[(2,7)] = [(7,2)]$ (Definition 2.2.1). Now suppose $[(2+c,\ 7+d)] = [(0,0)]$. By Sets and Functions, Theorem 5.2.1 (b) and Definition 2.1.1, $(2+c)+0 = (7+d)+0$, so $2+c = 7+d = 2+(5+d)$. Hence $c+2 = (5+d)+2$ by Theorem 1.2.3 (b), and $c = 5+d$ by cancellation (Proposition 1.3.1 (b)). Then $c+2 = (5+d)+2 = d+7$, that is, $(c,d) \sim (7,2)$, and $[(c,d)] = [(7,2)]$ (Sets and Functions, Theorem 5.2.1 (b)). Proposition 2.3.2 (a) below proves the same uniqueness in every commutative ring, from (F1)–(F7) and (F9) alone.

What the definition does not list must be derived, and the derivations work in every commutative ring at once.

Proposition 2.3.2. Algebra in a Commutative Ring

Let R be a commutative ring, and let $a, b \in R$.

(a) The elements 0 and 1 of (F3) and (F7) are unique, the element $-a$ of (F4) is unique, and $-(-a) = a$.
(b) $a \cdot 0 = 0$.
(c) $(-a)b = a(-b) = -(ab)$ and $(-a)(-b) = ab$.
(d) $(a+b)^2 = a^2 + 2ab + b^2$, where $x^2 \coloneqq xx$ and $2 \coloneqq 1+1$.

Idea. To show that something equals 0, or $-(ab)$, show that it satisfies the **defining equation** of that element and let uniqueness, part (a), finish.

Proof.

(a) Suppose 0 and $0'$ both satisfy (F3). Then $0' = 0'+0 = 0+0' = 0$, by (F3) for 0, (F1), and (F3) for $0'$. Likewise, if 1 and $1'$ both satisfy (F7), then $1' = 1' \cdot 1 = 1 \cdot 1' = 1$, by

(F7) and (F6). Next, suppose $a+b=0$ and $a+c=0$. Then

$$\begin{aligned} b &= b+0 = b+(a+c) = (b+a)+c \\ &= (a+b)+c = 0+c = c+0 = c, \end{aligned}$$

by (F3), the second hypothesis, (F2), (F1), the first hypothesis, (F1) and (F3). So $-a$ is the **only** element whose sum with a is 0. Finally, $(-a)+a = a+(-a) = 0$ by (F1) and (F4), so a satisfies the defining equation of $-(-a)$, and uniqueness gives $-(-a) = a$.

(b) Let $z := a \cdot 0$. By (F3) and (F9), $z = a(0+0) = a\cdot 0 + a\cdot 0 = z+z$. Adding $-z$ to both sides,

$$0 = z+(-z) = (z+z)+(-z) = z+\big(z+(-z)\big) = z+0 = z,$$

by (F4), the equation $z = z+z$, (F2), (F4) and (F3). So $a\cdot 0 = 0$.

(c) By (F6) and (F9), $ab+(-a)b = b(a+(-a)) = b\cdot 0 = 0$, the last step by (b). So $(-a)b$ satisfies the defining equation of $-(ab)$, and (a) gives $(-a)b = -(ab)$. By (F6) also $a(-b) = (-b)a = -(ba) = -(ab)$. Hence $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$, using the first identity, the second, and (a).

(d) By (F9) and (F6), $(a+b)(a+b) = (a+b)a + (a+b)b = a^2 + ba + ab + b^2$, and $ba+ab = ab\cdot 1 + ab\cdot 1 = ab(1+1) = 2ab$ by (F6), (F7) and (F9). Regrouping by (F1) and (F2) gives $(a+b)^2 = a^2 + 2ab + b^2$.

■

Every law of Proposition 2.3.2 could also be checked in $\mathbb{Z}$ by the method of Example (Method: reduce to the level below). The next example puts the two routes side by side.

Example. One law, two routes

Prove that $(-x)y = -(xy)$ for all $x, y \in \mathbb{Z}$ in two ways: (a) by reducing to the level below; (b) from (F1)–(F7) and (F9) alone.

Solution.

(a) Let $x = [(a,b)]$ and $y = [(c,d)]$. By Definition 2.2.1, $-x = [(b,a)]$, so

$$(-x)y = [(bc+ad,\ bd+ac)];$$

and $xy = [(ac+bd,\ ad+bc)]$, so $-(xy) = [(ad+bc,\ ac+bd)]$. The two pairs are equal, since $bc+ad = ad+bc$ and $bd+ac = ac+bd$ by commutativity of $+$ in $\mathbb{N}$ (Theorem 1.2.3 (b)). So the classes are equal.

(b) $\mathbb{Z}$ is a commutative ring (Theorem 2.3.1), so Proposition 2.3.2 (c), with $a = x$ and $b = y$, gives $(-x)y = -(xy)$.

Route (a) needs nothing beyond the definitions of the operations, but it proves the law only for $\mathbb{Z}$. Route (b) proves it in every commutative ring at once, which is why Proposition 2.3.2 is stated for rings.

The argument of Proposition 2.3.2 (a), with multiplication in place of addition, shows that in

a **field** a non-zero a has only one inverse: if $ab = 1$ and $ac = 1$, then $b = b \cdot 1 = b(ac) = (ba)c = (ab)c = 1 \cdot c = c$, by (F7), (F5) and (F6). So a^{-1}, and with it a/b, names a single element. And Proposition 2.3.2 (b) explains the zero ring: if $1 = 0$ in a commutative ring, then $a = a \cdot 1 = a \cdot 0 = 0$ for every a, so the ring is $\{0\}$.

Quick check. In a field, let $a \neq 0$. Show that $(a^{-1})^{-1} = a$.

Solution. First, $a^{-1} \neq 0$: otherwise $1 = aa^{-1} = a \cdot 0 = 0$, by (F8) and Proposition 2.3.2 (b), against $1 \neq 0$ in a field. So a^{-1} has an inverse by (F8). And $a^{-1}a = aa^{-1} = 1$ by (F6), so a is an inverse of a^{-1}. By the uniqueness of inverses just proved, it is the inverse: $(a^{-1})^{-1} = a$.

2.3.3 Exercises

2.3.3.1 A. Check your understanding

Exercise. A1

(a) State (F4) with its quantifiers, and say why the zero ring satisfies it.

(b) Name the move by which Theorem 2.3.1 was proved, and say which row of its table needed $\sim$ rather than equality of pairs.

(c) True or false, with a reason: in every commutative ring, $ab = 0$ implies $a = 0$ or $b = 0$.

(d) Prove from (F1)–(F7), (F9) and Proposition 2.3.2 that $(a+b)(a-b) = a^2 - b^2$ in every commutative ring, where $x^2 := xx$.

Solution.

(a) For every $a \in R$ there is an element $-a \in R$ with $a+(-a) = 0$. In $\{0\}$ the only element is 0, and $-0 := 0$ works, since $0 + 0 = 0$.

(b) **Reduce to the level below** (Example (Method: reduce to the level below)). Only (F4) needed $\sim$: the pair $(a + b, b + a)$ is related to $(0, 0)$, but it is not that pair.

(c) False. Give $\mathbb{Z} \times \mathbb{Z}$ the componentwise operations $(x, y) + (u, v) := (x + u,\ y + v)$ and $(x, y)(u, v) := (xu,\ yv)$. It is a commutative ring, with zero $(0, 0)$, one $(1, 1)$ and $-(x, y) = (-x, -y)$: each of (F1)–(F7) and (F9) holds in each coordinate by Theorem 2.3.1. In it $(1, 0)(0, 1) = (1 \cdot 0,\ 0 \cdot 1) = (0, 0)$ by Proposition 2.3.2 (b) and (F6), while $(1, 0) \neq (0, 0)$ and $(0, 1) \neq (0, 0)$, since $1 \neq 0$ in $\mathbb{Z}$. (The integers modulo 6 of the warning after Definition 2.3.1, with $[2][3] = [0]$, illustrate the same, with their ring laws taken for granted.)

(d) By (F9), and then (F6) and (F9) on each summand ($(a+b)a = a(a+b)$, $(a+b)(-b) = (-b)(a+b)$),

$$\begin{aligned}(a+b)(a+(-b)) &= (a+b)a + (a+b)(-b)\\ &= (aa+ab) + ((-b)a + (-b)b).\end{aligned}$$

By Proposition 2.3.2 (c), $(-b)a = -(ba) = -(ab)$, the last step by (F6), and $(-b)b = -(bb)$. Regrouping by (F1) and (F2), the sum is

$$a^2 + (ab + (-(ab))) + (-(b^2)) = a^2 + 0 + (-(b^2)) = a^2 - b^2,$$

by (F4) and (F3), and the definition $u - v := u + (-v)$.

2.3.3.2 B. Practice

Exercise. B1: Three rows in detail

Let $x = [(a,b)]$, $y = [(c,d)]$, $z = [(e,f)]$ be integers. Verify in detail, naming the law of $\mathbb{N}$ used at each step, the rows (F9), (F4) and (F6) of the table in the proof of Theorem 2.3.1.

Solution. All operations are those of Definition 2.2.1, and "Law" refers to Theorem 1.2.3.

(F9). Since $y + z = [(c+e, d+f)]$, $x(y+z) = [(a(c+e)+b(d+f),\ a(d+f)+b(c+e))]$, whose coordinates are $ac + ae + bd + bf$ and $ad + af + bc + be$ by Law (d). And $xy + xz = [((ac+bd)+(ae+bf),\ (ad+bc)+(af+be))]$, whose coordinates are the same sums by Laws (b), (c). So both sides are the class of one pair.

(F4). $x + (-x) = [(a+b, b+a)]$, and $(a+b, b+a) \sim (0,0)$ means $(a+b)+0 = (b+a)+0$ (Definition 2.1.1), true by Definition 1.2.1 (a) and Law (b). So $x + (-x) = [(0,0)]$ by Sets and Functions, Theorem 5.2.1 (b).

(F6). $yx = [(ca+db, cb+da)]$. By Law (f), $ca = ac$, $db = bd$, $cb = bc$, $da = ad$, and by Law (b), $bc + ad = ad + bc$; so the pair is $(ac+bd, ad+bc)$, the pair of xy.

Exercise. B2: Prove or disprove

Prove or disprove.

(a) For all $x, y \in \mathbb{Z}$, $-(x+y) = (-x) + (-y)$.

(b) In every commutative ring, $a \cdot a = a$ implies $a = 0$ or $a = 1$.

(c) The set $\mathbb{N}\times\mathbb{N}$ itself, with $(a,b)+(c,d) := (a+c,\ b+d)$ and $(a,b)\cdot(c,d) := (ac+bd,\ ad+bc)$, is a commutative ring.

(d) In every commutative ring, $a + b = a$ implies $b = 0$.

Solution.

(a) **True**, by reducing to the level below. Let $x = [(a,b)]$ and $y = [(c,d)]$. By Definition 2.2.1, $-(x+y) = -[(a+c,\ b+d)] = [(b+d,\ a+c)]$, and $(-x) + (-y) = [(b,a)] + [(d,c)] = [(b+d,\ a+c)]$. The pairs are equal, so the classes are.

(b) **False.** In the commutative ring $\mathbb{Z} \times \mathbb{Z}$ with componentwise operations (Exercise A1 (c)), $(1,0)(1,0) = (1\cdot 1,\ 0\cdot 0) = (1,0)$ by (F7) and Proposition 2.3.2 (b); and $(1,0)$ is neither the zero $(0,0)$ nor the one $(1,1)$, since $1 \neq 0$ in $\mathbb{Z}$. (In the integers modulo 6, ring laws taken for granted as in the warning after Definition 2.3.1, $[3][3] = [9] = [3]$ illustrates the same.)

(c) **False:** (F4) fails. A zero $z = (z_1, z_2)$ would satisfy $(a,b)+z = (a,b)$ for all pairs; taking $(a,b) = (0,0)$ gives $(0,0)+z = (0+z_1,\ 0+z_2) = z$ by Theorem 1.2.3 (a), so $z = (0,0)$. But $(1,0) + (x,y) = (1+x,\ y) \neq (0,0)$ for every pair (x,y), since $1 + x = x + 1 \neq 0$ (Theorem 1.2.3 (b), Corollary 1.2.2). So $(1,0)$ has no negative. Passing to classes is what supplies negatives.

(d) **True.** Add $-a$:

$$b = 0 + b = ((-a) + a) + b = (-a) + (a + b) = (-a) + a = 0,$$

by (F1) and (F3), (F1) and (F4), (F2), the hypothesis, and (F1) and (F4) again.

2.3.3.3 C. Going deeper

Exercise. C1: A ring in disguise

On $\mathbb{Z}$ define $x \oplus y := x + y - 1$ and $x \odot y := x + y - xy$.

(a) Find the elements that play the parts of 0 and 1 in (F3) and (F7), and the negative of x in (F4).

(b) Prove (F9) for $\oplus$, $\odot$ directly.

(c) Let $\varphi(x) := 1 - x$. Show that $\varphi(x \oplus y) = \varphi(x) + \varphi(y)$, $\varphi(x \odot y) = \varphi(x)\varphi(y)$ and $\varphi(\varphi(x)) = x$, and use this to prove (F5) for $\odot$ without expanding.

Solution (Hint 1). In (a), solve $x \oplus e = x$ for e.

Solution (Hint 2). In (c), apply φ to both sides of (F5) and use that φ is injective.

Solution. All computations below are in the commutative ring $\mathbb{Z}$ (Theorem 2.3.1), with the rules of Proposition 2.3.2 (a)–(c). We also use $-(u + v) = (-u) + (-v)$: both sides are negatives of $u + v$, by (F1)–(F4), and negatives are unique (Proposition 2.3.2 (a)).

(a) The zero is 1: $x \oplus 1 = x + 1 - 1 = x$. The one is 0: $x \odot 0 = x + 0 - x \cdot 0 = x$. The negative of x is $2 - x$: $x \oplus (2 - x) = x + (2 - x) - 1 = 1$, the zero.

(b) On the left,

$$\begin{aligned} x \odot (y \oplus z) &= x + (y + z - 1) - x(y + z - 1) \\ &= 2x + y + z - 1 - xy - xz. \end{aligned}$$

On the right, $(x \odot y) \oplus (x \odot z) = (x + y - xy) + (x + z - xz) - 1$, which regroups to the same expression.

(c) The three identities:

$$\begin{aligned} 1 - (x + y - 1) &= (1 - x) + (1 - y), \\ 1 - (x + y - xy) &= (1 - x)(1 - y), \\ 1 - (1 - x) &= x. \end{aligned}$$

(For the second, expand the right side: $1 - y - x + xy$.) The third makes φ injective: if $\varphi(x) = \varphi(x')$, then $x = \varphi(\varphi(x)) = \varphi(\varphi(x')) = x'$. Now

$$\begin{aligned} \varphi((x \odot y) \odot z) &= (\varphi(x)\varphi(y))\varphi(z) \\ &= \varphi(x)(\varphi(y)\varphi(z)) = \varphi(x \odot (y \odot z)), \end{aligned}$$

by the second identity twice, (F5) in $\mathbb{Z}$, and the second identity twice more. Injectivity gives $(x \odot y) \odot z = x \odot (y \odot z)$.

The level below is now $\mathbb{Z}$ itself, and φ carries every law of $\mathbb{Z}$ across to $\oplus$, $\odot$.

Marking scheme. (10 points) - 3: (a) finds the zero 1, the one 0 and the negative $2 - x$, each checked by a computation. - 3: (b) expands both sides to one expression, with the laws of $\mathbb{Z}$ used. - 1: (c) verifies the three identities for φ. - 1: (c) shows that φ is injective from $\varphi(\varphi(x)) = x$, or otherwise uses $\varphi(\varphi(x)) = x$ to recover arguments. - 2: (c) carries (F5) across φ and concludes, by injectivity or by applying φ once more ($\varphi \circ \varphi = \mathrm{id}_{\mathbb{Z}}$). Fatal errors (0 for the step): assuming the laws for $\oplus$, $\odot$ in (b); applying φ and concluding equality of the arguments without injectivity. A reader who proves (F5) for $\odot$ by direct expansion earns the 2 points of the last step only if the expansion is complete and correct, and loses none of the others.

2.4 Ordered Rings and the Rules for Inequalities

Section 2.3 made $\mathbb{Z}$ a commutative ring and proved the rules for equations from (F1)–(F9). Inequalities need the same treatment. In school, $-3 < -2$ is read off a number line; here there is no line, only classes of pairs. This section lists four order axioms, shows that an order is the same thing as a set of positive elements, orders $\mathbb{Z}$, and derives the rules for inequalities in every ordered ring. Each rule will come from one observation: $a < b$ is a statement about the single element $b - a$.

2.4.1 Ordered rings

The integers carry an order tied to the operations: adding the same number to both sides keeps an inequality, and a product of positives is positive. We want every rule for inequalities to follow from a short list of axioms, as for equations. The list is the order axioms (O1)–(O4) of *Analysis on the Real Line*, Chapter 1, with "field" relaxed to "commutative ring".

An ordered ring is a ring with a strict order that adding respects, in which a product of positives is positive.

Definition 2.4.1. Ordered Ring, Ordered Field

An **ordered commutative ring** is a commutative ring R (Definition 2.3.1) together with a relation $<$ on R such that, **for all** $a, b, c \in R$:

(O1) (trichotomy) **exactly one** of $a < b$, $a = b$, $b < a$ holds;

(O2) (transitivity) if $a < b$ and $b < c$, then $a < c$;

(O3) (addition) if $a < b$, then $a + c < b + c$;

(O4) (multiplication) if $0 < a$ and $0 < b$, then $0 < ab$.

An **ordered field** is a field that is an ordered commutative ring. We write $b > a$ for $a < b$, and $a \leq b$ for "$a < b$ or $a = b$" (likewise $b \geq a$). An element a is **positive** if $a > 0$, **negative** if $a < 0$, and **non-negative** if $a \geq 0$.

In words: any two elements compare in exactly one way, so $a < a$ never holds; comparisons chain; we may add the same element to both sides; and (O4), the only clause about multiplication, speaks only of **positive** elements.

The degenerate case: the zero ring $\{0\}$ with the **empty** relation is an ordered commutative ring. Only the middle option of (O1) holds, and (O2)–(O4) are vacuous. There $1 = 0$, so any rule concluding $1 > 0$ must assume $1 \neq 0$.

Checking (O1)–(O4) directly means handling pairs of elements. It is easier to say which elements are positive and let $a < b$ mean that $b - a$ is positive. The next proposition says which "positive sets" give an order, and that every order arises so. We call the move **order from a positive set**.

Proposition 2.4.1. Order from a Positive Set

Let R be a commutative ring and $P \subseteq R$ a subset such that:

(P1) if $a, b \in P$, then $a + b \in P$ and $ab \in P$;

(P2) **for every** $a \in R$, **exactly one** of $a \in P$, $a = 0$, $-a \in P$ holds.

Define $a < b$ to mean $b - a \in P$. Then $(R, <)$ is an ordered commutative ring (Definition 2.4.1), and $P = \{a \in R : a > 0\}$.

Conversely, if $(R, <)$ is an ordered commutative ring, then $P := \{a \in R : a > 0\}$ satisfies (P1) and (P2), and for all $a, b \in R$, $a < b$ holds if and only if $b - a \in P$.

The labels (P1), (P2) are local; they are not the clauses of a partition (*Sets and Functions*, Section 5.2) or Peano's axioms (Chapter 1).

Idea. Each order axiom is a statement about one difference. (O1) is (P2) applied to $b - a$, since $-(b - a) = a - b$. (O2) is closure under sums, since $(c - b) + (b - a) = c - a$. (O3) holds since $(b + c) - (a + c) = b - a$. (O4) is closure under products. The algebra is uniqueness of negatives.

Proof. We first record three identities in R, for $a, b, c \in R$:

$$-(b - a) = a - b, \qquad (c - b) + (b - a) = c - a,$$
$$(b + c) - (a + c) = b - a.$$

For the first, $(b - a) + (a - b) = (b + (-b)) + ((-a) + a) = 0$ by (F1), (F2), (F4), and Proposition 2.3.2 (a) gives the identity. The second is regrouping by (F1)–(F4). For the third, $(a + c) + ((-a) + (-c)) = 0$ likewise, so $-(a + c) = (-a) + (-c)$ by Proposition 2.3.2 (a), and the c's cancel. Also $-0 = 0$ (as $0 + 0 = 0$), so $a - 0 = a$; and $b - a = 0$ if and only if $b = a$: one direction is (F4), and $b - a = 0$ gives $b = (b - a) + a = 0 + a = a$.

Now let P satisfy (P1), (P2), and let $a < b$ mean $b - a \in P$.

(O1) Apply (P2) to $b - a$: exactly one of $b - a \in P$, $b - a = 0$, $-(b - a) \in P$ holds. By the facts above, these say $a < b$, $a = b$ and $a - b \in P$, that is, $b < a$. So exactly one of the three holds.

(O2) If $b - a \in P$ and $c - b \in P$, then $c - a = (c - b) + (b - a) \in P$ by (P1).

(O3) If $b-a \in P$, then $(b+c)-(a+c) = b-a \in P$.

(O4) Since $a-0=a$, we have $0<a$ if and only if $a \in P$. So $0<a$ and $0<b$ give $a,b \in P$, hence $ab \in P$ by (P1), that is, $0<ab$. The same equivalence says $P = \{a \in R : a > 0\}$.

Conversely, let $(R,<)$ be an ordered commutative ring and $P := \{a : a > 0\}$. If $a<b$, then $a+(-a) < b+(-a)$ by (O3), that is, $0 < b-a$. If $0<b-a$, then $0+a < (b-a)+a = b$ by (O3), that is, $a<b$. This is the last claim. For (P1), let $a,b>0$. Then $0+b<a+b$ by (O3), so $0<b<a+b$ and $a+b>0$ by (O2); and $ab>0$ by (O4). For (P2), let $a \in R$. By (O1) exactly one of $0<a$, $a=0$, $a<0$ holds, and by the last claim $a<0$ holds exactly when $0-a=-a$ is in P. This proves the proposition.

■

So an order on a commutative ring is the same thing as a positive set. For $\mathbb{Z}$, the natural one is the classes $[(a,b)]$ with $b<a$. Section 2.1 warned that "$b<a$" is a statement about **one** representative; we now check that it is a statement about the class.

2.4.2 The order of the integers

Definition 2.4.2. Order on the Integers

Let $S := \{(a,b) \in \mathbb{N}\times\mathbb{N} : b<a\}$, and

$$P_{\mathbb{Z}} := \{[(a,b)] : (a,b) \in S\} \subseteq \mathbb{Z}.$$

For $x,y \in \mathbb{Z}$, define $x<y$ to mean $y-x \in P_{\mathbb{Z}}$.

Well-definedness. By Theorem 2.2.1 (c), with $X = \mathbb{N}\times\mathbb{N}$, it suffices that $(a,b)\sim(a',b')$ and $b<a$ imply $b'<a'$. Write $a = b+k$ with $k \neq 0$ (Proposition 1.3.1 (a)). Then $a+b' = b+a'$ reads $(b'+k)+b = a'+b$ (Theorem 1.2.3 (b), (c)), and canceling b (Proposition 1.3.1 (b)) gives $a' = b'+k$, so $b'<a'$. Hence, for **every** pair,

$$[(a,b)] \in P_{\mathbb{Z}} \iff b<a.$$

So "positive" may be tested on any representative, as Section 2.1's warning required.

Examples. $[(5,2)] \in P_{\mathbb{Z}}$, since $2<5$; this is the integer 3. $[(2,5)] \notin P_{\mathbb{Z}}$ and $[(4,4)] \notin P_{\mathbb{Z}}$, since neither $5<2$ nor $4<4$. To compare $x=[(3,5)]$ and $y=[(4,1)]$: $y-x = [(4,1)]+[(5,3)] = [(9,4)]$, and $4<9$, so $x<y$, as "$-2<3$" leads us to expect. The degenerate pair $(0,0)$ is not in S, so $0 \notin P_{\mathbb{Z}}$: zero is not positive.

Quick check. Is $[(7,10)] < [(2,4)]$?

Solution. Compute $[(2,4)]-[(7,10)] = [(2,4)]+[(10,7)] = [(12,11)]$ by Definition 2.2.1. Since $11<12$, the difference lies in $P_{\mathbb{Z}}$, so yes: $[(7,10)] < [(2,4)]$, which is $-3<-2$.

Recall from Section 2.2 that $\mathbb{Z}_{>0} := \{n \in \mathbb{N} : n \neq 0\}$, a set of natural numbers, inside $\mathbb{Z}$ through the identification $n = \iota(n)$. The next theorem shows that it is exactly the set of positive integers.

Theorem 2.4.2. The Integers Form an Ordered Ring

$\mathbb{Z}$, with the operations of Definition 2.2.1 and the relation $<$ of Definition 2.4.2, is an ordered commutative ring, and its positive elements are exactly $P_{\mathbb{Z}}$. Moreover:

(a) for $m, n \in \mathbb{N}$, $\iota(m) < \iota(n)$ if and only if $m < n$;

(b) $\mathbb{N} = \{x \in \mathbb{Z} : x \geq 0\}$, and $\mathbb{Z}_{>0} = \{x \in \mathbb{Z} : x > 0\}$;

(c) (**discreteness**) there is no $k \in \mathbb{Z}$ with $0 < k < 1$; hence, for $m, n \in \mathbb{Z}$ with $m < n$, $m + 1 \leq n$.

Idea. By Proposition 2.4.1 it is enough to check (P1) and (P2) for $P_{\mathbb{Z}}$, each a fact about $\mathbb{N}$, reached by reducing to the level below (Example (Method: reduce to the level below)). For products, write $a = b + k$, $c = d + l$: the product's two coordinates differ by exactly $kl \neq 0$. (P2) is trichotomy in $\mathbb{N}$. Parts (a)–(c) follow from the positive classes being the $[(k, 0)]$ with $k \neq 0$.

Proof. By Theorem 2.3.1, $\mathbb{Z}$ is a commutative ring. We check (P1) and (P2) of Proposition 2.4.1 for $P = P_{\mathbb{Z}}$, using the criterion $[(a, b)] \in P_{\mathbb{Z}} \iff b < a$.

(P1), sums. Let $x = [(a, b)]$ and $y = [(c, d)]$ with $b < a$ and $d < c$. Then $x+y = [(a+c, b+d)]$, and

$$b + d < a + d < a + c$$

by Proposition 1.3.1 (c) twice (with commutativity), so $b + d < a + c$ by transitivity (Theorem 1.3.2 (a)). Hence $x + y \in P_{\mathbb{Z}}$.

(P1), products. With the same x, y, write $a = b + k$ and $c = d + l$ with $k, l \neq 0$ (Proposition 1.3.1 (a)). Then $xy = [(ac + bd, ad + bc)]$, and expanding by Theorem 1.2.3,

$$ac + bd = (b + k)(d + l) + bd = bd + bl + kd + kl + bd,$$
$$ad + bc = (b + k)d + b(d + l) = bd + kd + bd + bl.$$

So $ac + bd = (ad + bc) + kl$. Since $k \neq 0$ and $l \neq 0$, $kl \neq 0$ by Proposition 1.3.1 (d), and therefore $ad + bc < ac + bd$ by Proposition 1.3.1 (a). Hence $xy \in P_{\mathbb{Z}}$.

(P2). Let $x = [(a, b)]$. Then $x \in P_{\mathbb{Z}}$ if and only if $b < a$. Next, $x = [(0, 0)]$ if and only if $(a, b) \sim (0, 0)$ (Sets and Functions, Theorem 5.2.1 (b)), that is, $a + 0 = b + 0$, that is, $a = b$. Finally, $-x = [(b, a)] \in P_{\mathbb{Z}}$ if and only if $a < b$. By trichotomy in $\mathbb{N}$ (Theorem 1.3.2 (a)), exactly one of $b < a$, $a = b$, $a < b$ holds. So exactly one of $x \in P_{\mathbb{Z}}$, $x = 0$, $-x \in P_{\mathbb{Z}}$ holds.

By Proposition 2.4.1, $(\mathbb{Z}, <)$ is an ordered commutative ring with positive set $P_{\mathbb{Z}}$.

(a) By Definition 2.2.1, $\iota(n) - \iota(m) = [(n, 0)] + [(0, m)] = [(n, m)]$, which lies in $P_{\mathbb{Z}}$ if and only if $m < n$.

(b) If $x = [(a, b)] > 0$, then $a = b + k$ with $k \neq 0$ (Proposition 1.3.1 (a)), and $x = [(k, 0)] = \iota(k)$ by Proposition 2.1.2 (a). Conversely, for $k \neq 0$, $\iota(k) = [(k, 0)]$ is positive since $0 < k$ (Theorem 1.3.2 (d)). So the positive integers are the $\iota(k)$ with $k \in \mathbb{Z}_{>0}$; adding $0 = \iota(0)$ gives the first equation.

(c) If $0 < k$, then $k = \iota(m)$ with $m \neq 0$ by (b), so $1 \leq m$ (Theorem 1.3.2 (d)) and $1 \leq k$ by (a); then $k < 1$ would give $1 < 1$ by (O2), or contradict (O1) directly. Now let

$m < n$. Then $n - m > 0$ (Proposition 2.4.1), so $n - m \geq 1$, and adding m by (O3) gives $n \geq 1 + m$. This proves the theorem.

■

With the identification of Section 2.2, (a) says that the order of $\mathbb{Z}$ **extends** the order of $\mathbb{N}$.

Two non-examples by minimal change, each breaking exactly one order axiom:

- Replace $<$ by $\leq$. (O2)–(O4) still hold, by the axioms for $<$ and the cases of equality (for (O4), $ab = 0$ when a factor is 0, by Proposition 2.3.2 (b)). But (O1) fails at $a = b = 0$: both "$0 \leq 0$" and "$0 = 0$" hold. The axioms describe a **strict** order.
- Reverse the order: $a \prec b$ means $b < a$. (O1)–(O3) hold, by the same axioms for $<$. But (O4) fails: $0 \prec -1$, since $0 - (-1) = 1 = [(1, 0)] \in P_{\mathbb{Z}}$ (Proposition 2.3.2 (a)), and $(-1)(-1) = 1$ (Proposition 2.3.2 (c)), yet $0 \prec 1$ is false, since $0 - 1 = [(0, 1)] \notin P_{\mathbb{Z}}$. So (O4) fixes which way round the order goes.

2.4.3 Rules for inequalities

The rules below hold in every ordered commutative ring: in $\mathbb{Z}$ now, and in $\mathbb{Q}$ once Section 3.2 orders it. One of them needs multiples first.

Multiples. Let R be a commutative ring and $a \in R$. By Theorem 1.2.1, with $X = R$, starting value 0 and $g(n, y) = y + a$ (and $n + 1 = \mathrm{S}(n)$, Definition 1.2.1 (c)), there is exactly one function $\mathbb{N} \to R$, written $n \mapsto n \cdot a$, with

$$0 \cdot a = 0, \qquad (n + 1) \cdot a = n \cdot a + a \quad \text{for every } n \in \mathbb{N}.$$

So $1 \cdot a = a$ and $2 \cdot a = a + a$. From here to the end of the section, a dot with a natural number on its left means this multiple, and the product of two elements of R is written by juxtaposition, ab. When $R = \mathbb{Z}$, the two readings agree, by Theorem 1.1.2 (b): $0 \cdot a = 0 = 0a$ by Proposition 2.3.2 (b) and (F6), and $n \cdot a = na$ gives $(n + 1) \cdot a = na + a = (n + 1)a$, since $(n + 1)a = a(n + 1) = an + a = na + a$ by (F6), then (F9) with (F7), then (F6) (the sum $n + 1$ is the same in $\mathbb{N}$ and $\mathbb{Z}$ by Proposition 2.2.3 (a)).

Proposition 2.4.3. Rules for Inequalities in an Ordered Ring

Let R be an ordered commutative ring, and let $a, b, c, d \in R$.

(a) $a < b$ if and only if $b - a > 0$, if and only if $-b < -a$.

(b) If $a \leq b$ and $c \leq d$, then $a + c \leq b + d$, and the conclusion is strict if one of the hypotheses is strict.

(c) If $a < b$ and $c > 0$, then $ac < bc$.

(d) If $a < b$ and $c < 0$, then $ac > bc$.

(e) If $a \neq 0$, then $a^2 > 0$. If $1 \neq 0$, then $1 > 0$.

(f) If $ab = 0$, then $a = 0$ or $b = 0$.

(g) If $a \geq 0$ and $b \geq 0$, then $a < b$ if and only if $a^2 < b^2$.

(h) For every $m, k \in \mathbb{N}$, $(m + k) \cdot a = m \cdot a + k \cdot a$. If $1 \neq 0$, then $n \cdot 1 > 0$ for every $n \in \mathbb{N}$ with $n \geq 1$, and $m \cdot 1 < n \cdot 1$ whenever $m, n \in \mathbb{N}$ and $m < n$.

(i) $\leq$ is a total order on R (Sets and Functions, Definition 5.3.1), and its strict part is $<$.

Idea (Scratch work). Every part is a comparison, and by Proposition 2.4.1 a comparison $a < b$ is the statement $b - a > 0$. Positive elements are closed under $+$ and $\cdot$. So the plan is always: write the goal as one difference, and build it from given positive elements. For (c):

Givens	Goal
$a < b$, $c > 0$	$ac < bc$
$b - a > 0$, $c > 0$ (Proposition 2.4.1)	$bc - ac > 0$
$(b - a)c > 0$ by (O4)	$bc - ac = (b - a)c$?
(F9), (F6), Proposition 2.3.2 (c)	yes: the gap closes

Part (d) is (c) with $-c$ in place of c; (e) and (f) apply (O4) to whichever of a, $-a$ is positive. We prove (i) first, since (b) and (g) use it.

Proof.

(i) Reflexivity is $a = a$. Antisymmetry: $a \leq b$, $b \leq a$ and $a \neq b$ give $a < b < a$, so $a < a$ by (O2), against (O1). Transitivity: if $a \leq b \leq c$, either an equality holds and we substitute, or $a < c$ by (O2); and $a < c$ whenever one step is strict. Totality is (O1). Strict part: $a < b$ implies $a \neq b$ by (O1), and $a \leq b$, $a \neq b$ give $a < b$.

(a) The first equivalence is the last claim of Proposition 2.4.1. For the second, $(-a)-(-b) = (-a) + b = b - a$ by Proposition 2.3.2 (a) and (F1), so $-b < -a$ also says $b - a > 0$.

(b) By (O3) (or equality) and (F1), $a + c \leq b + c \leq b + d$; conclude by (i).

(c) By (a), $b - a > 0$. By (O4), $(b - a)c > 0$. By (F9), (F6) and Proposition 2.3.2 (c), $(b - a)c = bc + (-a)c = bc - ac$. So $bc - ac > 0$, and $ac < bc$ by (a).

(d) By (a), $c < 0$ gives $0 - c = -c > 0$. By (c) applied with $-c$, $a(-c) < b(-c)$, that is, $-(ac) < -(bc)$ by Proposition 2.3.2 (c), and then $bc < ac$ by (a) together with $-(-x) = x$.

(e) If $a > 0$, then $a^2 = aa > 0$ by (O4). If $a < 0$, then $-a > 0$ as in (d), and $a^2 = (-a)(-a) > 0$ by Proposition 2.3.2 (c) and (O4). By (O1) one of the two cases occurs when $a \neq 0$. If $1 \neq 0$, then $1 = 1^2 > 0$.

(f) Contrapositive: let $a, b \neq 0$. By (O1) and (a), some $a' \in \{a, -a\}$ and $b' \in \{b, -b\}$ are positive, so $a'b' > 0$ by (O4). By Proposition 2.3.2 (c), $a'b'$ is ab or $-(ab)$; if $ab = 0$, both are 0, against (O1). So $ab \neq 0$.

(g) Let $a, b \geq 0$. ($\Rightarrow$) If $a < b$, then $b > 0$ by (i), and (c) gives $ab < b^2$. Also $a^2 \leq ab$: by (c) if $a > 0$, and by Proposition 2.3.2 (b) if $a = 0$. So $a^2 < b^2$ by (i). ($\Leftarrow$) If $a^2 < b^2$ but not $a < b$, then $b = a$ or $b < a$ by (O1), giving $a^2 = b^2$ or, by ($\Rightarrow$), $b^2 < a^2$; both contradict (O1).

(h) Fix $m \in \mathbb{N}$. We prove the sum law by Theorem 1.1.2 (b) on k, for $P(k)$: "$(m+k) \cdot a = m \cdot a + k \cdot a$". The base case is $(m + 0) \cdot a = m \cdot a + 0 \cdot a$, by Definition 1.2.1 (a), (F3) and $0 \cdot a = 0$. For the step, $P(k)$ gives $(m + (k + 1)) \cdot a = (m + k) \cdot a + a = (m \cdot a + k \cdot a) + a = m \cdot a + (k + 1) \cdot a$, by associativity in $\mathbb{N}$ (Theorem 1.2.3 (c)) with the definition of multiples, $P(k)$, and (F2) with the definition of multiples again.

Now let $1 \neq 0$. By Theorem 1.1.2 (b) on n, for $Q(n)$: "$(n + 1) \cdot 1 > 0$": the base is $1 \cdot 1 = 0 + 1 = 1 > 0$ by (e), and the step is $(n + 2) \cdot 1 = (n + 1) \cdot 1 + 1 > 0$ by (b) and (e).

Every $n \geq 1$ in $\mathbb{N}$ is $j+1$ for some j (Proposition 1.3.1 (a)), so $n \cdot 1 > 0$. Finally, if $m < n$, then $n = m + k$ with $k \neq 0$ (Proposition 1.3.1 (a)), so $k \geq 1$ (Theorem 1.3.2 (d)), and by the sum law and (b),

$$n \cdot 1 = m \cdot 1 + k \cdot 1 > m \cdot 1 + 0 = m \cdot 1.$$

This proves the proposition.

■

The school rules for inequalities between integers are now theorems about $\mathbb{Z}$. The hypothesis $1 \neq 0$ in (e) and (h) cannot be dropped: in the zero ring with the empty order, neither $1 > 0$ nor $1 \cdot 1 > 0$ holds.

Example. Method: turn the comparison into one positive element

Template. To prove an inequality in an ordered ring: (1) rewrite the goal $u < v$ as $v - u > 0$ (Proposition 2.4.1); (2) express $v - u$ by the ring laws as a sum or product of elements known to be positive (or non-negative); (3) conclude by closure, (P1), or by Proposition 2.4.3 (b), (c); (4) translate back.

When to use it. Any rule for inequalities; comparisons in $\mathbb{Z}$ and, later, in every ordered field.

The usual failure. Multiplying by an element whose sign is unknown; treating $\leq$ as $<$ in (O4); using a picture of the number line as a reason.

Warning. (O4) and (c) are about multiplying by something **positive**. The tempting rule "if $a < b$ and $c < d$, then $ac < bd$" is false in $\mathbb{Z}$: $-3 < -1$ and $-3 < -1$, yet $(-3)(-3) = 9$ and $(-1)(-1) = 1$, and $9 > 1$.

With both smaller sides non-negative, the method gives a correct rule.

Example. Multiplying two inequalities

Let R be an ordered commutative ring, and let $a, b, c, d \in R$ with $0 \leq a < b$ and $0 \leq c < d$. Prove that $ac < bd$.

Solution. By the method, the goal is $bd - ac > 0$. Write the difference through elements of known sign:

$$bd - ac = b(d - c) + (b - a)c.$$

Indeed, $b(d - c) = bd - bc$ and $(b - a)c = bc - ac$, by (F9), (F6) and Proposition 2.3.2 (c); adding, the terms $-bc$ and bc cancel by (F1), (F2), (F4) and (F3), leaving $bd - ac$.

Now the signs. From $0 \leq a$ and $a < b$ we get $0 < b$ (Proposition 2.4.3 (i)), and $d - c > 0$ since $c < d$ (Proposition 2.4.3 (a)). So $b(d - c) > 0$ by (O4). Likewise $b - a > 0$. If $c > 0$, then $(b - a)c > 0$ by (O4); if $c = 0$, then $(b - a)c = 0$ by Proposition 2.3.2 (b). Either way $(b - a)c \geq 0$.

A positive element plus a non-negative one is positive: by Proposition 2.4.3 (b), with the strict hypothesis $0 < b(d - c)$, we get $0 = 0 + 0 < b(d - c) + (b - a)c = bd - ac$, where $0 + 0 = 0$

is (F3). By Proposition 2.4.3 (a), $ac < bd$.

The hypotheses $a \geq 0$ and $c \geq 0$ are exactly what make the two summands non-negative; the warning shows what goes wrong without them.

Quick check. Which positive element shows that $a < b$ gives $a + a < b + b$?

Solution. The element $b - a$, taken twice: $(b + b) - (a + a) = (b - a) + (b - a)$ by the ring laws (regrouping by (F1), (F2), with $-(a + a) = (-a) + (-a)$ as in the proof of Proposition 2.4.1). It is a sum of two positive elements, so it is positive by (P1) for the positive set (Proposition 2.4.1, converse), and $a + a < b + b$ by Proposition 2.4.3 (a).

Quick check. Is $\mathbb{Z}$ a field?

Solution. No: $1 + 1$ is non-zero (it is $> 1 > 0$, where $1 > 0$ by Proposition 2.4.3 (e), as $1 \neq 0$ in $\mathbb{Z}$ by Theorem 2.3.1) and has no inverse. Suppose $(1 + 1)x = 1$. Since $(1 + 1)x = x(1 + 1) = x + x$ by (F6), (F9) and (F7), this says $x + x = 1$. If $x \leq 0$, then $x + x \leq 0 < 1$ by Proposition 2.4.3 (b), (e). If $x > 0$, then $x \geq 1$ by discreteness (Theorem 2.4.2 (c)), so $x + x \geq 1 + 1 > 1$ by Proposition 2.4.3 (b), (e). Both contradict $x + x = 1$ by (O1).

2.4.4 Least elements, and a ring that cannot be ordered

The natural numbers are well ordered (Theorem 1.3.3); the integers are not, since $x - 1 < x$ for every x. But a set of integers bounded below can be shifted into $\mathbb{N}$.

Proposition 2.4.4. Least and Greatest Elements of Sets of Integers

Let $S \subseteq \mathbb{Z}$ be **non-empty**. If S has a lower bound $b \in \mathbb{Z}$ (that is, $b \leq s$ for every $s \in S$), then S has a least element. If S has an upper bound $c \in \mathbb{Z}$, then S has a greatest element.

Idea. This is Sets and Functions, Corollary 3.1.5 on the constructed $\mathbb{Z}$, by the same shift into $\mathbb{N}$, with Theorem 1.3.3 in place of Sets and Functions, Theorem 3.1.3.

Proof. Let b be a lower bound and $T := \{s - b : s \in S\}$, a non-empty set. For $s \in S$, $s \geq b$ gives $s - b \geq 0$ by (O3), so $T \subseteq \{x \in \mathbb{Z} : x \geq 0\} = \mathbb{N}$ by Theorem 2.4.2 (b). By Theorem 1.3.3, T has an element m with $m \leq t$ for every $t \in T$, in the order of $\mathbb{N}$ and so in that of $\mathbb{Z}$ (Theorem 2.4.2 (a)). Then $m = s_0 - b$ for some $s_0 \in S$, so $m + b = s_0 \in S$; and for $s \in S$, $s - b \geq m$ gives $s \geq m + b$ by (O3). So $m + b$ is least in S.

If c is an upper bound, then $-S := \{-s : s \in S\}$ has the lower bound $-c$ (Proposition 2.4.3 (a)), so by the first part it has a least element, which is $-s_1$ for some $s_1 \in S$, by the definition of $-S$. For $s \in S$, $-s_1 \leq -s$ gives $s \leq s_1$, by Proposition 2.4.3 (a), or by $-(-s) = s$ (Proposition 2.3.2 (a)) when $-s_1 = -s$. So s_1 is greatest in S. This proves the proposition.

■

Both hypotheses are needed: $\mathbb{Z}$ has no least element, and neither has $\emptyset$, which every integer bounds. The rules of Proposition 2.4.3 also show that some rings carry no order at all.

Example. A field with two elements that cannot be ordered

Let $R := \{0, 1\}$, two distinct elements, with addition and multiplication given by the tables

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Check that R satisfies (F1)–(F9), so that it is a field, and show that no relation $<$ on R makes it an ordered commutative ring.

Solution. The axioms. (F1), (F6): both tables are symmetric. (F3), (F7): the row of 0 in the first table and of 1 in the second repeat the headings. (F4): $-0 = 0$ and $-1 = 1$, since $1 + 1 = 0$. (F8): 1 has inverse 1, since the second table gives the product of 1 and 1 as 1. And $1 \neq 0$.

The three laws with three variables can be checked over the eight triples (a, b, c), or by cases. From the tables, $0x = 0$ and $1x = x$, and $x + y = 1$ exactly when exactly one of x, y is 1. (F2): both sides are 1 exactly when an odd number of a, b, c (one or three) are 1. (F5): both sides are 1 exactly when $a = b = c = 1$. (F9): if $a = 0$, both sides are 0; if $a = 1$, both sides are $b + c$. So R is a field.

No order. If $<$ made R an ordered commutative ring, then, as $1 \neq 0$, Proposition 2.4.3 (h) would give $2 \cdot 1 > 0$. But $2 \cdot 1 = 1 \cdot 1 + 1 = (0 + 1) + 1 = 0$ by the definition of multiples, and $0 > 0$ is forbidden by (O1).

The argument uses only $1 \neq 0$ and $n \cdot 1 = 0$ for some $n \geq 1$. So **no commutative ring with $1 \neq 0$ in which $n \cdot 1 = 0$ for some $n \in \mathbb{N}$, $n \geq 1$, can be ordered**. The integers modulo $n \geq 2$ of *Sets and Functions*, Section 5.2 are such rings (their ring laws taken for granted, as in the warning after Definition 2.3.1), since $n \cdot [1] = [n] = [0]$; *Algebraic Preliminaries*, Chapter 1 rebuilds them on the constructed $\mathbb{Z}$.

2.4.5 Exercises

2.4.5.1 A. Check your understanding

Exercise. A1

Let R be an ordered commutative ring.

(a) Here is a proof that if $a < 0 < b$, then $ab < a^2 + b^2$. Name the axiom, rule or result that justifies each of the five steps. (1) $ab < 0b$. (2) $0b = 0$. (3) $a^2 > 0$ and $b^2 > 0$. (4) $a^2 + b^2 > 0 + 0 = 0$. (5) $ab < a^2 + b^2$.

(b) True or false: "R has no zero divisors." Give a reason.

(c) True or false: "$1 > 0$ in every ordered commutative ring." Give a reason.

(d) Use the method of Example (Method: turn the comparison into one positive element) to show: if $a < b$ and $c \leq 0$, then $ac \geq bc$.

Solution.

(a) (a) Proposition 2.4.3 (c), with $a < 0$ and $b > 0$. (2) Proposition 2.3.2 (b) and (F6). (3) Proposition 2.4.3 (e), as $a, b \neq 0$ by (O1). (4) Proposition 2.4.3 (b), strict, and (F3). (5) (O2).

(b) True, by Proposition 2.4.3 (f); a general commutative ring may have zero divisors (the warning after Definition 2.3.1).

(c) False: in the zero ring with the empty relation, $1 = 0$. It is true when $1 \neq 0$ (Proposition 2.4.3 (e)).

(d) By the method, compute the difference $ac - bc$. By (F9), (F6) and Proposition 2.3.2 (c), $ac - bc = (a - b)c$; and $a - b = -(b - a)$ (the first identity in the proof of Proposition 2.4.1), so $(a - b)c = (-(b - a))c = (b - a)(-c)$ by Proposition 2.3.2 (c). Here $b - a > 0$ by Proposition 2.4.3 (a). If $c < 0$, then $-c > 0$ (Proposition 2.4.3 (a), as $0 - c = -c$), and the product is > 0 by (O4). If $c = 0$, then $-c = 0$ and the product is 0 by Proposition 2.3.2 (b). So $ac - bc \geq 0$: either $bc < ac$ by Proposition 2.4.3 (a), or $ac - bc = 0$, that is, $ac = bc$. In both cases $bc \leq ac$, that is, $ac \geq bc$.

2.4.5.2 B. Practice

Exercise. B1: The wrong positive set

Let $P' := \{[(a, b)] : a < b\} \subseteq \mathbb{Z}$. Show that membership in P' does not depend on the representative, that P' satisfies (P2) of Proposition 2.4.1, and that it does not satisfy (P1). Which part of (P1) fails?

Solution. Independence of representatives. If $(a, b) \sim (a', b')$ and $b = a + k$ with $k \neq 0$, then $b' + a = a + b' = b + a' = (a + k) + a' = (a' + k) + a$ (Theorem 1.2.3 (b), (c)), so $b' = a' + k$ by cancellation (Proposition 1.3.1 (b)), and $a' < b'$ by Proposition 1.3.1 (a). By Theorem 2.2.1 (c), $[(a, b)] \in P'$ if and only if $a < b$.

(P2). For $x = [(a, b)]$: $x \in P'$ iff $a < b$; $x = 0$ iff $a = b$ (as in the proof of Theorem 2.4.2); $-x \in P'$ iff $b < a$. By trichotomy in $\mathbb{N}$ (Theorem 1.3.2 (a)), exactly one holds.

(P1) fails for products. $[(0, 1)] \in P'$, but, by Definition 2.2.1 with $a = c = 0$ and $b = d = 1$ in $[(ac + bd,\ ad + bc)]$,

$$[(0, 1)][(0, 1)] = [(1, 0)],$$

and $[(1, 0)] \notin P'$, since $1 < 0$ is false. The sum half of (P1) holds, as in the proof of Theorem 2.4.2. So P', the negative integers, fails only closure under products; it would

define the reversed order, which breaks exactly (O4).

Exercise. B2: Prove or disprove

Prove or disprove.

(a) In every ordered commutative ring, $a < b$ implies $a^2 < b^2$.

(b) In every ordered commutative ring with $1 \neq 0$, $a^2 + 1 > 0$.

(c) In $\mathbb{Z}$, if $x > 0$ and $xy = x$, then $y = 1$.

(d) In every ordered commutative ring, $a < b$ and $c < d$ imply $a - d < b - c$.

(e) In $\mathbb{Z}$, $x < y$ implies $x + 1 < y$.

Solution.

(a) **False.** In $\mathbb{Z}$, $-3 < 1$, but $(-3)^2 = 9$ and $1^2 = 1$, and $9 > 1$. The true version needs $a \geq 0$ (Proposition 2.4.3 (g)).

(b) **True.** First, $a^2 \geq 0$: if $a = 0$, then $a^2 = 0 \cdot 0 = 0$ by Proposition 2.3.2 (b); otherwise $a^2 > 0$ by Proposition 2.4.3 (e). Next, $1 > 0$ by Proposition 2.4.3 (e), since $1 \neq 0$. By Proposition 2.4.3 (b), with the strict hypothesis $0 < 1$, $0 = 0 + 0 < a^2 + 1$, using (F3).

(c) **True.** By (F9), (F7) and Proposition 2.3.2 (c), $x(y - 1) = xy + x(-1) = xy - x$, and $xy - x = x - x = 0$ by the hypothesis and (F4). Since $x > 0$, $x \neq 0$ by (O1). So $y - 1 = 0$ by Proposition 2.4.3 (f), and $y = (y - 1) + 1 = 1$ by (F1)–(F4).

(d) **True.** From $c < d$, Proposition 2.4.3 (a) gives $-d < -c$. Adding this to $a < b$ by Proposition 2.4.3 (b), strict, gives $a + (-d) < b + (-c)$, that is, $a - d < b - c$.

(e) **False.** Take $x = 0$ and $y = 1$: then $x < y$ (Proposition 2.4.3 (e), as $1 \neq 0$ in $\mathbb{Z}$ by Theorem 2.3.1), but $x + 1 = 1$, and $1 < 1$ is ruled out by (O1). Discreteness gives only $x + 1 \leq y$ (Theorem 2.4.2 (c)).

2.4.5.3 C. Going deeper

Exercise. C1: Ordered rings with $1 \neq 0$ are infinite

(a) Prove the sum law of Proposition 2.4.3 (h) in full: for every ordered commutative ring R, every $a \in R$ and all $m, k \in \mathbb{N}$, $(m + k) \cdot a = m \cdot a + k \cdot a$. State the property you induct on and justify every equality.

(b) Deduce that every ordered commutative ring with $1 \neq 0$ is infinite (Sets and Functions, Definition 4.1.3).

(c) The set $\{0, 1, 2\}$ with addition and multiplication "modulo 3", given by the tables below, is a commutative ring with $1 \neq 0$ (you may take this for granted). Show that it cannot be ordered.

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

Solution (Hint 1). For (b): what does (h) say about $k \mapsto k \cdot 1$?

Solution (Hint 2). Suppose R has m elements; consider $k \mapsto k \cdot 1$ on $\{1, \dots, m+1\}$.

Solution.

(a) Fix R, $a \in R$, $m \in \mathbb{N}$, and use Theorem 1.1.2 (b) on k, for $P(k)$: "$(m+k)\cdot a = m\cdot a + k\cdot a$". *Base:* $(m+0)\cdot a = m\cdot a = m\cdot a + 0 = m\cdot a + 0\cdot a$, by Definition 1.2.1 (a), (F3) and $0\cdot a = 0$. *Step:* assume $P(k)$. Then

$$\begin{aligned}(m+(k+1))\cdot a &= ((m+k)+1)\cdot a = (m+k)\cdot a + a \\ &= (m\cdot a + k\cdot a) + a = m\cdot a + (k\cdot a + a) \\ &= m\cdot a + (k+1)\cdot a,\end{aligned}$$

by associativity in $\mathbb{N}$ (Theorem 1.2.3 (c)), the recursion for multiples, $P(k)$, (F2), and the recursion again. So $P(k+1)$ holds, and $P(k)$ holds for every k.

(b) Let $1 \neq 0$ in R, and suppose R has m elements for some $m \in \mathbb{N}$ (Sets and Functions, Definition 4.1.2). Let $f\colon \{1, \dots, m+1\} \to R$, $f(k) \coloneqq k \cdot 1$. If $k < l$, then $k \cdot 1 < l \cdot 1$ by Proposition 2.4.3 (h), so $k \cdot 1 \neq l \cdot 1$ by (O1): f is injective. But $\{1, \dots, m+1\}$ has $m+1 > m$ elements (by its identity map), so Sets and Functions, Corollary 4.1.3 (a) forbids an injection into R. This contradiction shows that R is infinite (Sets and Functions, Definition 4.1.3).

(c) By the definition of multiples and the tables, $1 \cdot 1 = 0 + 1 = 1$, $2 \cdot 1 = 1 + 1 = 2$ and $3 \cdot 1 = 2 + 1 = 0$. An order would give $3 \cdot 1 > 0$ by Proposition 2.4.3 (h), since $1 \neq 0$, that is, $0 > 0$, against (O1). (Alternatively, (b) applies to this finite ring.)

Marking scheme. (10 points) - 1: (a) states the induction property with m and a fixed and k the induction variable. - 2: (a) base case, citing $m + 0 = m$, $0 \cdot a = 0$ and (F3). - 2: (a) inductive step, with associativity in $\mathbb{N}$, the recursion, the hypothesis and (F2) each named. - 2: (b) shows that $k \mapsto k \cdot 1$ is injective, citing (h) and (O1). - 2: (b) derives the contradiction from Sets and Functions, Corollary 4.1.3 (a) with $m + 1 > m$ elements (or any correct count). - 1: (c) computes $3 \cdot 1 = 0$ and concludes. Fatal errors (0 for the step): inducting on m and k at once without fixing one; using $1 > 0$ without $1 \neq 0$; in (b), "the multiples are all different, so there are infinitely many" with no finiteness result cited (Sets and Functions, Corollary 4.1.3, Sets and Functions, Proposition 4.1.6 with $\mathbb{N}$ infinite, or equivalent). A route for (b) composing the inverse of a bijection $\{1, \dots, m\} \to R$ with f earns the same 4 points. So does a route that shows $k \mapsto k \cdot 1$ is an injection $\mathbb{N} \to R$ (2 points, as above), so that its image is an infinite subset of R, and concludes with Sets and Functions, Proposition 4.1.6 (2 points).

2.5 Division with Remainder

Two facts about the integers of school, taken as known, were proved in the previous volume: every integer is even or odd, and not both (*Sets and Functions*, Section 1.1, Sets and Functions, Theorem 1.1.2), and division with remainder (*Sets and Functions*, Section 3.2, Sets and Functions, Theorem 3.2.3). The integers are now the set of classes built in Section 2.1. Is $[(0, 7)]$, the

class that school calls -7, still odd, and could some integer be even and odd at once? Nothing proved so far says. This section answers by proving division with remainder again, on the constructed $\mathbb{Z}$: the proof is that of *Sets and Functions*, Section 3.2, with each school fact replaced by a result of this chapter. Divisibility comes first, since uniqueness of the remainder rests on it.

Throughout, $\mathbb{Z}$ is the ordered ring of Theorem 2.4.2, with $\mathbb{N} \subseteq \mathbb{Z}$ and $\mathbb{Z}_{>0} = \{n \in \mathbb{Z} : n > 0\}$ (Theorem 2.4.2 (b)), and $a - b \coloneqq a + (-b)$. Numerals $2 \coloneqq 1 + 1$, $3 \coloneqq 2 + 1$, and so on, are the natural numbers of those names, as in Section 2.2; a sum or product of numerals may be computed in $\mathbb{N}$, because the embedding of Proposition 2.2.3 preserves $+$ and $\cdot$, and a sign is moved by Proposition 2.3.2 (c), as in $(-4) \cdot 5 = -(4 \cdot 5) = -20$.

2.5.1 Divisibility on the constructed integers

The definition of divisibility needs nothing that the ring $\mathbb{Z}$ lacks. We restate it, and the definition of a prime, from *Sets and Functions*, Section 1.1, with $\mathbb{Z}$ now the ring of Chapter 2.

Divisibility (Sets and Functions, Definition 1.1.2). Let $a, b \in \mathbb{Z}$. Then a **divides** b, written $a \mid b$, if **there exists** $k \in \mathbb{Z}$ with $b = ka$. Then a is a **divisor** of b, and b is a **multiple** of a. We write $a \nmid b$ when a does not divide b.

Prime (Sets and Functions, Definition 1.1.3). An integer p is **prime** if $p \geq 2$ and the only **positive** divisors of p are 1 and p. A **prime factor** of an integer n is a prime p with $p \mid n$.

Divisibility is a question about existence: is b some integer times a?

Some instances. $3 \mid -12$, since $-12 = (-4) \cdot 3$. $-4 \mid 0$, since $0 = 0 \cdot (-4)$. Even $0 \mid 0$ holds, since $0 = 1 \cdot 0$: the definition asks for **some** k, and here every k works. On the other hand $5 \nmid 12$. That is a statement that **no** $k \in \mathbb{Z}$ has $12 = 5k$, and to prove it we need the theorem on division with remainder below.

The rules below are short consequences of the ring laws and the order.

Proposition 2.5.1. Rules for Divisibility in $\mathbb{Z}$

Let $a, b, c, x, y \in \mathbb{Z}$.

(a) $a \mid 0$, $1 \mid a$ and $a \mid a$. If $0 \mid a$, then $a = 0$.
(b) If $a \mid b$ and $b \mid c$, then $a \mid c$.
(c) If $a \mid b$ and $a \mid c$, then $a \mid bx + cy$.
(d) If $a \mid b$, then $(-a) \mid b$ and $a \mid (-b)$.
(e) If $a \mid b$, $a > 0$ and $b > 0$, then $a \leq b$.
(f) If $a, b \in \mathbb{N}$, $a \mid b$ and $b \mid a$, then $a = b$.
(g) If $a \mid 1$, then $a = 1$ or $a = -1$.
(h) If $a \neq 0$, then **exactly one** of a, $-a$ lies in $\mathbb{Z}_{>0}$.

Idea. Parts (a)–(d) are identities in the commutative ring $\mathbb{Z}$: write each hypothesis as an equation $b = ka$ and rearrange. Part (e) is the one that needs the order. If $b = ka$ with $a, b > 0$, the factor k cannot be zero or negative, so by discreteness it is at least 1, and multiplying a by something at least 1 cannot make it smaller (the method of Example (Method: turn the comparison into one positive element)). Parts (f) and (g) then follow

from (e), and (h) is (P2) for the positives of $\mathbb{Z}$.

Proof. We use the ring laws (F1)–(F7) and (F9) of $\mathbb{Z}$ (Theorem 2.3.1) without further mention. Parts (a)–(d) are one-line identities and (h) restates (P2); their proofs are written out in the fold. After it come (e), which needs the order and discreteness, and the two parts built on it, (f) and (g).

Second reading · *Parts (a)–(d) and (h)*

(a) $0 = a \cdot 0 = 0 \cdot a$ by Proposition 2.3.2 (b) and (F6), so $a \mid 0$ with $k = 0$. Also $a = a \cdot 1$ by (F7), so $1 \mid a$ with $k = a$; and $a = 1 \cdot a$, so $a \mid a$ with $k = 1$. If $0 \mid a$, then $a = k \cdot 0 = 0$ for some $k \in \mathbb{Z}$, by Proposition 2.3.2 (b).

(b) Let $b = ka$ and $c = lb$ with $k, l \in \mathbb{Z}$. Then $c = l(ka) = (lk)a$ by (F5), so $a \mid c$.

(c) Let $b = ka$ and $c = la$. Then

$$bx + cy = (ka)x + (la)y = (kx)a + (ly)a = (kx + ly)a$$

by (F5), (F6) and (F9), and $kx + ly \in \mathbb{Z}$. So $a \mid bx + cy$.

(d) Let $b = ka$. By Proposition 2.3.2 (c), $(-k)(-a) = ka = b$, so $(-a) \mid b$; and $(-k)a = -(ka) = -b$, so $a \mid (-b)$.

(e) Let $a \neq 0$. By the converse part of Proposition 2.4.1, applied to the ordered ring $\mathbb{Z}$, the set $\{x \in \mathbb{Z} : x > 0\}$ satisfies (P2), so exactly one of $a > 0$, $-a > 0$ holds. That set is $\mathbb{Z}_{>0}$ by Theorem 2.4.2 (b).

(e) Let $b = ka$ with $k \in \mathbb{Z}$, $a > 0$ and $b > 0$. We show first that $k > 0$. If $k = 0$, then $b = 0 \cdot a = 0$ by Proposition 2.3.2 (b), against $b > 0$. If $k < 0$, then $ka < 0 \cdot a = 0$ by Proposition 2.4.3 (c) (multiplying $k < 0$ by $a > 0$), again against $b > 0$. So $k > 0$, and by discreteness (Theorem 2.4.2 (c), with $0 < k$) $1 \leq k$. If $k = 1$, then $b = a$. If $1 < k$, then $k - 1 > 0$ by Proposition 2.4.3 (a), so $(k - 1)a > 0 \cdot a = 0$ by Proposition 2.4.3 (c), that is, $b - a = ka + (-1)a = (k - 1)a > 0$ (Proposition 2.3.2 (c), (F7) and (F9)), and $a < b$ by Proposition 2.4.3 (a). In both cases $a \leq b$.

(f) Let $a, b \in \mathbb{N}$ with $a \mid b$ and $b \mid a$. If $a = 0$, then $0 \mid b$, so $b = 0 = a$ by (a). If $b = 0$, then $a = 0$ in the same way. Otherwise $a, b \in \mathbb{N}$ are non-zero, hence positive (Theorem 2.4.2 (b)), and (e) gives $a \leq b$ and $b \leq a$. Since $\leq$ is a total order (Proposition 2.4.3 (i)), it is antisymmetric, and $a = b$.

(g) Let $a \mid 1$, say $1 = ka$. Then $a \neq 0$: otherwise $1 = k \cdot 0 = 0$, against $1 \neq 0$ in $\mathbb{Z}$ (Theorem 2.3.1). By (h), one of a, $-a$ is positive; call it e. Then $e \mid 1$, by (d) if $e = -a$. Since $1 > 0$ (Proposition 2.4.3 (e)), part (e) gives $e \leq 1$; and $e > 0$ gives $1 \leq e$ by discreteness. So $e = 1$. If $e = a$, then $a = 1$; if $e = -a$, then $a = -(-a) = -1$ by Proposition 2.3.2 (a). This proves the proposition.

■

Part (f) is the antisymmetry of divisibility on $\mathbb{N}$, Sets and Functions, Proposition 5.1.1 of *Sets and Functions*, Section 5.1, now proved on $\mathbb{N} \subseteq \mathbb{Z}$ for the constructed integers. On all of $\mathbb{Z}$ it

fails: $2 \mid -2$ and $-2 \mid 2$, by (d), yet $2 \neq -2$.

Quick check. Which hypothesis of (e) fails for $a = 3$, $b = 0$, and is the conclusion $a \leq b$ true there?

Solution. $3 \mid 0$ by (a), and $a = 3 > 0$, but $b = 0$ is not positive. The conclusion $3 \leq 0$ is false. Every integer divides 0, so no bound of the form $a \leq b$ can hold for $b = 0$; that is the reason for the hypothesis $b > 0$.

2.5.2 Division with remainder

Now the substitute for division. Given $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$, we want to write a as a multiple of n plus something small, and we want "small" to make the answer unique. The obvious attempt subtracts n from a until what is left lies in $\{0, \ldots, n-1\}$, adding n instead when a is negative. That is a process; the theorem replaces it by a least element.

Theorem 2.5.2. Division with Remainder in $\mathbb{Z}$

Let $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$.

(a) There are **unique** $q, r \in \mathbb{Z}$ with

$$a = qn + r \quad \text{and} \quad 0 \leq r < n.$$

The integers q and r are the **quotient** and the **remainder** of a on division by n.

Idea. This is Sets and Functions, Theorem 3.2.3, and the proof is that volume's: the method of Sets and Functions, Example (Method: the least element), with S the non-negative integers of the form $a - kn$, and Proposition 2.4.4 in place of the school least-element principle. The least one, r, lies below n, or else $r - n$ would be a smaller one. Uniqueness is Sets and Functions, Example (Method: existence and uniqueness): two remainders differ by a multiple of n and by less than n, and Proposition 2.5.1 (e) forbids a positive multiple of n below n. What is new is only that every step is now a theorem about the constructed $\mathbb{Z}$.

Proof. Read the proof of Sets and Functions, Theorem 3.2.3 with these replacements:

- the closure of $\mathbb{Z}$ under $+$, $-$, $\cdot$ (Sets and Functions, Proposition 3.1.2 (a)) $\to$ Theorem 2.3.1, with the identities $a - an = a(1-n)$ and $r - n = a - (q+1)n$ from the ring laws and Proposition 2.3.2;
- "a positive integer is at least 1" (Sets and Functions, Proposition 3.1.2 (b)) $\to$ discreteness, Theorem 2.4.2 (c);
- Sets and Functions, Corollary 3.1.5 $\to$ Proposition 2.4.4;
- the step "a positive multiple of n is at least n" $\to$ Proposition 2.5.1 (e);
- the school rules for inequalities, and cancelling $n \neq 0$ from $qn = q'n \to$ Proposition 2.4.3.

With these, the proof of *Sets and Functions*, Section 3.2 goes through step for step on the

constructed $\mathbb{Z}$; the fold below writes it out.

↻ **Second reading** · *The proof, line by line on the constructed $\mathbb{Z}$*

(a) *Existence.* Let

$$R := \{a - kn : k \in \mathbb{Z}\} \cap \{x \in \mathbb{Z} : x \geq 0\}.$$

We show $R \neq \emptyset$. If $a \geq 0$, take $k = 0$: $a - 0 \cdot n = a - 0 = a \geq 0$, by (F6), Proposition 2.3.2 (b) and $-0 = 0$ ($0 + 0 = 0$ and Proposition 2.3.2 (a)). If $a < 0$, take $k = a$: then $a - an = a + a(-n) = a(1 - n)$ by Proposition 2.3.2 (c), (F7) and (F9). Since $n > 0$, discreteness (Theorem 2.4.2 (c)) gives $1 \leq n$. If $n = 1$, then $a(1 - n) = a \cdot 0 = 0$ by Proposition 2.3.2 (b). If $1 < n$, then $-n < -1$ by Proposition 2.4.3 (a), so $1 - n < 0$ by (O3); multiplying $a < 0$ by $1 - n < 0$ reverses the inequality (Proposition 2.4.3 (d)), so $a(1 - n) > 0 \cdot (1 - n) = 0$. In every case R has an element.

The set $R \subseteq \mathbb{Z}$ is non-empty and has the lower bound 0, so by Proposition 2.4.4 it has a least element r. Write $r = a - qn$ with $q \in \mathbb{Z}$. Then $a = qn + r$ and $r \geq 0$.

Suppose $r \geq n$. Then $r - n \geq 0$: if $r = n$, then $r - n = 0$ by (F4); if $r > n$, then $r - n > 0$ by Proposition 2.4.3 (a). Also $r - n = a - qn - n = a + (-1)(qn) + (-1)n = a + (-1)((q+1)n) = a - (q + 1)n$, since $-x = (-1)x$ for every x (Proposition 2.3.2 (c) and (F7)) and by (F9), so $r - n \in R$. But $-n < -0 = 0$ (Proposition 2.4.3 (a), since $n > 0$), so $r - n < r$ by (O3). This contradicts the choice of r as least. Hence $r < n$.

Uniqueness. Let $a = qn + r = q'n + r'$ with $q, q', r, r' \in \mathbb{Z}$, $0 \leq r < n$ and $0 \leq r' < n$. Suppose $r \neq r'$. Exchanging (q, r) and (q', r') changes neither the hypotheses nor the goal, so we may suppose $r' < r$. Adding $-r' - qn$ to both sides of $qn + r = q'n + r'$ gives

$$r - r' = q'n - qn = (q' - q)n,$$

by (F1)–(F4), and by Proposition 2.3.2 (c) with (F9) for the last step, so $n \mid r - r'$. Here $r - r' > 0$ by Proposition 2.4.3 (a), and $n > 0$, so Proposition 2.5.1 (e) gives $n \leq r - r'$. But adding $r - r'$ to $0 \leq r'$ gives $r - r' \leq (r - r') + r' = r$ (Proposition 2.4.3 (b)), and $r < n$; so $r - r' < n$, a contradiction. Hence $r = r'$. Then $qn = q'n$ (add $-r$ to both sides of $qn + r = q'n + r$), so $(q - q')n = qn - q'n = 0$ by Proposition 2.3.2 (c) and (F9), and since $n \neq 0$, Proposition 2.4.3 (f) gives $q - q' = 0$. Hence $q = (q - q') + q' = q'$, by (F1)–(F4).

■

For $n = 0$ the statement fails: no r satisfies $0 \leq r < 0$. Exercise B1 handles $n < 0$.

Uniqueness is what turns the theorem into a test for divisibility: $n \mid a$ **if and only if the remainder of a on division by n is** 0. Indeed, if $a = kn$, then $a = kn + 0$ is a division with remainder, and by uniqueness it is the only one; conversely $a = qn + 0$ says $n \mid a$. For instance $12 = 2 \cdot 5 + 2$ with $0 \leq 2 < 5$, so the remainder is $2 \neq 0$, and $5 \nmid 12$.

Quick check. Find the quotient and the remainder of $a = -23$ on division by $n = 4$.

Solution. $q = -6$ and $r = 1$: $(-6) \cdot 4 + 1 = -24 + 1 = -23$, and $0 \leq 1 < 4$. By uniqueness this is the only such pair. The equation $-23 = (-5) \cdot 4 - 3$ is also true, but -3 is not a remainder, since it fails $0 \leq r < 4$; the warning below says why that matters.

Warning. "The remainder" means the r with $0 \leq r < n$. The equation $-17 = (-3) \cdot 5 - 2$ is true, but -2 is **not** the remainder of -17 on division by 5; the remainder is 3. If negative "remainders" were allowed, uniqueness would fail at once: $-17 = (-3)\cdot 5 + (-2) = (-4)\cdot 5 + 3$.

2.5.3 Even and odd

Recall that an integer a is **even** if $a = 2k$ for some $k \in \mathbb{Z}$, and **odd** if $a = 2k + 1$ for some $k \in \mathbb{Z}$ (Sets and Functions, Definition 1.1.1, as in Sets and Functions, Theorem 1.1.2), now read on the constructed $\mathbb{Z}$. With $n = 2$, the theorem answers the opening question.

Corollary 2.5.3. Even and Odd Integers

(a) Every $a \in \mathbb{Z}$ is even or odd, and **not both**. More precisely, either $a = 2q$ for exactly one $q \in \mathbb{Z}$ and a is not of the form $2k + 1$, or $a = 2q + 1$ for exactly one $q \in \mathbb{Z}$ and a is not of the form $2k$.

(b) If $a \in \mathbb{Z}$ and a^2 is even, then a is even.

Idea. Divide by 2. The remainder lies in $\{0, 1\}$, by discreteness, and it is unique, so each integer has exactly one of the two forms. For (b), square the odd form and read off the remainder.

Proof.

(a) Since $2 = 1 + 1 > 0$, the theorem with $n = 2$ gives unique $q, r \in \mathbb{Z}$ with $a = 2q + r$ and $0 \leq r < 2$. Then $r \in \{0, 1\}$: if $r \neq 0$, then $0 < r$, so $1 \leq r$ by discreteness (Theorem 2.4.2 (c)); and if moreover $1 < r$, then $2 = 1 + 1 \leq r$ by discreteness again, against $r < 2$. If $a = 2k$ and $a = 2k' + 1$ for some $k, k' \in \mathbb{Z}$, then $(k, 0)$ and $(k', 1)$ would be two pairs as in the theorem, and $0 \neq 1$ contradicts its uniqueness. The same uniqueness shows that q is the only integer with $a = 2q + r$.

(b) Suppose a is not even. By (a), $a = 2q + 1$ with $q \in \mathbb{Z}$. Then

$$a^2 = (2q + 1)^2 = 4q^2 + 4q + 1 = 2(2q^2 + 2q) + 1,$$

by Proposition 2.3.2 (d) and (F5), (F6), (F9), so a^2 is odd, and by (a) it is not even. This proves (b) by contraposition.

■

So $-7 = (-4) \cdot 2 + 1$ is odd and not even, as school says, and now by proof. Part (a) is Sets and Functions, Theorem 1.1.2 on the constructed $\mathbb{Z}$, and part (b) is Logic and Proof, Theorem 2.1.3, whose proof used (a) and the computation above as school facts. For $a \in \mathbb{N}$, (a)

is also Proposition 1.3.4 (d) of Chapter 1.

Remark (Looking ahead). *Algebraic Preliminaries*, Chapter 1 builds on this section: greatest common divisors and Bézout's identity, primes and unique factorization, and arithmetic modulo n, all on the constructed $\mathbb{Z}$. Chapter 3 of this volume builds the rationals as classes of fractions, and its integer part of a/n is the quotient found here.

2.5.4 Exercises

2.5.4.1 A. Check your understanding

Exercise. A1

(a) Find the quotient and the remainder on division by $n = 6$ of $a = 47$, of $a = -47$ and of $a = 0$.

(b) Where is $n > 0$ used in the existence half of Theorem 2.5.2 (a)?

(c) True or false, with a reason: $0 \mid 0$.

(d) True or false, with a reason: for all $a, b \in \mathbb{Z}$, if $a \mid b$ and $b \mid a$, then $a = b$.

Solution.

(a) $47 = 7 \cdot 6 + 5$, so $(q, r) = (7, 5)$. $-47 = (-8) \cdot 6 + 1$, since $-48 + 1 = -47$, so $(q, r) = (-8, 1)$. $0 = 0 \cdot 6 + 0$, so $(q, r) = (0, 0)$. In each case $0 \leq r < 6$, so by uniqueness these are the quotient and remainder. Note $(-8, 1)$, not $(-7, -5)$.

(b) Twice. For $a < 0$, the element $a - an = a(1 - n)$ of R is non-negative because $1 - n \leq 0$, which comes from $n \geq 1$, that is, from $n > 0$ and discreteness. And the contradiction "$r - n < r$" needs $-n < 0$, that is, $n > 0$.

(c) True: $0 = 1 \cdot 0$, so $k = 1$ witnesses $0 \mid 0$ (Proposition 2.5.1 (a)).

(d) False: $2 \mid -2$ and $-2 \mid 2$ (Proposition 2.5.1 (d)), but $2 \neq -2$. The statement is true for $a, b \in \mathbb{N}$ (Proposition 2.5.1 (f)).

2.5.4.2 B. Practice

Exercise. B1: Negative divisors

Let $a, n \in \mathbb{Z}$ with $n < 0$. Prove that there are unique $q, r \in \mathbb{Z}$ with $a = qn + r$ and $0 \leq r < -n$.

Solution. Since $n < 0$, $-n > 0$ by Proposition 2.4.3 (a), so $-n \in \mathbb{Z}_{>0}$.

Existence. By Theorem 2.5.2 (a) applied to a and $-n$, there are $q_0, r \in \mathbb{Z}$ with $a = q_0(-n) + r$ and $0 \leq r < -n$. Let $q := -q_0$. Then $q_0(-n) = (-q_0)n = qn$ by Proposition 2.3.2 (c), so $a = qn + r$.

Uniqueness. Let $a = qn + r = q'n + r'$ with $0 \leq r, r' < -n$. By Proposition 2.3.2 (c), $qn = (-q)(-n)$ and $q'n = (-q')(-n)$, so $a = (-q)(-n) + r = (-q')(-n) + r'$ are two divisions of a by $-n$. The uniqueness in Theorem 2.5.2 (a) gives $r = r'$ and $-q = -q'$, hence

$q = -(-q) = -(-q') = q'$ by Proposition 2.3.2 (a).

Exercise. B2: Three in a row

Let $a \in \mathbb{Z}$. Prove that **exactly one** of a, $a+1$, $a+2$ is divisible by 3.

Solution. At least one. By Theorem 2.5.2 (a), $a = 3q + r$ with $0 \leq r < 3$, and $r \in \{0, 1, 2\}$ by discreteness, as in the proof of Corollary 2.5.3. If $r = 0$, then $3 \mid a$. If $r = 1$, then $a + 2 = 3q + 3 = 3(q + 1)$. If $r = 2$, then $a + 1 = 3q + 3 = 3(q + 1)$.

At most one. Suppose $3 \mid a + i$ and $3 \mid a + j$ with $0 \leq i < j \leq 2$. By Proposition 2.5.1 (c) (with $x = 1$, $y = -1$), $3 \mid (a + j) - (a + i) = j - i$, and $j - i \in \{1, 2\}$ is positive. By Proposition 2.5.1 (e), $3 \leq j - i \leq 2$, which is false. So no two of the three are divisible by 3.

(Equivalently: the remainders of a, $a + 1$, $a + 2$ on division by 3 are r, and then the other two of $0, 1, 2$, and uniqueness of remainders allows only one of them to be 0.)

2.5.4.3 C. Going deeper

Exercise. C1: Exit check: remainders of squares

(a) Prove that for every $a \in \mathbb{Z}$, the remainder of a^2 on division by 4 is 0 or 1.

(b) Deduce that no integer whose remainder on division by 4 is 3 is a sum $x^2 + y^2$ of two squares of integers.

Name the result behind each step.

Solution.

(a) By Corollary 2.5.3 (a), $a = 2q$ or $a = 2q + 1$ for some $q \in \mathbb{Z}$. In the first case $a^2 = (2q)(2q) = 4q^2 = q^2 \cdot 4 + 0$, by the ring laws (F5), (F6) and $4 = 2 \cdot 2$, computed in $\mathbb{N}$. In the second,

$$a^2 = (2q + 1)^2 = 4q^2 + 4q + 1 = (q^2 + q) \cdot 4 + 1,$$

by Proposition 2.3.2 (d) and the ring laws (F5), (F6), (F9). Since 4 is a non-zero natural number, $4 > 0$ (Theorem 2.4.2 (b)), and both remainders, 0 and 1, satisfy $0 \leq r < 4$, so by the uniqueness in Theorem 2.5.2 (a) they are the remainders of a^2 on division by 4.

(b) Let $x, y \in \mathbb{Z}$. By (a), $x^2 = 4k + s$ and $y^2 = 4l + t$ with $k, l \in \mathbb{Z}$ and $s, t \in \{0, 1\}$. Then

$$x^2 + y^2 = 4(k + l) + (s + t), \qquad s + t \in \{0, 1, 2\},$$

by (F1), (F2) and (F9). So $0 \leq s + t \leq 2 < 4$, and by uniqueness $s + t$ is the remainder of $x^2 + y^2$ on division by 4. It is not 3, so no sum of two squares has remainder 3.

Marking scheme. (10 points)

- 2: splits into the two cases $a = 2q$, $a = 2q + 1$, citing Corollary 2.5.3 (a).

- 3: expands both squares and writes each in the form $q' \cdot 4 + r$, with the identities justified by the ring laws.
- 2: checks $0 \le r < 4$ and invokes uniqueness to conclude that r is **the** remainder.
- 3: for (b), adds the two forms, checks $0 \le s + t \le 2 < 4$, and concludes by uniqueness (or by Proposition 2.5.1 (e) applied to the difference) that the remainder is $s + t \neq 3$.

Fatal errors (0 for the step): calling a value outside $0 \le r < 4$ a remainder; concluding in (b) that the remainder is $s + t$ without checking $0 \le s + t < 4$ or an equivalent divisibility argument.

A proof that divides a by 4 instead, with four cases $a = 4m + j$, $j \in \{0, 1, 2, 3\}$, earns the first 2 points for the case split (citing Theorem 2.5.2 (a) and discreteness) and the rest as above.

2.5.4.4 R. Review

Exercise. R1

Let $f\colon \mathbb{N} \to \mathbb{N}$ satisfy $f(0) = 0$ and $f(n+1) = f(n) + 2n + 1$ for every $n \in \mathbb{N}$. Prove that $f(n) = n \cdot n$ for every n.

Solution. By Theorem 1.2.1, with $g(n, x) = x + 2n + 1$, exactly one such f exists, so the question is about a definite function. We prove $f(n) = n \cdot n$ by induction on n (Theorem 1.1.2).

Base. $f(0) = 0 = 0 \cdot 0$, by Definition 1.2.1 (b).

Step. Suppose $f(n) = n \cdot n$. Then $f(n+1) = (n \cdot n + 2n) + 1$. On the other side, with $\mathrm{S}(n) = n + 1$ (Definition 1.2.1 (c)),

$$\begin{aligned}(n+1)(n+1) &= (n+1) \cdot n + (n+1) \\ &= (n \cdot n + n) + (n+1) = (n \cdot n + (n+n)) + 1,\end{aligned}$$

by Definition 1.2.1 (b) applied to $(n+1) \cdot \mathrm{S}(n)$, then Theorem 1.2.3 (e) for $(n+1) \cdot n = n \cdot n + n$, and Theorem 1.2.3 (c) to regroup. Finally $2n = (1+1)n = 1 \cdot n + 1 \cdot n = n + n$, by $1 + 1 = 2$ (Definition 1.2.1 (c)) and Theorem 1.2.3 (d), (e). So $f(n+1) = (n+1)(n+1)$, and the claim holds for every n.

This is Chapter 1's move of fixing everything else and inducting on the recursion variable, as in the proof of Theorem 1.2.3.

Exercise. R2

Decide which of the following maps $\mathbb{Z} \to \mathbb{Z}$ are injective and which are surjective:

(a) $x \mapsto -x$;
(b) $x \mapsto 2x$;
(c) $x \mapsto x^2$.

Solution.

(a) Bijective. Since $-(-x) = x$ (Proposition 2.3.2 (a)), equal values $-x = -y$ give $x = -(-x) = -(-y) = y$, and every y is the value at $-y$.

(b) Injective, not surjective. If $2x = 2y$, then $2(x - y) = 2x - 2y = 0$ by (F9) and Proposition 2.3.2 (c); and $2 \neq 0$, since $2 = 1+1 > 0$. So $x - y = 0$ by Proposition 2.4.3 (f), and $x = y$. Not surjective: $1 = 2x$ would make 1 even, but $1 = 2 \cdot 0 + 1$ is odd, and no integer is both (Corollary 2.5.3 (a)).

(c) Neither. Not injective: $(-1)^2 = 1 = 1^2$ (Proposition 2.3.2 (c)), and $-1 \neq 1$, since $-1 < 0 < 1$ (shown below) gives $-1 < 1$ by (O2), and (O1) forbids equality. Not surjective: $x^2 \geq 0$ for every x ($x^2 = 0$ for $x = 0$ by Proposition 2.3.2 (b), and $x^2 > 0$ otherwise by Proposition 2.4.3 (e)), while $-1 < 0$ ($1 > 0$ by Proposition 2.4.3 (e), so $-1 < -0 = 0$ by Proposition 2.4.3 (a)). So -1 is not a value.

The methods are those of Sets and Functions, Example (Method: proving injectivity) and Sets and Functions, Example (Method: proving surjectivity): for injectivity, start from equal values and reach equal inputs; for surjectivity, solve for the input, or show that a target is never reached.

Exercise. R3

Prove that for every $a \in \mathbb{Z}$: if $3 \mid a^2$, then $3 \mid a$.

Solution. By contraposition, as in Logic and Proof, Theorem 2.1.3. Suppose $3 \nmid a$. Since 3 is a non-zero natural number, $3 > 0$ (Theorem 2.4.2 (b)), so division by 3 (Theorem 2.5.2 (a)) gives $a = 3q + r$ with $0 \leq r < 3$, so $r \in \{0, 1, 2\}$ by discreteness (Theorem 2.4.2 (c)), as in the proof of Corollary 2.5.3; and $r \neq 0$, since otherwise $a = 3q$ and $3 \mid a$. By Proposition 2.3.2 (d) and the ring laws,

$$\begin{aligned} r = 1: &\quad a^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1, \\ r = 2: &\quad a^2 = 9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1. \end{aligned}$$

In both cases $0 \leq 1 < 3$, so the remainder of a^2 on division by 3 is $1 \neq 0$, and $3 \nmid a^2$ by the divisibility test after Theorem 2.5.2. This proves the contrapositive.

Chapter 3

The Rationals

Chapter 2 built $\mathbb{Z}$, an ordered commutative ring in which $2x = 1$ has no solution. This chapter supplies the missing quotients by the move that built $\mathbb{Z}$: a pair (p, q) with $q > 0$ records "p divided by q" without performing it, two pairs are identified when $ps = rq$, and a rational number is a class of such pairs. Addition and multiplication are defined on representatives and checked to be well defined, and each field law becomes an identity in $\mathbb{Z}$, already proved. *Sets and Functions*, Section 3.1 took $\mathbb{Q} = \{m/n : m \in \mathbb{Z},\ n \in \mathbb{Z}_{>0}\}$ as a definition; here it is a theorem about the constructed set. The order comes from one choice: a class is positive when a representative has a positive numerator. The Archimedean property then follows from the construction, with no supremum anywhere.

The last section asks what $\mathbb{Q}$ still lacks. No rational squares to 2, and the set of rationals with square below 2 is bounded above but has no supremum in $\mathbb{Q}$. A sequence of rationals, $2, 3/2, 17/12, \ldots$, closes in on the gap from above. Chapters 4 and 5 turn sequences of this kind into the real numbers.

What you need. Chapter 2, all of it, and from Chapter 1 only induction, recursion and well-ordering on $\mathbb{N}$; *Sets and Functions*, Chapters 2, 3 and 5, chiefly Section 5.2; a few results of *Logic and Proof*. From Chapter 2: classes of pairs and the lemma on rules defined on classes (Definition 2.1.1, Theorem 2.2.1), the operations of $\mathbb{Z}$ and the copy of $\mathbb{N}$ in it (Definition 2.2.1, Proposition 2.2.3); commutative rings and fields with their algebra (Definition 2.3.1, Theorem 2.3.1, Proposition 2.3.2), from Section 2.3; ordered rings, orders from a positive set, the order of $\mathbb{Z}$, the rules for inequalities and least elements of sets of integers bounded below (Definition 2.4.1, Proposition 2.4.1, Definition 2.4.2, Theorem 2.4.2, Proposition 2.4.3, Proposition 2.4.4), from Section 2.4; division with remainder and parity (Theorem 2.5.2, Corollary 2.5.3), from Section 2.5. From Chapter 1: Corollary 1.4.3, by which induction (Logic and Proof, Axiom 4.2.1) and the recursion theorem (Sets and Functions, Theorem 3.3.1) apply to $\mathbb{N}$, and well-ordering (Theorem 1.3.3), which Section 3.2 contrasts with $\mathbb{Q}_{>0}$. From *Sets and Functions*: equivalence relations, classes and quotient sets (Sets and Functions, Definition 5.2.1, Sets and Functions, Definition 5.2.2, Sets and Functions, Definition 5.2.4, Sets and Functions, Theorem 5.2.1) and the method box for well-definedness (Sets and Functions, Example (Method: checking that a definition on classes is well defined)), from Section 5.2; upper bounds and suprema (Sets and Functions, Definition 5.3.2, Sets and Functions, Definition 5.3.5), from Section 5.3; functions and their equality (Sets and Functions, Definition 2.1.1, Sets and Functions, Proposition 2.1.1), from Chapter 2; and the smallest-counterexample method with its instance on lowest terms (Sets and Functions, Example (Method: the smallest counterexample), Sets and Functions, Lemma 3.2.4), from Chapter 3. Review exercise R2 uses the countability results of *Sets and Functions*, Chapter 4 (Sets and Functions, Theorem 4.2.6, Sets and Functions, Corollary 4.2.7, Sets and Functions, Proposition 4.2.3, Sets and Functions, Proposition 4.2.4, Sets and Functions, Theorem 4.2.8).

From *Logic and Proof*: the negation laws (Logic and Proof, Proposition 2.5.5, Logic and Proof, Proposition 6.2.1, Logic and Proof, Proposition 6.2.3, Logic and Proof, Proposition 6.2.2), for Exercise A1 of Section 3.2 and review exercise R3, and the catalogue of what a proof may not do (Logic and Proof, Example (What is not allowed)), for the find-the-error exercises. Section 3.3 compares its proof with the parity proof that $\sqrt{2}$ is irrational (Logic and Proof, Theorem 2.2.1) and does not use it.

Time. About 15 hours on the core track (all of A, one B and one C per section, the exit checks); about 25 hours for everything.

3.0.1 Roadmap

- **The Rationals as Classes of Fractions.** Pairs (p, q) with $q > 0$ identified by cross-multiplication, $\mathbb{Q}$ as their quotient set, a sum and a product checked on representatives (the second argument free by symmetry), rules that look like functions and are not, the proof that $\mathbb{Q}$ is a field, and the copy of $\mathbb{Z}$ inside it that turns the class $[(p, q)]$ into the quotient p/q.
- **The Rationals as an Ordered Field; the Archimedean Property.** The positive rationals as the classes with a positive numerator, and why negative denominators would break this; $\mathbb{Q}$ an ordered field with the comparison formula $p/q \leq r/s \iff ps \leq rq$; reciprocals reversing order and the midpoint of two rationals; every rational below a natural number, proved from the construction, with its form $1/n < \varepsilon$; and the integer part $\lfloor x \rfloor$.
- **The Gaps in the Rationals.** No rational squares to 2, proved by descent on a least denominator rather than by parity; the set $\{x \in \mathbb{Q} : x^2 < 2\}$, bounded above with no supremum in $\mathbb{Q}$, each candidate refuted by an explicit rational; and the rational sequence $2, 3/2, 17/12, 577/408, \ldots$ that closes in on the gap from above. The review exercises return to Chapter 2 and to *Sets and Functions* and *Logic and Proof*.

3.0.2 Named moves

The chapter introduces no new method box; the first four moves below were taught earlier and are run here on new data, and the last is new to this chapter.

- **Check on representatives, one argument at a time** (Sets and Functions, Example (Method: checking that a definition on classes is well defined), with Theorem 2.2.1). Change one representative with the other fixed; each change is one identity at the level below. A formula symmetric in its two inputs gets the second change free. (Section 3.1)
- **Reduce to the level below** (Section 2.3). Write a law about classes on representatives; the two sides become the same pair, or two related pairs, by an identity in $\mathbb{Z}$. (Section 3.1)
- **Order from a positive set** (Proposition 2.4.1). Say which elements are positive, check (P1) and (P2), and read off every rule for inequalities. (Section 3.2)
- **Descent on a least denominator** (Sets and Functions, Example (Method: the smallest counterexample)). Take a counterexample with the least denominator and build one with a smaller denominator. (Section 3.3)
- **Refute a supremum candidate by an explicit rational.** Given a rational s claimed to be the supremum, produce from s an element of the set above it or an upper bound below it. (Section 3.3, the scratch work for Theorem 3.3.2)

3.0.3 Threads

- **Classes of pairs.** Section 2.1 made integers from pairs of natural numbers, Section 3.1 makes rationals from pairs of integers, and Chapter 5 makes real numbers from classes of sequences of rationals, by the same route.
- **The integer part.** Section 3.2 defines $\lfloor x \rfloor$; for $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$, $\lfloor a/n \rfloor$ is the quotient of Section 2.5 (Exercise B3 of Section 3.2), and Section 4.2 builds the integer part in every Archimedean ordered field.
- $1/n < \varepsilon$**.** The corollary of Section 3.2 is the form of the Archimedean property that the limit arguments of Chapters 4 and 5 use, with $\varepsilon \in \mathbb{Q}_{>0}$.
- **The sequence** $2, 3/2, 17/12, \ldots$ Section 3.3 computes it; Section 4.6 proves it Cauchy in $\mathbb{Q}$ with no limit in $\mathbb{Q}$; Chapter 5 opens with it, as the first sequence of rationals that should converge and does not in $\mathbb{Q}$.

3.0.4 Summary

A rational number is a class of pairs $(p, q) \in \mathbb{Z} \times \mathbb{Z}_{>0}$ under $ps = rq$ (Definition 3.1.1), and the sum and product defined on representatives are well defined (Proposition 3.1.2). With them $\mathbb{Q}$ is a field (Theorem 3.1.3) containing a copy of $\mathbb{Z}$, and every rational is m/n with $m \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$ (Proposition 3.1.4). Declaring a class positive when its numerator is positive makes $\mathbb{Q}$ an ordered field whose order extends that of $\mathbb{Z}$ (Definition 3.2.1, Theorem 3.2.1). Every rational lies below a natural number (Theorem 3.2.3), so for each $\varepsilon \in \mathbb{Q}_{>0}$ some $1/n$ is smaller (Corollary 3.2.4), and every rational has an integer part (Proposition 3.2.5). No rational squares to 2 (Theorem 3.3.1), and $\{x \in \mathbb{Q} : x^2 < 2\}$ is bounded above with no supremum in $\mathbb{Q}$ (Theorem 3.3.2).

What is open: $\mathbb{Q}$ is an Archimedean ordered field with gaps. Can the gaps be filled, and by what? Chapter 4 develops limits in an ordered field to say what "closing in on a gap" means, and Chapter 5 fills the gaps.

3.0.5 Terms

- **denominator, numerator**: q and p in a representative (p, q) of a rational number (Definition 3.1.1).
- **integer part**: $\lfloor x \rfloor$, the one integer m with $m \leq x < m + 1$ (Proposition 3.2.5).
- **order on** $\mathbb{Q}$: $x < y$ when $y - x$ is positive (Definition 3.2.1).
- **positive rational**: a class $[(p, q)]$ with $p > 0$; the set $P_{\mathbb{Q}} = \mathbb{Q}_{>0}$ (Definition 3.2.1).
- **rational number**: a class $[(p, q)]$ of pairs in $\mathbb{Z} \times \mathbb{Z}_{>0}$ under $ps = rq$ (Definition 3.1.1).

3.1 The Rationals as Classes of Fractions

Chapter 2 built $\mathbb{Z}$ and made it an ordered commutative ring, but not a field: $2x = 1$ has no solution in $\mathbb{Z}$ (Quick check in Section 2.4). This section repairs that defect with the move of Section 2.1, used a second time. There a pair (a, b) recorded "$a - b$" without performing it; here a pair (p, q) records "p divided by q" without performing it. We define $\mathbb{Q}$ as a set of classes of such pairs, give it addition and multiplication through representatives, check that they are well defined, and prove that $\mathbb{Q}$ is a field containing a copy of $\mathbb{Z}$.

3.1.1 Fractions as pairs

Before reading on. The pairs $(1,2)$ and $(2,4)$ are different elements of $\mathbb{Z}\times\mathbb{Z}_{>0}$, yet both should name one half. Which condition on two pairs (p,q) and (r,s) says "same quotient", using only operations that $\mathbb{Z}$ already has?

At school, $\frac{1}{2}$, $\frac{2}{4}$ and $\frac{3}{6}$ are "the same number", and the test for sameness is cross-multiplication: $\frac{p}{q}=\frac{r}{s}$ when $ps=rq$. The test is the useful part. It mentions only multiplication of integers, which we have, and not division, which we do not. So we keep the test and let a rational number **be** what the equivalent fractions have in common: the class of all of them.

A rational number is a quotient you have not computed yet: the set of all fractions that would compute to it.

Signs need a decision. The fractions $\frac{1}{-2}$ and $\frac{-1}{2}$ name the same number, and allowing both would put a sign in two places. We require the **denominator to be positive**, so that a sign can sit only in the numerator. This is the convention *Sets and Functions*, Section 5.2 already used for its fractions, and Section 3.2 below will profit from it: there, the order of $\mathbb{Q}$ is read off a single inequality between integers. Recall that $\mathbb{Z}_{>0}=\{n\in\mathbb{Z}:n>0\}$, by Theorem 2.4.2 (b).

On $\mathbb{Z}\times\mathbb{Z}_{>0}$, let $(p,q)\sim(r,s)$ mean $ps=rq$. We reuse the symbol $\sim$ of Section 2.1; which relation is meant is always clear from the pairs, natural numbers there and an integer with a positive integer here.

Proposition 3.1.1. Cross-Multiplication Is an Equivalence Relation

The relation $\sim$ on $\mathbb{Z}\times\mathbb{Z}_{>0}$ given by

$$(p,q)\sim(r,s)\iff ps=rq$$

is an equivalence relation (Sets and Functions, Definition 5.2.1).

Idea. Reflexivity and symmetry are just reflexivity and symmetry of $=$: with this form of the test, $(p,q)\sim(p,q)$ reads $pq=pq$. For transitivity we are given $ps=rq$ and $ru=ts$, and want $pu=tq$. Multiply the first equation by u and use the second to replace ru: this gives $(pu)s=(tq)s$. We cannot divide by s, but $\mathbb{Z}$ has no zero divisors, so a factor $s\neq 0$ can be canceled.

Proof. Let $(p,q),(r,s),(t,u)\in\mathbb{Z}\times\mathbb{Z}_{>0}$. All computations are in the commutative ring $\mathbb{Z}$ (Theorem 2.3.1).

(Eq1) $pq=pq$, so $(p,q)\sim(p,q)$.

(Eq2) If $ps=rq$, then $rq=ps$, so $(r,s)\sim(p,q)$.

(Eq3) Let $ps=rq$ and $ru=ts$. Then, using (F5), (F6) and the two hypotheses,

$$(pu)s=(ps)u=(rq)u=(ru)q=(ts)q=(tq)s.$$

Hence $(pu-tq)s=(pu)s-(tq)s=0$, by (F9) and Proposition 2.3.2 (c). Since $s\in\mathbb{Z}_{>0}$, $s>0$, so $s\neq 0$ by (O1). By Proposition 2.4.3 (f), $pu-tq=0$, that is, $pu=tq$. So $(p,q)\sim(t,u)$. This proves the proposition.

■

The hypothesis that denominators are non-zero is used once, for the **middle** pair (r, s), and it is needed there. On $\mathbb{Z} \times \mathbb{N}$, with the same formula, $(1, 0) \sim (0, 0)$ (since $1 \cdot 0 = 0 \cdot 0$) and $(0, 0) \sim (1, 1)$ (since $0 \cdot 1 = 1 \cdot 0$), but $(1, 0) \not\sim (1, 1)$ (since $1 \cdot 1 \neq 1 \cdot 0$). The pair $(0, 0)$ is related to everything, and transitivity collapses through it. *Sets and Functions*, Section 5.2 met the same failure in an exercise; it is why denominators must be non-zero.

Definition 3.1.1. The Rational Numbers

Let $\sim$ be the equivalence relation of Proposition 3.1.1 on $\mathbb{Z} \times \mathbb{Z}_{>0}$. The set of **rational numbers** is the quotient set (Sets and Functions, Definition 5.2.4)

$$\mathbb{Q} \coloneqq (\mathbb{Z} \times \mathbb{Z}_{>0})/\!\sim.$$

A **rational number** is an element of $\mathbb{Q}$: a class $[(p, q)]$ with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$. We call p a **numerator** and q a **denominator** of the representative (p, q).

In words: a rational number is a **set of pairs**, all the pairs (r, s) with positive s that cross-multiply with (p, q). It is not a pair, and it is not a number with a sign and a bar. Numerators and denominators belong to representatives, not to the class: $(1, 2)$ and $(2, 4)$ represent the same rational number with different denominators. Nothing in the definition needs checking beyond Proposition 3.1.1, which makes the quotient set meaningful. By Sets and Functions, Theorem 5.2.1, every pair lies in exactly one class, and $[(p, q)] = [(r, s)]$ holds exactly when $ps = rq$.

Examples.

- $[(1, 2)] = \{(k, 2k) : k \in \mathbb{Z}_{>0}\}$. Indeed, $(r, s) \sim (1, 2)$ means $2r = s$. Then $2r > 0$, and $r \leq 0$ would give $2r = r + r \leq 0$ (Proposition 2.4.3 (b)), so $r > 0$. Conversely each $(k, 2k)$ with $k > 0$ has $2k > 0$ and $k \cdot 2 = 2k \cdot 1$. Its elements with denominator at most 6 are $(1, 2)$, $(2, 4)$, $(3, 6)$.
- $[(-2, 1)] = \{(-2s, s) : s \in \mathbb{Z}_{>0}\}$, since $(r, s) \sim (-2, 1)$ means $r = -2s$. This is the class that will be the integer -2.
- The degenerate case: $[(0, 1)] = \{(0, s) : s \in \mathbb{Z}_{>0}\}$, since $(r, s) \sim (0, 1)$ means $r \cdot 1 = 0 \cdot s = 0$. Every fraction with numerator 0 lies in this one class, whatever its denominator; it will be the zero of $\mathbb{Q}$.
- $[(2, 3)]$ contains $(4, 6)$ and $(6, 9)$, since $2 \cdot 6 = 4 \cdot 3$ and $2 \cdot 9 = 6 \cdot 3$. It does not contain $(-4, -6)$, which is not even an element of $\mathbb{Z} \times \mathbb{Z}_{>0}$. Describing the whole class needs lowest terms, proved in *Algebraic Preliminaries*, Chapter 1.

Why positive denominators? Non-zero denominators of either sign give a set in bijection with $\mathbb{Q}$, with matching classes (Exercise C3). The difference appears with the order: with positive denominators, "$[(p, q)]$ is positive" can mean "$p > 0$", one inequality in $\mathbb{Z}$; Section 3.2 shows what breaks otherwise.

Warning. Do not read $[(p, q)]$ as the number p/q. There is no division in $\mathbb{Z}$, and $\mathbb{Q}$ has no multiplication yet, so "p/q" means nothing at this point. The class is a set of pairs, and any statement made with one representative, such as "the denominator of $[(1, 2)]$ is 2", must be checked against all of them (here it fails: $(2, 4)$ is also in the class). The fraction notation becomes honest once $\mathbb{Z}$ is placed inside $\mathbb{Q}$, in Proposition 3.1.4 and the identification that

follows it.

Quick check. Is $(3,4) \sim (-6,-8)$ in $\mathbb{Z} \times \mathbb{Z}_{>0}$?

Solution. The question is malformed: $-8 \notin \mathbb{Z}_{>0}$, so $(-6,-8)$ is not an element of $\mathbb{Z} \times \mathbb{Z}_{>0}$, and $\sim$ says nothing about it.

3.1.2 Adding and multiplying classes

The school rules $\frac{p}{q} + \frac{r}{s} = \frac{ps+rq}{qs}$ and $\frac{p}{q} \cdot \frac{r}{s} = \frac{pr}{qs}$ are a **heuristic** for the definition, nothing more; the definition below does not depend on them, and the proofs never use them.

Definition 3.1.2. Operations on the Rationals

For $[(p,q)], [(r,s)] \in \mathbb{Q}$, define

$$\begin{aligned} [(p,q)] + [(r,s)] &\coloneqq [(ps+rq,\ qs)], \\ [(p,q)] \cdot [(r,s)] &\coloneqq [(pr,\ qs)], \\ -[(p,q)] &\coloneqq [(-p,\ q)]. \end{aligned}$$

The right-hand sides are classes of elements of $\mathbb{Z} \times \mathbb{Z}_{>0}$: since $q > 0$ and $s > 0$, $qs > 0$ by (O4), so $qs \in \mathbb{Z}_{>0}$ by Theorem 2.4.2 (b). Each formula is computed from **chosen representatives**, so the definition is legitimate only once we know that other representatives give the same class. That is the next proposition.

Proposition 3.1.2. The Operations on the Rationals Are Well Defined

Let $X \coloneqq \mathbb{Z} \times \mathbb{Z}_{>0}$, with $\sim$ as in Proposition 3.1.1. The maps

$$\begin{aligned} g_+((p,q),(r,s)) &\coloneqq (ps+rq,\ qs), \\ g_\cdot((p,q),(r,s)) &\coloneqq (pr,\ qs) \end{aligned}$$

from $X \times X$ to X satisfy the hypotheses of Theorem 2.2.1 (b), and $f(p,q) \coloneqq (-p,q)$ satisfies the hypothesis of Theorem 2.2.1 (a). Hence $+$, $\cdot$ and $-$ of Definition 3.1.2 are functions $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$, $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ and $\mathbb{Q} \to \mathbb{Q}$.

Idea. This is Sets and Functions, Example (Method: checking that a definition on classes is well defined), run one argument at a time as in Section 2.2: by Theorem 2.2.1 (b), it suffices to change one pair with the other fixed. Each change is one identity in $\mathbb{Z}$ that uses the hypothesis $pq' = p'q$ once. Both formulas are symmetric in the two pairs, so the second argument costs nothing.

Proof. All computations are in the commutative ring $\mathbb{Z}$ (Theorem 2.3.1), and we reorder products by (F5), (F6) without comment.

Multiplication, first argument. Let $(p,q)\sim(p',q')$, so $pq'=p'q$, and let $(r,s)\in X$. We need $(pr,qs)\sim(p'r,q's)$, that is, $(pr)(q's)=(p'r)(qs)$. Indeed

$$(pr)(q's)=(pq')(rs)=(p'q)(rs)=(p'r)(qs).$$

Addition, first argument. Let $pq'=p'q$. We need $(ps+rq,qs)\sim(p's+rq',q's)$, that is, $(ps+rq)(q's)=(p's+rq')(qs)$. By (F9),

$$(ps+rq)(q's)=(pq')s^2+rqq's,$$
$$(p's+rq')(qs)=(p'q)s^2+rqq's,$$

and the right-hand sides are equal because $pq'=p'q$.

Second argument, by symmetry. For pairs $x,y\in X$, $g_+(y,x)=g_+(x,y)$ and $g.(y,x)=g.(x,y)$ as pairs, by (F1) and (F6) in $\mathbb{Z}$: for instance $(rq+ps,\ sq)=(ps+rq,\ qs)$. So if $y\sim y'$, the first-argument case, applied with the pair y in first place, gives $g(x,y)=g(y,x)\sim g(y',x)=g(x,y')$, for $g=g_+$ and $g=g.$.

Negation. If $pq'=p'q$, then $(-p)q'=-(pq')=-(p'q)=(-p')q$ by Proposition 2.3.2 (c), so $(-p,q)\sim(-p',q')$.

So g_+ and $g.$ satisfy the hypotheses of Theorem 2.2.1 (b): for a change of both arguments, $g(x,y)\sim g(x',y)\sim g(x',y')$, and transitivity joins the two steps, which is what the proof of that lemma does. By its conclusion, there is exactly one function $\mathbb{Q}\times\mathbb{Q}\to\mathbb{Q}$ with $[x]+[y]=[g_+(x,y)]$, and likewise for $\cdot$. By Theorem 2.2.1 (a), $-[(p,q)]\coloneqq[(-p,q)]$ is a function $\mathbb{Q}\to\mathbb{Q}$. This proves the proposition.

■

Quick check. Which rules define a function $\mathbb{Q}\to\mathbb{Q}$: (a) $[(p,q)]\mapsto[(p+q,\ q)]$; (b) $[(p,q)]\mapsto[(p,\ q+1)]$?

Solution.

(a) Yes: it is $x\mapsto x+1$, since $[(p,q)]+[(1,1)]=[(p\cdot1+1\cdot q,\ q\cdot1)]=[(p+q,\ q)]$ by Definition 3.1.2. (b) No: $[(1,2)]=[(2,4)]$, but $(1,3)\not\sim(2,5)$, as $1\cdot5\neq2\cdot3$.

Not every formula in representatives passes this test, and the failures are instructive.

Example. Formulas that ignore the class

Decide which of the following rules define a function on $\mathbb{Q}$.

(a) $[(p,q)]\mapsto p+q\in\mathbb{Z}$.
(b) $[(p,q)]\mapsto p\in\mathbb{Z}$.

(c) $[(p,q)] \mapsto [(p^2,q^2)] \in \mathbb{Q}$.

Solution.

(a) Not a function. $[(1,2)] = [(2,4)]$, since $1 \cdot 4 = 2 \cdot 2$, yet the rule gives $1+2=3$ for one representative and $2+4=6$ for the other. A function has one value at each argument (Sets and Functions, Definition 2.1.1), and this rule offers two at the same class.

(b) Not a function, for the same pair: the outputs are 1 and 2.

(c) A function. First, $q^2 > 0$ by Proposition 2.4.3 (e), as $q \neq 0$, so $(p^2,q^2) \in \mathbb{Z} \times \mathbb{Z}_{>0}$. If $pq' = p'q$, then

$$p^2(q')^2 = (pq')^2 = (p'q)^2 = (p')^2q^2,$$

by (F5), (F6), so $(p^2,q^2) \sim (p'^2,q'^2)$, and Theorem 2.2.1 (a) applies. The function is $x \mapsto x \cdot x$, since $[(p,q)] \cdot [(p,q)] = [(p^2,q^2)]$ by Definition 3.1.2.

The tempting argument for (a), "x is p/q, so p and q are determined by x", fails at the word "is": (p,q) is **a** representative, and $(2p,2q)$ is another. A rule through representatives is a function only once every representative is shown to give the same value, as in (c).

3.1.3 The rationals form a field

The ring axioms for $\mathbb{Q}$ are checked by the move of Section 2.3, **reduce to the level below**: each side of an axiom, written on representatives, is the class of an explicit pair in $\mathbb{Z} \times \mathbb{Z}_{>0}$, and the two pairs are compared by an identity in $\mathbb{Z}$. One small observation recurs: for every $(p,q) \in \mathbb{Z} \times \mathbb{Z}_{>0}$ and $k \in \mathbb{Z}_{>0}$,

$$(p,q) \sim (kp,kq), \quad \text{since } p(kq) = (kp)q;$$

here $kq > 0$ by (O4), so $(kp,kq) \in \mathbb{Z} \times \mathbb{Z}_{>0}$ by Theorem 2.4.2 (b), and the relation makes sense. Scaling both entries by the same positive integer does not change the class.

Theorem 3.1.3. The Rationals Form a Field

$\mathbb{Q}$, with the operations of Definition 3.1.2, is a field (Definition 2.3.1), with zero $[(0,1)]$ and one $[(1,1)]$. A rational number $[(p,q)]$ is zero if and only if $p = 0$. If $p > 0$, its inverse is $[(q,p)]$; if $p < 0$, its inverse is $[(-q,-p)]$.

Idea. Eight laws are identities between pairs of integers, checked in a table as for $\mathbb{Z}$. As there, (F4) gives pairs that are related rather than equal; here (F9) does too, and the scaling observation is what relates them. The genuinely new axiom is (F8): inverting $[(p,q)]$ should swap numerator and denominator, and the sign of p decides whether the swapped pair is allowed.

Proof. By Proposition 3.1.2, the operations are functions. Let $x,y,z \in \mathbb{Q}$; by Sets and Functions, Definition 5.2.4, $x = [(p,q)]$, $y = [(r,s)]$, $z = [(t,u)]$ with $p,r,t \in \mathbb{Z}$ and $q,s,u \in \mathbb{Z}_{>0}$. As in Section 2.3, "equal" means that the two pairs are the same pair of integers, by the laws (F1)–(F9) of $\mathbb{Z}$ (Theorem 2.3.1); "related" means that they are related by $\sim$, so

that the classes agree by Sets and Functions, Theorem 5.2.1 (b).

Axiom	left side, as a pair	right side, as a pair	why the classes agree
(F1)	$(ps + rq, qs)$	$(rq + ps, sq)$	equal
(F2)	$((ps+rq)u+tqs,\ qsu)$	$(psu + (ru + ts)q,\ qsu)$	equal: both first entries are $psu + qru + qst$
(F3) $x + [(0,1)] = x$	$(p \cdot 1 + 0 \cdot q,\ q \cdot 1)$	(p, q)	equal, by Proposition 2.3.2 (b)
(F4) $x + (-x) = [(0,1)]$	$(pq + (-p)q,\ q^2) = (0, q^2)$	$(0,1)$	related: $0 \cdot 1 = 0 \cdot q^2$
(F5)	$((pr)t,\ (qs)u)$	$(p(rt),\ q(su))$	equal
(F6)	(pr, qs)	(rp, sq)	equal
(F7) $x \cdot [(1,1)] = x$	$(p \cdot 1,\ q \cdot 1)$	(p, q)	equal
(F9)	$(p(ru + ts),\ q(su))$	$((pr)(qu) + (pt)(qs),\ (qs)(qu))$	related: the right pair is the left pair scaled by q

In (F4), $pq + (-p)q = pq - pq = 0$ by Proposition 2.3.2 (c), and $0 \cdot 1 = 0 = 0 \cdot q^2$ by Proposition 2.3.2 (b). In (F9), the right side is $xy + xz = [(pr, qs)] + [(pt, qu)]$; its first entry is $q(pru + pts) = q \cdot p(ru + ts)$ and its second is $q \cdot q(su)$, so the scaling observation above, with $k = q$, relates it to the left pair.

The zero. $[(p, q)] = [(0, 1)]$ if and only if $p \cdot 1 = 0 \cdot q$, that is, $p = 0$.

$1 \neq 0$. $[(1, 1)] = [(0, 1)]$ would need $1 = 0$ in $\mathbb{Z}$, which Theorem 2.3.1 excludes.

(F8). Let $x = [(p, q)] \neq 0$, so $p \neq 0$. If $p > 0$, then $p \in \mathbb{Z}_{>0}$, so $(q, p) \in \mathbb{Z} \times \mathbb{Z}_{>0}$, and

$$[(p, q)] \cdot [(q, p)] = [(pq,\ qp)] = [(1, 1)],$$

since $(pq) \cdot 1 = 1 \cdot (qp)$. If $p < 0$, then $-p > 0$ by Proposition 2.4.3 (a), so $(-q, -p) \in \mathbb{Z} \times \mathbb{Z}_{>0}$, and

$$\begin{aligned}[(p, q)] \cdot [(-q, -p)] &= [(p(-q),\ q(-p))] \\ &= [(-(pq),\ -(pq))] = [(1, 1)],\end{aligned}$$

by Proposition 2.3.2 (c) and $(-(pq)) \cdot 1 = 1 \cdot (-(pq))$. By (O1), one of the two cases occurs. So every non-zero element has an inverse, and $\mathbb{Q}$ is a field. This proves the theorem.

■

Two rows, (F4) and (F9), are "related" rather than "equal"; (F9) needs the scaling observation because the sum multiplies the denominators qs and qu. The case split in (F8) is the price of positive denominators: $[(q, p)]$ is not a class when $p < 0$.

Quick check. Compute $[(1, 2)] + [(1, 3)]$ and $[(-3, 4)]^{-1}$, and check the second answer by multiplying.

Solution. $[(1,2)]+[(1,3)] = [(1\cdot3+1\cdot2,\ 2\cdot3)] = [(5,6)]$. For $[(-3,4)]$ the numerator is negative, so the inverse is $[(-4,3)]$. Check: $[(-3,4)]\cdot[(-4,3)] = [(12,12)]$, and $(12,12)\sim(1,1)$ since $12\cdot1 = 1\cdot12$.

3.1.4 The integers inside the rationals

As $\mathbb{N}$ sat inside $\mathbb{Z}$ through ι (Proposition 2.2.3), $\mathbb{Z}$ sits inside $\mathbb{Q}$ through the classes with denominator 1.

Proposition 3.1.4. The Integers Embed in the Rationals

Let $j\colon \mathbb{Z}\to\mathbb{Q}$, $j(m) \coloneqq [(m,1)]$. Then j is injective, and for all $m,n\in\mathbb{Z}$,

$$j(m+n) = j(m)+j(n), \qquad j(mn) = j(m)j(n), \qquad j(-m) = -j(m),$$

with $j(0)=0$ and $j(1)=1$. Moreover, writing $a/b \coloneqq ab^{-1}$ in the field $\mathbb{Q}$:

(a) for $p\in\mathbb{Z}$ and $q\in\mathbb{Z}_{>0}$, $[(p,q)] = j(p)\cdot j(q)^{-1} = j(p)/j(q)$;

(b) every rational number is $j(p)/j(q)$ for some $p\in\mathbb{Z}$ and $q\in\mathbb{Z}_{>0}$, and for $p,r\in\mathbb{Z}$ and $q,s\in\mathbb{Z}_{>0}$, $j(p)/j(q) = j(r)/j(s)$ if and only if $ps = rq$;

(c) $\mathbb{Q} = \{j(m)/j(n) : m\in\mathbb{Z},\ n\in\mathbb{Z}_{>0}\}$.

Idea. The classes with denominator 1 add and multiply exactly as their numerators do, by the formulas of Definition 3.1.2 with $q=s=1$. For (a), $[(1,q)]$ is the inverse of $[(q,1)]$, and multiplying it by $[(p,1)]$ gives $[(p,q)]$; (b) and (c) are (a) read through Sets and Functions, Theorem 5.2.1.

Proof. Injective. If $j(m)=j(n)$, then $(m,1)\sim(n,1)$ by Sets and Functions, Theorem 5.2.1 (b), that is, $m\cdot1 = n\cdot1$, so $m=n$.

Operations. By Definition 3.1.2, $j(m)+j(n) = [(m\cdot1+n\cdot1,\ 1\cdot1)] = [(m+n,1)] = j(m+n)$, $j(m)j(n) = [(mn,1)] = j(mn)$, and $-j(m) = [(-m,1)] = j(-m)$. Also $j(0) = [(0,1)]$ and $j(1) = [(1,1)]$ are the zero and one of $\mathbb{Q}$ (Theorem 3.1.3).

(a) Since $q>0$, $j(q) = [(q,1)] \neq 0$, and by Theorem 3.1.3 its inverse is $[(1,q)]$ (the case of a positive numerator; inverses are unique, Section 2.3). Hence

$$\begin{aligned} j(p)\cdot j(q)^{-1} &= [(p,1)]\cdot[(1,q)] \\ &= [(p\cdot1,\ 1\cdot q)] = [(p,q)]. \end{aligned}$$

(b) Every element of $\mathbb{Q}$ is $[(p,q)]$ for some $(p,q)\in\mathbb{Z}\times\mathbb{Z}_{>0}$ (Sets and Functions, Definition 5.2.4), which is $j(p)/j(q)$ by (a). By (a) again, $j(p)/j(q) = j(r)/j(s)$ means $[(p,q)] = [(r,s)]$, which holds if and only if $ps = rq$ (Sets and Functions, Theorem 5.2.1 (b)).

(c) Each $j(m)/j(n)$ lies in $\mathbb{Q}$, and every element of $\mathbb{Q}$ has this form by (b). This proves the proposition.

■

Identification. From now on we write m for $j(m)$, as Section 2.2 wrote n for $\iota(n)$, and regard $\mathbb{Z}$ as a subset of $\mathbb{Q}$; together with Section 2.2, $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$. This is a convenient abuse, made honest by j: it is injective and carries 0, 1, $+$, $\cdot$ and $-$ of $\mathbb{Z}$ to those of $\mathbb{Q}$, so a computation with integers gives the same result read in $\mathbb{Z}$ or in $\mathbb{Q}$. With this convention, (a)–(c) read:

$$[(p,q)] = \frac{p}{q}, \qquad \frac{p}{q} = \frac{r}{s} \iff ps = rq,$$
$$\mathbb{Q} = \left\{ \frac{m}{n} : m \in \mathbb{Z},\ n \in \mathbb{Z}_{>0} \right\}.$$

The last equation is the description of $\mathbb{Q}$ that *Sets and Functions*, Section 3.1 took as a definition; it is now a theorem about the constructed set. The fraction bar is division in the field $\mathbb{Q}$, and the school rules for fractions, such as $\frac{p}{q} + \frac{r}{s} = \frac{ps+rq}{qs}$, are now the formulas of Definition 3.1.2 read through (a). The order of $\mathbb{Q}$, and the fact that j respects it, are Section 3.2's business.

Quick check. Is j surjective? Decide whether $1/2 \in j(\mathbb{Z})$.

Solution. No. If $[(m,1)] = [(1,2)]$, then $2m = 1$ in $\mathbb{Z}$, which is impossible: $1 = 2 \cdot 0 + 1$ is odd, so it is not of the form $2m$ (Corollary 2.5.3 (a); it is also the Quick check "Is $\mathbb{Z}$ a field?" of Section 2.4). So $1/2 \notin j(\mathbb{Z})$: $\mathbb{Z}$ is a proper subset of $\mathbb{Q}$.

3.1.5 Exercises

3.1.5.1 A. Check your understanding

Exercise. A1

(a) In Definition 3.1.2, why is $qs \in \mathbb{Z}_{>0}$?

(b) Where does the proof of transitivity in Proposition 3.1.1 use that a denominator is non-zero, and which denominator is it?

(c) Find the inverse of $[(-3,5)]$ in $\mathbb{Q}$, as a class with positive denominator.

(d) True or false: $j(2)^{-1} = [(1,2)]$. Give a reason.

Solution.

(a) $q, s \in \mathbb{Z}_{>0}$ means $q > 0$ and $s > 0$ (Theorem 2.4.2 (b)); then $qs > 0$ by (O4), so $qs \in \mathbb{Z}_{>0}$, again by Theorem 2.4.2 (b).

(b) In the last step: from $(pu - tq)s = 0$ it concludes $pu - tq = 0$ by Proposition 2.4.3 (f), which needs $s \neq 0$. Here s is the denominator of the **middle** pair (r, s).

(c) The numerator -3 is negative, so by Theorem 3.1.3 the inverse is $[(-5,3)]$. Check: $[(-3,5)] \cdot [(-5,3)] = [(15,15)] = [(1,1)]$, since $15 \cdot 1 = 1 \cdot 15$.

(d) True. $j(2) = [(2,1)]$ has positive numerator, so its inverse is $[(1,2)]$ by Theorem 3.1.3; directly, $[(2,1)] \cdot [(1,2)] = [(2,2)] = [(1,1)]$.

3.1.5.2 B. Practice

Exercise. B1: Three rows in detail

Let $x = [(p, q)]$, $y = [(r, s)]$, $z = [(t, u)]$ be rational numbers. Verify in detail, naming the law of $\mathbb{Z}$ used at each step, the rows (F9), (F4) and (F2) of the table in the proof of Theorem 3.1.3.

Solution. All operations are those of Definition 3.1.2; (F1)–(F9) refer to the laws of $\mathbb{Z}$ (Theorem 2.3.1).

(F9). $y+z = [(ru+ts,\ su)]$, so $x(y+z) = [(p(ru+ts),\ q(su))] = [(pru+pts,\ qsu)]$ by (F9) and (F5). On the other side, $xy = [(pr, qs)]$ and $xz = [(pt, qu)]$, so

$$\begin{aligned} xy + xz &= [((pr)(qu) + (pt)(qs),\ (qs)(qu))] \\ &= [(q(pru + pts),\ q(qsu))], \end{aligned}$$

by (F5), (F6) and (F9). The pairs $(A, B) := (pru + pts, qsu)$ and (qA, qB) are related, since $A(qB) = (qA)B$ by (F5), (F6). By Sets and Functions, Theorem 5.2.1 (b), $x(y+z) = xy+xz$.

(F4). $-x = [(-p, q)]$, so $x + (-x) = [(pq + (-p)q,\ qq)]$. By Proposition 2.3.2 (c), $(-p)q = -(pq)$, so the numerator is $pq + (-(pq)) = 0$ by (F4). The pair $(0, q^2)$ is related to $(0, 1)$, since $0 \cdot 1 = 0 = 0 \cdot q^2$ (Proposition 2.3.2 (b)). So $x + (-x) = [(0, 1)]$.

(F2). $x + y = [(ps + rq, qs)]$, so

$$\begin{aligned} (x + y) + z &= [((ps + rq)u + t(qs),\ (qs)u)] \\ &= [(psu + qru + qst,\ qsu)], \end{aligned}$$

by (F9), (F5), (F6). And $y + z = [(ru + ts, su)]$, so

$$\begin{aligned} x + (y + z) &= [(p(su) + (ru + ts)q,\ q(su))] \\ &= [(psu + qru + qst,\ qsu)], \end{aligned}$$

by the same laws. The pairs are equal, so $(x + y) + z = x + (y + z)$.

Exercise. B2: Addition is well defined

Prove directly, by computing both changes and without the symmetry argument of Proposition 3.1.2, that the addition of Definition 3.1.2 is well defined. The first half is done for you.

Change the first argument. Let $(p, q) \sim (p', q')$, that is, $pq' = p'q$, and fix (r, s). Then

$$\begin{aligned} (ps + rq)(q's) &= (pq')s^2 + rqq's \\ &= (p'q)s^2 + rqq's = (p's + rq')(qs), \end{aligned}$$

so $(ps + rq, qs) \sim (p's + rq', q's)$.

Name the move, then complete it.

Solution. The move is **check on representatives, one argument at a time**: having changed the first argument with the second fixed, change the second with the first fixed, and join the two changes by transitivity.

Change the second argument. Let $(r,s) \sim (r',s')$, that is, $rs' = r's$, and fix (p,q). Then, by (F5), (F6), (F9) in $\mathbb{Z}$,

$$\begin{aligned}(ps+rq)(qs') &= pqss' + (rs')q^2 \\ &= pqss' + (r's)q^2 = (ps'+r'q)(qs),\end{aligned}$$

so $(ps+rq, qs) \sim (ps'+r'q, qs')$.

Join. Let $(p,q) \sim (p',q')$ and $(r,s) \sim (r',s')$. Write g for the sum formula on representatives. The first step, with (r,s) fixed, gives $g((p,q),(r,s)) \sim g((p',q'),(r,s))$; the second, with (p',q') fixed, gives $g((p',q'),(r,s)) \sim g((p',q'),(r',s'))$. By transitivity (Proposition 3.1.1), $g((p,q),(r,s)) \sim g((p',q'),(r',s'))$, so the classes $[g((p,q),(r,s))]$ and $[g((p',q'),(r',s'))]$ are equal (Sets and Functions, Theorem 5.2.1 (b)). The two steps are the hypotheses of Theorem 2.2.1 (b); the join is the step its proof makes, and its conclusion gives addition as a function $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.

Exercise. B3: Prove or disprove

Prove or disprove each statement.

(a) The rule $[(p,q)] \mapsto [(p+2q,\ 2q)]$ defines a function $\mathbb{Q} \to \mathbb{Q}$.
(b) The rule $[(p,q)] \mapsto [(p,\ q^2)]$ defines a function $\mathbb{Q} \to \mathbb{Q}$.
(c) If $x, y \in \mathbb{Q}$ and $xy = 0$, then $x = 0$ or $y = 0$.
(d) Every non-zero rational number has a representative with numerator 1.

Solution.

(a) **True.** Since $q > 0$, $2q = q + q > 0$ (Proposition 2.4.3 (b)), so $(p+2q, 2q) \in \mathbb{Z} \times \mathbb{Z}_{>0}$ by Theorem 2.4.2 (b). By Theorem 2.2.1 (a) it suffices that $(p,q) \sim (p',q')$ gives $(p+2q, 2q) \sim (p'+2q', 2q')$. By (F5), (F6), (F9) in $\mathbb{Z}$,

$$\begin{aligned}(p+2q)(2q') &= 2pq' + 4qq', \\ (p'+2q')(2q) &= 2p'q + 4qq',\end{aligned}$$

and the right-hand sides are equal since $pq' = p'q$. (The function is $x \mapsto [(p,2q)] + [(1,1)]$, half of x plus one, by Definition 3.1.2.)

(b) **False.** $[(1,1)] = [(2,2)]$, since $1 \cdot 2 = 2 \cdot 1$, but the rule gives $[(1,1)]$ for one representative and $[(2,4)]$ for the other, and $(1,1) \not\sim (2,4)$ because $1 \cdot 4 \neq 2 \cdot 1$.

(c) **True.** Let $x = [(p,q)]$ and $y = [(r,s)]$. Then $xy = [(pr, qs)]$, and by Theorem 3.1.3 this class is zero only if $pr = 0$. So $p = 0$ or $r = 0$ by Proposition 2.4.3 (f) in $\mathbb{Z}$, that is, $x = 0$ or $y = 0$, again by Theorem 3.1.3. (The statement holds in every field: if $x \neq 0$, multiply $xy = 0$ by x^{-1}.)

(d) **False.** Take $2 = [(2,1)]$. If $(1,s) \sim (2,1)$ with $s \in \mathbb{Z}_{>0}$, then $1 \cdot 1 = 2s$, and $2s = 1$ has

no solution in $\mathbb{Z}$ (Corollary 2.5.3 (a), as in the Quick check after the Identification). So no representative of 2 has numerator 1.

3.1.5.3 C. Going deeper

Exercise. C1: The least denominator

(a) Show that the rule "$[(p, q)] \mapsto q$" does not define a function $\mathbb{Q} \to \mathbb{Z}_{>0}$.

(b) Prove that there is exactly one function $N\colon \mathbb{Q} \to \mathbb{Z}_{>0}$ such that, for every $x \in \mathbb{Q}$, $N(x)$ is the least $q \in \mathbb{Z}_{>0}$ for which $x = [(p, q)]$ for some $p \in \mathbb{Z}$.

(c) Compute $N(6/4)$, justifying that no smaller denominator is possible.

Solution (Hint 1). For (b), fix x and look at the set of all denominators of representatives of x. Which result of Section 2.4 gives a least element?

Solution (Hint 2). Let $D_x := \{q \in \mathbb{Z}_{>0} : (p, q) \in x \text{ for some } p \in \mathbb{Z}\}$. It is non-empty because a class contains its representative.

Solution.

(a) $[(1, 2)] = [(2, 4)]$, since $1 \cdot 4 = 2 \cdot 2$, but the rule gives 2 for one representative and 4 for the other. A function has one value at each argument (Sets and Functions, Definition 2.1.1).

(b) For $x \in \mathbb{Q}$, let $D_x := \{q \in \mathbb{Z}_{>0} : (p, q) \in x \text{ for some } p \in \mathbb{Z}\}$. Note that $x = [(p, q)]$ holds if and only if $(p, q) \in x$. If $x = [(p, q)]$, then $(p, q) \in x$ by Sets and Functions, Theorem 5.2.1 (a). Conversely, write $x = [(p_0, q_0)]$ (Sets and Functions, Definition 5.2.4); then $(p, q) \in x$ means $(p, q) \sim (p_0, q_0)$ (Sets and Functions, Definition 5.2.2), so $[(p, q)] = [(p_0, q_0)] = x$ by Sets and Functions, Theorem 5.2.1 (b). So D_x is the set of denominators allowed in the definition of $N(x)$.
D_x has a least element. By Sets and Functions, Definition 5.2.4, $x = [(p_0, q_0)]$ for some pair, and $(p_0, q_0) \in x$ by Sets and Functions, Theorem 5.2.1 (a), so $q_0 \in D_x$ and D_x is non-empty. Every element of D_x is positive, so 0 is an integer lower bound. By Proposition 2.4.4, D_x has a least element, and it is unique, by antisymmetry of $\leq$ (Proposition 2.4.3 (i)).
Existence and uniqueness of N. Let N be the function whose graph is $\{(x, \min D_x) : x \in \mathbb{Q}\} \subseteq \mathbb{Q} \times \mathbb{Z}_{>0}$. Each x has exactly one partner $\min D_x$, so N is a function (Sets and Functions, Definition 2.1.1), and it has the required property. Any function with the property must take the value $\min D_x$ at every x, so it equals N (Sets and Functions, Proposition 2.1.1). Unlike the rule in (a), N is defined from the class x itself, not from a chosen representative, so there is nothing to check for well-definedness.

(c) $6/4 = [(6, 4)]$ by Proposition 3.1.4 (a). A pair (p, q) lies in this class if and only if $4p = 6q$.

- $q = 1$ is impossible: it needs $4p = 6$. By discreteness (Theorem 2.4.2 (c)), $p \leq 1$ or $p \geq 2$. If $p \leq 1$, then $4p \leq 4 < 6$; if $p \geq 2$, then $4p \geq 8 > 6$, by Proposition 2.4.3 (c) (or equality). Either way $4p \neq 6$, by (O1).

- $q = 2$ works: $4 \cdot 3 = 12 = 6 \cdot 2$, so $(3, 2) \in [(6, 4)]$.

Since 1 is the only element of $\mathbb{Z}_{>0}$ below 2 (discreteness), $N(6/4) = 2$.

Marking scheme. (10 points) - 2: (a) exhibits two representatives of one class with different denominators. - 2: (b) defines the set of denominators of x from the class, and shows it non-empty. - 2: (b) gets a least element from Proposition 2.4.4, with the lower bound named. - 2: (b) explains why this defines exactly one function (one value per class; any such function has those values). - 2: (c) shows $4p = 6$ has no integer solution and exhibits $(3, 2)$. Fatal errors (0 for the step): defining $N([(p, q)])$ as "q divided by a common factor" (the formula uses a representative, and lowest terms come only in *Algebraic Preliminaries*, Chapter 1); in (c), "$4p = 6$ has no solution since $p = 3/2$" (uses division in $\mathbb{Q}$ to decide a question about $\mathbb{Z}$ without the discreteness argument).

Exercise. C2: Find the error

Here is a proof that multiplication on $\mathbb{Q}$ is well defined. Is the conclusion true? Is the proof correct? If not, name the first line that fails, the error, and give a counterexample to the step.

> (1) Let $(p, q) \sim (p', q')$, and let $(r, s) \in \mathbb{Z} \times \mathbb{Z}_{>0}$. (2) Then $[(p, q)] = [(p', q')]$. (3) So $[(p, q)] \cdot [(r, s)] = [(p', q')] \cdot [(r, s)]$. (4) That is, $[(pr, qs)] = [(p'r, q's)]$, so the product does not depend on the first representative. (5) The second argument is handled in the same way.

Solution. The conclusion is **true**: it is Proposition 3.1.2. The proof is **not** correct.

The failing line is (3). Line (2) is correct: it is (1) restated by Sets and Functions, Theorem 5.2.1 (b), and says that (p, q) and (p', q') name the same class. Line (3) then multiplies that class by $[(r, s)]$, using the multiplication of $\mathbb{Q}$. But that multiplication is exactly what is being shown to exist: until it is known to be well defined, "$[(p, q)] \cdot [(r, s)]$" has no meaning, or rather means "$[(pr, qs)]$, computed from the representative (p, q)", and whether this depends on the representative is the claim. The line cites the goal, which is circular: entry (N9) of Logic and Proof, Example (What is not allowed).

Counterexample to the step. The inference "$[(p, q)] = [(p', q')]$, so $F(p, q) = F(p', q')$" fails for formulas F in representatives. With $F(p, q) \coloneqq p + q$, the same lines "prove" that $[(p, q)] \mapsto p+q$ is a function on $\mathbb{Q}$, since "$p+q$ is a function of the class $[(p, q)]$". It is not: $(1, 2) \sim (2, 4)$ and $1 + 2 \neq 2 + 4$ (Example (Formulas that ignore the class) (a)).

Repair. Replace (2)–(3) by an identity in $\mathbb{Z}$: from $pq' = p'q$, $(pr)(q's) = (pq')(rs) = (p'q)(rs) = (p'r)(qs)$, so $(pr, qs) \sim (p'r, q's)$. Do the same for the second argument, and conclude with Theorem 2.2.1 (b). Lines (1), (4) and (5) are correct as stated, once (3) is replaced.

Marking scheme. (10 points) - 2: says that the conclusion is true and that the proof is not correct. - 3: names line (3) as the failing line and explains that it uses the multiplication of $\mathbb{Q}$, which is being defined. - 2: names the error as circular, entry (N9). - 2: gives a counterexample to the step (any formula in representatives that is not constant on classes, such as $p + q$). - 1: gives the repair through the integer identity $(pr)(q's) = (p'r)(qs)$. Fatal errors (0 for the step): declaring the conclusion false; blaming line (1) or (2), which only

restate the hypothesis.

Exercise. C3: Non-zero denominators

Let $Y := \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, and for $(p, q), (r, s) \in Y$ let $(p, q) \approx (r, s)$ mean $ps = rq$. Show that $\approx$ is an equivalence relation on Y, and that

$$I : \mathbb{Q} \to Y/\!\approx, \qquad I([(p, q)]) := [(p, q)]_{\approx},$$

is a well-defined bijection.

Solution. Equivalence relation. The proof of Proposition 3.1.1 works on Y with two changes: the pairs are taken in Y instead of $\mathbb{Z}\times\mathbb{Z}_{>0}$, and the line deriving $s \neq 0$ from $s > 0$ is dropped, since in Y the middle denominator s is non-zero by definition. That is the only property of the denominators the proof uses.

Well defined. Since $q > 0$, $q \neq 0$ by (O1), so $(p, q) \in Y$ and the inclusion is a function. Apply Theorem 2.2.1 (a) to the inclusion $(p, q) \mapsto (p, q)$ from $\mathbb{Z} \times \mathbb{Z}_{>0}$ to Y: if $(p, q) \sim (p', q')$, then $pq' = p'q$, which says $(p, q) \approx (p', q')$.

Injective. If $[(p, q)]_{\approx} = [(p', q')]_{\approx}$, then $(p, q) \approx (p', q')$ by Sets and Functions, Theorem 5.2.1 (b), so $pq' = p'q$, so $(p, q) \sim (p', q')$ and $[(p, q)] = [(p', q')]$.

Surjective. Let $[(p, q)]_{\approx} \in Y/\!\approx$, with $q \neq 0$. By (O1), $q > 0$ or $q < 0$. If $q > 0$, the class is $I([(p, q)])$. If $q < 0$, then $-q > 0$ (Proposition 2.4.3 (a)), so $-q \in \mathbb{Z}_{>0}$ by Theorem 2.4.2 (b), and $(-p, -q) \in \mathbb{Z} \times \mathbb{Z}_{>0}$; and $(p, q) \approx (-p, -q)$, since $p(-q) = -(pq) = (-p)q$ by Proposition 2.3.2 (c); so $[(p, q)]_{\approx} = [(-p, -q)]_{\approx}$ by Sets and Functions, Theorem 5.2.1 (b), and the class is $I([(-p, -q)])$.

So I is a bijection: the two constructions have "the same" rational numbers, matched by I.

3.2 The Rationals as an Ordered Field; the Archimedean Property

Section 3.1 made $\mathbb{Q}$ a field (Theorem 3.1.3). It did not give $\mathbb{Q}$ an order, and the school number line cannot supply one. We order $\mathbb{Q}$ the way Section 2.4 ordered $\mathbb{Z}$: name the positive elements, check (P1) and (P2) of Proposition 2.4.1, and read off every rule for inequalities. Then we prove, from the construction and with no supremum anywhere, that every rational lies below a natural number; Chapters 4 and 5 use its form $1/n < \varepsilon$ at every turn. Finally every rational gets an integer part.

3.2.1 Which fractions are positive

Throughout, a rational is written as in Section 3.1. By Proposition 3.1.4 (b), every $x \in \mathbb{Q}$ is p/q for some $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$, where p/q is the class $[(p, q)]$, and $p/q = r/s$ if and only if $ps = rq$. The integers sit inside $\mathbb{Q}$ as $m = m/1$, and the natural numbers inside the integers.

The order of $\mathbb{Z}$ came from its positive set $P_{\mathbb{Z}}$ (Definition 2.4.2), and Proposition 2.4.1 turned

closure under $+$ and $\cdot$, together with a trichotomy, into all of (O1)–(O4). We do the same for $\mathbb{Q}$, so the whole question is: which fractions should count as positive? A first guess from school is "numerator and denominator have the same sign". But in $\mathbb{Q}$ the denominator is always positive. That was the decision recorded with Definition 3.1.1, and here it pays: the sign of a fraction is the sign of its numerator.

A fraction with a positive denominator is positive exactly when its numerator is.

Definition 3.2.1. The Order on $\mathbb{Q}$

Let

$$P_{\mathbb{Q}} := \{[(p,q)] : (p,q) \in \mathbb{Z} \times \mathbb{Z}_{>0} \text{ and } p > 0\}.$$

For $x, y \in \mathbb{Q}$, we write $x < y$ if $y - x \in P_{\mathbb{Q}}$. As in Definition 2.4.1, $x > y$ means $y < x$, and $x \leq y$ means $x < y$ or $x = y$.

In words: a rational is in $P_{\mathbb{Q}}$ if it **has a** representative (p,q) with $p > 0$, the inequality being the order of $\mathbb{Z}$; and $x < y$ means that the difference $y - x$, computed in the field $\mathbb{Q}$, lies in $P_{\mathbb{Q}}$.

Well-definedness. The definition says "has a representative with $p > 0$". To use it we want more: that **every** representative of a class in $P_{\mathbb{Q}}$ has a positive numerator, so that the test can be run on whichever representative is at hand. This is the situation of Theorem 2.2.1 (c), with $X = \mathbb{Z} \times \mathbb{Z}_{>0}$ and $S = \{(p,q) \in X : p > 0\}$. Its hypothesis asks: if $(p,q) \sim (p',q')$ and $p > 0$, is $p' > 0$? It is, and the reason is the sign of the denominators.

Suppose $pq' = p'q$ with $q, q' \in \mathbb{Z}_{>0}$ and $p > 0$. Since $q, q' > 0$ (Theorem 2.4.2 (b)), (O4) gives $pq' > 0$, so $p'q > 0$. If $p' = 0$, then $p'q = 0$ (Proposition 2.3.2 (b)); if $p' < 0$, then $p'q < 0 \cdot q = 0$ by Proposition 2.4.3 (c), since $q > 0$. Both contradict $p'q > 0$ by trichotomy, so $p' > 0$. Now Theorem 2.2.1 (c) gives, for every $(p,q) \in \mathbb{Z} \times \mathbb{Z}_{>0}$,

$$[(p,q)] \in P_{\mathbb{Q}} \iff p > 0. \tag{3.2.1}$$

Examples. The class $3/4 = 6/8$ lies in $P_{\mathbb{Q}}$, with both numerators positive, as Eq. 3.2.1 says it must; $-6/8$ does not. Zero does not: $0 = 0/7$, and $0 > 0$ fails. That matters for (P2), which asks for exactly one of $x \in P_{\mathbb{Q}}$, $x = 0$, $-x \in P_{\mathbb{Q}}$.

Non-example by minimal change. Allow negative denominators: let $Y := \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ with the same cross-multiplication, written $\approx$. It is still an equivalence relation, since the proof of Proposition 3.1.1 used only $s \neq 0$. Now $(1,1) \approx (-1,-1)$, and the test "the numerator is positive" passes on one representative and fails on the other: the hypothesis of Theorem 2.2.1 (c) fails, at the step "$q > 0$" of the check above. The comparison of two fractions breaks the same way. For $(0,1)$ and $(1,1)$, the test "$ps < rq$" asks whether $0 < 1$, which is true; for $(0,1)$ and $(-1,-1)$ it asks whether $0 < -1$, which is false, because multiplying by a negative denominator reverses an inequality (Proposition 2.4.3 (d)).

Warning. Comparing numerators alone is not comparing fractions. The rationals $1/2$ and $2/5$ have $1 < 2$, yet $1/2 > 2/5$, since $1/2 - 2/5 = 1/10 \in P_{\mathbb{Q}}$. Only after bringing both to the same positive denominator do the numerators decide: that is the comparison formula below.

Quick check. Which of $[(-2,3)]$, $[(0,5)]$, $[(4,6)]$ lie in $P_{\mathbb{Q}}$? Why can one not ask whether $[(2,-3)]$ lies in $P_{\mathbb{Q}}$?

Solution. By Eq. 3.2.1, only $[(4,6)]$ does: its numerator 4 is positive, while $-2 > 0$ and $0 > 0$ both fail. The symbol $[(2,-3)]$ names no element of $\mathbb{Q}$: the denominator -3 is not in $\mathbb{Z}_{>0}$, so $(2,-3)$ is not in $\mathbb{Z} \times \mathbb{Z}_{>0}$ and has no class. (The rational one might mean is $-2/3 = [(-2,3)]$, which is not in $P_{\mathbb{Q}}$.)

3.2.2 The rationals form an ordered field

The payoff is that the order of $\mathbb{Q}$ has every property of the order of $\mathbb{Z}$, together with a formula that compares two fractions with one integer inequality.

Theorem 3.2.1. The Rationals Form an Ordered Field

With the order of Definition 3.2.1, $\mathbb{Q}$ is an ordered field (Definition 2.4.1), and $P_{\mathbb{Q}} = \{x \in \mathbb{Q} : x > 0\}$. Moreover:

(a) *(The comparison formula.)* For $p, r \in \mathbb{Z}$ and $q, s \in \mathbb{Z}_{>0}$,

$$\frac{p}{q} < \frac{r}{s} \iff ps < rq;$$

in particular $p/q > 0$ if and only if $p > 0$.

(b) For $m, n \in \mathbb{Z}$, $j(m) < j(n)$ if and only if $m < n$.

Idea. Proposition 2.4.1 reduces everything to (P1) and (P2), and Eq. 3.2.1 turns each into a statement about numerators in $\mathbb{Z}$: the level below, as in Section 2.3. For $p, r > 0$, a sum $p/q + r/s$ has numerator $ps + rq$, positive **because s and q are**. The comparison formula is the same computation run on a difference. *Before reading on: where is $q > 0$ used?*

Proof. By Theorem 3.1.3, $\mathbb{Q}$ is a field, in particular a commutative ring. We check (P1) and (P2) of Proposition 2.4.1 for $P = P_{\mathbb{Q}}$.

(P1) Let $x, y \in P_{\mathbb{Q}}$, say $x = [(p,q)]$ and $y = [(r,s)]$ with $q, s \in \mathbb{Z}_{>0}$. By Eq. 3.2.1, $p > 0$ and $r > 0$; and $q, s > 0$ by Theorem 2.4.2 (b). By Definition 3.1.2,

$$x + y = [(ps + rq,\ qs)], \qquad xy = [(pr,\ qs)],$$

with $qs \in \mathbb{Z}_{>0}$. By (O4) in $\mathbb{Z}$, $ps > 0$ and $rq > 0$, so $ps + rq > 0 + 0 = 0$ by Proposition 2.4.3 (b); and $pr > 0$ by (O4). By Eq. 3.2.1, $x + y \in P_{\mathbb{Q}}$ and $xy \in P_{\mathbb{Q}}$.

(P2) Let $x = [(p,q)] \in \mathbb{Q}$. Then $-x = [(-p,q)]$ (Definition 3.1.2). By Theorem 3.1.3, the zero of $\mathbb{Q}$ is $[(0,1)]$, and $x = 0$ if and only if $p = 0$. By Eq. 3.2.1, $x \in P_{\mathbb{Q}}$ if and only if $p > 0$, and $-x \in P_{\mathbb{Q}}$ if and only if $-p > 0$, which by Proposition 2.4.3 (a) means $p < 0$. By (O1) in $\mathbb{Z}$, exactly one of $p > 0$, $p = 0$, $p < 0$ holds, so exactly one of $x \in P_{\mathbb{Q}}$, $x = 0$, $-x \in P_{\mathbb{Q}}$ holds.

By Proposition 2.4.1, the relation $x < y \iff y - x \in P_{\mathbb{Q}}$, which is the relation of Definition 3.2.1, satisfies (O1)–(O4), and $P_{\mathbb{Q}} = \{x : x > 0\}$. A field that is an ordered commutative ring is an ordered field (Definition 2.4.1).

(a) Let $p, r \in \mathbb{Z}$ and $q, s \in \mathbb{Z}_{>0}$. By Definition 3.1.2 and Proposition 2.3.2 (c),

$$\frac{r}{s} - \frac{p}{q} = [(r, s)] + [(-p, q)] = [(rq + (-p)s,\ sq)] = [(rq - ps,\ qs)].$$

So $p/q < r/s$ holds if and only if this class lies in $P_{\mathbb{Q}}$, that is (by Eq. 3.2.1, since $qs \in \mathbb{Z}_{>0}$), if and only if $rq - ps > 0$, which by Proposition 2.4.3 (a) means $ps < rq$. For the special case, compare $0 = 0/1$ with p/q: the formula gives $0/1 < p/q$ if and only if $0 \cdot q < p \cdot 1$, that is, $p > 0$.

(b) $j(m) = m/1$ and $j(n) = n/1$ (Proposition 3.1.4), so by (a), $j(m) < j(n)$ if and only if $m \cdot 1 < n \cdot 1$, that is, $m < n$.

This proves the theorem.

■

The answer to the prompt: three times. In the well-definedness check, q and q' carry the sign from p to p'; in (P1), $ps, rq > 0$; and in (a), the difference has denominator $qs > 0$. The non-example above shows the first and the third failing without it.

Since $p/q = r/s$ exactly when $ps = rq$ (Proposition 3.1.4 (b)), the comparison formula has a version with $\leq$: for $q, s \in \mathbb{Z}_{>0}$,

$$\frac{p}{q} \leq \frac{r}{s} \iff ps \leq rq. \tag{3.2.2}$$

By (b) and Theorem 2.4.2 (a), a comparison of two integers means the same in $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$, so we never say where it is made. Every rule of Proposition 2.4.3 holds in $\mathbb{Q}$, and $\mathbb{Q}_{>0} \coloneqq \{x \in \mathbb{Q} : x > 0\}$ is $P_{\mathbb{Q}}$.

One rule is new with fields, because rings have no inverses: taking reciprocals reverses the order of positive elements.

Proposition 3.2.2. Reciprocals Reverse Order

Let F be an ordered field and $a, b \in F$.

(a) If $a > 0$, then $a^{-1} > 0$.

(b) If $0 < a < b$, then $0 < b^{-1} < a^{-1}$, that is, $0 < 1/b < 1/a$.

Idea. For (a), turn a^{-1} into a product we can sign: $a^{-1} = a \cdot (a^{-1})^2$, a positive element times a non-zero square. For (b), multiply $a < b$ by the positive element $a^{-1}b^{-1}$: it cancels a on the left and b on the right, and swaps which reciprocal stands on which side.

Proof.

(a) Let $a > 0$. Then $a \neq 0$, so a^{-1} exists, and $a^{-1} \neq 0$ by the Quick check after the

uniqueness of inverses in Section 2.3. By Proposition 2.4.3 (e), $(a^{-1})^2 > 0$. Now

$$a \cdot (a^{-1})^2 = (aa^{-1})a^{-1} = 1 \cdot a^{-1} = a^{-1},$$

and the left side is a product of two positive elements, so $a^{-1} > 0$ by (O4).

(b) Let $0 < a < b$. Then $b > 0$ by (O2), so $a^{-1} > 0$ and $b^{-1} > 0$ by (a), and $c := a^{-1}b^{-1} > 0$ by (O4). Multiplying $a < b$ by $c > 0$ (Proposition 2.4.3 (c)) gives

$$b^{-1} = (aa^{-1})b^{-1} = ac < bc = (bb^{-1})a^{-1} = a^{-1}.$$

Since $1/b = 1 \cdot b^{-1} = b^{-1}$ (Definition 2.3.1), this is $0 < 1/b < 1/a$.

■

Quick check. Find non-zero $a, b \in \mathbb{Q}$ with $a < b$ and $1/a < 1/b$. Which hypothesis of Proposition 3.2.2 (b) fails?

Solution. Take $a = -1$ and $b = 1$. Then $1/a = -1$, since $(-1)(-1) = 1$, and $1/a = -1 < 1 = 1/b$. The hypothesis $0 < a$ fails: (b) is about positive elements only.

3.2.3 Between two rationals

In $\mathbb{Z}$ every integer has a next one: there is no integer strictly between m and $m+1$ (Theorem 2.4.2 (c)). In $\mathbb{Q}$ the opposite holds. Let $2 := 1+1$. By Proposition 2.4.3 (e) and (b), $1 > 0$ and $2 > 0$, so $1/2 > 0$ by Proposition 3.2.2 (a). Now let $a < b$ in $\mathbb{Q}$. Adding a, and then b, by (O3),

$$a + a < a + b < b + b,$$

and $a+a = (1+1)a = 2a$, $b+b = 2b$ by distributivity. Multiplying by $1/2 > 0$ (Proposition 2.4.3 (c)) gives

$$a < \frac{a+b}{2} < b.$$

So between two distinct rationals there is always a third, their **midpoint**. Applied with $a = 0$: for every $x > 0$, also $0 < x/2 < x$. Hence $\mathbb{Q}_{>0}$ has **no least element**. In particular, well-ordering is a property of $\mathbb{N}$ (Theorem 1.3.3) that $\mathbb{Q}_{>0}$ does not share.

Warning. Between any two rationals lies another, but that does not mean every rational has a neighbor "right next to it". It means the opposite: no rational x has a next one, since any candidate $y > x$ is beaten by the midpoint of x and y. Phrases like "the rational just after 0" name nothing.

Quick check. Find the midpoint of $1/3$ and $1/2$, and confirm with the comparison formula that it lies strictly between them.

Solution. $1/3 + 1/2 = 5/6$ by Definition 3.1.2 ($[(1 \cdot 2 + 1 \cdot 3,\, 6)]$), and $(5/6) \cdot (1/2) = 5/12$. By Theorem 3.2.1 (a): $1/3 < 5/12$ because $1 \cdot 12 = 12 < 15 = 5 \cdot 3$, and $5/12 < 1/2$ because $5 \cdot 2 = 10 < 12 = 1 \cdot 12$.

3.2.4 The Archimedean property

Here is a statement so familiar that it seems not to need proof: every rational number is smaller than some natural number. For the real numbers it is a genuine theorem. *Analysis on the Real Line*, Chapter 1 obtains it from the least upper bound axiom, and there are ordered fields in which $\mathbb{N}$ is bounded above. For the constructed $\mathbb{Q}$ it is true for a simpler reason: every rational is a quotient of integers, and integers are as discrete as natural numbers.

Theorem 3.2.3. The Rationals Are Archimedean

For every $x \in \mathbb{Q}$ there is $n \in \mathbb{N}$ with $x < n$.

Try it first (ten minutes): given $x = p/q$ with $q \in \mathbb{Z}_{>0}$, which natural number beats it? Do not forget negative p.

Idea. The first candidate is $p + 1$, on the grounds that dividing by $q \geq 1$ "makes things smaller", so $p/q \leq p < p + 1$. An attempt meets two obstacles. Dividing by $q \geq 1$ makes a **negative** number larger: $-3/2 > -3$. And $p+1$ is not a natural number when $p \leq -2$. Both disappear if we split on the sign of p. A fraction with $p \leq 0$ is at most 0, and 1 beats it. For $p > 0$ the school intuition is right and provable: $q \geq 1$ gives $p \leq pq$, which the comparison formula reads as $p/q \leq p$.

Proof. Let $x \in \mathbb{Q}$. By Proposition 3.1.4 (b), $x = p/q$ with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$.

Case 1: $p \leq 0$. By Eq. 3.2.2, $p/q \leq 0/1$, since $p \cdot 1 \leq 0 \cdot q$. So $x \leq 0 < 1$ (Proposition 2.4.3 (e)), and $n := 1 \in \mathbb{N}$ works.

Case 2: $p > 0$. Then $p \in \mathbb{N}$ by Theorem 2.4.2 (b). Since $q > 0$, discreteness (Theorem 2.4.2 (c), with $0 < q$) gives $1 \leq q$. Multiplying by $p > 0$ (Proposition 2.4.3 (c), or equality if $q = 1$) gives $p \cdot 1 \leq p \cdot q$. By Eq. 3.2.2 this says $p/q \leq p/1 = p$. Finally $p < p + 1$ by (O3), since $0 < 1$. So $x \leq p < p + 1$, and $n := p + 1$ lies in $\mathbb{N}$, because $\mathbb{N}$ is closed under addition. Hence $x < n$.

In both cases $x < n$ for some $n \in \mathbb{N}$, as claimed.

■

The n found depends on the representative, and any one will do. The proof uses only $x = p/q$, the comparison formula, and the ordered ring $\mathbb{Z}$ (discreteness, $\mathbb{N} = \{x \in \mathbb{Z} : x \geq 0\}$); no supremum appears. For the real numbers the same conclusion needs completeness.

The form in which later chapters of this volume use the property is about reciprocals: $1/n$ eventually falls below any positive rational.

Corollary 3.2.4. Small Reciprocals in $\mathbb{Q}$

For every $\varepsilon \in \mathbb{Q}_{>0}$ there is $n \in \mathbb{Z}_{>0}$ with $1/n < \varepsilon$. Moreover every $m \in \mathbb{Z}$ with $m \geq n$ then also satisfies $1/m < \varepsilon$. In symbols,

$$\forall \varepsilon \in \mathbb{Q}_{>0} \;\; \exists n \in \mathbb{Z}_{>0} \;\; \forall m \in \mathbb{Z}_{>0} \;\; (m \geq n \Rightarrow 1/m < \varepsilon).$$

Proof. Let $\varepsilon \in \mathbb{Q}_{>0}$. By Proposition 3.2.2 (a), $1/\varepsilon > 0$. By Theorem 3.2.3 there is $n \in \mathbb{N}$ with $1/\varepsilon < n$. Then $n > 0$, so $n \in \mathbb{Z}_{>0}$ (Theorem 2.4.2 (b)). By Proposition 3.2.2 (b) with $0 < 1/\varepsilon < n$,

$$\frac{1}{n} < \frac{1}{1/\varepsilon} = \varepsilon,$$

where $1/(1/\varepsilon) = \varepsilon$ is $(a^{-1})^{-1} = a$ for $a = \varepsilon$, the Quick check after the uniqueness of inverses in Section 2.3. If $m \geq n$, then either $m = n$, or $0 < n < m$ and $1/m < 1/n$ by Proposition 3.2.2 (b); in both cases $1/m \leq 1/n < \varepsilon$.

■

The order of the quantifiers is the point: ε is given first, and n is chosen after it and **depends on** ε (Exercise A1 (e) swaps them). Chapter 4 uses the corollary in exactly this form, with $\varepsilon \in \mathbb{Q}_{>0}$.

3.2.5 The integer part

By Theorem 3.2.3, every rational lies below an integer. With the least-element property of $\mathbb{Z}$, it pins every rational between two consecutive integers.

Proposition 3.2.5. Integer Part of a Rational

For every $x \in \mathbb{Q}$ there is **exactly one** $m \in \mathbb{Z}$ with $m \leq x < m + 1$. It is called the **integer part** of x and written $\lfloor x \rfloor$.

Idea. The integer just above x should be the least integer exceeding x. So look at the set of integers that exceed x: Theorem 3.2.3 makes it non-empty, and applied to $-x$ it bounds the set below, so Proposition 2.4.4 hands us its least element. One less than that is the integer part. Uniqueness is discreteness: two candidates one unit apart cannot both fit below x and above $x - 1$.

Proof. Let $x \in \mathbb{Q}$, and let $S := \{k \in \mathbb{Z} : x < k\}$. By Theorem 3.2.3 there is $n \in \mathbb{N} \subseteq \mathbb{Z}$ with $x < n$, so $S \neq \emptyset$. Applying Theorem 3.2.3 to $-x$ gives $n' \in \mathbb{N}$ with $-x < n'$, that is, $-n' < x$ (Proposition 2.4.3 (a)). Every $k \in S$ then has $-n' < x < k$, so the integer $-n'$ is a lower bound of S. By Proposition 2.4.4, S has a least element l. Let $m := l - 1 \in \mathbb{Z}$, so that $l = m + 1$.

Existence. $x < l = m + 1$, since $l \in S$. And $m \notin S$: otherwise $l \leq m = l - 1 < l$, so $l < l$, against (O1). So $x < m$ fails, and $m \leq x$ by totality (Proposition 2.4.3 (i)).

Uniqueness. Let $m, m' \in \mathbb{Z}$ both satisfy the condition, and suppose $m < m'$. By discreteness (Theorem 2.4.2 (c)), $m + 1 \leq m'$. Then $m' \leq x < m + 1 \leq m'$, so $m' < m'$, against (O1). By symmetry $m' < m$ is impossible too, so $m = m'$ by (O1).

■

The comparisons in this proof mix integers and rationals ($x < k$); by Theorem 3.2.1 (b) the two orders agree on $\mathbb{Z}$. For $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$, $\lfloor a/n \rfloor$ is the quotient in Section 2.5's division of a by n (Exercise B3).

Quick check. Compute $\lfloor -7/3 \rfloor$. Then find the least $n \in \mathbb{Z}_{>0}$ with $1/n < 3/1000$.

Solution. $\lfloor -7/3 \rfloor = -3$: by Eq. 3.2.2, $-3 = -3/1 \leq -7/3$ because $(-3) \cdot 3 = -9 \leq -7$, and $-7/3 < -2$ because $-7 < -6 = (-2) \cdot 3$ (Theorem 3.2.1 (a)). Not -2: the rule "drop the fraction" gives $\lfloor x \rfloor$ for $x \geq 0$, but not for negative non-integers such as $-7/3$.

For $n \in \mathbb{Z}_{>0}$, the comparison formula gives $1/n < 3/1000$ if and only if $1 \cdot 1000 < 3n$. Since $3 \cdot 333 = 999 < 1000$ and $3 \cdot 334 = 1002 > 1000$, the least such n is 334 (for $n \leq 333$, $3n \leq 999$).

3.2.6 Exercises

3.2.6.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Theorem 3.2.1 and about Corollary 3.2.4.

(a) Where in the proof (including the well-definedness check before it) is $q > 0$ used? What goes wrong without it?

(b) Why does (P2) need "**exactly** one" of $x \in P_{\mathbb{Q}}$, $x = 0$, $-x \in P_{\mathbb{Q}}$, and not just "at least one"?

(c) Run the comparison formula on $-2/3$ and $-3/5$: which is larger?

(d) State the idea of the proof in two sentences.

(e) Write Corollary 3.2.4 (its first sentence) in symbols. Then write the version with the two quantifiers swapped, negate it, and decide which of the two versions is true.

Solution.

(a) Three times: in the well-definedness check, $q, q' > 0$ turn $p > 0$ into $p' > 0$; in (P1), $ps > 0$ and $rq > 0$ need $s > 0$ and $q > 0$; in (a), the difference $r/s - p/q$ has denominator $qs > 0$, so its sign is the sign of its numerator. Without it, the non-example after Definition 3.2.1 shows both failures: $(1, 1) \approx (-1, -1)$ gives two verdicts for one class, and "$ps < rq$" gives $0 < 1$ for $(0, 1), (1, 1)$ but $0 < -1$ for $(0, 1), (-1, -1)$.

(b) "At least one" would allow a non-zero x with both x and $-x$ positive, and then $x < 0$ and $0 < x$ at once, against (O1). The "at most one" half is what makes the order

antisymmetric; Proposition 2.4.1 uses it to get trichotomy (O1).

(c) By Theorem 3.2.1 (a), $-2/3 < -3/5$ if and only if $(-2)\cdot 5 < (-3)\cdot 3$, that is, $-10 < -9$, which is true. So $-3/5$ is the larger.

(d) By Proposition 2.4.1 it is enough to show that the positive set is closed under $+$ and $\cdot$ and satisfies the trichotomy (P2). Each of these, run on representatives with positive denominators, is a statement about integer numerators, which holds because $\mathbb{Z}$ is an ordered ring.

(e) The corollary: $\forall \varepsilon \in \mathbb{Q}_{>0}\ \exists n \in \mathbb{Z}_{>0}\ \ 1/n < \varepsilon$. Swapped: $\exists n \in \mathbb{Z}_{>0}\ \forall \varepsilon \in \mathbb{Q}_{>0}\ \ 1/n < \varepsilon$, "one n works for every ε". By Logic and Proof, Proposition 6.2.3 (the bounded form of Logic and Proof, Proposition 6.2.1), applied twice, its negation is

$$\forall n \in \mathbb{Z}_{>0}\ \ \exists \varepsilon \in \mathbb{Q}_{>0}\ \ \neg(1/n < \varepsilon).$$

This negation is true: given $n \in \mathbb{Z}_{>0}$, take $\varepsilon := 1/n$, which lies in $\mathbb{Q}_{>0}$ by Proposition 3.2.2 (a), and $1/n < 1/n$ fails by (O1). So the swapped version is false, and the corollary (proved above) is true. The n must be allowed to depend on ε.

3.2.6.2 B. Practice

Exercise. B1: A rational of the form a + 1/n

(a) Let $a, b \in \mathbb{Q}$ with $a < b$. Prove that there is $n \in \mathbb{Z}_{>0}$ with $a < a + 1/n < b$, and that every $m \in \mathbb{Z}$ with $m \geq n$ works too.

(b) Find the least such n for $a = 1/3$ and $b = 1/2$.

Solution.

(a) Let $\varepsilon := b - a$. Then $\varepsilon > 0$ by Proposition 2.4.3 (a), so Corollary 3.2.4 gives $n \in \mathbb{Z}_{>0}$ with $1/n < \varepsilon$, and $1/m < \varepsilon$ for every $m \geq n$. Fix such an m (for instance $m = n$). Adding a (O3): $a + 1/m < a + (b - a) = b$. And $1/m > 0$ by Proposition 3.2.2 (a), since $m > 0$; adding a gives $a < a + 1/m$.

(b) Here $b - a = 1/2 - 1/3 = 1/6$, and for $n \in \mathbb{Z}_{>0}$ the comparison formula gives $1/n < 1/6$ if and only if $6 < n$. So the least n is 7, and $a + 1/7 = 1/3 + 1/7 = 10/21$. Check: $1/3 < 10/21$ since $21 < 30$, and $10/21 < 1/2$ since $20 < 21$. For $n = 6$, $a + 1/6 = 1/2 = b$, which is not strictly below b.

Exercise. B2

Compute $\lfloor 22/7 \rfloor$, $\lfloor -22/7 \rfloor$, $\lfloor 0 \rfloor$ and $\lfloor -1/100 \rfloor$, and check each against the defining inequalities of Proposition 3.2.5.

Solution. We use Eq. 3.2.2 and the comparison formula with denominators 1 for the integers.

- $\lfloor 22/7 \rfloor = 3$: $3 \leq 22/7$ since $3 \cdot 7 = 21 \leq 22$, and $22/7 < 4$ since $22 < 28 = 4 \cdot 7$.
- $\lfloor -22/7 \rfloor = -4$: $-4 \leq -22/7$ since $-28 \leq -22$, and $-22/7 < -3$ since $-22 < -21$.
- $\lfloor 0 \rfloor = 0$: $0 \leq 0 < 1$.
- $\lfloor -1/100 \rfloor = -1$: $-1 \leq -1/100$ since $-100 \leq -1$, and $-1/100 < 0$ since $-1 < 0$.

The first two show that $\lfloor -x \rfloor \neq -\lfloor x \rfloor$ in general: $\lfloor 22/7 \rfloor = 3$, but $\lfloor -22/7 \rfloor = -4$, not -3.

Exercise. B3: The quotient is the integer part

Let $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$, and let q, r be the quotient and remainder of Theorem 2.5.2 (a), so $a = qn + r$ with $0 \leq r < n$. Prove that $q = \lfloor a/n \rfloor$, and hence $r = a - n\lfloor a/n \rfloor$.

Solution. From the hypothesis, $0 \leq r < n$; adding qn (Proposition 2.4.3 (b)) gives

$$qn \leq a < qn + n.$$

These are inequalities between integers, and by Theorem 3.2.1 (b) and Proposition 3.1.4 they hold in $\mathbb{Q}$ as well, with the same sums and products. In $\mathbb{Q}$, $n > 0$, so $n^{-1} > 0$ (Proposition 3.2.2 (a)). Multiplying by n^{-1} preserves strict inequalities (Proposition 2.4.3 (c)) and equalities, so

$$q = qn \cdot n^{-1} \leq a \cdot n^{-1} < (qn + n) \cdot n^{-1} = q + 1.$$

That is, $q \leq a/n < q + 1$, with $q \in \mathbb{Z}$. By the uniqueness in Proposition 3.2.5, $q = \lfloor a/n \rfloor$, and then $r = a - qn = a - n\lfloor a/n \rfloor$.

Exercise. B4: Prove or disprove

Prove each true statement and refute each false one with a counterexample.

(a) If $x, y \in \mathbb{Q}$ and $0 < x < y$, then $1/x - 1/y < y - x$.

(b) For every $x \in \mathbb{Q}_{>0}$ there is $n \in \mathbb{Z}_{>0}$ with $1/n < x < n$.

(c) Every non-empty $S \subseteq \mathbb{Q}$ that has a positive lower bound has a least element.

(d) For every $x \in \mathbb{Q}$, $\lfloor x + 1 \rfloor = \lfloor x \rfloor + 1$.

Solution.

(a) **False.** Take $x = 1/2$ and $y = 1$. Then $0 < x < y$, but $1/x - 1/y = 2 - 1 = 1$ and $y - x = 1/2$, and $1 > 1/2$.

(b) **True.** By Theorem 3.2.3 there is $n_1 \in \mathbb{N}$ with $x < n_1$. By Corollary 3.2.4 there is $n_2 \in \mathbb{Z}_{>0}$ with $1/m < x$ for every $m \in \mathbb{Z}$ with $m \geq n_2$. Let n be the larger of n_1 and n_2 (totality, Proposition 2.4.3 (i)). Then $n \geq n_2 > 0$, so $n \in \mathbb{Z}_{>0}$ and $1/n < x$; and $x < n_1 \leq n$.

(c) **False.** The set $S := \{x \in \mathbb{Q} : 1 < x\}$ is non-empty ($2 \in S$) and has the positive lower bound 1. For $x \in S$, the midpoint $(1+x)/2$ lies strictly between 1 and x (the midpoint paragraph of this section), so it is in S and smaller than x. No element of S is least. A positive lower bound does not help: $\mathbb{Q}$ is not $\mathbb{Z}$.

(d) **True.** Let $m := \lfloor x \rfloor$, so $m \leq x < m+1$. Adding 1 (O3) gives $m+1 \leq x+1 < (m+1)+1$, with $m + 1 \in \mathbb{Z}$. By the uniqueness in Proposition 3.2.5, $\lfloor x + 1 \rfloor = m + 1$.

3.2.6.3 C. Going deeper

Exercise. C1: Exit check: $\mathbb{Q}$ is Archimedean

Prove, from the construction of $\mathbb{Q}$ and the results of this chapter before Theorem 3.2.3 (and not from any fact about the real numbers), that for every $x \in \mathbb{Q}$ there is $n \in \mathbb{N}$ with $x < n$; and deduce that for every $\varepsilon \in \mathbb{Q}_{>0}$ there is $n \in \mathbb{Z}_{>0}$ with $1/n < \varepsilon$.

Solution. The theorem. Let $x \in \mathbb{Q}$. By Proposition 3.1.4 (b), $x = p/q$ with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$.

If $p \leq 0$, then $p \cdot 1 \leq 0 \cdot q$, so $x = p/q \leq 0/1 = 0$ by the comparison formula (Theorem 3.2.1 (a), with equality by Proposition 3.1.4 (b)). Hence $x \leq 0 < 1$, and $n = 1$ works.

If $p > 0$, then $p \in \mathbb{N}$ (Theorem 2.4.2 (b)), and $q \geq 1$ by discreteness (Theorem 2.4.2 (c)), since $q > 0$. Multiplying $1 \leq q$ by $p > 0$ (Proposition 2.4.3 (c), or equality if $q = 1$) gives $p \leq pq$, which by the comparison formula says $p/q \leq p/1$. So $x \leq p < p + 1$, and $n := p + 1 \in \mathbb{N}$ works.

The corollary. Let $\varepsilon \in \mathbb{Q}_{>0}$ be given. Then $1/\varepsilon > 0$ (Proposition 3.2.2 (a)), and the theorem gives $n \in \mathbb{N}$ with $1/\varepsilon < n$; in particular $n > 0$, so $n \in \mathbb{Z}_{>0}$. By Proposition 3.2.2 (b), $1/n < 1/(1/\varepsilon) = \varepsilon$. The n depends on ε, which was fixed first.

Marking scheme. (10 points) - 1: writes $x = p/q$ with $p \in \mathbb{Z}$, $q \in \mathbb{Z}_{>0}$, citing the representation of rationals. - 2: the case $p \leq 0$: $x \leq 0 < 1$, justified by the comparison formula. - 3: the case $p > 0$: $q \geq 1$ by discreteness of $\mathbb{Z}$, and $p/q \leq p$ by the comparison formula (or an equivalent inequality argument from the ordered-field rules). - 1: exhibits $n \in \mathbb{N}$ (for instance $p + 1$) and checks $x < n$. - 3: the corollary: ε fixed first, $n > 1/\varepsilon$ from the theorem, reciprocals reverse order. Fatal errors (0 for the step): no separate treatment of $p < 0$ (there "$p/q \leq p$" fails once $q \geq 2$, as $-3/2 > -3$ shows, and $p + 1 \notin \mathbb{N}$ once $p \leq -2$); using a supremum, the real numbers, or "the decimal expansion of x". A route through Proposition 3.2.5 is circular here (its proof uses the theorem) and earns nothing for the theorem's 7 points; the corollary's 3 points, deduced from the theorem as stated, can still be earned.

Exercise. C2: The comparison, on representatives

The comparison formula of Theorem 3.2.1 (a) was proved through the positive set. Prove directly that the relation "$ps < rq$" on pairs in $\mathbb{Z} \times \mathbb{Z}_{>0}$ does not depend on the representatives: if $(p, q) \sim (p', q')$ and $(r, s) \sim (r', s')$, then $ps < rq$ if and only if $p's' < r'q'$.

Here is the start.

> We change one argument at a time. *First argument:* let $(p, q) \sim (p', q')$ and $ps < rq$; we must show $p's < rq'$.
>
> ...

Name the move, then complete it.

Solution. The move is **check on representatives, one argument at a time**: change the first pair with the second fixed, then the second with the first fixed, and join the two by transitivity of $\iff$.

First argument. Let $pq' = p'q$ and $ps < rq$. Multiply by $q' > 0$ (Proposition 2.4.3 (c)): $psq' < rqq'$. The left side is $(pq')s = (p'q)s = (p's)q$, and the right side is $(rq')q$. So $(p's)q < (rq')q$. If $p's \geq rq'$, then multiplying by $q > 0$ would give $(p's)q \geq (rq')q$ (Proposition 2.4.3 (c), or equality), against (O1). So $p's < rq'$. Since $\sim$ is symmetric, the same argument with the roles of (p, q) and (p', q') exchanged gives the converse: $ps < rq \iff p's < rq'$.

Second argument. Let $rs' = r's$, and fix (p', q'). If $p's < rq'$, multiply by $s' > 0$: $p'ss' < rq's'$, and $rs' = r's$ turns the right side into $(r's)q' = (r'q')s$. So $(p's')s < (r'q')s$, and canceling $s > 0$ as above gives $p's' < r'q'$. By symmetry again, $p's < rq' \iff p's' < r'q'$.

Join. Chaining the two equivalences,

$$ps < rq \iff p's < rq' \iff p's' < r'q'.$$

This proves the claim. By Theorem 3.2.1 (a) both sides also mean $p/q < r/s$; the point of the exercise is that the integer relation alone is already independent of the representatives, and positivity of q, q', s, s' is used at every multiplication and cancellation.

Marking scheme. (10 points) - 2: names the move (check on representatives, one argument at a time, joined by transitivity of $\iff$). - 3: first argument: multiplies $ps < rq$ by $q' > 0$, uses $pq' = p'q$ to reach $(p's)q < (rq')q$, cancels $q > 0$ (by contradiction via trichotomy), and gets the converse from symmetry of $\sim$. - 3: second argument: the same with $s' > 0$, $rs' = r's$ and cancellation of $s > 0$, with the converse by symmetry. - 2: chains $ps < rq \iff p's < rq' \iff p's' < r'q'$ and says where positivity of q, q', s, s' is used. Fatal errors (0 for the step): multiplying or canceling by a denominator without using its sign; proving only one direction of an equivalence without invoking symmetry; using $p/q < r/s$ (the comparison formula of Theorem 3.2.1 (a)) instead of working on the integer relation.

Exercise. C3: Find the error

Here is an argument for Theorem 3.2.3. Is the conclusion true? Is the argument correct? If not, name the first line that fails, say what makes it illegal, and give a counterexample to the step.

> (1) Let $x \in \mathbb{Q}$, and write $x = p/q$ with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$. (2) Then $q \geq 1$, by discreteness of $\mathbb{Z}$. (3) So $x = p/q \leq p$, since dividing by a number at least 1 cannot increase it. (4) Hence $n := p + 1$ is a natural number with $n > x$.

Solution. The conclusion is **true**: it is Theorem 3.2.3. The argument is **not** correct.

The failing line is (3). It cites the general "fact" that $p/q \leq p$ whenever $q \geq 1$. That is neither proved in this chapter nor true, so the line rests on nothing: entry (N9) of Logic and Proof, Example (What is not allowed). What is true, and what the proof of the theorem uses, is the special case $p \geq 0$.

Counterexample to the step. $p = -3$, $q = 2$: then $q \geq 1$, but $x = -3/2 > -3 = p$. Indeed the comparison formula gives $-3/1 < -3/2$, because $(-3) \cdot 2 = -6 < -3 = (-3) \cdot 1$. There

is no counterexample to the conclusion.

A second failing line. Line (4) also fails for $p \leq -2$: then $p+1 \leq -1$, which is not in $\mathbb{N} = \{n \in \mathbb{Z} : n \geq 0\}$ (Theorem 2.4.2 (b)). For $p = -1$, $n = 0$ happens to work, because $x < 0$; but the stated reason "$x \leq p$" still fails there once $q \geq 2$.

Repair. Split on the sign of p, as the proof of Theorem 3.2.3 does: for $p \leq 0$, $x \leq 0 < 1$; for $p > 0$, line (3) is correct, with the comparison formula as its reason ($p \cdot 1 \leq p \cdot q$).

Marking scheme. (10 points) - 2: says that the conclusion is true and the argument is not correct. - 3: names line (3), and the unproved general claim "$p/q \leq p$ for $q \geq 1$" (entry (N9)) as what makes it illegal. - 2: gives a counterexample to the step with $p < 0$ and $q \geq 2$, checked by the comparison formula. - 1: notes that line (4) fails as well for $p \leq -2$. - 2: repairs the argument by a case split on the sign of p. Fatal errors (0 for the step): declaring the conclusion false; offering a counterexample to the conclusion.

Exercise. C4: Division from the integer part

Let $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$. Prove the existence half of Theorem 2.5.2 (a) directly from Proposition 3.2.5, without the least-element argument of the proof in Section 2.5: put $q := \lfloor a/n \rfloor$ and $r := a - qn$, and show that $0 \leq r < n$. Say where $n > 0$ is used.

Solution. By Proposition 3.2.5, $q := \lfloor a/n \rfloor \in \mathbb{Z}$ satisfies

$$q \leq a/n < q+1$$

in $\mathbb{Q}$, where $a/n = a \cdot n^{-1}$ and $n \neq 0$ in $\mathbb{Q}$. Since $n > 0$, multiplying by n preserves the strict inequality (Proposition 2.4.3 (c)) and the equality case, so

$$qn \leq (a \cdot n^{-1}) \cdot n = a < (q+1)n = qn + n.$$

Subtracting qn (that is, adding $-qn$, (O3)) gives $0 \leq a - qn < n$. Here $r := a - qn$ is an integer, and by Theorem 3.2.1 (b) and Proposition 3.1.4 (j preserves sums and products, so $j(a) - j(q)j(n) = j(a - qn)$) the inequalities $0 \leq r < n$, proved in $\mathbb{Q}$, hold in $\mathbb{Z}$. So $a = qn + r$ with $0 \leq r < n$.

The hypothesis $n > 0$ is used when multiplying by n: for $n < 0$, multiplication reverses inequalities (Proposition 2.4.3 (d)), and the same q would give $qn \geq a > qn + n$, so $n < r \leq 0$ instead. (This exercise is practice with a result already proved; nothing later rests on it.)

Marking scheme. (10 points) - 2: states $q \leq a/n < q+1$ from Proposition 3.2.5. - 3: multiplies by $n > 0$, treating both the strict inequality and the equality case. - 2: subtracts qn to get $0 \leq a - qn < n$. - 1: moves the inequalities back to $\mathbb{Z}$ for the integer $r = a - qn$. - 2: says where $n > 0$ is used, with the reversed inequalities when $n < 0$. Fatal errors (0 for the step): using the existence half of Theorem 2.5.2 or its least-element proof (circular here); multiplying the inequalities by n without using $n > 0$.

3.3 The Gaps in the Rationals

The rationals now have everything Section 3.2 asked of them: they form an ordered field (Theorem 3.2.1), every rational lies below a natural number (Theorem 3.2.3), and no two rationals are neighbors, since their midpoint lies between them. Is anything missing? School geometry suggests an answer. The diagonal of a square with side 1 has a length whose square is 2, and no fraction seems to measure it. That picture is motivation only. This section proves the two facts behind it from the construction: no rational squares to 2, and the set of rationals whose square is below 2 has no least upper bound in $\mathbb{Q}$. We also build a sequence of rationals that closes in on the gap from above without ever landing in it. A later chapter of this volume fills the gap.

3.3.1 No rational squares to two

Throughout, $x^2 := x \cdot x$, and $2 := 1 + 1$, $4 = 2 \cdot 2$ and so on are the natural numbers of these names, read in $\mathbb{Q}$ through the identifications of Sections 2.2 and 3.1; their sums, products and comparisons may be computed in $\mathbb{N}$ (Proposition 2.2.3, Proposition 3.1.4, Theorem 2.4.2 (a), Theorem 3.2.1 (b)).

Logic and Proof, Chapter 2 proved by parity that $\sqrt{2}$ is irrational (Logic and Proof, Theorem 2.2.1), for the school rationals. Parity is available for the constructed $\mathbb{Z}$ too (Corollary 2.5.3), and Exercise C1 allows it. Here we take a route that needs only the order and the least elements of $\mathbb{Z}$, and that adapts to other non-squares (Exercise C2).

Theorem 3.3.1. No Rational Squares to Two

There is no $x \in \mathbb{Q}$ with $x^2 = 2$.

Idea. Work backwards from a supposed square root $x = p/q$. This is the smallest-counterexample method (Sets and Functions, Example (Method: the smallest counterexample)) run on denominators, as in Sets and Functions, Lemma 3.2.4; here no lowest terms are needed, only a least denominator. From $x^2 = 2$ we get $2 - x = x^2 - x = x(x - 1)$, so

$$x = \frac{2 - x}{x - 1} = \frac{2 - p/q}{p/q - 1} = \frac{2q - p}{p - q}.$$

Since $1 < x < 2$ (as $1^2 < 2 < 2^2$), the new denominator $p - q$ is positive and smaller than q. So no fraction for x can have a **least** positive denominator, while Proposition 2.4.4 says some fraction must. The written proof checks the descent with integers only: it never divides, and it verifies $(2q - p)^2 = 2(p - q)^2$ by expanding. *Before reading on: using only integers, why is the new denominator $q' := p - q$ positive, and why is $q' < q$?*

Proof. Suppose, for a contradiction, that $x \in \mathbb{Q}$ and $x^2 = 2$. Then $x \neq 0$, since $0^2 = 0$ (Proposition 2.3.2 (b)) and $2 > 0$. By (O1), $x > 0$ or $x < 0$; let $y := x$ in the first case and $y := -x$ in the second. Then $y > 0$ (Proposition 2.4.3 (a)) and $y^2 = x^2 = 2$, because $(-x)(-x) = xx$ (Proposition 2.3.2 (c)).

By Proposition 3.1.4 (b), $y = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$, and $a > 0$ by Theorem 3.2.1 (a). By Definition 3.1.2, $y^2 = a^2/b^2$, and $2 = 2/1$, so $y^2 = 2$ says $a^2 \cdot 1 = 2b^2$ (Proposition 3.1.4

(b)). Hence the set

$$D := \{q \in \mathbb{Z}_{>0} : p^2 = 2q^2 \text{ for some } p \in \mathbb{Z}_{>0}\}$$

contains b, so it is non-empty, and 0 is a lower bound of it. By Proposition 2.4.4, D has a least element q. Choose $p \in \mathbb{Z}_{>0}$ with $p^2 = 2q^2$.

$q < p$. Suppose $p \leq q$. Since $p, q \geq 0$, Proposition 2.4.3 (g) gives $p^2 < q^2$ if $p < q$, and $p^2 = q^2$ if $p = q$; so $p^2 \leq q^2$, that is, $q^2 + q^2 \leq q^2$, and adding $-q^2$ (O3) gives $q^2 \leq 0$. But $q \neq 0$, so $q^2 > 0$ by Proposition 2.4.3 (e). This contradicts (O1), so $p \leq q$ fails, and $q < p$ by totality (Proposition 2.4.3 (i)).

$p < 2q$. Suppose $2q \leq p$. Since $2q, p \geq 0$, (g) (or equality, if $2q = p$) gives $(2q)^2 \leq p^2$, that is, $2q^2 + 2q^2 = 4q^2 \leq 2q^2$, so $2q^2 \leq 0$. But $2q^2 > 0$ by (O4). As before, $p < 2q$.

Descent. Let $p' := 2q - p$ and $q' := p - q$, integers. By Proposition 2.4.3 (a), $p < 2q$ gives $p' > 0$, and $q < p$ gives $q' > 0$; so $p', q' \in \mathbb{Z}_{>0}$ (Theorem 2.4.2 (b)). Adding $-q$ to $p < 2q$ (O3) gives $q' = p - q < q$. Expanding with Proposition 2.3.2 (c) and (d),

$$\begin{aligned} p'^2 - 2q'^2 &= (4q^2 - 4pq + p^2) - (2p^2 - 4pq + 2q^2) \\ &= 2q^2 - p^2 = 0. \end{aligned}$$

So $p'^2 = 2q'^2$ with $p' \in \mathbb{Z}_{>0}$, that is, $q' \in D$. But $q' < q$, and q is the least element of D. This contradiction shows that no $x \in \mathbb{Q}$ has $x^2 = 2$.

■

The answer to the prompt: $q' = p - q > 0$ because $q < p$, and $q' < q$ because $p < 2q$. These are the two inequalities "$1 < x < 2$" of the Idea, written for integers. The hypothesis $p^2 = 2q^2$ is used in each of the three steps, and the minimality of q only once, in the final contradiction.

Quick check. Run the descent on $p = 2$, $q = 1$, a pair with $p^2 = 4q^2$. Which step fails, and why must it?

Solution. The step $p < 2q$ fails: $2q \leq p$ now gives $4q^2 \leq p^2 = 4q^2$, no contradiction, and indeed $p = 2q$. Then $p' = 2q - p = 0$ is not positive and $q' = p - q = 1$ is not below q (and $p'^2 - 2q'^2 = -2$). A descent must fail here, since $2^2 = 4$.

3.3.2 The set below the gap

Sets and Functions, Section 5.3 stated, without proof, that the set of positive rationals with square below 2 has no least upper bound in $\mathbb{Q}$, and pointed out that the ambient set decides whether a supremum exists. We now prove it. By Proposition 2.4.3 (i) and Theorem 3.2.1, $\leq$ is a total order on $\mathbb{Q}$, so $(\mathbb{Q}, \leq)$ is a partially ordered set and the words of *Sets and Functions*, Section 5.3 apply: upper bounds (Sets and Functions, Definition 5.3.2) and a supremum, an upper bound s with (S1) $x \leq s$ for every x in the set and (S2) $s \leq u$ for every upper bound u (Sets and Functions, Definition 5.3.5).

Theorem 3.3.2. A Bounded Set of Rationals with No Rational Supremum

Let

$$A := \{x \in \mathbb{Q} : x^2 < 2\}, \qquad A^+ := \{x \in A : x > 0\} = \{x \in \mathbb{Q} : x > 0,\ x^2 < 2\}.$$

(a) A is non-empty and bounded above in $(\mathbb{Q}, \leq)$, but A has no supremum in $\mathbb{Q}$.

(b) A and A^+ have the same upper bounds in $\mathbb{Q}$. Hence A^+ is also non-empty and bounded above, and has no supremum in $\mathbb{Q}$.

Try it first (ten minutes): a rational $s > 0$ is offered as $\sup A$. Find a rational that proves the offer wrong, once in the case $s^2 < 2$ and once in the case $s^2 > 2$.

Idea (Scratch work). Suppose a rational s is the supremum. Unwind what that gives and what would refute it.

Givens	Goal
s is a supremum of A: (S1) and (S2)	a contradiction
$1 \in A$, so $s \geq 1 > 0$; $s^2 \neq 2$ by Theorem 3.3.1	split by trichotomy: $s^2 < 2$ or $s^2 > 2$
$s^2 < 2$	an element of A above s, against (S1)
$s^2 > 2$	an upper bound of A below s, against (S2)

If $s^2 < 2$, we need an element of A above s. The obvious try is $s + h$ with $h > 0$ small, but $(s+h)^2 = s^2 + 2sh + h^2$ has an h^2 in it, and taming that needs a separate size estimate on h. If $s^2 > 2$, we need a smaller upper bound, and a guess $s - h$ runs into the same algebra.

A cleaner idea finds the new rational in one formula. If $s^2 > 2$, then $2/s < s$ and $(2/s)^2 = 4/s^2 < 2$: the gap lies between $2/s$ and s. Their average

$$t(s) := \frac{1}{2}\left(s + \frac{2}{s}\right) = \frac{s^2 + 2}{2s}$$

is smaller than s, and its square is still above 2; that is the smaller upper bound. If $s^2 < 2$, apply the same trick from the other side: $t(s)$ overshoots, so $u(s) := 2/t(s) = 4s/(s^2+2)$ undershoots, and it lands above s. Each claim is an identity in the field $\mathbb{Q}$, checked by clearing denominators.

Proof.

(a) *Non-empty:* $1^2 = 1 < 2$, so $1 \in A$. *Bounded above:* 2 is an upper bound. Indeed, let $x \in A$, and suppose $x > 2$. Then $0 \leq 2 < x$, so $4 = 2^2 < x^2$ by Proposition 2.4.3 (g), and $2 < 4$, so $2 < x^2$, against $x^2 < 2$ by (O1). So $x \leq 2$, by totality.
Two maps. For $s \in \mathbb{Q}_{>0}$, we have $2s > 0$ by (O4), and $s^2 + 2 > 0$ by Proposition 2.4.3 (e) and (b); both are non-zero. Let

$$t(s) := \frac{s^2 + 2}{2s}, \qquad u(s) := \frac{4s}{s^2 + 2}.$$

Both are rationals, and both are positive, by Proposition 3.2.2 (a) and (O4). They satisfy

$$\begin{aligned} t(s)^2 - 2 &= \frac{(s^2-2)^2}{4s^2}, & s - t(s) &= \frac{s^2-2}{2s}, \\ u(s) - s &= \frac{s(2-s^2)}{s^2+2}, & 2 - u(s)^2 &= \frac{2(s^2-2)^2}{(s^2+2)^2}. \end{aligned} \tag{3.3.1}$$

Each identity is checked by clearing its non-zero denominator.

Second reading · *Checking the four identities*

Multiply both sides of each identity by its denominator, which is non-zero, so that the multiplication can be undone with the inverse. For the first, $2s \cdot t(s) = s^2 + 2$, so $4s^2(t(s)^2-2) = (s^2+2)^2 - 8s^2 = s^4 - 4s^2 + 4 = (s^2-2)^2$. For the second, $2s(s-t(s)) = 2s^2 - (s^2+2) = s^2 - 2$. For the third, $(s^2+2)(u(s)-s) = 4s - s^3 - 2s = s(2-s^2)$. For the fourth, $(s^2+2)^2(2-u(s)^2) = 2(s^2+2)^2 - 16s^2 = 2(s^2-2)^2$.

No supremum. Suppose, for a contradiction, that $s \in \mathbb{Q}$ is a supremum of A. By (S1) and $1 \in A$, $s \geq 1 > 0$, so $s \in \mathbb{Q}_{>0}$. By Theorem 3.3.1, $s^2 \neq 2$, so by (O1), $s^2 < 2$ or $s^2 > 2$. In both cases $s^2 - 2 \neq 0$, so $(s^2-2)^2 > 0$ by Proposition 2.4.3 (e).
Case 1: $s^2 < 2$. Then $2 - s^2 > 0$, and the third identity of Eq. 3.3.1 exhibits $u(s) - s$ as a product of positive elements (with Proposition 3.2.2 (a) for the reciprocal), so $u(s) > s$. The fourth gives $2 - u(s)^2 > 0$, so $u(s)^2 < 2$ and $u(s) \in A$. So A has an element $u(s) > s$, and s is not an upper bound of A, against (S1).
Case 2: $s^2 > 2$. The first identity gives $t(s)^2 - 2 > 0$, so $t(s)^2 > 2$. Then $t(s)$ is an upper bound of A: if some $x \in A$ had $x > t(s)$, then $0 < t(s) < x$ would give $2 < t(s)^2 < x^2$ by Proposition 2.4.3 (g), against $x^2 < 2$; so $x \leq t(s)$ for every $x \in A$. By (S2), $s \leq t(s)$. But the second identity, with $s^2 - 2 > 0$ and $2s > 0$, gives $s - t(s) > 0$, that is, $t(s) < s$. Then $t(s) < s \leq t(s)$, so $t(s) < t(s)$, against (O1).
Both cases are impossible, so A has no supremum in $\mathbb{Q}$.

(b) Every upper bound of A bounds $A^+ \subseteq A$. Conversely, if u bounds A^+, then $u \geq 1 > 0$, since $1 \in A^+$; and an $x \in A$ is either positive, so $x \in A^+$ and $x \leq u$, or $x \leq 0 < u$. So A and A^+ have the same upper bounds. Conditions (S1) and (S2) mention only the set of upper bounds, so by (a), A^+ has no supremum. It contains 1 and is bounded above by 2.

■

Quick check. Compute $u(1/2)$ and $t(3)$, and say what each refutes.

Solution. $u(1/2) = 2/(9/4) = 8/9$, and $(8/9)^2 = 64/81 < 2$, so $8/9 \in A$ lies above $1/2$: $1/2$ is not an upper bound. $t(3) = 11/6$, and $(11/6)^2 = 121/36 > 2$ ($121 > 72$), so $11/6$ is an upper bound below 3: 3 is not the least.

The set A^+ is the set of *Sets and Functions*, Section 5.3, so (b) is the statement left unproved there. *Analysis on the Real Line*, Chapter 1 proves it again and then shows that A^+ has a supremum in the real numbers: that is the sense of "gap".

Warning. "A has no supremum in $\mathbb{Q}$, so A is unbounded" mixes up the two conditions a supremum needs. A is bounded above, by 2 for instance; what fails is (S2): among the rational upper bounds there is no least one, because every upper bound s is beaten by the smaller upper bound $t(s)$.

Quick check. In $\mathbb{Q}$, is $3/2$ the supremum of $\{x \in \mathbb{Q} : x^2 < 9/4\}$? Which step of the proof of (a) fails for this set?

Solution. Yes. (S1): $x > 3/2 > 0$ gives $x^2 > 9/4$ (Proposition 2.4.3 (g)), so every element of the set is at most $3/2$. (S2): if $v < 3/2$ bounded the set, let w be the larger of v and 1, and c the midpoint of w and $3/2$ (Section 3.2); then $0 < c < 3/2$, so $c^2 < 9/4$ and c is in the set, and $c > w \geq v$. The step that fails is "$s^2 \neq 2$": here $s^2 = 9/4$ is the target value, the third case of the trichotomy, which neither t nor u refutes.

3.3.3 A rational sequence aimed at the gap

The map t does more than refute a candidate. Started at an upper bound, it produces a smaller one, and it can be applied again. Iterating it gives a sequence of rationals that approaches the gap from above.

Example. A Rational Sequence Aimed at the Gap

For $s \in \mathbb{Q}_{>0}$ let $t(s) := (s^2 + 2)/(2s)$. Define $x_0 := 2$ and $x_{n+1} := t(x_n)$ for $n \in \mathbb{N}$, and let $e_n := x_n^2 - 2$. Show that this defines a sequence with $x_n \in \mathbb{Q}_{>0}$ for every $n \in \mathbb{N}$, and that for every $n \in \mathbb{N}$:

(i) $e_n > 0$, that is, $x_n^2 > 2$;
(ii) $e_{n+1} = e_n^2/(4x_n^2) \leq e_n^2/8$;
(iii) $x_n - x_{n+1} = e_n/(2x_n) > 0$, so $x_{n+1} < x_n$.

Compute $x_0, \ldots, x_3$ and $e_0, \ldots, e_3$.

Solution. The sequence. For $s \in \mathbb{Q}_{>0}$, $s^2 + 2 > 0$ and $2s > 0$, so $t(s) \in \mathbb{Q}_{>0}$ by (O4) and Proposition 3.2.2 (a): t maps $\mathbb{Q}_{>0}$ to $\mathbb{Q}_{>0}$. The recursion theorem (Sets and Functions, Theorem 3.3.1, which holds on $\mathbb{N}$ by Corollary 1.4.3 (b)), with $X = \mathbb{Q}_{>0}$, starting value 2 and $g(n, s) := t(s)$, gives exactly one function $n \mapsto x_n$ from $\mathbb{N}$ to $\mathbb{Q}_{>0}$ with $x_0 = 2$ and $x_{n+1} = t(x_n)$. So every x_n is a positive rational, with no further argument.

(i). We use Logic and Proof, Axiom 4.2.1 (on $\mathbb{N}$ by Corollary 1.4.3 (b)) for $P(n)$: $e_n > 0$. Base $P(0)$: $e_0 = 2^2 - 2 = 2 > 0$. Step: suppose $e_n > 0$. By the first identity of Eq. 3.3.1 with $s = x_n$,

$$e_{n+1} = t(x_n)^2 - 2 = \frac{e_n^2}{4x_n^2}.$$

Here $e_n^2 > 0$ by Proposition 2.4.3 (e), since $e_n \neq 0$, and $1/(4x_n^2) > 0$ by Proposition 3.2.2 (a); so $e_{n+1} > 0$ by (O4). This proves (i) for every n.

(ii). The equation was just shown, for every n. By (i), $x_n^2 > 2$, so $4x_n^2 > 4 \cdot 2 = 8$ by Proposition 2.4.3 (c). From $0 < 8 < 4x_n^2$, Proposition 3.2.2 (b) gives $1/(4x_n^2) < 1/8$, and multiplying by $e_n^2 > 0$ (Proposition 2.4.3 (c)) gives $e_{n+1} < e_n^2/8$; in particular $e_{n+1} \leq e_n^2/8$.

(iii). By the second identity of Eq. 3.3.1 with $s = x_n$, $x_n - x_{n+1} = e_n/(2x_n)$, a product of the positive elements e_n and $1/(2x_n)$; so it is positive, and $x_{n+1} < x_n$ by Proposition 2.4.3 (a).

The first terms. By the definition of t,

$$x_1 = \frac{4+2}{4} = \frac{3}{2}, \quad x_2 = \frac{9/4+2}{3} = \frac{17}{12}, \quad x_3 = \frac{289/144+2}{17/6} = \frac{577}{408}.$$

The errors are $e_0 = 2$, $e_1 = 9/4 - 2 = 1/4$, $e_2 = 289/144 - 2 = 1/144$ and $e_3 = 577^2/408^2 - 2 = 1/166464$, since $577^2 = 332929 = 2 \cdot 166464 + 1$ and $408^2 = 166464$. As a check on (ii): $e_1 = 2^2/(4 \cdot 4) = 1/4$, $e_2 = (1/16)/9 = 1/144$, and $e_3 = (1/144)^2/(4 \cdot 289/144) = 1/(144 \cdot 1156) = 1/166464$.

For orientation only (decimals are not part of this volume's toolkit), here are the terms in school decimals:

n	x_n	$e_n = x_n^2 - 2$	x_n, roughly
0	2	2	2
1	3/2	1/4	1.5
2	17/12	1/144	1.41667
3	577/408	1/166464	1.414216

The terms decrease, each is an upper bound of A (Case 2 of the proof above), and by (ii) the error is roughly squared at each step: $e_3 = 1/166464$. Yet no rational L can be "the number they approach": the squares would approach L^2, the errors force $L^2 = 2$, and Theorem 3.3.1 forbids that. This is a picture, not a proof, since limits in $\mathbb{Q}$ are not defined yet. Chapter 4 defines them, with $\varepsilon \in \mathbb{Q}_{>0}$, and proves that this sequence is Cauchy in $\mathbb{Q}$ with no limit in $\mathbb{Q}$, citing (i)–(iii).

Quick check. Is $3/2$ an upper bound of $A = \{x \in \mathbb{Q} : x^2 < 2\}$? Is $17/12$? Is $4/3$?

Solution. The first two are. Each is positive, and $(3/2)^2 = 9/4 > 2$ because $9 > 8$, while $(17/12)^2 = 289/144 > 2$ because $289 > 288$ (the comparison formula, Theorem 3.2.1 (a)). A positive rational b with $b^2 > 2$ bounds A: if $x \in A$ had $x > b$, then $x^2 > b^2 > 2$ by Proposition 2.4.3 (g), against $x^2 < 2$.

But $4/3$ is not, although it is positive: $(4/3)^2 = 16/9 < 2$ because $16 < 18$, so the hypothesis $b^2 > 2$ fails, and indeed $7/5 \in A$, as $(7/5)^2 = 49/25 < 2$ ($49 < 50$), while $7/5 > 4/3$ ($21 > 20$).

Chapters 2 and 3 built two number systems by one method. From $\mathbb{N}$ to $\mathbb{Z}$: pairs (a, b), identified when $a + d = b + c$, so that subtraction becomes possible. From $\mathbb{Z}$ to $\mathbb{Q}$: pairs (p, q) with $q > 0$,

identified when $ps = rq$, so that division becomes possible. Chapter 5 takes the third step, from $\mathbb{Q}$ to the real numbers, by the same route: the representatives are sequences of rationals like the one above, identified when their difference tends to 0.

3.3.4 Exercises

3.3.4.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Theorem 3.3.2.

(a) Where is Theorem 3.3.1 used, and what would go wrong without it?

(b) Run the proof on $s = 3/2$ and on $s = 7/5$: which case applies, which rational does the proof produce, and why does it refute s?

(c) Let F be an ordered field containing an element $r > 0$ with $r^2 = 2$, and let $A_F := \{x \in F : x^2 < 2\}$. Which step of the proof fails for A_F, and what happens to $t(r)$ and $u(r)$?

(d) State the idea of the proof in two sentences.

Solution.

(a) Once, right after a supposed supremum s is fixed: it excludes $s^2 = 2$, so that trichotomy leaves the two cases $s^2 < 2$ and $s^2 > 2$. Without it there is a third case, in which neither map helps: $(s^2 - 2)^2 = 0$, so the identities give $u(s) = s$ and $t(s) = s$.

(b) For $s = 3/2$: $s^2 = 9/4 > 2$ (as $9 > 8$), so Case 2. The proof produces $t(3/2) = (9/4 + 2)/3 = 17/12$, an upper bound of A, since $(17/12)^2 = 289/144 > 2$; and $17/12 < 3/2$, since $17 \cdot 2 = 34 < 36 = 3 \cdot 12$. So $3/2$ is not the **least** upper bound. For $s = 7/5$: $s^2 = 49/25 < 2$ (as $49 < 50$), so Case 1. The proof produces

$$u(7/5) = \frac{28/5}{49/25 + 2} = \frac{28/5}{99/25} = \frac{140}{99},$$

which lies in A, since $140^2 = 19600 < 19602 = 2 \cdot 99^2$, and exceeds $7/5$, since $140 \cdot 5 = 700 > 693 = 7 \cdot 99$. So $7/5$ is not an upper bound.

(c) The step "$s^2 \neq 2$": in F the candidate $s = r$ satisfies $r^2 = 2$, and it falls in neither case. Indeed r is the supremum of A_F in F: it is an upper bound (if $x > r > 0$, then $x^2 > 2$), and an upper bound $v < r$ would have $v > 0$ (as $1 \in A_F$) and $v^2 < 2$, and then $u(v) \in A_F$ with $u(v) > v$, as in Case 1. The maps fix r: $t(r) = (2+2)/(2r) = 2/r = r$, since $r \cdot r = 2$, and $u(r) = 4r/4 = r$.

(d) Suppose a rational s were the supremum; since no rational squares to 2, either $s^2 < 2$ or $s^2 > 2$. The explicit rationals $u(s)$ and $t(s)$ then give, respectively, an element of the set above s or an upper bound below s, so s fails (S1) or (S2).

3.3.4.2 B. Practice

Exercise. B1: Put the proof in order

Here are lines of a proof of Theorem 3.3.1, shuffled, together with two lines that do not belong. Write the lines of a correct proof in a correct order, and say what is wrong with each

line you leave out.

(i) Set $p' := 2q - p$ and $q' := p - q$; both are positive integers, by the two inequalities between p and q.

(ii) Write $x = a/b$ with $a, b \in \mathbb{Z}_{>0}$; then $a^2 = 2b^2$.

(iii) Since $p^2 = 2q^2$, we have $p < q$.

(iv) Since $q' = p - q < q$, this contradicts the choice of q; so no rational squares to 2.

(v) Suppose $x \in \mathbb{Q}$ has $x^2 = 2$; replacing x by $-x$ if necessary, take $x > 0$.

(vi) $p < 2q$, since $p \geq 2q$ would give $p^2 \geq 4q^2 > 2q^2$.

(vii) Choose such an x with the least positive value.

(viii) Among all pairs (p, q) of positive integers with $p^2 = 2q^2$, choose one with q least (Proposition 2.4.4; b is such a q).

(ix) Expanding, $p'^2 - 2q'^2 = 2q^2 - p^2 = 0$, so $p'^2 = 2q'^2$.

(x) $q < p$, since $p \leq q$ would give $2q^2 = p^2 \leq q^2$, that is, $q^2 \leq 0$.

Solution. An admissible order is

$$\text{(v)},\ \text{(ii)},\ \text{(viii)},\ \text{(x)},\ \text{(vi)},\ \text{(i)},\ \text{(ix)},\ \text{(iv)}.$$

Lines (x) and (vi) may be exchanged, since neither uses the other. Otherwise the order is forced by what each line uses: (ii) needs the $x > 0$ of (v); (viii) needs the pair (a, b) of (ii), which makes the set of such q non-empty; (x) and (vi) are about the pair fixed in (viii), and if they came before it, the pair might change afterwards; (i) needs both inequalities; (ix) needs the definitions in (i); and (iv) needs $q' \in \mathbb{Z}_{>0}$ with $p'^2 = 2q'^2$, from (i) and (ix), and $q' < q$, which comes from (vi).

Line (vii) is legal but does no work. There is only one positive rational x with $x^2 = 2$: if $0 < x < y$ had $x^2 = y^2 = 2$, then Proposition 2.4.3 (g) would give $2 = x^2 < y^2 = 2$, against (O1). So "the least" just picks out the x of (v) and gives nothing new. The descent does not produce a smaller x: it keeps the number and shrinks its denominator, $p'/q' = p/q$, so choosing x least never leads to a contradiction. What must be least is the denominator q, as in (viii).

Line (iii) fails. Its conclusion is false: from $p^2 = 2q^2$ and $q > 0$ we get $p^2 > q^2$, hence $q < p$ by Proposition 2.4.3 (g), the opposite of (iii). It also clashes with (x).

Exercise. B2: Prove or disprove

Prove or disprove each statement.

(a) There is no $x \in \mathbb{Q}$ with $x^2 = 8$.

(b) For all $x, y \in \mathbb{Q}$, if $x^2 < y^2$, then $x < y$.

(c) If $s \in \mathbb{Q}$, $s > 0$ and $s^2 > 2$, then $(2/s)^2 < 2$.

(d) If $s \in \mathbb{Q}$, $s > 0$ and $s^2 < 2$, then $(s + 1/10)^2 < 2$.

Solution.

(a) **True.** If $x^2 = 8$, then $y := x/2$ has $y^2 = x^2/4 = 2$ (Definition 3.1.2), against Theorem 3.3.1.

(b) **False:** $x = 1$, $y = -2$ give $x^2 = 1 < 4 = y^2$, but $1 > -2$. (Rule (g) of Proposition 2.4.3 needs $0 \le x$ and $0 \le y$.)

(c) **True.** $(2/s)^2 = 4/s^2$. From $s^2 > 2 > 0$, Proposition 3.2.2 (b) gives $1/s^2 < 1/2$, and multiplying by $4 > 0$ (Proposition 2.4.3 (c)) gives $4/s^2 < 2$. (So the map $s \mapsto 2/s$ carries upper bounds of A^+ into A^+; the proof's $u(s)$ improves on it.)

(d) **False:** $s = 7/5$ has $s^2 = 49/25 < 2$ ($49 < 50$), but $s + 1/10 = 3/2$ and $(3/2)^2 = 9/4 > 2$ ($9 > 8$). A fixed step can overshoot; the proof's $u(s)$ is chosen from s.

Exercise. B3

Compute x_4 of Example (A Rational Sequence Aimed at the Gap), as a fraction of integers, and $e_4 = x_4^2 - 2$. Check your value of e_4 against (ii) of the example. Hence x_4 is an upper bound of $\{x \in \mathbb{Q} : x^2 < 2\}$ below x_3.

Solution. With $x_3 = 577/408$,

$$x_4 = t(x_3) = \frac{577^2 + 2 \cdot 408^2}{2 \cdot 577 \cdot 408} = \frac{332929 + 332928}{470832} = \frac{665857}{470832}.$$

Then $665857^2 = 443365544449$ and $470832^2 = 221682772224$, and $443365544449 - 2 \cdot 221682772224 = 1$, so

$$e_4 = \frac{1}{221682772224}.$$

By (ii), $e_4 = e_3^2/(4x_3^2) = 1/(166464 \cdot 4 \cdot 332929)$, and $4 \cdot 166464 \cdot 332929 = 221682772224$, which agrees. Since $x_4 > 0$ and $e_4 > 0$, x_4 bounds the set (the argument of the Quick check after the table), and $x_4 < x_3$ by (iii).

3.3.4.3 C. Going deeper

Exercise. C1: Exit check: the gap at $\sqrt{2}$

Prove that no rational number squares to 2, and that the set $\{x \in \mathbb{Q} : x^2 < 2\}$ has no supremum in $\mathbb{Q}$.

Solution. No rational squares to 2. Suppose $x \in \mathbb{Q}$ with $x^2 = 2$. Then $x \neq 0$, and replacing x by $-x$ if needed ($(-x)^2 = x^2$), $x > 0$. Write $x = a/b$ with $a, b \in \mathbb{Z}_{>0}$ (Proposition 3.1.4 (b), Theorem 3.2.1 (a)); then $a^2 = 2b^2$. Among all $q \in \mathbb{Z}_{>0}$ for which some $p \in \mathbb{Z}_{>0}$ has $p^2 = 2q^2$, take the least (Proposition 2.4.4; the set contains b and is bounded below by 0), with its p. Then $q < p$ (else $p^2 \le q^2$ by Proposition 2.4.3 (g), so $q^2 \le 0$, against $q^2 > 0$) and $p < 2q$ (else $4q^2 \le p^2 = 2q^2$, so $2q^2 \le 0$). So $p' := 2q - p$ and $q' := p - q$ are positive integers with $q' < q$, and $p'^2 - 2q'^2 = 2q^2 - p^2 = 0$ by expanding. This contradicts the choice of q.

Bounded above. 2 bounds the set A: $x > 2$ would give $x^2 > 4 > 2$ (Proposition 2.4.3 (g)). And $1 \in A$.

No supremum. Suppose $s \in \mathbb{Q}$ is a supremum of A. As $1 \in A$, $s \geq 1 > 0$, and $s^2 \neq 2$ by the first part.

If $s^2 < 2$, let $u := 4s/(s^2+2) \in \mathbb{Q}_{>0}$. By the third and fourth identities of Eq. 3.3.1, $u - s > 0$ and $2 - u^2 > 0$. So $u \in A$ and $u > s$: s is not an upper bound.

If $s^2 > 2$, let $t := (s^2+2)/(2s) \in \mathbb{Q}_{>0}$. By the first and second identities of Eq. 3.3.1, $t^2 - 2 > 0$ and $s - t > 0$. So $t^2 > 2$, and t bounds A (an $x \in A$ with $x > t > 0$ would have $x^2 > t^2 > 2$); and $t < s$. So s is not the least upper bound.

By trichotomy one of the two cases holds, and each contradicts the choice of s. So A has no supremum in $\mathbb{Q}$.

Marking scheme. (10 points) - 4: no rational squares to 2: a least denominator, or another correct route (1); $q < p < 2q$, or the parity step (2); the contradiction (1). - 1: shows $1 \in A$, so that a supremum s satisfies $s \geq 1 > 0$ (an upper bound of A is optional). - 2: the case $s^2 < 2$: an explicit element of A larger than s, with both properties verified. - 2: the case $s^2 > 2$: an explicit smaller rational, verified to be an upper bound of A. - 1: concludes by trichotomy, with $s^2 = 2$ excluded by the first part. Fatal errors (0 for the step): assuming that $\sqrt{2}$ exists (in $\mathbb{Q}$ or in a larger field) and comparing s with it; "$s^2 < 2$, so $s + \varepsilon \in A$ for small ε" with no ε exhibited and checked. A correct parity route (every integer is $2m$ or $2m+1$, and not both, Corollary 2.5.3 (a)) earns the same 4 points.

Exercise. C2: Which integers have a rational square root?

For which $n \in \mathbb{Z}_{>0}$ does the set $A_n := \{x \in \mathbb{Q} : x^2 < n\}$ have a supremum in $\mathbb{Q}$? Prove your answer.

Solution (Hint 1). First find $k \in \mathbb{N}$ with $k^2 \leq n < (k+1)^2$, and treat $n = k^2$ on its own. For $k^2 < n$, adapt both proofs of this section: the descent, and then the two maps.

Solution (Hint 2). Let $T := \{m \in \mathbb{Z} : m \geq 0,\ n < m^2\}$. It contains $n+1$, it is bounded below by 0, and $0 \notin T$; so its least element is $k+1$ for some $k \in \mathbb{N}$. In the descent for $k^2 < n$, try $q' := p - kq$.

Solution. **Answer:** exactly when $n = k^2$ for some $k \in \mathbb{N}$, and then the supremum is k.

Finding k. Let $T := \{m \in \mathbb{Z} : m \geq 0,\ n < m^2\}$. Since $n+1 \geq 1$, $(n+1)^2 \geq n+1 > n$ (Proposition 2.4.3 (c)), so $n+1 \in T$; 0 is a lower bound of T; and $0 \notin T$, as $0^2 = 0 < n$. By Proposition 2.4.4, T has a least element l. Since $0 \notin T$ and $l \geq 0$, we have $l > 0$, so $l \geq 1$ by discreteness (Theorem 2.4.2 (c)); then $l - 1 \geq 0$ by (O3), so $k := l - 1 \in \mathbb{N}$ by Theorem 2.4.2 (b). Then $n < (k+1)^2$, and $k \notin T$ with $k \geq 0$ gives $k^2 \leq n$. So either $n = k^2$ or $k^2 < n < (k+1)^2$.

Case $n = k^2$. Here $k \geq 1$, since $n \geq 1$. We show $k = \sup A_n$. (S1): if $x \in A_n$ had $x > k$, then $x > k \geq 0$ would give $x^2 > k^2 = n$ (Proposition 2.4.3 (g)), against $x^2 < n$; so $x \leq k$. (S2): let v be an upper bound of A_n, and suppose $v < k$. Let $w := v$ if $v \geq 0$ and $w := 0$

otherwise; then $v \leq w$ and $0 \leq w < k$. Let c be the midpoint of w and k (Section 3.2): $w < c < k$. Then $c > 0$ and $c^2 < k^2 = n$ by (g), so $c \in A_n$, and $c > w \geq v$, so v is not an upper bound. Hence $k \leq v$ for every upper bound v, and k is the supremum.

Case $k^2 < n < (k+1)^2$. Then $n \geq 2$, since $n = 1 = 1^2$ is the other case; so $1 \in A_n$.

(i) *No rational squares to* n. Suppose $x^2 = n$ with $x \in \mathbb{Q}$; as in Theorem 3.3.1, we may take $x > 0$, and then $p^2 = nq^2$ for some $p, q \in \mathbb{Z}_{>0}$, with q least among all such denominators (Proposition 2.4.4). Multiplying $k^2 < n < (k+1)^2$ by $q^2 > 0$ gives $(kq)^2 < p^2 < ((k+1)q)^2$, so $kq < p < (k+1)q$ by (g). Let

$$p' \coloneqq nq - kp, \qquad q' \coloneqq p - kq.$$

Then $0 < q' < q$. For p', expand:

$$q \cdot p' = nq^2 - kpq = p^2 - kpq = p(p - kq) > 0$$

by (O4); since $q > 0$, $p' \leq 0$ would give $qp' \leq 0$ (Proposition 2.4.3 (c)), so $p' > 0$. Finally, expanding,

$$p'^2 - nq'^2 = (k^2 - n)(p^2 - nq^2) = 0,$$

so q' is a smaller admissible denominator, a contradiction.

(ii) *No supremum.* The proof of Theorem 3.3.2 (a) goes through with n in place of 2, using (i) where it used Theorem 3.3.1, and A_n contains 1 and is bounded above by n. For $s \in \mathbb{Q}_{>0}$, let

$$t(s) \coloneqq \frac{s^2 + n}{2s}, \qquad u(s) \coloneqq \frac{2ns}{s^2 + n}.$$

Clearing denominators as before gives the four identities

$$t(s)^2 - n = \frac{(s^2 - n)^2}{4s^2}, \qquad s - t(s) = \frac{s^2 - n}{2s},$$
$$u(s) - s = \frac{s(n - s^2)}{s^2 + n}, \quad n - u(s)^2 = \frac{n(s^2 - n)^2}{(s^2 + n)^2}.$$

So A_n has a supremum in $\mathbb{Q}$ exactly when n is the square of a natural number.

Marking scheme. (10 points) - 2: produces $k \in \mathbb{N}$ with $k^2 \leq n < (k+1)^2$, from a least element. - 2: the case $n = k^2$: checks (S1) and (S2) for k. - 3: the case $k^2 < n$: no rational squares to n, by a descent with a least denominator (or another correct route), including a proof that the new numerator is positive. - 2: the two maps (or any explicit rationals) and the two cases, as in the proof for 2. - 1: states the answer: exactly the squares of natural numbers. Fatal errors (0 for the step): using a square root of n, or the floor of one, as if it existed; deducing $nq - kp > 0$ from $kq < p < (k+1)q$ alone, which fails for $n = 5$, $k = 2$, $q = 10$, $p = 28$.

3.3.4.4 R. Review

Exercise. R1

Recall that an integer is a class $[(a, b)]$ of pairs of natural numbers, with $(a, b) \sim (c, d)$ if and only if $a + d = b + c$. For $(a, b) \in \mathbb{N} \times \mathbb{N}$, let $f(a, b) \coloneqq (a, b)$ if $b \leq a$, and $f(a, b) \coloneqq (b, a)$ if

$a < b$. Does $[(a,b)] \mapsto [f(a,b)]$ define a function $\mathbb{Z} \to \mathbb{Z}$? Prove your answer, and say which integer it assigns to $x \geq 0$ and to $x < 0$.

Solution. **Yes.** By Theorem 2.2.1 (a), with $X = Y = \mathbb{N} \times \mathbb{N}$, it suffices to show that $(a,b) \sim (c,d)$ implies $f(a,b) \sim f(c,d)$ (the representative check, Sets and Functions, Example (Method: checking that a definition on classes is well defined)). Let $a + d = b + c$, and $x := [(a,b)] = [(c,d)]$.

The cases match. $x \in P_{\mathbb{Z}}$ if and only if $b < a$, whichever representative is used (the well-definedness paragraph after Definition 2.4.2), and $x = 0$ if and only if $a = b$, since $(a,b) \sim (0,0)$ means $a = b$. So $b \leq a \iff (x \in P_{\mathbb{Z}}$ or $x = 0) \iff d \leq c$.

Hence either $b \leq a$ and $d \leq c$, and then $f(a,b) = (a,b) \sim (c,d) = f(c,d)$; or $a < b$ and $c < d$, and then $f(a,b) = (b,a)$, $f(c,d) = (d,c)$, and $b + c = a + d$ says $(b,a) \sim (d,c)$.

Since $x \geq 0$ if and only if $b \leq a$, and $-[(a,b)] = [(b,a)]$ (Definition 2.2.1), the function sends x to x if $x \geq 0$ and to $-x$ if $x < 0$: it is the familiar $x \mapsto |x|$.

Exercise. R2

Show that the set $\mathbb{Q}$, as constructed in this chapter, is countable.

Solution. The tools are the results on countable sets of *Sets and Functions*, Chapter 4, which run on $\mathbb{N}$ by the citing rule after Corollary 1.4.3 (group (i) for Sets and Functions, Proposition 4.2.3 and Sets and Functions, Proposition 4.2.4; group (ii), citable in review exercises, for Sets and Functions, Theorem 4.2.6 and Sets and Functions, Corollary 4.2.7 (a)).

$\mathbb{Z}$ is countable. $\mathbb{N} \times \mathbb{N}$ is countable (Sets and Functions, Theorem 4.2.6). The map $a \mapsto [a]$, from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\!\sim$, is surjective, since every class is the class of each of its elements (Sets and Functions, Definition 5.2.4). By Sets and Functions, Proposition 4.2.4, $\mathbb{Z}$ is countable.

$\mathbb{Z} \times \mathbb{Z}_{>0}$ is countable. $\mathbb{Z}_{>0} \subseteq \mathbb{Z}$ is countable by Sets and Functions, Proposition 4.2.3, and the product of two countable sets is countable (Sets and Functions, Corollary 4.2.7 (a)).

$\mathbb{Q}$ is countable. The map $(p,q) \mapsto [(p,q)]$, from $\mathbb{Z} \times \mathbb{Z}_{>0}$ to $\mathbb{Q}$, is surjective (Sets and Functions, Definition 5.2.4), so $\mathbb{Q}$ is countable by Sets and Functions, Proposition 4.2.4.

The conclusion is Sets and Functions, Theorem 4.2.8 for the constructed $\mathbb{Q}$; that theorem, about the school rationals, is not used.

Exercise. R3

Let $A := \{x \in \mathbb{Q} : x^2 < 2\}$ and $s \in \mathbb{Q}$. Write "s is a supremum of A in $\mathbb{Q}$" in symbols, with bounded quantifiers. Then write its negation, moving $\neg$ inward as far as it goes and naming the law used at each step. The proof that A has no supremum in $\mathbb{Q}$ has two cases; which half of the negation does each case establish, and with which witness?

Solution. By Sets and Functions, Definition 5.3.5, "s is a supremum of A" is the conjunction of (S1) and (S2):

$$(\forall x \in A \;\; x \leq s) \;\wedge\; \forall v \in \mathbb{Q}\; ((\forall x \in A \;\; x \leq v) \to s \leq v).$$

Its negation is, by Logic and Proof, Proposition 2.5.5 (d),

$$\neg(\forall x \in A \;\; x \leq s) \;\vee\; \neg\forall v \in \mathbb{Q}\; ((\forall x \in A \;\; x \leq v) \to s \leq v).$$

By Logic and Proof, Proposition 6.2.3 (a), the first disjunct becomes $\exists x \in A\; \neg(x \leq s)$, that is, $\exists x \in A \;\; s < x$ (totality of $\leq$, Proposition 2.4.3 (i)). By Logic and Proof, Proposition 6.2.3 (a) and then Logic and Proof, Proposition 6.2.2, the second becomes

$$\exists v \in \mathbb{Q}\; ((\forall x \in A \;\; x \leq v) \wedge \neg(s \leq v)),$$

that is, there is an upper bound v of A with $v < s$. In words: s fails to be the supremum when **some element of A lies above s**, or **some upper bound lies below s**.

In the proof of Theorem 3.3.2, Case 1 ($s^2 < 2$) establishes the first disjunct, with witness $x = u(s) = 4s/(s^2+2)$. Case 2 ($s^2 > 2$) establishes the second, with witness $v = t(s) = (s^2+2)/(2s)$. (The proof itself needs no negation law: it derives a contradiction from (S1) in Case 1 and from (S2) in Case 2.)

Exercise. R4

Is there a rational number x with $x^3 = 2$, where $x^3 := x^2 \cdot x$? Prove your answer.

Solution. **No.** The tool is descent on the denominator, as in Theorem 3.3.1, run with parity.

Parity. Every integer is $2m$ or $2m+1$ for some $m \in \mathbb{Z}$, and not both (Corollary 2.5.3 (a)). An odd cube is odd:

$$(2m+1)^3 = 2(4m^3 + 6m^2 + 3m) + 1.$$

Descent. Suppose $x \in \mathbb{Q}$ and $x^3 = 2$. Then $x > 0$: $x = 0$ gives $x^3 = 0$, and $x < 0$ gives $x^2 > 0$ and $x^3 = x^2 \cdot x < 0$ (Proposition 2.4.3 (e), (c)). So $x = p/q$ with $p, q \in \mathbb{Z}_{>0}$ and $p^3 = 2q^3$ (Proposition 3.1.4 (b), Definition 3.1.2). Take q least among such denominators (Proposition 2.4.4), with its p. Since $p^3 = 2q^3$ has the even form, p is not odd (its cube would be odd), so $p = 2m$ with $m \in \mathbb{Z}$, and $m > 0$ because $p > 0$ ($m \leq 0$ would give $2m \leq 0$). Then $2q^3 = 8m^3$, and canceling the non-zero factor 2 (Proposition 2.4.3 (f)), $q^3 = 2(2m^3)$. So q is not odd either: $q = 2l$ with $l \in \mathbb{Z}_{>0}$. Then $8m^3 = 2 \cdot 8l^3$, so $m^3 = 2l^3$, and $l < l + l = q$. So l is a smaller admissible denominator, with $m \in \mathbb{Z}_{>0}$: a contradiction.

Why the look-alike fails. The substitution $p' := 2q - p$, $q' := p - q$ of the square-root proof came from $x = (2-x)/(x-1)$, which used $x^2 = 2$. For cubes the same algebra fails: expanding,

$$(2q-p)^3 - 2(p-q)^3 = -3p^3 + 12p^2q - 18pq^2 + 10q^3,$$

which is not a constant multiple of $p^3 - 2q^3$ (it has a p^2q term), so this identity gives no way to derive $p'^3 = 2q'^3$ from $p^3 = 2q^3$, and the descent has no smaller pair to move to. Parity, by contrast, works for every power: "even cube, so even base" is the same argument for squares and cubes.

Chapter 4

Ordered Fields and Sequences in an Ordered Field

In $\mathbb{Q}$ the numbers $2, 3/2, 17/12, 577/408, \ldots$ crowd together, and their squares crowd toward 2; yet Chapter 3 proved that no rational squares to 2 (Theorem 3.3.1). Chapter 5 will make the missing number out of sequences like this one. Before it can, we must say what "crowd together" means when the only yardsticks are rationals, and prove the rules for such sequences without presupposing the number they are aiming at. This chapter does that once, for an **arbitrary ordered field** F, with every tolerance ε taken from F itself. It adds to the inequality rules of Chapter 2 what limits need (halving, fractions, the absolute value and the triangle inequality), places a copy of $\mathbb{N}$ and of $\mathbb{Q}$ inside every ordered field, and names the Archimedean property, which $\mathbb{Q}$ has and an ordered field need not have. Then come limits (uniqueness, boundedness, the limit laws, limits and order) and Cauchy sequences (convergent implies Cauchy, Cauchy implies bounded, sums and products, and the sign of a Cauchy sequence that is not null). Chapter 5 applies all of it with $F = \mathbb{Q}$.

The chapter ends at the gap it was written for. The sequence above is Cauchy in $\mathbb{Q}$ and has no limit in $\mathbb{Q}$, so in $\mathbb{Q}$ "Cauchy" does not imply "convergent". If you have met limits in a course on $\mathbb{R}$, most statements here will look familiar. What is new is that they are proved in any ordered field, from the axioms alone. Watch where the Archimedean property is needed (each use names it), and note that completeness never appears: no supremum, no monotone convergence theorem and no Bolzano–Weierstrass theorem is used, as Section 4.6 checks.

What you need. Chapters 1–3, chiefly Sections 2.3, 2.4, 3.2 and 3.3; *Logic and Proof*, Chapters 2–8, chiefly Chapter 6; and *Sets and Functions*, Chapters 1–5. From Chapter 2, Sections 2.3 and 2.4: commutative rings and fields, ordered rings and fields and the rules for inequalities (Definition 2.3.1, Definition 2.4.1, Proposition 2.3.2, Proposition 2.4.3, Proposition 2.4.1), the integers as an ordered ring with least elements of sets bounded below (Theorem 2.3.1, Theorem 2.4.2, Proposition 2.4.4), and the two-element field that cannot be ordered (Example (A field with two elements that cannot be ordered)). From Sections 2.2, 3.1 and 3.2: the copies of $\mathbb{N}$ in $\mathbb{Z}$ and of $\mathbb{Z}$ in $\mathbb{Q}$ (Proposition 2.2.3, Proposition 3.1.4), $\mathbb{Q}$ as an ordered field with reciprocals reversing order (Theorem 3.2.1, Proposition 3.2.2), and the Archimedean property of $\mathbb{Q}$, its form $1/N < \varepsilon$, and the integer part (Theorem 3.2.3, Corollary 3.2.4, Proposition 3.2.5), whose method review exercise R1 of Section 4.6 reuses. From Section 3.3: no rational squares to 2, and the Newton sequence with the term-wise facts proved there (Theorem 3.3.1, Example (A Rational Sequence Aimed at the Gap)), which Sections 4.3, 4.5 and 4.6 use; and the set of rationals with square below 2, which has no rational supremum (Theorem 3.3.2), named in Section 4.6 only. From Chapter 1: the citing rule of Section 1.4 (Corollary 1.4.3), by which induction

(Logic and Proof, Axiom 4.2.1) and the recursion theorem (Sets and Functions, Theorem 3.3.1, Theorem 1.2.1) apply to $\mathbb{N}$; the arithmetic and order of $\mathbb{N}$ (Definition 1.2.1, Theorem 1.2.3, Theorem 1.3.2, Proposition 1.3.1); and the powers of Section 1.3 (Proposition 1.3.4), which Section 4.2 compares with powers in F and Exercise B2 of Section 4.6 uses as indices. From *Logic and Proof*, Chapter 6: the negation laws (Logic and Proof, Proposition 6.2.1, Logic and Proof, Proposition 6.2.3, Logic and Proof, Proposition 6.2.2), used to negate the definitions of limit, Cauchy sequence and Archimedean field, and the statement "eventually keeps one sign" in review exercise R3; and the argument that numbers arbitrarily close are equal (Logic and Proof, Proposition 6.2.5), which Section 4.3 reruns in F. From *Logic and Proof*, Chapter 2, De Morgan's laws (Logic and Proof, Proposition 2.5.5), which R3 also uses; from its Chapter 8, the catalogue of what a proof may not do (Logic and Proof, Example (What is not allowed)), for the find-the-error exercises C2 of Section 4.3 and C3 of Section 4.5. From *Sets and Functions*: sequences as functions (Sets and Functions, Definition 2.1.1), from Chapter 2; total orders (Sets and Functions, Definition 5.3.1) and finite maxima (Sets and Functions, Proposition 3.3.8), from Sections 5.3 and 3.3. A few items reach further. Exercise C1 of Section 4.2 uses finite and infinite sets (Sets and Functions, Definition 4.1.2, Sets and Functions, Definition 4.1.3, Sets and Functions, Corollary 4.1.3) from *Sets and Functions*, Section 4.1. Section 4.6 names suprema (Sets and Functions, Definition 5.3.5) without using them. One example, the non-Archimedean field $\mathbb{Q}(t)$ of Section 4.2, uses school algebra of polynomials, which *Algebraic Preliminaries*, Chapter 4 builds; it appears only in examples, remarks, warnings and exercises, and no proof of a numbered result cites it. Nothing from *Analysis on the Real Line* is used, and no fact about $\mathbb{R}$.

Time. About 25 hours on the core track (all of A, one B and one C per section, the exit checks); about 40 hours for everything.

4.0.1 Roadmap

- **Working in an Ordered Field.** The standing setup of an arbitrary ordered field F, the fraction rules of a field and the absence of zero divisors, halving a tolerance and the midpoint, comparing reciprocals and multiplying non-negative inequalities, and the absolute value with the triangle and reverse triangle inequalities, ending in the form $|x - a| < \varepsilon \iff a - \varepsilon < x < a + \varepsilon$ that the rest of the chapter uses.
- **The Rationals inside an Ordered Field; the Archimedean Property.** The elements $n \cdot 1_F$ and powers in F, which add, multiply and compare as in $\mathbb{N}$; the unique embedding ι_F of $\mathbb{Q}$ into every ordered field, strictly increasing and order-reflecting; the Archimedean property as a property of F, with its reciprocal form $1/(n \cdot 1_F) < \varepsilon$; the integer part and the density of the rationals in an Archimedean field; and $\mathbb{Q}(t)$, an ordered field that is not Archimedean.
- **Sequences and Limits in an Ordered Field.** Sequences indexed from 0 and "eventually", the limit with tolerances from F and its negation, first limits in $\mathbb{Q}$ and the divergence of $((-1)^n)$ in every ordered field, null and bounded sequences, $1/(n+1) \to 0$ as a restatement of the Archimedean property, changing finitely many terms, uniqueness of limits, and the boundedness of convergent sequences.
- **The Algebra and Order of Limits.** Null times bounded is null, terms near a non-zero limit stay away from 0, the limit laws for sums, multiples, products and quotients, and weak inequalities passing to the limit, with the squeeze; no result of the section uses the Archimedean property.
- **Cauchy Sequences.** Terms close to one another instead of to a point, convergent sequences Cauchy and Cauchy sequences bounded, sums and products of Cauchy sequences, the sign theorem (a Cauchy sequence that is not null eventually stays at least some $c > 0$ above 0, or at least c below it), sign stability under null differences, reciprocals, and a

sequence of rationals whose steps shrink to 0 and which is not Cauchy.

- **A Cauchy Sequence with No Rational Limit.** The Newton sequence for $\sqrt{2}$ is decreasing with overshoot $e_n \leq 1/(4n)$, Cauchy in $\mathbb{Q}$, and convergent to no rational, since a limit L would satisfy $L^2 = 2$; the section then lists what the chapter did not prove (suprema, monotone convergence, Bolzano–Weierstrass), each of which fails in $\mathbb{Q}$, and the review exercises return to the integer part of a rational, to induction, and to the negation laws of *Logic and Proof*, Chapter 6.

4.0.2 Named moves

- **Split ε.** To make a sum of k small terms smaller than ε, give each term the tolerance ε/k, and combine the finitely many thresholds by a maximum. It proves the sum law in Section 4.4 and the sum of two Cauchy sequences in Section 4.5; Exercise B3 of Section 4.4 and Exercise C1 of Section 4.5 leave part of it to you.
- **Bound, then shrink.** In a product, replace the factor you cannot control by a bound fixed in advance, then make the other factor small enough to absorb the bound. The bound comes from the whole sequence, before ε is named. It proves the product law in Section 4.4 and the product of two Cauchy sequences in Section 4.5.
- **Fix the bad ε.** "Not null" says: some ε is missed beyond every threshold. It hands you one ε, to keep for the rest of the proof; ask the Cauchy condition for a threshold with $\varepsilon/2$ first, and only then take a term beyond it that misses ε. This is the sign theorem of Section 4.5.
- **Locate the missing limit.** Suppose a limit L exists, and use the limit laws and uniqueness to find an equation that L must satisfy; if the field has no solution, there is no limit. In Section 4.6 the equation is $L^2 = 2$.

4.1 Working in an Ordered Field

Section 2.4 proved the rules for inequalities from the order axioms, and Section 3.2 added the one rule that needs division: reciprocals reverse order. A limit argument asks for a few more. It halves a tolerance, so it needs $0 < \varepsilon/2 < \varepsilon$; it subtracts fractions, so it needs $1/b - 1/d = (d - b)/(bd)$; and above all it measures how far apart two numbers are, so it needs the absolute value and the triangle inequality. Each of these is a few lines from the rules we have. This section proves them once, in an arbitrary ordered field, and the rest of the chapter uses them on nearly every page.

4.1.1 The standing setup

Throughout this chapter, F is an **ordered field** (Definition 2.4.1): a field with a relation $<$ satisfying (O1)–(O4). The rationals are one, by Theorem 3.2.1, and for most of the chapter they are the one that matters: the next chapter applies these results first with $F = \mathbb{Q}$, to build the real numbers from Cauchy sequences of rationals, and then again with $F = \mathbb{R}$, once $\mathbb{R}$ is shown to be an ordered field. We state results for a general F because the proofs never use anything about $\mathbb{Q}$ beyond (F1)–(F9) and (O1)–(O4), and saying so on the page makes it checkable.

From here on we use the field rules and the rules for inequalities of Sections 2.3, 2.4 and 3.2 without naming them, as in school algebra. A step is cited only where it needs care: multiplying or dividing an inequality, which needs the sign of the factor; taking reciprocals, which reverses order (Proposition 3.2.2); and a weak inequality, which hides an equality case. Such a step cites

its rule once in each proof. In a field $1 \neq 0$, so $1 > 0$ (Proposition 2.4.3 (e)). We write

$$F_{>0} := \{x \in F : x > 0\}, \qquad F_{\geq 0} := \{x \in F : x \geq 0\},$$

so that $\mathbb{Q}_{>0}$ of Section 3.2 is the case $F = \mathbb{Q}$. Recall that $a/b := ab^{-1}$ for $b \neq 0$, so $1/b = b^{-1}$, and that $2 := 1 + 1$ and $x^2 := xx$ (Proposition 2.3.2 (d)).

4.1.2 Fractions

School algebra adds fractions by cross-multiplying and subtracts reciprocals by a common denominator. Both are identities in any field, and the second is how we will compare $1/b_n$ with $1/b$ in Section 4.4. The only point needing care is that a product of non-zero elements is non-zero, so that bd may stand in a denominator.

Lemma 4.1.1. Fraction Rules in a Field

Let F be a field (Definition 2.3.1), and let $a, b, c, d \in F$ with $b \neq 0$ and $d \neq 0$.

(a) b^{-1} is the **only** element y with $by = 1$, and $(b^{-1})^{-1} = b$. Moreover $bd \neq 0$ and $(bd)^{-1} = b^{-1}d^{-1}$.

(b) $\dfrac{a}{b} + \dfrac{c}{d} = \dfrac{ad + bc}{bd}$ and $\dfrac{a}{b} \cdot \dfrac{c}{d} = \dfrac{ac}{bd}$.

(c) $\dfrac{1}{b} - \dfrac{1}{d} = \dfrac{d - b}{bd}$.

(d) For **all** $x, y \in F$: if $xy = 0$, then $x = 0$ or $y = 0$.

Idea. Everything rests on one move from Section 2.3: to identify an element, show it satisfies the **defining equation** of the element you want and let uniqueness finish. The product $b^{-1}d^{-1}$ satisfies the defining equation of $(bd)^{-1}$; once bd is known to be a legal denominator, (b) and (c) are bookkeeping.

Proof.

(a) Suppose $by = 1$. Then

$$y = 1 \cdot y = (b^{-1}b)y = b^{-1}(by) = b^{-1} \cdot 1 = b^{-1}.$$

This is the uniqueness argument of Proposition 2.3.2 (a), run with multiplication. Next, $b^{-1} \neq 0$: otherwise $1 = bb^{-1} = b \cdot 0 = 0$ by Proposition 2.3.2 (b), against $1 \neq 0$. So $(b^{-1})^{-1}$ exists, and since $b^{-1}b = bb^{-1} = 1$, the element b satisfies its defining equation; uniqueness gives $(b^{-1})^{-1} = b$. Now

$$(bd)(b^{-1}d^{-1}) = (bb^{-1})(dd^{-1}) = 1 \cdot 1 = 1.$$

If $bd = 0$, the left side would be $0 \cdot (b^{-1}d^{-1}) = 0$ by Proposition 2.3.2 (b), so $1 = 0$, which is false in a field. Hence $bd \neq 0$, and the display says that $b^{-1}d^{-1}$ satisfies the defining equation of $(bd)^{-1}$; by uniqueness, $(bd)^{-1} = b^{-1}d^{-1}$.

(b) By (a), $bd \neq 0$, so both right-hand sides are defined. For the product, by (a),

$$\frac{a}{b} \cdot \frac{c}{d} = (ab^{-1})(cd^{-1}) = (ac)(b^{-1}d^{-1}) \\ = (ac)(bd)^{-1} = \frac{ac}{bd}.$$

For the sum, first $\frac{ad}{bd} = (ad)(b^{-1}d^{-1}) = (ab^{-1})(dd^{-1}) = \frac{a}{b}$, and likewise $\frac{bc}{bd} = \frac{c}{d}$. Then distributivity gives

$$\frac{a}{b} + \frac{c}{d} = (ad)(bd)^{-1} + (bc)(bd)^{-1} \\ = (ad + bc)(bd)^{-1} = \frac{ad + bc}{bd}.$$

(c) Apply the sum rule of (b) with $a = 1$ and $c = -1$. Since $(-1)d^{-1} = -(d^{-1})$ by Proposition 2.3.2 (c), the left side is $1/b + (-1)/d = 1/b - 1/d$, and the numerator on the right is $1 \cdot d + b(-1) = d - b$.

(d) Let $xy = 0$, and suppose $x \neq 0$. Then $y = 1 \cdot y = (x^{-1}x)y = x^{-1}(xy) = x^{-1} \cdot 0 = 0$. So $x = 0$ or $y = 0$. This proves the lemma.

■

In an ordered field, (d) is also Proposition 2.4.3 (f). The point of (d) here is that it needs **no** order: a field has no zero divisors. The commutative ring of integers modulo 6, in the warning after Definition 2.3.1, has them ($[2][3] = [0]$ with $[2] \neq [0] \neq [3]$), so by (d) it is not a field.

4.1.3 Halving and comparing

The commonest move in a limit proof is to give each of two terms half the tolerance. That needs $\varepsilon/2$ to be positive, smaller than ε, and to add up to ε with itself. The next proposition collects these facts with two comparisons we will need as often.

Proposition 4.1.2. Halves and Comparisons in an Ordered Field

Let F be an ordered field.

(a) $\leq$ is a total order on F (Sets and Functions, Definition 5.3.1), and its strict part is $<$.

(b) $2 := 1 + 1 > 0$. For every $\varepsilon \in F_{>0}$,

$$0 < \frac{\varepsilon}{2} < \varepsilon \qquad \text{and} \qquad \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

For all $a, b \in F$ with $a < b$, $a < \frac{a+b}{2} < b$.

(c) For $a, b \in F_{>0}$: if $a \leq b$, then $\frac{1}{b} \leq \frac{1}{a}$.

(d) For $a, b, c, d \in F$: if $0 \leq a \leq b$ and $0 \leq c \leq d$, then $ac \leq bd$.

Idea. Each part is one or two of the rules of Sections 2.4 and 3.2 in sequence. The only care needed is with $\leq$, which hides an "or": each non-strict hypothesis splits into an equality case and a strict case, and the strict rules apply only to the second.

Proof.

(a) This is Proposition 2.4.3 (i).

(b) Since $1 > 0$, (O3) gives $2 = 1 + 1 > 0 + 1 = 1$, and $2 > 0$ by (O2). So $2 \neq 0$, and $2^{-1} > 0$ by Proposition 3.2.2 (a). For $\varepsilon > 0$, $\varepsilon/2 = \varepsilon \cdot 2^{-1} > 0$ by (O4), and

$$\begin{aligned}\frac{\varepsilon}{2} + \frac{\varepsilon}{2} &= \varepsilon \cdot 2^{-1} \cdot 1 + \varepsilon \cdot 2^{-1} \cdot 1 \\ &= (\varepsilon \cdot 2^{-1})(1 + 1) = \varepsilon(2^{-1} \cdot 2) = \varepsilon.\end{aligned}$$

Adding $\varepsilon/2$ to $0 < \varepsilon/2$ by (O3) gives $\varepsilon/2 < \varepsilon/2 + \varepsilon/2 = \varepsilon$. For the midpoint, let $a < b$. By (O3), $a + a < a + b$ and $a + b < b + b$, and $a + a = a(1 + 1) = 2a$, $b + b = 2b$. Multiplying by $2^{-1} > 0$ (Proposition 2.4.3 (c)) gives $a < (a + b)/2 < b$, since $(2a)2^{-1} = a(2 \cdot 2^{-1}) = a$.

(c) If $a = b$, then $1/b = 1/a$. Otherwise $0 < a < b$, and $1/b < 1/a$ by Proposition 3.2.2 (b). In both cases $1/b \leq 1/a$.

(d) First, $ac \leq bc$: if $a = b$ or $c = 0$, the two sides are equal (for $c = 0$ both are 0); otherwise $a < b$ and $c > 0$, and $ac < bc$ by Proposition 2.4.3 (c). Next, $bc \leq bd$, by the same argument with the roles of the factors exchanged: $c \leq d$, and $b \geq 0$ since $0 \leq a \leq b$ (transitivity, part (a)). By transitivity of $\leq$ (part (a)), $ac \leq bd$.

■

Part (b) is used whenever a proof splits a tolerance: the sum of two quantities, each below $\varepsilon/2$, is below ε. The midpoint is the same fact read for two points: between any two elements of F lies a third. Part (d) is how a product of two small, or two bounded, quantities is estimated.

Warning. The non-negativity in (d) cannot be dropped. In $\mathbb{Q}$, take $a = c = -2$ and $b = d = 0$: then $a \leq b$ and $c \leq d$, but $ac = 4$ and $bd = 0$, and $4 \leq 0$ is false. Before multiplying two inequalities, check that both sides are non-negative.

4.1.4 Absolute value

To say that x is close to a, we want one element of F measuring the gap between them, whichever of the two is larger. In school this is $|x - a|$, and $|x|$ is sometimes written $\sqrt{x^2}$. That formula needs a square-root function, and an ordered field has none in general: no rational squares to 2 (Theorem 3.3.1). The square x^2 has the two square roots x and $-x$, and picking the non-negative one is already a definition by cases. So we define the absolute value by cases, using only the order.

The absolute value of x is x with its sign made non-negative.

Definition 4.1.1. Absolute Value in an Ordered Field

Let F be an ordered field. For $x \in F$, the **absolute value** of x is

$$|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

For $x, y \in F$, the element $|x - y|$ is the **distance** from x to y.

The two cases do not overlap and cover F, by trichotomy (O1), so $|x|$ is a well-defined element of F, and $x \mapsto |x|$ is a function $F \to F$.

Examples. In $\mathbb{Q}$, $|3/4| = 3/4$ and $|-3/4| = 3/4$, the second because $-3/4 < 0$ and $-(-3/4) = 3/4$. The distance from $1/3$ to $1/2$ is $|1/3 - 1/2| = |-1/6| = 1/6$. The degenerate case is $|0| = 0$. Part (a) below shows that 0 is the only element with absolute value 0, so the distance from x to y is 0 exactly when $x = y$.

↻ **Second reading** · *Why not x^2?*

A non-example by minimal change. Replace the definition by $x \mapsto x^2$. This map is also non-negative, zero only at 0, and multiplicative, so it passes (a) and the product rule of (b) below. It fails the triangle inequality (e): in $\mathbb{Q}$, with $x = y = 1$, the "size" of $x + y$ would be $2^2 = 4$, while the sizes of x and y add up to $1 + 1 = 2$. A measure of size must not grow faster than its argument, and that is the clause the square breaks.

Proposition 4.1.3. Rules for the Absolute Value

Let F be an ordered field, and let $x, y, z, r \in F$.

(a) $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$.
(b) $|-x| = |x|$, $|xy| = |x||y|$ and $|x|^2 = x^2$; if $x \neq 0$, then $|1/x| = 1/|x|$.
(c) $-|x| \leq x \leq |x|$.
(d) $|x| \leq r$ if and only if $-r \leq x \leq r$; and $|x| < r$ if and only if $-r < x < r$.
(e) *(Triangle inequality.)* $|x + y| \leq |x| + |y|$, and $|x - z| \leq |x - y| + |y - z|$.
(f) *(Reverse triangle inequality.)* $||x| - |y|| \leq |x - y|$.

Idea. Parts (a)–(d) are case checks on the sign of x (and of y). The triangle inequality then needs **no** cases: by (c), x and y are each trapped between $\pm$ their absolute values; adding the two traps traps $x + y$ between $\pm(|x| + |y|)$, and (d) reads that as the inequality. The reverse triangle inequality is the triangle inequality applied to $x = (x - y) + y$ and to $y = (y - x) + x$.

Proof.

(a) If $x \geq 0$, then $|x| = x \geq 0$. If $x < 0$, then $|x| = -x > 0$ by Proposition 2.4.3 (a), since $-0 = 0$. So $|x| \geq 0$ always, and $|x| = 0$ only in the first case with $x = 0$; conversely $|0| = 0$.

(b) *Negation.* If $x > 0$, then $-x < 0$ (Proposition 2.4.3 (a)), so $|-x| = -(-x) = x = |x|$. If $x = 0$, both sides are 0. If $x < 0$, then $-x > 0$, so $|-x| = -x = |x|$.
Product. If $x \geq 0$ and $y \geq 0$, then $xy \geq 0$ (by (O4), or Proposition 2.3.2 (b) if a factor is 0), so $|xy| = xy = |x||y|$. If $x \geq 0$ and $y < 0$, then $-y > 0$ and $|x||y| = x(-y) = -(xy)$; and $-(xy) \geq 0$ by the first case, so $xy \leq 0$ and $|xy| = -(xy)$ (if $xy = 0$, both readings give 0). The case $x < 0 \leq y$ is the same with the roles exchanged. If $x < 0$ and $y < 0$,

then $xy = (-x)(-y) > 0$ by (O4), so $|xy| = xy = (-x)(-y) = |x||y|$.
Square. $|x|^2$ is xx or $(-x)(-x)$, and both are x^2.
Reciprocal. Let $x \neq 0$. By the product rule, $|x|\,|1/x| = |x \cdot x^{-1}| = |1| = 1$, and $|x| \neq 0$ by (a). So $|1/x|$ satisfies the defining equation of $|x|^{-1}$, and Lemma 4.1.1 (a) gives $|1/x| = 1/|x|$.

(c) If $x \geq 0$, then $-|x| = -x \leq 0 \leq x = |x|$, using Proposition 2.4.3 (a) for $-x \leq 0$. If $x < 0$, then $-|x| = -(-x) = x$, and $x < 0 < -x = |x|$. In both cases $-|x| \leq x \leq |x|$.

(d) ($\Rightarrow$) Let $|x| \leq r$. By (c), $x \leq |x| \leq r$. By (c) and (b) applied to $-x$, $-x \leq |-x| = |x| \leq r$, and negating reverses the inequality (Proposition 2.4.3 (a), or equality), so $-r \leq x$. ($\Leftarrow$) Let $-r \leq x \leq r$. If $x \geq 0$, then $|x| = x \leq r$. If $x < 0$, then $|x| = -x \leq r$, by negating $-r \leq x$. The strict version is the same argument with $<$ in place of $\leq$ at the steps that involve r: for instance $x \leq |x| < r$ gives $x < r$ by transitivity.

(e) By (c), $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. Adding (Proposition 2.4.3 (b)),

$$-(|x| + |y|) \leq x + y \leq |x| + |y|,$$

the left end because $-(u+v) = (-u)+(-v)$. By (d) with $r := |x|+|y|$, $|x+y| \leq |x|+|y|$. For the second form, apply the first to $x - z = (x - y) + (y - z)$.

(f) By (e), $|x| = |(x - y) + y| \leq |x - y| + |y|$, so $|x| - |y| \leq |x - y|$ by (O3) (or equality). Exchanging x and y, and using $|y - x| = |-(x - y)| = |x - y|$ from (b), gives $|y| - |x| \leq |x - y|$. So $-|x - y| \leq -(|y| - |x|)$ by Proposition 2.4.3 (a) (or equality), and $-(|y| - |x|) = |x| - |y|$, since $-(u + v) = (-u) + (-v)$; that is, $-|x - y| \leq |x| - |y|$. By (d), $||x| - |y|| \leq |x - y|$. This proves the proposition.

■

The form of (d) that the rest of the chapter uses is about a distance. Apply the strict half of (d) to $x - a$ with $r = \varepsilon$, and add a throughout; adding $-a$ goes back, so the step is reversible:

$$|x - a| < \varepsilon \iff a - \varepsilon < x < a + \varepsilon.$$

In words: x lies within ε of a exactly when it lies strictly between $a - \varepsilon$ and $a + \varepsilon$. In Section 4.3 the role of x is played by the terms a_n of a sequence and a is a candidate limit.

Example. In $\mathbb{Q}$, which x satisfy $|x - 1/2| < 1/4$? By the display above, exactly those with $1/2 - 1/4 < x < 1/2 + 1/4$, that is, $1/4 < x < 3/4$. So $2/3$ qualifies and $3/4$ does not: the inequality is strict, and $|3/4 - 1/2| = 1/4$.

Quick check. Is $|a| - |b| \leq |a - b|$ for all $a, b \in F$? Is $|a - b| \leq |a| - |b|$ for all $a, b \in F$?

Solution. The first: yes. By (f) and (c), $|a| - |b| \leq ||a| - |b|| \leq |a - b|$. The second: no. In $\mathbb{Q}$, take $a = 0$ and $b = 1$: then $|a - b| = |-1| = 1$, while $|a| - |b| = -1$, and $1 \leq -1$ is false.

Quick check. For which $x \in \mathbb{Q}$ do both $|x - 1| < 2$ and $|x + 1| < 2$ hold?

Solution. By the display above, $|x-1|<2$ says $-1<x<3$, and $|x+1|=|x-(-1)|<2$ says $-3<x<1$. Both hold exactly when $-1<x<1$.

The triangle inequality is the tool of the whole chapter. In its second form it says that a detour through a third point y cannot shorten the way from x to z. Run backwards, it turns two small distances, from x to y and from y to z, into one small distance from x to z. That is the move of almost every proof in Sections 4.3–4.5.

4.1.5 Exercises

4.1.5.1 A. Check your understanding

Exercise. A1

Let F be an ordered field.

(a) Here is a proof that $0<1/2<1$ in F: (1) $1>0$; (2) $2=1+1>1>0$; (3) $0<1/2<1/1=1$. Name the rule behind each of the three steps.

(b) True or false in every ordered field: "if $a<b$, then $a^2<b^2$". Give a reason, or a counterexample in $\mathbb{Q}$.

(c) True or false in every ordered field: "if $|a|<|b|$, then $a<b$". Give a reason, or a counterexample in $\mathbb{Q}$.

(d) True or false in every ordered field: "$x^2+1>0$ for every x". Give a reason, or a counterexample in $\mathbb{Q}$.

Solution.

(a) (a) Proposition 2.4.3 (e), since $1\neq 0$ in a field. (2) $1+1>0+1=1$ by (O3) (or Proposition 2.4.3 (b), strict), then $2>0$ by (O2). (3) Proposition 3.2.2 (b), applied to $0<1<2$; and $1/1=1\cdot 1^{-1}=1$, since 1 is its own inverse ($1\cdot 1=1$, and Lemma 4.1.1 (a)).

(b) False. In $\mathbb{Q}$, $-2<1$, but $(-2)^2=4>1=1^2$. The true statement needs $a,b\geq 0$: Proposition 2.4.3 (g).

(c) False. In $\mathbb{Q}$, $a=1$ and $b=-2$ give $|a|=1<2=|b|$, but $a=1>-2=b$.

(d) True. By Proposition 2.4.3 (e), $x^2>0$ if $x\neq 0$, and $x^2=0$ if $x=0$ (Proposition 2.3.2 (b)); so $x^2\geq 0$. Also $1>0$. Hence $x^2+1>0+0=0$ by Proposition 2.4.3 (b), strict in the second summand.

4.1.5.2 B. Practice

Exercise. B1: Two companions of (c) and (d)

Let F be an ordered field.

(a) Prove that if $0\leq a<b$ and $0<c\leq d$, then $ac<bd$.

(b) Prove that for $a,b\in F_{>0}$, $a\leq b$ **if and only if** $1/b\leq 1/a$.

Name the rule behind each step.

Solution.

(a) Since $0 \le a < b$, $b > 0$ by Proposition 2.4.3 (i). First, $ac < bc$: $a < b$ and $c > 0$, so Proposition 2.4.3 (c) applies. Next, $bc \le bd$: if $c = d$ the two sides are equal; otherwise $c < d$ and $b > 0$, so $cb < db$ by Proposition 2.4.3 (c), that is, $bc < bd$ by (F6). Combining $ac < bc \le bd$ by Proposition 2.4.3 (i) gives $ac < bd$.

(b) ($\Rightarrow$) is Proposition 4.1.2 (c). ($\Leftarrow$) Let $1/b \le 1/a$. Both are positive, by Proposition 3.2.2 (a). Applying Proposition 4.1.2 (c) to the positive elements $1/b \le 1/a$ gives $1/(1/a) \le 1/(1/b)$, and $1/(1/a) = (a^{-1})^{-1} = a$, $1/(1/b) = b$ by Lemma 4.1.1 (a). So $a \le b$.

Exercise. B3: Two more triangle inequalities

Prove that for all a, b, c in an ordered field,

(a) $|a + b + c| \le |a| + |b| + |c|$;
(b) $||a| - |b|| \le |a + b|$.

Solution.

(a) By Proposition 4.1.3 (e) twice, $|(a + b) + c| \le |a + b| + |c| \le (|a| + |b|) + |c|$, the second step by adding $|c|$ to $|a+b| \le |a|+|b|$ (Proposition 2.4.3 (b)). Conclude by transitivity (Proposition 2.4.3 (i)).

(b) Apply Proposition 4.1.3 (f) to a and $-b$: $||a| - |-b|| \le |a-(-b)|$. By Proposition 4.1.3 (b), $|-b| = |b|$, and $a - (-b) = a + b$ by Proposition 2.3.2 (a). This is the claim.

4.1.5.3 C. Going deeper

Exercise. C1: A field that cannot be ordered

(a) Prove that in an ordered field no x satisfies $x^2 = -1$.
(b) Deduce that if a field contains an element i with $i^2 = -1$, then no relation $<$ makes it an ordered field.

Solution (Hint 1). Which sign does a square have, and which sign does -1 have?

Solution (Hint 2). Show first that $x^2 \ge 0$ for every x, splitting on whether $x = 0$.

Solution.

(a) Let F be an ordered field and $x \in F$. If $x = 0$, then $x^2 = 0$ (Proposition 2.3.2 (b)); if $x \ne 0$, then $x^2 > 0$ (Proposition 2.4.3 (e)). So $x^2 \ge 0$. On the other hand $1 > 0$

(Proposition 2.4.3 (e), as $1 \neq 0$ in a field), so $-1 < 0$ by Proposition 2.4.3 (a). If $x^2 = -1$, then $0 \leq x^2 = -1 < 0$, so $0 < 0$ by Proposition 2.4.3 (i), against (O1).

(b) If some relation $<$ made the field an ordered field, then (a) would forbid $i^2 = -1$. So no such relation exists.

The complex numbers of *Analytic Preliminaries*, Chapter 1 contain such an i, so they form a field that cannot be ordered. Compare Example (A field with two elements that cannot be ordered), where the obstacle was $1 + 1 = 0$; here it is that -1 is a square.

Marking scheme. (10 points) - 4: shows $x^2 \geq 0$ for every x (cases $x = 0$, $x \neq 0$), or rules out $x = 0$ for the x with $x^2 = -1$ and then shows $x^2 > 0$, giving the rule for each step. - 2: shows $-1 < 0$ from $1 > 0$ (from $1 \neq 0$ and Proposition 2.4.3 (e), or the standing setup of Section 4.1) and from negation reversing order. - 2: derives the contradiction with trichotomy (O1) or irreflexivity. - 2: (b) states that any order would make (a) apply, so none exists. Fatal errors: using strictness on $x^2 > 0$ without ruling out $x = 0$; asserting $1 > 0$ with no reason; using $\mathbb{R}$ or $\sqrt{}$.

Exercise. C2: Maximum and minimum through the absolute value

For a, b in an ordered field F, let $\max\{a, b\}$ be b if $a \leq b$ and a otherwise, and $\min\{a, b\}$ be a if $a \leq b$ and b otherwise.

(a) Prove that

$$\max\{a, b\} = \frac{a + b + |a - b|}{2}, \qquad \min\{a, b\} = \frac{a + b - |a - b|}{2}.$$

(b) Deduce that $|x| = \max\{x, -x\}$ for every $x \in F$.

Solution.

(a) *Case* $a \leq b$. Then $a - b \leq 0$ (Proposition 2.4.3 (a), or equality), so $|a - b| = b - a$: by the second clause of Definition 4.1.1 if $a - b < 0$, and because both sides are 0 if $a = b$. Hence

$$\frac{a + b + |a - b|}{2} = \frac{a + b + b - a}{2} = \frac{2b}{2} = b = \max\{a, b\},$$

and likewise $\dfrac{a + b - (b - a)}{2} = \dfrac{2a}{2} = a = \min\{a, b\}$, using $b + b = 2b$ and $(2b)2^{-1} = b$ as in the proof of Proposition 4.1.2 (b).
Case $b < a$. Then $a - b > 0$, so $|a - b| = a - b$, and the two formulas give $(2a)/2 = a = \max\{a, b\}$ and $(2b)/2 = b = \min\{a, b\}$.

(b) By (a) with $a = x$ and $b = -x$: $\max\{x, -x\} = (x + (-x) + |2x|)/2 = |2x|/2$. By Proposition 4.1.3 (b), $|2x| = |2||x| = 2|x|$, since $2 > 0$. So $\max\{x, -x\} = |x|$.

4.2 The Rationals inside an Ordered Field; the Archimedean Property

In an ordered field F we can form 1, $1+1$, $1+1+1$, and so on. They look like the natural numbers $1, 2, 3, \ldots$, but they are elements of F, and two questions need answers before the next section can use them as thresholds. Do they behave like natural numbers, adding, multiplying and comparing as $\mathbb{N}$ does? And does every element of F lie below one of them? For $\mathbb{Q}$, Section 3.2 answered the second question with yes (Theorem 3.2.3). This section answers the first for every ordered field, and goes further: a whole copy of $\mathbb{Q}$ sits inside F, in exactly one way. The second question has the answer no in general. Fields where the answer is yes are called **Archimedean**, and in them every element has an integer part and the rationals are dense.

4.2.1 The natural numbers in an ordered field

Section 2.4 defined, in any commutative ring R and for any $a \in R$, the multiples $n \cdot a$ for $n \in \mathbb{N}$ (just before Proposition 2.4.3): $0 \cdot a = 0$ and $(n+1) \cdot a = n \cdot a + a$. With $a = 1$ they are the elements $0, 1, 1+1, \ldots$ of the opening paragraph. We give them a name, together with powers, which this chapter needs for estimates such as $2^k \geq k+1$.

Definition 4.2.1. Natural Numbers and Powers in an Ordered Field

Let F be an ordered field, with unit element 1_F.

(a) For $n \in \mathbb{N}$, $n \cdot 1_F \in F$ is the multiple of 1_F of Section 2.4: the map $\mathbb{N} \to F$, $n \mapsto n \cdot 1_F$, is the unique function with

$$0 \cdot 1_F = 0, \qquad (n+1) \cdot 1_F = n \cdot 1_F + 1_F \quad \text{for every } n \in \mathbb{N}.$$

(b) For $x \in F$, the **powers** x^k ($k \in \mathbb{N}$) are given by Sets and Functions, Theorem 3.3.1, read on $\mathbb{N} = \omega$ through Corollary 1.4.3 (b), with $X = F$, starting value 1_F and step $(k, y) \mapsto yx$:

$$x^0 := 1_F, \qquad x^{k+1} := x^k x \quad \text{for every } k \in \mathbb{N}.$$

The subscript in 1_F is there because two fields will often be in play, F and $\mathbb{Q}$, each with its own 1; where only one field is in play the subscript is dropped. For natural numbers n, k, the power $n^k \in \mathbb{N}$ is the one of Section 1.3, defined (just before Proposition 1.3.4) by the same recursion with $X = \mathbb{N}$: $n^0 = 1$ and $n^{k+1} = n^k n$. So 2^k may mean a natural number or an element of F; part (c) of the next proposition says the two readings match.

Examples. $1 \cdot 1_F = 0 \cdot 1_F + 1_F = 0 + 1_F = 1_F$, and $2 \cdot 1_F = 1_F + 1_F$, the element written 2 in Section 4.1. Also $x^1 = 1_F x = x$ and $x^2 = xx$, which is the square of Proposition 2.3.2 (d). The degenerate case: $0 \cdot 1_F = 0$, and $0^0 = 1_F$ by the first equation, a convention the recursion imposes and nothing below questions.

Second reading · *Without the order, the copy of $\mathbb{N}$ collapses*

Keep the field axioms and drop the order. In the two-element field of Example (A field with two elements that cannot be ordered), the same recursion gives $2 \cdot 1 = 1 + 1 = 0 = 0 \cdot 1$. So the map $n \mapsto n \cdot 1$ is not injective there, and the elements $0 \cdot 1, 1 \cdot 1, 2 \cdot 1, \ldots$ are not a copy of $\mathbb{N}$.

The next proposition shows that the order axioms prevent this.

Proposition 4.2.1. The Natural Numbers inside an Ordered Field

Let F be an ordered field, and let $m, n, k \in \mathbb{N}$.

(a) $(m+n)\cdot 1_F = m\cdot 1_F + n\cdot 1_F$ and $(mn)\cdot 1_F = (m\cdot 1_F)(n\cdot 1_F)$.

(b) If $m < n$, then $m\cdot 1_F < n\cdot 1_F$. Hence $n \mapsto n\cdot 1_F$ is injective, and $n\cdot 1_F \geq 1_F > 0$ whenever $n \geq 1$.

(c) $(n^k)\cdot 1_F = (n\cdot 1_F)^k$. In particular $(2^k)\cdot 1_F = (2\cdot 1_F)^k$, the k-th power in F of $2 = 1+1$.

(d) In $\mathbb{Q}$, $n\cdot 1_{\mathbb{Q}} = j(\iota(n))$, the image of n under the embeddings Proposition 2.2.3 and Proposition 3.1.4. With the identifications $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$, this says $n\cdot 1_{\mathbb{Q}} = n$.

Idea. The sum law in (a) and part (b) are rules for multiples (Proposition 2.4.3 (h)); the product law and (c) are inductions following the recursion defining the left side. The product law is the only step with content: multiplying by $n+1$ is multiplying by n and adding one more copy, and the sum law and distributivity turn that into the same step on the right. Part (d) compares two functions on $\mathbb{N}$ that satisfy the same recursion, and the uniqueness half of the recursion theorem makes them equal.

Proof.

(a) The sum law is Proposition 2.4.3 (h) with $R = F$ and $a = 1_F$. For the product law, fix m and use Logic and Proof, Axiom 4.2.1 (through Corollary 1.4.3 (b)) on n, for $P(n)$: "$(mn)\cdot 1_F = (m\cdot 1_F)(n\cdot 1_F)$". *Base:* $m\cdot 0 = 0$ by Definition 1.2.1 (b), so the left side is $0\cdot 1_F = 0$, and the right side is $(m\cdot 1_F)\cdot 0 = 0$ by Proposition 2.3.2 (b). *Step:* assume $P(n)$. By Definition 1.2.1 (b) and (c), $m(n+1) = mn + m$. Hence

$$\begin{aligned}(m(n+1))\cdot 1_F &= (mn)\cdot 1_F + m\cdot 1_F = (m\cdot 1_F)(n\cdot 1_F) + (m\cdot 1_F)1_F \\ &= (m\cdot 1_F)(n\cdot 1_F + 1_F) = (m\cdot 1_F)((n+1)\cdot 1_F),\end{aligned}$$

by the sum law, $P(n)$, distributivity, and the recursion of (a) in Definition 4.2.1. So $P(n+1)$ holds, and $P(n)$ holds for every n.

(b) Since $1_F \neq 0$ in a field, Proposition 2.4.3 (h) gives $m\cdot 1_F < n\cdot 1_F$ whenever $m < n$. If $m \neq n$, then $m < n$ or $n < m$ by trichotomy in $\mathbb{N}$ (Theorem 1.3.2 (a)), so $m\cdot 1_F \neq n\cdot 1_F$ by (O1): the map is injective. Finally $1\cdot 1_F = 1_F$ (the examples above), and for $n > 1$, $1_F = 1\cdot 1_F < n\cdot 1_F$; so $n\cdot 1_F \geq 1_F$ for every $n \geq 1$, and $1_F > 0$ by Proposition 2.4.3 (e).

(c) Fix n and induct on k. *Base:* $n^0 = 1$, so $(n^0)\cdot 1_F = 1\cdot 1_F = 1_F = (n\cdot 1_F)^0$. *Step:* if $(n^k)\cdot 1_F = (n\cdot 1_F)^k$, then, since $n^{k+1} = n^k n$,

$$(n^{k+1})\cdot 1_F = ((n^k)\cdot 1_F)(n\cdot 1_F) = (n\cdot 1_F)^k(n\cdot 1_F) = (n\cdot 1_F)^{k+1},$$

by the product law of (a), the hypothesis, and Definition 4.2.1 (b). The special case is $n = 2$, with $2\cdot 1_F = 1_F + 1_F$.

Part (d) is proved in the fold below.

■

↻ **Second reading** · *Part (d): the two copies of $\mathbb{N}$ in $\mathbb{Q}$ agree*

(d) Let $f(n) := j(\iota(n)) \in \mathbb{Q}$. The zero and one of $\mathbb{Z}$ are $\iota(0)$ and $\iota(1)$ (Theorem 2.3.1), and j sends them to the zero and one of $\mathbb{Q}$ (Proposition 3.1.4). So $f(0) = 0$, and, since ι and j preserve sums (Proposition 2.2.3 (a), Proposition 3.1.4),

$$f(n+1) = j(\iota(n) + \iota(1)) = j(\iota(n)) + j(\iota(1)) = f(n) + 1_{\mathbb{Q}}.$$

So f and $n \mapsto n \cdot 1_{\mathbb{Q}}$ both satisfy the recursion of Definition 4.2.1 (a) with $F = \mathbb{Q}$, and by the uniqueness in Sets and Functions, Theorem 3.3.1 (through Corollary 1.4.3 (b)) they are equal. This proves the proposition.

So the order axioms do what the two-element field of Example (A field with two elements that cannot be ordered) (the fold before the proposition) lacks: by (b), the elements $n \cdot 1_F$ are all different and increase with n, and by (a) they add and multiply as natural numbers do. In $\mathbb{Q}$, (d) says that nothing new has happened: $n \cdot 1_{\mathbb{Q}}$ is n. When only one field is in play, we will sometimes write n for $n \cdot 1_F$ in a general ordered field too; the statements of this section keep $n \cdot 1_F$, because there the difference between $n \in \mathbb{N}$ and its image in F is the point.

Quick check. In an ordered field F, write $(3 \cdot 1_F)(4 \cdot 1_F) - 5 \cdot 1_F$ as a single multiple of 1_F, and say why it is not 0.

Solution. By the product law of Proposition 4.2.1 (a), $(3 \cdot 1_F)(4 \cdot 1_F) = 12 \cdot 1_F$, and by the sum law $12 \cdot 1_F = 7 \cdot 1_F + 5 \cdot 1_F$, since $12 = 7 + 5$ in $\mathbb{N}$. So the element is $7 \cdot 1_F$, and $7 \cdot 1_F \geq 1_F > 0$ by part (b).

4.2.2 A copy of the rationals in every ordered field

The copy of $\mathbb{N}$ extends. Negatives exist in F, so the integers have images too, and inverses of non-zero elements exist, so fractions have images. The next theorem says that the result is a copy of $\mathbb{Q}$, compatible with $+$, $\cdot$ and $<$, and that there is no other way to place $\mathbb{Q}$ inside F. Later chapters of this volume use this map to compare an arbitrary ordered field with the rationals.

Every ordered field contains a copy of $\mathbb{Q}$, and only one.

Theorem 4.2.2. The Rationals Embed in Every Ordered Field

Let F be an ordered field. There is **exactly one** map $\iota_F \colon \mathbb{Q} \to F$ such that, for all $x, y \in \mathbb{Q}$,

$$\iota_F(1) = 1_F,$$
$$\iota_F(x+y) = \iota_F(x) + \iota_F(y), \qquad \iota_F(xy) = \iota_F(x)\,\iota_F(y).$$

It satisfies $\iota_F(n) = n \cdot 1_F$ for every $n \in \mathbb{N}$. It is injective and **strictly increasing** ($x < y$ implies $\iota_F(x) < \iota_F(y)$), and it **reflects order**: $\iota_F(x) < \iota_F(y)$ implies $x < y$.

Idea. Uniqueness tells us how to build ι_F. A map respecting 1 and + must send $n = 1+\cdots+1$ to $n \cdot 1_F$; respecting negatives, it must send $-n$ to $-(n \cdot 1_F)$; and respecting products, it must send p/q to $(p \cdot 1_F)(q \cdot 1_F)^{-1}$. So we take that formula as the definition, and the work is to check it. **Step roadmap.** ① Extend $n \mapsto n \cdot 1_F$ to $\mathbb{Z}$ and check that it respects $+$, $\cdot$ and negation. ② Define ι_F on fractions and check that the value does not depend on the fraction chosen. ③ Check $+$ and $\cdot$ with the fraction rules (①–③ are in the fold below). ④ Check the order. ⑤ Uniqueness: run the opening argument as a proof.

The map. By Proposition 3.1.4 (b), every $x \in \mathbb{Q}$ is p/q with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$. Extend $n \mapsto n \cdot 1_F$ to $\mathbb{Z}$ by $(-n) \cdot 1_F := -(n \cdot 1_F)$, and set

$$\iota_F\left(\frac{p}{q}\right) := (p \cdot 1_F)(q \cdot 1_F)^{-1} = \frac{p \cdot 1_F}{q \cdot 1_F}.$$

Here $q \geq 1$, so $q \cdot 1_F \geq 1_F > 0$ (Proposition 4.2.1 (b)), and the inverse exists. Steps 1–3 check that this is a well-defined map respecting 1, $+$ and $\cdot$.

↻ **Second reading** · *Steps 1–3: the integers, the fractions, sums and products*

Step 1: the integers. By Proposition 2.2.3 (c), read through the identification $\mathbb{N} \subseteq \mathbb{Z}$, every $m \in \mathbb{Z}$ is n or $-n$ for some $n \in \mathbb{N}$. Define

$$m \cdot 1_F := \begin{cases} n \cdot 1_F & \text{if } m = n \text{ with } n \in \mathbb{N}, \\ -(n \cdot 1_F) & \text{if } m = -n \text{ with } n \in \mathbb{N}. \end{cases}$$

This is well defined. Within one case, n is determined by m: directly in the first case, and in the second because $-n = -n'$ gives $n = n'$ by Proposition 2.3.2 (a). The cases overlap only at 0: if $n = -n'$ with $n, n' \in \mathbb{N}$, then $n + n' = 0$, and $n, n' \geq 0$ (Theorem 2.4.2 (b)); if $n' > 0$, then $n + n' > n + 0 \geq 0$ by Proposition 2.4.3 (b), against $n + n' = 0$, so $n' = 0$ and then $n = 0$. At $m = 0$ both cases give 0, since $-0 = 0$.

We check three rules, for $m, m' \in \mathbb{Z}$.

Negation: $(-m) \cdot 1_F = -(m \cdot 1_F)$. If $m = n \in \mathbb{N}$, this is the second case of the definition. If $m = -n$, then $-m = n$ by Proposition 2.3.2 (a), and $(-m) \cdot 1_F = n \cdot 1_F = -(-(n \cdot 1_F)) = -(m \cdot 1_F)$.

Sums: $(m + m') \cdot 1_F = m \cdot 1_F + m' \cdot 1_F$. If $m, m' \in \mathbb{N}$, this is Proposition 4.2.1 (a). We write out the mixed case $m = n$, $m' = -n'$ with $n, n' \in \mathbb{N}$. If $n' \leq n$, then $n = n' + k$ for some $k \in \mathbb{N}$ (Proposition 1.3.1 (a)), so $m + m' = (n' + k) - n' = k$ in $\mathbb{Z}$, and

$$\begin{aligned} m \cdot 1_F + m' \cdot 1_F &= (n' + k) \cdot 1_F - n' \cdot 1_F \\ &= n' \cdot 1_F + k \cdot 1_F - n' \cdot 1_F = k \cdot 1_F, \end{aligned}$$

by Proposition 4.2.1 (a). If $n < n'$, then $n' = n + k$, $m + m' = -k$, and the same computation gives $n \cdot 1_F - (n + k) \cdot 1_F = -(k \cdot 1_F) = (-k) \cdot 1_F$. The case $m = -n$, $m' = n'$ follows by (F1). If $m = -n$ and $m' = -n'$, then $m + m' = -(n + n')$, and $-((n + n') \cdot 1_F) = -(n \cdot 1_F) - (n' \cdot 1_F)$, by the sum law and uniqueness of negatives.

Products: $(mm')\cdot 1_F = (m\cdot 1_F)(m'\cdot 1_F)$. If $m, m' \in \mathbb{N}$, this is Proposition 4.2.1 (a). We write out $m = -n$, $m' = n'$: then $mm' = -(nn')$ by Proposition 2.3.2 (c), so

$$\begin{aligned}(mm')\cdot 1_F &= -((nn')\cdot 1_F) = -((n\cdot 1_F)(n'\cdot 1_F))\\ &= (-(n\cdot 1_F))(n'\cdot 1_F) = (m\cdot 1_F)(m'\cdot 1_F),\end{aligned}$$

by negation, Proposition 4.2.1 (a), and Proposition 2.3.2 (c) in F. The two remaining sign patterns are the same with signs: $(-n)(-n') = nn'$ in $\mathbb{Z}$ and $(-(n\cdot 1_F))(-(n'\cdot 1_F)) = (n\cdot 1_F)(n'\cdot 1_F)$ in F, both by Proposition 2.3.2 (c), and $m = n$, $m' = -n'$ follows from the case written out by (F6).

Step 2: the definition on $\mathbb{Q}$. As above, $q\cdot 1_F \neq 0$. Define ι_F with the formula of the paragraph above. The value does not depend on the fraction chosen: if $p/q = r/s$ with $r \in \mathbb{Z}$ and $s \in \mathbb{Z}_{>0}$, then $ps = rq$ (Proposition 3.1.4 (b)), so $(p\cdot 1_F)(s\cdot 1_F) = (r\cdot 1_F)(q\cdot 1_F)$ by the product rule of Step 1. Multiplying both sides by $(q\cdot 1_F)^{-1}(s\cdot 1_F)^{-1}$ and using (F5), (F6), (F8) gives $(p\cdot 1_F)(q\cdot 1_F)^{-1} = (r\cdot 1_F)(s\cdot 1_F)^{-1}$. So ι_F is a function $\mathbb{Q}\to F$. For $m \in \mathbb{Z}$, $m = m/1$, so $\iota_F(m) = (m\cdot 1_F)(1_F)^{-1} = m\cdot 1_F$; in particular $\iota_F(n) = n\cdot 1_F$ for $n \in \mathbb{N}$, and $\iota_F(1) = 1_F$.

Step 3: sums and products. Let $x = p/q$ and $y = r/s$ as in Step 2. In the field $\mathbb{Q}$, Lemma 4.1.1 (b) gives

$$x + y = \frac{ps + rq}{qs}, \qquad xy = \frac{pr}{qs},$$

with $qs \in \mathbb{Z}_{>0}$ by (O4) in $\mathbb{Z}$. By the definition, Step 1, and Lemma 4.1.1 (b) in F,

$$\begin{aligned}\iota_F(x+y) &= \frac{(p\cdot 1_F)(s\cdot 1_F) + (r\cdot 1_F)(q\cdot 1_F)}{(q\cdot 1_F)(s\cdot 1_F)}\\ &= \frac{p\cdot 1_F}{q\cdot 1_F} + \frac{r\cdot 1_F}{s\cdot 1_F} = \iota_F(x) + \iota_F(y),\end{aligned}$$

and in the same way $\iota_F(xy) = \dfrac{(p\cdot 1_F)(r\cdot 1_F)}{(q\cdot 1_F)(s\cdot 1_F)} = \iota_F(x)\,\iota_F(y)$.

Proof. Steps 1–3 (the fold above) show that ι_F is well defined, satisfies the three equations, and $\iota_F(n) = n\cdot 1_F$ for $n \in \mathbb{N}$.

Step 4: order. Let $x < y$ in $\mathbb{Q}$. Then $y - x > 0$, and writing $y - x = p/q$ with $q \in \mathbb{Z}_{>0}$, the comparison formula (Theorem 3.2.1 (a)) gives $p > 0$, so $p \in \mathbb{Z}_{>0}$ (Theorem 2.4.2 (b)). By Proposition 4.2.1 (b), $p\cdot 1_F > 0$ and $q\cdot 1_F > 0$; by Proposition 3.2.2 (a) and (O4), $\iota_F(y - x) > 0$. By Step 3, $\iota_F(y) = \iota_F(x) + \iota_F(y - x) > \iota_F(x)$. So ι_F is strictly increasing. If $x \neq y$, then $x < y$ or $y < x$, and the images differ by (O1): ι_F is injective. If $\iota_F(x) < \iota_F(y)$, then neither $x = y$ nor $y < x$ can hold, since they would give $\iota_F(x) = \iota_F(y)$ or $\iota_F(y) < \iota_F(x)$, against (O1); so $x < y$ by trichotomy.

Step 5: uniqueness. Let $\psi\colon \mathbb{Q}\to F$ satisfy the three equations of the statement. Then $\psi(0) = \psi(0+0) = \psi(0) + \psi(0)$, and adding $-\psi(0)$ gives $\psi(0) = 0$. By Logic and Proof, Axiom 4.2.1 (through Corollary 1.4.3 (b)), $\psi(n) = n\cdot 1_F$ for every $n \in \mathbb{N}$: the base is $\psi(0) = 0 = 0\cdot 1_F$, and $\psi(n+1) = \psi(n) + \psi(1) = n\cdot 1_F + 1_F = (n+1)\cdot 1_F$. For $x \in \mathbb{Q}$, $\psi(x) + \psi(-x) = \psi(0) = 0$, so $\psi(-x) = -\psi(x)$; hence $\psi(m) = m\cdot 1_F$ for every $m \in \mathbb{Z}$. For $q \in \mathbb{Z}_{>0}$, $\psi(q)\,\psi(q^{-1}) = \psi(1) = 1_F$, so $\psi(q^{-1}) = (q\cdot 1_F)^{-1}$ by Lemma 4.1.1 (a). Finally $p/q = pq^{-1}$ (Proposition 3.1.4 (a)), so

$$\psi\Big(\frac{p}{q}\Big) = \psi(p)\,\psi(q^{-1}) = (p\cdot 1_F)(q\cdot 1_F)^{-1} = \iota_F\Big(\frac{p}{q}\Big).$$

So $\psi = \iota_F$. This proves the theorem.

■

For $F = \mathbb{Q}$ the identity satisfies the three equations, so $\iota_{\mathbb{Q}} = \mathrm{id}$. In general we may read "$\mathbb{Q} \subseteq F$", as $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ was read in Chapters 2 and 3; the statements below keep ι_F visible.

Remark. Uniqueness has a consequence that later chapters of this volume use when comparing two ordered fields. Let L be another ordered field and $\chi\colon F \to L$ a map with $\chi(1_F) = 1_L$ that respects $+$ and $\cdot$. Then $\chi \circ \iota_F\colon \mathbb{Q} \to L$ satisfies the three equations of the theorem, so $\chi \circ \iota_F = \iota_L$: such a map fixes the rationals.

4.2.3 The Archimedean property

Section 3.2 proved that every rational lies below a natural number (Theorem 3.2.3). By Proposition 4.2.1 (d), that theorem says: for every $x \in \mathbb{Q}$ there is $n \in \mathbb{N}$ with $x < n \cdot 1_{\mathbb{Q}}$. This statement makes sense in any ordered field, and it deserves a name, because it can fail.

No element of the field lies above all the natural numbers.

Definition 4.2.2. Archimedean Ordered Field

An ordered field F is **Archimedean** if **for every** $x \in F$ **there is** $n \in \mathbb{N}$ with $x < n \cdot 1_F$. In symbols,

$$\forall x \in F \quad \exists n \in \mathbb{N} \quad x < n \cdot 1_F.$$

In words: whatever element of F we pick, some natural number, read in F, lies strictly above it. The n may depend on x; it is chosen after x.

Examples. $\mathbb{Q}$ is Archimedean: this is Theorem 3.2.3, by Proposition 4.2.1 (d). The constructed real numbers of the next chapter are Archimedean too (proved there). A non-example, $\mathbb{Q}(t)$, comes at the end of this section: there the element t lies above every $n \cdot 1$.

The name recalls the ancient axiom about lengths that Exercise B1 states: some multiple $(n \cdot 1_F)y$ of any $y > 0$ exceeds any given x.

The form in which the next section uses the property is about reciprocals, as in Corollary 3.2.4, now in an arbitrary Archimedean ordered field.

Proposition 4.2.3. Small Reciprocals in an Archimedean Field

In an Archimedean ordered field F, **for every** $\varepsilon \in F_{>0}$ **there is** $N \in \mathbb{Z}_{>0}$ such that $1/(n \cdot 1_F) < \varepsilon$ **for every** $n \in \mathbb{N}$ with $n \geq N$. In symbols,

$$\forall \varepsilon \in F_{>0} \quad \exists N \in \mathbb{Z}_{>0} \quad \forall n \in \mathbb{N} \quad \Big(n \geq N \Rightarrow \frac{1}{n \cdot 1_F} < \varepsilon\Big).$$

Proof. Let $\varepsilon \in F_{>0}$. Then $1/\varepsilon > 0$ by Proposition 3.2.2 (a). By Definition 4.2.2 there is $N \in \mathbb{N}$ with $1/\varepsilon < N \cdot 1_F$; it depends on ε. Then $N \neq 0$, since $0 \cdot 1_F = 0 < 1/\varepsilon$, so $N \in \mathbb{Z}_{>0}$. Let $n \geq N$. Then $n \cdot 1_F \geq N \cdot 1_F$ (Proposition 4.2.1 (b), or equality), so $0 < 1/\varepsilon < n \cdot 1_F$,

and Proposition 3.2.2 (b) gives

$$\frac{1}{n \cdot 1_F} < \frac{1}{1/\varepsilon} = \varepsilon.$$

■

The order of the quantifiers is as in Section 3.2: ε first, then N, depending on ε, and then **every** $n \geq N$ at once. That last clause is what Section 4.3 will call "eventually".

Quick check. Find $n \in \mathbb{N}$ with $n > 355/113$ in $\mathbb{Q}$. Then find the least $n \in \mathbb{Z}_{>0}$ with $1/n < 7/500$.

Solution. By the comparison formula (Theorem 3.2.1 (a)), $355/113 < n/1$ if and only if $355 < 113n$. Since $113 \cdot 3 = 339 < 355$ and $113 \cdot 4 = 452 > 355$, $n = 4$ works (and is the least such n).

For $n \in \mathbb{Z}_{>0}$, $1/n < 7/500$ if and only if $500 < 7n$. Since $7 \cdot 71 = 497 < 500$ and $7 \cdot 72 = 504 > 500$, the least such n is 72; every $n \geq 72$ works too, as Proposition 4.2.3 promises.

4.2.4 Integer part and density

In $\mathbb{Q}$, the Archimedean property gave every rational an integer part (Proposition 3.2.5). About $\mathbb{Q}$ the proof used only Archimedes, applied to x and to $-x$; the rest (least elements and discreteness of $\mathbb{Z}$, totality of the order) holds in any ordered field. So it runs in any Archimedean field, with ι_F carrying the integers into F.

Proposition 4.2.4. Integer Part in an Archimedean Field

Suppose that the ordered field F is Archimedean. Then for every $x \in F$ there is **exactly one** $m \in \mathbb{Z}$ with

$$\iota_F(m) \leq x < \iota_F(m+1).$$

Idea. The integer just above x should be the least integer k with $x < \iota_F(k)$. Archimedes applied to x shows that such k exist; applied to $-x$, it bounds them below, so Proposition 2.4.4 supplies a least one. The comparisons happen in F, and ι_F reflecting order (Theorem 4.2.2) carries them back to $\mathbb{Z}$.

Proof. Let $x \in F$ and $S := \{k \in \mathbb{Z} : x < \iota_F(k)\}$. By Definition 4.2.2 there is $n \in \mathbb{N}$ with $x < n \cdot 1_F = \iota_F(n)$ (Theorem 4.2.2), so $n \in S$ and $S \neq \emptyset$. Applied to $-x$, it gives $n' \in \mathbb{N}$ with $-x < \iota_F(n')$, that is, $\iota_F(-n') = -\iota_F(n') < x$ (Proposition 2.4.3 (a); ι_F respects negatives, as in Step 5 of the proof of Theorem 4.2.2). For $k \in S$, $\iota_F(-n') < x < \iota_F(k)$, so $-n' < k$ because ι_F reflects order. Thus $-n'$ is a lower bound of S, and by Proposition 2.4.4 S has a least element l. Let $m := l - 1$, so $l = m + 1$.

Existence. $x < \iota_F(m+1)$, since $l \in S$. And $m \notin S$, since $m < l$ and l is least. So $x < \iota_F(m)$

fails, and $\iota_F(m) \le x$ by totality (Proposition 4.1.2 (a)).

Uniqueness. Let $m, m' \in \mathbb{Z}$ both satisfy the condition, and suppose $m < m'$. By discreteness (Theorem 2.4.2 (c)), $m+1 \le m'$, so $\iota_F(m+1) \le \iota_F(m')$ (Theorem 4.2.2). Then

$$x < \iota_F(m+1) \le \iota_F(m') \le x,$$

so $x < x$, against (O1). By symmetry $m' < m$ is impossible, so $m = m'$.

■

For $F = \mathbb{Q}$, $\iota_{\mathbb{Q}}$ is the identity and this is Proposition 3.2.5. The constructed real numbers of the next chapter will use it with F the new field, where m is an integer approximant of a number not yet known to be anything else.

Between two rationals there is always a third, their midpoint (Section 3.2). In an Archimedean field more is true: between any two elements, however close, lies a **rational**.

Theorem 4.2.5. The Rationals Are Dense in an Archimedean Field

In an Archimedean ordered field F, for all $x, y \in F$ with $x < y$ there is $q \in \mathbb{Q}$ with $x < \iota_F(q) < y$.

Idea. Look for q with denominator n: we need an integer k with $nx < k < ny$, after scaling by n. Integers are spaced 1 apart, so such a k exists once the scaled gap $n(y-x)$ exceeds 1; that is where Archimedes enters, through Proposition 4.2.3. The integer to take is the one just above nx, which the integer part supplies.

Proof. Since $y - x > 0$, Proposition 4.2.3 gives $n \in \mathbb{Z}_{>0}$ with $1/(n \cdot 1_F) < y - x$. Let $c \coloneqq n \cdot 1_F$, which is positive (Proposition 4.2.1 (b)). Multiplying by $c > 0$ (Proposition 2.4.3 (c)) gives $1_F < c(y-x) = cy - cx$, and adding cx, $cx + 1_F < cy$. Let m be the integer part of cx (Proposition 4.2.4), so that $\iota_F(m) \le cx < \iota_F(m+1)$. Then

$$cx < \iota_F(m+1) = \iota_F(m) + 1_F \le cx + 1_F < cy,$$

by Theorem 4.2.2. Multiplying by $c^{-1} > 0$ (Proposition 3.2.2 (a)) gives $x < \iota_F(m+1)c^{-1} < y$. Let $q \coloneqq (m+1)/n \in \mathbb{Q}$. Since $c = \iota_F(n)$ and ι_F respects products, $\iota_F(q)\, c = \iota_F(m+1)$, so $\iota_F(q) = \iota_F(m+1)c^{-1}$, and $x < \iota_F(q) < y$.

■

Applied again, the theorem shows that an interval of an Archimedean field contains infinitely many rationals (Exercise C1). Chapters 5 and 6 use the theorem with F a field of real numbers (Section 5.4 for the constructed reals).

4.2.5 A field that is not Archimedean

Section 3.2 proved the Archimedean property of $\mathbb{Q}$ from the construction of $\mathbb{Q}$: every rational is a quotient of integers. Nothing in (F1)–(F9) and (O1)–(O4) forces it. The standard witness

uses polynomials, which this volume has not built; *Algebraic Preliminaries*, Chapter 4 builds them as formal polynomials. We use the school algebra of polynomials in this one example, and say so. No proof of a numbered result in this chapter cites it; it serves only as a counterexample, in prose, warnings and exercises.

Example. A Non-Archimedean Ordered Field

Let $\mathbb{Q}(t)$ be the set of quotients p/q of polynomials p, q in an indeterminate t with rational coefficients, $q \neq 0$, where $p/q = r/s$ means $ps = rq$, with the usual sum and product of fractions. Call p/q with $p \neq 0$ **positive** if the leading coefficients of p and q have the same sign. With this positive set, $\mathbb{Q}(t)$ is an ordered field; we state this and do not verify it. Show that $t > n \cdot 1$ for every $n \in \mathbb{N}$, so that $\mathbb{Q}(t)$ is **not** Archimedean, and that $0 < 1/t < 1/(n \cdot 1)$ for every $n \geq 1$.

Solution. In $\mathbb{Q}(t)$ the element $n \cdot 1$ is the constant polynomial n, by the recursion of Definition 4.2.1 (a) and the addition of constants. The difference $t - n \cdot 1 = (t - n)/1$ has numerator $t - n$, with leading coefficient 1, and denominator 1, with leading coefficient 1. The signs agree, so $t - n \cdot 1$ is positive, that is, $n \cdot 1 < t$ (Proposition 2.4.1). This holds for every $n \in \mathbb{N}$, so $x = t$ has no n with $x < n \cdot 1$: $\mathbb{Q}(t)$ is not Archimedean.

Next, $1/t$ has numerator 1 and denominator t, both with leading coefficient 1, so $1/t > 0$. For $n \geq 1$, Lemma 4.1.1 (c) gives

$$\frac{1}{n \cdot 1} - \frac{1}{t} = \frac{t - n}{nt},$$

whose numerator has leading coefficient 1 and whose denominator nt has leading coefficient $n > 0$. So the difference is positive, and $1/t < 1/(n \cdot 1)$.

Warning. The Archimedean property does not follow from the ordered-field axioms: $\mathbb{Q}(t)$ satisfies (F1)–(F9) and (O1)–(O4) and fails it. So in a general ordered field the move "take $\varepsilon = 1/n$" does not reach every ε: in $\mathbb{Q}(t)$, the tolerance $\varepsilon = 1/t$ is below every $1/n$. Each later use of "$1/n$ eventually drops below ε" must name the Archimedean property.

The density theorem needs its hypothesis: in $\mathbb{Q}(t)$ no rational lies strictly between 0 and $1/t$.

↻ **Second reading** · *Why no rational lies between 0 and 1/t*

Suppose $q \in \mathbb{Q}$ had $0 < \iota_{\mathbb{Q}(t)}(q) < 1/t$, with $\iota_{\mathbb{Q}(t)}$ the embedding of Theorem 4.2.2. Then $q > 0$, because $\iota_{\mathbb{Q}(t)}$ reflects order. By Proposition 3.2.2 (b) and Lemma 4.1.1 (a), $t = 1/(1/t) < 1/\iota_{\mathbb{Q}(t)}(q) = \iota_{\mathbb{Q}(t)}(1/q)$. By Theorem 3.2.3, $1/q < n$ for some $n \in \mathbb{N}$, so $\iota_{\mathbb{Q}(t)}(1/q) < \iota_{\mathbb{Q}(t)}(n) = n \cdot 1$, and $t < n \cdot 1$, against the example.

4.2.6 Exercises

4.2.6.1 A. Check your understanding

Exercise. A1

(a) Write "F is **not** Archimedean" in symbols, moving the negation inside by Logic and Proof, Proposition 6.2.1.

(b) Give two different elements of $\mathbb{Q}(t)$ (Example (A Non-Archimedean Ordered Field)) that lie above every $n \cdot 1$.

(c) In an ordered field, can $n \cdot 1_F = 0$ for some $n \geq 1$? Give a reason.

(d) What is $\iota_{\mathbb{Q}}$? Which part of Theorem 4.2.2 tells you?

Solution.

(a) The definition is $\forall x \in F\ \exists n \in \mathbb{N}\ \ x < n \cdot 1_F$. By Logic and Proof, Proposition 6.2.1, in its bounded form Logic and Proof, Proposition 6.2.3, applied twice, its negation is $\exists x \in F\ \forall n \in \mathbb{N}\ \ \neg(x < n \cdot 1_F)$, and by totality (Proposition 4.1.2 (a)) $\neg(x < n \cdot 1_F)$ is $n \cdot 1_F \leq x$:

$$\exists x \in F\ \ \forall n \in \mathbb{N}\ \ n \cdot 1_F \leq x.$$

(b) t, as in Example (A Non-Archimedean Ordered Field), and $t+1$, since $t+1 > t > n \cdot 1$.

(c) No. By Proposition 4.2.1 (b), $n \cdot 1_F \geq 1_F > 0$ for $n \geq 1$. (The two-element field of Example (A field with two elements that cannot be ordered) has $2 \cdot 1 = 0$, and it cannot be ordered.)

(d) The identity map of $\mathbb{Q}$: it satisfies $\mathrm{id}(1) = 1$ and respects $+$ and $\cdot$, and the theorem says **exactly one** map does.

4.2.6.2 B. Practice

Exercise. B1: Archimedes' form

Suppose the ordered field F is Archimedean, and take $x \in F$ and $y \in F_{>0}$. Prove that there is $n \in \mathbb{N}$ with $(n \cdot 1_F)\, y > x$.

Solution. Since $y > 0$, $y^{-1} > 0$ (Proposition 3.2.2 (a)). By Definition 4.2.2 applied to x/y, there is $n \in \mathbb{N}$ with $x/y < n \cdot 1_F$. Multiplying by $y > 0$ (Proposition 2.4.3 (c)) gives $(xy^{-1})y < (n \cdot 1_F)y$, and $(xy^{-1})y = x(y^{-1}y) = x$ by (F5), (F6), (F8). Hence $(n \cdot 1_F)y > x$.

Exercise. B2: Smaller than every reciprocal

(a) Let F be Archimedean, and let $x \in F$ satisfy $0 \leq x < 1/(n \cdot 1_F)$ for every $n \in \mathbb{N}$ with $n \geq 1$. Prove that $x = 0$.

(b) Give an element $x \neq 0$ of $\mathbb{Q}(t)$ (Example (A Non-Archimedean Ordered Field)) with this property.

Solution.

(a) Suppose, for a contradiction, that $x \neq 0$. Then $x > 0$, since $x \geq 0$. By Proposition 4.2.3 with $\varepsilon := x$, there is $N \in \mathbb{Z}_{>0}$ with $1/(N \cdot 1_F) < x$. The hypothesis with $n = N \geq 1$ gives $x < 1/(N \cdot 1_F)$. By (O2), $x < x$, against (O1). So $x = 0$.

(b) $x = 1/t$: by Example (A Non-Archimedean Ordered Field), $0 < 1/t < 1/(n \cdot 1)$ for every $n \geq 1$, and $1/t \neq 0$. So part (a) fails without the Archimedean hypothesis.

Exercise. B3: Rationals with a power of two as denominator

(a) Prove that $2^k \geq k + 1$ in $\mathbb{Q}$ for every $k \in \mathbb{N}$.

(b) Let $a, b \in \mathbb{Q}$ with $a < b$. Prove that there are $m \in \mathbb{Z}$ and $k \in \mathbb{N}$ with $a < (m+1)/2^k < b$.

(c) Carry out the construction of (b) for $a = 1/3$, $b = 1/2$, taking k least with $1/(k+1) < b - a$.

Hint: follow the proof of Theorem 4.2.5, with 2^k in place of n.

Solution.

(a) Read 2^k as $(2^k) \cdot 1_{\mathbb{Q}} = (2 \cdot 1_{\mathbb{Q}})^k$ (Proposition 4.2.1 (c), (d)). We use Logic and Proof, Axiom 4.2.1 (through Corollary 1.4.3 (b)) on k for $P(k)$: "$(2^k) \cdot 1_{\mathbb{Q}} \geq (k+1) \cdot 1_{\mathbb{Q}}$". *Base:* $2^0 = 1 = 0 + 1$, so both sides are $1 \cdot 1_{\mathbb{Q}}$. *Step:* assume $P(k)$. Then

$$\begin{aligned}(2^{k+1}) \cdot 1_{\mathbb{Q}} &= ((2^k) \cdot 1_{\mathbb{Q}})(2 \cdot 1_{\mathbb{Q}}) \geq ((k+1) \cdot 1_{\mathbb{Q}})(2 \cdot 1_{\mathbb{Q}}) \\ &= (2k+2) \cdot 1_{\mathbb{Q}} \geq (k+2) \cdot 1_{\mathbb{Q}},\end{aligned}$$

by Proposition 4.2.1 (a) and $2^{k+1} = 2^k \cdot 2$; then Proposition 4.1.2 (d), all factors being non-negative by Proposition 4.2.1 (b); then (a) again with $(k+1)2 = 2k+2$ (Theorem 1.2.3); and finally Proposition 4.2.1 (b), since $k + 2 \leq 2k + 2$. This is $P(k+1)$.

(b) By Proposition 4.2.3 with $F = \mathbb{Q}$ and $\varepsilon := b - a > 0$, there is $k \in \mathbb{N}$ with $1/(k+1) < b - a$ (take $k + 1 \geq N$). By (a) and Proposition 4.1.2 (c), $1/2^k \leq 1/(k+1) < b - a$. Let $m := \lfloor 2^k a \rfloor$ (Proposition 3.2.5), so that $m \leq 2^k a < m+1$. Then $2^k a < m+1 \leq 2^k a + 1$, so

$$a < \frac{m+1}{2^k} \leq a + \frac{1}{2^k} < b,$$

dividing by $2^k > 0$ (Proposition 2.4.3 (c), Proposition 3.2.2 (a)) and using $1/2^k < b - a$ with (O3).

(c) $b - a = 1/6$. For $k \in \mathbb{N}$, $1/(k+1) < 1/6$ if and only if $6 < k + 1$, so the least k is 6, with $2^6 = 64$. Then $2^6 a = 64/3$, and $\lfloor 64/3 \rfloor = 21$, since $21 \cdot 3 = 63 \leq 64 < 66 = 22 \cdot 3$. So $(m+1)/2^k = 22/64 = 11/32$. Check: $1/3 < 11/32$ since $32 < 33$, and $11/32 < 1/2$ since $22 < 32$. (The bound $1/2^k \leq 1/(k+1)$ is far from sharp: $k = 3$ already gives $3/8$.)

4.2.6.3 C. Going deeper

Exercise. C1: Many rationals, and rational approximation

Throughout, F is an Archimedean ordered field.

(a) Let $x < y$ in F. Prove that the set $S := \{q \in \mathbb{Q} : x < \iota_F(q) < y\}$ is infinite (Sets and Functions, Definition 4.1.3).

(b) Let $x \in F$ and $n \in \mathbb{Z}_{>0}$. Prove that there is $q \in \mathbb{Q}$ with $\iota_F(q) \le x < \iota_F(q + 1/n)$.

Solution (Hint 1). For (a), apply Theorem 4.2.5 twice; for (b), apply Proposition 4.2.4 to a multiple of x.

Solution (Hint 2). For (a): let $r, s \in \mathbb{Q}$ with $x < \iota_F(r) < \iota_F(s) < y$; it is enough to find infinitely many rationals strictly between r and s. For (b): let m be the integer part of $(n \cdot 1_F)x$.

Solution.

(a) By Theorem 4.2.5 there is $s \in \mathbb{Q}$ with $x < \iota_F(s) < y$, and, applied to $x < \iota_F(s)$, there is $r \in \mathbb{Q}$ with $x < \iota_F(r) < \iota_F(s)$. Then $r < s$, since ι_F reflects order (Theorem 4.2.2). For $k \in \mathbb{Z}_{>0}$, let

$$s_k := r + \frac{s - r}{k + 1} \in \mathbb{Q}.$$

Since $k+1 > 1 > 0$, $0 < 1/(k+1) < 1$ (Proposition 3.2.2), so $0 < (s-r)/(k+1) < s-r$ (Proposition 2.4.3 (c)), and $r < s_k < s$ by (O3). As ι_F is strictly increasing, $x < \iota_F(r) < \iota_F(s_k) < \iota_F(s) < y$, so $s_k \in S$. If $k < l$, then $1/(l+1) < 1/(k+1)$ (Proposition 3.2.2 (b), Proposition 4.2.1 (b)), so $s_l < s_k$: the map $k \mapsto s_k$ is injective. Suppose S were finite, with m elements for some $m \in \mathbb{N}$ (Sets and Functions, Definition 4.1.2). The map $\{1, \dots, m+1\} \to S$, $k \mapsto s_k$, would be injective, against Sets and Functions, Corollary 4.1.3 (a). So S is infinite.

(b) Let $c := n \cdot 1_F > 0$ (Proposition 4.2.1 (b)) and let m be the integer part of cx (Proposition 4.2.4): $\iota_F(m) \le cx < \iota_F(m+1)$. Multiplying by $c^{-1} > 0$ (Proposition 3.2.2 (a), Proposition 2.4.3 (c), with equality allowed on the left) gives $\iota_F(m)c^{-1} \le x < \iota_F(m+1)c^{-1}$. Let $q := m/n$. Since $c = \iota_F(n)$ and ι_F respects products, $\iota_F(m)c^{-1} = \iota_F(m/n) = \iota_F(q)$ and $\iota_F(m+1)c^{-1} = \iota_F((m+1)/n) = \iota_F(q + 1/n)$, using $(m+1)/n = m/n + 1/n$ (Lemma 4.1.1 (b) in $\mathbb{Q}$). So $\iota_F(q) \le x < \iota_F(q + 1/n)$.

Marking scheme. (10 points) - 2: (a) obtains two rationals $r < s$ inside the interval, by density twice, with $r < s$ justified (order reflected). - 2: (a) exhibits infinitely many distinct rationals between r and s (any explicit injective family, or an induction producing one more each time), each checked to lie strictly between. - 2: (a) concludes that S is infinite from an injection, citing a finiteness result (pigeonhole or subsets of finite sets). - 2: (b) applies the integer part to $(n \cdot 1_F)x$. - 2: (b) divides by $n \cdot 1_F > 0$ and identifies the bounds as $\iota_F(q)$, $\iota_F(q + 1/n)$. Fatal errors (0 for the step): using density in an arbitrary ordered field (the Archimedean hypothesis must be named); "infinitely many because the construction never stops" with no finiteness argument; dividing by $n \cdot 1_F$ without its sign.

Exercise. C2: Where the proof for the rationals breaks

In $\mathbb{Q}(t)$ (Example (A Non-Archimedean Ordered Field)):

(a) Show that the set $\{n \cdot 1 : n \in \mathbb{N}\}$ is bounded above.

(b) The proof of Theorem 3.2.3 shows that every rational lies below a natural number. Which step of that proof has no analogue for the element t of $\mathbb{Q}(t)$? Prove that it has none.

Solution.

(a) By Example (A Non-Archimedean Ordered Field), $n \cdot 1 < t$ for every $n \in \mathbb{N}$, so t is an upper bound.

(b) The first step: "write $x = p/q$ with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$". The rest of the proof (the case split on the sign of p, discreteness of $\mathbb{Z}$, the comparison formula) works for any element of the form $\iota_F(p)/\iota_F(q)$ of an ordered field F, in particular of $\mathbb{Q}(t)$. But t is not of this form. Indeed, $\iota_{\mathbb{Q}(t)}(p)/\iota_{\mathbb{Q}(t)}(q) = \iota_{\mathbb{Q}(t)}(p/q)$ by Theorem 4.2.2; if $t = \iota_{\mathbb{Q}(t)}(r)$ with $r \in \mathbb{Q}$, then Theorem 3.2.3 gives $n \in \mathbb{N}$ with $r < n$, so $t = \iota_{\mathbb{Q}(t)}(r) < \iota_{\mathbb{Q}(t)}(n) = n \cdot 1$, against (a). So t is not the image of any rational, and the proof has nothing to start from.

Marking scheme. (10 points) - 2: (a), t is an upper bound, cited. - 3: (b) names the representation $x = p/q$ as the step that fails. - 3: (b) shows that $t = \iota(r)$ would give $t < n \cdot 1$ by Theorem 3.2.3. - 2: (b) the contradiction with (a). Fatal errors (0 for the step): blaming the comparison formula or discreteness.

4.3 Sequences and Limits in an Ordered Field

In $\mathbb{Q}$ the fractions $1/2, 2/3, 3/4, \ldots$ "tend to 1". Sections 4.1 and 4.2 supplied the tools to say what that means without leaving $\mathbb{Q}$: an absolute value to measure distance, and the Archimedean property to make $1/(n+1)$ small. This section defines the limit of a sequence in an arbitrary ordered field, with tolerances taken from that field. It then proves that a sequence has at most one limit, and that a convergent sequence is bounded. Chapter 5 applies these results with $F = \mathbb{Q}$, at a stage when no other number system is available. On the way it isolates the method behind the limit proofs of the chapter: estimate the distance, find a threshold, then write the proof forwards.

Throughout, F is an ordered field (Definition 2.4.1), and $F_{>0} = \{x \in F : x > 0\}$ is its set of positive elements, as in Section 4.1; for $F = \mathbb{Q}$ this is $\mathbb{Q}_{>0}$. As agreed after Proposition 4.2.1, we write n for $n \cdot 1_F$, so $1/(n+1) := 1/((n+1) \cdot 1_F)$ exists and is positive (Proposition 4.2.1 (b), Proposition 3.2.2 (a)).

4.3.1 Sequences

A list $1, 1/2, 1/3, \ldots$ is a rule that assigns a term to each position, and a rule of that kind already has a name: a function on $\mathbb{N}$.

Definition 4.3.1. Sequence; Eventually

A **sequence** in F is a function $a\colon \mathbb{N} \to F$ (Sets and Functions, Definition 2.1.1). Its value at n is written a_n and called the n-th **term**; the sequence is written $(a_n)_{n\in\mathbb{N}}$, or (a_n), and $\{a_n : n \in \mathbb{N}\}$ is its set of values.

A property $P(n)$ of natural numbers holds **eventually** if **there is** $N_0 \in \mathbb{N}$ such that $P(n)$ holds **for every** $n \in \mathbb{N}$ with $n \geq N_0$.

Examples.

- In $\mathbb{Q}$: $a_n := 1/(n+1)$, with terms $1, 1/2, 1/3, \ldots$.
- In any F: $a_n := n$, that is, $n \cdot 1_F$, with terms $0, 1, 2, \ldots$; and $a_n := (-1)^n$, with the powers of Definition 4.2.1, with terms $1, -1, 1, \ldots$. The second sequence has infinitely many terms but only two values: its set of values is $\{1, -1\}$. A sequence is ordered and may repeat; its set of values forgets both.
- The degenerate case: a **constant sequence** $a_n := c$ for a fixed $c \in F$. It matters because every definition below must give the expected answer on it.
- A sequence may be defined by recursion (Sets and Functions, Theorem 3.3.1, read on $\mathbb{N} = \omega$ as Theorem 1.2.1 through Corollary 1.4.3 (b)). The Newton sequence of Example (A Rational Sequence Aimed at the Gap), $x_0 = 2$ and $x_{n+1} = (x_n^2+2)/(2x_n)$, is a sequence in $\mathbb{Q}$; Section 4.6 studies it.

Non-example by minimal change. The formula $n \mapsto 1/n$ defines no sequence: at $n = 0$ it asks for $1/0$, and 0 has no inverse. Everything else about the first example survives the change, and only "a function on all of $\mathbb{N}$" fails. This is why the book's sequences start at $n = 0$ and why we write $1/(n+1)$.

"Eventually" absorbs any finite beginning. In $\mathbb{Q}$, "$1/(n+1) < 1/10$" holds eventually, from $N_0 = 10$ on, although it fails for $n \leq 9$; "$(-1)^n = 1$" does not hold eventually, since it fails at every odd n. Two eventual properties hold together eventually: if $P(n)$ holds for $n \geq N_0$ and $Q(n)$ for $n \geq N_1$, then both hold for $n \geq \max\{N_0, N_1\}$, the larger of the two numbers, which exists by trichotomy in $\mathbb{N}$ (Theorem 1.3.2 (a)). The same goes for any finite number of properties, taking the larger number one comparison at a time.

↻ **Second reading** · *Infinitely many properties*

For infinitely many properties it fails: "$1/(n+1) < 1/(k+1)$" holds eventually for each $k \in \mathbb{N}$, from $N_0 = k+1$ on, but no single N_0 serves every k, since $k = N_0$ fails at $n = N_0$.

4.3.2 The limit of a sequence

We want a precise sense in which $1/2, 2/3, 3/4, \ldots$ "tends to 1" in $\mathbb{Q}$. The first idea is "the terms get closer and closer to 1". Here is why it fails. With $a_n := (n+1)/(n+2)$,

$$1 - a_n = \frac{1}{n+2}, \qquad 2 - a_n = \frac{n+3}{n+2} = 1 + \frac{1}{n+2}.$$

Both distances shrink at every step, so the terms get "closer and closer" to 2 as well. What singles out 1 is that the distance to 1 falls below **any** tolerance we name, and stays below it,

while the distance to 2 never falls below 1. The tolerances must come from F itself, since in $\mathbb{Q}$ there is nothing else to measure with.

Every tolerance in F is eventually met.

Definition 4.3.2. Limit of a Sequence in an Ordered Field

Let (a_n) be a sequence in F and $a \in F$. The sequence **converges to** a in F, written $a_n \to a$, if **for every** $\varepsilon \in F_{>0}$ **there exists** $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ **for every** $n \in \mathbb{N}$ with $n \geq N$. In symbols,

$$\forall \varepsilon \in F_{>0} \;\; \exists N \in \mathbb{N} \;\; \forall n \in \mathbb{N} \;\; \big(n \geq N \Rightarrow |a_n - a| < \varepsilon\big).$$

Then a is **a limit** of (a_n). The sequence is **convergent** if it converges to some $a \in F$, and **divergent** otherwise.

In words (we say **a** limit until Theorem 4.3.4): the first clause hands us an arbitrary tolerance ε, a positive element of F. The second says that some index N can be found, and it may depend on ε. The third says that from N on, **every** term lies within ε of a: the inequality holds **eventually**. The absolute value is that of Definition 4.1.1, and by Proposition 4.1.3 (d) the condition on each term reads

$$|a_n - a| < \varepsilon \iff a - \varepsilon < a_n < a + \varepsilon.$$

The domain of ε is part of the definition. For $F = \mathbb{Q}$, ε is a **rational** number, and "$a_n \to a$ in $\mathbb{Q}$" mentions nothing outside $\mathbb{Q}$.

Negating the definition, quantifier by quantifier (Logic and Proof, Proposition 6.2.3 for the bounded quantifiers, Logic and Proof, Proposition 6.2.2 for the implication), and using that $\neg(x < \varepsilon)$ means $x \geq \varepsilon$ in the totally ordered F (Proposition 4.1.2 (a)): $a_n \not\to a$ means

$$\exists \varepsilon \in F_{>0} \;\; \forall N \in \mathbb{N} \;\; \exists n \in \mathbb{N} \;\; \big(n \geq N \wedge |a_n - a| \geq \varepsilon\big).$$

One bad tolerance, and beyond every threshold a term that misses it. A sequence is divergent when this holds for **every** $a \in F$.

Example. First Limits

Decide, from Definition 4.3.2, which of these converge, and to what.

(a) A constant sequence $a_n = c$ in any ordered field F.
(b) $a_n = 1/(n+1)$ in $\mathbb{Q}$.
(c) $a_n = (n+1)/(n+2)$ in $\mathbb{Q}$.
(d) $a_n = (-1)^n$ in any ordered field F.

Solution.

(a) $a_n \to c$. Let $\varepsilon \in F_{>0}$ and take $N := 0$. For every n, $|a_n - c| = |0| = 0 < \varepsilon$. Here N does not depend on ε at all: the degenerate case.

(b) $a_n \to 0$. The field $\mathbb{Q}$ is Archimedean (Definition 4.2.2, by Theorem 3.2.3). Let $\varepsilon \in \mathbb{Q}_{>0}$. By Proposition 4.2.3 there is $N \in \mathbb{Z}_{>0}$ with $1/m < \varepsilon$ for every $m \geq N$. For $n \geq N$ also $n + 1 \geq N$, so $|a_n - 0| = 1/(n+1) < \varepsilon$, where the absolute value drops because

$1/(n+1) > 0$. The N depends on ε, and Archimedes is what supplies it.

(c) $a_n \to 1$. By the computation above the definition, $|a_n - 1| = 1/(n+2)$. Given $\varepsilon \in \mathbb{Q}_{>0}$, take the N of (b). For $n \geq N$, $n+2 > n+1 \geq N$, so $1/(n+2) < \varepsilon$.

(d) Divergent, in **every** ordered field. By the recursion for powers, $(-1)^{n+1} = (-1)^n(-1) = -(-1)^n$ (Proposition 2.3.2 (c)), and $|(-1)^n| = 1$ for every n, by induction with Proposition 4.1.3 (b). Hence

$$|a_{n+1} - a_n| = |-2a_n| = 2|a_n| = 2$$

for every n. Suppose $a_n \to a$ for some $a \in F$. Take $\varepsilon := 1$ and its N, and let $n := N$, so that n and $n+1$ are both $\geq N$. The three-point triangle inequality (Proposition 4.1.3 (e)) gives

$$2 = |a_{n+1} - a_n| \leq |a_{n+1} - a| + |a - a_n| < 1 + 1 = 2,$$

so $2 < 2$, against (O1). So no $a \in F$ is a limit. The bad tolerance is $\varepsilon = 1$: two consecutive terms are 2 apart and cannot both lie within 1 of one point.

Quick check. For $a_n = 1/(n+1)$ in $\mathbb{Q}$ and $\varepsilon = 1/100$, which $N \in \mathbb{N}$ satisfy "$|a_n| < \varepsilon$ for every $n \geq N$"?

Solution. Exactly the $N \geq 100$. By the comparison formula (Theorem 3.2.1 (a)), $1/(n+1) < 1/100$ if and only if $100 < n+1$, that is, $n \geq 100$. So every $N \geq 100$ works, and an $N \leq 99$ fails at $n = N$ itself, where $1/(N+1) \geq 1/100$. The definition asks only for **some** working N; any larger one works too.

Each clause of the definition is needed; the fold below shows what goes wrong when one is changed.

Warning. $a_n \to a$ does **not** say that some term equals a: $1/(n+1) \to 0$ in $\mathbb{Q}$ (Example (First Limits) (b)), and no term is 0. Nor does it say that $|a_n - a|$ decreases: $a_n := (1+(-1)^n)/(n+1)$ in $\mathbb{Q}$ has $0 \leq a_n \leq 2/(n+1)$, so $|a_n| < \varepsilon$ as soon as $1/(n+1) < \varepsilon/2$, and it tends to 0; yet its terms run $2, 0, 2/3, 0, 2/5, \ldots$, up and down, hitting 0 at $n = 1, 3, 5, \ldots$.

↻ **Second reading** · *Why this definition, clause by clause*

Non-example by minimal change. Keep the sequence $a_n = (n+1)/(n+2)$ and change the candidate limit from 1 to 2. The distance $|a_n - 2| = 1 + 1/(n+2)$ still shrinks at every step, and still the definition fails, at its first clause: for $\varepsilon := 1$, every term has $|a_n - 2| > 1$, so no N exists. This is the negation displayed above, with $\varepsilon = 1$ and $n = N$ for every N.

- *Strict or not.* Replacing $|a_n - a| < \varepsilon$ by $|a_n - a| \leq \varepsilon$ gives the same notion: given ε, apply the weak form to $\varepsilon/2$, and $\varepsilon/2 < \varepsilon$ (Proposition 4.1.2 (b)).
- *Every $n \geq N$, not some.* With "some $n \geq N$" the sequence $(-1)^n$ would "converge" both to 1 and to -1: beyond any N there are terms equal to each.
- *The order of the quantifiers.* With $\exists N$ before $\forall \varepsilon$, the condition says that $a_n = a$ for every

$n \geq N$: for each such n, $|a_n - a|$ lies below every element of $F_{>0}$, so it is not itself positive, and it is 0 by Proposition 4.1.3 (a). (Exercise A1 (b) asks for the details.) Then $1/(n+1)$ would have no limit in $\mathbb{Q}$.

- *Tolerances from F.* In a field with more elements than $\mathbb{Q}$, "every rational tolerance is eventually met" can be strictly weaker than "every tolerance in F is eventually met". The fold after Proposition 4.3.1 ("Rational tolerances are not enough in $\mathbb{Q}(t)$") shows this happen.

Two kinds of sequence get their own names, because they recur throughout the rest of the chapter.

Definition 4.3.3. Null Sequence

A sequence (a_n) in F is **null** if $a_n \to 0$.

Definition 4.3.4. Bounded Sequence

A sequence (a_n) in F is **bounded** if **there exists** $M \in F_{>0}$ with $|a_n| \leq M$ **for every** $n \in \mathbb{N}$. Such an M is a **bound** for (a_n).

Examples. In $\mathbb{Q}$, $(1/(n+1))$ is null and bounded by 1; in every F, $((-1)^n)$ is bounded by 1 and not null (Example (First Limits) (d)); a constant sequence c is bounded by $|c|+1$, which is positive even when $c = 0$. **Non-example.** In $\mathbb{Q}$ the sequence (n) is not bounded: for any $M \in \mathbb{Q}_{>0}$, Archimedes (Theorem 3.2.3) gives $n \in \mathbb{N}$ with $M < n = |n|$. The bound is required to be positive only for convenience, so that we may divide by it; any $M \geq 0$ that works can be replaced by $M+1$.

4.3.3 Proving a limit from the definition

In Example (First Limits) the distance $|a_n - a|$ was $1/(n+1)$ or $1/(n+2)$, and Archimedes supplied N at once. Usually the distance is messier, and the threshold has to be found before the proof can be written.

Example. A limit found by working backwards

Prove from Definition 4.3.2 that $a_n := \dfrac{n^2+1}{2n^2+3} \to \dfrac{1}{2}$ in $\mathbb{Q}$.

Idea (Scratch work). The goal has the shape $\forall \varepsilon\, \exists N\, \forall n \geq N$. So the proof starts "Let $\varepsilon \in \mathbb{Q}_{>0}$", names an N, and checks every $n \geq N$. Finding N is the work, done backwards from the distance. One fraction:

$$\frac{n^2+1}{2n^2+3} - \frac{1}{2} = \frac{-1}{2(2n^2+3)},$$

so the distance is $1/(2(2n^2+3))$. Solving $1/(2(2n^2+3)) < \varepsilon$ for n exactly would need a square root. But no proof needs the least N, only some N; so we replace the distance by something larger and simpler. Since $2(2n^2+3) - (n+1) = 4n^2 - n + 5 > 0$ for every n, the

distance is at most $1/(n+1)$, and "$1/(n+1) < \varepsilon$ eventually" is Archimedes.

Givens	Goal
$\varepsilon \in \mathbb{Q}_{>0}$	N with $\lvert a_n - \frac{1}{2}\rvert < \varepsilon$ for every $n \geq N$
$\lvert a_n - \frac{1}{2}\rvert = \frac{1}{2(2n^2+3)}$	N with $\frac{1}{2(2n^2+3)} < \varepsilon$ for $n \geq N$
$\frac{1}{2(2n^2+3)} \leq \frac{1}{n+1}$ for every n	N with $\frac{1}{n+1} < \varepsilon$ for $n \geq N$
Proposition 4.2.3: $N \in \mathbb{Z}_{>0}$ with $1/m < \varepsilon$ for $m \geq N$	$n \geq N$ gives $n+1 \geq N$: the gap is closed

The proof runs the table upwards: given ε, choose N, check $n \geq N$.

Solution. Let $\varepsilon \in \mathbb{Q}_{>0}$. By Proposition 4.2.3 ($\mathbb{Q}$ is Archimedean, Theorem 3.2.3) choose $N \in \mathbb{Z}_{>0}$ with $1/m < \varepsilon$ for every $m \geq N$. Bring the difference to one fraction:

$$a_n - \frac{1}{2} = \frac{2(n^2+1) - (2n^2+3)}{2(2n^2+3)} = \frac{-1}{2(2n^2+3)},$$

so the distance is $1/(2(2n^2+3))$. We bound it by $1/(n+1)$. For $n \geq 1$, $n^2 = n \cdot n \geq n \cdot 1 = n$ (Proposition 4.1.2 (d), applied to $0 \leq 1 \leq n$ and $0 \leq n \leq n$), so

$$2(2n^2+3) - (n+1) = 4n^2 - n + 5 \geq 3n + 5 > 0;$$

for $n = 0$ the left side is $5 > 0$. So $0 < n+1 < 2(2n^2+3)$, and Proposition 4.1.2 (c) gives $1/(2(2n^2+3)) \leq 1/(n+1)$. For $n \geq N$, $n+1 \geq N$, so $\lvert a_n - \frac{1}{2}\rvert \leq 1/(n+1) < \varepsilon$.

Archimedes entered only in choosing N; the algebra of the distance holds in every ordered field.

The proof has a shape worth fixing as a method.

Example. Method: proving a limit from the definition

Template. (1) Bring the distance $\lvert a_n - a\rvert$ to one fraction. (2) Bound it, for all $n \geq n_0$, by $C/(n+1)$ with $C \in F_{>0}$, enlarging numerators and shrinking denominators. (3) If F is Archimedean, take N' from Proposition 4.2.3 with $1/m < \varepsilon/C$ for $m \geq N'$, and let $N := \max\{N', n_0\}$. (4) Write forwards: let $\varepsilon \in F_{>0}$; let N be as in (3); let $n \geq N$; then $\lvert a_n - a\rvert \leq C/(n+1) < \varepsilon$.

When to use it. The candidate limit is known, the distance has a formula, and F is Archimedean (as $\mathbb{Q}$ is, Theorem 3.2.3); in a general F, step (3) needs another source of thresholds (Section 4.4).

The usual failure. Writing the backward computation as the proof: that proves the converse, entry (N5) of Logic and Proof, Example (What is not allowed). Choosing N before ε, or letting it depend on n.

Example. A limit with an oscillating numerator

Prove from the definition that $a_n = \dfrac{2n + (-1)^n}{n+1} \to 2$ in $\mathbb{Q}$.

Solution. The steps are numbered as in Example (Method: proving a limit from the definition).

(1) $a_n - 2 = \dfrac{(-1)^n - 2}{n+1}$.

(2) $(-1)^n$ is 1 or -1 (Example (First Limits) (d)), so $|(-1)^n - 2|$ is 1 or 3, and $|a_n - 2| \le 3/(n+1)$ for every n ($n_0 = 0$, $C = 3$).

(3) Let $\varepsilon \in \mathbb{Q}_{>0}$. Then $\varepsilon/3 > 0$; choose $N \in \mathbb{Z}_{>0}$ with $1/m < \varepsilon/3$ for every $m \ge N$ (Proposition 4.2.3).

(4) For $n \ge N$, $n+1 \ge N$, so $|a_n - 2| \le 3/(n+1) < 3 \cdot \varepsilon/3 = \varepsilon$.

The distance itself goes up and down: $1, 3/2, 1/3, 3/4, 1/5, \ldots$. Step (2) replaced it by $3/(n+1)$, which decreases, and only the bound had to be pushed below ε.

Quick check. In the proof just given, take $\varepsilon = 1/10$. Which N does step (3) produce if N is taken least? What is the least N that works in the definition itself?

Solution. $1/m < 1/30$ exactly when $m \ge 31$, so step (3) gives $N = 31$. In the definition: at even n the distance $1/(n+1)$ is below $1/10$ once $n \ge 10$; at odd n the distance $3/(n+1)$ is below $1/10$ exactly when $n + 1 > 30$. At $n = 29$ the distance is $3/30 = 1/10$, not below it; from $n = 30$ on every term qualifies. So the least N is 30, and the proof's larger N costs nothing.

4.3.4 Archimedes and the sequence 1/(n + 1)

In $\mathbb{Q}$, $1/(n+1) \to 0$ came straight from Archimedes. In a general ordered field, the two statements turn out to be the same.

Proposition 4.3.1. Archimedes as a Null Sequence

The sequence $\big(1/((n+1)\cdot 1_F)\big)_{n\in\mathbb{N}}$ is null in F if and only if F is Archimedean.

Idea. One direction is Proposition 4.2.3 read as a limit. For the other, a field that is not Archimedean has an element x above every $n \cdot 1_F$; then every $1/(n+1)$ is at least $1/x$, and $1/x$ is a tolerance that is never met.

Proof. ($\Leftarrow$) Let F be Archimedean and $\varepsilon \in F_{>0}$. By Proposition 4.2.3 there is $N \in \mathbb{Z}_{>0}$

(depending on ε) with $1/(m \cdot 1_F) < \varepsilon$ for every $m \geq N$. For $n \geq N$ also $n+1 \geq N$, so

$$\left|\frac{1}{(n+1)\cdot 1_F} - 0\right| = \frac{1}{(n+1)\cdot 1_F} < \varepsilon,$$

where the absolute value drops because the term is positive. So the sequence is null.

($\Rightarrow$) We prove the contrapositive. Suppose F is not Archimedean. Negating Definition 4.2.2 (Logic and Proof, Proposition 6.2.3, and Proposition 4.1.2 (a) for $\neg(x < n \cdot 1_F) \iff x \geq n \cdot 1_F$), there is $x \in F$ with $x \geq n \cdot 1_F$ for every $n \in \mathbb{N}$. In particular $x \geq 1 \cdot 1_F = 1 > 0$, so $\varepsilon := 1/x \in F_{>0}$ (Proposition 3.2.2 (a)). For every $n \in \mathbb{N}$, $0 < (n+1)\cdot 1_F \leq x$ (Proposition 4.2.1 (b)), so Proposition 4.1.2 (c) gives

$$\left|\frac{1}{(n+1)\cdot 1_F} - 0\right| = \frac{1}{(n+1)\cdot 1_F} \geq \frac{1}{x} = \varepsilon.$$

So for this ε no N works, even with $n = N$, and the sequence is not null.

■

So in $\mathbb{Q}(t)$, the field of Example (A Non-Archimedean Ordered Field), which is not Archimedean, $1/(n+1)$ does not tend to 0: the tolerance $1/t$ is never met.

Second reading · *Rational tolerances are not enough in $\mathbb{Q}(t)$*

The same example shows why tolerances must come from F. In $\mathbb{Q}(t)$, every **rational** tolerance is eventually met by $1/(n+1)$. Write ι for the embedding ι_F of Theorem 4.2.2. Since ι respects products, $\iota(1) = 1$ and $\iota(n+1) = (n+1)\cdot 1$, we have $((n+1)\cdot 1)\cdot \iota(1/(n+1)) = \iota(1) = 1$, so the term $1/((n+1)\cdot 1)$ is $\iota(1/(n+1))$. Given $q \in \mathbb{Q}_{>0}$, Example (First Limits) (b) gives N with $1/(n+1) < q$ in $\mathbb{Q}$ for every $n \geq N$, and ι, being strictly increasing, carries this to $\iota(1/(n+1)) < \iota(q)$. Yet the sequence is not null in $\mathbb{Q}(t)$. For $F = \mathbb{Q}$ the two readings coincide, because there every tolerance is rational.

4.3.5 Changing finitely many terms

Whether a sequence converges, and to what, depends only on its terms from some point on. Two forms of this are used constantly: altering finitely many terms, and dropping the first k.

Proposition 4.3.2. Only the Tail Matters

Let (a_n) and (b_n) be sequences in F, and $a \in F$.

(a) If there is $N_0 \in \mathbb{N}$ with $a_n = b_n$ for every $n \geq N_0$, then $a_n \to a$ if and only if $b_n \to a$; in particular (a_n) is null if and only if (b_n) is; and (a_n) is bounded if and only if (b_n) is.

(b) For $k \in \mathbb{N}$, the shifted sequence $(a_{n+k})_{n\in\mathbb{N}}$ converges to a if and only if (a_n) does.

Idea. For limits, a threshold for one sequence becomes a threshold for the other once it is pushed past N_0, or moved by k. For bounds, the finitely many terms before N_0 are absorbed

by a finite maximum.

Proof.

(a) *Limits.* Suppose $a_n \to a$, and let $\varepsilon \in F_{>0}$. Choose N with $|a_n - a| < \varepsilon$ for every $n \geq N$, and let $N' \coloneqq \max\{N, N_0\}$. For $n \geq N'$, $b_n = a_n$ and $|b_n - a| = |a_n - a| < \varepsilon$. So $b_n \to a$. The hypothesis is symmetric in the two sequences, so the converse follows by exchanging them. The case $a = 0$ is the statement about null sequences.

↻ **Second reading** · *The bound part of (a)*

Bounds. Suppose $|a_n| \leq M$ for every n, with $M \in F_{>0}$. If $N_0 = 0$, the two sequences are equal. Otherwise let

$$M' \coloneqq \max\{|b_0|, \ldots, |b_{N_0-1}|, M\},$$

the maximum of finitely many elements of F, which exists by Sets and Functions, Proposition 3.3.8 (its proof works in any ordered field, as *Sets and Functions*, Section 5.3 remarks). Then $M' \geq M > 0$. For $n < N_0$, $|b_n| \leq M'$ because $|b_n|$ is one of the listed elements; for $n \geq N_0$, $|b_n| = |a_n| \leq M \leq M'$. So M' bounds (b_n). The converse is the same with the sequences exchanged.

(b) Suppose $a_n \to a$, and let $\varepsilon \in F_{>0}$, with N as in the definition. For $n \geq N$, also $n + k \geq N$, so $|a_{n+k} - a| < \varepsilon$. Hence $(a_{n+k})_n$ converges to a, with the same N.

Conversely, suppose $a_{n+k} \to a$, and let $\varepsilon \in F_{>0}$, with N such that $|a_{m+k} - a| < \varepsilon$ for every $m \geq N$. Let $n \geq N + k$. Then $n = m + k$ for the natural number $m \coloneqq n - k$, and $m \geq N$ (Proposition 1.3.1 (a), (c)). So $|a_n - a| = |a_{m+k} - a| < \varepsilon$, and $N + k$ is a threshold for (a_n). This proves the proposition.

■

The shift clause is about convergence only; it is used to start a sequence later, for instance to read a formula such as $1/n$, which needs $n \geq 1$, as the shift by 1 of a sequence indexed from 0.

One more reformulation moves every limit to 0.

Proposition 4.3.3. Limits Through Null Sequences

For a sequence (a_n) in F and $a \in F$: $a_n \to a$ if and only if the sequence $(a_n - a)$ is null.

Proof. For every n, $|(a_n - a) - 0| = |a_n - a|$. So the two conditions of Definition 4.3.2, for (a_n) with limit a and for $(a_n - a)$ with limit 0, say word for word the same thing, with the same N for each ε.

■

4.3.6 A sequence has at most one limit

If "the terms are eventually within every tolerance of a" is to name **the** number the sequence tends to, it must single out one a. It does.

Theorem 4.3.4. Uniqueness of Limits

Let (a_n) be a sequence in F and $a, b \in F$. If $a_n \to a$ and $a_n \to b$, then $a = b$.

Try it first (ten minutes): suppose $a_n \to a$ and $a_n \to b$ with $a \neq b$. Which ε makes the two tails collide?

Idea. The natural first choice is $\varepsilon := |a - b|$, positive because $a \neq b$; it is the choice that worked in Logic and Proof, Proposition 6.2.5. Beyond both thresholds a term a_n is within $|a - b|$ of a and of b, and the triangle inequality through a_n yields only $|a - b| < 2|a - b|$: true, and no contradiction. The two tolerances **add**. So give each half: with $\varepsilon := |a - b|/2$, the same chain ends in $|a - b| < |a - b|$.

Proof. Let $a_n \to a$ and $a_n \to b$, and suppose, for a contradiction, that $a \neq b$. Then $a - b \neq 0$, so $|a - b| > 0$ (Proposition 4.1.3 (a)), and

$$\varepsilon := \frac{|a - b|}{2} \in F_{>0}$$

by Proposition 4.1.2 (b); it depends only on a and b. Since $a_n \to a$, there is N_1 with $|a_n - a| < \varepsilon$ for every $n \geq N_1$; since $a_n \to b$, there is N_2 with $|a_n - b| < \varepsilon$ for every $n \geq N_2$. Let $n := \max\{N_1, N_2\}$. Then both inequalities hold at this n, and the three-point triangle inequality (Proposition 4.1.3 (e)), with $|a - a_n| = |a_n - a|$ by part (b) of the same proposition, gives

$$|a - b| \leq |a - a_n| + |a_n - b| < \varepsilon + \varepsilon = |a - b|.$$

So $|a - b| < |a - b|$, against (O1). This contradiction shows $a = b$.

■

Since a convergent sequence has exactly one limit, we may speak of **the** limit and write it $\lim a_n$, or $\lim_{n\to\infty} a_n$. So $\lim a_n = a$ means the same as $a_n \to a$. In $\mathbb{Q}$, Example (First Limits) gives $\lim 1/(n+1) = 0$ and $\lim (n+1)/(n+2) = 1$, while $\lim (-1)^n$ names nothing.

4.3.7 Convergent sequences are bounded

A convergent sequence cannot run off: from some index on, its terms are trapped near the limit, and only finitely many terms come before that index.

Proposition 4.3.5. Convergent Sequences Are Bounded

Every convergent sequence in F is bounded.

Idea. Use the definition once, with the tolerance 1: from some N on, $|a_n| \le |a_n - a| + |a| < 1 + |a|$. The terms before N are finitely many, so one of them, or $|a| + 1$, is the largest absolute value in sight.

Proof. Let $a_n \to a$. Applying Definition 4.3.2 with $\varepsilon := 1 \in F_{>0}$ gives $N \in \mathbb{N}$ with $|a_n - a| < 1$ for every $n \ge N$. For such n, the triangle inequality (Proposition 4.1.3 (e), with $x = a_n - a$ and $y = a$) gives

$$|a_n| = |(a_n - a) + a| \le |a_n - a| + |a| < 1 + |a|.$$

If $N = 0$, let $M := |a| + 1$. Otherwise let

$$M := \max\{|a_0|, \dots, |a_{N-1}|, |a| + 1\},$$

which exists by Sets and Functions, Proposition 3.3.8 in the totally ordered F (as in the proof of Proposition 4.3.2). In both cases $M \ge |a| + 1 \ge 1 > 0$, by Proposition 4.1.3 (a), so $M \in F_{>0}$. For $n < N$, $|a_n| \le M$ because $|a_n|$ is one of the listed elements; for $n \ge N$, $|a_n| < |a| + 1 \le M$. So M bounds (a_n).

■

The bound M depends on the whole sequence, through a, N and the first N terms, but on no tolerance: it is one element of F, fixed once. Section 4.4 uses exactly that, in the product of two limits. The converse fails: $((-1)^n)$ is bounded by 1 and divergent (Example (First Limits) (d)).

4.3.8 Exercises

4.3.8.1 A. Check your understanding

Exercise. A1

(a) Write "$a_n \not\to a$" in symbols, and name the law used at each step.

(b) What does $\exists N \in \mathbb{N}\ \forall \varepsilon \in F_{>0}\ \forall n \in \mathbb{N}\ (n \ge N \Rightarrow |a_n - a| < \varepsilon)$ say about (a_n)? Does $(1/(n+1))$ satisfy it in $\mathbb{Q}$, with $a = 0$?

(c) For $a_n = (2n+1)/(n+1)$ in $\mathbb{Q}$, find the least N that works in Definition 4.3.2 for the limit 2 and $\varepsilon = 1/50$.

Solution.

(a) Start from $\neg\forall \varepsilon \in F_{>0}\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \ge N \Rightarrow |a_n - a| < \varepsilon)$. The bounded negation laws (Logic and Proof, Proposition 6.2.3 (a), then (b), then (a)) move $\neg$ past the three quantifiers, turning each $\forall$ into $\exists$ and the $\exists$ into $\forall$. That leaves $\neg(n \ge N \Rightarrow |a_n - a| < \varepsilon)$, which is $n \ge N \wedge \neg(|a_n - a| < \varepsilon)$ by Logic and Proof, Proposition 6.2.2. Finally $\neg(|a_n - a| < \varepsilon)$ is $|a_n - a| \ge \varepsilon$, by trichotomy in F (Proposition 4.1.2 (a)). The result:

$$\exists \varepsilon \in F_{>0}\ \ \forall N \in \mathbb{N}\ \ \exists n \in \mathbb{N}\ \ (n \ge N \wedge |a_n - a| \ge \varepsilon).$$

(b) It says that $a_n = a$ for every $n \ge N$: the sequence is eventually constant, equal to a. Indeed, fix $n \ge N$. Then $|a_n - a| < \varepsilon$ for every $\varepsilon \in F_{>0}$. If $a_n \ne a$, then

$\varepsilon := |a_n - a|$ is in $F_{>0}$ (Proposition 4.1.3 (a)) and gives $|a_n - a| < |a_n - a|$, against (O1); so $a_n = a$. Conversely, a sequence with $a_n = a$ for every $n \geq N$ satisfies the formula, since $|a_n - a| = |0| = 0 < \varepsilon$. The sequence $(1/(n+1))$ does not satisfy it: no term equals 0. It converges to 0, but its N must depend on ε. The quantifier order is the whole difference.

(c) $a_n - 2 = (2n + 1 - 2(n+1))/(n+1) = -1/(n+1)$, so $|a_n - 2| = 1/(n+1)$. By the comparison formula, $1/(n+1) < 1/50$ if and only if $n + 1 > 50$, that is, $n \geq 50$. So $N = 50$ works, and every $N \leq 49$ fails at $n = 49$ (which is $\geq N$), where $|a_{49} - 2| = 1/50$. The least N is 50.

4.3.8.2 B. Practice

Exercise. B1: Limits from the definition

Prove from Definition 4.3.2, following Example (Method: proving a limit from the definition), that in $\mathbb{Q}$:

(a) $\dfrac{3n+2}{n+4} \to 3$;

(b) $\dfrac{n+5}{n^2+1} \to 0$.

Say where the Archimedean property is used.

Solution.

(a) (a) $\dfrac{3n+2}{n+4} - 3 = \dfrac{-10}{n+4}$, so the distance is $10/(n+4)$. (2) $0 < n+1 < n+4$, so $1/(n+4) < 1/(n+1)$ (Proposition 3.2.2 (b)), and the distance is $< 10/(n+1)$. (3) Given $\varepsilon \in \mathbb{Q}_{>0}$, choose $N \in \mathbb{Z}_{>0}$ with $1/m < \varepsilon/10$ for $m \geq N$ (Proposition 4.2.3). (4) For $n \geq N$, $n + 1 \geq N$, so the distance is $< 10/(n+1) < \varepsilon$.

(b) (a) The distance is $(n+5)/(n^2+1)$. (2) For $n \geq 1$: $n + 5 \leq 6n$; $n^2 < n^2 + 1$ gives $1/(n^2+1) < 1/n^2$; so the distance is $< 6n/n^2 = 6/n$, and $6/n \leq 12/(n+1)$ because $n + 1 \leq 2n$. So the distance is $< 12/(n+1)$ for $n \geq 1$ ($n_0 = 1$; at $n = 0$ the term is 5). (3) Choose $N \in \mathbb{Z}_{>0}$ with $1/m < \varepsilon/12$ for $m \geq N$; then $N \geq 1 = n_0$. (4) For $n \geq N$: $n \geq 1$ and $n + 1 \geq N$, so the distance is $< 12/(n+1) < \varepsilon$.

Where Archimedes is used: in step (3) only, both times.

Exercise. B2

Prove that if $a_n \to a$ in F, then $|a_n| \to |a|$. Show by an example that the converse fails, and prove that it holds when $a = 0$.

Solution. Let $\varepsilon \in F_{>0}$, and choose N with $|a_n - a| < \varepsilon$ for $n \geq N$. By the reverse triangle inequality (Proposition 4.1.3 (f)), for $n \geq N$,

$$||a_n| - |a|| \leq |a_n - a| < \varepsilon.$$

So $|a_n| \to |a|$, with the same N.

The converse fails for $a_n = (-1)^n$ and $a = 1$: $|a_n| = 1$ for every n, a constant sequence, so $|a_n| \to 1 = |1|$ (Example (First Limits) (a)); but (a_n) converges to nothing (part (d) of that example).

For $a = 0$ the converse holds: $||a_n| - |0|| = ||a_n|| = |a_n| = |a_n - 0|$, by Proposition 4.1.3 (a), since $|a_n| \geq 0$. So the two conditions coincide: a sequence is null if and only if its absolute values form a null sequence.

Exercise. B3: Bounded but not convergent, in $\mathbb{Q}(t)$

In the field $\mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field), show that the sequence $(n \cdot 1)_{n \in \mathbb{N}}$ is bounded and does not converge. Is (n) bounded in $\mathbb{Q}$?

Solution. Bounded. By Example (A Non-Archimedean Ordered Field), $t > n \cdot 1$ for every $n \in \mathbb{N}$. Also $n \cdot 1 \geq 0$ (Proposition 4.2.1 (b), and $0 \cdot 1 = 0$), so $|n \cdot 1| = n \cdot 1 < t$. And $t > 1 \cdot 1 = 1 > 0$, so $M := t \in \mathbb{Q}(t)_{>0}$ is a bound.

Not convergent. Suppose $n \cdot 1 \to a$ for some $a \in \mathbb{Q}(t)$. Take $\varepsilon := 1/2$, which is positive (Proposition 4.1.2 (b)), and its N. With $n := N$, both n and $n + 1$ are $\geq N$, and $(n + 1) \cdot 1 - n \cdot 1 = 1$ by Definition 4.2.1 (a). The three-point triangle inequality gives

$$1 = |(n + 1) \cdot 1 - n \cdot 1| \leq |(n + 1) \cdot 1 - a| + |a - n \cdot 1| < \tfrac{1}{2} + \tfrac{1}{2} = 1,$$

against (O1). So the sequence diverges.

In $\mathbb{Q}$. No: by the non-example after Definition 4.3.4, Archimedes puts some n above any proposed bound. The same sequence is bounded in one field and unbounded in another. Boundedness is a property of the sequence **in** F, like convergence.

Exercise. B4: Prove or disprove

Let F be an ordered field and (a_n), (b_n) sequences in F. Prove each true statement from Definition 4.3.2 and refute each false one with a counterexample.

(a) If $a_n \to a$, then $(a_{2n})_{n \in \mathbb{N}}$ converges to a.
(b) If $(a_{2n})_n$ and $(a_{2n+1})_n$ both converge, then (a_n) converges.
(c) If (a_n) is null and $|b_n| \leq |a_n|$ for every n, then (b_n) is null.
(d) If $a_n \to a$, then $(a_{n+1} - a_n)$ is null.
(e) If $(a_n b_n)$ is null, then (a_n) or (b_n) is null.
(f) In $F = \mathbb{Q}$: if $a_n \to a$ and every a_n is an integer, then there is N with $a_n = a_N$ for every $n \geq N$.

Solution.

(a) **True.** Given ε, take the N of $a_n \to a$. For $n \geq N$, $2n = n + n \geq n \geq N$, so $|a_{2n} - a| < \varepsilon$.
(b) **False.** Take $a_n = (-1)^n$ in $\mathbb{Q}$. Then $(-1)^{2n} = 1$ and $(-1)^{2n+1} = -1$ for every n (induction, from $(-1)^{k+2} = (-1)^k$), so the two subsequences are constant, with limits

1 and -1; and (a_n) diverges (Example (First Limits) (d)).

(c) **True.** Given ε, take N with $|a_n| < \varepsilon$ for $n \geq N$; then $|b_n - 0| \leq |a_n| < \varepsilon$ for $n \geq N$.

(d) **True.** Given ε, take N with $|a_n - a| < \varepsilon/2$ for $n \geq N$. For $n \geq N$ also $n+1 \geq N$, and $|a_{n+1} - a_n| \leq |a_{n+1} - a| + |a - a_n| < \varepsilon$.

(e) **False.** In $\mathbb{Q}$ take $a_n = (1 + (-1)^n)/2$ and $b_n = (1 - (-1)^n)/2$. Since $(-1)^n$ is 1 or -1 (Example (First Limits) (d)), their values are $1, 0, 1, \ldots$ and $0, 1, 0, \ldots$, and $a_n b_n = 0$ for every n: the constant 0, which is null. Neither factor is null: for $\varepsilon = 1$ and any N, one of $N, N+1$ is even and the other odd, and there $|a_n| = 1$, resp. $|b_n| = 1$.

(f) **True.** Take N with $|a_n - a| < 1/2$ for $n \geq N$. For $n \geq N$, $k := a_n - a_N$ is an integer with $|k| \leq |a_n - a| + |a - a_N| < 1$. If $k > 0$, then $k \geq 1$ by discreteness (Theorem 2.4.2 (c); the order of $\mathbb{Q}$ agrees with that of $\mathbb{Z}$ on integers, by the comparison formula of Theorem 3.2.1 (a) with denominators 1), against $|k| < 1$; if $k < 0$, the same with $-k$. So $k = 0$, that is, $a_n = a_N$.

4.3.8.3 C. Going deeper

Exercise. C1: A positive limit keeps the terms positive

(a) Prove that if $a_n \to a$ in F and $a > 0$, then $a_n > a/2$ eventually; in particular $a_n > 0$ eventually.

(b) Give a sequence in $\mathbb{Q}$ with $a_n > 0$ for every n whose limit is not positive. What does this say about the converse of (a)?

Solution.

(a) Let $\varepsilon := a/2$, which lies in $F_{>0}$ by Proposition 4.1.2 (b), since $a > 0$. Choose N with $|a_n - a| < a/2$ for every $n \geq N$. By Proposition 4.1.3 (d), for such n,

$$a_n > a - \frac{a}{2} = \frac{a}{2} > 0,$$

where $a - a/2 = a/2$ because $a/2 + a/2 = a$ (Proposition 4.1.2 (b)). So $a_n > a/2$ for every $n \geq N$.

(b) $a_n = 1/(n+1)$: every term is positive, and the limit is 0 (Example (First Limits) (b)). So "every term positive" does not give "limit positive": the strict inequality is lost in the limit. What survives is the weak inequality $a \geq 0$, proved in Section 4.4.

Exercise. C2: Find the error

Here is an argument for Proposition 4.3.5. Is the conclusion true? Is the argument correct? If not, name the line that fails, say what makes it illegal, and give a counterexample to the step. Then repair the argument.

> (1) Let $a_n \to a$. (2) Take $\varepsilon := 1$ in the definition of the limit, and let N be the index it provides. (3) Then for every $n \in \mathbb{N}$, $|a_n| \leq |a_n - a| + |a| < 1 + |a|$. (4) So $M := 1 + |a|$ is a bound for (a_n).

Solution. The conclusion is **true**: it is Proposition 4.3.5. The argument is **not** correct.

The failing line is (3). The first inequality is the triangle inequality and is fine. The second uses $|a_n - a| < 1$ "for every $n \in \mathbb{N}$". Line (2) supplies it only for $n \geq N$: it is the bounded universal $\forall n \in \mathbb{N}\,(n \geq N \Rightarrow |a_n - a| < 1)$. Instantiating it at n needs the line $n \geq N$ (the bounded universal applies only to $n \geq N$), and for $n < N$ no such line is available. Line (3) cites a line that is not there: entry (N9) of Logic and Proof, Example (What is not allowed).

Counterexample to the step. In $\mathbb{Q}$, let $a_0 := 5$ and $a_n := 0$ for $n \geq 1$, with $a = 0$. Then $a_n \to 0$, and $N = 1$ is a legitimate choice in line (2) for $\varepsilon = 1$, since $|a_n - 0| = 0 < 1$ for $n \geq 1$. At $n = 0$, line (3) claims $|a_0| < 1 + |0| = 1$, but $|a_0| = 5$; and the $M = 1$ of line (4) is not a bound. No counterexample to the conclusion exists: 5 is a bound.

Repair. Keep line (3) for $n \geq N$ only, and take $M := \max\{|a_0|, \ldots, |a_{N-1}|, |a|+1\}$, the finite maximum of the proof of Proposition 4.3.5.

Marking scheme. (10 points) - 2: says that the conclusion is true and the argument is not correct. - 3: names line (3) and the use of $|a_n - a| < 1$ for $n < N$, where the hypothesis does not apply. - 3: gives a convergent sequence and a legitimate N for which the step's inequality fails at some $n < N$, and checks it. - 2: repairs the argument by a bound that also covers the finitely many terms before N (a finite maximum, or a finite sum of $|a_0|, \ldots, |a_{N-1}|$ and $|a|+1$). Fatal errors (0 for the step): declaring the conclusion false; offering a counterexample to the conclusion; blaming the triangle inequality.

4.4 The Algebra and Order of Limits

So far every limit has been found by hand: we computed the distance from a_n to the candidate limit and bounded it, as in Example (First Limits) and Example (A limit found by working backwards). This section proves the rules that make most such work unnecessary. Sums, multiples, products and quotients of convergent sequences converge to the sum, multiple, product and quotient of the limits; limits respect $\leq$; and a sequence trapped between two sequences with a common limit has that limit too. Everything holds in an arbitrary ordered field F. No result of this section uses the Archimedean property; only the examples in $\mathbb{Q}$ do, through Example (First Limits) (b) or the non-example after Definition 4.3.4.

Throughout, F is an ordered field and $F_{>0}$ its set of positive elements, as in Section 4.3.

4.4.1 Two tools

The first tool says that a small factor wins against a factor that merely stays bounded.

Proposition 4.4.1. Null Times Bounded Is Null

Let (a_n) be a null sequence in F, and let (b_n) be a sequence with $|b_n| \leq M$ for every n, where $M \in F_{>0}$. Then $(a_n b_n)$ is null.

Proof. Let $\varepsilon \in F_{>0}$. Then $\varepsilon/M \in F_{>0}$ (Proposition 3.2.2 (a) and (O4)). Since (a_n) is null,

there is N with $|a_n| < \varepsilon/M$ for every $n \geq N$. For such n, by Proposition 4.1.3 (b),

$$|a_n b_n - 0| = |a_n||b_n| \leq |a_n| M < \frac{\varepsilon}{M} M = \varepsilon,$$

where the first inequality multiplies $|b_n| \leq M$ by $|a_n| \geq 0$ (Proposition 4.1.2 (d)), and the second multiplies $|a_n| < \varepsilon/M$ by $M > 0$ (Proposition 2.4.3 (c)).

■

For instance, $((-1)^n/(n+1))$ is null in $\mathbb{Q}$: it is $1/(n+1)$, null by Example (First Limits) (b), times $(-1)^n$, bounded by 1. The bound M is fixed before ε and enters the tolerance ε/M. That is the pattern of the product law below.

The second tool is the fact behind every division by b_n: if $b_n \to b \neq 0$, the terms eventually stay away from 0, by a margin we can name.

Lemma 4.4.2. Terms Near a Non-zero Limit

Let $b_n \to b$ in F with $b \neq 0$. Then there is $N_0 \in \mathbb{N}$ with $|b_n| > |b|/2$ for every $n \geq N_0$. If $b > 0$, then moreover $b_n > b/2$ for every $n \geq N_0$.

Proof. Since $b \neq 0$, $|b| > 0$ (Proposition 4.1.3 (a)), so $\varepsilon := |b|/2 \in F_{>0}$ (Proposition 4.1.2 (b)). Choose N_0 (depending on (b_n) and b) with $|b_n - b| < |b|/2$ for every $n \geq N_0$. For such n, the reverse triangle inequality (Proposition 4.1.3 (f)), together with $x \leq |x|$ and $|-x| = |x|$ (parts (c), (b)), gives

$$|b| - |b_n| \leq \big||b| - |b_n|\big| \leq |b - b_n| = |b_n - b| < \frac{|b|}{2},$$

so $|b_n| > |b| - |b|/2 = |b|/2$, adding $|b_n| - |b|/2$ to both sides (Proposition 2.4.3 (b)). If $b > 0$, then $|b| = b$, and Proposition 4.1.3 (d) turns $|b_n - b| < b/2$ into $b - b/2 < b_n$, that is, $b_n > b/2$.

■

In particular $b_n \neq 0$ for every $n \geq N_0$, since $|b_n| > |b|/2 > 0$. The lemma needs $b \neq 0$. In $\mathbb{Q}$, $b_n := (1 + (-1)^n)/(n+1)$ tends to 0 (the warning in Section 4.3), yet $b_n = 0$ at every odd n, so for $b = 0$ no N_0 has $|b_n| > |b|/2 = 0$ for every $n \geq N_0$.

4.4.2 The limit laws

Theorem 4.4.3. Limit Laws

Let $a_n \to a$ and $b_n \to b$ in F, and let $c \in F$. Then:

(a) $a_n + b_n \to a + b$;
(b) $ca_n \to ca$; in particular $-a_n \to -a$ and $a_n - b_n \to a - b$;
(c) $a_n b_n \to ab$;
(d) if $b \neq 0$ and $b_n \neq 0$ for every n, then $a_n/b_n \to a/b$.

Idea. Each part keeps the shape of Example (Method: proving a limit from the definition) (estimate the distance, then choose the threshold and write forwards), but the threshold comes from the hypotheses $a_n \to a$, $b_n \to b$ instead of from Archimedes, and the estimate bounds the distance by $|a_n - a|$, $|b_n - b|$ through split ε or bound-then-shrink. Work backwards from each target distance. For (a), $|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b|$: a sum of two terms, each of which we can make small, so we give each half the tolerance. For (c), write

$$a_n b_n - ab = a_n(b_n - b) + b(a_n - a).$$

The factors $b_n - b$ and $a_n - a$ can be made small, but a_n cannot. It is, however, bounded (Proposition 4.3.5), so we replace $|a_n|$ by a bound M and ask $|b_n - b|$ to be smaller than $\varepsilon/(2M)$. For (d), it is enough to treat $1/b_n$ and then use (c). *Before reading on: why must M be fixed before ε?*

Proof.

(a) Let $\varepsilon \in F_{>0}$. Then $\varepsilon/2 \in F_{>0}$ (Proposition 4.1.2 (b)). Since $a_n \to a$, choose N_1 (depending on ε) with $|a_n - a| < \varepsilon/2$ for every $n \geq N_1$; since $b_n \to b$, choose N_2 with $|b_n - b| < \varepsilon/2$ for every $n \geq N_2$. Let $N := \max\{N_1, N_2\}$. For $n \geq N$, both estimates hold, and the triangle inequality (Proposition 4.1.3 (e)) gives

$$\begin{aligned} |(a_n + b_n) - (a + b)| &= |(a_n - a) + (b_n - b)| \\ &\leq |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

(b) If $c = 0$, then $ca_n = 0 = ca$ for every n (Proposition 2.3.2 (b)), a constant sequence, which converges to 0 (Example (First Limits) (a)). Let $c \neq 0$, so $|c| > 0$ (Proposition 4.1.3 (a)), and let $\varepsilon \in F_{>0}$. Then $\varepsilon/|c| \in F_{>0}$ (Proposition 3.2.2 (a) and (O4)). Choose N with $|a_n - a| < \varepsilon/|c|$ for every $n \geq N$. For such n, by distributivity and Proposition 4.1.3 (b),

$$|ca_n - ca| = |c(a_n - a)| = |c||a_n - a| < |c| \frac{\varepsilon}{|c|} = \varepsilon,$$

multiplying by $|c| > 0$ (Proposition 2.4.3 (c)). With $c = -1$, and $(-1)x = -x$ (Proposition 2.3.2 (c)), this gives $-a_n \to -a$. Then (a), applied to (a_n) and $(-b_n)$, gives $a_n - b_n = a_n + (-b_n) \to a + (-b) = a - b$.

(c) By Proposition 4.3.5 there is $M \in F_{>0}$ with $|a_n| \leq M$ for every $n \in \mathbb{N}$. We fix this M now; it depends on the sequence (a_n) only. Now let $\varepsilon \in F_{>0}$. Since $M > 0$ and $|b| + 1 \geq 1 > 0$, the tolerances $\varepsilon/(2M)$ and $\varepsilon/(2(|b|+1))$ are positive (Proposition 4.1.2 (b), (O4), Proposition 3.2.2 (a)). Since $b_n \to b$, choose N_1 with $|b_n - b| < \varepsilon/(2M)$ for every $n \geq N_1$. Since $a_n \to a$, choose N_2 with $|a_n - a| < \varepsilon/(2(|b|+1))$ for every $n \geq N_2$. N_1 depends on ε and M, N_2 on ε and b (and both on the given sequences); neither depends on n. Let $N := \max\{N_1, N_2\}$, and let $n \geq N$. Using the identity of the Idea, the triangle inequality and the product rule (Proposition 4.1.3 (e), (b)),

$$\begin{aligned} |a_n b_n - ab| &\leq |a_n||b_n - b| + |b||a_n - a| \\ &\leq M|b_n - b| + (|b| + 1)|a_n - a| \\ &< M \frac{\varepsilon}{2M} + (|b| + 1) \frac{\varepsilon}{2(|b| + 1)} = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

The second line uses Proposition 4.1.2 (d), all factors being non-negative (and $|b| \le |b|+1$, by Proposition 2.4.3 (b)); the third multiplies the two estimates by the positive elements M and $|b|+1$ and adds them (Proposition 2.4.3 (b)). So $a_n b_n \to ab$.

(d) First we show $1/b_n \to 1/b$. By Lemma 4.4.2 there is N_0 with $|b_n| > |b|/2$ for every $n \ge N_0$. For such n, Lemma 4.1.1 (c) and Proposition 4.1.3 (b) give

$$\Big|\frac{1}{b_n} - \frac{1}{b}\Big| = \Big|\frac{b - b_n}{b_n b}\Big| = \frac{|b_n - b|}{|b_n||b|}.$$

Multiplying $|b_n| > |b|/2$ by $|b| > 0$ gives $|b_n||b| > |b|^2/2 > 0$, so $1/(|b_n||b|) < 2/|b|^2$ (Proposition 3.2.2 (b)). Multiplying by $|b_n - b| \ge 0$,

$$\Big|\frac{1}{b_n} - \frac{1}{b}\Big| \le \frac{2}{|b|^2}\,|b_n - b| \qquad (n \ge N_0).$$

Now let $\varepsilon \in F_{>0}$. Then $\varepsilon|b|^2/2 \in F_{>0}$ ($|b|^2 > 0$ by Proposition 2.4.3 (e), then (O4) and Proposition 4.1.2 (b)); choose N_1 with $|b_n - b| < \varepsilon|b|^2/2$ for every $n \ge N_1$, and let $N := \max\{N_0, N_1\}$. For $n \ge N$, the last display and multiplication of $|b_n - b| < \varepsilon|b|^2/2$ by $2/|b|^2 > 0$ gives $|1/b_n - 1/b| < (2/|b|^2)(\varepsilon|b|^2/2) = \varepsilon$. So $1/b_n \to 1/b$. Finally $a_n/b_n = a_n \cdot (1/b_n)$, and (c) gives $a_n/b_n \to a \cdot (1/b) = a/b$. This proves the theorem.

■

The answer to the prompt: the tolerance $\varepsilon/(2M)$ contains M, so M must exist before that tolerance is written down. And M must not depend on n: it has to bound $|a_n|$ for **every** $n \ge N$ at once, including indices not yet chosen. That is why it comes from Proposition 4.3.5, a single bound for the whole sequence, and not from "$|a_n|$ is some element of F". The $+1$ in $|b|+1$ is there because b may be 0, and $\varepsilon/(2|b|)$ would then mean nothing.

Part (a) is the move we call **split** ε: to make a sum of k small terms smaller than ε, give each term the tolerance ε/k, and combine the finitely many thresholds by a maximum. Part (c) is **bound, then shrink**: replace the factor you cannot control by a bound fixed in advance, then make the other factor small.

Second reading · *The product law from the two tools*

Alternatively, (c) follows from the earlier tools without new estimates. By Proposition 4.3.3, $(b_n - b)$ and $(a_n - a)$ are null; $a_n(b_n - b)$ is null by Proposition 4.4.1, with the bound M; $b(a_n - a)$ is null by (b); so their sum $a_n b_n - ab$ is null by (a), and Proposition 4.3.3 again gives $a_n b_n \to ab$. The bound M is still fixed first: it is the hypothesis of Proposition 4.4.1.

In (d), the hypothesis "$b_n \ne 0$ for every n" is there only so that a_n/b_n is defined for every n. By Lemma 4.4.2, $b_n \ne 0$ holds eventually anyway once $b \ne 0$. If some early b_n vanish, the quotient is a sequence from some N_0 on, and Proposition 4.3.2 (b) applies the law to the shifted sequences (a_{n+N_0}) and (b_{n+N_0}).

Warning. Part (d) needs $b \ne 0$, even when every $b_n \ne 0$. In $\mathbb{Q}$, $a_n = 1/(n+1)$ and $b_n = 1/(n+1)^2$ are both null (the first by Example (First Limits) (b), the second by (c),

since $b_n = a_n a_n$), with every $b_n \neq 0$, and $a_n/b_n = n{+}1$. Since $n{+}1 > n$, the sequence $(n{+}1)$ is not bounded (the non-example after Definition 4.3.4), and so has no limit (Proposition 4.3.5). The "law" $\lim(a_n/b_n) = 0/0$ names nothing.

Quick check. Find $\lim(3n^2 - n)/(n^2 + 5)$ in $\mathbb{Q}$, citing a law, or another result, for each step.

Solution. The limit is 3. Let $u_0 := 0$ and $u_n := 1/n$ for $n \geq 1$. The shifted sequence $(u_{n+1})_n = (1/(n{+}1))_n$ is null (Example (First Limits) (b); this is where Archimedes enters), so (u_n) is null by Proposition 4.3.2 (b). By Theorem 4.4.3: $3 - u_n \to 3 - 0 = 3$ (the constant 3 converges to 3 by Example (First Limits) (a); then (b) and (a)); $u_n^2 \to 0$ ((c)); $1 + 5u_n^2 \to 1$ ((b), (a)). Every $1 + 5u_n^2 \geq 1 > 0$: $u_n^2 \geq 0$ (Proposition 2.4.3 (e) if $u_n \neq 0$, and $u_0^2 = 0$), so $5u_n^2 \geq 0$ (Proposition 4.1.2 (d)), and $1 + 5u_n^2 \geq 1$ by Proposition 2.4.3 (b). Also the limit $1 \neq 0$, so (d) gives

$$c_n := \frac{3 - u_n}{1 + 5u_n^2} \to \frac{3}{1} = 3.$$

For $n \geq 1$, multiplying numerator and denominator by n^2 shows $c_n = (3n^2 - n)/(n^2 + 5)$. The two sequences agree from $N_0 = 1$ on (at $n = 0$ they differ: 3 versus 0), so Proposition 4.3.2 (a) gives the limit 3.

4.4.3 Limits and order

Inequalities between terms pass to the limits, provided we accept that a strict inequality may become an equality.

Proposition 4.4.4. Limits Preserve Weak Inequalities

Let $a_n \to a$ and $b_n \to b$ in F, and let $N_0 \in \mathbb{N}$.

(a) If $a_n \leq b_n$ for every $n \geq N_0$, then $a \leq b$.

(b) In particular, for $c \in F$: if $a_n \leq c$ for every $n \geq N_0$, then $a \leq c$; and if $a_n \geq c$ for every $n \geq N_0$, then $a \geq c$.

(c) *(Squeeze.)* Let (c_n) be a sequence in F. If $a = b$ and $a_n \leq c_n \leq b_n$ for every $n \geq N_0$, then $c_n \to a$.

Idea. For (a), suppose instead that $b < a$. The midpoint $(a + b)/2$ separates them, and from some index on a_n lies above it and b_n below it, contradicting $a_n \leq b_n$. The tolerance that achieves this is half the gap, $(a - b)/2$. For (c), from some index on a_n and b_n both lie in the window $(a - \varepsilon, a + \varepsilon)$, and c_n is trapped between them.

Proof.

(a) Suppose, for a contradiction, that $b < a$. Let

$$\varepsilon := \frac{a-b}{2},$$

which lies in $F_{>0}$ by Proposition 2.4.3 (a) and Proposition 4.1.2 (b). Choose N_1 (depending on ε) with $|a_n - a| < \varepsilon$ for every $n \geq N_1$, and N_2 with $|b_n - b| < \varepsilon$ for every $n \geq N_2$. Let $n := \max\{N_0, N_1, N_2\}$. By Proposition 4.1.3 (d), at this n,

$$a_n > a - \varepsilon = \frac{a+b}{2} = b + \varepsilon > b_n.$$

This contradicts $a_n \leq b_n$, which holds because $n \geq N_0$. So $b < a$ is false, and $a \leq b$ by trichotomy (O1).

(b) A constant sequence c converges to c (Example (First Limits) (a)). Apply (a) to (a_n) and the constant sequence c for the first claim, and to the constant sequence c and (a_n) for the second.

(c) Let $\varepsilon \in F_{>0}$. Choose N_1 with $|a_n - a| < \varepsilon$ for $n \geq N_1$ and N_2 with $|b_n - a| < \varepsilon$ for $n \geq N_2$ (recall $b = a$), and let $N := \max\{N_0, N_1, N_2\}$. For $n \geq N$, Proposition 4.1.3 (d) gives

$$a - \varepsilon < a_n \leq c_n \leq b_n < a + \varepsilon,$$

so $|c_n - a| < \varepsilon$ by the same part (d). Hence $c_n \to a$.

■

The two midpoint equalities in (a) are field algebra: $a - (a-b)/2 = (2a - a + b)/2 = (a+b)/2$, and likewise $b + (a-b)/2 = (a+b)/2$. Chapter 5 uses the squeeze (c) repeatedly, mostly with $F = \mathbb{Q}$ to show that a sequence is null. In the squeeze (c), the middle sequence is not assumed to converge: its convergence is the conclusion.

Warning. Strict inequalities do not pass to the limit. In $\mathbb{Q}$, $1/(n+1) > 0$ for every n, and the limit is 0, not positive (Example (First Limits) (b)). From $a_n < b_n$ for every n, (a) gives only $a \leq b$.

Quick check. Let $a_n \to a$ in F with $|a_n| \leq 3$ for every n. Show $|a| \leq 3$.

Solution. By Proposition 4.1.3 (d), $-3 \leq a_n \leq 3$ for every n. Part (b) of Proposition 4.4.4, used twice, gives $-3 \leq a \leq 3$, and (d) again gives $|a| \leq 3$.

4.4.4 Exercises

4.4.4.1 A. Check your understanding

Exercise. A1

These questions are about the proof of the product law, Theorem 4.4.3 (c).

(a) Why is M fixed before ε, and what does it depend on?

(b) Which hypothesis supplies N_1, which supplies N_2, and which supplies M?

(c) Why does the proof use $|b|+1$ rather than $|b|$?

(d) State the idea of the proof in two sentences.

(e) Run the proof in $\mathbb{Q}$ on $a_n = 1 + 1/(n+1)$, $b_n = 2$ and $\varepsilon = 1/10$: give a valid M, the least N_1, the least N_2, and N.

Solution.

(a) The tolerance $\varepsilon/(2M)$ is built from M, so M must be available first. It depends only on the sequence (a_n): it bounds $|a_n|$ for every n, independently of ε and of n. A "bound" depending on n would be useless, since the tolerance for $|b_n - b|$ must be named before n is.

(b) N_1 comes from $b_n \to b$, applied with the tolerance $\varepsilon/(2M)$; N_2 from $a_n \to a$, with the tolerance $\varepsilon/(2(|b|+1))$; M also from $a_n \to a$, through Proposition 4.3.5.

(c) b may be 0, and then $\varepsilon/(2|b|)$ is undefined. Since $|b| + 1 \geq 1 > 0$, the tolerance $\varepsilon/(2(|b|+1))$ always exists, and $|b| \leq |b| + 1$ keeps the estimate valid.

(d) Write $a_n b_n - ab = a_n(b_n - b) + b(a_n - a)$, so that each term has a factor tending to 0. The other factor of the first term is not small, but it is bounded by a constant M fixed in advance, so making $|b_n - b| < \varepsilon/(2M)$ and $|a_n - a| < \varepsilon/(2(|b|+1))$ makes each term smaller than $\varepsilon/2$.

(e) Here $a = 1$ and $b = 2$. Since $0 < 1/(n+1) \leq 1$, $|a_n| \leq 2$ for every n, so $M = 2$ is valid. Then $\varepsilon/(2M) = 1/40$, and $|b_n - b| = 0 < 1/40$ for every n: the least N_1 is 0. Next $\varepsilon/(2(|b|+1)) = (1/10)/6 = 1/60$, and $|a_n - a| = 1/(n+1) < 1/60$ if and only if $n + 1 > 60$, that is, $n \geq 60$: the least N_2 is 60. So $N = \max\{0, 60\} = 60$. Check at $n = 60$: $a_{60}b_{60} - ab = 2(1 + 1/61) - 2 = 2/61 < 1/10$.

4.4.4.2 B. Practice

Exercise. B1

Find the following limits in $\mathbb{Q}$ by the limit laws, naming the result used at each step.

(a) $\dfrac{n+2}{3n+1}$.

(b) $\dfrac{n^2+1}{(n+1)^2}$.

(c) $\Big(\dfrac{1}{n+1} - 2\Big)\Big(3 + \dfrac{(-1)^n}{n+1}\Big)$.

Hint: write each term through $u_n := 1/(n+1)$.

Solution. Let $u_n := 1/(n+1)$, so $u_n \to 0$ (Example (First Limits) (b)), and constant sequences converge to their value (part (a) of that example). All laws cited are parts of Theorem 4.4.3.

(a) With $m := n + 1 \geq 1$: $(n+2)/(3n+1) = (m+1)/(3m-2) = (1+u_n)/(3-2u_n)$, dividing numerator and denominator by m. Now $1 + u_n \to 1$ by (a), and $3 - 2u_n \to 3$

by (b) and (a). Since $0 < u_n \leq 1$, $3 - 2u_n \geq 1 > 0$ for every n, and the limit $3 \neq 0$, so (d) gives the limit $1/3$.

(b) With $m = n + 1$: $(n^2 + 1)/(n + 1)^2 = ((m - 1)^2 + 1)/m^2 = 1 - 2u_n + 2u_n^2$. By (c), $u_n^2 \to 0$; by (b) and (a), $1 - 2u_n + 2u_n^2 \to 1 - 0 + 0 = 1$. No quotient law is needed.

(c) $u_n - 2 \to -2$ by (b) and (a). The sequence $(-1)^n u_n$ is null by Proposition 4.4.1, since u_n is null and $|(-1)^n| = 1$ (Example (First Limits) (d)); so $3 + (-1)^n u_n \to 3$ by (a). By (c), the product tends to $(-2) \cdot 3 = -6$.

In each part Archimedes enters once, through $u_n \to 0$; every other step is a law valid in any ordered field.

Exercise. B2: Put the proof in order

Here are lines of the proof of Proposition 4.4.4 (a), shuffled, together with two lines that do not belong. Recall the hypotheses: $a_n \to a$, $b_n \to b$, and $a_n \leq b_n$ for every $n \geq N_0$. Write the seven lines of a correct proof in a correct order, say which other orders are also admissible, and say why each of the other two lines fails.

(i) Let $n := \max\{N_0, N_1, N_2\}$.
(ii) Let $\varepsilon := a - b$, which lies in $F_{>0}$.
(iii) Choose N_2 with $|b_n - b| < \varepsilon$ for every $n \geq N_2$.
(iv) Suppose, for a contradiction, that $b < a$.
(v) This contradicts $a_n \leq b_n$; so $b < a$ is false, and $a \leq b$ by trichotomy.
(vi) Let $\varepsilon := (a - b)/2$, which lies in $F_{>0}$ by Proposition 4.1.2.
(vii) Let $n := N_1$.
(viii) Then $a_n > a - \varepsilon = (a + b)/2 = b + \varepsilon > b_n$.
(ix) Choose N_1 with $|a_n - a| < \varepsilon$ for every $n \geq N_1$.

Solution. An admissible order is

$$\text{(iv)},\ \text{(vi)},\ \text{(ix)},\ \text{(iii)},\ \text{(i)},\ \text{(viii)},\ \text{(v)}.$$

The only other admissible order exchanges (ix) and (iii), which use only ε and one hypothesis each. Nothing else moves: (vi) uses the assumption $b < a$ of (iv) to make ε positive; (ix) and (iii) use ε; (i) uses N_1 and N_2; (viii) uses the two estimates at this n, and needs $n \geq N_1$ and $n \geq N_2$; (v) uses (viii) and $n \geq N_0$, and closes the assumption of (iv).

Line (ii) is an ε of the wrong size. It is positive, but with it (viii) becomes $a_n > a - (a - b) = b$ and $b_n < b + (a - b) = a$, and "$a_n > b$ and $b_n < a$" is no contradiction with $a_n \leq b_n$: for instance $a_n = b_n = (a + b)/2$ satisfies all three at once. The two windows of half-width $a - b$ around a and b overlap; half-width $(a - b)/2$ makes them touch only at the midpoint, which neither open window contains.

Line (vii) chooses an index that depends on the wrong thing. With $n := N_1$, the estimate for a_n holds, but nothing makes $n \geq N_2$ or $n \geq N_0$: the estimate for b_n and the hypothesis $a_n \leq b_n$ may both fail at this n. The index must lie beyond **all three** thresholds, which the maximum in (i) ensures.

Exercise. B3: Close to a convergent sequence

Let $a_n \to a$ in F, and let (b_n) be a sequence in F such that $(b_n - a_n)$ is null. Prove that $b_n \to a$. Here is the start of a proof.

> Let $\varepsilon \in F_{>0}$. Then $\varepsilon/2 \in F_{>0}$. Since $a_n \to a$, choose N_1 with $|a_n - a| < \varepsilon/2$ for every $n \geq N_1$. Since $(b_n - a_n)$ is null, choose N_2 with $|b_n - a_n| < \varepsilon/2$ for every $n \geq N_2$. Let $N \coloneqq \max\{N_1, N_2\}$.
>
> ...

Name the move, then complete it.

Solution. The move is **split** ε, with two terms: each gets $\varepsilon/2$, and the two thresholds are combined by a maximum.

Completion. Let $n \geq N$. Then $n \geq N_1$ and $n \geq N_2$, and the three-point triangle inequality (Proposition 4.1.3 (e)) gives

$$|b_n - a| \leq |b_n - a_n| + |a_n - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon \in F_{>0}$ was arbitrary, $b_n \to a$. (The same result also follows from the laws: $b_n = (b_n - a_n) + a_n \to 0 + a$ by Theorem 4.4.3 (a). The exercise is the law's proof, run once more.)

4.4.4.3 C. Going deeper

Exercise. C1: Exit check: the product law

Prove, from Definition 4.3.2, that if $a_n \to a$ and $b_n \to b$ in an ordered field F, then $a_n b_n \to ab$. You may use that convergent sequences are bounded (Proposition 4.3.5), the rules for the absolute value (Proposition 4.1.3), Proposition 4.3.3 and Proposition 4.4.1, but not Theorem 4.4.3.

Solution. By Proposition 4.3.5 there is $M \in F_{>0}$ with $|a_n| \leq M$ for every $n \in \mathbb{N}$. It depends only on (a_n), and we fix it before any tolerance is named.

Let $\varepsilon \in F_{>0}$. Since $|b| + 1 \geq 1 > 0$, both $\varepsilon/(2M)$ and $\varepsilon/(2(|b| + 1))$ lie in $F_{>0}$. By $b_n \to b$ there is N_1 with $|b_n - b| < \varepsilon/(2M)$ for every $n \geq N_1$; by $a_n \to a$ there is N_2 with $|a_n - a| < \varepsilon/(2(|b| + 1))$ for every $n \geq N_2$. N_1 depends on ε and M, N_2 on ε and b (and both on the given sequences); neither depends on n. Let $N \coloneqq \max\{N_1, N_2\}$.

Let $n \geq N$. Since

$$a_n b_n - ab = a_n(b_n - b) + b(a_n - a),$$

the triangle inequality and $|xy| = |x||y|$ give

$$\begin{aligned} |a_n b_n - ab| &\leq |a_n||b_n - b| + |b||a_n - a| \\ &\leq M|b_n - b| + (|b| + 1)|a_n - a| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

where the second line uses $0 \leq |a_n| \leq M$ and $0 \leq |b| \leq |b| + 1$ (Proposition 4.1.2 (d)), and

the third multiplies the two estimates by $M > 0$ and $|b| + 1 > 0$. So for every $\varepsilon \in F_{>0}$ there is $N \in \mathbb{N}$ with $|a_n b_n - ab| < \varepsilon$ for every $n \geq N$: $a_n b_n \to ab$.

Marking scheme. (10 points) - 2: fixes a bound $M \in F_{>0}$ for one of the two factor sequences (say $(|a_n|)$; the mirror decomposition with $(|b_n|)$ is equally good), justified (convergent sequences are bounded), before ε. - 2: writes $a_n b_n - ab$ as a sum of two terms, each with a factor tending to 0. - 3: chooses N_1, N_2 independent of n (N_1 from ε and M, N_2 from ε and b), with the case $b = 0$ handled, and N as their maximum. - 2: carries out the estimate with the triangle inequality and the product rule for $|\cdot|$. - 1: concludes with the quantifiers in the order $\forall \varepsilon\, \exists N\, \forall n \geq N$. Fatal errors (0 for the step): a threshold or tolerance that depends on n; the bound "$|a_n| \leq |a| + 1$ for every n" without the finite part for $n < N$; dividing by $|b|$ without $b \neq 0$. A route through Proposition 4.4.1 earns the same points, split as: bound fixed, 2; decomposition into two terms, 2; each product null by Proposition 4.4.1 (the constant factor bounded by $|b| + 1$, or by $|b|$ with $b = 0$ handled separately), 3; sum of two null sequences proved from the definition, 2; null shift (Proposition 4.3.3) to conclude, 1. Citing Theorem 4.4.3 for the sum or for $b(a_n - a)$ gets 0 for that step. A three-term route ($a_n b_n - ab = (a_n - a)(b_n - b) + a(b_n - b) + b(a_n - a)$, with a tolerance such as $\delta = \min\{1, \varepsilon/(1 + |a| + |b|)\}$ chosen from ε, a and b only) earns the same points: decomposition 2; the choice of δ, independent of n and with $\delta \leq 1$ controlling the product term (in place of the bound M), 2; a threshold N for both sequences at tolerance δ (the case $a = 0$ or $b = 0$ covered by the 1 in the denominator) 3; the estimate 2; conclusion 1.

Exercise. C2: Non-negative terms and reciprocals

Let $a_n \to a$ in F, with $a_n \geq 0$ for every n.

(a) Prove that $a \geq 0$.

(b) Suppose moreover $a > 0$. Prove that there is $N_0 \in \mathbb{N}$ with $a_n > 0$ for every $n \geq N_0$, and that the sequence $(1/a_{n+N_0})_{n \in \mathbb{N}}$ converges to $1/a$.

(c) Give an example in $\mathbb{Q}$ where $a_n > 0$ for every n and $(1/a_n)$ does not converge.

Solution (Hint 1). For (a), compare with a constant sequence. For (b), which result keeps terms away from a non-zero limit?

Solution (Hint 2). For (b), start: "By Lemma 4.4.2, since $a > 0$, there is N_0 with $a_n > a/2$ for every $n \geq N_0$."

Solution.

(a) Since $a_n \geq 0$ for every $n \geq 0$, Proposition 4.4.4 (b) with $c = 0$ gives $a \geq 0$.

(b) Since $a > 0$, Lemma 4.4.2 gives N_0 with $a_n > a/2 > 0$ for every $n \geq N_0$. The shifted sequence $(a_{n+N_0})_n$ converges to a by Proposition 4.3.2 (b), and each of its terms is non-zero. By Theorem 4.4.3 (d), applied to the constant sequence 1 and to $(a_{n+N_0})_n$, whose limit a is non-zero, $1/a_{n+N_0} \to 1/a$.

(c) $a_n = 1/(n+1)$: every term is positive, and $1/a_n = n + 1$ is not bounded in $\mathbb{Q}$: for any

$M \in \mathbb{Q}_{>0}$, the non-example after Definition 4.3.4 gives n with $M < n < n+1 = |1/a_n|$. So it does not converge (Proposition 4.3.5). The hypothesis $a > 0$ of (b) fails here: $a = 0$.

Marking scheme. (10 points) - 3: (a), by comparison with the constant sequence 0 (or by contradiction with $\varepsilon = -a/2$). - 3: (b), an index N_0 beyond which a_n is bounded away from 0, from $a > 0$. - 2: (b), $1/a_{n+N_0} \to 1/a$, either by the quotient law (Theorem 4.4.3 (d)) applied to the shift, with its convergence to a and $a \neq 0$ checked, or directly from $|1/a_{n+N_0} - 1/a| \leq (2/a^2)|a_{n+N_0} - a|$, which uses $a_{n+N_0} > a/2$. - 2: (c), an example with the failure of convergence justified. Fatal errors (0 for the step): concluding $a > 0$ in (a) from $a_n > 0$; applying the quotient law without checking that the limit is non-zero.

4.5 Cauchy Sequences

Every convergence proof so far began with the limit in hand: to show $a_n \to a$ we estimated $|a_n - a|$, and the limit laws of Section 4.4 compute a new limit only from limits already known. That is no help for the sequence of Chapter 3 that closes in on the gap at $\sqrt{2}$ (Example (A Rational Sequence Aimed at the Gap)), whose terms $2, 3/2, 17/12, 577/408, \ldots$ visibly settle down while no rational is available to be their limit. This section names a property that can be checked from the terms alone: eventually they are all close **to each other**. We prove the facts about it that Chapter 5 needs: convergent sequences have it, sequences with it are bounded and closed under sums and products, and, when such a sequence does not tend to 0, its terms eventually keep one sign and stay away from 0.

Throughout, F is an ordered field (Definition 2.4.1), sequences are indexed by $\mathbb{N}$ (Definition 4.3.1), and ε ranges over $F_{>0}$.

4.5.1 Terms that agree with one another

A limit relates a sequence to a point of F; boundedness (Definition 4.3.4) is a property of the sequence alone. We want a second property of the sequence alone, saying "the terms settle down" without saying where.

In $\mathbb{Q}$ the Newton terms satisfy $17/12 - 577/408 = 1/408$, and the later terms are closer still to one another (Section 4.6 proves it). The way to say "settle down" without a target is to compare terms with terms: however small a tolerance is named, from some index on, any two terms differ by less than it.

The terms eventually agree with one another to within any tolerance.

Definition 4.5.1. Cauchy Sequence

A sequence $(a_n)_{n\in\mathbb{N}}$ in F is a **Cauchy sequence in** F (or **Cauchy**) if **for every** $\varepsilon \in F_{>0}$ **there exists** $N \in \mathbb{N}$ such that $|a_m - a_n| < \varepsilon$ **for all** $m, n \in \mathbb{N}$ with $m \geq N$ and $n \geq N$. In symbols:

$$\forall \varepsilon \in F_{>0} \;\; \exists N \in \mathbb{N} \;\; \forall m \in \mathbb{N} \;\; \forall n \in \mathbb{N}$$
$$((m \geq N \wedge n \geq N) \Rightarrow |a_m - a_n| < \varepsilon).$$

In words: the tolerance is arbitrary and named first; the threshold comes after it, so it may depend on ε and on nothing else; and **every** pair of indices beyond the threshold, near or far

apart, gives terms within ε of each other. Compared with Definition 4.3.2, the point a has been replaced by a second index m.

The tolerances are the elements of $F_{>0}$, so the field is part of the definition: for $F = \mathbb{Q}$, "Cauchy in $\mathbb{Q}$" is a statement about rationals only, as Chapter 5 needs. Since $|a_m - a_n| = |a_n - a_m|$, the roles of m and n are symmetric.

Idea. Negate as in *Logic and Proof*, Chapter 6: each bounded quantifier flips (Logic and Proof, Proposition 6.2.3), the implication becomes a conjunction (Logic and Proof, Proposition 6.2.2), and $\neg(x < \varepsilon)$ is $x \geq \varepsilon$. "(a_n) is not Cauchy" reads

$$\exists \varepsilon \in F_{>0} \;\; \forall N \in \mathbb{N} \;\; \exists m, n \in \mathbb{N} \;\; \big(m \geq N \wedge n \geq N \wedge |a_m - a_n| \geq \varepsilon\big).$$

One tolerance is beaten beyond every threshold.

Examples.

A constant sequence. Let $a_n = c$ for every n. Then $|a_m - a_n| = |0| = 0 < \varepsilon$ for all m, n, so $N = 0$ works for every ε. This degenerate case matters: Chapter 5 represents each rational c by its constant sequence.

The reciprocals in $\mathbb{Q}$. Let $a_n = 1/(n+1)$ in $\mathbb{Q}$. Given $\varepsilon \in \mathbb{Q}_{>0}$, Corollary 3.2.4 gives $N \in \mathbb{Z}_{>0}$ with $1/N < \varepsilon$. Let $m, n \geq N$. Then $m + 1 > N$ and $n + 1 > N$, so $0 < 1/(m+1) < 1/N$ and $0 < 1/(n+1) < 1/N$ (Proposition 3.2.2 (b)). Two elements of $(0, 1/N)$ differ by less than $1/N$, so $|a_m - a_n| < 1/N < \varepsilon$. The threshold depends on ε only: for $\varepsilon = 1/100$, $N = 101$ works.

Sequences with a limit. By the first result of the next subsection, the convergent sequences of Section 4.3 (Example (First Limits)) are all examples. The definition earns its keep on sequences with no limit available, which Section 4.6 supplies.

Non-example by a minimal change. Change the constant sequence 1 at every odd index: let $a_n = 1$ for even n and $a_n = 0$ for odd n. The sequence is still bounded, by 1, and every two terms still differ by at most 1. What fails is "for every ε": take $\varepsilon = 1$. Whatever N is offered, the indices N and $N + 1$ are both $\geq N$, one even and one odd, and their terms differ by $1 \not< 1$. So the negation above holds with $\varepsilon = 1$, $m = N$, $n = N + 1$.

Why this definition. Each clause is there for a reason.

- *Every pair, not only neighbors.* Asking only that $|a_{n+1} - a_n| < \varepsilon$ for $n \geq N$ is too weak: the example at the end of this section has consecutive differences tending to 0 and is not Cauchy.
- *The threshold after the tolerance.* With the quantifiers swapped, "$\exists N \; \forall \varepsilon \; \forall m, n \geq N$", the condition forces $|a_m - a_n| < \varepsilon$ for every ε, hence $a_m = a_n$ for all $m, n \geq N$ (the argument of Theorem 4.3.4): only sequences that are eventually constant would qualify.
- *Tolerances from the field.* $\mathbb{Q}(t)$ has the tolerance $1/t$, below every positive rational; Exercise C2 of Section 4.6 shows a sequence of rationals that is Cauchy in $\mathbb{Q}$ and not in $\mathbb{Q}(t)$.

Warning. The definition asks for **one** N that serves **every** pair $m, n \geq N$. A proof that checks only some pairs, say each n against $n + 2$, proves nothing: in the non-example above, $a_{n+2} = a_n$ for every n, so those pairs pass every test, and the sequence still fails.

Payoff. The tail-change principle extends at once. If $a_n = b_n$ for every $n \geq N_0$ and (a_n) is Cauchy, then for $\varepsilon \in F_{>0}$ and its threshold N for (a_n), every $m, n \geq \max\{N, N_0\}$ give $|b_m - b_n| = |a_m - a_n| < \varepsilon$. So (b_n) is Cauchy, with thresholds $\max\{N, N_0\}$, as in Proposition 4.3.2.

4.5.2 Convergent sequences are Cauchy, and Cauchy sequences are bounded

The first result says the new property is no stronger than convergence: it is a necessary condition, one we can test without knowing the limit.

Proposition 4.5.1. Convergent Sequences Are Cauchy

If $a_n \to a$ in F, then (a_n) is a Cauchy sequence in F.

Idea. Two terms that are each within $\varepsilon/2$ of a are within ε of each other: insert the point a between them with the triangle inequality, and spend half the tolerance on each side. This is split ε, as in the uniqueness proof of Section 4.3, with the point a between the two terms; the threshold comes from the hypothesis, as in the limit laws.

Proof. Let $\varepsilon \in F_{>0}$. Since $a_n \to a$, there is $N \in \mathbb{N}$ (depending on ε) with $|a_n - a| < \varepsilon/2$ for every $n \geq N$. Let $m, n \geq N$. By the three-point triangle inequality (Proposition 4.1.3 (e)) with a in the middle,

$$|a_m - a_n| \leq |a_m - a| + |a - a_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so (a_n) is Cauchy.

■

The converse is the question of the chapter. In $\mathbb{Q}$ it fails (Section 4.6). Chapter 5 builds the field $\mathbb{R}$, and *Analysis on the Real Line*, Chapter 4 proves that it holds there.

Quick check. Is the sequence $\left((-1)^n/(n+1)\right)_{n\in\mathbb{N}}$ Cauchy in $\mathbb{Q}$? Answer without estimating $|a_m - a_n|$.

Solution. Yes. The sequence $(1/(n+1))$ is null in $\mathbb{Q}$ (Example (First Limits)), and $((-1)^n)$ is bounded by 1, since $|(-1)^n| = 1$ (Example (First Limits) (d)). So the product is null by Proposition 4.4.1. A null sequence converges, to 0, so it is Cauchy by Proposition 4.5.1.

The second result is the Cauchy version of Proposition 4.3.5. There the tail was trapped near the limit a; here no limit is available, so we trap the tail near one of its own terms.

Proposition 4.5.2. Cauchy Sequences Are Bounded

Every Cauchy sequence in F is bounded.

Idea. Apply the definition once, with the tolerance 1. It gives an index N beyond which every term is within 1 of the term a_N, a base point we can name. The finitely many terms before N are handled by a finite maximum, as for convergent sequences.

Proof. Let (a_n) be Cauchy in F. Since $1 > 0$, there is $N \in \mathbb{N}$ with $|a_m - a_n| < 1$ for all $m, n \geq N$. For $n \geq N$, taking $m = N$ and using the triangle inequality (Proposition 4.1.3 (e)),

$$|a_n| = |(a_n - a_N) + a_N| \leq |a_n - a_N| + |a_N| < 1 + |a_N|.$$

Let

$$M := \max\{|a_0|, \ldots, |a_{N-1}|, |a_N| + 1\},$$

a finite maximum (Sets and Functions, Proposition 3.3.8, whose proof works in any totally ordered set; for $N = 0$ the set is $\{|a_0| + 1\}$). Then $M \geq |a_N| + 1 \geq 1 > 0$, so $M \in F_{>0}$. For $n < N$, $|a_n| \leq M$ because $|a_n|$ is in the set; for $n \geq N$, $|a_n| < |a_N| + 1 \leq M$. So $|a_n| \leq M$ for every $n \in \mathbb{N}$, and (a_n) is bounded (Definition 4.3.4).

■

The bound M depends on the whole sequence, not on any tolerance. That is what makes it usable in the next proof: it can be fixed before ε is named.

4.5.3 Sums and products of Cauchy sequences

Chapter 5 will add and multiply real numbers by adding and multiplying Cauchy sequences of rationals term by term. For that, sums and products of Cauchy sequences must again be Cauchy. The limit laws (Theorem 4.4.3) do not apply, since they need limits; the proofs below copy their estimates with a second index in place of the limit.

Theorem 4.5.3. Sums and Products of Cauchy Sequences

Let (a_n) and (b_n) be Cauchy sequences in F, and let $c \in F$.

(a) Every constant sequence and every null sequence in F is Cauchy.

(b) The sequences $(a_n + b_n)$, $(c\,a_n)$, $(-a_n)$ and $(a_n - b_n)$ are Cauchy.

(c) The sequence $(a_n b_n)$ is Cauchy.

Idea.

(a) is Proposition 4.5.1. For (b), split ε: give each of (a_n) and (b_n) the tolerance $\varepsilon/2$ and take the larger threshold. For (c), the difference $a_n b_n - a_m b_m$ splits as $a_n(b_n - b_m) + b_m(a_n - a_m)$. The factors a_n and b_m do not get small, but they are bounded (Proposition 4.5.2). So **bound, then shrink**: replace each factor we cannot control by a bound fixed in advance, then make the other factor small enough to absorb the

bound.

Proof.

(a) A constant sequence (c) converges to c, and a null sequence converges to 0 (Example (First Limits), Definition 4.3.3). Both are Cauchy by Proposition 4.5.1.

(b) *Sums.* Let $\varepsilon \in F_{>0}$. Since (a_n) is Cauchy, there is N_1 with $|a_m - a_n| < \varepsilon/2$ for all $m, n \geq N_1$; since (b_n) is Cauchy, there is N_2 with $|b_m - b_n| < \varepsilon/2$ for all $m, n \geq N_2$. Let $N := \max\{N_1, N_2\}$, which depends only on ε. For $m, n \geq N$, we have $(a_m + b_m) - (a_n + b_n) = (a_m - a_n) + (b_m - b_n)$, so the triangle inequality gives

$$\begin{aligned} |(a_m + b_m) - (a_n + b_n)| &\leq |a_m - a_n| + |b_m - b_n| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Multiples. The proof of Theorem 4.4.3 (b) carries over with a_m in place of a: for $c \neq 0$, the tolerance $\varepsilon/|c|$ for (a_n) gives $|ca_m - ca_n| = |c|\,|a_m - a_n| < \varepsilon$ for all $m, n \geq N$; for $c = 0$ it is the constant sequence 0, Cauchy by (a).
Negatives and differences. Since $-a_n = (-1)a_n$, $(-a_n)$ is the multiple with $c = -1$. Then $a_n - b_n = a_n + (-b_n)$ is the sum of two Cauchy sequences.

(c) By Proposition 4.5.2, fix $A, B \in F_{>0}$ with $|a_n| \leq A$ and $|b_n| \leq B$ for every $n \in \mathbb{N}$; they depend on the two sequences only. Now let $\varepsilon \in F_{>0}$. Then $\varepsilon/(2A)$ and $\varepsilon/(2B)$ are positive (Proposition 3.2.2 (a)), so there are N_1 with $|b_n - b_m| < \varepsilon/(2A)$ for all $m, n \geq N_1$, and N_2 with $|a_n - a_m| < \varepsilon/(2B)$ for all $m, n \geq N_2$. Let $N := \max\{N_1, N_2\}$, depending on ε, A, B, and let $m, n \geq N$. Expanding the right side shows

$$a_n b_n - a_m b_m = a_n(b_n - b_m) + b_m(a_n - a_m).$$

By the triangle inequality and $|xy| = |x||y|$, then Proposition 4.1.2 (d) applied to $0 \leq |a_n| \leq A$ and to $0 \leq |b_m| \leq B$,

$$\begin{aligned} |a_n b_n - a_m b_m| &\leq |a_n|\,|b_n - b_m| + |b_m|\,|a_n - a_m| \\ &\leq A|b_n - b_m| + B|a_n - a_m| \\ &< A \cdot \frac{\varepsilon}{2A} + B \cdot \frac{\varepsilon}{2B} = \varepsilon, \end{aligned}$$

where the strict step multiplies by $A > 0$ and $B > 0$ (Proposition 2.4.3 (c)) and adds. So $(a_n b_n)$ is Cauchy.

■

The order of choices in (c) is the whole proof: A and B are chosen before ε, from the sequences alone, so that $\varepsilon/(2A)$ is one fixed tolerance. Exercise C3 shows what goes wrong when the bound is skipped.

Remark. Chapter 5 adds and multiplies classes of Cauchy sequences of rationals termwise; clauses (b) and (c) with $F = \mathbb{Q}$ keep the termwise results Cauchy.

4.5.4 The sign of a Cauchy sequence that is not null

A null sequence is the Cauchy analogue of 0. Chapter 5 must decide, for each Cauchy sequence of rationals that is not null, whether it represents a positive or a negative number, and it needs an inverse for it. Both rest on one fact: such a sequence eventually keeps one sign, at a definite distance from 0.

Theorem 4.5.4. The Sign of a Non-Null Cauchy Sequence

Let (a_n) be a Cauchy sequence in F that is not null. Then there are $c \in F_{>0}$ and $N \in \mathbb{N}$ such that either

$$a_n \geq c \text{ for every } n \geq N, \qquad \text{or} \qquad a_n \leq -c \text{ for every } n \geq N.$$

The two alternatives do not both hold.

Try it first (ten minutes): write "(a_n) is not null" in symbols. Which ε does it hand you, and what does the Cauchy condition do with it?

Idea. "Not null" is the negation of Definition 4.3.3:

$$\exists \varepsilon \in F_{>0} \;\; \forall N \in \mathbb{N} \;\; \exists n \in \mathbb{N} \;\; \big(n \geq N \wedge |a_n| \geq \varepsilon\big).$$

The obvious attempt reads this as "$|a_n| \geq \varepsilon$ for all large n", and that is not what it says: it promises, beyond each threshold, **one** term at least ε from 0, and between such terms others might come near 0 or change sign. The Cauchy condition is what forbids that wandering. So **fix the bad** ε and keep it for the rest of the proof. Ask the Cauchy condition for a threshold N_0 with tolerance $\varepsilon/2$, and only then ask the negation for one term a_{n_0}, beyond N_0, with $|a_{n_0}| \geq \varepsilon$. Every later term is within $\varepsilon/2$ of a_{n_0}, so it has the sign of a_{n_0} and size at least $\varepsilon/2$. *Before reading on: where is $n_0 \geq N_0$ used?*

Proof. Negating Definition 4.3.3 (Logic and Proof, Proposition 6.2.3, Logic and Proof, Proposition 6.2.2) gives $\varepsilon \in F_{>0}$ such that for every $N \in \mathbb{N}$ some $n \geq N$ has $|a_n| \geq \varepsilon$. Fix such an ε.

Since (a_n) is Cauchy, there is $N_0 \in \mathbb{N}$ (depending only on the sequence, through ε) with

$$|a_m - a_n| < \frac{\varepsilon}{2} \quad \text{for all } m, n \geq N_0.$$

Applying the negation with $N := N_0$ gives $n_0 \in \mathbb{N}$ with $n_0 \geq N_0$ and $|a_{n_0}| \geq \varepsilon$. So $a_{n_0} \geq \varepsilon$ or $a_{n_0} \leq -\varepsilon$, according to the sign of a_{n_0} (Definition 4.1.1).

Let $n \geq N_0$. Since both n and n_0 are $\geq N_0$, $|a_n - a_{n_0}| < \varepsilon/2$, that is, $-\varepsilon/2 < a_n - a_{n_0} < \varepsilon/2$ (Proposition 4.1.3 (d)).

Case 1: $a_{n_0} \geq \varepsilon$. Adding a_{n_0} to the left inequality, $a_n > a_{n_0} - \varepsilon/2 \geq \varepsilon - \varepsilon/2 = \varepsilon/2$.

Case 2: $a_{n_0} \leq -\varepsilon$. Adding a_{n_0} to the right inequality, $a_n < a_{n_0} + \varepsilon/2 \leq -\varepsilon + \varepsilon/2 = -\varepsilon/2$.

Let $c := \varepsilon/2$ and $N := N_0$. In Case 1, $a_n \geq c$ for every $n \geq N$; in Case 2, $a_n \leq -c$ for every

$n \geq N$.

Finally, both alternatives cannot hold, even with different constants and thresholds: if $a_n \geq c$ for $n \geq N$ and $a_n \leq -c'$ for $n \geq N'$, with $c, c' \in F_{>0}$, then $n := \max\{N, N'\}$ gives $0 < c \leq a_n \leq -c' < 0$, a contradiction. This proves the theorem.

■

The answer to the prompt: $n_0 \geq N_0$ is used once, to apply the Cauchy estimate to the pair n, n_0. An n_0 taken before N_0 is known might lie below N_0 and say nothing about the tail.

Both hypotheses are needed; the witnesses are in $\mathbb{Q}$.

- *Not null.* The sequence $(1/(n+1))$ is Cauchy (the examples after Definition 4.5.1) and null. For any $c \in \mathbb{Q}_{>0}$ and any N, Corollary 3.2.4 gives an index $n \geq N$ with $0 < 1/(n+1) < c$, so neither alternative holds.
- *Cauchy.* The sequence $((-1)^n)$ is not null, since $|(-1)^n| = 1 \geq 1$ for every n. Its terms alternate between 1 and -1, so neither alternative holds for any $c > 0$ and N: a_N and a_{N+1} have opposite signs.

Quick check. For $a_n = (n-3)/(n+1)$ in $\mathbb{Q}$, which alternative of Theorem 4.5.4 holds? Give a valid c and N.

Solution. $a_n = 1 - 4/(n+1)$. For $n \geq 7$, $n+1 \geq 8$, so $0 < 1/(n+1) \leq 1/8$ (Proposition 3.2.2 (b), or equality at $n = 7$), hence $4/(n+1) \leq 1/2$ and $a_n \geq 1/2$: the first alternative, with $c = 1/2$, $N = 7$. (At $n = 6$, $a_6 = 3/7 < 1/2$, so this N is least for this c; other pairs, such as $c = 1/5$, $N = 4$, work too.)

4.5.5 Sign stability and reciprocals

Two smaller results complete the toolkit: the sign found by Theorem 4.5.4 survives a null change of the sequence, and a sequence that stays away from 0 can be inverted.

Proposition 4.5.5. Sign Stability under Null Differences

Let (a_n) and (b_n) be sequences in F such that $(a_n - b_n)$ is null. If $c \in F_{>0}$, $N \in \mathbb{N}$, and $a_n \geq c$ for every $n \geq N$, then there is $N' \in \mathbb{N}$ with $b_n \geq c/2$ for every $n \geq N'$.

Proof. Since $(a_n - b_n)$ is null, there is N_1 (depending on c) with $|a_n - b_n| < c/2$ for every $n \geq N_1$. Let $N' := \max\{N, N_1\}$ and $n \geq N'$. Then $a_n - b_n < c/2$, so

$$b_n > a_n - \frac{c}{2} \geq c - \frac{c}{2} = \frac{c}{2}.$$

In particular $b_n \geq c/2$.

■

Neither sequence needs to be Cauchy here. Chapter 5 uses the proposition to show that being eventually $\geq c$ for some $c > 0$ does not depend on which of two sequences with null difference represents a real number.

Proposition 4.5.6. Reciprocals of Cauchy Sequences

Let (a_n) be a Cauchy sequence in F, and let $c \in F_{>0}$ and $N \in \mathbb{N}$ be such that $|a_n| \geq c$ for every $n \geq N$. Define

$$b_n := \begin{cases} 1/a_n & \text{if } n \geq N, \\ 1 & \text{if } n < N. \end{cases}$$

Then (b_n) is a Cauchy sequence in F, and $a_n b_n = 1$ for every $n \geq N$.

Idea. The difference of reciprocals is $(a_m - a_n)/(a_n a_m)$; its numerator can be made small, and the denominator is at least c^2 in absolute value. So $|b_n - b_m| \leq |a_m - a_n|/c^2$, and the tolerance $c^2\varepsilon$ for (a_n) gives the tolerance ε for (b_n).

Proof. For $n \geq N$, $|a_n| \geq c > 0$, so $a_n \neq 0$, and $b_n = a_n^{-1}$ is defined, with $a_n b_n = 1$. For $m, n \geq N$, Lemma 4.1.1 (c) gives

$$b_n - b_m = \frac{1}{a_n} - \frac{1}{a_m} = \frac{a_m - a_n}{a_n a_m}.$$

So $|b_n - b_m| = |a_m - a_n| \cdot (1/(|a_n|\,|a_m|))$. From $0 < c \leq |a_n|$ and $0 < c \leq |a_m|$, Proposition 4.1.2 (d) gives $c^2 \leq |a_n|\,|a_m|$, and then Proposition 4.1.2 (c) gives $1/(|a_n|\,|a_m|) \leq 1/c^2$. Multiplying by $|a_m - a_n| \geq 0$ (Proposition 4.1.2 (d), as $1/(|a_n|\,|a_m|) > 0$) gives

$$|b_n - b_m| \leq \frac{|a_m - a_n|}{c^2} \quad \text{for all } m, n \geq N.$$

Now let $\varepsilon \in F_{>0}$. Then $c^2\varepsilon > 0$, so there is N_1 (depending on ε and c) with $|a_m - a_n| < c^2\varepsilon$ for all $m, n \geq N_1$. Let $N' := \max\{N, N_1\}$. For $m, n \geq N'$, multiplying by $1/c^2 > 0$ (Proposition 3.2.2 (a), Proposition 2.4.3 (c)),

$$|b_n - b_m| \leq \frac{|a_m - a_n|}{c^2} < \frac{c^2\varepsilon}{c^2} = \varepsilon.$$

So (b_n) is Cauchy. The values $b_n = 1$ for $n < N$ play no part; any values would do, by the tail-change remark after Definition 4.5.1.

■

The hypothesis "$|a_n| \geq c$" cannot be weakened to "$a_n \neq 0$". In $\mathbb{Q}$, $a_n = 1/(n+1)$ is Cauchy and never 0, but $1/a_n = n + 1$ is not bounded (for each $M \in \mathbb{Q}_{>0}$, Theorem 3.2.3 gives an n with $n + 1 > M$), so it is not Cauchy, by Proposition 4.5.2. Combined with Theorem 4.5.4, which supplies c and N for every Cauchy sequence that is not null, this proposition is how Chapter 5 inverts a non-zero real number.

4.5.6 Small steps do not make a Cauchy sequence

The commonest wrong proof that a sequence is Cauchy looks only at neighboring terms.

Warning. "If $a_{n+1} - a_n \to 0$, then (a_n) is Cauchy" is **false**. The next example gives a sequence of rationals whose steps shrink to 0 and whose terms do not settle down at all.

Example. Shrinking Steps, Terms That Do Not Settle

Here is an argument for the false claim in the warning.

"Let $\varepsilon \in F_{>0}$. Since $a_{n+1} - a_n \to 0$, choose N with $|a_{k+1} - a_k| < \varepsilon$ for every $k \geq N$. Let $m > n \geq N$. Then $a_m - a_n$ is the sum of the consecutive differences $a_{k+1} - a_k$ for $k = n, \ldots, m-1$, and each of them is smaller than ε in absolute value. So $|a_m - a_n| < \varepsilon$, and (a_n) is Cauchy."

In $\mathbb{Q}$, let $H_0 := 0$ and $H_{n+1} := H_n + 1/(n+1)$ for $n \in \mathbb{N}$. Show that $H_{n+1} - H_n \to 0$, that $H_{2n} - H_n \geq 1/2$ for every $n \geq 1$, and that (H_n) is not Cauchy. Find the line of the argument that fails.

Solution. The sequence. By Sets and Functions, Theorem 3.3.1 (read on $\mathbb{N} = \omega$ as Theorem 1.2.1, by Corollary 1.4.3 (b)), with $X = \mathbb{Q}$, starting value 0 and $g(n, s) := s + 1/(n+1)$, there is exactly one sequence (H_n) in $\mathbb{Q}$ with $H_0 = 0$ and $H_{n+1} = H_n + 1/(n+1)$. Its first terms are $0, 1, 3/2, 11/6, 25/12$.

Shrinking steps. $H_{n+1} - H_n = 1/(n+1)$, and $(1/(n+1))$ is null in $\mathbb{Q}$ (Example (First Limits)).

Long stretches. Fix $n \geq 1$. We show, by Logic and Proof, Axiom 4.2.1 on k (through Corollary 1.4.3 (b)), that

$$H_{n+k} - H_n \geq \frac{k}{n+k} \quad \text{for every } k \in \mathbb{N}.$$

Base $P(0)$: both sides are 0. Step: assume it for k. Since $n + k \geq 1$, we have $0 < n + k < n + k + 1$, so $0 < 1/(n+k+1) < 1/(n+k)$ (Proposition 3.2.2 (b)). With $0 \leq k \leq k$, Proposition 4.1.2 (d) gives $k/(n+k+1) \leq k/(n+k)$. Hence, by the induction hypothesis,

$$\begin{aligned} H_{n+k+1} - H_n &= (H_{n+k} - H_n) + \frac{1}{n+k+1} \\ &\geq \frac{k}{n+k+1} + \frac{1}{n+k+1} = \frac{k+1}{n+k+1}. \end{aligned}$$

Taking $k = n$ gives $H_{2n} - H_n \geq n/(2n) = 1/2$.

Not Cauchy. We prove the negation displayed after Definition 4.5.1 with $\varepsilon = 1/2$. Let $N \in \mathbb{N}$, and put $n := N + 1$ and $m := 2n$. Then $m, n \geq N$, $n \geq 1$, and $|H_m - H_n| = H_{2n} - H_n \geq 1/2$. So no threshold works for the tolerance $1/2$.

The failing line. "Each of them is smaller than ε ... so $|a_m - a_n| < \varepsilon$." A sum of $m - n$ terms, each of absolute value below ε, is only below $(m-n)\varepsilon$ (triangle inequality, Proposition 4.1.3 (e), applied $m - n - 1$ times). The number of summands $m - n$ is not fixed: m is chosen

after N, and the definition allows it to be as far beyond n as we like. To repair the line one would need each step below $\varepsilon/(m-n)$, a tolerance that depends on m and n; no single N is promised for it.

Section 4.6 goes further: its Exercise B2 shows $H_{2^k} \geq 1 + k/2$ for every $k \in \mathbb{N}$, so (H_n) even grows without bound.

4.5.7 Exercises

4.5.7.1 A. Check your understanding

Exercise. A1

These questions are about Theorem 4.5.4 and its proof.

(a) Write "(a_n) is not null" in symbols, with every quantifier and its domain.

(b) The proof asks the Cauchy condition for the tolerance $\varepsilon/2$. Why not ε itself?

(c) Where is the inequality $n_0 \geq N_0$ used?

(d) Run the proof in $\mathbb{Q}$ on $a_n = 1/(n+1) - 1$ with $\varepsilon = 1/2$: check that this ε witnesses "not null", give the least N_0 that the Cauchy condition with tolerance $1/4$ allows, choose n_0, and state the c and N the proof produces.

Solution.

(a) $\exists \varepsilon \in F_{>0}\ \forall N \in \mathbb{N}\ \exists n \in \mathbb{N}\ (n \geq N \wedge |a_n| \geq \varepsilon)$: one tolerance and, beyond every threshold, some term at least that far from 0.

(b) With tolerance ε, Case 1 would give only $a_n > a_{n_0} - \varepsilon \geq \varepsilon - \varepsilon = 0$: every later term is positive, but no positive c with $a_n \geq c$ comes out, and the estimate $a_n > 0$ alone does not rule out terms approaching 0. Any tolerance strictly smaller than ε leaves a positive margin; $\varepsilon/2$ is the simplest.

(c) In the line "since both n and n_0 are $\geq N_0$, $|a_n - a_{n_0}| < \varepsilon/2$": the Cauchy estimate holds only for pairs of indices beyond N_0. It is the reason N_0 is chosen first and the negation is applied **at** N_0.

(d) Here $|a_n| = 1 - 1/(n+1) = n/(n+1)$, which is 0 at $n = 0$ and at least $1/2$ for every $n \geq 1$ (as $2n \geq n+1$). So beyond any N the index $n = N+1$ has $|a_n| \geq 1/2$: $\varepsilon = 1/2$ witnesses "not null". Next, $|a_m - a_n| = |1/(m+1) - 1/(n+1)|$. For $m, n \geq 3$, both reciprocals lie in $(0, 1/4]$, so their difference is below $1/4$, as in the examples after Definition 4.5.1. For $N_0 = 2$ the pair $m = 2$, $n = 11$ gives $1/3 - 1/12 = 1/4 \not< 1/4$. So the least N_0 is 3. Take $n_0 = 3 \geq N_0$: $a_3 = -3/4$, $|a_3| = 3/4 \geq 1/2$, Case 2. The proof gives $a_n < a_3 + 1/4 = -1/2 \leq -1/4$ for $n \geq 3$, and it states the result as $c = \varepsilon/2 = 1/4$, $N = 3$: $a_n \leq -1/4$ for every $n \geq 3$. ($n_0 = 0$ is not allowed: it is below N_0, and $|a_0| = 0 < 1/2$.)

4.5.7.2 B. Practice

Exercise. B1

Determine which of the following sequences in $\mathbb{Q}$ are Cauchy. Justify your answers, and for

the first one work directly from Definition 4.5.1.

(a) $a_n = (n+1)/(n+2)$.
(b) $a_n = (-1)^n$.
(c) $a_n = n$.

Solution.

(a) Cauchy. For every $k \in \mathbb{N}$, $(k+1)/(k+2) = 1 - 1/(k+2)$, so $|a_m - a_n| = |1/(n+2) - 1/(m+2)|$. Let $\varepsilon \in \mathbb{Q}_{>0}$, and use Corollary 3.2.4 to choose $N \in \mathbb{Z}_{>0}$ with $1/N < \varepsilon$. For $m, n \geq N$, both $1/(m+2)$ and $1/(n+2)$ lie in $(0, 1/N)$, since $m+2, n+2 > N$ (Proposition 3.2.2 (b)). Two elements of $(0, 1/N)$ differ by less than $1/N$ (as in the examples after the definition), so $|a_m - a_n| < 1/N < \varepsilon$.

(b) Not Cauchy. Take $\varepsilon = 1$. For any N, the indices N and $N+1$ have $a_{N+1} = -a_N$ and $|a_N| = 1$, so $|a_N - a_{N+1}| = |2a_N| = 2 \geq 1$.

(c) Not Cauchy. If it were, it would be bounded (Proposition 4.5.2): $|n| \leq M$ for some $M \in \mathbb{Q}_{>0}$ and every n. By Theorem 3.2.3 there is $n \in \mathbb{N}$ with $M < n = |n|$, a contradiction. (Archimedes is used here; in the non-Archimedean field $\mathbb{Q}(t)$ of Section 4.2 the sequence $(n \cdot 1)$ is bounded, by t, and the boundedness argument says nothing.)

Exercise. B2

Let (a_n) and (b_n) be Cauchy sequences in F. Prove directly from Definition 4.5.1 that $(a_n - b_n)$ is Cauchy, naming the rule behind each step.

Solution. Let $\varepsilon \in F_{>0}$. By Proposition 4.1.2 (b), $\varepsilon/2 \in F_{>0}$. Since (a_n) is Cauchy, choose N_1 with $|a_m - a_n| < \varepsilon/2$ for all $m, n \geq N_1$; since (b_n) is Cauchy, choose N_2 with $|b_m - b_n| < \varepsilon/2$ for all $m, n \geq N_2$. Let $N := \max\{N_1, N_2\}$ (a maximum of two natural numbers; it depends only on ε). Let $m, n \geq N$; then $m, n \geq N_1$ and $m, n \geq N_2$. By the ring rules (Proposition 2.3.2),

$$(a_m - b_m) - (a_n - b_n) = (a_m - a_n) + (-(b_m - b_n)).$$

By the triangle inequality (Proposition 4.1.3 (e)) and $|-x| = |x|$ (Proposition 4.1.3 (b)),

$$|(a_m - b_m) - (a_n - b_n)| \leq |a_m - a_n| + |b_m - b_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

adding strict inequalities by Proposition 2.4.3 (b) and using Proposition 4.1.2 (b) for the equality. Since ε was arbitrary, $(a_n - b_n)$ is Cauchy.

Exercise. B3

Let $a_n = 2 - 1/(n+1)$ in $\mathbb{Q}$. Show that (a_n) is Cauchy and not null, and find $c \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ as in Theorem 4.5.4. Then run the proof of the theorem with $\varepsilon = 1$ and the least admissible N_0: which c and N does it produce?

Solution. Cauchy. (a_n) is the difference of the constant sequence 2 and the Cauchy sequence $(1/(n+1))$ (examples after Definition 4.5.1), so it is Cauchy by Theorem 4.5.3 (a), (b).

Not null, and c, N directly. For every n, $0 < 1/(n+1) \leq 1$, so $a_n \geq 1$. Then $|a_n| \geq 1$ for every n, so $\varepsilon = 1$ witnesses "not null", and $c = 1$, $N = 0$ satisfy the first alternative. (Any smaller positive c, for instance $c = 1/2$, works with $N = 0$ as well.)

The proof run. With $\varepsilon = 1$, the Cauchy condition is asked for tolerance $1/2$. Here $|a_m - a_n| = |1/(n+1) - 1/(m+1)|$; for $m, n \geq 1$ both reciprocals lie in $(0, 1/2]$, so the difference is below $1/2$, while $N_0 = 0$ fails ($m = 0$, $n = 1$: $1 - 1/2 = 1/2 \not< 1/2$). So $N_0 = 1$. Take $n_0 = 1$: $a_1 = 3/2 \geq 1$, Case 1. The proof gives $c = 1/2$, $N = 1$: $a_n > 3/2 - 1/2 = 1 \geq 1/2$ for every $n \geq 1$.

4.5.7.3 C. Going deeper

Exercise. C1

Let (a_n) be a Cauchy sequence in F and (b_n) a sequence in F such that $(a_n - b_n)$ is null. Prove that (b_n) is Cauchy.

Here is the beginning of a proof. *Name the move, then complete it.*

"Let $\varepsilon \in F_{>0}$. For all $m, n \in \mathbb{N}$, the triangle inequality (Proposition 4.1.3 (e)), applied twice, gives

$$|b_n - b_m| \leq |b_n - a_n| + |a_n - a_m| + |a_m - b_m|.$$

..."

Solution. The move is **split** ε, now into three parts, one for each term on the right: two come from the null difference, one from the Cauchy condition.

Completion. Let $3 := 1 + 1 + 1 = 3 \cdot 1_F$; then $3 > 0$ (Proposition 4.2.1 (b)), so $\varepsilon/3 \in F_{>0}$ (Proposition 3.2.2 (a), (O4)), and $\varepsilon/3 + \varepsilon/3 + \varepsilon/3 = 3 \cdot (\varepsilon/3) = \varepsilon$: by (F5)–(F9), with $3 \neq 0$. Since $(a_n - b_n)$ is null, choose N_1 with $|a_n - b_n| < \varepsilon/3$ for every $n \geq N_1$. Since (a_n) is Cauchy, choose N_2 with $|a_n - a_m| < \varepsilon/3$ for all $m, n \geq N_2$. Let $N := \max\{N_1, N_2\}$, which depends only on ε. For $m, n \geq N$, using $|b_n - a_n| = |a_n - b_n|$ (Proposition 4.1.3 (b)),

$$|b_n - b_m| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,$$

adding three strict inequalities (Proposition 2.4.3 (b)). So (b_n) is Cauchy.

The split $\varepsilon/4$, $\varepsilon/2$, $\varepsilon/4$, which needs only halving, works as well.

Exercise. C2: Exit check: the sign theorem

Let (a_n) be a Cauchy sequence in an ordered field F that is not null. Prove that there are $c \in F_{>0}$ and $N \in \mathbb{N}$ such that either $a_n \geq c$ for every $n \geq N$, or $a_n \leq -c$ for every $n \geq N$.

Solution. "Null" means $\forall \varepsilon \in F_{>0}\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \Rightarrow |a_n| < \varepsilon)$. Its negation (Logic and Proof, Proposition 6.2.3, Logic and Proof, Proposition 6.2.1 (a), Logic and Proof, Proposition 6.2.2, then (O1)) is

$$\exists \varepsilon \in F_{>0}\ \ \forall N \in \mathbb{N}\ \ \exists n \in \mathbb{N}\ \ \big(n \geq N \wedge |a_n| \geq \varepsilon\big).$$

Fix such an ε; it stays fixed from here on.

Apply the Cauchy condition with the tolerance $\varepsilon/2 \in F_{>0}$ (Proposition 4.1.2 (b)): there is N_0 with $|a_m - a_n| < \varepsilon/2$ for all $m, n \geq N_0$. Apply the negation at N_0: there is $n_0 \geq N_0$ with $|a_{n_0}| \geq \varepsilon$. By the definition of $|\cdot|$, $a_{n_0} \geq \varepsilon$ or $a_{n_0} \leq -\varepsilon$.

Let $n \geq N_0$. As $n, n_0 \geq N_0$, $-\varepsilon/2 < a_n - a_{n_0} < \varepsilon/2$ (Proposition 4.1.3 (d)). If $a_{n_0} \geq \varepsilon$, then $a_n > a_{n_0} - \varepsilon/2 \geq \varepsilon/2$. If $a_{n_0} \leq -\varepsilon$, then $a_n < a_{n_0} + \varepsilon/2 \leq -\varepsilon/2$. With $c := \varepsilon/2$ and $N := N_0$, the first case gives $a_n \geq c$ and the second $a_n \leq -c$, for every $n \geq N$.

Marking scheme. (10 points) - 2: negates "null" correctly, with the quantifiers in order, and fixes one ε. - 2: applies the Cauchy condition with a fixed tolerance δ such that $0 < \delta < \varepsilon$ (for example $\delta = \varepsilon/2$) to get N_0. - 2: obtains $n_0 \geq N_0$ with $|a_{n_0}| \geq \varepsilon$ from the negation applied **at** N_0. - 3: derives a one-signed bound $a_n \geq c$ or $a_n \leq -c$, with some $c \in F_{>0}$ (for example $c = \varepsilon - \delta$), for every $n \geq N_0$, in both sign cases. - 1: names c and N and concludes. A proof by contraposition (assume neither alternative holds for any c, N; show the sequence null using the Cauchy condition at tolerance $\varepsilon/2$ and, beyond N_0, a term below $\varepsilon/2$ and a term above $-\varepsilon/2$) earns the same points, step for step: 2 for the negated conclusion, 2 for the Cauchy threshold, 2 for choosing the witnesses beyond N_0, 3 for the two-sided bound, 1 for concluding. A complete proof by contradiction likewise earns the same $2+2+2+3+1$. Fatal errors (0 for the step): an n_0 not known to satisfy $n_0 \geq N_0$, or "$|a_n| \geq \varepsilon$ for all large n" read off "not null" alone.

Exercise. C3: Find the error

Here is an argument that the product of two Cauchy sequences (a_n), (b_n) in F is Cauchy.

"For all m, n, $|a_n b_n - a_m b_m| \leq |a_n|\,|b_n - b_m| + |b_m|\,|a_n - a_m|$. Given $\varepsilon \in F_{>0}$, choose N such that $|b_n - b_m| < \varepsilon/(2|a_n|)$ and $|a_n - a_m| < \varepsilon/(2|b_m|)$ for all $m, n \geq N$. Then for $m, n \geq N$, $|a_n b_n - a_m b_m| < \varepsilon/2 + \varepsilon/2 = \varepsilon$."

Is the conclusion true? Does the argument prove it? If not, name the line that fails, say why, and repair it.

Solution. The conclusion is true: it is Theorem 4.5.3 (c).

The argument does not prove it. The failing line is "choose N such that $|b_n - b_m| < \varepsilon/(2|a_n|)$ ... for all $m, n \geq N$". The tolerances $\varepsilon/(2|a_n|)$ and $\varepsilon/(2|b_m|)$ depend on n and m, which are introduced **after** N. The definition of Cauchy supplies a threshold for one fixed tolerance $\varepsilon' \in F_{>0}$, named before the threshold; here the threshold is asked to serve a tolerance that changes with the very indices it controls. This is a witness depending on variables introduced later, entry (N2) of Logic and Proof, Example (What is not allowed).

A counterexample to the step. The step gets one N for all the tolerances $\varepsilon/(2|a_n|)$ from their being positive. In $\mathbb{Q}$ take the Cauchy sequence $b_n = 1/(n+1)$, the never-zero (but

unbounded) $a_n = n + 1$, and $\varepsilon = 1$. For any N, $n = N$ and $m = 2N + 1$ give

$$|b_n - b_m| = \frac{1}{N+1} - \frac{1}{2N+2} = \frac{1}{2(N+1)} = \frac{\varepsilon}{2|a_n|},$$

which is not smaller: no N works. Inside the exercise (a_n) is Cauchy, hence bounded, but the argument never uses that; the repair does.

Repair. Fix bounds first: by Proposition 4.5.2, choose $A, B \in F_{>0}$ with $|a_n| \le A$ and $|b_n| \le B$ for every n, before ε is named. Then the fixed tolerances $\varepsilon/(2A) \le \varepsilon/(2|a_n|)$ (when $a_n \neq 0$) and $\varepsilon/(2B)$ give thresholds N_1, N_2, and $N := \max\{N_1, N_2\}$ works, as in the proof of Theorem 4.5.3 (c).

4.6 A Cauchy Sequence with No Rational Limit

Section 4.5 proved that every convergent sequence is Cauchy (Proposition 4.5.1). In $\mathbb{Q}$ the converse fails, and the failure sits exactly at the hole Chapter 3 found at $\sqrt{2}$: no rational squares to 2 (Theorem 3.3.1). This section proves that the rational sequence of Chapter 3 that closes in on that hole (Example (A Rational Sequence Aimed at the Gap)) is Cauchy in $\mathbb{Q}$ and converges to no rational. Chapter 5 starts from this sequence and builds the number it is aiming at. The section ends by listing what the chapter has not proved, and why, in $\mathbb{Q}$, it could not have.

Throughout this section the field is $F = \mathbb{Q}$: every tolerance ε is a positive rational, and "converges" means "converges in $\mathbb{Q}$" (Definition 4.3.2).

4.6.1 The sequence again

Recall the sequence of Example (A Rational Sequence Aimed at the Gap). For $s \in \mathbb{Q}_{>0}$ let $t(s) := (s^2+2)/(2s)$, the average of s and $2/s$; let $x_0 := 2$ and $x_{n+1} := t(x_n)$, and let $e_n := x_n^2 - 2$, the amount by which x_n^2 overshoots 2. The first terms are

$$x_0 = 2, \quad x_1 = \frac{3}{2}, \quad x_2 = \frac{17}{12}, \quad x_3 = \frac{577}{408},$$

with

$$e_0 = 2, \quad e_1 = \frac{1}{4}, \quad e_2 = \frac{1}{144}, \quad e_3 = \frac{1}{166464}.$$

Chapter 3 proved, for every $n \in \mathbb{N}$, the following facts, and we cite them from Example (A Rational Sequence Aimed at the Gap) without proving them again:

$$x_n \in \mathbb{Q}_{>0}, \qquad e_n > 0 \ \ (\text{so } x_n^2 > 2),$$
$$e_{n+1} = \frac{e_n^2}{4x_n^2} \le \frac{e_n^2}{8}, \qquad x_n - x_{n+1} = \frac{e_n}{2x_n} > 0.$$

The second line says two things. The overshoot is roughly squared at each step, so it shrinks very fast; and the terms decrease. Section 4.5 asks for more than decrease: any two late terms must be close. The overshoot e_n measures how close, because every later term lies between $2/x_n$ and x_n, an interval of length e_n/x_n.

For the proof we do not need the true speed of e_n. A crude bound is enough, as long as it can be pushed below every rational tolerance:

$$e_1 = \frac{1}{4}, \quad e_2 = \frac{1}{144} \le \frac{1}{8}, \quad e_3 = \frac{1}{166464} \le \frac{1}{12};$$

that is, $e_n \le 1/(4n)$ for $n = 1, 2, 3$, and the theorem proves it for every $n \ge 1$. Archimedes then turns $1/(4n)$ into anything below a given ε.

4.6.2 The theorem

Theorem 4.6.1. The Newton Sequence Is Cauchy with No Rational Limit

Let $(x_n)_{n\in\mathbb{N}}$ and $e_n = x_n^2 - 2$ be as in Example (A Rational Sequence Aimed at the Gap). Then $x_n > 1$ for every $n \in \mathbb{N}$, $x_m \le x_n$ whenever $m \ge n$, and $e_n \le 1/(4n)$ for every $n \ge 1$. The sequence (x_n) is a Cauchy sequence in $\mathbb{Q}$, and there is **no** $L \in \mathbb{Q}$ with $x_n \to L$.

The first three clauses are part of the statement, rather than steps hidden in the proof, so that Chapter 5 can cite them.

Idea. Cauchy. This keeps the shape of Example (Method: proving a limit from the definition), with a second index in place of the limit: bound $|x_m - x_n|$ by an expression in n alone, then let Archimedes push it below ε. For $m \ge n$, the later term x_m lies below x_n, since the terms decrease, and above $2/x_n$, since $x_m > 2/x_m \ge 2/x_n$. So $0 \le x_n - x_m \le x_n - 2/x_n = e_n/x_n \le e_n$. With $e_n \le 1/(4n)$, every two terms beyond N differ by at most $1/(4N)$, and Archimedes makes $1/N$ smaller than any rational ε.

No rational limit. **Locate the missing limit**: suppose $x_n \to L$ in $\mathbb{Q}$ and find out what L would have to satisfy. The limit laws turn $x_n \to L$ into $x_n^2 \to L^2$. But $x_n^2 = 2 + e_n \to 2$, because e_n is squeezed between 0 and $1/(4n)$. Limits are unique, so $L^2 = 2$, and no rational satisfies that. *Before reading on: which law turns $x_n \to L$ into $x_n^2 \to L^2$, and applied to which two sequences?*

Proof. We prove six facts, then the two claims. Induction is Logic and Proof, Axiom 4.2.1, read on $\mathbb{N} = \omega$ through Corollary 1.4.3 (b).

(i), (ii), (iv) in part: cited. By Example (A Rational Sequence Aimed at the Gap), for every $n \in \mathbb{N}$: (i) $x_n \in \mathbb{Q}_{>0}$, $e_n > 0$ and $x_n^2 > 2$; (ii) $e_{n+1} = e_n^2/(4x_n^2) \le e_n^2/8$; and $x_{n+1} < x_n$.

(iii) $x_n > 1$. Suppose $x_n \le 1$. Since $x_n > 0$ by (i), either $x_n < 1$, and then $x_n^2 < 1$ by Proposition 2.4.3 (g), or $x_n = 1$ and $x_n^2 = 1$. Either way $x_n^2 \le 1 < 2$, a contradiction with $x_n^2 > 2$. So $x_n > 1$, by totality (Proposition 2.4.3 (i)).

(iv) $x_m \le x_n$ *for* $m \ge n$. Fix n. By induction on k, $x_{n+k} \le x_n$ for every $k \in \mathbb{N}$: the base $P(0)$ is $x_n \le x_n$, and if $x_{n+k} \le x_n$, then $x_{n+k+1} < x_{n+k} \le x_n$. Every $m \ge n$ in $\mathbb{N}$ is $n + k$ for some $k \in \mathbb{N}$ (Proposition 1.3.1 (a)), so $x_m \le x_n$.

(v) $e_n \le 1/(4n)$ *for* $n \ge 1$. By induction on k, for $P(k)$: $e_{k+1} \le 1/(4(k+1))$. Base $P(0)$: $e_1 = 1/4$. Step: assume $P(k)$, and let $n := k + 1 \ge 1$, so $0 < e_n \le 1/(4n) \le 1/4$ (by (i), $P(k)$, and Proposition 4.1.2 (c)). Then, by (ii) and Proposition 4.1.2 (d) (multiplying by the

non-negative factors $e_n/8$ and $1/32$),

$$e_{n+1} \le \frac{e_n \cdot e_n}{8} \le \frac{e_n}{8} \cdot \frac{1}{4} = \frac{e_n}{32} \le \frac{1}{128n} \le \frac{1}{4(n+1)},$$

where the last step is Proposition 4.1.2 (c), since $0 < 4n + 4 \le 128n$ for $n \ge 1$. This is $P(k+1)$.

(vi) For $m \ge n \ge 1$: $0 \le x_n - x_m \le 1/(4n)$. By (iv), $x_m \le x_n$, so $x_n - x_m \ge 0$. By (i), $x_m^2 > 2$; multiplying by $1/x_m > 0$ (Proposition 3.2.2 (a), Proposition 2.4.3 (c)) gives $x_m > 2/x_m$. From $0 < x_m \le x_n$, Proposition 4.1.2 (c) gives $1/x_n \le 1/x_m$, so $2/x_n \le 2/x_m < x_m$ (Proposition 4.1.2 (d)). Hence, by negation and addition (Proposition 2.4.3 (a), (b)),

$$x_n - x_m \le x_n - \frac{2}{x_n} = \frac{x_n^2 - 2}{x_n} = \frac{e_n}{x_n} \le e_n \le \frac{1}{4n}.$$

Here $e_n/x_n \le e_n$ because $0 < 1 < x_n$ by (iii), so $1/x_n < 1$ (Proposition 3.2.2 (b)), and $e_n > 0$, so multiplying by e_n preserves the inequality (Proposition 2.4.3 (c)); the last step is (v).

Cauchy. Let $\varepsilon \in \mathbb{Q}_{>0}$. By Corollary 3.2.4 there is $N \in \mathbb{Z}_{>0}$ with $1/N < \varepsilon$; it depends only on ε. Let $m, n \ge N$, and let $k := \min\{m, n\} \ge N \ge 1$ and $l := \max\{m, n\}$. By (vi) with the pair $l \ge k$,

$$|x_m - x_n| = x_k - x_l \le \frac{1}{4k} \le \frac{1}{4N} \le \frac{1}{N} < \varepsilon,$$

where the equality holds because $x_k - x_l \ge 0$ by (vi), so $|x_k - x_l| = x_k - x_l$ by the definition of the absolute value, and $|x_m - x_n| = |x_k - x_l|$ since $|-y| = |y|$ (Proposition 4.1.3 (b)), and $1/(4k) \le 1/(4N) \le 1/N$ is Proposition 4.1.2 (c) applied to $N \le 4N \le 4k$. So (x_n) is Cauchy in $\mathbb{Q}$.

No rational limit. Suppose, for a contradiction, that $x_n \to L$ for some $L \in \mathbb{Q}$. By the product law, Theorem 4.4.3 (c), applied with both sequences equal to (x_n), $x_n^2 \to L^2$.

Next, (e_n) is null. We use the squeeze, Proposition 4.4.4 (c), with $N_0 = 1$, the lower sequence constantly 0 (which tends to 0, Example (First Limits) (a)) and the upper sequence $(1/(n+1))_{n \in \mathbb{N}}$. The upper sequence is defined for every $n \in \mathbb{N}$, and it is null in $\mathbb{Q}$ by Proposition 4.2.3, as in Example (First Limits). For $n \ge 1$ we have $n + 1 \le 4n$ (as $1 \le 3n$), so by (i), (v) and Proposition 4.1.2 (c),

$$0 < e_n \le \frac{1}{4n} \le \frac{1}{n+1} \qquad (n \ge 1).$$

(At $n = 0$ the middle term is undefined and $e_0 = 2$; the squeeze needs the inequalities only from $N_0 = 1$ on.) So $e_n \to 0$.

Then $x_n^2 = 2 + e_n$ for every n, and the sum law, Theorem 4.4.3 (a), with the constant sequence 2 (Example (First Limits) (a)) gives $x_n^2 \to 2 + 0 = 2$. The sequence (x_n^2) therefore tends to L^2 and to 2, and by Theorem 4.3.4, $L^2 = 2$. This contradicts Theorem 3.3.1. Hence no $L \in \mathbb{Q}$ has $x_n \to L$.

■

The answer to the prompt: the product law, Theorem 4.4.3 (c), applied to $a_n = b_n = x_n$. It is the only step that uses the hypothesis $x_n \to L$; the rest of the second half is about e_n, which we control without any limit, and then uniqueness compares the two limits of (x_n^2).

Archimedes is used twice, both times in $\mathbb{Q}$, where Theorem 3.2.3 provides it: for the Cauchy threshold, through Corollary 3.2.4, and for the upper sequence $(1/(n+1))$ of the squeeze, through Proposition 4.2.3.

Remark. The sequence is Cauchy and not null, since $|x_n| = x_n > 1$ for every n, so Theorem 4.5.4 applies to it. Its proof, run with $\varepsilon = 1$, gives $c = 1/2$; directly, $x_n \geq 1$ for every n. Chapter 5 turns the class of (x_n) into a positive real number whose square is 2.

Warning. "Cauchy" does not mean "convergent to a limit we have not named yet". In $\mathbb{Q}$ there is nothing to name: by Theorem 4.6.1, the statement "every Cauchy sequence of rationals converges in $\mathbb{Q}$" is **false**, not merely unproved. A proof that says "(a_n) is Cauchy, so let a be its limit" is wrong in $\mathbb{Q}$, whatever it goes on to do.

Quick check. Give an upper bound and a positive lower bound for the terms x_n in $\mathbb{Q}$, with reasons.

Solution. Upper bound 2: $x_n \leq x_0 = 2$ for every n, by clause (iv) of the proof with $n = 0$. Lower bound 1: $x_n > 1$ for every n, by (iii). The upper bound is attained, at x_0; the lower bound 1 is crude (every term also exceeds $2/x_1 = 4/3$, by the argument of (vi), and $x_0 = 2 > 4/3$), but any positive lower bound answers the question.

Quick check. For $\varepsilon = 1/100$, which N does the Cauchy half of the proof give, if N is taken least?

Solution. The proof takes $N \in \mathbb{Z}_{>0}$ with $1/N < 1/100$, that is $N > 100$. The least is $N = 101$; any $N \geq 101$ serves. (Because of the factor 4 in $1/(4N)$, a smaller N would do, but the proof does not ask for it.)

4.6.3 What this chapter did not prove

Looking back over the chapter, three familiar tools of real analysis are absent, and their absence is checkable.

- **No result asserts convergence without naming the limit.** Each result of the chapter that concludes $a_n \to a$ has a in its hypotheses or in its statement: the examples of Section 4.3 (Example (First Limits)), the tail-change rules (Proposition 4.3.2), the limit laws (Theorem 4.4.3) and the squeeze (Proposition 4.4.4 (c)). The results of Section 4.5, about sequences whose limit is unknown, conclude "Cauchy", "bounded", or a sign, never "convergent".
- **No supremum.** No result uses the supremum of a subset of F (Sets and Functions, Definition 5.3.5), so nothing here rests on a least upper bound property.
- **No monotone convergence theorem, and no Bolzano–Weierstrass theorem.** Nei-

ther is stated or used.

In $\mathbb{Q}$ each of the three would fail. A least upper bound property fails by Section 3.3: the set of rationals with square below 2 is non-empty and bounded above, with no rational supremum (Theorem 3.3.2 (a)). A monotone convergence theorem fails by the sequence of this section: (x_n) is decreasing (clause (iv) of the proof), bounded (the Quick check above) and Cauchy, and it has no limit in $\mathbb{Q}$. And so does Bolzano–Weierstrass, which would give the bounded sequence (x_n) a subsequence converging in $\mathbb{Q}$. Indeed, let $n_0 < n_1 < n_2 < \cdots$ be indices and suppose $x_{n_k} \to L \in \mathbb{Q}$ as $k \to \infty$. By induction, $n_k \geq k$ for every k (if $n_k \geq k$, then $n_{k+1} > n_k \geq k$, so $n_{k+1} \geq k+1$). So for $k \geq 1$, $0 < e_{n_k} \leq 1/(4n_k) \leq 1/(4k) \leq 1/(k+1)$, and the argument of the proof, run on the sequence $(x_{n_k})_{k\in\mathbb{N}}$, gives $L^2 = 2$ again.

The reader who knows real analysis will have met all three as theorems about $\mathbb{R}$. They are true there because of an axiom about $\mathbb{R}$, completeness, that no ordered field satisfies automatically, and this chapter never assumed it. Chapter 5 builds, from Cauchy sequences of rationals, a field in which every non-empty subset that is bounded above has a supremum: a complete ordered field. Chapter 6 proves that any two complete ordered fields are copies of each other, by exactly one order-preserving field isomorphism. *Analysis on the Real Line*, Chapter 1 takes the axioms of a complete ordered field as its starting point, and its Chapter 4 revisits this chapter's sequences, now in $\mathbb{R}$, and proves that every Cauchy sequence of real numbers converges.

4.6.4 Exercises

4.6.4.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Theorem 4.6.1.

(a) Which result gives $x_n^2 \to L^2$ from $x_n \to L$, applied to which two sequences?

(b) Why does the proof need the bound $e_n \leq 1/(4n)$, and not only "$e_n > 0$ and (e_n) is decreasing"?

(c) Where is the Archimedean property used?

(d) Check fact (vi) at $n = 2$, $m = 3$: compute $x_2 - x_3$ and compare it with e_2/x_2, e_2 and $1/(4 \cdot 2)$.

(e) State the idea of the proof that (x_n) has no rational limit, in two sentences.

Solution.

(a) The product law, Theorem 4.4.3 (c), with $a_n = b_n = x_n$ and $a = b = L$.

(b) The proof must show that (e_n) is null, that is, below every rational ε from some index on. A sequence can be positive and decreasing without being null: $1 + 1/(n+1)$ decreases and stays above 1. The bound $1/(4n) \leq 1/(n+1)$ ties e_n to a sequence known to be null, and the same bound gives the Cauchy threshold.

(c) Twice: for the Cauchy threshold, by Corollary 3.2.4 (an N with $1/N < \varepsilon$); and for the upper sequence $(1/(n+1))$ of the squeeze, null by Proposition 4.2.3.

(d) $x_2 - x_3 = 17/12 - 577/408 = (17 \cdot 34 - 577)/408 = (578 - 577)/408 = 1/408$. Next, $e_2/x_2 = (1/144) \cdot (12/17) = 1/204$. So

$$\frac{1}{408} \leq \frac{1}{204} \leq \frac{1}{144} \leq \frac{1}{8},$$

which is the chain $x_2 - x_3 \le e_2/x_2 \le e_2 \le 1/(4 \cdot 2)$ of (vi).

(e) A rational limit L of (x_n) would make x_n^2 tend to L^2, by the product law. But $x_n^2 = 2 + e_n$ tends to 2, since e_n is squeezed to 0, so uniqueness of limits gives $L^2 = 2$, which no rational satisfies.

4.6.4.2 B. Practice

Exercise. B1

With (x_n) as in Theorem 4.6.1, prove that the sequence $(x_n - 2/x_n)$ is null in $\mathbb{Q}$, that $(2/x_n)$ is Cauchy in $\mathbb{Q}$, and that $(2/x_n)$ has no limit in $\mathbb{Q}$.

Solution. Null. For every n, $x_n - 2/x_n = (x_n^2 - 2)/x_n = e_n/x_n$, and as in step (vi) of the proof, $0 < e_n/x_n \le e_n \le 1/(4n) \le 1/(n+1)$ for $n \ge 1$ (using $x_n > 1$ and $e_n \le 1/(4n)$ from the theorem, and $e_n > 0$ from Example (A Rational Sequence Aimed at the Gap) (i)). By the squeeze (Proposition 4.4.4 (c)) with $N_0 = 1$, lower sequence 0 and upper sequence $(1/(n+1))$, null in $\mathbb{Q}$ (Example (First Limits)), $x_n - 2/x_n \to 0$.

Cauchy. $2/x_n = x_n - (x_n - 2/x_n)$ is the difference of a Cauchy sequence (Theorem 4.6.1) and a null one, which is Cauchy by Theorem 4.5.3 (a); so it is Cauchy by Theorem 4.5.3 (b).

No rational limit. Suppose $2/x_n \to L \in \mathbb{Q}$. Then $x_n = 2/x_n + (x_n - 2/x_n) \to L + 0 = L$ by the sum law (Theorem 4.4.3 (a)), against Theorem 4.6.1.

The terms $2/x_n$ approach the gap from below ($2/x_n < x_m$ for all m, and $(2/x_n)^2 = 4/x_n^2 < 2$), while the x_n approach it from above; their differences are null, so in Chapter 5 the two sequences will represent the same real number.

Exercise. B2

Let (H_n) be the sequence of Example (Shrinking Steps, Terms That Do Not Settle): $H_0 = 0$ and $H_{n+1} = H_n + 1/(n+1)$ in $\mathbb{Q}$. Prove that $H_{2^k} \ge 1 + k/2$ for every $k \in \mathbb{N}$, and hence that (H_n) is not bounded.

Solution. The powers 2^k of the natural number 2 are those of Section 1.3 (defined just before Proposition 1.3.4 by Theorem 1.2.1), with $2^0 = 1$ and $2^{k+1} = 2^k \cdot 2 = 2^k + 2^k$.

We use Logic and Proof, Axiom 4.2.1 on k. Base $P(0)$: $H_{2^0} = H_1 = H_0 + 1/1 = 1 \ge 1 + 0/2$. Step: assume $H_{2^k} \ge 1 + k/2$. The example shows $H_{2n} - H_n \ge 1/2$ for every $n \ge 1$; with $n = 2^k \ge 1$ (by induction, $2^k \ge 1$) it gives $H_{2^{k+1}} \ge H_{2^k} + 1/2 \ge 1 + (k+1)/2$.

Not bounded. Let $M \in \mathbb{Q}_{>0}$. By Theorem 3.2.3 there is $k \in \mathbb{N}$ with $2M < k$. Then $H_{2^k} \ge 1 + k/2 > M$, and $H_{2^k} > 0$, so $|H_{2^k}| > M$. No M bounds the sequence.

So (H_n) is not Cauchy for a second reason: a Cauchy sequence is bounded (Proposition 4.5.2).

4.6.4.3 C. Going deeper

Exercise. C1: Exit check: a Cauchy sequence with no rational limit

Let $y_0 := 2$ and $y_{n+1} := (y_n^2 + 3)/(2y_n)$, a sequence in $\mathbb{Q}_{>0}$, and let $d_n := y_n^2 - 3$.

(a) Prove that no rational number squares to 3.

(b) Prove that (y_n) is a Cauchy sequence in $\mathbb{Q}$ and that it converges to no element of $\mathbb{Q}$.

Solution (Hint 1). For (b), copy facts (i)–(vi) of the proof of Theorem 4.6.1 with 3 in place of 2. Since only the case 2 was proved in Chapter 3, prove the analogues of (i), (ii) and the decrease yourself, as Example (A Rational Sequence Aimed at the Gap) does for 2. For (a), run the descent of Theorem 3.3.1.

Solution (Hint 2). First verify the identities $d_{n+1} = d_n^2/(4y_n^2)$ and $y_n - y_{n+1} = d_n/(2y_n)$ by clearing denominators. In the descent for (a), from $p^2 = 3q^2$ show $q < p < 2q$, and try $p' := 3q - p$, $q' := p - q$.

Solution.

(a) Suppose $x \in \mathbb{Q}$ with $x^2 = 3$. Then $x \neq 0$, and replacing x by $-x$ if needed, $x > 0$; write $x = a/b$ with $a, b \in \mathbb{Z}_{>0}$ (Proposition 3.1.4 (b), Theorem 3.2.1 (a)), so $a^2 = 3b^2$. The set of $q \in \mathbb{Z}_{>0}$ for which some $p \in \mathbb{Z}_{>0}$ has $p^2 = 3q^2$ contains b and is bounded below by 0, so it has a least element q (Proposition 2.4.4); fix such a p.
$q < p$: if $p \leq q$, then $p^2 \leq q^2$ (Proposition 2.4.3 (g), or equality), so $3q^2 \leq q^2$ and $2q^2 \leq 0$, against $2q^2 > 0$ (Proposition 2.4.3 (e), (O4)). *$p < 2q$:* if $2q \leq p$, then $4q^2 \leq p^2 = 3q^2$, so $q^2 \leq 0$, against $q^2 > 0$. By totality, $q < p < 2q$.
Let $p' := 3q - p$ and $q' := p - q$. Then $q' > 0$ since $q < p$; $q' < q$ since $p < 2q$; and $p' > 0$ since $p < 2q < 3q$. Expanding,

$$\begin{aligned} p'^2 - 3q'^2 &= (9q^2 - 6pq + p^2) - (3p^2 - 6pq + 3q^2) \\ &= 6q^2 - 2p^2 = 2(3q^2 - p^2) = 0. \end{aligned}$$

So q' is an admissible denominator smaller than q, a contradiction. No rational squares to 3.

(b) *Positivity and the two identities.* For $s \in \mathbb{Q}_{>0}$, $(s^2+3)/(2s) \in \mathbb{Q}_{>0}$ ($s^2+3 > 0$, $2s > 0$, Proposition 3.2.2 (a), (O4)), so the recursion (Sets and Functions, Theorem 3.3.1, read on ω through Corollary 1.4.3 (b)) defines (y_n) in $\mathbb{Q}_{>0}$. With $u(s) := (s^2 + 3)/(2s)$, clearing the non-zero denominators $4s^2$ and $2s$,

$$u(s)^2 - 3 = \frac{(s^2+3)^2 - 12s^2}{4s^2} = \frac{(s^2-3)^2}{4s^2}, \qquad s - u(s) = \frac{s^2 - 3}{2s}.$$

So $d_{n+1} = d_n^2/(4y_n^2)$ and $y_n - y_{n+1} = d_n/(2y_n)$. By induction, $d_n > 0$: $d_0 = 1$, and $d_n > 0$ gives $d_n^2 > 0$, so $d_{n+1} > 0$. Hence $y_n^2 > 3$, $4y_n^2 > 12$, and $d_{n+1} \leq d_n^2/12$ (Proposition 3.2.2 (b)). Also $y_n - y_{n+1} > 0$.

Values. $y_1 = 7/4$, $y_2 = 97/56$; $d_0 = 1$, $d_1 = 49/16 - 3 = 1/16$, $d_2 = 9409/3136 - 3 = 1/3136$, and indeed $d_2 = (1/16)^2/(4 \cdot 49/16) = 1/3136$.

Monotone, and $y_n > 1$. As in (iii) and (iv): $y_n > 1$, since $0 < y_n \leq 1$ would give $y_n^2 \leq 1 < 3$; and $y_m \leq y_n$ for $m \geq n$, by induction from $y_{n+1} < y_n$.

A bound on d_n. $d_n \leq 1/(16n)$ for $n \geq 1$, by induction: $d_1 = 1/16$; if $0 < d_n \leq 1/(16n) \leq 1/16$, then $d_{n+1} \leq d_n \cdot d_n/12 \leq d_n/192 \leq 1/(3072n) \leq 1/(16(n+1))$, since $16n+16 \leq 3072n$ for $n \geq 1$.

Close terms. For $m \geq n \geq 1$: $y_m \leq y_n$, and $y_m > 3/y_m \geq 3/y_n$, so

$$0 \leq y_n - y_m \leq y_n - \frac{3}{y_n} = \frac{d_n}{y_n} \leq d_n \leq \frac{1}{16n},$$

using $y_n > 1$.

Cauchy. Given $\varepsilon \in \mathbb{Q}_{>0}$, Corollary 3.2.4 gives $N \in \mathbb{Z}_{>0}$ with $1/N < \varepsilon$. For $m, n \geq N$, with $k \coloneqq \min\{m, n\} \geq N$, $|y_m - y_n| \leq 1/(16k) \leq 1/N < \varepsilon$.

No rational limit. If $y_n \to L \in \mathbb{Q}$, then $y_n^2 \to L^2$ (Theorem 4.4.3 (c)). For $n \geq 1$, $0 < d_n \leq 1/(16n) \leq 1/(n+1)$, so $d_n \to 0$ by the squeeze (Proposition 4.4.4 (c), with $N_0 = 1$ and the null upper sequence $(1/(n+1))$), and $y_n^2 = 3 + d_n \to 3$ (Theorem 4.4.3 (a)). By Theorem 4.3.4, $L^2 = 3$, against (a).

Marking scheme. (10 points) - 2: (a), a descent with a least denominator (Proposition 2.4.4), with the inequalities $q < p < 2q$ and the positivity of p', q' justified (another correct route, such as divisibility by 3 with a least denominator, earns the same). - 1: positivity of y_n and d_n, and the recursion identity for d_{n+1}. - 2: a comparison of y_n and y_m for $m \geq n$: monotonicity together with $y_m \geq 3/y_n$, or (factorization route, as in R4) $|y_n - y_m| = |d_n - d_m|/(y_n + y_m)$ with $y_n + y_m > 2$, since $y_n > 1$ follows from $y_n^2 > 3$. - 2: an explicit bound on d_n (or on $y_n - y_m$) tending to 0, with Archimedes giving N for each rational ε. - 1: Cauchy concluded with the quantifiers in order. - 2: no rational limit, by the product law, uniqueness of limits and (a). Fatal errors (0 for the step): assuming that $\sqrt{3}$ exists in $\mathbb{Q}$ or in a larger field and comparing with it; invoking a monotone convergence theorem. A route by factorization as in R4 earns the next 2 for showing $d_n \to 0$ (by an explicit bound with Archimedes, or by the squeeze), and the 1 for concluding that (d_n) is Cauchy by Proposition 4.5.1 and transferring this to (y_n) with the quantifiers in order. A route through the steps $y_n - y_{n+1}$ and a geometric bound earns the middle 2 + 1 only if the sum of the steps from n to m is bounded explicitly, independently of m (the example at the end of Section 4.5 shows why).

Exercise. C2: Cauchy depends on the field

In the ordered field $\mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field), consider the sequence $a_n \coloneqq 1/((n+1) \cdot 1)$, whose terms are the rationals $1/(n+1)$. Explain why (a_n) is **not** a Cauchy sequence in $\mathbb{Q}(t)$, although it is Cauchy in $\mathbb{Q}$.

Solution. In $\mathbb{Q}$, $(1/(n+1))$ is Cauchy (Section 4.5, the examples after Definition 4.5.1).

In $\mathbb{Q}(t)$, take the tolerance $\varepsilon \coloneqq 1/t$, which is positive (the example shows $0 < 1/t$). Let $N \in \mathbb{N}$, and take $m \coloneqq N$, $n \coloneqq N+1$. Then

$$|a_m - a_n| = \frac{1}{N+1} - \frac{1}{N+2} = \frac{1}{(N+1)(N+2)},$$

by Lemma 4.1.1 (c), a positive rational. The natural number $k := (N+1)(N+2)$ satisfies $k \cdot 1 < t$, because $t - k \cdot 1$ has leading coefficient $1 > 0$ (Example (A Non-Archimedean Ordered Field)). Since $0 < k \cdot 1 < t$, Proposition 3.2.2 (b) gives $1/t < 1/(k \cdot 1) = |a_m - a_n|$. So for the tolerance $1/t$ no threshold N works, and (a_n) is not Cauchy in $\mathbb{Q}(t)$.

The terms are the same in both fields; what changed is the set of tolerances. $\mathbb{Q}(t)$ has the positive element $1/t$, below every positive rational, and the rational differences $1/((N+1)(N+2))$ never get below it. Being Cauchy is a property of a sequence **in a field**, not of its terms alone. (The example $\mathbb{Q}(t)$ uses school polynomial algebra; nothing else in the book depends on this exercise.)

4.6.4.4 R. Review

Exercise. R1

Let $x \in \mathbb{Q}$. Prove that there is a sequence (k_n) of integers such that $k_n/(n+1) \leq x$ for every $n \in \mathbb{N}$ and $k_n/(n+1) \to x$ in $\mathbb{Q}$.

Solution. The tool is the integer part of a rational (Proposition 3.2.5). For $n \in \mathbb{N}$, the rational $(n+1)x$ has exactly one integer k_n with

$$k_n \leq (n+1)x < k_n + 1,$$

so $n \mapsto k_n := \lfloor (n+1)x \rfloor$ is a sequence of integers; uniqueness makes it a definition, with no choices. Multiplying by $1/(n+1) > 0$ (Proposition 3.2.2 (a), Proposition 2.4.3 (c); the case of equality is trivial) gives

$$\frac{k_n}{n+1} \leq x < \frac{k_n}{n+1} + \frac{1}{n+1},$$

which is the first claim. Adding $-k_n/(n+1)$ (Proposition 2.4.3 (b)), $0 \leq x - k_n/(n+1) < 1/(n+1)$, so $|k_n/(n+1) - x| = x - k_n/(n+1) < 1/(n+1)$ (Definition 4.1.1, Proposition 4.1.3 (b)). Let $\varepsilon \in \mathbb{Q}_{>0}$. Since $1/(n+1) \to 0$ in $\mathbb{Q}$ (Example (First Limits) (b)), there is $N \in \mathbb{N}$ with $1/(n+1) < \varepsilon$ for every $n \geq N$, and then $|k_n/(n+1) - x| < \varepsilon$ for every $n \geq N$. So $k_n/(n+1) \to x$.

The same proof runs in any Archimedean ordered field, with Proposition 4.2.4 in place of the integer part of a rational and Proposition 4.3.1 for $1/(n+1) \to 0$.

Exercise. R2

Let F be an ordered field. Define $r_0 := 1$ and $r_{n+1} := r_n/2 + 1$ for $n \in \mathbb{N}$. Prove that $1 \leq r_n < 2$ and $r_n < r_{n+1}$ for every $n \in \mathbb{N}$.

Solution. The tools are definition by recursion and proof by induction (Sets and Functions, Theorem 3.3.1, Logic and Proof, Axiom 4.2.1), both read on $\mathbb{N} = \omega$ through Corollary 1.4.3 (b). Here $2 := 1 + 1 > 0$ (Proposition 4.1.2 (b)), so $2 \neq 0$ and $x/2$ is defined for every $x \in F$. The recursion theorem with $X = F$, starting value 1 and rule $g(n, x) := x/2 + 1$ gives exactly one sequence (r_n) in F satisfying the two equations. Its first terms are $r_1 = 3/2$ and $r_2 = 7/4$.

The bounds. We prove $P(n)$: "$1 \le r_n < 2$" by induction on n. *Base* $P(0)$*:* $r_0 = 1$, and $1 = 2/2 < 2$ by Proposition 4.1.2 (b) with $\varepsilon = 2$. *Step:* assume $1 \le r_n < 2$. Then $r_n > 0$, so $r_n/2 > 0$ (Proposition 3.2.2 (a), (O4)). Multiplying $r_n < 2$ by $1/2 > 0$ gives $r_n/2 < 2/2 = 1$ (Proposition 2.4.3 (c)). Adding 1 (Proposition 2.4.3 (b)),

$$1 = 0 + 1 < \frac{r_n}{2} + 1 = r_{n+1} < 1 + 1 = 2.$$

So $1 \le r_{n+1} < 2$, which is $P(n+1)$.

The increase. Let $n \in \mathbb{N}$. By $P(n)$, $r_n > 0$, so $r_n = r_n/2 + r_n/2$ (Proposition 4.1.2 (b) with $\varepsilon = r_n$), and $r_n/2 < 1$ as in the step. Hence

$$r_{n+1} - r_n = \left(\frac{r_n}{2} + 1\right) - \left(\frac{r_n}{2} + \frac{r_n}{2}\right) = 1 - \frac{r_n}{2} > 0,$$

and $r_n < r_{n+1}$ by Proposition 2.4.3 (a).

Exercise. R3

Say that a sequence (a_n) in an ordered field F **eventually keeps one sign** if there are $c \in F_{>0}$ and $N \in \mathbb{N}$ such that either $a_n \ge c$ for every $n \ge N$, or $a_n \le -c$ for every $n \ge N$. Write this in symbols, and form its negation by moving $\neg$ inward one law at a time, naming each law. Then use the negation to show that $a_n := (-1)^n/(n+1)$ in $\mathbb{Q}$ does not eventually keep one sign, and explain why this does not contradict the theorem that a Cauchy sequence that is not null eventually keeps one sign.

Solution. "(a_n) eventually keeps one sign" is

$$\exists c \in F_{>0}\ \exists N \in \mathbb{N}\ \Big(\forall n \in \mathbb{N}\ (n \ge N \to a_n \ge c)\ \vee\ \forall n \in \mathbb{N}\ (n \ge N \to a_n \le -c)\Big).$$

Negating, with the laws for bounded quantifiers (Logic and Proof, Proposition 6.2.3), De Morgan's laws (Logic and Proof, Proposition 2.5.5) and negated implication (Logic and Proof, Proposition 6.2.2):

1. $\forall c \in F_{>0}\ \neg\exists N \in \mathbb{N}\ (\dots)$, by (b) of the bounded law;
2. $\forall c \in F_{>0}\ \forall N \in \mathbb{N}\ \neg(\forall n\,(\dots) \vee \forall n\,(\dots))$, by (b) again;
3. $\forall c \in F_{>0}\ \forall N \in \mathbb{N}\ (\neg\forall n\,(\dots) \wedge \neg\forall n\,(\dots))$, by De Morgan (a) (and (b) for the converse);
4. $\forall c \in F_{>0}\ \forall N \in \mathbb{N}\ (\exists n \in \mathbb{N}\ \neg(n \ge N \to a_n \ge c) \wedge \exists n \in \mathbb{N}\ \neg(n \ge N \to a_n \le -c))$, by (a) of the bounded law, twice;
5. $\forall c \in F_{>0}\ \forall N \in \mathbb{N}\ (\exists n \in \mathbb{N}\ (n \ge N \wedge \neg(a_n \ge c)) \wedge \exists n \in \mathbb{N}\ (n \ge N \wedge \neg(a_n \le -c)))$, by Logic and Proof, Proposition 6.2.2, twice;
6. and $\neg(a_n \ge c)$ is $a_n < c$, $\neg(a_n \le -c)$ is $a_n > -c$, by totality (Proposition 4.1.2 (a)).

So the negation says: **for every** $c \in F_{>0}$ and **every** $N \in \mathbb{N}$, some $n \ge N$ has $a_n < c$, and some $n \ge N$ has $a_n > -c$.

The example. Here $(-1)^n$ is the power in $\mathbb{Q}$ (Definition 4.2.1), so $(-1)^{n+1} = -(-1)^n$ (Proposition 2.3.2 (c)), and by induction $(-1)^n$ is 1 or -1 for every n. Let $c \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$. Of $(-1)^N$ and $(-1)^{N+1} = -(-1)^N$, one is 1 and the other -1; call p the index among $N, N+1$ with $(-1)^p = 1$, and q the other. Both are $\ge N$, and $1/(p+1) > 0$, $1/(q+1) > 0$ (Proposition 3.2.2 (a)). So $a_q = -1/(q+1) < 0 < c$ and $a_p = 1/(p+1) > 0 > -c$ (Proposition 2.4.3

(a)). This is the negation, so (a_n) does not eventually keep one sign.

No contradiction. $|a_n| = |(-1)^n| \cdot |1/(n+1)| = 1/(n+1)$ (Proposition 4.1.3 (b)), and $(1/(n+1))$ is null in $\mathbb{Q}$ (Example (First Limits) (b)). So for each $\varepsilon \in \mathbb{Q}_{>0}$ the threshold that serves $(1/(n+1))$ gives $|a_n - 0| < \varepsilon$: the sequence (a_n) is null. It is Cauchy (Theorem 4.5.3 (a)), but the theorem (Theorem 4.5.4) assumes "not null", so it says nothing about (a_n). The example shows that this hypothesis cannot be dropped.

Exercise. R4

Let (a_n) be a sequence in $\mathbb{Q}$ with $a_n > 0$ for every n and $a_n^2 \to 2$ in $\mathbb{Q}$. Is (a_n) a Cauchy sequence in $\mathbb{Q}$? Prove your answer.

Solution. **Yes.** The tool is the factorization $a_n^2 - a_m^2 = (a_n - a_m)(a_n + a_m)$, which turns closeness of the squares into closeness of the terms once the factor $a_n + a_m$ is bounded below by a positive constant.

A lower bound. By Lemma 4.4.2 applied to $a_n^2 \to 2 > 0$, there is N_0 with $a_n^2 > 2/2 = 1$ for every $n \geq N_0$. For such n, $a_n > 1$: otherwise $0 < a_n \leq 1$ would give $a_n^2 \leq 1$ (Proposition 2.4.3 (g)). So $a_n + a_m > 2$ for $m, n \geq N_0$, and $1/(a_n + a_m) < 1/2$ (Proposition 3.2.2 (b)).

The estimate. For $m, n \geq N_0$, dividing the factorization by $a_n + a_m \neq 0$,

$$|a_n - a_m| = \frac{|a_n^2 - a_m^2|}{a_n + a_m} \leq \frac{|a_n^2 - a_m^2|}{2}.$$

The sequence (a_n^2) converges, so it is Cauchy (Proposition 4.5.1). Given $\varepsilon \in \mathbb{Q}_{>0}$, choose N_1 with $|a_n^2 - a_m^2| < 2\varepsilon$ for all $m, n \geq N_1$, and let $N := \max\{N_0, N_1\}$. For $m, n \geq N$, $|a_n - a_m| < 2\varepsilon/2 = \varepsilon$. So (a_n) is Cauchy.

Why the look-alike tools fail. The limit laws would need a limit of (a_n), and there is none in $\mathbb{Q}$: $a_n \to L$ would give $a_n^2 \to L^2$, so $L^2 = 2$, against Theorem 3.3.1. Theorem 4.5.4 and Proposition 4.5.6 presuppose that (a_n) is Cauchy, which is what is asked. The Newton sequence of Theorem 4.6.1 is one such (a_n), so the question is not empty.

Chapter 5

Constructing the Real Numbers from Cauchy Sequences

Chapter 4 ended at a hole. The Newton sequence $2, 3/2, 17/12, 577/408, \ldots$ of Chapter 3 is Cauchy in $\mathbb{Q}$, and no rational is its limit (Theorem 4.6.1). Its terms crowd around a place where $\mathbb{Q}$ has no element. This chapter does not search for a number in the hole. It **takes the sequence itself as the number**. Many sequences crowd around the same hole: the decimal truncations $1, 7/5, 141/100, \ldots$ approach it from below and never meet the Newton sequence. So two Cauchy sequences of rationals must count as one number when their difference dies out. That is an equivalence relation, and *Sets and Functions*, Section 5.2 knows what to do with one: take its classes. *A real number is a Cauchy sequence of rationals, up to a null difference.* The chapter defines the sum, product and order of these classes on representatives and proves each well defined. The laws of arithmetic are read off the laws of $\mathbb{Q}$, one index at a time. The order needs a new idea, because comparing terms does not compare classes. The chapter then places $\mathbb{Q}$ inside the new field, proves it Archimedean with $\mathbb{Q}$ dense in it, and finds there a positive number whose square is 2. Until the field exists, every tolerance, margin and threshold is rational, so nothing about real numbers is assumed on the way to building them.

Then comes completeness. A set of real numbers that is bounded above is trapped by halving intervals with rational ends; the ends form a Cauchy sequence of rationals, which is a real number, and that number is the least upper bound. So the constructed $\mathbb{R}$ is a complete ordered field. Chapter 6 shows that any two complete ordered fields are isomorphic, and *Axiomatic Set Theory* builds one from Dedekind cuts. *Analysis on the Real Line*, Chapter 1 takes the axioms of a complete ordered field as its starting point, and this chapter gives them a model. *Analytic Preliminaries*, Chapter 5 builds the complex numbers on this $\mathbb{R}$. *Analysis on the Real Line*, Chapter 5 repeats the construction for an arbitrary metric space.

What you need. Chapter 4 above all, applied with $F = \mathbb{Q}$ and, once Section 5.3 has made $\mathbb{R}$ an ordered field, with $F = \mathbb{R}$; Chapters 2 and 3; Chapter 1's arithmetic and order of $\mathbb{N}$; *Sets and Functions*, Chapters 2–6, chiefly Section 5.2; and *Logic and Proof*, Chapters 4 and 6. From Chapter 4: the rules of an ordered field and the absolute value (Proposition 4.1.2, Lemma 4.1.1, Definition 4.1.1, Proposition 4.1.3); the copy ι_F of $\mathbb{Q}$ in an ordered field (Theorem 4.2.2); the Archimedean property in its reciprocal form, the integer part and density (Proposition 4.2.3, Proposition 4.2.4, Theorem 4.2.5); null sequences, tail changes, the limit laws, null times bounded, and limits preserving $\leq$ (Definition 4.3.3, Proposition 4.3.2, Theorem 4.4.3, Proposition 4.4.1, Proposition 4.4.4); Cauchy sequences, their boundedness and algebra, the sign theorem and reciprocals (Definition 4.5.1, Proposition 4.5.2, Theorem 4.5.3, Theorem 4.5.4, Proposition 4.5.6); and the Newton sequence with no rational limit (Theorem 4.6.1). From

Chapters 2 and 3: the lemma on rules defined on classes (Theorem 2.2.1); commutative rings, fields and ordered fields (Definition 2.3.1, Definition 2.4.1, Proposition 2.4.3); the order from a positive set (Proposition 2.4.1), used here a third time; least elements of sets of integers (Proposition 2.4.4); the Archimedean property of $\mathbb{Q}$ (Theorem 3.2.3); and the Newton sequence and the gaps (Example (A Rational Sequence Aimed at the Gap), Theorem 3.3.1, Theorem 3.3.2). From Chapter 1: the arithmetic of $\mathbb{N}$ and its laws (Definition 1.2.1, Theorem 1.2.3), its order and how the order meets the arithmetic (Theorem 1.3.2, Proposition 1.3.1), parity of powers (Proposition 1.3.4), and the citing rule by which induction and recursion apply to $\mathbb{N}$ (Corollary 1.4.3 (b)). From *Sets and Functions*: equivalence classes, quotient sets and the partition theorem (Sets and Functions, Definition 5.2.2, Sets and Functions, Definition 5.2.4, Sets and Functions, Theorem 5.2.1), with the method of checking a rule on representatives (Sets and Functions, Example (Method: checking that a definition on classes is well defined)); the recursion theorem (Sets and Functions, Theorem 3.3.1); suprema, with their uniqueness and the greatest element as supremum (Sets and Functions, Definition 5.3.5, Sets and Functions, Proposition 5.3.1); and the axiom of choice (Sets and Functions, Axiom 6.1.1), named only to show that Section 5.5 does not use it. From *Logic and Proof*: induction (Logic and Proof, Axiom 4.2.1) and the negation of bounded quantifiers (Logic and Proof, Proposition 6.2.3). Some exercises also draw on the non-Archimedean field $\mathbb{Q}(t)$ of Chapter 4, the fractions of Chapter 3, countability from *Sets and Functions*, Chapter 4, and the catalogue of what a proof may not do from *Logic and Proof*, Chapter 8. Nothing from *Analysis on the Real Line* is used, and no fact about $\mathbb{R}$ is assumed.

Time. About 25 hours on the core track (all of A, one B and one C per section, the exit checks); about 35 hours for everything.

5.0.1 Roadmap

- **Sequences That Should Converge.** The Newton sequence and the decimal truncations, defined inside $\mathbb{Q}$ by a rule, as two sequences aimed at one hole; the set $\mathcal{C}$ of Cauchy sequences of rationals; the relation $a \sim b$ when $a - b$ is null, proved an equivalence relation; the truncations equivalent to the Newton sequence; and finite changes and shifts, which never change the class.
- **The Real Numbers and Their Arithmetic.** $\mathbb{R}$ defined as the quotient set $\mathcal{C}/\!\sim$; sum, product and negative defined on representatives and proved well defined; the laws of a commutative ring read on representatives, one index at a time; and inverses through the sign theorem of Chapter 4, after the failed termwise reciprocal.
- **Positive Classes and the Order.** Why termwise comparison defines no order on classes; positive sequences, eventually above a rational margin that is named before the index; positivity as a property of the class, which makes $\mathbb{R}$ an ordered field; and termwise $\leq$ passing to classes, which strict $<$ does not.
- **The Rationals Inside the Reals.** The map $\kappa(q) = [(q)_n]$, which preserves the operations and the order and is Chapter 4's embedding $\iota_{\mathbb{R}}$; $\mathbb{R}$ Archimedean, with $\mathbb{Q}$ dense in it by Chapter 4's theorem; and the class of the Newton sequence, a positive real number whose square is 2.
- **Least Upper Bounds.** Complete ordered fields; the bound $2^k \geq k + 1$ in any ordered field; bisection with rational ends, whose right ends form a Cauchy sequence of rationals naming the supremum, so that $\mathbb{R}$ is complete; the bisection and the Newton sequence naming one number; and why no axiom of choice was needed.

5.0.2 Named moves

The first move was taught in *Sets and Functions*; the other two are method boxes of this chapter, cited from then on.

- **Representatives, then independence** (Sets and Functions, Example (Method: checking that a definition on classes is well defined)). To define an operation or a property on classes, state it on representatives, then show that the class of the answer survives replacing each input by an equivalent one. It proves the sum, product and negative well defined in Section 5.2 and positivity a property of the class in Section 5.3. The termwise reciprocal of Section 5.2 and the termwise order of Section 5.3 fail it. Exercise B2 of Section 5.2 and Exercise C1 of Section 5.3 leave the last step, and then the last half, to you.
- **Name the margin before the index** (Example (Method: name the margin before the index)). To prove that a Cauchy sequence of rationals is positive, or that $[a] < [b]$, first name a rational margin δ from data that do not depend on n; only then ask the given sequences for thresholds at tolerances computed from δ, and take the largest. (Section 5.3; used again in the proof that $\mathbb{R}$ is Archimedean in Section 5.4 and in Steps 4 and 5 of the completeness proof in Section 5.5; Exercise B4 (d) of Section 5.3 and Exercise C3 of Section 5.4.)
- **Bisect and keep the invariant** (Example (Method: bisect and keep the invariant)). Start with a point that fails a test and one that passes it; halve by a rule that keeps one end on each side; prove the invariants and the length $D/2^k$ by induction; make the lengths small by the Archimedean property; and spend one invariant on each half of the conclusion. (Section 5.5; Exercises C1 and C2 of Section 5.5.)

5.0.3 Threads

- **The Newton class.** Section 3.3 computes the Newton sequence and Section 4.6 proves it Cauchy with no rational limit; Section 5.1 shows the decimal truncations equivalent to it; Section 5.2 takes its class as a first real number; Section 5.4 proves that this class is positive with square 2 (Example (The Class of the Newton Sequence)); the worked example of Section 5.5 finds the same number by bisection; and Chapter 6 returns to it when it proves that every non-negative element of a complete ordered field has a square root.
- **Archimedes.** Chapter 3 proves the Archimedean property of $\mathbb{Q}$ from the construction; Section 4.2 defines it for any ordered field and derives the integer part and density; Section 5.4 proves it for $\mathbb{R}$; Step 3 of the completeness proof in Section 5.5 uses it in $\mathbb{Q}$, and density, which came from it in $\mathbb{R}$, supplies the rational ends; Chapter 6 shows that a complete ordered field is always Archimedean.
- **Choice.** *Sets and Functions*, Chapter 6 states the axiom of choice and asks every proof to name its rules; Section 5.1 defines the truncations by a rule, with no choices made; and the example on representatives in Section 5.5 shows that the completeness proof picks finitely many objects and defines the rest by recursion.

5.0.4 Summary

The set $\mathcal{C}$ of Cauchy sequences of rationals (Definition 5.1.1) carries the relation $a \sim b$, "$a - b$ is null" (Definition 5.1.2), which is an equivalence relation (Proposition 5.1.1), and a real number is one of its classes (Definition 5.2.1). The sum, product and negative defined on representatives are well defined (Theorem 5.2.1), the laws of $\mathbb{Q}$ applied index by index make $\mathbb{R}$ a commutative ring (Theorem 5.2.2), and the sign theorem of Chapter 4 supplies inverses, so $\mathbb{R}$ is a field (Theorem 5.2.3). A Cauchy sequence of rationals is positive when its terms eventually stay above one rational margin (Definition 5.3.1); positivity belongs to the class (Proposition 5.3.1), and the positive classes make $\mathbb{R}$ an ordered field (Theorem 5.3.3), in which termwise $\leq$ passes to classes (Lemma 5.3.4). The map κ places a copy of $\mathbb{Q}$ in $\mathbb{R}$ that respects the operations and the order (Theorem 5.4.1); $\mathbb{R}$ is Archimedean (Theorem 5.4.2), $\mathbb{Q}$ is dense in it (Theorem 5.4.3), and the class of the Newton sequence is a positive real number whose square is 2 (Example (The

Class of the Newton Sequence)). Bisection with rational ends shows that every non-empty set of real numbers that is bounded above has a least upper bound, so $\mathbb{R}$ is a complete ordered field (Theorem 5.5.2), and the proof uses no axiom of choice (Example (Picking Representatives Is a Use of Choice)).

What is open: is this the only complete ordered field? Other constructions give other sets, the cuts of *Axiomatic Set Theory* among them, and Chapter 6 proves that any two complete ordered fields are linked by exactly one order-preserving field isomorphism. The sequence form of completeness, that every Cauchy sequence of real numbers converges, is proved in *Analysis on the Real Line*, Chapter 4.

5.0.5 Terms

- **Cauchy sequences of rationals**: the set $\mathcal{C}$ of sequences $\mathbb{N} \to \mathbb{Q}$ that are Cauchy in $\mathbb{Q}$ (Definition 5.1.1).
- **Cauchy-complete**: said of an ordered field in which every Cauchy sequence converges (Definition 5.5.1).
- **complete ordered field**: an ordered field in which every non-empty subset that is bounded above has a supremum (Definition 5.5.1).
- **copy of $\mathbb{Q}$ in $\mathbb{R}$**: $\kappa(q) = [(q)_n]$, the class of the constant sequence (Definition 5.4.1).
- **equivalent** (Cauchy sequences): $a \sim b$ when $a - b$ is null (Definition 5.1.2).
- **integer part** (in $\mathbb{R}$): the one $m \in \mathbb{Z}$ with $\kappa(m) \le x < \kappa(m+1)$ (Proposition 4.2.4 with $F = \mathbb{R}$).
- **positive** (Cauchy sequence): from some index on, every term is at least one fixed positive rational (Definition 5.3.1).
- **real number**: a class $[a]$ of Cauchy sequences of rationals under $\sim$; $\mathbb{R} = \mathcal{C}/\!\sim$ (Definition 5.2.1).
- **representative**: any $b \in [a]$ (Definition 5.2.1).
- **supremum**: the least upper bound of a set, as defined in *Sets and Functions* (Sets and Functions, Definition 5.3.5).

5.1 Sequences That Should Converge

Chapter 4 ended with a rational sequence that is Cauchy in $\mathbb{Q}$ and converges to no rational (Theorem 4.6.1). Its terms crowd around a place where $\mathbb{Q}$ has a hole. This chapter builds the number that belongs in the hole, and the first step is to notice that many different sequences crowd around the same hole. This section sets up the raw material, the set $\mathcal{C}$ of Cauchy sequences of rationals, and the relation that will decide when two of them aim at the same number: their difference dies out. Nothing about real numbers is assumed. Every tolerance, bound and threshold in this section is a rational number.

5.1.1 Two sequences aimed at one hole

The first sequence is the Newton sequence of Chapter 3 (Example (A Rational Sequence Aimed at the Gap)): $x_0 := 2$ and

$$x_{n+1} := \frac{x_n^2 + 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n} \qquad (n \in \mathbb{N}),$$

with error $e_n := x_n^2 - 2$. Its first terms are $2, 3/2, 17/12, 577/408$. We shall use the following facts, each proved earlier, for every $n \in \mathbb{N}$:

(NS1) $x_n \in \mathbb{Q}_{>0}$, and $e_n > 0$, that is, $x_n^2 > 2$ (Example (A Rational Sequence Aimed at the Gap));

(NS2) $x_n > 1$, and $e_n \leq 1/(4n)$ when $n \geq 1$ (the first clauses of the statement of Theorem 4.6.1);

(NS3) (x_n) is a Cauchy sequence in $\mathbb{Q}$, and there is no $L \in \mathbb{Q}$ with $x_n \to L$ (Theorem 4.6.1).

From (NS1) and (NS2), $0 < x_n^2 - 2 \leq 1/(4n)$ for every $n \geq 1$. At $n = 0$ the expression $1/(4n)$ is not defined, and indeed $x_0^2 - 2 = 2$ is not small.

The second sequence approaches the hole from below, by decimal truncation. To define it inside $\mathbb{Q}$, without the school decimals, we need powers of ten.

Powers of ten. For $k \in \mathbb{N}$, 10^k is the power in the ordered field $\mathbb{Q}$ of Definition 4.2.1 (b), with $10^0 = 1$ and $10^{k+1} = 10^k \cdot 10$. (Here $10 = 10 \cdot 1_{\mathbb{Q}}$, by the identification $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ of Chapters 2 and 3 and Proposition 4.2.1 (d).) Two inductions, in the fold below, show that $10^k \in \mathbb{Z}$ and $10^k \geq k + 1$. In particular $10^k \geq 1 > 0$, so $1/10^k > 0$ (Proposition 3.2.2 (a)), and Proposition 4.1.2 (c), applied to $0 < k + 1 \leq 10^k$, gives $1/10^k \leq 1/(k + 1)$. Together,

$$0 < \frac{1}{10^k} \leq \frac{1}{k+1} \qquad (k \in \mathbb{N}). \tag{5.1.1}$$

The inequality is not strict at $k = 0$, where both sides are 1. We write $1/10^k$ throughout; negative powers have not been defined.

↻ **Second reading** · *Two facts about powers of ten*

Both are proved by induction (Logic and Proof, Axiom 4.2.1, read on $\mathbb{N}$ through Corollary 1.4.3 (b)) with base $P(0)$.

- $10^k \in \mathbb{Z}$**.** Base: $10^0 = 1 \in \mathbb{Z}$. Step: if $10^k \in \mathbb{Z}$, then $10^{k+1} = 10^k \cdot 10$ is a product of two integers, and products of integers computed in $\mathbb{Q}$ are integers (Proposition 3.1.4: $j(m)j(n) = j(mn)$).
- $10^k \geq k + 1$**.** Base: $10^0 = 1 = 0 + 1$. Step: suppose $10^k \geq y$, where $y := k + 1$. Then $y \geq 1$ (Proposition 4.2.1 (b)), and multiplying by $10 > 0$ gives $10^{k+1} = 10^k \cdot 10 \geq 10y$ (by Proposition 2.4.3 (c) if $10^k > y$; trivially if $10^k = y$). Moreover $10y - (y + 1) = 9y - 1 \geq 9 - 1 > 0$. So $10^{k+1} > y + 1 = k + 2$, where $(k + 1) + 1 = k + 2$ is arithmetic in $\mathbb{N}$ (Theorem 1.2.3), carried into $\mathbb{Q}$ by the sum law of Proposition 4.2.1 (a).

The truncations. For $k \in \mathbb{N}$, let m_k be the **largest** natural number m with

$$m^2 \leq 2 \cdot (10^k)^2,$$

and let $d_k := m_k/10^k$. In school terms, d_k is $\sqrt{2}$ cut off after k decimal places; but the definition mentions only integers and the rational 10^k.

That such a largest m exists, and is unique, is a fact about sets of integers (Proposition 2.4.4), checked in the fold below. Because exactly one number is described for each k, $k \mapsto d_k$ is a function $\mathbb{N} \to \mathbb{Q}$ (Sets and Functions, Definition 2.1.1), with no choices made.

Second reading · *Why the largest* m *exists*

Let

$$S_k := \{m \in \mathbb{Z} : m \geq 0 \text{ and } m^2 > 2 \cdot (10^k)^2\}.$$

It is bounded below by 0, and it is non-empty: $2 \cdot 10^k$ is a non-negative integer, and $(2 \cdot 10^k)^2 = 4(10^k)^2 > 2(10^k)^2$ because $(10^k)^2 > 0$. By Proposition 2.4.4, S_k has a least element m^*. Since $0^2 = 0 \leq 2(10^k)^2$, $0 \notin S_k$, so $m^* \geq 1$ by discreteness (Theorem 2.4.2 (c)), hence $m_k := m^* - 1 \geq 0$ is a natural number (Theorem 2.4.2 (b)). It is not in S_k, being smaller than the least element, so $m_k^2 \leq 2(10^k)^2$. Every natural number $m > m_k$ satisfies $m \geq m^*$ (discreteness again), so $m^2 \geq (m^*)^2 > 2(10^k)^2$ (by Proposition 2.4.3 (g) if $m > m^*$; trivially if $m = m^*$). So m_k is the largest natural number with $m^2 \leq 2(10^k)^2$, and there is exactly one.

The first values are:

k	0	1	2	3	4	5	6
m_k	1	14	141	1414	14142	141421	1414213

For instance $m_1 = 14$, because $14^2 = 196 \leq 200 = 2 \cdot 10^2$ and $15^2 = 225 > 200$. So $d_0 = 1$, $d_1 = 7/5$, $d_2 = 141/100$, $d_3 = 707/500$.

Different sequences, one intended number. The two sequences never meet: $x_k^2 > 2 \geq d_k^2$ for every k, so $x_k \neq d_k$. They approach from opposite sides, and their differences $x_1 - d_1 = 1/10$, $x_2 - d_2 = 1/150$, $x_3 - d_3 = 11/51000$ shrink fast. Both "should" converge to the same number, and $\mathbb{Q}$ has no such number. The plan of the chapter is to make the sequences themselves into the numbers, and for that we must first say when two sequences count as one.

5.1.2 Cauchy sequences of rationals

The raw material is every sequence that behaves as if it had a limit.

Definition 5.1.1. Cauchy Sequences of Rationals

$\mathcal{C}$ is the set of all sequences $a\colon \mathbb{N} \to \mathbb{Q}$ (Definition 4.3.1, $F = \mathbb{Q}$) that are Cauchy sequences in $\mathbb{Q}$ (Definition 4.5.1): **for every** $\varepsilon \in \mathbb{Q}_{>0}$ **there is** $N \in \mathbb{N}$ with $|a_m - a_n| < \varepsilon$ **for all** $m, n \geq N$. In symbols, $a \in \mathcal{C}$ means

$$\forall \varepsilon \in \mathbb{Q}_{>0} \;\; \exists N \in \mathbb{N} \;\; \forall m \in \mathbb{N} \;\; \forall n \in \mathbb{N} \\ \big((m \geq N \wedge n \geq N) \Rightarrow |a_m - a_n| < \varepsilon\big).$$

A sequence a of rationals is **null** if $a_n \to 0$ in $\mathbb{Q}$ (Definition 4.3.3, $F = \mathbb{Q}$).

We write a sequence as a single letter, $a = (a_n)_{n \in \mathbb{N}}$, and do arithmetic on sequences **termwise**: $a + b := (a_n + b_n)_n$, $ab := (a_n b_n)_n$, $-a := (-a_n)_n$ and $a - b := (a_n - b_n)_n$. For $q \in \mathbb{Q}$, $(q)_n$ is the constant sequence with every term q.

Nothing is new here except the name: this is Chapter 4's definition with $F = \mathbb{Q}$, so every tolerance ε is **rational**. That is the point. The definition can be read by someone who knows only $\mathbb{Q}$, and it will be used to build a larger field without assuming anything about it.

Examples. Every constant sequence $(q)_n$ is in $\mathcal{C}$, with $N = 0$ for every ε. The sequence $(1/(n+1))_n$ is null (Example (First Limits) (b)), so it is in $\mathcal{C}$ (Theorem 4.5.3 (a)). The Newton sequence (x_n) is in $\mathcal{C}$, by (NS3), and is not null, since $x_n > 1$ for every n. The sequence $(n)_n$ is not in $\mathcal{C}$: a Cauchy sequence is bounded (Proposition 4.5.2), and for every $M \in \mathbb{Q}_{>0}$ some $n \in \mathbb{N}$ has $n > M$ (Theorem 3.2.3).

The toolkit from Chapter 4, with $F = \mathbb{Q}$. This chapter uses the following facts so often that we list them once. Each was proved in Chapter 4 for an arbitrary ordered field; here $F = \mathbb{Q}$.

- Constant sequences and null sequences are in $\mathcal{C}$, and $\mathcal{C}$ is closed under $+$, $-$, $\cdot$ and negatives (Theorem 4.5.3).
- Sums, negatives and constant multiples of null sequences are null (Theorem 4.4.3 (a), (b)).
- A null sequence times a bounded sequence is null (Proposition 4.4.1).
- Every member of $\mathcal{C}$ is bounded (Proposition 4.5.2).
- Changing finitely many terms changes no limit, and so no nullity (Proposition 4.3.2 (a)).
- A sequence squeezed, from some index on, between two sequences with the same limit has that limit too (Proposition 4.4.4 (c)); in particular one squeezed between $(0)_n$, or $(-u_n)_n$, and a null $(u_n)_n$ is null.

5.1.3 When two sequences name the same number

The Newton sequence and the truncations should name one number, yet they share no term. "Eventually equal" is therefore too strict a test. What the two sequences do share is visible in the differences above: $x_k - d_k$ shrinks to nothing. In $\mathbb{Q}$ we cannot compare the two sequences through a common limit, since there is none; but their difference is again a sequence of rationals, and whether it tends to 0 is a question $\mathbb{Q}$ can answer.

Two Cauchy sequences are equivalent when their difference dies out.

Definition 5.1.2. Equivalent Cauchy Sequences

For $a, b \in \mathcal{C}$, we write $a \sim b$, and say that a is **equivalent** to b, if the sequence $a - b = (a_n - b_n)_n$ is null. In symbols, $a \sim b$ means

$$\forall \varepsilon \in \mathbb{Q}_{>0} \;\; \exists N \in \mathbb{N} \;\; \forall n \in \mathbb{N} \;\; (n \geq N \Rightarrow |a_n - b_n| < \varepsilon).$$

In words: whatever rational tolerance is named, **from some index on**, the two sequences agree to within it **at the same index**. The threshold N comes after ε, so it may depend on ε. The comparison is always between a_n and b_n with the same n; nothing is said about a_m against b_n. The relation is defined only on $\mathcal{C}$: both sequences must be Cauchy before we ask whether they are equivalent.

Examples.

- *Constant sequences (the degenerate case).* For $p, q \in \mathbb{Q}$, $(p)_n \sim (q)_n$ if and only if $p = q$. The difference is the constant $(p - q)_n$, which converges to $p - q$ (Example (First Limits) (a)); by uniqueness of limits (Theorem 4.3.4) it is null exactly when $p - q = 0$. So on constant sequences $\sim$ is plain equality, as it must be if $(q)_n$ is to stand for q.
- *A null sequence and zero.* $(1/(n+1))_n \sim (0)_n$: the difference is $(1/(n+1))_n$, null by Example (First Limits) (b). No term of the first sequence is 0.
- *The truncations and the Newton sequence.* $(d_k) \sim (x_k)$; this is Example (The Truncations and the Newton Sequence Are Equivalent) below.

Non-example by a minimal change. Shift the null sequence of the second example up by one: $b_n := 1 + 1/(n+1)$. It is still in $\mathcal{C}$ (a constant plus a null sequence, Theorem 4.5.3 (a), (b)), and $b - (1)_n = (1/(n+1))_n$ is still null, so $b \sim (1)_n$. But $b \not\sim (0)_n$: the clause that fails is "for every ε". With $\varepsilon = 1$, every n has $|b_n - 0| = 1 + 1/(n+1) \geq 1$, so no threshold exists.

Why this definition.

- *Why "null", and not "same limit".* Two sequences with a common limit in $\mathbb{Q}$ have null difference (Theorem 4.4.3 (b)). Conversely, if $a \to L$ and $b \to L'$ with $a - b$ null, then $a - b$ tends to both $L - L'$ and 0, so $L = L'$ (Theorem 4.3.4). So for convergent sequences the new relation says exactly "same limit". But it is stated without mentioning any limit, so it applies equally to (x_k) and (d_k), which have none in $\mathbb{Q}$.
- *Why not "eventually equal".* Then (x_k) and (d_k) would name different numbers, and so would $(1/(n+1))_n$ and $(0)_n$. Sequences that agree ever more closely, without ever agreeing exactly, would never be identified.
- *Why only on $\mathcal{C}$.* On all sequences of rationals the same relation would not respect products, and Section 5.2 needs it to. For instance $(1/(n+1))_n$ and $(0)_n$ have null difference, but multiplying both termwise by $(n+1)_n$, which is not Cauchy, gives $(1)_n$ and $(0)_n$, whose difference is not null. Inside $\mathcal{C}$ every multiplier is bounded, and that is what saves the product.

It must be checked that $\sim$ really sorts sequences into groups, that is, that it is an equivalence relation in the sense of *Sets and Functions*, Section 5.2.

Proposition 5.1.1. Equivalence of Cauchy Sequences Is an Equivalence Relation

The relation $\sim$ of Definition 5.1.2 is an equivalence relation on $\mathcal{C}$ (Sets and Functions, Definition 5.2.1).

Idea. Each of the three clauses becomes a statement about one difference sequence. Reflexivity looks at $a - a$, the zero sequence. Symmetry looks at $b - a$, the negative of $a - b$. Transitivity writes $a - c$ as the sum $(a - b) + (b - c)$ of two null sequences. *Before reading on: which fact from Chapter 4 does transitivity use, and where is the triangle inequality hiding?*

Proof. Let $a, b, c \in \mathcal{C}$.

(Eq1) $a - a = (0)_n$, which is null (Example (First Limits) (a)). So $a \sim a$.

(Eq2) Suppose $a \sim b$. For every n, $b_n - a_n = -(a_n - b_n)$ (since $(a_n - b_n) + (b_n - a_n) = 0$ by (F1)–(F4), and negatives are unique, Proposition 2.3.2 (a)), so $b - a = -(a - b)$, which is null by Theorem 4.4.3 (b) ($-0 = 0$). So $b \sim a$.

(Eq3) Suppose $a \sim b$ and $b \sim c$. For every n,

$$a_n - c_n = (a_n - b_n) + (b_n - c_n),$$

so $a - c = (a - b) + (b - c)$, a sum of two null sequences. By the sum rule Theorem 4.4.3 (a), it is null ($0 + 0 = 0$). So $a \sim c$.

This proves that $\sim$ is an equivalence relation on $\mathcal{C}$.

■

The answer to the prompt: transitivity uses the sum rule of the limit laws, Theorem 4.4.3 (a), and the triangle inequality is inside its proof, where $|(a_n - b_n) + (b_n - c_n)| \le |a_n - b_n| + |b_n - c_n|$ splits the tolerance into two halves. Written out directly: give $a - b$ and $b - c$ the tolerance $\varepsilon/2$ each, and take the larger threshold.

By Sets and Functions, Theorem 5.2.1, $\mathcal{C}$ now falls apart into disjoint classes, and two Cauchy sequences lie in the same class exactly when their difference is null. Section 5.2 takes these classes as the real numbers.

5.1.4 The truncations and the Newton sequence

The two sequences of the hook are the first test of the definition.

Three facts about the truncations. The solution below rests on three properties of (d_k), proved in the fold that follows. For all $k \in \mathbb{N}$:

(T1) (d_k) is non-decreasing: $d_k \le d_j$ whenever $j \ge k$; in particular $d_k \ge d_0 = 1$.
(T2) $d_k \le d_j < d_k + 1/10^k$ whenever $j \ge k$, so that, with Eq. 5.1.1,

$$0 \le d_j - d_k < \frac{1}{10^k} \le \frac{1}{k+1} \qquad (j \ge k). \tag{5.1.2}$$

(T3) $0 \le 2 - d_k^2 < 5/10^k$.

Second reading · *Proofs of the three facts*

(T1): (d_k) is non-decreasing. Let $k \in \mathbb{N}$. By the choice of m_k and the defining equation $10^{k+1} = 10^k \cdot 10$,

$$(10m_k)^2 = 100\, m_k^2 \le 200\, (10^k)^2 = 2\, (10^k \cdot 10)^2 = 2\, (10^{k+1})^2.$$

So $10m_k$ is a natural number whose square is at most $2(10^{k+1})^2$, and m_{k+1} is the largest such: $10m_k \le m_{k+1}$. Dividing by $10^{k+1} > 0$ (Proposition 3.2.2 (a), and Proposition 2.4.3 (c) if $10m_k < m_{k+1}$; trivially if they are equal),

$$d_{k+1} = \frac{m_{k+1}}{10^{k+1}} \ge \frac{10m_k}{10^k \cdot 10} = \frac{m_k}{10^k} = d_k.$$

By Logic and Proof, Axiom 4.2.1 on i, with base $P(0)$ ($d_k \le d_k$) and the step just proved, $d_k \le d_{k+i}$ for every $i \in \mathbb{N}$. Every $j \ge k$ is $k+i$ for some $i \in \mathbb{N}$ (Proposition 1.3.1 (a)), so $d_k \le d_j$ whenever $j \ge k$. In particular $d_k \ge d_0 = 1$ for every k ($m_0 = 1$, since $1^2 \le 2 < 2^2$). No law of exponents was used, only the defining equation of 10^{k+1}.

(T2): $d_k \le d_j < d_k + 1/10^k$ for $j \ge k$. The lower bound is (T1). For the upper bound, $m_j^2 \le 2(10^j)^2$ gives $d_j^2 \le 2$. Since $m_k + 1 > m_k$, the maximality of m_k gives $(m_k+1)^2 > 2(10^k)^2$, that is,

$$\left(d_k + \frac{1}{10^k}\right)^2 = \frac{(m_k+1)^2}{(10^k)^2} > 2.$$

If we had $d_j \ge d_k + 1/10^k$, then, both sides being non-negative, $d_j^2 \ge (d_k + 1/10^k)^2 > 2$ (Proposition 2.4.3 (g)), against $d_j^2 \le 2$. By trichotomy, $d_j < d_k + 1/10^k$. With Eq. 5.1.1,

$$0 \le d_j - d_k < \frac{1}{10^k} \le \frac{1}{k+1} \qquad (j \ge k).$$

(T3): $0 \le 2 - d_k^2 < 5/10^k$**.** The first inequality is $d_k^2 \le 2$. For the second, expand the first display in the proof of (T2) (Proposition 2.3.2 (d)):

$$2 < d_k^2 + \frac{2d_k}{10^k} + \frac{1}{10^k} \cdot \frac{1}{10^k}.$$

Here $d_k < 2$, since $0 \le d_k$ and $d_k^2 \le 2 < 4 = 2^2$ (Proposition 2.4.3 (g)); and $(1/10^k)(1/10^k) \le 1/10^k$, since $0 < 1/10^k \le 1$ by Eq. 5.1.1, so that Proposition 4.1.2 (d) applies with $0 \le 1/10^k \le 1/10^k$ and $0 \le 1/10^k \le 1$. Multiplying $d_k < 2$ by $2/10^k > 0$ (Proposition 2.4.3 (c)) and adding the two bounds (Proposition 2.4.3 (b)), $2 - d_k^2 < 4/10^k + 1/10^k = 5/10^k$.

Example. The Truncations and the Newton Sequence Are Equivalent

Show that (d_k) is non-decreasing, that $(d_k) \in \mathcal{C}$, and that $(d_k) \sim (x_k)$.

Solution. (d_k) *is non-decreasing, and in* $\mathcal{C}$. The first claim is (T1), and $(d_k) \in \mathcal{C}$ by (T2). Now let $\varepsilon \in \mathbb{Q}_{>0}$. By Corollary 3.2.4 there is $n_\varepsilon \in \mathbb{Z}_{>0}$ with $1/m < \varepsilon$ for every integer $m \ge n_\varepsilon$. Let $N := n_\varepsilon$, which depends on ε only. For $j, k \ge N$, say with $j \ge k$ (the other case is symmetric, as $|d_j - d_k| = |d_k - d_j|$), Eq. 5.1.2 gives $|d_j - d_k| < 1/(k+1) < \varepsilon$, since $k + 1 > N$. So $(d_k) \in \mathcal{C}$.

$(d_k) \sim (x_k)$. Let $k \ge 1$. By (NS2), $x_k > 1$, and $d_k \ge 1$ by (T1), so $x_k + d_k > 1 + 1 > 1 + 0 = 1$ (Proposition 2.4.3 (b), with $1 > 0$ by (e)). Since $|x_k - d_k| \ge 0$, Proposition 4.1.2 (d) gives

$$\begin{aligned} |x_k - d_k| &\le |x_k - d_k|(x_k + d_k) = |x_k^2 - d_k^2| \\ &\le |x_k^2 - 2| + |2 - d_k^2| \le \frac{1}{4k} + \frac{5}{10^k}, \end{aligned}$$

by Proposition 4.1.3 (b) and (e) (with $(x_k - d_k)(x_k + d_k) = x_k^2 - d_k^2$ by distributivity and Proposition 2.3.2 (c), and $|x_k + d_k| = x_k + d_k$ as $x_k + d_k > 1 > 0$), the bound $0 < x_k^2 - 2 \le 1/(4k)$ from (NS1)–(NS2), and (T3). Since $4k - (k+1) = 3k - 1 > 0$, $1/(4k) \le 1/(k+1)$ (Proposition 4.1.2 (c)). Define, for **every** $k \in \mathbb{N}$,

$$u_k := \frac{1}{k+1} + \frac{5}{10^k}.$$

Then $-u_k \le x_k - d_k \le u_k$ for every $k \ge 1$ (Proposition 4.1.3 (d)). The sequence (u_k) is null: $(1/(k+1))$ is null (Example (First Limits) (b)); $(1/10^k)$ is null by the squeeze Proposition 4.4.4 (c) between $(0)_k$ and $(1/(k+1))$, using Eq. 5.1.1; $(5/10^k)$ is null by Theorem 4.4.3 (b) with $c = 5$; and the sum of two null sequences is null (Theorem 4.4.3 (a)). So $(-u_k)$ is null too, by (b) with $c = -1$. The squeeze Proposition 4.4.4 (c), with $N_0 = 1$, lower sequence $(-u_k)$ and upper sequence (u_k), shows that $(x_k - d_k)$ is null. Both sequences are in $\mathcal{C}$ (by (NS3) and (T2)), so $(x_k) \sim (d_k)$, and $(d_k) \sim (x_k)$ by symmetry.

The bound $|x_k - d_k| \le 1/(4k) + 5/10^k$ is crude but honest: at $k = 1$ it gives $|x_1 - d_1| \le 1/4 + 5/10 = 3/4$, while in fact $x_1 - d_1 = 3/2 - 7/5 = 1/10$.

The upper sequence (u_k) is defined for every k, including $k = 0$, where $1/(4k)$ is not; the squeeze only needs the inequality from $N_0 = 1$ on. One consequence is worth recording at once: (d_k) **has no limit in** $\mathbb{Q}$ **either.** If $d_k \to L \in \mathbb{Q}$, then $x_k = d_k + (x_k - d_k) \to L + 0 = L$ by the sum rule, against (NS3). So the truncations, like the Newton sequence, are Cauchy and aim at

nothing in $\mathbb{Q}$, and Example (The Truncations and the Newton Sequence Are Equivalent) says they aim at the same nothing.

5.1.5 Finite changes, and shifts

A class will contain far more than the two sequences of the hook. The next example shows how loose the relation is at the start of a sequence.

Example. Finite Changes and Shifts

(a) Let $a \in \mathcal{C}$, let $b\colon \mathbb{N} \to \mathbb{Q}$, and suppose there is $N_0 \in \mathbb{N}$ with $a_n = b_n$ for every $n \geq N_0$. Show that $b \in \mathcal{C}$ and $a \sim b$.

(b) Let $a \in \mathcal{C}$, and let $s \coloneqq (a_{n+1})_{n\in\mathbb{N}}$ be the shifted sequence. Show that $s \in \mathcal{C}$ and $s \sim a$.

Solution.

(a) $b \in \mathcal{C}$ by the tail-change remark after Definition 4.5.1 (Section 4.5). For $n \geq N_0$, $a_n - b_n = 0$. So $a - b$ agrees with the null sequence $(0)_n$ from N_0 on, and is null by Proposition 4.3.2 (a). Hence $a \sim b$.

(b) Let $\varepsilon \in \mathbb{Q}_{>0}$, and let N be a threshold for a with tolerance ε (Definition 5.1.1). For $m, n \geq N$, also $m+1, n+1 \geq N$, so $|s_m - s_n| = |a_{m+1} - a_{n+1}| < \varepsilon$: $s \in \mathcal{C}$, with the same thresholds. For $n \geq N$, apply the Cauchy condition to the pair $(n+1, n)$, both indices $\geq N$:

$$|s_n - a_n| = |a_{n+1} - a_n| < \varepsilon.$$

So $s - a$ is null, and $s \sim a$.

By (a), a representative of a class may begin with **any** finite list of rationals: $(5, -3, 7, 1/4, 1/5, 1/6, \dots)$, which is $1/(n+1)$ from $n = 3$ on, is equivalent to $(1/(n+1))_n$, and so to $(0)_n$. Part (b) uses the Cauchy condition with the two indices $n+1$ and n; a sequence of rationals that is not Cauchy can fail it (for $((-1)^n)_n$, every $|a_{n+1} - a_n|$ is 2).

Warning. **Equivalent is not "eventually equal", and not "close at the start".** $(1/(n+1))_n \sim (0)_n$, though no term of the first sequence is 0. And $(0)_n$ and $(1-1/(n+1))_n$ have the same first term, 0, but are **not** equivalent: their difference $(0)_n - (1-1/(n+1))_n$ converges to -1 (Theorem 4.4.3 (a), (b) with Example (First Limits) (a), (b)), so it is not null, by uniqueness of limits (Theorem 4.3.4). Only the behavior far out counts, and it counts at every tolerance.

Quick check. Let $a \coloneqq (1, 0, 1, 0, \dots)$, with $a_n = 1$ for even n and $a_n = 0$ for odd n; and let $b \coloneqq (1, 0, 1/2, 0, 1/3, 0, \dots)$, with $b_{2k} = 1/(k+1)$ and $b_{2k+1} = 0$. Which of them lies in $\mathcal{C}$? For the one that does, find a constant sequence equivalent to it.

Solution. $a \notin \mathcal{C}$. The negation of the Cauchy condition (moving $\neg$ inward as in Logic and

Proof, Proposition 6.2.3) reads

$$\exists \varepsilon \in \mathbb{Q}_{>0} \quad \forall N \in \mathbb{N} \quad \exists m, n \geq N \quad |a_m - a_n| \geq \varepsilon,$$

and it holds with $\varepsilon = 1/2$: for any N, the indices $m = N$ and $n = N + 1$ have one even and one odd (Proposition 1.3.4 (d)), so $|a_m - a_n| = 1 \geq 1/2$.

$b \in \mathcal{C}$, and $b \sim (0)_n$. Indeed b is null: given $\varepsilon \in \mathbb{Q}_{>0}$, Corollary 3.2.4 gives $K \in \mathbb{Z}_{>0}$ with $1/m < \varepsilon$ for every integer $m \geq K$. For $n \geq 2K$, n is odd or of the form $2k$ (Proposition 1.3.4 (d)), and in the second case $k \geq K$. So either n is odd and $b_n = 0 < \varepsilon$, or $n = 2k$ with $k \geq K$ and $b_n = 1/(k+1) < \varepsilon$. So b is null, hence in $\mathcal{C}$ (Theorem 4.5.3 (a)), and $b - (0)_n = b$ is null. The pattern "a non-zero term every other index" is the same in a and b; what differs is whether those terms die out.

The relation is in place, and it is an equivalence relation. The next section takes its classes as the real numbers and gives them an arithmetic.

5.1.6 Exercises

5.1.6.1 A. Check your understanding

Exercise. A1

(a) For each sequence of rationals, decide whether it lies in $\mathcal{C}$ and whether it is null, with a one-line reason: $((-1)^n/(n+1))_n$; $(n)_n$; $(1 + 1/(n+1))_n$; $(d_k)_k$; $((-1)^n)_n$.

(b) Write "$a \not\sim b$", for $a, b \in \mathcal{C}$, in symbols, with no $\neg$ in front of a quantifier.

(c) True or false: if $a \sim b$, then $a_n = b_n$ for at least one n. Give a reason.

Solution.

(a)
- $((-1)^n/(n+1))$: in $\mathcal{C}$ and null. Its absolute value is $1/(n+1)$ (Proposition 4.1.3 (b), $|(-1)^n| = 1$), which is below any $\varepsilon \in \mathbb{Q}_{>0}$ from some index on (Example (First Limits) (b)); null sequences are in $\mathcal{C}$ (Theorem 4.5.3 (a)).
- (n): neither. It is not bounded (Theorem 3.2.3), so it is not Cauchy (Proposition 4.5.2), and not null (a null sequence is convergent, hence bounded, Proposition 4.3.5).
- $(1 + 1/(n+1))$: in $\mathcal{C}$, not null. It converges to 1 (Theorem 4.4.3 (a)), so it is Cauchy (Proposition 4.5.1), and its limit is not 0 (Theorem 4.3.4).
- (d_k): in $\mathcal{C}$ (Example (The Truncations and the Newton Sequence Are Equivalent)), not null: $d_k \geq 1$ for every k, so the tolerance $\varepsilon = 1$ has no threshold.
- $((-1)^n)$: neither. Consecutive terms differ by 2, so $\varepsilon = 1$ has no Cauchy threshold; and it is not null, since every term has absolute value 1.

(b) By Logic and Proof, Proposition 6.2.3 and Logic and Proof, Proposition 6.2.2, reading $\neg(x < \varepsilon)$ as $x \geq \varepsilon$:

$$\exists \varepsilon \in \mathbb{Q}_{>0} \quad \forall N \in \mathbb{N} \quad \exists n \in \mathbb{N} \quad (n \geq N \wedge |a_n - b_n| \geq \varepsilon).$$

One rational tolerance is beaten at indices beyond every threshold.

(c) False. $(1/(n+1))_n \sim (0)_n$, and $1/(n+1) \neq 0$ for every n.

5.1.6.2 B. Practice

Exercise. B1: Equivalent or not

(a) Show that $(1/(n+1))_n \sim (1/(n+2))_n$ and that both are equivalent to $(0)_n$.

(b) Determine which constant sequences $(c)_n$, $c \in \mathbb{Q}$, are equivalent to $(3 + 1/(n+1))_n$. Justify your answer.

Solution.

(a) Both sequences are null: the first by Example (First Limits) (b); the second is the first shifted by one, $(a_{n+1})_n$ with $a_n = 1/(n+1)$, so it is null by Proposition 4.3.2 (b). Null sequences lie in $\mathcal{C}$ (Theorem 4.5.3 (a)), and a null sequence a satisfies $a \sim (0)_n$, since $a - (0)_n = a$. By symmetry and transitivity (Proposition 5.1.1), $(1/(n+1)) \sim (0)_n \sim (1/(n+2))$ gives $(1/(n+1)) \sim (1/(n+2))$.

(b) Only $c = 3$. The difference $(3+1/(n+1))-(c)_n = (3-c)+1/(n+1)$ converges to $3-c$, by the sum rule (Theorem 4.4.3 (a)) with the constant $3 - c$ and the null $(1/(n+1))$. By uniqueness of limits (Theorem 4.3.4) it is null if and only if $3 - c = 0$. Hence $(3+1/(n+1)) \sim (c)_n$ exactly when $c = 3$; this is consistent with the degenerate example after Definition 5.1.2, since two constants equivalent to one sequence are equivalent to each other, hence equal.

Exercise. B2: Null differences preserve Cauchy

(a) Let $a \in \mathcal{C}$, and let $b\colon \mathbb{N} \to \mathbb{Q}$ be a sequence such that $a - b$ is null. Prove that $b \in \mathcal{C}$ and $b \sim a$.

(b) Hence show that $(2/x_n)_n \in \mathcal{C}$ and $(2/x_n)_n \sim (x_n)_n$, where (x_n) is the Newton sequence.

Solution.

(a) For every n, $b_n = a_n - (a_n - b_n)$, so $b = a - (a - b)$. Here $a \in \mathcal{C}$, and $a - b$ is null, hence in $\mathcal{C}$ (Theorem 4.5.3 (a)); the difference of two members of $\mathcal{C}$ is in $\mathcal{C}$ (Theorem 4.5.3 (b)). So $b \in \mathcal{C}$. Then $a \sim b$ by definition, and $b \sim a$ by symmetry.

(b) By (NS1), $x_n > 0$, so $2/x_n$ is defined. For every n,

$$x_n - \frac{2}{x_n} = \frac{x_n^2 - 2}{x_n} = \frac{e_n}{x_n}.$$

By (NS1) and (NS2), $e_n > 0$ and $x_n > 1$, so $0 < 1/x_n < 1$ (Proposition 3.2.2) and $0 < e_n/x_n < e_n$ (Proposition 2.4.3 (c)). For $n \geq 1$, $e_n \leq 1/(4n) \leq 1/(n+1)$. So $0 \leq x_n - 2/x_n \leq 1/(n+1)$ for every $n \geq 1$, and the squeeze Proposition 4.4.4 (c), with $N_0 = 1$ and the null sequence $(1/(n+1))$ (Example (First Limits) (b)), shows that $(x_n - 2/x_n)$ is null. Part (a), with $a = (x_n) \in \mathcal{C}$ (NS3) and $b = (2/x_n)$, gives the claim. The terms $2/x_n$ satisfy $(2/x_n)^2 = 4/x_n^2 < 2$, so this sequence approaches the hole from below, as the truncations do.

Exercise. B3: Every other Newton term

Show that $(x_{2n})_n \sim (x_n)_n$ for the Newton sequence. *Hint: imitate Example (Finite Changes and Shifts) (b).*

Solution. First, $(x_{2n})_n \in \mathcal{C}$. Let $\varepsilon \in \mathbb{Q}_{>0}$, and let N be a threshold for (x_n) with tolerance ε (NS3). For $m, n \geq N$, also $2m \geq m \geq N$ and $2n \geq N$, so $|x_{2m} - x_{2n}| < \varepsilon$.

Second, the difference is null. For $n \geq N$, the two indices $2n$ and n are both $\geq N$, so $|x_{2n} - x_n| < \varepsilon$. So $(x_{2n} - x_n)_n$ is null, with the same thresholds as (x_n), and $(x_{2n}) \sim (x_n)$.

The Cauchy condition was used with two indices that are far apart ($2n$ and n). This is exactly where "every pair of indices beyond N" is needed, and not only neighbors.

5.1.6.3 C. Going deeper

Exercise. C1: Subsequences

Let $a \in \mathcal{C}$, and let $k \mapsto n_k$ be a function $\mathbb{N} \to \mathbb{N}$ with $n_k < n_{k+1}$ for every $k \in \mathbb{N}$ (for instance $n_k = 2k$, or any sequence of indices defined by Sets and Functions, Theorem 3.3.1 with this property).

(a) Prove that $n_k \geq k$ for every k, that $(a_{n_k})_{k \in \mathbb{N}} \in \mathcal{C}$, and that $(a_{n_k})_k \sim a$.

(b) Deduce: if $a, b \in \mathcal{C}$ and $a_{n_k} = b_{n_k}$ for every k, then $a \sim b$. Compare with the warning after Example (Finite Changes and Shifts): why does this not say that sequences which are "close at the start" are equivalent?

Solution (Hint 1). For (a), the indices n_k run ahead of k. Once you know $n_k \geq k$, the Cauchy condition applies to the pair (n_k, k).

Solution (Hint 2). For (a), prove $n_k \geq k$ by Logic and Proof, Axiom 4.2.1 on k; the step uses $n_{k+1} > n_k \geq k$.

Solution.

(a) $n_k \geq k$. By Logic and Proof, Axiom 4.2.1 on k, base $P(0)$: $n_0 \geq 0$ (Theorem 1.3.2 (d)). Step: if $n_k \geq k$, then $n_{k+1} > n_k \geq k$, so $k < n_{k+1}$; by Theorem 1.3.2 (b), $n_{k+1} < k+1$ would give $n_{k+1} \leq k$, so by trichotomy $n_{k+1} \geq k+1$.
In $\mathcal{C}$. Let $\varepsilon \in \mathbb{Q}_{>0}$, and let N be a threshold for a with tolerance ε. For $j, k \geq N$, $n_j \geq j \geq N$ and $n_k \geq k \geq N$, so $|a_{n_j} - a_{n_k}| < \varepsilon$.
Equivalent to a. For $k \geq N$, both $n_k \geq N$ and $k \geq N$, so $|a_{n_k} - a_k| < \varepsilon$. So $(a_{n_k} - a_k)_k$ is null, with the same thresholds, and $(a_{n_k})_k \sim a$.

(b) By (a), $(a_{n_k})_k \sim a$ and $(b_{n_k})_k \sim b$. The two subsequences are the same sequence, since $a_{n_k} = b_{n_k}$ for every k. So $a \sim (a_{n_k})_k = (b_{n_k})_k \sim b$, using symmetry for the first step, and transitivity (Proposition 5.1.1) gives $a \sim b$.

The hypothesis of (b) is that the sequences agree at **infinitely many** indices, $n_0 < n_1 < \cdots$, arbitrarily far out. In the warning, $(0)_n$ and $(1-1/(n+1))_n$ agree only at $n = 0$. Agreement far out, along a whole subsequence, forces equivalence for Cauchy sequences; agreement at finitely many places says nothing. (For sequences that are not Cauchy, (b) fails: $(1, 0, 1, 0, \dots)$ and $(1)_n$ agree at every even index, and their difference is not null.)

Marking scheme. (10 points)

- 2: proves $n_k \geq k$ for every k, justifying the passage from $n_{k+1} > n_k \geq k$ to the next whole number.
- 3: shows $(a_{n_k}) \in \mathcal{C}$, with one threshold that serves all $j, k \geq N$ and with $n_j, n_k \geq N$ justified from $n_k \geq k$.
- 3: shows $(a_{n_k} - a_k)_k$ is null by applying the Cauchy condition to the pair (n_k, k).
- 2: deduces (b) by symmetry and transitivity of $\sim$.

Fatal error (0 for the step): a threshold that depends on k.

5.2 The Real Numbers and Their Arithmetic

Section 5.1 sorted the Cauchy sequences of rationals by a relation, $a \sim b$ when $a - b$ is null, and proved it an equivalence relation (Proposition 5.1.1). This section takes the classes as the new numbers. It defines their sum, product and negative on representatives, proves that the definitions do not depend on the representatives, and shows that the result is a field. The pattern has been met three times: the integers modulo n in *Sets and Functions*, Section 5.2, the integers in Chapter 2, and the rationals in Chapter 3. Only the computation behind the check is new: products need Cauchy sequences to be bounded. The order comes in Section 5.3.

5.2.1 The real numbers as classes

The Newton sequence and the decimal truncations are different sequences that aim at one number (Example (The Truncations and the Newton Sequence Are Equivalent)). Rather than search $\mathbb{Q}$ for that number, we let the whole group of sequences aiming at it **be** the number.

A real number is a Cauchy sequence of rationals, up to a null difference.

Definition 5.2.1. The Real Numbers

The set of **real numbers** is the quotient set (Sets and Functions, Definition 5.2.4)

$$\mathbb{R} := \mathcal{C}/\sim$$

of the set $\mathcal{C}$ of Cauchy sequences of rationals (Definition 5.1.1) by the relation $\sim$ of Definition 5.1.2. Its elements are the classes $[a] = \{b \in \mathcal{C} : b \sim a\}$ for $a \in \mathcal{C}$ (Sets and Functions, Definition 5.2.2), and any $b \in [a]$ is a **representative** of $[a]$.

In words: a real number is a set of Cauchy sequences of rationals, namely all those equivalent to one of them. By Sets and Functions, Theorem 5.2.1 (b), for all $a, b \in \mathcal{C}$,

$$[a] = [b] \iff a \sim b \iff a - b \text{ is null.}$$

This is the only tool we have, so far, for deciding when two real numbers are equal: compare representatives, and ask whether their difference is null.

Examples.

- *The Newton class.* By Example (The Truncations and the Newton Sequence Are Equivalent), $[(x_n)_n] = [(d_k)_k]$: one real number, with two quite different representatives. By Example (Finite Changes and Shifts) (b), $(x_{n+1})_n$ is a third.
- *The class of a constant.* For $q \in \mathbb{Q}$, the class $[(q)_n]$ contains $(q)_n$, and also $(q+1/(n+1))_n$, which lies in $\mathcal{C}$ by Theorem 4.5.3 (a), (b) and is equivalent to $(q)_n$, since the difference $(1/(n+1))_n$ is null (Example (First Limits) (b)). Different constants give different classes, because constant sequences are equivalent only when they are equal (the degenerate example after Definition 5.1.2). Section 5.4 uses these classes to place a copy of $\mathbb{Q}$ inside $\mathbb{R}$.
- *Zero.* $[(0)_n]$ is the set of all null sequences in $\mathcal{C}$: $b \sim (0)_n$ means that $b - (0)_n = b$ is null. Every null sequence lies in $\mathcal{C}$ (Theorem 4.5.3 (a)). So $(1/(n+1))_n$ (Example (First Limits) (b)), $((-1)^n/(n+1))_n$ (null times bounded, as $|(-1)^n| = 1$; Proposition 4.4.1) and every sequence that is eventually 0 (null by Proposition 4.3.2 (a)) are representatives of it.

What has been done, and what has not. The symbol $\mathbb{R}$ now names one definite set, and every fact about it must be proved from the definition. At this moment $\mathbb{R}$ has no addition, no multiplication and no order, and it does not contain $\mathbb{Q}$: the rational number 2 is not a class of sequences. This section builds the arithmetic.

5.2.2 Arithmetic on representatives

To add two real numbers, add representatives term by term; likewise for products and negatives. The terms are rationals, where the arithmetic is known.

Definition 5.2.2. Sum, Product and Negative of Real Numbers

For $a, b \in \mathcal{C}$, let

$$[a] + [b] := [a + b], \qquad [a]\,[b] := [ab], \qquad -[a] := [-a],$$

where $a + b$, ab and $-a$ are the termwise sum, product and negative. Let $0_{\mathbb{R}} := [(0)_n]$ and $1_{\mathbb{R}} := [(1)_n]$.

The right-hand sides are classes of members of $\mathcal{C}$, because $\mathcal{C}$ is closed under termwise sums, products and negatives and contains the constants (Theorem 4.5.3). That makes the right-hand sides **meaningful**. It does not yet make the definition **legitimate**. Each formula takes a class and reads off a representative; if two representatives of the same class gave different answers, the "operation" would assign two values to one input, and so would not be a function (Sets and Functions, Definition 2.1.1).

This danger is real, and it has been met before. In *Sets and Functions*, Section 5.2, the arithmetic of the integers modulo n was defined by the same kind of formula and survived the check (Sets and Functions, Proposition 5.2.5), while the rule $[a]_6 \mapsto [a]_4$ failed it (Sets and Functions, Example (A rule on classes that is not a function)). Chapters 2 and 3 made the same check for the integers and the rationals, through the lemma Theorem 2.2.1, which says that a rule on classes is a function as soon as equivalent inputs give equivalent outputs. Here the check is the same, and only the computation behind it is new.

Representatives, then independence. To define an operation on classes, define it on representatives; then show that the output class survives replacing each input by an equivalent one. This is the method of Sets and Functions, Example (Method: checking that a definition on classes is well defined).

For sums this is a one-line computation. For products it is not, and the next theorem leans on Chapter 4's result that Cauchy sequences are bounded (Proposition 4.5.2).

Theorem 5.2.1. The Operations on Real Numbers Are Well Defined

Let $a, a', b, b' \in \mathcal{C}$ with $a \sim a'$ and $b \sim b'$. Then

$$a + b \sim a' + b', \qquad ab \sim a'b', \qquad -a \sim -a'.$$

Hence the rules of Definition 5.2.2 define functions $+\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, $\cdot\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and $-\colon \mathbb{R} \to \mathbb{R}$.

Idea. For the product we must show that $a_n b_n - a'_n b'_n$ dies out, knowing only that $a_n - a'_n$ and $b_n - b'_n$ do. Replace one factor at a time: subtract and add the mixed term $a_n b'_n$, so that

$$a_n b_n - a'_n b'_n = a_n(b_n - b'_n) + (a_n - a'_n)\, b'_n.$$

Each summand is a null sequence times another sequence. A null sequence times an **unbounded** one need not be null, so we need the multipliers (a_n) and (b'_n) bounded; they are, because they are Cauchy. Then each summand is null, and so is the sum. Passing to classes reads the result one argument at a time (Example (Method: check on representatives, one argument at a time)).

Proof. Let $a, a', b, b' \in \mathcal{C}$ with $a \sim a'$ and $b \sim b'$; so $a - a'$ and $b - b'$ are null (Definition 5.1.2).

Sum. For every n, $(a_n + b_n) - (a'_n + b'_n) = (a_n - a'_n) + (b_n - b'_n)$, by the field axioms of $\mathbb{Q}$. So $(a + b) - (a' + b')$ is the sum of two null sequences, which is null by Theorem 4.4.3 (a). Hence $a + b \sim a' + b'$.

Negative. For every n, $(-a_n) - (-a'_n) = (-a_n) + a'_n = -(a_n - a'_n)$: the first step is $-(-a'_n) = a'_n$, and for the second, $((-a_n) + a'_n) + (a_n - a'_n) = 0$ by the field axioms of $\mathbb{Q}$, so $(-a_n) + a'_n$ is the negative of $a_n - a'_n$ by its uniqueness (both by Proposition 2.3.2 (a)). So $(-a) - (-a') = -(a - a')$, which is null by Theorem 4.4.3 (b). Hence $-a \sim -a'$.

Product. For every n, by the field axioms of $\mathbb{Q}$,

$$a_n b_n - a'_n b'_n = a_n(b_n - b'_n) + (a_n - a'_n)\, b'_n. \tag{5.2.1}$$

The sequence a is in $\mathcal{C}$, so it is bounded (Proposition 4.5.2): there is $M \in \mathbb{Q}_{>0}$ with $|a_n| \le M$ for every n. Since $b - b'$ is null, the sequence $\big((b_n - b'_n)\, a_n\big)_n$ is null by Proposition 4.4.1, and it is the first summand of Eq. 5.2.1, by commutativity in $\mathbb{Q}$. Likewise $b' \in \mathcal{C}$ is bounded, by some $M' \in \mathbb{Q}_{>0}$ (Proposition 4.5.2), and $a - a'$ is null, so the second summand $\big((a_n - a'_n)\, b'_n\big)_n$ is null (Proposition 4.4.1). By the sum rule (Theorem 4.4.3 (a)), $ab - a'b'$ is null. Hence $ab \sim a'b'$.

Functions. The maps $(a, b) \mapsto a + b$ and $(a, b) \mapsto ab$ are functions $\mathcal{C} \times \mathcal{C} \to \mathcal{C}$, and $a \mapsto -a$ is a function $\mathcal{C} \to \mathcal{C}$ (Theorem 4.5.3). Taking $b' = b$ above ($b \sim b$ by reflexivity) shows that each binary map respects $\sim$ in its first argument; taking $a' = a$ shows the same for the second. These are the hypotheses of Theorem 2.2.1 (b), and $-a \sim -a'$ is the hypothesis of Theorem 2.2.1 (a). So the three rules define functions $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$, $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and $\mathbb{R} \to \mathbb{R}$, as claimed.

■

The proof used two different bounds: one for a, which multiplies $b_n - b'_n$, and one for b', which multiplies $a_n - a'_n$. A bound for b would be of no use, since b appears nowhere in Eq. 5.2.1 as a multiplier. Every Cauchy sequence is bounded, so both bounds exist; this is where the restriction of $\sim$ to $\mathcal{C}$ pays (compare the "Why only on $\mathcal{C}$" paragraph of Section 5.1).

So $\mathbb{R}$ has a sum, a product and a negative. For example, $[(x_n)] + [(1)_n] = [(x_n + 1)_n]$, and the answer does not change if (x_n) is replaced by (d_k): the theorem says $(x_n + 1) \sim (d_n + 1)$.

5.2.3 The laws of arithmetic

The operations exist; now the laws. Each law of a field is an equation, and each side is computed in $\mathbb{R}$ through representatives. So the law holds in $\mathbb{R}$ as soon as it holds for sequences, and a law for sequences is the law in $\mathbb{Q}$, applied at each index.

Theorem 5.2.2. The Real Numbers Form a Commutative Ring

$(\mathbb{R}, +, \cdot)$, with the operations of Definition 5.2.2, is a commutative ring (Definition 2.3.1) with zero $0_{\mathbb{R}}$ and one $1_{\mathbb{R}}$: it satisfies (F1)–(F7) and (F9), the additive inverse of $[a]$ is $-[a]$, and $0_{\mathbb{R}} \neq 1_{\mathbb{R}}$.

Idea. Two moves. A law for termwise operations is an equation between functions on $\mathbb{N}$, checked one index at a time in $\mathbb{Q}$. A law for classes moves to representatives by Definition 5.2.2, uses the law for sequences, and moves back. For $0_{\mathbb{R}} \neq 1_{\mathbb{R}}$: the constant 1 is not null.

Proof.

Claim. $\mathcal{C}$, with the termwise operations, is a commutative ring with zero $(0)_n$, one $(1)_n$, and $-a$ the additive inverse of a.

Proof. By Theorem 4.5.3, the termwise sum and product of two members of $\mathcal{C}$, the termwise negative of a member, and the constants $(0)_n$, $(1)_n$ lie in $\mathcal{C}$; so $+$ and $\cdot$ are functions $\mathcal{C} \times \mathcal{C} \to \mathcal{C}$. With the witnesses $(0)_n$, $-a$ and $(1)_n$ for (F3), (F4) and (F7), each of (F1)–(F7) and (F9) asserts an equality of sequences, that is, of functions $\mathbb{N} \to \mathbb{Q}$, so it holds if and only if it holds at every index n (Sets and Functions, Proposition 2.1.1). At the index n it is the same clause in $\mathbb{Q}$, with a_n, b_n, c_n in place of a, b, c, and $\mathbb{Q}$ is a field (Theorem 3.2.1). For example, (F9) at n reads $a_n(b_n + c_n) = a_n b_n + a_n c_n$, and (F4) reads $a_n + (-a_n) = 0$.

■

Every element of $\mathbb{R}$ is some $[a]$ (Sets and Functions, Definition 5.2.4), so let $a, b, c \in \mathcal{C}$. By

Definition 5.2.2, applied once for each operation on each side, $([a]+[b])+[c]=[(a+b)+c]$ and $[a]+([b]+[c])=[a+(b+c)]$; the sequences agree by (F2) in $\mathcal{C}$ (the claim), so the classes are equal. Distributivity goes the same way:

$$\begin{aligned}[a]([b]+[c]) &= [a]\,[b+c] = [a(b+c)] \\ &= [ab+ac] = [ab]+[ac] = [a][b]+[a][c],\end{aligned}$$

where the step from the first line to the second is (F9) in $\mathcal{C}$. For (F4): $[a]+(-[a]) = [a+(-a)] = [(0)_n] = 0_{\mathbb{R}}$, so the additive inverse of $[a]$ is $-[a]$. Every other clause of (F1)–(F7) and (F9) is proved in the same way: move to representatives, use the clause in $\mathcal{C}$, move back. Both moves are Definition 5.2.2; what made them legitimate is Theorem 5.2.1, which turned the rules into functions on $\mathbb{R}$.

Finally, $0_{\mathbb{R}} \neq 1_{\mathbb{R}}$: otherwise $(1)_n \sim (0)_n$ (Sets and Functions, Theorem 5.2.1 (b)), that is, $(1)_n$ is null; but for $\varepsilon = 1/2$ every term has $|1-0| = 1 \geq 1/2$. This proves the theorem.

■

In the language of *Algebraic Preliminaries*, Chapter 1, the null sequences form an ideal $\mathcal{N}$ of the ring $\mathcal{C}$, and $\mathbb{R}$ is the quotient ring $\mathcal{C}/\mathcal{N}$; the proof above is that volume's quotient theorem, done by hand for one ring.

5.2.4 Inverses

A commutative ring is not yet a field. The missing clause is (F8): every $x \neq 0_{\mathbb{R}}$ needs an inverse. The obvious idea is to invert term by term, and it fails; the example after the theorem shows how. What saves it is a fact from Chapter 4: a Cauchy sequence that is not null stays **a fixed distance** away from 0 from some index on (Theorem 4.5.4).

Theorem 5.2.3. The Real Numbers Form a Field

Every $x \in \mathbb{R}$ with $x \neq 0_{\mathbb{R}}$ has a multiplicative inverse: there is $y \in \mathbb{R}$ with $xy = 1_{\mathbb{R}}$. Hence $\mathbb{R}$, with the operations of Definition 5.2.2, is a field (Definition 2.3.1).

Idea. Write $x = [a]$. Since $x \neq 0_{\mathbb{R}}$, a is not null, so from some index N on its terms are at least $c > 0$ in absolute value. Invert a_n for $n \geq N$, and put anything harmless, say 1, before N; Chapter 4 showed the result is again Cauchy (Proposition 4.5.6). The product with a is 1 from N on, and the first N terms do not matter. *Before reading on: the inverse is built from one representative a. Why does the class $[b]$ not depend on which representative of x we started from?*

Proof. Let $x \in \mathbb{R}$ with $x \neq 0_{\mathbb{R}}$. By Sets and Functions, Definition 5.2.4, $x = [a]$ for some $a \in \mathcal{C}$; fix one such a. The sequence a is not null: otherwise $a \sim (0)_n$, and $x = [(0)_n] = 0_{\mathbb{R}}$ by Sets and Functions, Theorem 5.2.1 (b).

By Theorem 4.5.4 (with $F = \mathbb{Q}$) there are $c \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ such that $a_n \geq c$ for every $n \geq N$, or $a_n \leq -c$ for every $n \geq N$. Fix one such pair (c, N). In either case $|a_n| \geq c$ for every $n \geq N$: if $a_n \geq c > 0$, then $|a_n| = a_n$; if $a_n \leq -c$, then $|a_n| = -a_n \geq c$ (Definition 4.1.1, Proposition 2.4.3 (a)).

By Proposition 4.5.6, the sequence b with $b_n := 1/a_n$ for $n \geq N$ and $b_n := 1$ for $n < N$ is in $\mathcal{C}$, and $a_n b_n = 1$ for every $n \geq N$. So the sequence $ab - (1)_n$ agrees with $(0)_n$ from N on, and it is null by Proposition 4.3.2 (a). Hence $ab \sim (1)_n$, and

$$x\,[b] = [a]\,[b] = [ab] = [(1)_n] = 1_{\mathbb{R}}.$$

So $y := [b]$ is an inverse of x, and (F8) holds. Together with Theorem 5.2.2, which gives (F1)–(F7), (F9) and $0_{\mathbb{R}} \neq 1_{\mathbb{R}}$, this shows that $\mathbb{R}$ is a field.

■

The answer to the prompt: **inverses are unique.** By the theorem, $\mathbb{R}$ is a field, so x has only one inverse (Lemma 4.1.1 (a) with $F = \mathbb{R}$); the reason is the one-line computation $y = y(xy') = (yx)y' = y'$ for two inverses y, y'. So starting from another representative a' of x, with its own c', N' and b', gives $[b'] = [b]$; we write x^{-1} for it.

Example. Termwise Reciprocals Are Not a Definition

A shorter route to inverses would be: for $x = [a] \neq 0_{\mathbb{R}}$, let $x^{-1} := [(1/a_n)_n]$. Explain why this is not a definition, and why $0_{\mathbb{R}}$ must be excluded in any case.

Solution. It need not make sense. The sequences $a := (0, 1, 1, 1, \dots)$ and $(1)_n$ agree from $n = 1$ on; a is in $\mathcal{C}$, since $|a_m - a_n| = 0$ for all $m, n \geq 1$ (threshold $N = 1$ for every tolerance), so the two sequences are equivalent (Example (Finite Changes and Shifts) (a)) and represent the same real number $1_{\mathbb{R}} \neq 0_{\mathbb{R}}$. With the representative a, the rule asks for $1/a_0 = 1/0$, which does not exist. A rule that cannot be applied to some representatives does not define a function on classes; it would need a statement of which representatives are allowed, and a proof that the answer does not depend on the allowed one. The proof of Theorem 5.2.3 supplies both at once: it inverts only from an index N on, and the uniqueness of inverses makes the class independent of all the choices.

Why $0_{\mathbb{R}}$ is excluded. The null sequence $(1/(n+1))_n$ represents $0_{\mathbb{R}}$ and has no zero term, so the termwise reciprocal $(n+1)_n$ exists. But it is not bounded (Theorem 3.2.3), so it is not in $\mathcal{C}$ (Proposition 4.5.2), and it represents no real number. No inverse of $0_{\mathbb{R}}$ is possible in any case, since $0_{\mathbb{R}} \cdot y = y \cdot 0_{\mathbb{R}} = 0_{\mathbb{R}} \neq 1_{\mathbb{R}}$ for every y ((F6), Proposition 2.3.2 (b), Theorem 5.2.2).

Warning. **A termwise formula is not an operation until it is checked on every representative.** "$[a]^{-1} := [(1/a_n)]$" fails on $(0, 1, 1, 1, \dots)$, a representative of $1_{\mathbb{R}}$, whose first term cannot be inverted. Every definition on $\mathbb{R}$ must say what it does to **each** representative of a class, and then prove that the class of the answer is the same for all of them.

Quick check. Find a representative of $[(1 + 1/(n+1))_n] \cdot [(2)_n] - [(2)_n]$, and name its class.

Solution. By Definition 5.2.2, with $x - y := x + (-y)$ in the ring $\mathbb{R}$, the real number is $[(1+1/(n+1)) \cdot 2] + [(-2)_n] = [(2+2/(n+1)-2)_n]$, so $(2/(n+1))_n$ represents it. This sequence

is null (Theorem 4.4.3 (b) with $c = 2$, applied to $(1/(n+1))$, which is null by Example (First Limits) (b)), so the class is $0_{\mathbb{R}}$. In other words, $[(1 + 1/(n+1))_n] \cdot [(2)_n] = [(2)_n]$. The first factor is $1_{\mathbb{R}}$, since $(1 + 1/(n+1))_n - (1)_n = (1/(n+1))_n$ is null, so $(1 + 1/(n+1))_n \sim (1)_n$; it is $1_{\mathbb{R}}$ represented by a sequence none of whose terms is 1.

$\mathbb{R}$ is now a field. It has no order yet: "positive" for a class of sequences needs care, and Section 5.3 defines it.

5.2.5 Exercises

5.2.5.1 A. Check your understanding

Exercise. A1

(a) Prove, directly from Definition 5.2.2 and the field axioms of $\mathbb{Q}$, that $([a][b])[c] = [a]([b][c])$ and $1_{\mathbb{R}}\,[a] = [a]$ for all $a, b, c \in \mathcal{C}$.

(b) In the product part of the proof of Theorem 5.2.1, which two of the four sequences a, a', b, b' must be bounded, and why is "b is bounded" not what Eq. 5.2.1 needs?

(c) Run that proof on $a = (2 + 1/(n+1))_n$, $a' = (2)_n$, $b = (x_n)_n$ (the Newton sequence) and $b' = (d_n)_n$ (the truncations): check that $a \sim a'$ and $b \sim b'$, write $a_n b_n - a'_n b'_n$ as in Eq. 5.2.1, and give a bound for each multiplier.

(d) State the idea of the proof of Theorem 5.2.1 in two sentences.

Solution.

(a) By Definition 5.2.2 twice, $([a][b])[c] = [ab][c] = [(ab)c]$ and $[a]([b][c]) = [a][bc] = [a(bc)]$. For every n, $(a_n b_n)c_n = a_n(b_n c_n)$ by (F5) in $\mathbb{Q}$, so $(ab)c = a(bc)$ as sequences (Sets and Functions, Proposition 2.1.1), and the two classes are equal. Likewise $1_{\mathbb{R}}[a] = [(1)_n][a] = [(1 \cdot a_n)_n] = [a]$, by (F6) and (F7) in $\mathbb{Q}$.

(b) a and b'. In Eq. 5.2.1, a_n multiplies the null $b_n - b'_n$, and b'_n multiplies the null $a_n - a'_n$; Proposition 4.4.1 needs the multiplier bounded. The sequence b appears in Eq. 5.2.1 only inside the null factor $b_n - b'_n$, so its bound is never used.

(c) $a - a' = (1/(n+1))_n$ is null (Example (First Limits) (b)), so $a \sim a'$ (both are in $\mathcal{C}$ by Theorem 4.5.3 (a), (b)); $b \sim b'$ is Example (The Truncations and the Newton Sequence Are Equivalent). The split is

$$\Big(2 + \frac{1}{n+1}\Big)x_n - 2d_n = \Big(2 + \frac{1}{n+1}\Big)(x_n - d_n) + \frac{1}{n+1}\,d_n.$$

Multipliers: $|2 + 1/(n+1)| \le 3$, since $0 < 1/(n+1) \le 1$; and $|d_n| \le 2$, since $1 = d_0 \le d_n$ by (T1), and $d_n^2 \le 2 < 4$ by (T3), so $d_n < 2$ (Proposition 2.4.3 (g)); (T1) and (T3) are those before Example (The Truncations and the Newton Sequence Are Equivalent). So the first summand is null (null times bounded by 3) and the second is null (null times bounded by 2).

(d) Changing both factors of a product at once is split into two changes, one factor at a time, by adding and subtracting a mixed term. Each change is a null sequence times a bounded one, because Cauchy sequences are bounded, so each is null, and so is their sum.

5.2.5.2 B. Practice

Exercise. B1: The negative, directly

Directly from Definition 5.2.2 and Theorem 2.2.1, prove that the rule $-[a] \coloneqq [-a]$ defines a function $\mathbb{R} \to \mathbb{R}$, and that $[a] + (-[a]) = 0_{\mathbb{R}}$ for every $a \in \mathcal{C}$. Deduce that $[a] - [b] = [a - b]$ for all $a, b \in \mathcal{C}$.

Solution. Well defined. $-a \in \mathcal{C}$ for $a \in \mathcal{C}$ (Theorem 4.5.3 (b)). Let $a \sim a'$. For every n, $(-a_n) - (-a'_n) = (-a_n) + a'_n = -(a_n - a'_n)$: the first step is $-(-a'_n) = a'_n$, and for the second, $((-a_n) + a'_n) + (a_n - a'_n) = 0$ by the field axioms of $\mathbb{Q}$, so $(-a_n) + a'_n$ is the negative of $a_n - a'_n$ by its uniqueness (both by Proposition 2.3.2 (a)), and $(-(a_n - a'_n))_n$ is null because $a - a'$ is (Theorem 4.4.3 (b)). So $-a \sim -a'$, and Theorem 2.2.1 (a) gives a function $\mathbb{R} \to \mathbb{R}$.

Inverse. $[a] + (-[a]) = [a] + [-a] = [a + (-a)] = [(a_n + (-a_n))_n] = [(0)_n] = 0_{\mathbb{R}}$, by Definition 5.2.2 and (F4) in $\mathbb{Q}$ at each index.

Differences. $[a] - [b] = [a] + (-[b]) = [a] + [-b] = [a + (-b)] = [a - b]$, by the definition $x - y \coloneqq x + (-y)$ in a ring and Definition 5.2.2.

Exercise. B2: The larger of two reals, faded

For $x, y \in \mathbb{Q}$ let $\max(x, y) \coloneqq x$ if $x \geq y$, and $\max(x, y) \coloneqq y$ otherwise. For $a, b \in \mathcal{C}$ consider the rule

$$g([a], [b]) \coloneqq [(\max(a_n, b_n))_n].$$

(a) Prove that $|\max(x, y) - \max(x', y')| \leq |x - x'| + |y - y'|$ for all $x, y, x', y' \in \mathbb{Q}$. *Hint:* $\max(x, y)$ *is one of* x, y, *and is* $\geq$ *both.*

(b) Deduce from (a) that $(\max(a_n, b_n))_n \in \mathcal{C}$ whenever $a, b \in \mathcal{C}$.

(c) Here is the start of a proof that the rule g is well defined. *Name the move, then complete it.*
"Let $a, a', b, b' \in \mathcal{C}$ with $a \sim a'$ and $b \sim b'$. By (b), the sequences $(\max(a_n, b_n))_n$ and $(\max(a'_n, b'_n))_n$ are in $\mathcal{C}$. Let $e_n \coloneqq \max(a_n, b_n) - \max(a'_n, b'_n)$. By (a), for every n,

$$0 \leq |e_n| \leq |a_n - a'_n| + |b_n - b'_n|.$$

⟨ blank ⟩"

Solution.

(a) Write $M \coloneqq \max(x, y)$ and $M' \coloneqq \max(x', y')$. Note $M \geq x$, $M \geq y$, and M is one of x, y; likewise for M'. Suppose first $M \geq M'$. If $M = x$, then $0 \leq M - M' \leq x - x' \leq |x - x'|$, since $M' \geq x'$ (Proposition 2.4.3 (b), Proposition 4.1.3 (c)). If $M = y$, then likewise $0 \leq M - M' \leq y - y' \leq |y - y'|$. Either way $|M - M'| = M - M' \leq |x - x'| + |y - y'|$. If $M' > M$, exchange the roles of (x, y) and (x', y'); the right-hand side is unchanged, as $|x' - x| = |x - x'|$.

(b) Let $\varepsilon \in \mathbb{Q}_{>0}$, and let N be a threshold with tolerance $\varepsilon/2$ for both a and b (the larger of the two thresholds). For $m, n \geq N$, (a) with $(x, y, x', y') = (a_m, b_m, a_n, b_n)$ gives $|\max(a_m, b_m) - \max(a_n, b_n)| \leq |a_m - a_n| + |b_m - b_n| < \varepsilon/2 + \varepsilon/2 = \varepsilon$.

(c) *The move:* representatives, then independence; here its last step, "a sequence squeezed between 0 and a null sequence is null, so the outputs are equivalent and the classes agree". *Completion:* The sequences $(|a_n - a'_n|)_n$ and $(|b_n - b'_n|)_n$ are null, since $||t| - 0| = |t - 0|$ for every $t \in \mathbb{Q}$ (Definition 4.3.3). So their sum is null (Theorem 4.4.3 (a)). By the squeeze Proposition 4.4.4 (c), with lower sequence $(0)_n$ and this upper sequence, $(|e_n|)_n$ is null; hence $(e_n)_n$ is null, by the same remark on $|e_n - 0|$. So $(\max(a_n, b_n))_n \sim (\max(a'_n, b'_n))_n$. Taking $b' = b$, and then $a' = a$, gives the two hypotheses of Theorem 2.2.1 (b), so g is a function $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$.

Exercise. B3: Zero divisors in $\mathcal{C}$, none in $\mathbb{R}$

(a) Find $a, b \in \mathcal{C}$, neither equal to the zero sequence, with $ab = (0)_n$.

(b) Let $a, b \in \mathcal{C}$ with ab null. Prove that a or b is null (use Lemma 4.1.1 (d) with $F = \mathbb{R}$).

(c) For your a, b of (a), which of $[a]$, $[b]$ is $0_{\mathbb{R}}$? Could both classes be different from $0_{\mathbb{R}}$?

Solution.

(a) $a := (1, 0, 0, \dots)$, $b := (0, 1, 1, \dots)$. Both are eventually constant, hence in $\mathcal{C}$ (Example (Finite Changes and Shifts) (a) with a constant sequence), and $a_n b_n = 0$ for every n.

(b) Since ab is null, $ab \sim (0)_n$, so $[a][b] = [ab] = 0_{\mathbb{R}}$ (Definition 5.2.2, Sets and Functions, Theorem 5.2.1 (b)). $\mathbb{R}$ is a field (Theorem 5.2.3), so Lemma 4.1.1 (d) gives $[a] = 0_{\mathbb{R}}$ or $[b] = 0_{\mathbb{R}}$, that is, $a \sim (0)_n$ or $b \sim (0)_n$: a or b is null.

(c) $[a] = 0_{\mathbb{R}}$, since a is eventually 0 (Proposition 4.3.2 (a)); $[b] = 1_{\mathbb{R}}$ (Example (Finite Changes and Shifts) (a)). The two classes cannot both differ from $0_{\mathbb{R}}$, by (b), since $ab = (0)_n$ is null. So the ring $\mathcal{C}$ has zero divisors and the field $\mathbb{R}$ has none: passing to classes sends every null sequence, such as a, to $0_{\mathbb{R}}$. (Contrast with (a): there ab is the zero sequence itself, and neither factor is.)

5.2.5.3 C. Going deeper

Exercise. C1: Exit check: multiplication is well defined

State precisely what must be proved for the rule $[a][b] := [ab]$ to define a function $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$, and prove it, including the fact that $ab \in \mathcal{C}$ for $a, b \in \mathcal{C}$.

Solution. What must be shown. (i) For $a, b \in \mathcal{C}$, the termwise product ab is in $\mathcal{C}$, so that $[ab]$ is a real number. (ii) If $a \sim a'$ and $b \sim b'$, then $ab \sim a'b'$. Taking $b' = b$ and then $a' = a$ in (ii), with $b \sim b$ and $a \sim a$ by reflexivity, gives the two one-argument hypotheses of Theorem 2.2.1 (b). By that theorem, (i) and (ii) give exactly one function $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ with $([a], [b]) \mapsto [ab]$; without (ii), one pair of classes could receive two values, and (Fn2) of Sets and Functions, Definition 2.1.1 would fail.

(i) Bound, then shrink. Let $a, b \in \mathcal{C}$. Both are bounded (Proposition 4.5.2); let $M \in \mathbb{Q}_{>0}$ be a bound for both (the larger of the two bounds). For all m, n,

$$\begin{aligned} |a_m b_m - a_n b_n| &= |a_m(b_m - b_n) + (a_m - a_n)b_n| \\ &\le M|b_m - b_n| + M|a_m - a_n|, \end{aligned}$$

by Proposition 4.1.3 (b), (e) and Proposition 4.1.2 (d). Let $\varepsilon \in \mathbb{Q}_{>0}$, and let N be a threshold with tolerance $\varepsilon/(2M)$ for both a and b. For $m, n \geq N$, the right side is less than $M \cdot \varepsilon/(2M) + M \cdot \varepsilon/(2M) = \varepsilon$ (Proposition 2.4.3 (b), (c), as $M > 0$). So $ab \in \mathcal{C}$.

(ii) The add-and-subtract decomposition. For every n,

$$a_n b_n - a'_n b'_n = a_n(b_n - b'_n) + (a_n - a'_n)b'_n.$$

The multipliers a and b' are in $\mathcal{C}$, hence bounded (Proposition 4.5.2). The sequences $b - b'$ and $a - a'$ are null. By Proposition 4.4.1, both summands are null, and by the sum rule (Theorem 4.4.3 (a)) so is $ab - a'b'$. Hence $ab \sim a'b'$.

Marking scheme. (10 points)

- 2: states what must be shown (the product lies in $\mathcal{C}$; equivalent inputs give equivalent outputs) and why this makes the rule a function.
- 2: proves $ab \in \mathcal{C}$, by "bound, then shrink" with the threshold's dependence on ε and M clear.
- 2: the add-and-subtract decomposition, or any identity writing $a_n b_n - a'_n b'_n$ as a sum of (null) $\times$ (sequence) terms.
- 2: boundedness of the two multipliers, justified by "Cauchy implies bounded".
- 2: concludes with null $\times$ bounded is null and null + null is null.

A proof that changes one factor at a time (first $ab \sim a'b$, then $a'b \sim a'b'$) earns the same points; the multipliers are then b and a'. Fatal errors (0 for the step): "both differences are small, so the product difference is small" with no bound on the multipliers; a bound for b used where b' is the multiplier.

Exercise. C2: No square is zero except zero

(a) Let a be a sequence of rationals such that $(a_n^2)_n$ is null. Prove that a is null.

(b) Deduce, without Theorem 5.2.3 (that is, without inverses), that for $x \in \mathbb{R}$, $x^2 = 0_{\mathbb{R}}$ implies $x = 0_{\mathbb{R}}$.

Solution.

(a) Let $\varepsilon \in \mathbb{Q}_{>0}$. Then $\varepsilon^2 \in \mathbb{Q}_{>0}$, so there is N with $|a_n^2| < \varepsilon^2$ for every $n \geq N$. For such n, $|a_n|^2 = a_n^2 = |a_n^2| < \varepsilon^2$ (Proposition 4.1.3 (b), and $a_n^2 \geq 0$ by Proposition 2.4.3 (e), or $a_n = 0$), and $|a_n| \geq 0$, $\varepsilon > 0$; so $|a_n| < \varepsilon$ by Proposition 2.4.3 (g). The threshold for ε is the threshold of (a_n^2) for ε^2. So a is null.

(b) Let $x = [a]$ with $a \in \mathcal{C}$. Then $x^2 = [a][a] = [(a_n^2)_n]$ by Definition 5.2.2. If $x^2 = 0_{\mathbb{R}} = [(0)_n]$, then $(a_n^2)_n \sim (0)_n$ (Sets and Functions, Theorem 5.2.1 (b)), that is, $(a_n^2)_n$ is null. By (a), a is null, so $a \sim (0)_n$ and $x = 0_{\mathbb{R}}$.

In a field this also follows from inverses: if $x \neq 0_{\mathbb{R}}$, then $x = x^{-1}x^2 = x^{-1}0_{\mathbb{R}} = 0_{\mathbb{R}}$, a contradiction. The direct proof shows the fact is already visible in the representatives, before any inverse exists. Exercise B3 is the companion fact for products: there the termwise statement fails in $\mathcal{C}$, and only the classes behave.

Marking scheme. (10 points)

- 3: chooses the tolerance ε^2 for (a_n^2) and states the threshold.
- 3: $|a_n|^2 = |a_n^2|$ and the step from squares to $|a_n| < \varepsilon$ by Proposition 2.4.3 (g), with the nonnegativity hypotheses.
- 2: (b), moving from $x^2 = 0_{\mathbb{R}}$ to (a_n^2) null via Definition 5.2.2.
- 2: concludes $x = 0_{\mathbb{R}}$.

Fatal error: using x^{-1}, which does not exist yet.

5.3 Positive Classes and the Order

Section 5.2 made $\mathbb{R}$ a field: its elements are classes of Cauchy sequences of rationals, added and multiplied through representatives. A field is not yet a number line. To compare two real numbers we need an order, and it must cooperate with the arithmetic: $\mathbb{R}$ should be an ordered field in the sense of Definition 2.4.1. Chapter 3 ordered $\mathbb{Q}$ in two steps. It first said which rationals are positive, and then let $a < b$ mean that $b - a$ is positive (Proposition 2.4.1). This section takes the same two steps for $\mathbb{R}$. All the work is in the first: deciding which classes are positive, in a way that does not depend on the representative.

Notation is that of Sections 5.1 and 5.2: $\mathcal{C}$, $\sim$, $[a]$, the operations of Definition 5.2.2, $0_{\mathbb{R}} = [(0)_n]$, $1_{\mathbb{R}} = [(1)_n]$, and $[a] - [b] = [a - b]$, used without comment. Until $\mathbb{R}$ has an order, every margin, tolerance and threshold is rational or natural: we may not measure with what we are building.

5.3.1 Comparing terms does not compare classes

The first idea is to compare two classes by comparing their terms. It fails, and the way it fails shows what the definition must contain.

Example. Termwise Order Is Not an Order on Classes

Consider the rule

> $[a] < [b]$ means that $a_n < b_n$ for every n from some index on.

Let $a_n := 0$ and $b_n := 1/(n+1)$ for $n \in \mathbb{N}$. Show that $[a] = [b]$. Then apply the rule to the classes $[a]$ and $[a]$ twice: once through the representatives a and a, once through a and b. Conclude that the rule does not define a relation on $\mathbb{R}$.

Solution. Both sequences are in $\mathcal{C}$: a constant sequence and a null sequence are Cauchy (Theorem 4.5.3 (a)), and $(1/(n+1))$ is null in $\mathbb{Q}$ (Example (First Limits) (b)). Termwise, $b - a = (1/(n+1))$, which is null, so $b \sim a$, and $[a] = [b]$ by Sets and Functions, Theorem 5.2.1 (b).

Through the representatives a and a: the rule asks whether $0 < 0$ from some index on, and $0 < 0$ never holds. So the rule says that $[a] < [a]$ is false.

Through the representatives a and b, which name the same two classes, since $b \in [b] = [a]$: for **every** n, $a_n = 0 < 1/(n+1) = b_n$. So the rule says that $[a] < [a]$ is true.

A relation on $\mathbb{R}$ must give one answer for each pair of classes. The rule gives two answers for the pair $(0_{\mathbb{R}}, 0_{\mathbb{R}})$, depending on the representatives, so it defines no relation on $\mathbb{R}$. Worse, the second answer, $0_{\mathbb{R}} < 0_{\mathbb{R}}$, is excluded by trichotomy (O1) in any ordered field.

The trouble is that the gap between a_n and b_n is allowed to shrink to nothing, and a gap that shrinks to nothing is exactly what $\sim$ ignores. A class that deserves to be called positive must have terms that stay above 0 by a **fixed** margin.

5.3.2 Positive sequences

We look for a property of a single sequence $a \in \mathcal{C}$ that says "the number a names is greater than 0". The example shows that "eventually $a_n > 0$" is too weak, because a null sequence can have it. The repair is to name the margin first.

Positive means: eventually above one fixed positive rational.

Definition 5.3.1. Positive Cauchy Sequence

A sequence $a \in \mathcal{C}$ is **positive** if **there exist** $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ such that $a_n \geq \delta$ **for every** $n \in \mathbb{N}$ with $n \geq N$. In symbols,

$$\exists \delta \in \mathbb{Q}_{>0} \;\; \exists N \in \mathbb{N} \;\; \forall n \in \mathbb{N} \;\; (n \geq N \Rightarrow a_n \geq \delta).$$

In words: the first quantifier names a margin δ, a positive rational, before anything else is named; the second names a threshold; the third says that **every** term from the threshold on is at least the margin. Because δ comes first, one margin serves all the late terms at once. The word "positive" describes the sequence as a name for a number, not its terms one by one: the terms before N may be negative, and, as the warning below shows, terms that are all > 0 are not enough.

Compare the weaker condition

$$\exists N \in \mathbb{N} \;\; \forall n \in \mathbb{N} \;\; (n \geq N \Rightarrow a_n > 0).$$

There each late term clears 0 by its own amount, which may shrink with n; in the definition one margin is named before n, and that order of the quantifiers is the whole point.

Examples.

A positive constant. For $q \in \mathbb{Q}_{>0}$, the constant sequence $(q)_n$ is positive, with $\delta = q$ and $N = 0$. This degenerate case is how Section 5.4 will see that the rational q, placed inside $\mathbb{R}$, is greater than 0.

The Newton sequence. Let (x_n) be the sequence of Example (A Rational Sequence Aimed at the Gap), which is in $\mathcal{C}$ by Theorem 4.6.1. The same theorem states that $x_n > 1$ for every $n \in \mathbb{N}$. So (x_n) is positive, with $\delta = 1$ and $N = 0$.

Late positivity. Let $a_n := 1 - 2/(n+1)$. It is in $\mathcal{C}$, as the difference of a constant sequence and a constant multiple of the null sequence $(1/(n+1))$ (Theorem 4.5.3 (a), (b)). Its first terms are $-1, 0, 1/3, 1/2$, so it is not positive "from the start". For $n \geq 3$, $n+1 \geq 4$, so $1/(n+1) \leq 1/4$ (Proposition 4.1.2 (c)), hence $2/(n+1) \leq 2/4 = 1/2$ (Proposition 4.1.2 (d), with $0 \leq 2 \leq 2$), and $a_n \geq 1/2$. So a is positive, with $\delta = 1/2$ and $N = 3$. No smaller N works with this δ, since $a_2 = 1/3 < 1/2$; the definition allows any finite stretch of bad terms.

Non-example by a minimal change. Keep every term positive, but let the margin shrink: $a_n := 1/(n+1)$.

Second reading · *Why this sequence is not positive*

Every term is > 0, so the weaker condition above holds with $N = 0$. What fails is the order of the quantifiers. The negation of the definition, by Logic and Proof, Proposition 6.2.3 and Logic and Proof, Proposition 6.2.2, reads

$$\forall \delta \in \mathbb{Q}_{>0} \;\; \forall N \in \mathbb{N} \;\; \exists n \in \mathbb{N} \;\; (n \geq N \wedge a_n < \delta),$$

and it holds: given δ and N, Corollary 3.2.4 gives $n_0 \in \mathbb{Z}_{>0}$ with $1/m < \delta$ for every $m \geq n_0$. With $n := \max\{N, n_0\}$, we have $n \geq N$, and $m := n + 1 \geq n_0$, so $a_n = 1/(n+1) < \delta$. Every margin is eventually undercut.

Why this definition. Each clause has a job.

- *The margin δ.* Without it, the definition would be the weaker condition, and Example (Termwise Order Is Not an Order on Classes) shows the result: $(1/(n+1))$ would be positive, while $(0)_n$, which names the same number $0_{\mathbb{R}}$, is not.
- *A rational margin.* The only positive numbers available are rationals. A real margin would use the order of $\mathbb{R}$, which is what we are defining.
- *The threshold N.* Changing finitely many terms does not change the class (Example (Finite Changes and Shifts)), so a definition meant for classes must ignore every finite stretch of terms. The late-positivity example needs $N = 3$.

Second reading · *Why $\geq$ and not $>$*

Either inequality would do: $a_n \geq \delta$ gives $a_n > \delta/2$, and $a_n > \delta$ gives $a_n \geq \delta$. We use $\geq$, the form in which Theorem 4.5.4 delivers its conclusion.

Warning. "Every term of a is positive" does **not** make a positive. The sequence $(1/(n+1))$ has every term > 0, and it is not positive (non-example above); its class is $0_{\mathbb{R}}$.

Quick check. Let $a \in \mathcal{C}$, with δ ranging over $\mathbb{Q}_{>0}$. Which of the two conditions

$$\text{(A) } \exists \delta \; \exists N \; \forall n \geq N \;\; a_n \geq \delta, \qquad \text{(B) } \exists N \; \forall n \geq N \; \exists \delta \;\; a_n \geq \delta$$

implies the other? Which of them says the same as "eventually $a_n > 0$"?

Solution.

(A) implies (B): the margin found once serves as the δ for every late n. (B) does not imply (A): $(1/(n+1))$ satisfies (B), with $\delta := 1/(n+1)$ chosen after n, and fails (A). (B) says the same as "eventually $a_n > 0$": if $a_n > 0$, take $\delta := a_n$; if $a_n \geq \delta > 0$, then $a_n > 0$.

So (A) is the definition of positive, and (B) is the weaker condition.

Payoff: positivity belongs to the class. A definition about sequences says something about a class only if equivalent sequences agree on it. The proof shows the method of this section in its simplest form.

Proposition 5.3.1. Positivity Is a Property of the Class

Let $a, a' \in \mathcal{C}$ with $a' \sim a$. If a is positive, then a' is positive.

Idea (Scratch work). The goal is an existence statement whose first quantifier is a margin. Name it first, from the data, before any index appears; then ask the given sequences for thresholds.

Givens	Goal
$a_n \geq \delta$ for $n \geq N$, with $\delta \in \mathbb{Q}_{>0}$; $a' - a$ null	$\exists \delta' \in \mathbb{Q}_{>0}\ \exists N'\ \forall n \geq N'\ \ a'_n \geq \delta'$
the margin comes first and may depend only on δ: afford to lose half, $\delta' \coloneqq \delta/2$	$\exists N'\ \forall n \geq N'\ \ a'_n \geq \delta/2$
nullity of $a' - a$ at the tolerance $\delta/2$: $a'_n - a_n$ is within $\delta/2$ of 0 for $n \geq N''$	the threshold $N' \coloneqq \max\{N, N''\}$
for $n \geq N'$: $a'_n > a_n - \delta/2 \geq \delta - \delta/2$	gap closed

Proof. Fix $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $a_n \geq \delta$ for every $n \geq N$. Then $\delta/2 \in \mathbb{Q}_{>0}$ (Proposition 4.1.2 (b)). Since $a' - a$ is null, fix N'' with $|a'_n - a_n| < \delta/2$ for every $n \geq N''$, and let $N' \coloneqq \max\{N, N''\}$. For $n \geq N'$, Proposition 4.1.3 (d) gives $-\delta/2 < a'_n - a_n$, so $a'_n > a_n - \delta/2 \geq \delta - \delta/2 = \delta/2$ (Proposition 2.4.3 (b), Proposition 4.1.2 (b)). So a' is positive, with margin $\delta/2$ and threshold N'. ■

This is Proposition 4.5.5 with $F = \mathbb{Q}$, written out because its order of choices is the method of this section.

The margin halves, and that is harmless: the definition asks only for **some** margin. Exercise C2 shows a tempting proof of this proposition that keeps a margin of 0, and so does not prove it.

Example. Method: name the margin before the index

Template. To prove that $b \in \mathcal{C}$ is positive, or that $[a] < [b]$: (1) name a margin $\delta \in \mathbb{Q}_{>0}$ from data that do not depend on n: a given margin, or a fraction (such as half) of a given margin or of a positive value the terms visibly approach; (2) only then ask the given sequences (Cauchy, null, positive) for thresholds at tolerances computed from δ; (3) take the largest of these finitely many thresholds and check that the terms of the sequence to be shown positive (b, or $b - a$ for $[a] < [b]$) are $\geq \delta$ beyond it.

When to use it. Whenever the goal reads $\exists \delta\ \exists N\ \forall n \geq N\ \ldots$: positivity, a strict inequality

between classes, a bound that must hold beyond one index.

The usual failure. Proving "$b_n > 0$ for all large n" and calling b_n the margin: it depends on n (the Quick check above; Exercise C2).

We can now name the set of positive real numbers. Let

$$P := \{[a] : a \in \mathcal{C} \text{ and } a \text{ is positive}\} \subseteq \mathbb{R}.$$

Chapters 2 and 3 built the positive sets $P_{\mathbb{Z}}$ and $P_{\mathbb{Q}}$; P is the same step for $\mathbb{R}$. For $x \in \mathbb{R}$:

$$x \in P \iff \text{some representative of } x \text{ is positive} \iff \text{every representative of } x \text{ is positive.}$$

The first equivalence is the definition of P. For the second, every class has a representative (Sets and Functions, Theorem 5.2.1 (a)), which gives "every ⇒ some". Conversely, if $x = [a]$ with a positive and $b \in x$, then $b \sim a$ (Sets and Functions, Definition 5.2.2), so b is positive by Proposition 5.3.1. Membership in P can therefore be tested on **any** representative we like. This is the well-definedness the termwise rule lacked.

5.3.3 The positive classes

Chapter 2's recipe needs two facts about P: it is closed under sum and product, and it splits $\mathbb{R}$ into three parts, P, $\{0_{\mathbb{R}}\}$ and the negatives of P. These are clauses (P1) and (P2) of Proposition 2.4.1.

Theorem 5.3.2. The Positive Real Numbers

(i) For every $x \in \mathbb{R}$, **exactly one** of $x \in P$, $x = 0_{\mathbb{R}}$, $-x \in P$ holds.

(ii) If $x \in P$ and $y \in P$, then $x + y \in P$ and $xy \in P$.

Idea. For (i), "at least one" is the sign theorem of Chapter 4: a Cauchy sequence that is not null is eventually above c or eventually below $-c$ (Theorem 4.5.4). A null representative gives $0_{\mathbb{R}}$. "At most one" holds because a sequence cannot be eventually above a positive margin and also eventually below a negative one, or null. For (ii), add or multiply the two margins and take the later of the two thresholds.

Proof. Let $x \in \mathbb{R}$, and write $x = [a]$ with $a \in \mathcal{C}$.

(i), *at least one.* Suppose first that a is null. Termwise $a - (0)_n = a$, so $a \sim (0)_n$ and $x = [(0)_n] = 0_{\mathbb{R}}$. Suppose next that a is not null. By Theorem 4.5.4 with $F = \mathbb{Q}$, there are $c \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ such that either $a_n \geq c$ for every $n \geq N$, or $a_n \leq -c$ for every $n \geq N$; fix them. In the first case a is positive, so $x \in P$. In the second case $-a_n \geq c$ for every $n \geq N$ (Proposition 2.4.3 (a), in each case of $\leq$), so $-a$ is positive and $-x = [-a] \in P$.

(i), *at most one.* We exclude the three pairs.

- $x \in P$ and $x = 0_{\mathbb{R}}$. Then $(0)_n$ is a representative of x, so it is positive, since every representative of an element of P is: some $\delta \in \mathbb{Q}_{>0}$ has $0 \geq \delta$. This contradicts $\delta > 0$ by (O1) in $\mathbb{Q}$.
- $-x \in P$ and $x = 0_{\mathbb{R}}$. Then $-x = [-(0)_n] = [(0)_n] = 0_{\mathbb{R}}$, and the previous case, applied to $-x$, excludes this.
- $x \in P$ and $-x \in P$. Then a and $-a$ are both positive: $a_n \geq \delta$ for $n \geq N$ and $-a_n \geq \delta'$

for $n \geq N'$, with $\delta, \delta' \in \mathbb{Q}_{>0}$. At $n := \max\{N, N'\}$, $\delta \leq a_n \leq -\delta'$ (Proposition 2.4.3 (a), in each case of $\leq$), so $0 < \delta \leq -\delta' < 0$, against (O1) in $\mathbb{Q}$.

(ii) Let $x, y \in P$, and write $x = [a]$, $y = [b]$. Both a and b are positive: there are $\delta_a, \delta_b \in \mathbb{Q}_{>0}$ and $N_a, N_b \in \mathbb{N}$ with $a_n \geq \delta_a$ for $n \geq N_a$ and $b_n \geq \delta_b$ for $n \geq N_b$. Let $N := \max\{N_a, N_b\}$ and $n \geq N$. Adding the two inequalities (Proposition 2.4.3 (b)),

$$a_n + b_n \geq \delta_a + \delta_b,$$

and $\delta_a + \delta_b > 0 + 0 = 0$ by the same rule. Multiplying, by Proposition 4.1.2 (d) applied to $0 \leq \delta_a \leq a_n$ and $0 \leq \delta_b \leq b_n$,

$$a_n b_n \geq \delta_a \delta_b,$$

and $\delta_a \delta_b > 0$ by (O4). So $a + b$ and ab are positive, with margins $\delta_a + \delta_b$ and $\delta_a \delta_b$ and the threshold N. Hence $x + y = [a + b] \in P$ and $xy = [ab] \in P$ (Definition 5.2.2).

■

5.3.4 The order of the reals

With P in hand the definition is the one Chapters 2 and 3 used for $\mathbb{Z}$ and $\mathbb{Q}$.

Definition 5.3.2. Order on the Reals

For $x, y \in \mathbb{R}$, $x < y$ means $y - x \in P$. As usual, $x \leq y$ means $x < y$ or $x = y$, and $y > x$ means $x < y$.

In terms of representatives: $[a] < [b]$ exactly when $b - a$ is positive, that is, when $b_n - a_n \geq \delta$ from some index on, for some margin $\delta \in \mathbb{Q}_{>0}$. The terms of b must eventually overtake those of a by a fixed rational amount. Since $0_{\mathbb{R}} - x = -x$ and $x - 0_{\mathbb{R}} = x$, we have $x > 0_{\mathbb{R}}$ if and only if $x \in P$, and $x < 0_{\mathbb{R}}$ if and only if $-x \in P$.

Theorem 5.3.3. The Reals Form an Ordered Field

With the operations of Definition 5.2.2 and the order of Definition 5.3.2, $\mathbb{R}$ is an ordered field (Definition 2.4.1), and its positive elements are exactly the elements of P.

Idea. Each order axiom is a statement about one difference, and Theorem 5.3.2 is exactly what Proposition 2.4.1 asks for. (O1) is (i) applied to $y - x$, since $-(y - x) = x - y$. (O2) is closure under sums, since $(z-y)+(y-x) = z-x$. (O3) holds because $(y+z)-(x+z) = y-x$. (O4) is closure under products.

Proof. By Theorem 5.2.3, $\mathbb{R}$ is a field, in particular a commutative ring (Definition 2.3.1). By Theorem 5.3.2, the subset $P \subseteq \mathbb{R}$ satisfies clause (P1) of Proposition 2.4.1 (part (ii)) and clause (P2) (part (i)). That proposition then says that the relation "$y - x \in P$", which is $x < y$ of Definition 5.3.2, makes $\mathbb{R}$ an ordered commutative ring with $P = \{x \in \mathbb{R} : x > 0_{\mathbb{R}}\}$. An ordered commutative ring that is a field is an ordered field (Definition 2.4.1).

■

So $\mathbb{R}$ is an ordered field, and every result that Chapters 2 and 4 proved for an arbitrary ordered field holds in it. We write $\mathbb{R}_{>0} := \{x \in \mathbb{R} : x > 0_{\mathbb{R}}\}$, the set $F_{>0}$ of Chapter 4 for $F = \mathbb{R}$; by the theorem, $\mathbb{R}_{>0} = P$. The rules for inequalities (Proposition 2.4.3), the rules for halves (Proposition 4.1.2), the absolute value (Definition 4.1.1, Proposition 4.1.3), and limits, null sequences and Cauchy sequences with tolerances in $\mathbb{R}_{>0}$ (Definition 4.3.2, Definition 4.5.1) are now available in $\mathbb{R}$. From here on they may be applied with $F = \mathbb{R}$. It pays to name the field at each such use, so that a tolerance in $\mathbb{Q}_{>0}$ is never confused with one in $\mathbb{R}_{>0}$. One consequence deserves a line. In any ordered field, $x \leq y$ holds if and only if $y < x$ fails: by (O1), exactly one of $x < y$, $x = y$, $y < x$ holds.

Example. A Margin Read Off the Limit

Let $a_n := (2n+1)/(n+3)$. Show that $a \in \mathcal{C}$ and $[a] > 1_{\mathbb{R}}$, and find a margin and a threshold.

Solution. Since $2n+1 = 2(n+3)-5$, we have $a_n = 2-5/(n+3)$. The sequence $(1/(n+3))_n$ is the shift by 2 of the null sequence $(1/(n+1))_n$ (Proposition 4.3.2 (b), Example (First Limits) (b)), so a is a constant minus a multiple of a null sequence and lies in $\mathcal{C}$ (Theorem 4.5.3 (a), (b)).

By Definition 5.3.2 we need $a - (1)_n$ to be positive. Its terms are

$$a_n - 1 = \frac{n-2}{n+3} = 1 - \frac{5}{n+3}.$$

Margin first. These terms approach 1, so we aim at half of it, $\delta := 1/2$, before naming any index.

Then the index. Multiplying by $2(n+3) > 0$, or by its reciprocal (Proposition 2.4.3 (c), with equality as the other case),

$$\frac{n-2}{n+3} \geq \frac{1}{2} \iff 2(n-2) \geq n+3 \iff n \geq 7.$$

So $\delta = 1/2$ and $N = 7$ work, and $[a] > 1_{\mathbb{R}}$. No smaller threshold works with this margin, since $a_6 - 1 = 4/9 < 1/2$.

Any margin $\delta < 1$ works with a later threshold; $\delta = 1$ fails, since every $a_n - 1 = 1-5/(n+3) < 1$. The tempting shortcut, "$a_n - 1 > 0$ for $n \geq 3$", is the weaker condition and proves nothing about the class.

5.3.5 Comparing classes through their terms

The order is defined through margins, but in practice we usually know a termwise inequality and want an inequality between classes. The next lemma is the bridge, and the example at the start of the section shows its limit.

Lemma 5.3.4. Termwise Inequalities Pass to Classes

Let $a, b \in \mathcal{C}$ and $N_0 \in \mathbb{N}$. If $a_n \leq b_n$ for every $n \geq N_0$, then $[a] \leq [b]$.

Idea. Suppose instead that $[b] < [a]$. Then $a - b$ is positive, so from some index on a_n exceeds b_n by a fixed margin; one index beyond both thresholds then contradicts $a_n \leq b_n$. By trichotomy, "not $[b] < [a]$" is "$[a] \leq [b]$".

Proof. Suppose that $[b] < [a]$. By Definition 5.3.2, $[a] - [b] = [a - b] \in P$, so $a - b$ is positive, since every representative of an element of P is (Proposition 5.3.1). Fix $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $a_n - b_n \geq \delta$ for every $n \geq N$. Let $n \coloneqq \max\{N, N_0\}$. Then $a_n - b_n \geq \delta > 0$. But $n \geq N_0$, so $a_n \leq b_n$, that is, $a_n - b_n \leq 0$ (Proposition 2.4.3 (a), in each case of $\leq$). These contradict (O1) in $\mathbb{Q}$. Hence $[b] < [a]$ is false. By (O1) in the ordered field $\mathbb{R}$ (Theorem 5.3.3), $[a] < [b]$ or $[a] = [b]$; that is, $[a] \leq [b]$.

■

Warning. The strict version of the lemma is **false**. In Example (Termwise Order Is Not an Order on Classes), $a_n = 0 < 1/(n+1) = b_n$ for **every** n, and yet $[a] = [b]$, so $[a] < [b]$ fails. A strict termwise inequality passes to classes only as $\leq$. To get $<$ between classes, show that the difference is eventually above a fixed margin.

Most uses of the lemma compare a class with a constant: if $a_n \leq q$ for every $n \geq N_0$, then $[a] \leq [(q)_n]$; and if $a_n \geq q$ for every $n \geq N_0$, then $[a] \geq [(q)_n]$.

Quick check. Let $a_n \coloneqq 1/(n+1) - 1/(n+1)^2$. Every term satisfies $a_n \geq 0$. Is a positive?

Solution. No. Here $a_n = n/(n+1)^2$, so $0 \leq a_n \leq 1/(n+1)$ for every n (as $n \leq n+1$). The lower sequence $(0)_n$ and the upper sequence $(1/(n+1))$ are null, so a is null by the squeeze, Proposition 4.4.4 (c) with $F = \mathbb{Q}$; being null, $a \in \mathcal{C}$ (Theorem 4.5.3 (a)). Then $[a] = 0_{\mathbb{R}}$, as in the proof of Theorem 5.3.2 (i), and $0_{\mathbb{R}} \notin P$ by the same part. So a is not positive, although no term is negative. Lemma 5.3.4 gives only $[a] \geq 0_{\mathbb{R}}$, which is true.

Since $\mathbb{R}$ is an ordered field, every $x \in \mathbb{R}$ has an absolute value $|x|$ (Definition 4.1.1 with $F = \mathbb{R}$); Exercise C1 shows that it can be computed on representatives, $|[a]| = [(|a_n|)_n]$.

5.3.6 Exercises

5.3.6.1 A. Check your understanding

Exercise. A1

In parts (a)–(d), decide for each sequence whether it is in $\mathcal{C}$ and whether it is positive; when it is positive, give a margin δ and a threshold N.

(a) $a_n = 1 - 1/(n+1)$.

(b) $a_n = 1/(n+1)$.

(c) $a_k = d_k - 1$, where (d_k) is the sequence of decimal truncations of Example (The Truncations and the Newton Sequence Are Equivalent).

(d) $a_n = (-1)^n$.

(e) Show from Definition 5.3.2 and Definition 5.3.1 that $1_{\mathbb{R}} > 0_{\mathbb{R}}$. Which result of Chapter 2 gives the same conclusion in every ordered field?

(f) In the proof of Proposition 5.3.1, why is the tolerance for $a' - a$ taken to be $\delta/2$ and not δ? What would the proof give with δ?

Solution.

(a) In $\mathcal{C}$ (a constant minus a null sequence, Theorem 4.5.3 (a), (b)), and positive. For $n \geq 1$, $n+1 \geq 2$, so $1/(n+1) \leq 1/2$ (Proposition 4.1.2 (c)) and $a_n \geq 1/2$. So $\delta = 1/2$, $N = 1$ work. ($N = 0$ fails for this δ: $a_0 = 0$.)

(b) In $\mathcal{C}$, being null, and not positive: this is the non-example after Definition 5.3.1.

(c) In $\mathcal{C}$, as the difference of $(d_k) \in \mathcal{C}$ (Example (The Truncations and the Newton Sequence Are Equivalent)) and a constant, and positive. The first truncations are $d_0 = 1$ and $d_1 = 7/5$ (14 is the largest natural number whose square is at most $2 \cdot 10^2 = 200$), and (d_k) is non-decreasing ((T1), stated before Example (The Truncations and the Newton Sequence Are Equivalent)). So $d_k \geq d_1 = 7/5$ for every $k \geq 1$, and $a_k \geq 2/5$. So $\delta = 2/5$, $N = 1$ work. ($N = 0$ fails for every $\delta > 0$: $a_0 = 0$.)

(d) Not in $\mathcal{C}$: for every N, the indices N and $N+1$ give terms differing by 2, so the tolerance 1 has no threshold. The question "positive?" does not arise; the definition applies only to members of $\mathcal{C}$.

(e) $1_{\mathbb{R}} - 0_{\mathbb{R}} = [(1)_n - (0)_n] = [(1)_n]$, and $(1)_n$ is positive with $\delta = 1$, $N = 0$. So $1_{\mathbb{R}} - 0_{\mathbb{R}} \in P$, that is, $0_{\mathbb{R}} < 1_{\mathbb{R}}$. In every ordered commutative ring with $1 \neq 0$, Proposition 2.4.3 (e) gives $1 > 0$; for $\mathbb{R}$, $1_{\mathbb{R}} \neq 0_{\mathbb{R}}$ holds by Theorem 5.2.2.

(f) With the tolerance δ the proof gives only $a'_n > a_n - \delta \geq 0$: terms above 0 with no margin, which is the weaker condition. The tolerance $\delta/2$ leaves the margin $\delta/2$, named before n. Any tolerance strictly between 0 and δ would do.

5.3.6.2 B. Practice

Exercise. B1

Let $a_n = 1 - 1/(n+1)$ and $b_n = x_n - 1$, with (x_n) the Newton sequence of Example (A Rational Sequence Aimed at the Gap). Find margins and thresholds for a and b, and state the margins and threshold that the proof of Theorem 5.3.2 (ii) produces for $a + b$ and ab.

Hint: for b, compare x_n with $7/5$ by squaring.

Solution. The example. For a, Exercise A1 (a) gives $\delta_a = 1/2$, $N_a = 1$. For b: $x_n > 0$ and $x_n^2 > 2$ for every n (Example (A Rational Sequence Aimed at the Gap)), while $(7/5)^2 = 49/25 < 2$. Both $7/5$ and x_n are ≥ 0, so Proposition 2.4.3 (g) turns $(7/5)^2 < x_n^2$ into $7/5 < x_n$. Here b is in $\mathcal{C}$, as the difference of (x_n), which is in $\mathcal{C}$ by Theorem 4.6.1, and a constant (Theorem 4.5.3 (a), (b)). From $7/5 < x_n$, $b_n = x_n - 1 > 2/5$ for every n, and $\delta_b = 2/5$, $N_b = 0$ work. The proof then gives the threshold $N = \max\{1, 0\} = 1$, the margin $1/2 + 2/5 = 9/10$ for $a + b$, and the margin $(1/2)(2/5) = 1/5$ for ab.

Exercise. B2

Let $a, b \in \mathcal{C}$. Prove that $[a] < [b]$ if and only if there are $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $b_n - a_n \geq \delta$ for every $n \geq N$.

Solution. By Definition 5.3.2, $[a] < [b]$ means $[b] - [a] \in P$. By Definition 5.2.2, $[b] - [a] = [b - a]$, where $b - a \in \mathcal{C}$ is the termwise difference (Theorem 4.5.3 (b)). An element of $\mathbb{R}$ lies in P if and only if one (equivalently, every) representative of it is positive, by the definition of P and Proposition 5.3.1. So $[a] < [b]$ holds if and only if $b - a$ is positive, which, by Definition 5.3.1, is the condition stated, since the n-th term of $b - a$ is $b_n - a_n$.

Exercise. B3

(a) Let $a \in \mathcal{C}$ with $0 \leq a_n \leq 1$ for every $n \in \mathbb{N}$. Prove that $0_{\mathbb{R}} \leq [a] \leq 1_{\mathbb{R}}$.

(b) Give an example of $a \in \mathcal{C}$ with $0 < a_n < 1$ for every n and $[a] = 1_{\mathbb{R}}$. What does your example show about part (a)?

Solution.

(a) The constant sequences $(0)_n$ and $(1)_n$ are in $\mathcal{C}$ (Theorem 4.5.3 (a)). By Lemma 5.3.4 with $N_0 = 0$, $(0)_n \leq a$ termwise gives $0_{\mathbb{R}} = [(0)_n] \leq [a]$, and $a \leq (1)_n$ termwise gives $[a] \leq [(1)_n] = 1_{\mathbb{R}}$.

(b) Let $a_n := 1 - 1/(n+2)$. For every n, $n + 2 \geq 2 > 1$, so $0 < 1/(n+2) < 1$ (Proposition 3.2.2 (b)), and $0 < a_n < 1$. The sequence $a - (1)_n = (-1/(n+2))$ is null: $(1/(n+2))$ is the shift by one of the null sequence $(1/(n+1))$ (Proposition 4.3.2 (b)), and the negative of a null sequence is null (Theorem 4.4.3 (b)). So $a = (1)_n + (-1/(n+2))_n$ is in $\mathcal{C}$, as the sum of a constant and a null sequence (Theorem 4.5.3 (a), (b)). Since $a - (1)_n$ is null, $a \sim (1)_n$ and $[a] = 1_{\mathbb{R}}$. The example shows that part (a) cannot be sharpened to strict inequalities: $a_n < 1$ for every n, and still $[a] = 1_{\mathbb{R}}$. This is the warning after Lemma 5.3.4 once more. (Any a with $0 < a_n < 1$ for all n and $1 - a_n$ null is a correct answer.)

Exercise. B4: Prove or disprove

Let $a, b \in \mathcal{C}$. Prove each statement or give a counterexample.

(a) If $a_n > 0$ for every n, then $[a] \geq 0_{\mathbb{R}}$.

(b) If a is positive and $b \sim a$, then $b_n > 0$ for every n.

(c) If $a \not\sim b$, then $a - b$ or $b - a$ is positive.

(d) If a is positive and $b_n \geq a_n - 1/(n+1)$ for every n, then b is positive.

(e) If a and b are positive, then $a - b$ or $b - a$ is positive.

Solution.

(a) **True.** The constant $(0)_n$ is in $\mathcal{C}$ (Theorem 4.5.3 (a)), and $0 \leq a_n$ for every n, so

Lemma 5.3.4 with $N_0 = 0$ gives $0_{\mathbb{R}} = [(0)_n] \leq [a]$.

(b) **False.** Let $b \coloneqq (-1, 1, 1, 1, \ldots)$, which agrees with $(1)_n$ from index 1 on. Then $b \in \mathcal{C}$ and $b \sim (1)_n$ (Example (Finite Changes and Shifts) (a)), and $(1)_n$ is positive, with $\delta = 1$, $N = 0$. But $b_0 = -1$.

(c) **True.** Since $a \not\sim b$, $[a] \neq [b]$ (Sets and Functions, Theorem 5.2.1 (b)), so $[a - b] = [a] - [b] \neq 0_{\mathbb{R}}$. By Theorem 5.3.2 (i), $[a - b] \in P$ or $-[a - b] = [b - a] \in P$. Every representative of an element of P is positive (Proposition 5.3.1), so $a - b$ or $b - a$ is positive.

(d) **True**, margin first. Fix $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $a_n \geq \delta$ for every $n \geq N$, and name the margin $\delta/2 \in \mathbb{Q}_{>0}$ for b. Only then the index: Corollary 3.2.4 gives $K \in \mathbb{Z}_{>0}$ with $1/m < \delta/2$ for every integer $m \geq K$. For $n \geq \max\{N, K\}$, $n + 1 > K$, so

$$b_n \geq a_n - \frac{1}{n+1} > \delta - \frac{\delta}{2} = \frac{\delta}{2}.$$

So b is positive, with margin $\delta/2$ and threshold $\max\{N, K\}$.

(e) **False.** Let $a = b = (1)_n$, both positive. Then $a - b = b - a = (0)_n$, which is not positive: no $\delta \in \mathbb{Q}_{>0}$ has $0 \geq \delta$.

5.3.6.3 C. Going deeper

Exercise. C1, used later

(a) Show that the rule $[a] \mapsto [(|a_n|)_n]$ is a well-defined function $\mathbb{R} \to \mathbb{R}$. Here is the beginning of a proof. *Name the move, then complete it.* "Let $a, a' \in \mathcal{C}$ with $a \sim a'$. The sequences $(|a_n|)$ and $(|a'_n|)$ are in $\mathcal{C}$: by the reverse triangle inequality in $\mathbb{Q}$ (Proposition 4.1.3 (f)), $||a_m| - |a_n|| \leq |a_m - a_n|$, so each has the thresholds of the original. . . ."

(b) Prove that this function is $x \mapsto |x|$: $|[a]| = [(|a_n|)_n]$ for every $a \in \mathcal{C}$.

Solution.

(a) *The move* is **representatives, then independence**: the rule is defined on a representative, so we must show that replacing a by an equivalent a' gives the same output class. Then each class has exactly one output, and the rule is a function (Sets and Functions, Definition 2.1.1).
Completion. Let $e_n \coloneqq |a_n| - |a'_n|$. By the reverse triangle inequality in $\mathbb{Q}$ (Proposition 4.1.3 (f)),

$$0 \leq |e_n| \leq |a_n - a'_n| \quad \text{for every } n \in \mathbb{N}.$$

The sequence $(a_n - a'_n)$ is null, since $a \sim a'$ (Definition 5.1.2); so is $(|a_n - a'_n|)$, because $||a_n - a'_n| - 0| = |a_n - a'_n - 0|$ and the two sequences pass the same tests of Definition 4.3.3. The squeeze Proposition 4.4.4 (c) in $\mathbb{Q}$, with lower sequence $(0)_n$ and upper sequence $(|a_n - a'_n|)$, shows that $(|e_n|)$ is null. Then (e_n) is null, for the same reason as before: $|e_n - 0| = ||e_n| - 0|$. So $(|a_n|) \sim (|a'_n|)$, and $[(|a_n|)_n] = [(|a'_n|)_n]$ (Sets and Functions, Theorem 5.2.1 (b)).
So the output depends only on the class $[a]$, and the rule defines a function $\mathbb{R} \to \mathbb{R}$.

(b) *Equality.* We use three facts. A class $[b]$ with b eventually equal to b' equals $[b']$ (Example (Finite Changes and Shifts)). A null sequence represents $0_{\mathbb{R}}$ (proof of Theorem 5.3.2 (i)). And, in the ordered field $\mathbb{R}$, $|x| = x$ for $x > 0_{\mathbb{R}}$, $|x| = -x$ for $x < 0_{\mathbb{R}}$, and $|0_{\mathbb{R}}| = 0_{\mathbb{R}}$

(Definition 4.1.1).

Case 1: a is null. Then $[a] = 0_{\mathbb{R}}$, so $|[a]| = 0_{\mathbb{R}}$. The sequence $(|a_n|)$ is null too, since $||a_n| - 0| = |a_n| = |a_n - 0|$ (Proposition 4.1.3 (a) and Definition 4.1.1), so the two sequences pass the same tests of Definition 4.3.3. Hence $[(|a_n|)_n] = 0_{\mathbb{R}} = |[a]|$.

Case 2: a is not null. By Theorem 4.5.4 with $F = \mathbb{Q}$, fix $c \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $a_n \geq c$ for every $n \geq N$, or with $a_n \leq -c$ for every $n \geq N$.

If $a_n \geq c$ for $n \geq N$, then a is positive, so $[a] \in P$ and $[a] > 0_{\mathbb{R}}$, whence $|[a]| = [a]$. For $n \geq N$, $a_n > 0$, so $|a_n| = a_n$. The sequences $(|a_n|)$ and a agree from N on, so $[(|a_n|)_n] = [a] = |[a]|$.

If $a_n \leq -c$ for $n \geq N$, then $-a$ is positive, so $-[a] = [-a] \in P$ and $[a] < 0_{\mathbb{R}}$, whence $|[a]| = -[a] = [-a]$. For $n \geq N$, $a_n < 0$, so $|a_n| = -a_n$. The sequences $(|a_n|)$ and $-a$ agree from N on, so $[(|a_n|)_n] = [-a] = |[a]|$.

Part (b) does **not** say that $|a_n| = a_n$ or $-a_n$ for all n; a representative of a positive number may have negative terms early on, as the late-positivity example shows. It says that the class of $(|a_n|)$ is the right one, and classes do not see early terms.

Marking scheme. Part (a), 10 points. - 2: names the move (representatives, then independence) and states the goal $(|a_n|) \sim (|a'_n|)$. - 3: proves $||a_n| - |a'_n|| \leq |a_n - a'_n|$ for every n, citing the reverse triangle inequality in $\mathbb{Q}$. - 3: from $a \sim a'$ and the inequality of the previous step, shows that $(|a_n| - |a'_n|)_n$ is null. - 2: concludes $[(|a_n|)_n] = [(|a'_n|)_n]$, so each class has one output and the rule is a function $\mathbb{R} \to \mathbb{R}$.

A direct ε–N proof of the third step, with no squeeze, earns its 3 points: given $\varepsilon \in \mathbb{Q}_{>0}$, take N for the null sequence $(a_n - a'_n)$; then $||a_n| - |a'_n|| \leq |a_n - a'_n| < \varepsilon$ for every $n \geq N$. Fatal errors (0 for the step): using $|x|$ on $\mathbb{R}$, or citing part (b); proving only that the two output sequences are Cauchy, without proving them equivalent. Part (b) is not marked.

Exercise. C2: Find the error

Here is an argument for Proposition 5.3.1.

"Let a be positive and $a' \sim a$. Fix $\delta \in \mathbb{Q}_{>0}$ and N with $a_n \geq \delta$ for every $n \geq N$. Since $a' - a$ is null, there is N' with $|a'_n - a_n| < \delta$ for every $n \geq N'$. So, for $n \geq \max\{N, N'\}$, $a'_n > a_n - \delta \geq 0$. Hence $a'_n > 0$ for all large n. Therefore a' is positive."

Is the conclusion true? Does the argument prove it? If not, name the line that fails, say why, give a counterexample to that step, and repair the argument.

Solution (Hint 1). Write out in symbols what "a' is positive" requires, and compare it with what the argument has when it reaches the last line.

Solution (Hint 2). "Positive" asks for one margin $\delta' \in \mathbb{Q}_{>0}$, chosen before n, with $a'_n \geq \delta'$ for all large n. Find the line where the argument should produce such a δ', and ask what it produces instead.

Solution. The conclusion is true: it is Proposition 5.3.1.

The middle lines are correct. With the tolerance δ and the strict inequality $|a'_n - a_n| < \delta$, we get $a'_n > a_n - \delta \geq \delta - \delta = 0$ for every $n \geq \max\{N, N'\}$. So the line "$a'_n > 0$ for all large n" is proved.

The failing line is the last: "Therefore a' is positive." What has been proved is

$$\exists N \in \mathbb{N} \;\; \forall n \geq N \;\; a'_n > 0,$$

while "positive" (Definition 5.3.1) is

$$\exists \delta \in \mathbb{Q}_{>0} \;\; \exists N \in \mathbb{N} \;\; \forall n \geq N \;\; a'_n \geq \delta.$$

To pass from the first to the second, one would take the margin $\delta := a'_n$, which depends on n, and then claim one margin for all n. That swaps a "for every n, there is a margin" for a "there is a margin for every n". It is the analogue, for a margin depending on n, of entry (N2) of Logic and Proof, Example (What is not allowed), where a threshold depending on ε is treated as uniform.

Counterexample to the step. The sequence $(1/(n+1))$ has $a'_n > 0$ for all n, so it satisfies the premise of the last line, and it is not positive (the non-example after Definition 5.3.1). So "eventually > 0" does not imply "positive". The argument reached a true conclusion through a step that is not valid.

Repair. Use the tolerance $\delta/2 \in \mathbb{Q}_{>0}$ (Proposition 4.1.2 (b)) instead of δ. Since $a' - a$ is null, choose N'' with $|a'_n - a_n| < \delta/2$ for every $n \geq N''$. For $n \geq \max\{N, N''\}$,

$$a'_n > a_n - \frac{\delta}{2} \geq \delta - \frac{\delta}{2} = \frac{\delta}{2}.$$

So $a'_n \geq \delta/2$ for every $n \geq \max\{N, N''\}$: one margin, $\delta/2$, fixed before n, and a' is positive. This is the proof of Proposition 5.3.1.

Marking scheme. (10 points) - 2: says the conclusion is true, citing Proposition 5.3.1. - 3: identifies the last line as the failing one and names the quantifier swap (a margin depending on n treated as one margin for all n). - 2: a counterexample to the step: a sequence whose terms are eventually > 0 and which is not positive, such as $(1/(n+1))$. - 3: a repair that produces one margin in $\mathbb{Q}_{>0}$ fixed before n (for instance the tolerance $\delta/2$ above; any tolerance strictly between 0 and δ works equally well). Fatal errors (0 for the step): blaming one of the valid middle lines; a repair whose margin still depends on n.

5.4 The Rationals Inside the Reals

Section 5.3 made $\mathbb{R}$ an ordered field. We built it to enlarge $\mathbb{Q}$, but $\mathbb{Q}$ is not yet inside it: a rational is a class of pairs of integers, a real number a class of sequences. This section places a copy of $\mathbb{Q}$ in $\mathbb{R}$, preserved by the operations and the order; proves that $\mathbb{R}$ is Archimedean, so that Chapter 4's theorems about Archimedean fields apply to it, density first; and meets the number that Chapter 4 promised, a positive real whose square is 2.

Notation is as in Section 5.3: $\mathcal{C}$ is the set of Cauchy sequences of rationals, $[a]$ is the class of $a \in \mathcal{C}$, $(q)_n$ is the constant sequence with value q, $P = \mathbb{R}_{>0}$ is the set of positive reals, and $x < y$

means $y - x \in P$ (Definition 5.3.2). By Theorem 5.3.3, the results of Chapters 2 and 4 for an arbitrary ordered field apply with $F = \mathbb{R}$; we say "in $\mathbb{R}$" where the context does not make the field clear.

5.4.1 A copy of the rationals

Section 5.2 already represented each rational q by the class of its constant sequence, as Chapter 4 said it would. We make that a map.

Definition 5.4.1. The Rationals in the Reals

For $q \in \mathbb{Q}$, let

$$\kappa(q) := [(q)_n] \in \mathbb{R},$$

the class of the constant sequence with value q. This defines a function $\kappa\colon \mathbb{Q} \to \mathbb{R}$.

The right side is in $\mathbb{R}$ because constant sequences are Cauchy (Theorem 4.5.3 (a)), and κ is a function because each q determines one constant sequence and that sequence one class. In this notation $0_{\mathbb{R}} = \kappa(0)$ and $1_{\mathbb{R}} = \kappa(1)$ (Definition 5.2.2). The letter κ is new on purpose: Chapter 2 used ι and j for $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q}$, and Chapter 4 wrote ι_F for the copy of $\mathbb{Q}$ in an ordered field F. Part (d) of the next theorem shows that κ is $\iota_{\mathbb{R}}$.

The theorem says that κ carries everything we know how to do with rationals into $\mathbb{R}$ without change: sums, products, negatives, reciprocals, the order and the absolute value.

Theorem 5.4.1. The Rationals Embed in the Reals

For all $q, s \in \mathbb{Q}$:

(a) $\kappa(q+s) = \kappa(q) + \kappa(s)$, $\kappa(qs) = \kappa(q)\kappa(s)$, $\kappa(0) = 0_{\mathbb{R}}$ and $\kappa(1) = 1_{\mathbb{R}}$;

(b) $q < s$ if and only if $\kappa(q) < \kappa(s)$; hence κ is injective;

(c) $\kappa(-q) = -\kappa(q)$, $\kappa(|q|) = |\kappa(q)|$, and, if $q \neq 0$, $\kappa(q) \neq 0_{\mathbb{R}}$ and $\kappa(1/q) = \kappa(q)^{-1}$;

(d) κ is the map $\iota_{\mathbb{R}}$ of Theorem 4.2.2 for the ordered field $F = \mathbb{R}$; in particular $\kappa(n) = n \cdot 1_{\mathbb{R}}$ for every $n \in \mathbb{N}$ (Definition 4.2.1).

Idea. Constant sequences add, multiply and negate termwise into constant sequences, so (a) and the first identity of (c) are the definitions of the operations. For (b), the difference of two constant sequences is constant, and a constant $s - q > 0$ is its own margin. The rest is algebra in a field: the absolute value by cases, reciprocals by uniqueness of inverses, and (d) by the uniqueness clause of Chapter 4's embedding theorem.

Proof.

(a) By Definition 5.2.2, $\kappa(q)+\kappa(s) = [(q)_n+(s)_n] = [(q+s)_n] = \kappa(q+s)$, since the termwise sum of two constant sequences is the constant sequence of the sum. The product is the same with $\cdot$ in place of $+$. The last two identities are the definitions of $0_{\mathbb{R}}$ and $1_{\mathbb{R}}$.
(c), first identity. By Definition 5.2.2, $-\kappa(q) = [-(q)_n] = [(-q)_n] = \kappa(-q)$.

(b) ($\Rightarrow$) Let $q < s$. By (a) and the first identity of (c), $\kappa(s) - \kappa(q) = \kappa(s) + \kappa(-q) = \kappa(s - q) = [(s - q)_n]$. The constant $s - q$ lies in $\mathbb{Q}_{>0}$ (Proposition 2.4.3 (a)), so the

constant sequence $(s-q)_n$ is positive, with margin $\delta = s-q$ and threshold $N=0$ (Definition 5.3.1). Hence $\kappa(s)-\kappa(q) \in P$, that is, $\kappa(q) < \kappa(s)$.

($\Leftarrow$) Let $\kappa(q) < \kappa(s)$. By trichotomy in $\mathbb{Q}$, $q<s$, $q=s$ or $s<q$. If $q=s$, then $\kappa(q)=\kappa(s)$; if $s<q$, then $\kappa(s)<\kappa(q)$ by ($\Rightarrow$). Either contradicts $\kappa(q)<\kappa(s)$ by (O1) in $\mathbb{R}$ (Theorem 5.3.3). So $q<s$.

Injectivity: if $q \neq s$, then $q<s$ or $s<q$, so $\kappa(q)<\kappa(s)$ or $\kappa(s)<\kappa(q)$, and in either case $\kappa(q)\neq\kappa(s)$ by (O1).

(c), the rest. If $q>0$, then $\kappa(q)>\kappa(0)=0_{\mathbb{R}}$ by (b), so $|\kappa(q)| = \kappa(q) = \kappa(|q|)$ (Definition 4.1.1, in $\mathbb{R}$ and in $\mathbb{Q}$). If $q=0$, both sides are $0_{\mathbb{R}}$. If $q<0$, then $\kappa(q)<0_{\mathbb{R}}$ by (b), so $|\kappa(q)| = -\kappa(q) = \kappa(-q) = \kappa(|q|)$. Now let $q\neq 0$. Then $\kappa(q)\neq\kappa(0)=0_{\mathbb{R}}$ by injectivity, and by (a),

$$\kappa(q)\,\kappa(1/q) = \kappa(q\cdot(1/q)) = \kappa(1) = 1_{\mathbb{R}}.$$

In the field $\mathbb{R}$ (Theorem 5.2.3), the inverse of $\kappa(q)$ is the only element with this property (Lemma 4.1.1 (a)), so $\kappa(1/q)=\kappa(q)^{-1}$.

(d) By Theorem 5.3.3, $\mathbb{R}$ is an ordered field, so Theorem 4.2.2 gives **exactly one** map $\iota_{\mathbb{R}}\colon \mathbb{Q}\to\mathbb{R}$ with $\iota_{\mathbb{R}}(1)=1_{\mathbb{R}}$ that preserves sums and products. By (a), κ is such a map, so $\kappa=\iota_{\mathbb{R}}$. The same theorem states that $\iota_{\mathbb{R}}(n)=n\cdot 1_{\mathbb{R}}$ for every $n\in\mathbb{N}$.

■

By (d), part (b) and the injectivity of κ also follow from Theorem 4.2.2 with $F=\mathbb{R}$; the direct proof of (b) shows a constant sequence as its own margin.

Two more identities follow from (a) and (c):

$$\kappa(q-s) = \kappa(q)+\kappa(-s) = \kappa(q)-\kappa(s),$$
$$\kappa(q/s) = \kappa(q)\,\kappa(1/s) = \kappa(q)/\kappa(s) \quad \text{for } s\neq 0.$$

So any computation with rationals, done in $\mathbb{Q}$ and then moved into $\mathbb{R}$ by κ, gives the same answer as the computation done in $\mathbb{R}$ on the images. Part (b) adds that comparisons agree too.

5.4.2 Archimedes in the reals

Chapter 4 showed that an ordered field need not be Archimedean: in $\mathbb{Q}(t)$ the element t is larger than every $n\cdot 1$ (Example (A Non-Archimedean Ordered Field)). So the Archimedean property of $\mathbb{R}$ has to be proved. It comes from $\mathbb{Q}$, through the fact that a Cauchy sequence is bounded.

Theorem 5.4.2. The Reals Are Archimedean

$\mathbb{R}$ is an Archimedean ordered field (Definition 4.2.2): for every $x\in\mathbb{R}$ there is $n\in\mathbb{N}$ with $x < n\cdot 1_{\mathbb{R}}$.

Idea. A representative of x is a Cauchy sequence of rationals, so all its terms lie below one rational bound M. Archimedes in $\mathbb{Q}$ puts a natural number above M; one more unit gives a fixed margin above every term, and a fixed margin is what positivity asks for (Example (Method: name the margin before the index)).

Proof. Let $x \in \mathbb{R}$, and write $x = [a]$ with $a \in \mathcal{C}$. By Proposition 4.5.2 with $F = \mathbb{Q}$, there is $M \in \mathbb{Q}_{>0}$ with $|a_k| \leq M$ for every $k \in \mathbb{N}$. By Theorem 3.2.3, there is $n \in \mathbb{N}$ with $M < n$. For every k,

$$(n+1) - a_k \geq (n+1) - M = (n - M) + 1 > 1,$$

since $a_k \leq |a_k| \leq M$ (Proposition 4.1.3 (c)), so $-M \leq -a_k$ and then $(n+1)-a_k \geq (n+1)-M$ (Proposition 2.4.3 (a), (b)), and $n - M > 0$. So the sequence $((n+1) - a_k)_k$ is positive, with margin 1 and threshold 0. It represents $\kappa(n+1) - x$, by Definition 5.2.2. Hence $x < \kappa(n+1)$, and $\kappa(n+1) = (n+1) \cdot 1_{\mathbb{R}}$ by Theorem 5.4.1 (d). Since $n + 1 \in \mathbb{N}$, this proves the theorem. ■

Every result of Chapter 4 about Archimedean ordered fields now applies with $F = \mathbb{R}$. Two of them are used at once.

The integer part in $\mathbb{R}$. By Proposition 4.2.4 with $F = \mathbb{R}$, and $\iota_{\mathbb{R}} = \kappa$ (Theorem 5.4.1 (d)), for every $x \in \mathbb{R}$ there is **exactly one** $m \in \mathbb{Z}$ with

$$\kappa(m) \leq x < \kappa(m+1).$$

We call m the **integer part** of x. For a rational q, the integer part of $\kappa(q)$ is the integer part $\lfloor q \rfloor$ of Proposition 3.2.5, because κ preserves and reflects $\leq$ (Theorem 5.4.1 (b)). The word "exactly" matters: an integer part is given by a rule, so a sequence of integer parts can be defined without choosing anything (Exercise B2).

5.4.3 Between two reals there is a rational

Chapter 4 proved that the rationals are dense in every Archimedean ordered field (Theorem 4.2.5), and $\mathbb{R}$ is one.

Theorem 5.4.3. The Rationals Are Dense in the Reals

Let $x, y \in \mathbb{R}$ with $x < y$. Then there is $q \in \mathbb{Q}$ with $x < \kappa(q) < y$.

Proof. By Theorem 5.4.2, $\mathbb{R}$ is an Archimedean ordered field, so Theorem 4.2.5 with $F = \mathbb{R}$ gives $q \in \mathbb{Q}$ with $x < \iota_{\mathbb{R}}(q) < y$; and $\iota_{\mathbb{R}} = \kappa$ by Theorem 5.4.1 (d). ■

Density can also be seen directly from the representatives, with no appeal to Archimedes: write $x = [a]$, $y = [b]$, go far out, and take the midpoint of two late terms. Exercise C3 carries this out; it shows that in this construction the rationals are dense because each real number is made of rationals.

Identification. From here on we write q for $\kappa(q)$ when $q \in \mathbb{Q}$, and $\mathbb{Q} \subseteq \mathbb{R}$. This is a convention about notation, of the kind Chapter 2 adopted twice, writing n for $\iota(n)$ and m for $j(m)$ (Proposition 2.2.3, Proposition 3.1.4). It is safe because of Theorem 5.4.1: κ is injective and preserves sums, products, negatives, reciprocals, the order and the absolute value, so a statement about rationals means the same whether it is read in $\mathbb{Q}$ or, through κ, in $\mathbb{R}$. It is not a set inclusion: the rational q is a class of pairs of integers, and $\kappa(q)$ is a class of sequences of rationals, a

different set. In this notation $\kappa(1/n)$ is written $1/n$, and Proposition 4.2.3 in $\mathbb{R}$ says that for every $\varepsilon \in \mathbb{R}_{>0}$ some $1/n$, $n \in \mathbb{Z}_{>0}$, lies below ε.

5.4.4 The square root of two

Chapter 4 ended with the Newton sequence (x_n) of Example (A Rational Sequence Aimed at the Gap), Cauchy in $\mathbb{Q}$ and with no limit in $\mathbb{Q}$ (Theorem 4.6.1). Its class is the number it was aiming at.

Example. The Class of the Newton Sequence

Let (x_n) be the Newton sequence of Example (A Rational Sequence Aimed at the Gap), and let $r := [(x_n)] \in \mathbb{R}$. Show that $r > 0$, that $r^2 = 2$, that $r \neq q$ for every $q \in \mathbb{Q}$, and that $r = [(d_k)]$, where (d_k) is the sequence of decimal truncations of Example (The Truncations and the Newton Sequence Are Equivalent).

Solution. The sequence (x_n) is in $\mathcal{C}$ by Theorem 4.6.1, so $r \in \mathbb{R}$. We use the facts $x_n > 1$ for every $n \in \mathbb{N}$ and $e_n \leq 1/(4n)$ for every $n \geq 1$, where $e_n := x_n^2 - 2$, stated in Theorem 4.6.1, and $e_n > 0$ for every $n \in \mathbb{N}$, proved in Example (A Rational Sequence Aimed at the Gap). The first values are

$$e_0 = 2, \quad e_1 = \frac{1}{4}, \quad e_2 = \frac{1}{144}, \quad e_3 = \frac{1}{166464}, \quad e_4 = \frac{1}{221682772224}.$$

$r > 0$. Since $x_n > 1$ for every n, the sequence (x_n) is positive, with margin 1 and threshold 0. So $r \in P$, that is, $r > 0$. By Lemma 5.3.4, also $r \geq 1$.

$r^2 = 2$. By Definition 5.2.2, $r^2 = r \cdot r = [(x_n^2)]$, and $2 = [(2)_n]$. Termwise, $(x_n^2) - (2)_n = (e_n)$. For $n \geq 1$,

$$0 < e_n \leq \frac{1}{4n} \leq \frac{1}{n+1},$$

where the last step is Proposition 4.1.2 (c), since $4n - (n+1) = 3n - 1 > 0$. The sequence $(1/(n+1))$ is defined for every $n \in \mathbb{N}$ and null in $\mathbb{Q}$ (Example (First Limits) (b)); so is $(0)_n$. The squeeze Proposition 4.4.4 (c) in $\mathbb{Q}$, with $N_0 = 1$, shows that (e_n) is null. (At $n = 0$ the bound $1/(4n)$ is undefined and $e_0 = 2$; the threshold $N_0 = 1$ skips it.) Hence $(x_n^2) \sim (2)_n$ and $r^2 = 2$.

r is not rational. Suppose $r = \kappa(q)$ for some $q \in \mathbb{Q}$. Then $\kappa(q^2) = \kappa(q)^2 = r^2 = \kappa(2)$ by Theorem 5.4.1 (a), so $q^2 = 2$ by injectivity. This contradicts Theorem 3.3.1. So $r \notin \kappa(\mathbb{Q})$.

$r = [(d_k)]$. By Example (The Truncations and the Newton Sequence Are Equivalent), $(d_k) \sim (x_k)$, so the classes are equal.

So $\mathbb{R}$ contains a positive element whose square is 2, and $\mathbb{Q}$ contains none. It is the number Chapter 4 promised, and the decimal truncations name the same number as the Newton sequence.

Warning. r is **not** "the limit of (x_n) in $\mathbb{Q}$": there is no such limit (Theorem 4.6.1), and nothing in this section changes that. r is a new element, a class of sequences. Whether the x_n converge to it is a question about $\mathbb{R}$, with tolerances in $\mathbb{R}_{>0}$; Exercise B3 answers it.

Quick check. Find a rational number strictly between $[(d_k)]$ and $[(d_k + 1/(k+1))]$.

Solution. There is none, and Theorem 5.4.3 does not apply. The two sequences differ by $(1/(k+1))$, which is null, so they are equivalent and the classes are equal. The hypothesis $x < y$ fails, and no number lies strictly between a number and itself.

5.4.5 Exercises

5.4.5.1 A. Check your understanding

Exercise. A1

(a) Show that $\kappa(q^k) = \kappa(q)^k$ for every $q \in \mathbb{Q}$ and $k \in \mathbb{N}$, the powers being those of Definition 4.2.1 in $\mathbb{Q}$ and in $\mathbb{R}$.

(b) Let r be as in Example (The Class of the Newton Sequence). Find the integer parts of r and of $10r$, and justify each from the terms d_k: use $d_1 = 7/5$, $d_2 = 141/100$, and the bounds (T1) and (T2) stated before Example (The Truncations and the Newton Sequence Are Equivalent).

(c) True or false: if $a \in \mathcal{C}$, $q \in \mathbb{Q}$ and $a_n < q$ for every n, then $[a] < q$. Give a reason or a counterexample.

(d) Is $\kappa\colon \mathbb{Q} \to \mathbb{R}$ surjective?

Solution.

(a) By induction on k (Logic and Proof, Axiom 4.2.1, read on $\mathbb{N}$ through Corollary 1.4.3 (b)), applied to the property "$\kappa(q^k) = \kappa(q)^k$". Base: $\kappa(q^0) = \kappa(1) = 1_{\mathbb{R}} = \kappa(q)^0$, by Definition 4.2.1 (b) in $\mathbb{Q}$ and in $\mathbb{R}$ and Theorem 5.4.1 (a). Step: if $\kappa(q^k) = \kappa(q)^k$, then

$$\kappa(q^{k+1}) = \kappa(q^k q) = \kappa(q^k)\,\kappa(q) = \kappa(q)^k\,\kappa(q) = \kappa(q)^{k+1},$$

by the defining equation of powers, Theorem 5.4.1 (a), and the induction hypothesis. In particular $\kappa(q^2) = \kappa(q)^2$, the step used in Example (The Class of the Newton Sequence).

(b) By (T1) and (T2), stated before Example (The Truncations and the Newton Sequence Are Equivalent), (d_k) is non-decreasing, and $d_j < d_k + 1/10^k$ whenever $j \geq k$. With $k = 2$: for every $j \geq 2$,

$$\frac{7}{5} = d_1 \leq d_j < d_2 + \frac{1}{100} = \frac{142}{100} = \frac{71}{50}.$$

By Lemma 5.3.4 with $N_0 = 2$, and $r = [(d_k)]$ (Example (The Class of the Newton Sequence)), $7/5 \leq r \leq 71/50$. Since $1 \leq 7/5$ and $71/50 < 2$, $1 \leq r < 2$, so the integer part of r is 1. Multiplying by $10 > 0$ in $\mathbb{R}$ (Proposition 2.4.3 (c), with equality allowed), $14 \leq 10r \leq 71/5 < 15$, so the integer part of $10r$ is 14.

(c) False. Take $a_n := q - 1/(n+1)$. Here $a \in \mathcal{C}$: $(q)_n$ is constant and $(1/(n+1))$ is null (Example (First Limits) (b)), so both are Cauchy (Theorem 4.5.3 (a)), and so is their difference (Theorem 4.5.3 (b)). Then $a_n < q$ for every n, and $a - (q)_n = (-1/(n+1))$

is null (Theorem 4.4.3 (b)), so $[a] = \kappa(q)$, and $[a] < \kappa(q)$ fails. The correct conclusion is $[a] \leq q$ (Lemma 5.3.4); see the warning after it.

(d) No. By Example (The Class of the Newton Sequence), $r \in \mathbb{R}$ is not $\kappa(q)$ for any $q \in \mathbb{Q}$.

5.4.5.2 B. Practice

Exercise. B1, used later: Rational tolerances suffice

Prove:

(a) For every $\varepsilon \in \mathbb{R}_{>0}$ there is $N \in \mathbb{Z}_{>0}$ with $\kappa(1/n) < \varepsilon$ for every $n \in \mathbb{N}$ with $n \geq N$.

(b) Let $(x_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}$ and $x \in \mathbb{R}$. Then $x_k \to x$ in $\mathbb{R}$ (Definition 4.3.2 with $F = \mathbb{R}$) if and only if, **for every** $n \in \mathbb{Z}_{>0}$, **there is** $K \in \mathbb{N}$ with $|x_k - x| < \kappa(1/n)$ **for every** $k \geq K$. Likewise, (x_k) is Cauchy in $\mathbb{R}$ (Definition 4.5.1 with $F = \mathbb{R}$) if and only if, for every $n \in \mathbb{Z}_{>0}$, there is $K \in \mathbb{N}$ with $|x_j - x_k| < \kappa(1/n)$ for all $j, k \geq K$.

Solution.

(a) Let $\varepsilon \in \mathbb{R}_{>0}$. Since $\mathbb{R}$ is Archimedean (Theorem 5.4.2), Proposition 4.2.3 in $\mathbb{R}$ gives $N \in \mathbb{Z}_{>0}$ with $1/(n \cdot 1_{\mathbb{R}}) < \varepsilon$ for every $n \geq N$. For such n, $n \neq 0$, $n \cdot 1_{\mathbb{R}} = \kappa(n)$ (Theorem 5.4.1 (d)) and $1/\kappa(n) = \kappa(1/n)$ (Theorem 5.4.1 (c)). So $\kappa(1/n) < \varepsilon$.

(b) ($\Rightarrow$) For $n \in \mathbb{Z}_{>0}$, $1/n > 0$ in $\mathbb{Q}$ (Proposition 3.2.2 (a)), so $\kappa(1/n) > \kappa(0) = 0_{\mathbb{R}}$ (Theorem 5.4.1 (b)): it is one of the tolerances of the definition, which then supplies K.

($\Leftarrow$) Let $\varepsilon \in \mathbb{R}_{>0}$. By (a), there is $N \in \mathbb{Z}_{>0}$ with $\kappa(1/N) < \varepsilon$. The hypothesis with $n := N$ gives K with $|x_k - x| < \kappa(1/N)$ for every $k \geq K$, and then $|x_k - x| < \varepsilon$ by (O2). So K serves for ε. The Cauchy version is the same argument with $|x_j - x_k|$ for $j, k \geq K$ in place of $|x_k - x|$.

In practice: to prove a limit in $\mathbb{R}$, fix $n \in \mathbb{Z}_{>0}$ and find a threshold for the rational tolerance $1/n$. The threshold may depend on n, and on nothing chosen later.

Exercise. B2, used later

Let $x \in \mathbb{R}$. For $k \in \mathbb{N}$, let m_k be the integer part of $(k+1)x$, and $q_k := m_k/(k+1) \in \mathbb{Q}$. Prove that $q_k \leq x < q_k + 1/(k+1)$ for every k, and that $q_k \to x$ in $\mathbb{R}$.

Solution. Fix $k \in \mathbb{N}$. By the definition of the integer part (after Theorem 5.4.2), $m_k \leq (k+1)x < m_k + 1$. Since $k+1 > 0$, also $1/(k+1) > 0$ (Proposition 3.2.2 (a)), and multiplying by it in $\mathbb{R}$ preserves both inequalities (Proposition 2.4.3 (c), with equality allowed on the left):

$$q_k = \frac{m_k}{k+1} \leq x < \frac{m_k + 1}{k+1} = q_k + \frac{1}{k+1}.$$

So $0 \leq x - q_k < 1/(k+1)$ for every k. In $\mathbb{R}$, the sequence $(1/(k+1))$, that is $(1/((k+1) \cdot 1_{\mathbb{R}}))$ by Theorem 5.4.1 (c), (d), is null, by Proposition 4.3.1 and Theorem 5.4.2. The squeeze Proposition 4.4.4 (c) in $\mathbb{R}$, with lower sequence (0), gives that $(x - q_k)$ is null. Then $(q_k - x)$ is null (Theorem 4.4.3 (b)), so $q_k \to x$ (Proposition 4.3.3).

Each q_k is given by a rule: the integer part of $(k+1)x$ is the **only** integer with the defining property, so nothing is chosen.

Exercise. B3, used later: Put the proof in order

Let $a \in \mathcal{C}$. We prove that $\kappa(a_k) \to [a]$ in $\mathbb{R}$: **a Cauchy sequence of rationals converges in $\mathbb{R}$ to its own class.** The lines below, written with the identification (q for $\kappa(q)$), contain the proof, shuffled, together with two lines that do not belong. Put the correct lines in order, and say what is wrong with each of the other two.

(i) so $|[a] - a_k| = [(|a_m - a_k|)_m]$, by Exercise (C1, used later) (b);
(ii) since $a \in \mathcal{C}$, fix N with $|a_m - a_k| < 1/(2n)$ for all $m, k \geq N$;
(iii) its terms are $< 1/(2n)$ for $m \geq N$, so $|[a] - a_k| < 1/(2n)$;
(iv) so $|[a] - a_k| \leq 1/(2n) < 1/n$, and $K = N$ serves;
(v) let $k \geq N$;
(vi) let $n \in \mathbb{Z}_{>0}$; by Exercise (B1, used later: Rational tolerances suffice) (b), it suffices to find K with $|[a]-a_k| < 1/n$ for every $k \geq K$, since $|a_k-[a]| = |[a]-a_k|$ (Proposition 4.1.3 (b));
(vii) $[a] - a_k$ is represented by $(a_m - a_k)_m$;
(viii) for each k, fix N_k with $|a_m - a_k| < 1/(2n)$ for every $m \geq N_k$;
(ix) its terms are $\leq 1/(2n)$ for $m \geq N$, so $|[a] - a_k| \leq 1/(2n)$, by Lemma 5.3.4.

Solution. Order: (vi), (ii), (v), (vii), (i), (ix), (iv). No other order is admissible: line (ii) introduces N, which line (v) needs; line (v) fixes the k that lines (vii), (i) and (ix) are about; line (i) rewrites the class that line (vii) names; line (ix) bounds that class; and line (iv) concludes from line (ix), so it comes last.

Line (iii) uses the strict version of Lemma 5.3.4, which is false: termwise $<$ passes to classes only as $\leq$ (Example (Termwise Order Is Not an Order on Classes)). Concretely, for $a_m = 1/2 - 1/(m+2)$, $n = 1$ and $k = 0$, every term $|a_m - a_0|$ is $< 1/2$, yet $|[a] - a_0| = 1/2$. That is why the proof aims at $1/(2n)$ and then uses $1/(2n) < 1/n$.

Line (viii) claims a threshold N_k for every k, and the Cauchy property does not supply one for an early k: a_k can be far from all later terms. For $a = (5, 0, 0, \dots) \in \mathcal{C}$, $n = 1$ and $k = 0$, $|a_m - a_0| = 5$ for every $m \geq 1$, so no N_0 exists. The Cauchy property gives $|a_m - a_k| < 1/(2n)$ only when both m and k are at least one N, which is why line (v) takes $k \geq N$. (If line (viii) did hold for every k, the remaining lines would give the bound for every k, so it is the claim that is unjustified, not the dependence on k.)

Why $1/(2n)$ and not $1/n$ directly? The Cauchy property gives $|a_m - a_k| < 1/n$, but Lemma 5.3.4 passes only the non-strict inequality to classes, so we would reach $|[a] - \kappa(a_k)| \leq \kappa(1/n)$, one symbol short of the definition. Halving the tolerance first leaves room for the strict inequality.

So every real number is a limit of rationals, namely of the terms of any of its representatives. Equivalent sequences have the same class, so their terms converge in $\mathbb{R}$ to the same number; by uniqueness of limits (Theorem 4.3.4 in $\mathbb{R}$), that number is the only limit they have. A real number is what all its representatives converge to.

5.4.5.3 C. Going deeper

Exercise. C1

(a) Let $x \in \mathbb{R}$ with $|x| \leq 1/n$ for every $n \in \mathbb{Z}_{>0}$. Prove that $x = 0$.

(b) Where does your proof use that $\mathbb{R}$ is Archimedean? Show that the statement of (a) fails in the ordered field $\mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field).

Solution.

(a) Suppose that $|x| > 0$. Since $\mathbb{R}$ is Archimedean (Theorem 5.4.2), Proposition 4.2.3 in $\mathbb{R}$ with $\varepsilon := |x|$ gives $N \in \mathbb{Z}_{>0}$ with $1/N < |x|$. The hypothesis with $n = N$ gives $|x| \leq 1/N$. These contradict (O1). So $|x| > 0$ is false. Since $|x| \geq 0$ (Proposition 4.1.3 (a)), trichotomy leaves $|x| = 0$, and then $x = 0$ by Proposition 4.1.3 (a).

(b) Proposition 4.2.3 applies to $\mathbb{R}$ because of Theorem 5.4.2. In $\mathbb{Q}(t)$, Example (A Non-Archimedean Ordered Field) shows $0 < 1/t < 1/(n \cdot 1)$ for every $n \geq 1$. So $x := 1/t$ satisfies $|x| = x \leq 1/(n \cdot 1)$ for every $n \geq 1$, and $x \neq 0$. The statement of (a) is a form of the Archimedean property, and it fails without it.

Exercise. C2

Let r be as in Example (The Class of the Newton Sequence). Prove that the set $\{x \in \mathbb{R} : x \notin \kappa(\mathbb{Q})\}$ of irrational real numbers is dense in $\mathbb{R}$: for all $x, y \in \mathbb{R}$ with $x < y$ there is an irrational z with $x < z < y$.

Solution (Hint 1). Use Theorem 5.4.3, and the irrational number r.

Solution (Hint 2). Let $q \in \mathbb{Q}$ with $x < q < y$ (Theorem 5.4.3). How small must r/k be for $q + r/k$ to stay below y?

Solution. Let $x < y$. By Theorem 5.4.3, fix $q \in \mathbb{Q}$ with $x < q < y$. Then $y - q > 0$, and $r > 0$ (Example (The Class of the Newton Sequence)), so $\varepsilon := (y - q)/r \in \mathbb{R}_{>0}$ (Proposition 3.2.2 (a), (O4)). By Proposition 4.2.3 in $\mathbb{R}$ (Archimedean by Theorem 5.4.2), there is $k \in \mathbb{Z}_{>0}$ with $1/k < \varepsilon$. Multiplying by $r > 0$ (Proposition 2.4.3 (c)), $r/k < y - q$. Let $z := q + r/k$. Since $r/k > 0$,

$$x < q < z < q + (y - q) = y.$$

If z were rational, $z = p$ with $p \in \mathbb{Q}$, then $r = k(p - q)$ would be rational, since κ preserves differences and products (the identities after Theorem 5.4.1, and its part (a)); this contradicts Example (The Class of the Newton Sequence). So z is irrational, and lies strictly between x and y.

The same argument works with r replaced by any positive irrational number. Between any two reals there are therefore both rationals and irrationals.

Exercise. C3: Density from the representatives

(a) Let $x = [a]$ and $y = [b]$ with $a, b \in \mathcal{C}$ and $x < y$. Prove, without Theorem 5.4.2 or Theorem 3.2.3, that there is $q \in \mathbb{Q}$ with $x < \kappa(q) < y$.

(b) Which hypothesis does your proof use that the proof of Theorem 5.4.3 in the text does not, and conversely?

Solution (Hint 1). Name the margin first: $b - a$ is positive. Then take the midpoint of a_N and b_N for one large N.

Solution (Hint 2). Fix δ and N_1 with $b_n - a_n \geq \delta$ for $n \geq N_1$, and thresholds N_2, N_3 for a and b at the tolerance $\delta/4$.

Solution.

(a) Write $x = [a]$ and $y = [b]$ with $a, b \in \mathcal{C}$. By Definition 5.3.2, $y - x = [b - a] \in P$, so $b - a$ is positive (Proposition 5.3.1). Fix $\delta \in \mathbb{Q}_{>0}$ and $N_1 \in \mathbb{N}$ with

$$b_n - a_n \geq \delta \quad \text{for every } n \geq N_1.$$

Then $\delta/4 = (\delta/2)/2 \in \mathbb{Q}_{>0}$ (Proposition 4.1.2 (b), twice). Since a and b are Cauchy in $\mathbb{Q}$, fix N_2 with $|a_m - a_n| < \delta/4$ for all $m, n \geq N_2$, and N_3 with $|b_m - b_n| < \delta/4$ for all $m, n \geq N_3$. Let

$$N := \max\{N_1, N_2, N_3\}, \qquad q := \frac{a_N + b_N}{2} \in \mathbb{Q}.$$

Let $n \geq N$. Expanding the right sides shows

$$q - a_n = (a_N - a_n) + \frac{b_N - a_N}{2},$$
$$b_n - q = (b_n - b_N) + \frac{b_N - a_N}{2}.$$

Since $n, N \geq N_2$, $a_N - a_n > -\delta/4$ (Proposition 4.1.3 (d)); since $N \geq N_1$, $b_N - a_N \geq \delta$, and multiplying by $1/2 > 0$ (Proposition 4.1.2 (b) with $\varepsilon = 1$) gives $(b_N - a_N)/2 \geq \delta/2$ (Proposition 2.4.3 (c), with equality allowed). Adding (Proposition 2.4.3 (b)),

$$q - a_n > -\frac{\delta}{4} + \frac{\delta}{2} = \frac{\delta}{4}.$$

In the same way, since $n, N \geq N_3$, $b_n - b_N > -\delta/4$, and $b_n - q > \delta/4$. So the sequences $(q - a_n)_n$ and $(b_n - q)_n$ are positive, each with margin $\delta/4$ and threshold N. By Definition 5.2.2 they represent $\kappa(q) - x$ and $y - \kappa(q)$. Hence $x < \kappa(q)$ and $\kappa(q) < y$. The rational q depends on the representatives a, b and on the constants fixed in the proof; nothing makes it the "best" rational, and a different run of the proof may give a different q.

(b) The proof in the text uses that $\mathbb{R}$ is Archimedean, through Theorem 4.2.5, and knows nothing of representatives. Your proof in (a) uses the Cauchy property of a and b and positivity of $b-a$, and no Archimedean property of $\mathbb{Q}$ or $\mathbb{R}$.

Marking scheme. (10 points) - 2: the margin δ from positivity of $b-a$, named before any index. - 3: Cauchy thresholds at a tolerance small enough for the estimates (relative to δ), and one N beyond every threshold used. - 3: both estimates $q-a_n>\delta/4$ and $b_n-q>\delta/4$ for every $n\geq N$. - 2: the estimates read as positivity of two representatives, of $\kappa(q)-x$ and $y-\kappa(q)$. Fatal errors (0 for the step): a midpoint of a_n and b_n with n varying; q compared only with a_N and b_N, not with every later term.

5.5 Least Upper Bounds

Section 5.4 made $\mathbb{R}$ an Archimedean ordered field with $\mathbb{Q}$ dense in it. *Analysis on the Real Line*, Chapter 1 starts from a property of sets: every non-empty set of real numbers that is bounded above has a least upper bound. In $\mathbb{Q}$ it fails (Theorem 3.3.2). This section names the property for an arbitrary ordered field and proves that the constructed $\mathbb{R}$ has it. The proof halves intervals with rational ends. Their right ends form a Cauchy sequence of rationals, which is a real number by definition, and that number is the least upper bound. The sequence form of completeness, that every Cauchy sequence of reals converges, is proved from this one in *Analysis on the Real Line*, Chapter 4.

Throughout, F is an ordered field (Definition 2.4.1). Its order $\leq$ is a total order (Proposition 4.1.2 (a)), so the words "upper bound", "bounded above" and "supremum" of *Sets and Functions*, Section 5.3 apply to subsets of F (Sets and Functions, Definition 5.3.2, Sets and Functions, Definition 5.3.5). When $F=\mathbb{R}$, we keep the identification of Section 5.4, writing q for $\kappa(q)$.

5.5.1 Complete ordered fields

Section 3.3 found a gap in $\mathbb{Q}$ without any sequence: the set $A=\{x\in\mathbb{Q}:x^2<2\}$ is non-empty and bounded above in $\mathbb{Q}$, and among its rational upper bounds there is no least one (Theorem 3.3.2). The upper bounds crowd down toward a place where $\mathbb{Q}$ has no element. A field without such gaps is one in which every set that could have a least upper bound does have one.

Every non-empty set that is bounded above has a least upper bound.

Definition 5.5.1. Complete Ordered Field

An ordered field F is **complete** if **every non-empty** subset $A\subseteq F$ that is **bounded above** in $(F,\leq)$ (Sets and Functions, Definition 5.3.2) has a supremum in F (Sets and Functions, Definition 5.3.5). In symbols, writing $\mathrm{ub}(u)$ for $\forall a\in A\ \ a\leq u$,

$$\forall A\subseteq F\ \ \big((A\neq\emptyset\wedge\exists u\in F\ \mathrm{ub}(u))\Rightarrow \\ \exists s\in F\ (\mathrm{ub}(s)\wedge\forall u\in F\ (\mathrm{ub}(u)\Rightarrow s\leq u))\big).$$

An ordered field in which every Cauchy sequence (Definition 4.5.1) converges is called **Cauchy-complete**.

In words: the hypothesis asks for a set with at least one element and at least one upper bound in F; the conclusion asks for an element s of F that is an upper bound, clause (S1), and lies

below every upper bound, clause (S2). The supremum need not belong to A.

Well-definedness. A set has at most one supremum, since a supremum is a least element of the set of upper bounds (*Sets and Functions*, Section 5.3, the remark after Sets and Functions, Definition 5.3.5, by Sets and Functions, Proposition 5.3.1). So "the supremum of A", written $\sup A$, names one element when it exists.

Examples.

The negative elements. In every ordered field, $A := \{x \in F : x < 0\}$ has supremum 0. It is non-empty ($-1 \in A$). The element 0 is an upper bound. If an upper bound b had $b < 0$, then $b < b/2 < 0$ (Proposition 4.1.2 (b)), so $b/2 \in A$ lies above b, which is impossible. So every upper bound b satisfies $b \geq 0$, and $\sup A = 0 \notin A$. No completeness was needed: some sets have suprema in every ordered field.

The rationals. $\mathbb{Q}$ is **not** complete: Theorem 3.3.2 exhibits a non-empty subset bounded above with no supremum in $\mathbb{Q}$.

The constructed reals. $\mathbb{R}$ is complete; this is the main theorem of the section.

↻ **Second reading** · *A degenerate case, and non-examples by minimal change*

A singleton. $\sup\{a\} = a$, in every ordered field, since a greatest element is the supremum (the same remark of *Sets and Functions*, Section 5.3). This degenerate case matters below: a set may be bounded above by its own element.

Drop "non-empty": the empty set is bounded above by every element of F, so its upper bounds are all of F, and F has no least element ($x - 1 < x$). Drop "bounded above": F itself has no upper bound ($x < x + 1$), so no supremum. With either clause removed, no ordered field at all would be complete.

Why this definition. It asks about suprema, not maxima: the set of negative elements has no greatest element in any ordered field, since $x < x/2 < 0$ for $x < 0$, and asking for maxima would make no field complete. It asks about sets, not sequences, the form *Analysis on the Real Line*, Chapter 1 takes as its axiom. "Complete" alone always means this property; the sequence property is "Cauchy-complete".

Warning. The supremum of a set is an upper bound **not necessarily in the set**, and "complete" is not "every non-empty bounded set has a **largest** element": the set of negative elements above has supremum 0 and no largest element. A proof that ends "so $\sup A \in A$" needs a reason, such as A being finite.

So $\mathbb{Q}$ is not complete. The rest of the section proves that $\mathbb{R}$ is.

5.5.2 Powers of two in an ordered field

The bisection below halves an interval at each step, so after k steps its length is the first length divided by 2^k. We need 2^k to grow at least as fast as $k + 1$, here in $\mathbb{Q}$; the induction is the one for 10^k in Section 5.1.

Lemma 5.5.1. A Lower Bound for Powers of Two

Let F be an ordered field, and let $(2 \cdot 1_F)^k$ be the power in F of Definition 4.2.1 (b). Then

$$(2 \cdot 1_F)^k \geq (k+1) \cdot 1_F \qquad \text{for every } k \in \mathbb{N}.$$

In particular $(2 \cdot 1_F)^k \geq 1_F > 0$.

Idea. Induction on k. Multiplying by 2 doubles the right side, $(k+1)+(k+1)$, and doubling gains at least 1, which is all the step needs.

Proof. We use Logic and Proof, Axiom 4.2.1, read on $\mathbb{N}$ through Corollary 1.4.3 (b), for $P(k)$: $(2 \cdot 1_F)^k \geq (k+1) \cdot 1_F$.

Base $P(0)$. $(2 \cdot 1_F)^0 = 1_F$ by Definition 4.2.1 (b), and $(0+1) \cdot 1_F = 1 \cdot 1_F = 0 \cdot 1_F + 1_F = 1_F$ by (a) of that definition, since $0 + 1 = 1$ (Definition 1.2.1 (c)). Equality holds.

Step. Assume $P(k)$, and let $y := (k+1) \cdot 1_F$. By Definition 4.2.1 (a) and Definition 1.2.1 (c), $2 \cdot 1_F = 1 \cdot 1_F + 1_F = 1_F + 1_F$, which is positive (Proposition 4.1.2 (b)). By Definition 4.2.1 (b), $P(k)$, and Proposition 2.4.3 (c) (or equality, if $P(k)$ holds with equality),

$$(2 \cdot 1_F)^{k+1} = (2 \cdot 1_F)^k (2 \cdot 1_F) \geq y(1_F + 1_F) = y + y,$$

the last step by distributivity (F9) and (F7). Since $k + 1 \geq 1$ (Theorem 1.3.2 (d)), Proposition 4.2.1 (b) gives $y \geq 1_F$, so $y + y \geq y + 1_F$ (Proposition 2.4.3 (b)). Finally $y + 1_F = ((k+1)+1) \cdot 1_F$ by Definition 4.2.1 (a). Hence $(2 \cdot 1_F)^{k+1} \geq ((k+1)+1) \cdot 1_F$, which is $P(k+1)$.

The last sentence of the lemma follows, since $(k+1) \cdot 1_F \geq 1_F > 0$ (Proposition 4.2.1 (b), as $k + 1 \geq 1$).

■

From here on, 2^k means $(2 \cdot 1_F)^k$. By Proposition 4.2.1 (c) it equals $(2^k) \cdot 1_F$, the multiple of 1_F by the natural number 2^k, so the two readings agree; the proofs below use only the power in F and the lemma. Since $2^k > 0$, the quotient $D/2^k$ is defined for every $D \in F$.

5.5.3 Bisection with rational ends

Section 3.3 found the gap of $\mathbb{Q}$ by improving each rational candidate. In $\mathbb{R}$ we trap the candidate instead. Start with a rational that is an upper bound of A and one that is not, and halve: keep the half whose right end is still an upper bound and whose left end is still not. The ends stay rational, and the intervals shrink, so the right ends form a Cauchy sequence of rationals: by definition, a real number. *Before reading on: which two properties of that number are needed, and which end of the intervals gives each?*

Theorem 5.5.2. The Constructed Reals Form a Complete Ordered Field

The set $\mathbb{R}$ of Definition 5.2.1, with the operations of Definition 5.2.2 and the order of Defi-

nition 5.3.2, is a complete ordered field. The map $\kappa\colon \mathbb{Q} \to \mathbb{R}$ of Definition 5.4.1 is injective, preserves $+$, $\cdot$ and $<$, and has dense image: between any two real numbers $x < y$ there is $\kappa(q)$ with $q \in \mathbb{Q}$.

Idea (Scratch work). A real number is a class of members of $\mathcal{C}$, so the supremum should arrive as the class of a Cauchy sequence of rationals. The bisection supplies two: right ends u_k, each an upper bound of A, and left ends l_k, none an upper bound, with the gap halved at each step. The invariants are proved by induction, finding the claim for k inside the claim for $k+1$ (Sets and Functions, Example (Method: finding $P(n)$ inside $P(n+1)$)).

Givens	Goal
$A \subseteq \mathbb{R}$ non-empty, bounded above	$s \in \mathbb{R}$ with (S1) and (S2)
a real number is a class $[(u_k)]$, $(u_k) \in \mathcal{C}$	rationals u_k, upper bounds, and l_k, not upper bounds, closing in
a step rule on the midpoint; induction keeps (I1)–(I3), $u_k - l_k = D/2^k$	$(u_k), (l_k) \in \mathcal{C}$, $(l_k) \sim (u_k)$: $s \coloneqq [(u_k)] = [(l_k)]$
$s < a \in A$: a rational q between; $q - u_k$ eventually $\geq \delta$	some $u_k < a$, against (I1): (S1)
$b < s$, b an upper bound: a rational q between; $l_k - q$ eventually $\geq \delta$	some $l_k > b$, against (I2): (S2)

Both clauses of the supremum are proved by contradiction, and each spends one invariant.

Proof. $\mathbb{R}$ is an ordered field by Theorem 5.3.3. Let $A \subseteq \mathbb{R}$ be non-empty and bounded above.

Step 1: rational ends. Fix $a_0 \in A$ and an upper bound M of A. By Theorem 5.4.3, fix $l_0, u_0 \in \mathbb{Q}$ with $a_0 - 1 < l_0 < a_0$ and $M < u_0 < M + 1$. Then l_0 is not an upper bound, since $a_0 \in A$ and $a_0 > l_0$; u_0 is one, since $a \leq M < u_0$ for every $a \in A$; and $D \coloneqq u_0 - l_0 \in \mathbb{Q}_{>0}$, since $l_0 < a_0 \leq M < u_0$. Each of these is one object, fixed once.

Step 2: the bisection. Define $g\colon \mathbb{N} \times (\mathbb{Q} \times \mathbb{Q}) \to \mathbb{Q} \times \mathbb{Q}$ by

$$g(k,(l,u)) \coloneqq \begin{cases} (l,c) & \text{if } c \text{ is an upper bound of } A, \\ (c,u) & \text{otherwise,} \end{cases} \qquad c \coloneqq \frac{l+u}{2} \in \mathbb{Q}.$$

For each input exactly one case applies, so g is a function, given by a rule. By Sets and Functions, Theorem 3.3.1 (read on $\mathbb{N}$ through Corollary 1.4.3 (b)), with $X = \mathbb{Q} \times \mathbb{Q}$ and starting value (l_0, u_0), there is exactly one sequence $k \mapsto (l_k, u_k)$ with $(l_{k+1}, u_{k+1}) = g(k,(l_k,u_k))$ for every k.

We prove by induction on k that, for every $k \in \mathbb{N}$:

(I1) u_k is an upper bound of A;

(I2) l_k is not an upper bound of A;

(I3) $u_k - l_k = D/2^k$, in $\mathbb{Q}$.

The base is Step 1, with $D/2^0 = D/1 = D$. Suppose (I1)–(I3) hold for k, and let $c \coloneqq (l_k + u_k)/2$. Then $c - l_k = u_k - c = (u_k - l_k)/2 = D/(2^k \cdot 2) = D/2^{k+1}$, by field arithmetic in $\mathbb{Q}$

(Lemma 4.1.1 (b)) and Definition 4.2.1 (b). If c is an upper bound, then $(l_{k+1}, u_{k+1}) = (l_k, c)$: the right end c is an upper bound, the left end l_k is not, by (I2), and $c - l_k = D/2^{k+1}$. Otherwise $(l_{k+1}, u_{k+1}) = (c, u_k)$: the left end c is not an upper bound, the right end u_k is, by (I1), and $u_k - c = D/2^{k+1}$. This proves (I1)–(I3) for $k+1$.

By (I3), $u_k - l_k = D/2^k > 0$, so $l_k < c < u_k$ (Proposition 4.1.2 (b)); in either case $l_k \le l_{k+1}$ and $u_{k+1} \le u_k$. By induction on i, $l_k \le l_{k+i}$ and $u_{k+i} \le u_k$ for every $i \in \mathbb{N}$, and every $j \ge k$ is $k+i$ for some i (Proposition 1.3.1 (a)). So for $j \ge k$, $l_k \le l_j < u_j \le u_k$, and therefore

$$\begin{aligned} 0 \le u_k - u_j \le \frac{D}{2^k}, \\ 0 \le l_j - l_k \le \frac{D}{2^k} \end{aligned} \qquad (j \ge k). \tag{5.5.1}$$

Step 3: a real number. Let $\varepsilon \in \mathbb{Q}_{>0}$. By Theorem 3.2.3 there is $n \in \mathbb{N}$ with $D/\varepsilon < n$. Let $k \ge n$. By Lemma 5.5.1 with $F = \mathbb{Q}$ and Proposition 4.2.1 (b), $2^k \ge k+1 > n > D/\varepsilon > 0$, so $1/2^k < \varepsilon/D$ (Proposition 3.2.2 (b)), and multiplying by $D > 0$ (Proposition 2.4.3 (c)) gives $D/2^k < \varepsilon$. So $(D/2^k)_k$ is null in $\mathbb{Q}$. For $j, k \ge n$, say $j \ge k$, Eq. 5.5.1 gives $|u_j - u_k| = u_k - u_j \le D/2^k < \varepsilon$ (Proposition 4.1.3 (b), Definition 4.1.1); so $(u_k) \in \mathcal{C}$. Then $(l_k) = (u_k - D/2^k)_k \in \mathcal{C}$ (Theorem 4.5.3 (a), (b)), and $(u_k - l_k)_k = (D/2^k)_k$ is null, so $(u_k) \sim (l_k)$ (Definition 5.1.2). Let $s := [(u_k)]$; then $s = [(l_k)]$ (Sets and Functions, Theorem 5.2.1 (b)).

Step 4: (S1). Let $a \in A$, and suppose $s < a$. By Theorem 5.4.3, fix $q \in \mathbb{Q}$ with $s < q < a$. By Definition 5.2.2, $q - s = [(q - u_k)_k]$, and $q - s \in P$ (Definition 5.3.2); so $(q - u_k)_k$ is positive, since every representative of an element of P is (Proposition 5.3.1). Fix $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $q - u_k \ge \delta$ for every $k \ge N$. Then $u_N \le q - \delta < q$, so $u_N < q < a$ (Theorem 5.4.1 (b)), and $a \le u_N$ fails, against (I1). So $s < a$ is false, and $a \le s$ by (O1). Hence s is an upper bound of A.

Step 5: (S2). Let b be an upper bound of A, and suppose $b < s$. Fix $q \in \mathbb{Q}$ with $b < q < s$. Now read s through the left ends: $s - q = [(l_k - q)_k] \in P$, so, as in Step 4, there are $\delta \in \mathbb{Q}_{>0}$ and $N \in \mathbb{N}$ with $l_k - q \ge \delta$ for every $k \ge N$; in particular $l_N > q > b$. By (I2), l_N is not an upper bound: some $a \in A$ has $a \not\le l_N$ (Logic and Proof, Proposition 6.2.3), that is, $a > l_N$ by totality (Proposition 4.1.2 (a)). Then $a > l_N > b$, against b being an upper bound. So $b < s$ is false, and $s \le b$ by (O1).

By (S1) and (S2), $s = \sup A$ (Sets and Functions, Definition 5.3.5). Since A was an arbitrary non-empty subset bounded above, $\mathbb{R}$ is complete. The statements about κ are Theorem 5.4.1 (a), (b) (injectivity is part of (b)) and Theorem 5.4.3.

■

The answer to the prompt: (S1) is read off the right ends and spends (I1); (S2) is read off the left ends and spends (I2); $(l_k) \sim (u_k)$ is what lets one number be read both ways. Archimedes enters twice, never as an assumption: the Archimedean property of $\mathbb{Q}$ (Chapter 3) makes $D/2^k$ small in Step 3, and density of $\mathbb{Q}$ in $\mathbb{R}$, which came from the Archimedean property of $\mathbb{R}$, supplies the rationals of Steps 1, 4 and 5. One more consequence: for $j \ge k$, $u_j \le u_k$ and $l_j \ge l_k$, so Lemma 5.3.4 gives $l_k \le s \le u_k$ for every k. The number s lies in every interval of the bisection.

Remark. Steps 2–5 use only that l_0 and u_0 are rational, that l_0 is not an upper bound of A and that u_0 is one (which forces $D > 0$); Step 1 merely shows that such a pair exists. So the

bisection may start from any such pair.

Example. Method: bisect and keep the invariant

Template. (1) Start with two points on opposite sides of a test (here: not an upper bound, an upper bound). (2) Define the step by a rule that applies the test to the midpoint and keeps one end on each side; the recursion theorem then builds the sequence with no choice. (3) Prove the invariants and the length $D/2^k$ by induction. (4) Make $D/2^k$ small, by Archimedes, to get Cauchy sequences. (5) Spend one invariant on each half of the conclusion.

When to use it. The point you want is the boundary between two kinds of points, and one test decides the kind of each point.

The usual failure. A step that "takes some point of A above the midpoint" with no rule. A starting left end that may itself be an upper bound ($l_0 = a_0$ for $A = \{a_0\}$). Asserting $1/2^k \to 0$ without Archimedes.

Quick check. Run Step 2 on $A := \{q \in \mathbb{Q} : q^2 < 2\}$, viewed as a subset of $\mathbb{R}$, from $l_0 = 0$, $u_0 = 2$ (0 is not an upper bound, as $1 \in A$; 2 is one). List (l_k, u_k) for $k = 0, 1, 2, 3$. This start was not obtained by Step 1; by the remark after the proof, it is allowed.

Solution. $(l_0, u_0) = (0, 2)$. Two tests decide the midpoints. An element $c > 0$ with $c^2 \geq 2$ is an upper bound of A: if some $q \in A$ had $q > c$, then $q^2 > c^2 \geq 2$ (Proposition 2.4.3 (g)), against $q^2 < 2$. An element exceeded by some member of A is not an upper bound; we use the member $7/5$, which is in A because $(7/5)^2 = 49/25 < 2$.

- $k = 0 \to 1$: $c = 1$; $7/5 \in A$ and $7/5 > 1$, so 1 is not an upper bound: $(l_1, u_1) = (1, 2)$.
- $k = 1 \to 2$: $c = 3/2$; $(3/2)^2 = 9/4 \geq 2$, so $3/2$ is an upper bound: $(l_2, u_2) = (1, 3/2)$.
- $k = 2 \to 3$: $c = 5/4$; $7/5 \in A$ and $7/5 > 5/4$, so $5/4$ is not an upper bound: $(l_3, u_3) = (5/4, 3/2)$.

So the pairs are $(0, 2)$, $(1, 2)$, $(1, 3/2)$, $(5/4, 3/2)$, and $u_3 - l_3 = 1/4 = D/2^3$ with $D = 2$, as (I3) says.

The bisection names $\sup A$ as a class. For this set a second name is already known: Section 5.4 built a real number whose square is 2 from the Newton sequence. The next example shows that the two names agree.

Example. Bisection Finds the Square Root of Two

Let A and (l_k, u_k) be as in the Quick check, and let r be the class of the Newton sequence (Example (The Class of the Newton Sequence)). Show that a rational $c \geq 0$ is an upper bound of A if and only if $c \geq r$. Deduce that $l_k < r \leq u_k$ for every k, and that $[(u_k)] = r$.

Solution. Recall $r > 0$ and $r^2 = 2$ (Example (The Class of the Newton Sequence)).

($\Leftarrow$) Let $c \geq r$, and suppose some $q \in A$ has $q > c$. Then $q > r > 0$, so $q^2 > r^2 = 2$ (Proposition 2.4.3 (g) in $\mathbb{R}$), against $q^2 < 2$. So c is an upper bound.

($\Rightarrow$) Let $0 \leq c < r$. By Theorem 5.4.3 there is $q \in \mathbb{Q}$ with $c < q < r$. Then $0 < q < r$ gives $q^2 < r^2 = 2$, so $q \in A$, and $q > c$: c is not an upper bound.

Every end is ≥ 0: $l_0 = 0$, the left ends never decrease, and $u_k > l_k$. So (I1) and (I2) read $r \leq u_k$ and $l_k < r$.

By Steps 3–5 of the proof of Theorem 5.5.2, which apply to this start by the remark after the proof, $[(u_k)] = \sup A$. And $r = \sup A$: it is an upper bound, by the ($\Leftarrow$) argument with c replaced by r; and if $t < r$, density gives $q \in \mathbb{Q}$ with $\max\{t, 0\} < q < r$, so $q \in A$ and $q > t$, and t is not an upper bound. A set has at most one supremum, so $[(u_k)] = r$.

So the bisection and the Newton sequence, two unrelated Cauchy sequences of rationals, name one real number, as the truncations did in Section 5.1.

The proof of Theorem 5.5.2 built two infinite sequences of rationals and chose nothing. The last example says what choosing would have meant.

Example. Picking Representatives Is a Use of Choice

A proof about a sequence (x_k) in $\mathbb{R}$ says: "for each k, pick a representative $a^{(k)} \in x_k$." What does that ask for? Does the proof of Theorem 5.5.2 make such picks?

Solution. Each x_k is a class, a non-empty subset of $\mathcal{C}$ (Sets and Functions, Definition 5.2.2). The family $(x_k)_{k\in\mathbb{N}}$ is a family of non-empty sets indexed by $\mathbb{N}$, and "pick $a^{(k)} \in x_k$ for each k" asks for a function $c\colon \mathbb{N} \to \mathcal{C}$ with $c(k) \in x_k$ for every k: a choice function for the family. Its existence, with no rule given, is what Sets and Functions, Axiom 6.1.1 provides. Fixing one object would give one representative of one x_k; it cannot give infinitely many in one step.

The proof of Theorem 5.5.2 fixes finitely many objects, one at a time: a_0, M, l_0, u_0; in Step 3, for the given ε, one n; and in Steps 4 and 5 one q, δ, N (and in Step 5 one $a \in A$). Each is fixed once and is ordinary logic. The infinitely many ends l_k, u_k are not picked: the recursion theorem defines them from a rule, which applies a test, "is c an upper bound?", to a point it has already computed. So Sets and Functions, Axiom 6.1.1 is not used. A proof that needs infinitely many objects must name the rule that gives them, or say that the axiom of choice is used and where (Sets and Functions, Example (Method: name the choice)).

5.5.4 What we built, and what comes next

The table collects what the chapter proved about $\mathbb{R}$, beside the status of the same property in $\mathbb{Q}$.

Property	In $\mathbb{R}$	In $\mathbb{Q}$
ordered field	yes: Theorem 5.3.3	yes: Theorem 3.2.1
Archimedean	yes: Theorem 5.4.2	yes: Theorem 3.2.3
an element whose square is 2	yes: Example (The Class of the Newton Sequence)	no: Theorem 3.3.1

Property	In $\mathbb{R}$	In $\mathbb{Q}$
complete	yes: Theorem 5.5.2	no: Theorem 3.3.2

The first two rows are shared; the last two are what the construction was for.

Several chapters build on this one. Chapter 6 proves that any two complete ordered fields are isomorphic, by exactly one order-preserving field isomorphism; so "the" real numbers are well defined, and no theorem about them depends on how they were built. *Axiomatic Set Theory* builds a second complete ordered field from Dedekind cuts. *Analysis on the Real Line*, Chapter 1 starts from the axioms of a complete ordered field; the theorem above gives those axioms a model, so they cannot contradict each other unless the theory that supplies $\mathbb{N}$ and sets already does (this volume's Chapter 1 takes $\mathbb{N}$ as a Peano system; *Axiomatic Set Theory* builds it from axioms for sets). *Analytic Preliminaries*, Chapter 5 builds the complex numbers on this $\mathbb{R}$. And *Analysis on the Real Line*, Chapter 5 runs the construction of this chapter for an arbitrary metric space, as its completion.

5.5.5 Exercises

5.5.5.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Theorem 5.5.2 and the definition of a complete ordered field.

(a) Where in the proof is (I1) used, and where (I2)?

(b) Why must l_0 not be an upper bound, and why might a_0 itself be one?

(c) Where does the proof use that $\mathbb{Q}$ is Archimedean, and where that $\mathbb{R}$ is?

(d) State the idea of the proof in two sentences.

(e) Write "F is complete" in symbols, and form its negation.

Solution.

(a) (I1) is used in Step 4, for (S1): a rational q between s and a larger $a \in A$ would put some right end u_N below a. (I2) is used in Step 5, for (S2): a left end l_N above a supposed smaller upper bound b yields an element of A above b. Both are carried by the induction of Step 2, which is what keeps them true.

(b) The left end must **not** be an upper bound, and a_0 might be one: for $A = \{a_0\}$, the element a_0 is an upper bound (even the supremum), and (I2) would fail at $k = 0$. Any $l_0 < a_0$ is safe, since $a_0 \in A$ lies above it, so it is never an upper bound.

(c) That $\mathbb{Q}$ is Archimedean: in Step 3, to make $D/2^k$ smaller than ε (Theorem 3.2.3). That $\mathbb{R}$ is: through density (Theorem 5.4.3, proved from Theorem 5.4.2), which supplies the rationals of Steps 1, 4 and 5.

(d) Halve an interval with rational ends by a rule that keeps the right end an upper bound and the left end not one; the ends form two equivalent Cauchy sequences of rationals, hence one real number s. A rational between s and a larger element of A would undercut a right end, and one between a smaller upper bound and s would be overtaken by a left end.

(e) With $\mathrm{ub}(u)$ for $\forall a \in A\ a \le u$, "complete" is the display in Definition 5.5.1. Its negation (Logic and Proof, Proposition 6.2.1, Logic and Proof, Proposition 6.2.3, Logic and Proof, Proposition 6.2.2, Logic and Proof, Proposition 2.5.5 (d), and $\neg(x \le y)$ as $y < x$ by (O1)):

$$\exists A \subseteq F\ \big(A \neq \emptyset \wedge \exists u \in F\ \mathrm{ub}(u) \wedge \forall s \in F \\ (\exists a \in A\ s < a\ \vee\ \exists u \in F\ (\mathrm{ub}(u) \wedge u < s))\big).$$

Some non-empty set bounded above defeats every candidate s, by an element above it or an upper bound below it.

5.5.5.2 B. Practice

Exercise. B1

Prove that every non-empty subset $A \subseteq \mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$ (Sets and Functions, Definition 5.3.5), and that $\inf A = -\sup(-A)$, where $-A := \{-a : a \in A\}$.

Solution. Let b be a lower bound of A. The set $-A$ is non-empty, and $-b$ is an upper bound of it: for $a \in A$, $b \le a$ gives $-a \le -b$ (Proposition 2.4.3 (a)). By Theorem 5.5.2, $s := \sup(-A)$ exists.

We show that $-s$ is the infimum of A. *Lower bound:* for $a \in A$, $-a \in -A$, so $-a \le s$ and $-s \le a$. *Greatest:* let c be a lower bound of A. As for b, $-c$ is an upper bound of $-A$, so $s \le -c$ by (S2), that is, $c \le -s$. Hence $\inf A = -s = -\sup(-A)$.

Exercise. B2

Let $A := \{1 - 1/(n+1) : n \in \mathbb{N}\} \subseteq \mathbb{R}$. Prove that $\sup A = 1$, and decide whether $1 \in A$.

Solution. Upper bound. $1/(n+1) > 0$ (Proposition 3.2.2 (a)), so $1 - 1/(n+1) < 1$ for every n.

Least. Let $b < 1$. Then $1 - b \in \mathbb{R}_{>0}$, and Proposition 4.2.3 in $\mathbb{R}$ (Theorem 5.4.2) gives $N \in \mathbb{Z}_{>0}$ with $1/N < 1 - b$. With $n := N$, the element $1 - 1/(N+1)$ of A satisfies $1/(N+1) < 1/N < 1 - b$, so $1 - 1/(N+1) > b$, and b is not an upper bound. By (O1), every upper bound u satisfies $u \ge 1$.

So $\sup A = 1$, and $1 \notin A$, since every element of A is < 1. The supremum exists here without Theorem 5.5.2: we exhibited it.

Exercise. B3: Prove or disprove

Let $A \subseteq \mathbb{R}$ be non-empty and bounded above, and let (l_k, u_k) and s be as in the proof of Theorem 5.5.2, from any rational start allowed by the remark after the proof. Prove each true statement and refute each false one with a counterexample.

(a) $a \le u_k$ for every $a \in A$ and every $k \in \mathbb{N}$.

(b) $s \in A$.

(c) If A has a largest element m, then $s = m$.

(d) Some l_k lies in A.

(e) If the step rule is changed to "(c, u) if $c \in A$, and (l, c) otherwise", the new s is still $\sup A$.

Solution.

(a) **True.** This is (I1): u_k is an upper bound of A.

(b) **False.** For $A = \{x \in \mathbb{R} : x < 0\}$, $s = \sup A = 0 \notin A$ (the example of the negative elements after Definition 5.5.1).

(c) **True.** A greatest element is the supremum (*Sets and Functions*, Section 5.3, the remark after Sets and Functions, Definition 5.3.5), so $m = \sup A$, and $s = \sup A$ by the theorem's proof; a set has at most one supremum, so $s = m$.

(d) **False.** For $A = \{0\}$, (I2) says that l_k is not an upper bound of $\{0\}$, that is, $l_k < 0$; so $l_k \notin A$ for every k.

(e) **False.** The changed rule still halves the interval and keeps l_{k+1}, u_{k+1} between l_k and u_k, so Steps 2 and 3 still give a real number $s = [(u_k)]$. Take $A = \{0, 1\}$, with the start $l_0 = -1/2$, $u_0 = 3/2$, one that Step 1 allows for $a_0 = 0$, $M = 1$ ($-1 < -1/2 < 0$, $1 < 3/2 < 2$). The midpoint $c = 1/2$ is not in A, so $(l_1, u_1) = (-1/2, 1/2)$. From then on every end lies in $[l_1, u_1]$, so $u_k \leq 1/2$ for every $k \geq 1$, and the new $s \leq 1/2 < 1 = \sup A$ (Lemma 5.3.4). The test must be the invariant itself.

5.5.5.3 C. Going deeper

Exercise. C1: Exit check: bisection on a dyadic grid

Let $A \subseteq \mathbb{R}$ be non-empty and bounded above. For $k \in \mathbb{N}$, let

$$U_k := \{m \in \mathbb{Z} : m/2^k \text{ is an upper bound of } A\}.$$

(a) Show that U_k has a least element m_k.

(b) Let $u_k := m_k/2^k$. Show that $u_{k+1} \leq u_k$ and that $u_k - 1/2^k$ is not an upper bound of A; deduce that $u_k - 1/2^k < u_j \leq u_k$ for $j \geq k$.

(c) Show that $(u_k) \in \mathcal{C}$, and that $s := [(u_k)]$ is the supremum of A. Name the lines that use an Archimedean property, and of which field.

Solution (Hint 1). Least elements of sets of integers (Chapter 2), and integer parts in $\mathbb{R}$.

Solution (Hint 2). For (a): fix $a_0 \in A$ and an upper bound M, and look at the integer parts of $2^k M$ and $2^k a_0$.

Solution. Integer parts exist in $\mathbb{R}$ by Proposition 4.2.4 with $F = \mathbb{R}$, which is Archimedean (Theorem 5.4.2).

(a) Fix $a_0 \in A$ and an upper bound M. Let p be the integer part of $2^k M$. Then $2^k M < p+1$, so $M < (p+1)/2^k$ (multiply by $1/2^k > 0$), and $(p+1)/2^k$ bounds A: $p+1 \in U_k$, and $U_k \neq \emptyset$. If $m \in U_k$, then $m/2^k \geq a_0$, so $m \geq 2^k a_0 \geq p'$, where p' is the integer part of $2^k a_0$; thus $p' \leq m$ in $\mathbb{Z}$ (Theorem 5.4.1 (b)), and p' is a lower bound of U_k. By Proposition 2.4.4, U_k has a least element m_k.

(b) $2m_k/2^{k+1} = m_k/2^k = u_k$ bounds A, so $2m_k \in U_{k+1}$ and $m_{k+1} \leq 2m_k$, that is, $u_{k+1} \leq u_k$. Since m_k is least, $m_k - 1 \notin U_k$: $(m_k - 1)/2^k = u_k - 1/2^k$ is not an upper bound. For $j \geq k$: $u_j \leq u_k$ by induction (Logic and Proof, Axiom 4.2.1; every $j \geq k$ is $k + i$, Proposition 1.3.1 (a)). And $u_j > u_k - 1/2^k$, since otherwise $u_k - 1/2^k \geq u_j$ would be an upper bound.

(c) By (b), $|u_j - u_k| = u_k - u_j < 1/2^k$ for $j \geq k$, and $(1/2^k)_k$ is null in $\mathbb{Q}$, as in Step 3 of the proof of Theorem 5.5.2 (Archimedes in $\mathbb{Q}$). So $(u_k) \in \mathcal{C}$. With $l_k := u_k - 1/2^k \in \mathbb{Q}$: $(l_k) \in \mathcal{C}$ and $(l_k) \sim (u_k)$, l_k is not an upper bound and u_k is one. Steps 4 and 5 of the proof used nothing else, so they apply word for word, and $s = \sup A$.

Archimedean lines: the integer parts in (a), for $\mathbb{R}$; $1/2^k \to 0$ in (c), for $\mathbb{Q}$; and the rationals q of Steps 4 and 5, reused in (c), come from density (Theorem 5.4.3), that is, from the Archimedean property of $\mathbb{R}$.

Marking scheme. (10 points) - 2: (a), U_k non-empty and bounded below, each with a reason. - 2: (b), $u_{k+1} \leq u_k$, and "$u_k - 1/2^k$ is not an upper bound" from leastness. - 2: $(u_k) \in \mathcal{C}$, with the Archimedean property of $\mathbb{Q}$ named (and that of $\mathbb{R}$ for the integer parts in (a)). - 2: (S1), from (I1) and the fact that s is the class (or limit) of the u_k. - 2: (S2), through $(l_k) \sim (u_k)$. Fatal errors (0 for the step): "choose an upper bound $m/2^k$" with no least element (no rule); $s \in A$ asserted.

Exercise. C2: The rationals are not complete, by bisection

Run the bisection of Step 2 inside $F = \mathbb{Q}$, on $A := \{q \in \mathbb{Q} : q^2 < 2\}$, with "upper bound" meaning upper bound in $\mathbb{Q}$, $(l_0, u_0) = (0, 2)$ and $D = 2$.

(a) Explain why (I1)–(I3) and Step 3 hold in $\mathbb{Q}$, so that (u_k) is Cauchy in $\mathbb{Q}$.

(b) Prove: if A had a supremum s in $\mathbb{Q}$, then $u_k \to s$ in $\mathbb{Q}$.

(c) Prove that (u_k) has no limit in $\mathbb{Q}$. *(Show $l_k^2 < 2 \leq u_k^2$.)*

(d) Conclude that A has no supremum in $\mathbb{Q}$. Why is "(u_k) is Cauchy with no rational limit" not enough on its own for this?

Solution (Hint 1). For (b), compare s with u_k and with l_k, using (I1), (I2), (S1) and (S2). For (c), the left ends are below some element of A.

Solution (Hint 2). For $u_k^2 \geq 2$ in (c): suppose $u_k^2 < 2$, and look for a rational h with $0 < h \leq 1$ and $(u_k + h)^2 < 2$.

Solution.

(a) $1 \in A$ as $1^2 < 2$, so 0 is not an upper bound, and 2 bounds A, as $q > 2 \geq 0$ gives $q^2 > 4$ (Proposition 2.4.3 (g)). By the remark after the proof, this start is allowed. Steps 2 and 3 use only rational arithmetic, Lemma 5.5.1 and Theorem 3.2.3, so (I1)–(I3) hold, $(D/2^k)$ is null in $\mathbb{Q}$, and (u_k) is Cauchy in $\mathbb{Q}$. Steps 4 and 5 need $\mathbb{R}$: the rational between s and a presupposes s.

(b) Suppose $s = \sup A$ in $\mathbb{Q}$. Each u_k is an upper bound (I1), so $s \leq u_k$ by (S2). Each l_k is not an upper bound (I2), so some $a \in A$ has $l_k < a \leq s$. Hence $0 \leq u_k - s < u_k - l_k = D/2^k$ for every k. By the squeeze, Proposition 4.4.4 (c), with lower sequence 0 and the null upper sequence $(D/2^k)$, $u_k - s \to 0$, so $u_k \to s$ (Proposition 4.3.3).

(c) *Left ends.* $l_k \geq l_0 = 0$, since the left ends are non-decreasing (Step 2). By (I2) some $a \in A$ has $a > l_k \geq 0$, so $l_k^2 < a^2 < 2$ (Proposition 2.4.3 (g)).
Right ends. $u_k > l_k \geq 0$. Suppose $u_k^2 < 2$, and let $h := \min\{1, (2 - u_k^2)/(2(2u_k + 1))\}$, a positive rational ($2u_k + 1 > 0$). Since $h \leq 1$, $h^2 \leq h$, and

$$\begin{aligned}(u_k + h)^2 &= u_k^2 + 2u_k h + h^2 \leq u_k^2 + (2u_k + 1)h \\ &\leq u_k^2 + \frac{2 - u_k^2}{2} < 2.\end{aligned}$$

So $u_k + h \in A$ and $u_k + h > u_k$, against (I1). Hence $u_k^2 \geq 2$.
No limit. Suppose $u_k \to L \in \mathbb{Q}$. Then $l_k = u_k - D/2^k \to L$ (Theorem 4.4.3 (b)), and by the product law (c), $u_k^2 \to L^2$ and $l_k^2 \to L^2$. By Proposition 4.4.4 (b), $u_k^2 \geq 2$ gives $L^2 \geq 2$, and $l_k^2 < 2$ gives $L^2 \leq 2$. So $L^2 = 2$, against Theorem 3.3.1.

(d) By (b) and (c), a supremum $s \in \mathbb{Q}$ would be a rational limit of (u_k), and there is none. So A has no supremum in $\mathbb{Q}$: $\mathbb{Q}$ is not complete. A Cauchy sequence with no rational limit shows only that $\mathbb{Q}$ is not **Cauchy**-complete; part (b) is the link between the two properties, and it uses the invariants of the bisection. Section 3.3 proved the same conclusion (Theorem 3.3.2 (a)) directly, by improving each rational candidate; here a sequence finds the gap.

Marking scheme. (10 points) - 2: (a), with the Archimedean property of $\mathbb{Q}$ cited (Step 3), and Steps 4–5 located as needing $s \in \mathbb{R}$ and density. - 3: (b), the two invariants used to get $l_k < s \leq u_k$, and the squeeze. - 3: (c), both bounds $l_k^2 < 2 \leq u_k^2$ justified (the second by a rational h), then the limit laws and Theorem 3.3.1. - 2: (d), the conclusion, with the difference between "not Cauchy-complete" and "not complete" stated. Fatal errors (0 for the step): using $\sqrt{2}$ as a rational or as a real number in (b)–(d); concluding (d) from (c) alone.

5.5.5.4 R. Review

Exercise. R1

Let F be an ordered field, (a_n) a sequence in F, and $L \in F$ with $L > 0$ and $a_n \to L$. Prove that $a_n \geq L/2$ for every sufficiently large n.

Solution. By Definition 4.3.2 with $\varepsilon := L/2 \in F_{>0}$ (Proposition 4.1.2 (b)), there is N with $|a_n - L| < L/2$ for every $n \geq N$. For such n, Proposition 4.1.3 (d) gives $-L/2 < a_n - L$, so $a_n > L - L/2 = L/2$; in particular $a_n \geq L/2$.

This is the second half of Lemma 4.4.2. It is also the shape of Theorem 4.5.4, in the case

where the limit is known: a sequence with a positive limit is eventually above a fixed positive element, which is what "positive" meant for Cauchy sequences in Section 5.3.

Exercise. R2

On $X := \mathbb{Z} \times \mathbb{Z}_{>0}$, let $(p, q) \sim (p', q')$ mean $pq' = p'q$. Show that the condition "$p \geq q$" on a pair gives a well-defined property of classes: if $(p, q) \sim (p', q')$, then $p \geq q$ if and only if $p' \geq q'$. Then show that "$p \geq 2$" does **not** give a well-defined property of classes.

Solution. These are the classes of Definition 3.1.1, and the move is **representatives, then independence**: a property stated on a representative is a property of the class only if equivalent representatives agree. By Theorem 2.2.1 (c), with $S := \{(p, q) \in X : p \geq q\}$, it suffices to check one direction (the relation is symmetric).

Let $pq' = p'q$ with $q, q' > 0$, and suppose $p \geq q$. Multiplying by $q' > 0$ (Proposition 2.4.3 (c), or equality), $pq' \geq qq'$, that is, $p'q \geq q'q$. If $p' < q'$, then $p'q < q'q$ by Proposition 2.4.3 (c), since $q > 0$, a contradiction by (O1). So $p' \geq q'$. In $\mathbb{Q}$, the property says $[(p, q)] \geq 1$, by the comparison formula of Theorem 3.2.1 (a).

"$p \geq 2$" is not well defined: $(1, 2) \sim (2, 4)$, since $1 \cdot 4 = 2 \cdot 2$, but $1 \not\geq 2$ while $2 \geq 2$. One class, two verdicts: the hypothesis of Theorem 2.2.1 (c) fails. (The test "$p \geq 1$" would pass, but only because for an integer it says $p > 0$ (Theorem 2.4.2 (c)), the positivity test of Definition 3.2.1.)

Exercise. R3

Prove that $\mathbb{R}$ is uncountable. For $s \in \{0, 1\}^{\mathbb{N}}$, consider the rationals

$$S_k(s) := \sum_{i=0}^{k} \frac{2s_i}{3^{i+1}} \qquad (k \in \mathbb{N}),$$

and show that $s \mapsto [(S_k(s))_k]$ is an injection $\{0, 1\}^{\mathbb{N}} \to \mathbb{R}$.

Solution. (i) The sums. The powers 3^i are those of Definition 4.2.1 in $\mathbb{Q}$, with $3^{i+1} = 3^i \cdot 3$. The sum is defined by Sets and Functions, Theorem 3.3.1 with $X = \mathbb{Q}$: $S_0(s) := 2s_0/3$ and $S_{k+1}(s) := S_k(s) + 2s_{k+1}/3^{k+2}$.

(ii) A tail bound. For $m \geq k$,

$$0 \leq S_m(s) - S_k(s) \leq \frac{1}{3^{k+1}} - \frac{1}{3^{m+1}}. \tag{5.5.2}$$

By Logic and Proof, Axiom 4.2.1 on $m - k$ (base $P(0)$, $m = k$: $0 \leq 0 \leq 0$): passing from m to $m + 1$ adds $2s_{m+1}/3^{m+2}$, which lies between 0 and $2/3^{m+2}$, and $-1/3^{m+1} + 2/3^{m+2} = -3/3^{m+2} + 2/3^{m+2} = -1/3^{m+2}$.

So $|S_m(s) - S_k(s)| < 1/3^{k+1}$ for $m \geq k$. Also $3^{k+1} \geq 2^{k+1} \geq (k + 2) \cdot 1$: the first by induction

($3^0 = 2^0$, and $3^{i+1} = 3^i \cdot 3 \geq 2^i \cdot 2$ by Proposition 4.1.2 (d), with $2^i > 0$), the second by Lemma 5.5.1 with $F = \mathbb{Q}$. So $1/3^{k+1} < 1/(k+1)$. Given $\varepsilon \in \mathbb{Q}_{>0}$, Corollary 3.2.4 gives $N \in \mathbb{Z}_{>0}$ with $1/n < \varepsilon$ for $n \geq N$; for $m, k \geq N$, say $m \geq k$, $|S_m(s) - S_k(s)| < 1/(k+1) < \varepsilon$. So $(S_k(s))_k \in \mathcal{C}$.

(iii) Different sequences give different classes. Let $s \neq t$. The set $\{i \in \mathbb{N} : s_i \neq t_i\}$ is non-empty and bounded below by 0, so it has a least element j (Proposition 2.4.4). Say $s_j = 1$ and $t_j = 0$. Since $s_i = t_i$ for $i < j$, induction gives $S_{j-1}(s) = S_{j-1}(t)$ when $j \geq 1$, so $S_j(s) - S_j(t) = 2/3^{j+1}$ (also when $j = 0$). For $k \geq j$, by Eq. 5.5.2 applied to s and to t,

$$\begin{aligned} S_k(s) - S_k(t) &= \frac{2}{3^{j+1}} + (S_k(s) - S_j(s)) - (S_k(t) - S_j(t)) \\ &\geq \frac{2}{3^{j+1}} + 0 - \frac{1}{3^{j+1}} = \frac{1}{3^{j+1}}. \end{aligned}$$

So $S(s) - S(t)$ is eventually $\geq \delta := 1/3^{j+1} \in \mathbb{Q}_{>0}$: it is positive (Definition 5.3.1), so $[S(t)] < [S(s)]$ (Definition 5.3.2, Definition 5.2.2), and in particular $[S(s)] \neq [S(t)]$. The map $\Phi\colon s \mapsto [(S_k(s))_k]$ is injective.

(iv) Uncountable. If $\mathbb{R}$ were countable, there would be an injection $g\colon \mathbb{R} \to \mathbb{N}$ (Sets and Functions, Theorem 4.2.2, (i) $\Rightarrow$ (ii)). Then $g \circ \Phi\colon \{0,1\}^{\mathbb{N}} \to \mathbb{N}$ would be injective (Sets and Functions, Theorem 2.2.1 (a)), so $\{0,1\}^{\mathbb{N}}$ would be countable ((ii) $\Rightarrow$ (i)), against Sets and Functions, Theorem 4.3.1.

The method is that of *Sets and Functions*, Chapter 4: an injection from a set known to be uncountable. (The route through Sets and Functions, Proposition 6.1.3 is not taken: its ($\Leftarrow$) direction uses choice.)

Exercise. R4

Let $x \in \mathbb{R}$. Prove that there is a **strictly increasing** sequence (q_k) of rationals with $q_k \to x$ in $\mathbb{R}$.

Solution. The tool is Sets and Functions, Theorem 3.3.1 with a step given by a rule, on the set $X := \{q \in \mathbb{Q} : q < x\}$. Integer parts (Proposition 4.2.4 with $F = \mathbb{R}$, Archimedean by Theorem 5.4.2; unique) supply the rule.

Start. Let p be the integer part of x, and $q_0 := p - 1$. Then $q_0 < p \leq x$, so $q_0 \in X$.

Step. For $k \in \mathbb{N}$ and $q \in X$, let $L := \max\{q, x - 1/(k+1)\}$, a real number with $L < x$. Let n be the integer part of $2/(x - L)$, plus 1: a positive integer (with r the integer part, $0 < 2/(x-L) < r + 1$ gives $r + 1 > 0$, so $r \geq 0$ and $n = r + 1 \geq 1$) with $n > 2/(x - L)$, so $2/n < x - L$ (Proposition 3.2.2 (b)). Let m be the integer part of nx, and put $g(k, q) := (m-1)/n$. From $m \leq nx < m + 1$, dividing by n,

$$L < x - \frac{2}{n} < \frac{m-1}{n} \leq x - \frac{1}{n} < x.$$

So $g(k, q) \in X$, and $g\colon \mathbb{N} \times X \to X$ is a function: each step is a formula, made unambiguous by the uniqueness of integer parts. By Sets and Functions, Theorem 3.3.1 there is exactly one sequence with $q_{k+1} = g(k, q_k)$.

Properties. By the display, $q_{k+1} > L \geq q_k$, so the sequence is strictly increasing; and

$x - 1/(k+1) \le L < q_{k+1} < x$, so $|q_{k+1} - x| < 1/(k+1)$. Given $\varepsilon \in \mathbb{R}_{>0}$, Proposition 4.2.3 in $\mathbb{R}$ (Theorem 5.4.2) gives $N \in \mathbb{Z}_{>0}$ with $1/N < \varepsilon$; for $k \ge N+1$, $|q_k - x| < 1/k \le 1/N < \varepsilon$ (Proposition 4.1.2 (c)). So $q_k \to x$.

Why the look-alikes fail. The terms of a representative of x converge to x (Exercise (B3, used later: Put the proof in order)) but need not increase. The approximants $m_k/(k+1)$ of Exercise (B2, used later) are $\le x$ but not always strictly increasing (for $x = 0$ they are all 0). Density (Theorem 5.4.3) only says that a rational exists; it is not a rule in the data (k, q), so it cannot serve as a recursion step. And "at each step take some rational between L and x" makes one pick for each k with no rule, an unflagged use of Sets and Functions, Axiom 6.1.1 (compare Example (Picking Representatives Is a Use of Choice)).

Chapter 6

The Complete Ordered Field Is Unique

Chapter 5 ended with one complete ordered field, whose elements are classes of Cauchy sequences of rationals. *Axiomatic Set Theory* builds another, whose elements are sets of rationals, and *Analysis on the Real Line*, Chapter 1 builds none: it assumes that a complete ordered field exists and calls it $\mathbb{R}$. Are these the same $\mathbb{R}$? The axioms of an ordered field alone do not decide it. The field $\mathbb{Q}(t)$ of rational functions of Chapter 4, ordered by leading coefficients, satisfies all of them and has an element above every natural number; the field $\mathbb{Q}(\sqrt{2})$ is even Archimedean, and still 3 has no square root in it. This chapter proves that completeness is the property that pins $\mathbb{R}$ down: any two complete ordered fields are linked by an order-preserving field isomorphism, and by only one. So "the" real numbers are well defined, and no theorem about them depends on how they were built.

The tool is one idea. In an Archimedean ordered field an element is determined by **the rationals below it**, and a map that respects sums, products, 1 and the order must send each element to one with the same rationals below it. In a complete field the supremum of the copies of those rationals is such an element. So from an Archimedean field into a complete one there is exactly one order-preserving field map, and each identity it satisfies is checked by comparing rationals below the two sides, with no arithmetic of suprema. When both fields are complete, maps run both ways, and the word "exactly" makes them inverse to each other without a computation. The same uniqueness leaves a complete ordered field no symmetry: a map of it to itself that respects only sums, products and 1 is the identity, because there the positive elements are the non-zero squares. Without the order, a field like $\mathbb{Q}(\sqrt{2})$ does have a symmetry that breaks it; that example belongs to *Algebraic Preliminaries*, Chapter 2, and this chapter only points to it. The chapter, the last of the volume's main line, ends by comparing the two ways to begin analysis, assuming the axioms or building a model, and by listing what *Analysis on the Real Line* takes from this volume.

What you need. Chapters 4 and 5 above all, with Chapters 2 and 3; *Sets and Functions*, Chapters 1, 2, 4 and 5, chiefly Section 5.3; and *Logic and Proof*, Chapters 2 and 6, with Chapters 5 and 7 for one remark. From Chapter 4 of this book: the rules of an ordered field and the absolute value (Proposition 4.1.2, Lemma 4.1.1, Definition 4.1.1, Proposition 4.1.3); powers and the one copy ι_F of $\mathbb{Q}$ in an ordered field, with its uniqueness clause (Definition 4.2.1, Theorem 4.2.2); the Archimedean property, density of $\mathbb{Q}$, and the non-Archimedean field $\mathbb{Q}(t)$ (Definition 4.2.2, Proposition 4.2.3, Theorem 4.2.5, Example (A Non-Archimedean Ordered Field)); and, in Section 6.3 and its exercises, limits and Cauchy sequences (Definition 4.3.2, Example (First Limits), Theorem 4.4.3, Proposition 4.4.4, Definition 4.5.1, Proposition 4.5.1,

Theorem 4.6.1). From Chapter 5: complete ordered fields (Definition 5.5.1); the Cauchy model $\mathbb{R}$, its classes and its order (Definition 5.1.1, Definition 5.1.2, Definition 5.2.1, Theorem 5.3.3, Lemma 5.3.4); the theorems that it is a complete ordered field (Theorem 5.5.2) and Archimedean (Theorem 5.4.2); the map $\kappa = \iota_{\mathbb{R}}$ (Definition 5.4.1, Theorem 5.4.1); and the class of the Newton sequence, a positive real number whose square is 2 (Example (The Class of the Newton Sequence)). From Chapter 3: $\mathbb{Q}$ as classes of pairs (Definition 3.1.1, Proposition 3.1.1), $\mathbb{Q}$ Archimedean (Theorem 3.2.3), reciprocals and the order (Proposition 3.2.2), no rational square equal to 2 (Theorem 3.3.1, Theorem 3.3.2), and the countability of $\mathbb{Q}$ (review exercise R2 of Section 3.3). From Chapter 2: the axioms (F1)–(F9) and (O1)–(O4) (Definition 2.3.1, Definition 2.4.1) and their rules (Proposition 2.3.2, Proposition 2.4.3). Chapter 1 supplies one analogy (Theorem 1.4.1) and, in Exercise C3 of Section 6.3, induction and the order arithmetic of $\mathbb{N}$ (Theorem 1.1.2, Proposition 1.3.1). From *Sets and Functions*: proving uniqueness, Section 1.4 (Sets and Functions, Example (Method: existence and uniqueness)); injective and bijective maps, compositions and two-sided inverses, Sections 2.2 and 2.4 (Sets and Functions, Definition 2.2.1, Sets and Functions, Definition 2.2.3, Sets and Functions, Theorem 2.2.1, Sets and Functions, Theorem 2.4.4, Sets and Functions, Example (Method: construct and verify)); countable sets, Section 4.2, for exercises of Section 6.3 (Sets and Functions, Theorem 4.2.2, Sets and Functions, Proposition 4.2.4); equivalence relations and well-definedness, Section 5.2 (Sets and Functions, Definition 5.2.1, Sets and Functions, Example (Method: checking that a definition on classes is well defined)); and upper bounds and suprema, Section 5.3 (Sets and Functions, Definition 5.3.2, Sets and Functions, Definition 5.3.5). From *Logic and Proof*: the contrapositive, Chapter 2 (Logic and Proof, Proposition 2.3.2); the negation of quantifiers, unbounded and bounded, Chapter 6 (Logic and Proof, Proposition 6.2.1, Logic and Proof, Proposition 6.2.3); and, for the remark on what an isomorphism carries in Section 6.3, first-order formulas and their invariance under isomorphism, Chapters 5 and 7 (Logic and Proof, Definition 5.2.2, Logic and Proof, Theorem 7.5.1).

Time. About 15 hours on the core track (all of A, one B and one C per section, the exit checks); about 25 hours for everything.

6.0.1 Roadmap

- **What Completeness Gives at Once.** Order-preserving field maps, which reflect the order and fix the rationals; the theorem that a complete ordered field is Archimedean, by a supremum of the copy of $\mathbb{N}$ that cannot exist; the rationals below an element, $\mathbb{Q}_F(x)$, which determine x when F is Archimedean (rationals separate); and square roots in a complete ordered field, built as a supremum and unique.
- **Building the Map from the Rationals Below.** The rationals below a sum, and below a product of positive elements, computed from those below each term; and the main theorem in its one-sided form: from an Archimedean ordered field F into a complete one L there is exactly one order-preserving field map, which sends x to the supremum in L of the copies of the rationals below x.
- **Two Complete Ordered Fields, One Isomorphism.** Any two complete ordered fields are isomorphic by exactly one order-preserving field isomorphism, with the inverse obtained from the uniqueness clause instead of computed; every complete ordered field a copy of the Cauchy model, and a Dedekind cut read as a set of rationals below; the automorphisms of a complete ordered field, of which there is one, the identity, even among the maps that respect only sums, products and 1; what an isomorphism carries (limits, as the worked case) and what it does not (a model whose zero is the empty set); and the two ways to begin analysis, with a table of what *Analysis on the Real Line*, Chapter 1 assumes and where this series supplies it. The review exercises return to Chapter 5's real numbers, to well-definedness on $\mathbb{Q}$ as a set of classes of pairs, and to countability and the density of

the elements of a complete ordered field that are not rational.

6.0.2 Named moves

- **Rationals separate** (Lemma 6.1.3). To show $a = b$ in an Archimedean ordered field, show that the same rationals lie below each; otherwise a rational would lie strictly between them. It is proved in Section 6.1 and then proves every equation between elements of the complete field in the main theorem of Section 6.2 (Steps 3, 5, 6 and 7); the exit check C1 of Section 6.2 runs it alone. In $\mathbb{Q}(t)$, where Archimedes fails, t and $t + 1$ have the same rationals below them (the fold in Section 6.1).
- **Construct a point as a supremum** (Example (Method: construct a point as a supremum)). To produce an element you cannot name, take the supremum of a well-chosen set and refute the two wrong cases of trichotomy, one with an element of the set above the supremum, the other with a smaller upper bound. It builds square roots in Section 6.1 and the map of Section 6.2. Exercise B2 of Section 6.2 (cube roots) leaves the last step to you, and Exercise C3 of Section 6.3 (a bounded increasing sequence converges to the supremum of its terms) the last half.
- **Uniqueness gives inverses** (Sets and Functions, Example (Method: existence and uniqueness)). If the only order-preserving field map $F \to F$ is the identity, then two maps $F \to L \to F$ compose to the identity without computing anything. It proves in Section 6.3 that the map between two complete ordered fields is bijective, and the same uniqueness clause shows there that a complete ordered field has no order-preserving automorphism except the identity; once square roots recover the order, also no field automorphism.

6.0.3 Threads

- **Rationals below an element.** The copy ι_F of $\mathbb{Q}$ in an ordered field and density (Chapter 4) are what let Section 6.1 define $\mathbb{Q}_F(x)$; Section 6.2 turns it into a map, and Section 6.3 into the uniqueness theorem.
- **Square roots.** Chapter 3 shows that 2 has no square root in $\mathbb{Q}$, Chapter 5 builds one in $\mathbb{R}$ (Example (The Class of the Newton Sequence)), and Section 6.1 proves that every positive element of a complete ordered field has one; Section 6.3 uses them to recover the order from the arithmetic.
- **Categoricity.** Section 1.4 proved that any two Peano systems are isomorphic by exactly one isomorphism; Section 6.3 proves the same for complete ordered fields, with the same shape: existence by a recursion or a supremum, uniqueness by induction or by rationals separate.
- **Two ways to begin analysis.** Section 1.5 says where $\mathbb{N}$ comes from, Chapter 5 builds $\mathbb{R}$ on it, and Section 6.3 compares that route with assuming the axioms, as *Analysis on the Real Line* does.

6.0.4 Summary

An order-preserving field map between ordered fields fixes the rationals and reflects the order (Definition 6.1.1, Proposition 6.1.1). A complete ordered field is Archimedean (Theorem 6.1.2), and in an Archimedean ordered field an element is determined by the set of rationals below it (Definition 6.1.2, Lemma 6.1.3). Every positive element of a complete ordered field has a square root (Proposition 6.1.4). From an Archimedean ordered field into a complete one there is exactly one order-preserving field map, which sends x to the supremum of the copies of the rationals below x (Theorem 6.2.3). Hence any two complete ordered fields are isomorphic by exactly one order-preserving field isomorphism (Theorem 6.3.1), every complete ordered field is a copy

of the Cauchy model (Corollary 6.3.2), and a complete ordered field has no order-preserving automorphism but the identity (Corollary 6.3.3), nor any field automorphism (Theorem 6.3.4).

What is open: $\mathbb{R}$ is now one structure, and nothing here says anything about its sequences, series or continuous functions. *Analysis on the Real Line* starts there, from the axioms this volume has shown to describe exactly one thing.

6.0.5 Terms

- **order-preserving field map**: a map $\chi\colon F \to L$ of ordered fields that respects sums, products and 1 and is strictly increasing (Definition 6.1.1).
- **rationals below an element**: $\mathbb{Q}_F(x) = \{q \in \mathbb{Q} : \iota_F(q) < x\}$ in an ordered field F (Definition 6.1.2).
- **complete ordered field**: an ordered field in which every non-empty set bounded above has a supremum (Definition 5.5.1).
- **isomorphic complete ordered fields**: linked by an order-preserving field isomorphism, which is unique (Theorem 6.3.1).

6.1 What Completeness Gives at Once

Chapter 5 ended with one complete ordered field, the classes of Cauchy sequences of rationals (Theorem 5.5.2). *Axiomatic Set Theory* builds another from Dedekind cuts, and *Analysis on the Real Line* simply assumes that one exists. This chapter asks whether these are the same field. Before that question can be asked we need a precise meaning for "the same": a map between ordered fields that respects everything. This section fixes that meaning, and then collects what completeness gives at once: the Archimedean property, a way to tell elements apart by the rationals below them, and square roots.

Standing setting. Throughout the chapter, F and L are ordered fields (Definition 2.4.1), so $\leq$ is a total order on each (Proposition 4.1.2 (a)) and upper bounds and suprema in the sense of *Sets and Functions*, Section 5.3 apply to their subsets (Sets and Functions, Definition 5.3.2, Sets and Functions, Definition 5.3.5). "Complete" is always the property of Definition 5.5.1: every non-empty subset that is bounded above has a supremum. We do not redefine it.

Each ordered field F carries its own copy of $\mathbb{Q}$: the map $\iota_F\colon \mathbb{Q} \to F$ of Theorem 4.2.2, the only map with $\iota_F(1) = 1_F$ that preserves sums and products. It is injective, strictly increasing and reflects order, and $\iota_F(n) = n \cdot 1_F$ for $n \in \mathbb{N}$. For the constructed reals, $\iota_{\mathbb{R}}$ is the map κ of Chapter 5 (Theorem 5.4.1 (d)). When only one field is in play, a proof may write q for $\iota_F(q)$, as Chapter 4 did; statements keep ι_F visible, and so does every argument that moves between two fields.

6.1.1 Maps that respect everything

To say that two ordered fields are "the same", we want a dictionary between them: a map that turns sums into sums, products into products, 1 into 1, and smaller into smaller. An isomorphism of Peano systems in Chapter 1 (Theorem 1.4.1) was such a dictionary for 0 and the successor; here the structure is larger.

A map that respects addition, multiplication, the unit and the order.

Definition 6.1.1. Order-Preserving Field Map

Let F and L be ordered fields. A map $\chi\colon F \to L$ is an **order-preserving field map** if, **for all** $a, b \in F$,

$$\chi(a+b) = \chi(a) + \chi(b), \qquad \chi(ab) = \chi(a)\,\chi(b), \qquad \chi(1_F) = 1_L,$$

and $a < b$ implies $\chi(a) < \chi(b)$. An **order-preserving field isomorphism** is an order-preserving field map that is **bijective** (Sets and Functions, Definition 2.2.3). An **order-preserving field automorphism** of F is an order-preserving field isomorphism $F \to F$.

In words: the first two equations say that computing in F and then translating gives the same answer as translating and then computing in L; the third pins down the unit; the last says that χ is **strictly** increasing.

Two consequences come free. A strictly increasing map is injective: if $a \neq b$, then $a < b$ or $b < a$ by (O1), and the images differ in the same way. And it **reflects order**: if $\chi(a) < \chi(b)$, then $a < b$. For $a = b$ would give $\chi(a) = \chi(b)$, and $b < a$ would give $\chi(b) < \chi(a)$; both contradict $\chi(a) < \chi(b)$ by (O1), so only $a < b$ is left. So an order-preserving field map satisfies $a < b \iff \chi(a) < \chi(b)$.

Examples.

The identity. id_F satisfies all four clauses trivially; it is an order-preserving field automorphism of F.

The copy of the rationals. For every ordered field F, $\iota_F\colon \mathbb{Q} \to F$ is an order-preserving field map, by Theorem 4.2.2. It is injective and usually not surjective: $\iota_{\mathbb{R}} = \kappa$ misses the class r of the Newton sequence, which is not $\kappa(q)$ for any rational q (Example (The Class of the Newton Sequence)).

Doubling. On $\mathbb{Q}$, $\chi(x) := 2x$ preserves sums and is strictly increasing, but $\chi(1) = 2 \neq 1$, and $\chi(1 \cdot 1) = 2 \neq 4 = \chi(1)\chi(1)$. It is not a field map: the clauses about multiplication and the unit fail, the additive ones survive.

↻ **Second reading** · *A degenerate case*

The zero map $\chi(a) := 0_L$ preserves sums and products, since $0+0=0$ and $0\cdot 0 = 0$. It fails the unit clause, because $1_L \neq 0_L$ in a field, and it fails strict increase. In Definition 6.1.1 the unit clause in fact follows from the other three: additivity gives $\chi(0) = \chi(0) + \chi(0)$, so $\chi(0) = 0$; strict increase then gives $\chi(1) > \chi(0) = 0$; and $\chi(1) = \chi(1)^2$ with $\chi(1) \neq 0$ forces $\chi(1) = 1$. It is stated anyway because Proposition 6.1.1, Theorem 6.3.4 and the remark after its proof assume sums, products and the unit but no order. Without the unit hypothesis, the zero map would be a counterexample to each of them.

Why this definition. Preserving sums, products and the unit is what makes χ a dictionary for the field operations; strict increase makes it a dictionary for the order. *Algebraic Preliminaries*, Chapter 2 meets a map of $\mathbb{Q}(\sqrt{2})$ to itself that respects sums, products and the unit but sends a positive element to a negative one, so the order clause does not follow from the others in general. Section 6.3 shows that for complete ordered fields it does.

Quick check. Which clauses of Definition 6.1.1 does the map $\chi\colon \mathbb{Q} \to \mathbb{Q}$, $\chi(x) \coloneqq -x$, satisfy?

Solution. Only the sum clause: $-(a+b) = (-a)+(-b)$. The unit clause fails, $\chi(1) = -1 \neq 1$, and so does the product clause, since $\chi(1 \cdot 1) = -1$ while $\chi(1)\chi(1) = (-1)(-1) = 1$; and the order is reversed, $0 < 1$ but $\chi(1) = -1 < 0 = \chi(0)$.

The first result about these maps needs no order hypothesis at all. A map that preserves sums, products and the unit cannot move a rational number.

Proposition 6.1.1. Field Maps Fix the Rationals

Let F and L be ordered fields, and let $\chi\colon F \to L$ satisfy $\chi(a+b) = \chi(a) + \chi(b)$, $\chi(ab) = \chi(a)\chi(b)$ for all $a, b \in F$, and $\chi(1_F) = 1_L$. Then

$$\chi \circ \iota_F = \iota_L, \qquad \text{that is,} \qquad \chi(\iota_F(q)) = \iota_L(q) \quad \text{for every } q \in \mathbb{Q}.$$

Proof. The composite $\chi \circ \iota_F\colon \mathbb{Q} \to L$ sends 1 to $\chi(1_F) = 1_L$, and for $x, y \in \mathbb{Q}$,

$$\chi(\iota_F(x+y)) = \chi(\iota_F(x) + \iota_F(y)) = \chi(\iota_F(x)) + \chi(\iota_F(y)),$$

by Theorem 4.2.2 for F and the hypothesis on χ; products are the same with $\cdot$ in place of $+$. So $\chi \circ \iota_F$ has the three properties that, by the uniqueness clause of Theorem 4.2.2 for L, only ι_L has. Hence $\chi \circ \iota_F = \iota_L$. ∎

This is the remark after Theorem 4.2.2, now with a label. Read with the identifications $q = \iota_F(q)$ and $q = \iota_L(q)$, it says $\chi(q) = q$: such a map **fixes the rationals**. If χ is also order-preserving, more follows. For $q \in \mathbb{Q}$ and $x \in F$, $\iota_F(q) < x$ holds if and only if $\chi(\iota_F(q)) < \chi(x)$, that is, if and only if $\iota_L(q) < \chi(x)$. The rationals below x in F are exactly the rationals below $\chi(x)$ in L. That observation drives the whole chapter.

6.1.2 A complete ordered field is Archimedean

Chapter 4 called F Archimedean if every element lies below some $n \cdot 1_F$ (Definition 4.2.2), and showed that this can fail: in $\mathbb{Q}(t)$ the element t lies above every $n \cdot 1$ (Example (A Non-Archimedean Ordered Field)). Chapter 5 proved it for the constructed $\mathbb{R}$ (Theorem 5.4.2) on the way to completeness. For the supremum form of completeness, no assumption is needed: it implies the Archimedean property. We write $\iota_F(\mathbb{N}) \coloneqq \{\iota_F(n) : n \in \mathbb{N}\} = \{n \cdot 1_F : n \in \mathbb{N}\}$.

Theorem 6.1.2. A Complete Ordered Field Is Archimedean

Let F be a complete ordered field. Then F is Archimedean (Definition 4.2.2).

Idea. If some element bounded $\iota_F(\mathbb{N})$ above, completeness would give a **least** upper bound s. Step down by 1: $s - 1$ is not an upper bound, so some $n \cdot 1_F$ is above it, and one more step, $(n+1) \cdot 1_F$, is above s. The set $\iota_F(\mathbb{N})$ is closed under adding 1, so it cannot have a least upper bound, and hence no upper bound at all.

Proof. Suppose that $\iota_F(\mathbb{N})$ is bounded above. It is non-empty, since $0 \cdot 1_F \in \iota_F(\mathbb{N})$, so by completeness it has a supremum s. Since $s - 1_F < s$, the element $s - 1_F$ is not an upper bound of $\iota_F(\mathbb{N})$: some $n \in \mathbb{N}$ has $n \cdot 1_F > s - 1_F$ (Logic and Proof, Proposition 6.2.3 (a), with totality, Proposition 4.1.2 (a)). Then $(n+1) \cdot 1_F = n \cdot 1_F + 1_F > s$ (Definition 4.2.1 (a) and (O3)), and $(n+1) \cdot 1_F \in \iota_F(\mathbb{N})$, against s being an upper bound. So $\iota_F(\mathbb{N})$ is not bounded above: for each $x \in F$, x is not an upper bound, and as before some $n \in \mathbb{N}$ has $x < n \cdot 1_F$. Hence F is Archimedean (Definition 4.2.2).

■

Both hypotheses of completeness were used: "non-empty" through $0 \cdot 1_F$, "bounded above" through the existence of s. The converse fails: $\mathbb{Q}$ is Archimedean (Theorem 3.2.3) and not complete (Theorem 3.3.2). And dropping completeness can lose the conclusion: $\mathbb{Q}(t)$ is an ordered field that is not Archimedean.

The proof never claimed that s is an element of $\iota_F(\mathbb{N})$; a supremum need not belong to its set. Exercise C2 shows how tempting that claim is.

Quick check. In a complete ordered field F, is $\iota_F(\mathbb{Z})$ bounded below?

Solution. No. If $b \leq \iota_F(k)$ for every $k \in \mathbb{Z}$, then $b \leq \iota_F(-n) = -\iota_F(n)$ for every $n \in \mathbb{N}$, taking $k = -n$ (ι_F preserves sums, hence negatives, Proposition 2.3.2 (a)), so $-b \geq \iota_F(n) = n \cdot 1_F$ (Proposition 2.4.3 (a)); so $-b$ bounds $\iota_F(\mathbb{N})$ above, against Theorem 6.1.2.

Warning. Completeness here is about **subsets**, not sequences. In this chapter "complete" is the supremum property of Definition 5.5.1, and the theorem above shows that it brings the Archimedean property with it. Do not replace it by Cauchy-completeness in the proofs below: a Cauchy-complete ordered field need not be Archimedean, a fact this book does not prove and *Analysis on the Real Line*, Chapter 10 returns to.

Every complete ordered field therefore has the consequences of the Archimedean property that Chapter 4 proved. The one we use most is density (Theorem 4.2.5): in an Archimedean ordered field F, for all $x < y$ there is $q \in \mathbb{Q}$ with $x < \iota_F(q) < y$. We cite it, not reprove it; Section 6.2 uses it in two fields at once.

Here is the Archimedean property at work on a specific supremum. The pattern has two halves: check the candidate is an upper bound, then show that anything smaller is not one.

Example. A Supremum Computed by Archimedes

Let F be a complete ordered field. Show that $\sup\{\iota_F(1-1/n) : n \in \mathbb{Z}_{>0}\} = 1_F$.

Solution. Let $A \coloneqq \{\iota_F(1-1/n) : n \in \mathbb{Z}_{>0}\}$.

Upper bound. For $n \in \mathbb{Z}_{>0}$, $1/n > 0$ in $\mathbb{Q}$, so $1 - 1/n < 1$, and $\iota_F(1-1/n) < \iota_F(1) = 1_F$ because ι_F is strictly increasing (Theorem 4.2.2). So 1_F is an upper bound of A.

Least. Let b be an upper bound of A and suppose $b < 1_F$. By Theorem 6.1.2 and Proposition 4.2.3 (with $\varepsilon \coloneqq 1_F - b > 0$, by Proposition 2.4.3 (a)) some $N \in \mathbb{Z}_{>0}$ has $1/(N \cdot 1_F) < 1_F - b$. Since ι_F preserves sums, hence negatives (Proposition 2.3.2 (a)), and products, hence inverses (Lemma 4.1.1 (a)), and $\iota_F(N) = N \cdot 1_F$,

$$\iota_F(1 - 1/N) = 1_F - \frac{1}{N \cdot 1_F} > b :$$

an element of A above b, which is impossible. So every upper bound is $\geq 1_F$, and $1_F = \sup A$ (Sets and Functions, Definition 5.3.5).

The supremum 1_F is not an element of A: every element is strictly below it. Completeness entered only through the Archimedean property; in an ordered field that is Archimedean but not complete, such as $\mathbb{Q}$ (Theorem 3.2.3), the same two halves give the same answer. Exercises B1 and B2 follow this pattern.

6.1.3 The rationals below an element

How can we tell two elements of a field apart? In an Archimedean field, by a rational between them. That suggests recording, for each element, which rationals lie below it.

An element is measured by the rationals below it.

Definition 6.1.2. The Rationals below an Element

Let F be an ordered field and $x \in F$. The **rationals below** x form the set

$$\mathbb{Q}_F(x) \coloneqq \{q \in \mathbb{Q} : \iota_F(q) < x\} \subseteq \mathbb{Q}.$$

In words: $\mathbb{Q}_F(x)$ is a set of **rational numbers**, not of elements of F; a rational q belongs to it when its copy $\iota_F(q)$ lies **strictly** below x.

Two properties hold in every ordered field and are used without comment. The set is **closed downward**: if $q \in \mathbb{Q}_F(x)$ and $p < q$, then $\iota_F(p) < \iota_F(q) < x$, so $p \in \mathbb{Q}_F(x)$. And if $x \leq y$, then $\mathbb{Q}_F(x) \subseteq \mathbb{Q}_F(y)$, by transitivity.

Examples.

A rational element. For $r \in \mathbb{Q}$, $\mathbb{Q}_F(\iota_F(r)) = \{q \in \mathbb{Q} : q < r\}$. Indeed $\iota_F(q) < \iota_F(r)$ holds if and only if $q < r$, because ι_F is strictly increasing and reflects order. In particular $r \notin \mathbb{Q}_F(\iota_F(r))$: the inequality is strict.

Example. The Rationals below the Square Root of Two

Let $r \in \mathbb{R}$ be the class of the Newton sequence, so that $r > 0$ and $r^2 = 2$ (Example (The Class of the Newton Sequence)). Show that

$$\mathbb{Q}_{\mathbb{R}}(r) = \{q \in \mathbb{Q} : q < 0\} \cup \{q \in \mathbb{Q} : q \geq 0 \text{ and } q^2 < 2\}.$$

Solution. Negative rationals lie below $r > 0$; for $q \geq 0$, compare squares.

↻ **Second reading** · *The computation*

Let $q \in \mathbb{Q}$, and recall $\iota_{\mathbb{R}} = \kappa$ (Theorem 5.4.1 (d)). If $q < 0$, then $\kappa(q) < \kappa(0) = 0 < r$, so $q \in \mathbb{Q}_{\mathbb{R}}(r)$. Let $q \geq 0$, so $\kappa(q) \geq 0$. Since also $r > 0$, Proposition 2.4.3 (g) in $\mathbb{R}$ gives

$$\kappa(q) < r \iff \kappa(q)^2 < r^2 = 2.$$

Now $\kappa(q)^2 = \kappa(q^2)$ and $2 = \kappa(2)$ (Theorem 5.4.1 (a)), and $\kappa(q^2) < \kappa(2)$ if and only if $q^2 < 2$ (Theorem 5.4.1 (b)). So for $q \geq 0$, $q \in \mathbb{Q}_{\mathbb{R}}(r)$ if and only if $q^2 < 2$. Together the two cases give the displayed set.

The set of Section 3.3, $\{x \in \mathbb{Q} : x^2 < 2\}$ (Theorem 3.3.2), is **not** $\mathbb{Q}_{\mathbb{R}}(r)$: it omits -2, whose square is 4. That set is not closed downward, and a rationals-below set always is.

Quick check. Let F be an ordered field and $x := \iota_F(3/2)$. Describe $\mathbb{Q}_F(x)$, and say whether $3/2 \in \mathbb{Q}_F(x)$.

Solution. By the first example, $\mathbb{Q}_F(x) = \{q \in \mathbb{Q} : q < 3/2\}$. It does not contain $3/2$, since $\iota_F(3/2) < \iota_F(3/2)$ is false by (O1).

The point of the definition is that in an Archimedean field, these sets lose no information.

Lemma 6.1.3. Rationals Separate

Let F be an Archimedean ordered field, and let $x, y \in F$. Then

$$x \leq y \iff \mathbb{Q}_F(x) \subseteq \mathbb{Q}_F(y).$$

Consequently $x = y \iff \mathbb{Q}_F(x) = \mathbb{Q}_F(y)$.

Proof. ($\Rightarrow$) This is the second property after Definition 6.1.2: if $\iota_F(q) < x$ and $x \leq y$, then $\iota_F(q) < y$.

($\Leftarrow$) We prove the contrapositive (Logic and Proof, Proposition 2.3.2). Suppose that $x \leq y$

fails. Then $y < x$ by totality (Proposition 4.1.2 (a)). By Theorem 4.2.5 there is $q \in \mathbb{Q}$ with $y < \iota_F(q) < x$. Then $q \in \mathbb{Q}_F(x)$, and $q \notin \mathbb{Q}_F(y)$, since $\iota_F(q) < y$ and $y < \iota_F(q)$ cannot both hold (O1). So $\mathbb{Q}_F(x) \not\subseteq \mathbb{Q}_F(y)$.

For the consequence: $x = y$ holds if and only if $x \leq y$ and $y \leq x$, by antisymmetry of the total order $\leq$; by the equivalence just proved, that is $\mathbb{Q}_F(x) \subseteq \mathbb{Q}_F(y)$ and $\mathbb{Q}_F(y) \subseteq \mathbb{Q}_F(x)$, that is, $\mathbb{Q}_F(x) = \mathbb{Q}_F(y)$.

■

This is the first named move of the chapter, **rationals separate**: to prove two elements of an Archimedean field equal, show that the same rationals lie below each. A rational strictly between them would otherwise exist. Section 6.2 uses it in four of the seven steps of its main proof (Steps 3, 5, 6 and 7), each time to prove that two elements of the complete field are equal.

↻ **Second reading** · *Without Archimedes the sets cannot tell elements apart*

The direction ($\Rightarrow$) and the remark that equal elements have equal sets hold in every ordered field. The direction ($\Leftarrow$) needs density, and density needs Archimedes. In the non-Archimedean field $\mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field) (school polynomial algebra, as flagged there; nothing below depends on it), every rational q satisfies $q < n$ for some $n \in \mathbb{N}$ (Theorem 3.2.3), and $n \cdot 1 < t$, so $\iota_{\mathbb{Q}(t)}(q) < t < t + 1$. Hence $\mathbb{Q}_{\mathbb{Q}(t)}(t) = \mathbb{Q} = \mathbb{Q}_{\mathbb{Q}(t)}(t + 1)$, although $t \neq t+1$. Both elements lie above all of the rationals, and the rationals cannot see the difference.

6.1.4 Constructing a point as a supremum

Some elements cannot be written down in advance. In $\mathbb{Q}$ the equation $x^2 = 2$ has no solution (Theorem 3.3.1); in a complete ordered field we expect one, but no formula names it. What we can name is the set of elements that should lie **below** it. Completeness turns that set into an element, its supremum, and the work is to prove that this element does what we want. We do it in full for square roots, extract the method, and run it again to find an element between two sets.

Proposition 6.1.4. Square Roots in a Complete Ordered Field

Let F be a complete ordered field and $a \in F$ with $a \geq 0$. Then there is **exactly one** $b \in F$ with $b \geq 0$ and $b^2 = a$.

Idea (Scratch work). The square root of $a > 0$ should be the top of $S := \{y \geq 0 : y^2 < a\}$. Two jobs before taking a supremum: $S \neq \emptyset$ ($0 \in S$) and an upper bound ($a + 1$: a larger y has $y^2 > y > a$). Put $s := \sup S$. The goal $s^2 = a$ is an equation, so we refute the two strict cases.

Givens	Goal
F complete, $a > 0$; $S = \{y \geq 0 : y^2 < a\}$ non-empty, bounded by $a + 1$	some $b \geq 0$ with $b^2 = a$

$s = \sup S$, $s \geq 0$	$s^2 = a$: rule out $s^2 < a$ and $s^2 > a$
case $s^2 < a$	an element of S **above** s, against (S1): $s + h$ with $(s+h)^2 < a$
case $s^2 > a$	an upper bound of S **below** s, against (S2): $s - h > 0$ with $(s-h)^2 \geq a$

If $s^2 < a$, there is room above s: we look for $h > 0$ with $(s+h)^2 < a$, which puts $s+h$ in S above its supremum. Working backwards, for $0 < h \leq 1$ we have $h^2 \leq h$, so

$$(s+h)^2 = s^2 + 2sh + h^2 \leq s^2 + h(2s+1),$$

and it is enough that $h(2s+1)$ be at most half the room $a - s^2$. So $h := \min\{1, (a - s^2)/(2(2s+1))\}$.

If $s^2 > a$, then s overshoots: we look for $h > 0$ with $(s-h)^2 \geq a$ still, which makes $s - h$ a smaller upper bound. Since $(s-h)^2 \geq s^2 - 2sh$, it is enough that $2sh = s^2 - a$, so $h := (s^2 - a)/(2s)$.

Both h depend on s and a only; the proof fixes them first and checks forwards.

Proof. Existence. If $a = 0$, then $b := 0$ works. Let $a > 0$, and let

$$S := \{y \in F : y \geq 0 \text{ and } y^2 < a\}.$$

Then $0 \in S$, since $0^2 = 0 < a$. And $a+1$ is an upper bound of S: if $y > a+1$, then $y > 1 > 0$, so $y^2 = y \cdot y > y \cdot 1 = y > a$ (Proposition 2.4.3 (c)), and $y \notin S$; so every $y \in S$ satisfies $y \leq a + 1$ by totality. Since F is complete, let $s := \sup S$. Since $0 \in S$, $s \geq 0$.

Case $s^2 < a$. Let $h := \min\{1, (a - s^2)/(2(2s+1))\}$. It is defined and positive, since $2s + 1 \geq 1 > 0$ and $a - s^2 > 0$ (Proposition 3.2.2 (a), (O4)). Since $0 < h \leq 1$, $h^2 \leq h$ (Proposition 2.4.3 (c)), so

$$\begin{aligned}(s+h)^2 = s^2 + 2sh + h^2 &\leq s^2 + h(2s+1) \\ &\leq s^2 + \frac{a - s^2}{2} < a,\end{aligned}$$

where the second inequality uses $h \leq (a - s^2)/(2(2s+1))$ and $2s + 1 > 0$, and the last uses $(a - s^2)/2 < a - s^2$ (Proposition 4.1.2 (b)). So $s + h \geq 0$ lies in S, and $s + h > s$, against s being an upper bound of S.

Case $s^2 > a$. Then $s \neq 0$, since $0^2 = 0 < a$; so $s > 0$. Let $h := (s^2 - a)/(2s)$. It is positive, since $s^2 - a > 0$ and $2s > 0$ (Proposition 3.2.2 (a), (O4)). Then $s - h = (s^2 + a)/(2s) > 0$ by the same reasons, since $s^2 + a > 0$; and

$$(s-h)^2 = s^2 - 2sh + h^2 = s^2 - (s^2 - a) + h^2 = a + h^2 > a.$$

We claim that $s - h$ is an upper bound of S. Let $y > s - h$. Then $y > 0$, and Proposition 2.4.3 (g) gives $y^2 > (s-h)^2 > a$, so $y \notin S$. Hence every element of S is $\leq s - h$. But $s - h < s$, against s being the **least** upper bound.

Both cases are impossible, so $s^2 = a$ by trichotomy (O1), and $b := s \geq 0$ works.

Uniqueness. Let $b, c \geq 0$ with $b^2 = c^2 = a$. Then $(b-c)(b+c) = b^2 - c^2 = 0$, so $b = c$ or $b + c = 0$ (Proposition 2.4.3 (f)). If $b + c = 0$, then $b = -c \leq 0 \leq b$, so $b = 0$ and $c = -b = 0$. In either case $b = c$.

■

For $a \geq 0$ in a complete ordered field we write $\sqrt{a}$ for this b. In the constructed reals, the class r of Example (The Class of the Newton Sequence) has $r > 0$ and $r^2 = 2$, so $r = \sqrt{2}$ by the uniqueness clause, and Example (The Rationals below the Square Root of Two) describes $\mathbb{Q}_{\mathbb{R}}(\sqrt{2})$.

Completeness was used once, to produce s; the rest is ordered-field arithmetic. Without completeness the statement fails: 2 has no square root in $\mathbb{Q}$, and *Algebraic Preliminaries*, Chapter 2 meets $\mathbb{Q}(\sqrt{2})$, an Archimedean field that still lacks square roots.

Example. Method: construct a point as a supremum

Template. To produce an element with a property P in a complete ordered field: (1) choose a set S of elements that should lie below it; (2) check that S is non-empty and bounded above; (3) let $s := \sup S$; (4) prove $P(s)$. An inequality $y \leq s$ for $y \in S$ is (S1). An inequality $s \leq u$ follows once u is shown to be an upper bound of S (S2). An equation is proved by refuting both strict cases: one by an element of S above s, the other by an upper bound of S below s.

When to use it. When the element has no formula but the elements below it can be described: a square root, an element between two sets, the value $\varphi(x)$ in Section 6.2, the limit of a bounded increasing sequence (Exercise C3 of Section 6.3). It is Sets and Functions, Example (Method: construct and verify) with the construction step done by completeness.

The usual failure. Taking the supremum before checking (2); assuming $s \in S$; choosing h that depends on something other than s and the data.

Example. An Element Between Two Sets

Let F be a complete ordered field, and let $A, B \subseteq F$ be non-empty with $a \leq b$ for every $a \in A$ and $b \in B$.

(a) Show that some $c \in F$ has $a \leq c \leq b$ for every $a \in A$ and $b \in B$.

(b) Show that if, in addition, for every $\varepsilon > 0$ there are $a \in A$ and $b \in B$ with $b - a < \varepsilon$, then c is the only such element.

Solution.

(a) *Choose S.* The element should lie above all of A: take $S := A$, non-empty. *An upper bound.* Fix $b_0 \in B$; every $a \in A$ has $a \leq b_0$. *Take the supremum* $c := \sup A$. Then $a \leq c$ for $a \in A$ by (S1), and for $b \in B$, b is an upper bound of A by hypothesis, so $c \leq b$ by (S2).

Second reading · *Only one such element*

(b) Let c' be another such element, and suppose $c \neq c'$. Take $\varepsilon := |c - c'| > 0$ and $a \in A$, $b \in B$ with $b - a < \varepsilon$. Both c and c' lie between a and b, so $-(b-a) \leq c - c' \leq b - a$, and $|c - c'| \leq b - a < \varepsilon$ (Proposition 4.1.3 (d)): a contradiction.

No equation had to be refuted case by case here: (S1) and (S2) gave the two inequalities directly. Section 6.3 meets a related picture: the rationals below a real number form a Dedekind cut.

Quick check. In $\mathbb{Q}$, let $A := \{p \in \mathbb{Q} : p^2 < 2\}$ and $B := \{p \in \mathbb{Q} : p > 0,\ p^2 > 2\}$. Every element of A is below every element of B. Is there a rational c with $a \leq c \leq b$ for all $a \in A$, $b \in B$?

Solution. No. Such a c is an upper bound of A. Every rational upper bound u of A lies in B: $u \geq 1 > 0$ since $1 \in A$; $u^2 \neq 2$ (Theorem 3.3.1); and $u^2 < 2$ would make u the largest element of A, hence a supremum of A in $\mathbb{Q}$. So $c \leq u$ for every upper bound u, and c would be a supremum of A in $\mathbb{Q}$, against Theorem 3.3.2 (a). The example needs completeness.

6.1.5 Exercises

6.1.5.1 A. Check your understanding

Exercise. A1

(a) True or false, with a reason: in **every** ordered field F, if $\mathbb{Q}_F(x) \neq \mathbb{Q}_F(y)$ then $x \neq y$. Is the converse true in every ordered field?

(b) Write "F is Archimedean" in symbols, and state its negation with the quantifiers moved to the front (Logic and Proof, Proposition 6.2.1).

(c) The eight lines (i)–(viii) below are about a complete ordered field F; six of them, in a suitable order, prove Theorem 6.1.2. Write an admissible order, and say why each of the other two lines is wrong.

(i) Then $\iota_F(n+1) = \iota_F(n) + 1 > s$, and $n + 1 \in \mathbb{N}$.
(ii) For every $n \in \mathbb{N}$, $\iota_F(n) > s - 1$.
(iii) The set $\iota_F(\mathbb{N})$ is non-empty ($0 \in \mathbb{N}$), so $s := \sup \iota_F(\mathbb{N})$ exists.
(iv) This contradicts that s is an upper bound of $\iota_F(\mathbb{N})$.
(v) Suppose that $\iota_F(\mathbb{N})$ is bounded above.
(vi) So $\iota_F(n) > s - 1$ for some $n \in \mathbb{N}$.
(vii) So $s \in \iota_F(\mathbb{N})$, say $s = \iota_F(m)$.
(viii) Since $s - 1 < s$, the element $s - 1$ is not an upper bound of $\iota_F(\mathbb{N})$.

Solution.

(a) True. It is the contrapositive (Logic and Proof, Proposition 2.3.2) of "if $x = y$ then $\mathbb{Q}_F(x) = \mathbb{Q}_F(y)$", which holds because equal elements have the same rationals below them. The converse, "if $x \neq y$ then $\mathbb{Q}_F(x) \neq \mathbb{Q}_F(y)$", is the direction ($\Leftarrow$) of Lemma 6.1.3, and it needs the Archimedean property: in $\mathbb{Q}(t)$, $t \neq t+1$ and both have every rational below them (the second-reading note after Lemma 6.1.3).

(b) $\forall x \in F\ \exists n \in \mathbb{N}\ \ x < n \cdot 1_F$. Negation: $\exists x \in F\ \forall n \in \mathbb{N}\ \ n \cdot 1_F \leq x$, using $\neg(x < y) \iff y \leq x$ (totality and (O1)). In words: some element lies above or at every $n \cdot 1_F$.

(c) (v), (iii), (viii), (vi), (i), (iv). No other order is admissible: each line uses the one before it ((iii) needs the bound of (v) for the supremum to exist, (viii) needs s, (vi) unpacks (viii), (i) uses the n of (vi), and (iv) is the contradiction with (i)). The contradiction in (iv) refutes (v), so $\iota_F(\mathbb{N})$ is not bounded above; unpacked with the negation laws, as at the end of the proof of Theorem 6.1.2, that is the statement of (b).

Line (vii) is wrong: a supremum need not belong to the set (in every ordered field, the set of negative elements has supremum 0, which is not negative: the first example after Definition 5.5.1). Line (ii) has the wrong quantifier: "$s - 1$ is not an upper bound" gives **some** n with $\iota_F(n) > s - 1$, by Logic and Proof, Proposition 6.2.3, not every n (and for $n = 0$ it may be false).

6.1.5.2 B. Practice

Exercise. B1

Let F be a complete ordered field. Prove that $\inf\{\iota_F(1/n) : n \in \mathbb{Z}_{>0}\} = 0$.

Solution. Let $T := \{\iota_F(1/n) : n \in \mathbb{Z}_{>0}\}$. For $n \in \mathbb{Z}_{>0}$, $1/n > 0$ in $\mathbb{Q}$, so $\iota_F(1/n) > 0$: 0 is a lower bound of T.

Let c be a lower bound of T, and suppose, for contradiction, that $c > 0$. By Theorem 6.1.2, F is Archimedean, so Proposition 4.2.3 (with $\varepsilon := c$, applied at $n = N$) gives $N \in \mathbb{Z}_{>0}$ with $1/(N \cdot 1_F) < c$. Since ι_F preserves products and $\iota_F(N) = N \cdot 1_F$, we have $\iota_F(1/N) \cdot (N \cdot 1_F) = \iota_F(1) = 1_F$, so $\iota_F(1/N) = 1/(N \cdot 1_F)$ by uniqueness of inverses (Lemma 4.1.1 (a)). So the element $\iota_F(1/N)$ of T lies below c, against c being a lower bound. Hence every lower bound c satisfies $c \leq 0$, and 0 is the greatest lower bound (Sets and Functions, Definition 5.3.5).

Exercise. B2

Let F be an Archimedean ordered field and $x \in F$. Prove that the set $\iota_F(\mathbb{Q}_F(x)) = \{\iota_F(q) : q \in \mathbb{Q}_F(x)\}$ has supremum x in F.

Solution. Upper bound. Every element $\iota_F(q)$ of the set satisfies $\iota_F(q) < x$ by the definition of $\mathbb{Q}_F(x)$.

Least. Let u be an upper bound, and suppose $u < x$. By Theorem 4.2.5 there is $q \in \mathbb{Q}$ with $u < \iota_F(q) < x$. Then $q \in \mathbb{Q}_F(x)$, so $\iota_F(q)$ is in the set and exceeds u, against u being an

upper bound. So $x \le u$ by totality, for every upper bound u. Hence $x = \sup \iota_F(\mathbb{Q}_F(x))$.

No completeness was used: the supremum exists because we exhibited it.

Exercise. B3: Prove or disprove

Prove each true statement and refute each false one. Throughout, F is a complete ordered field and rationals stand for their copies $\iota_F(q)$.

(a) Every $a > 0$ in F has exactly two elements b with $b^2 = a$.
(b) In every Archimedean ordered field, every element $a \ge 0$ has a square root.
(c) If $S \subseteq F$ is non-empty and bounded above, then $\sup S \in S$.
(d) For every $a > 0$ in F there is $x > 0$ with $x^2 + x = a$.
(e) If $S \subseteq F$ is non-empty, $c \in F$, and $y < c$ for every $y \in S$, then $\sup S < c$.

Solution.

(a) **True.** Let $r := \sqrt{a}$ (Proposition 6.1.4). Then $r \neq 0$, since $a \neq 0$; so $r > 0 > -r$ (Proposition 2.4.3 (a)) and $r \neq -r$, and both square to a. Conversely, let $b^2 = a$. If $b \ge 0$, then $b = r$ by the uniqueness clause. If $b < 0$, then $-b > 0$ and $(-b)^2 = b^2 = a$, so $-b = r$ and $b = -r$. So the two elements are r and $-r$.

(b) **False.** $\mathbb{Q}$ is Archimedean (Theorem 3.2.3), and 2 has no square root in it (Theorem 3.3.1).

(c) **False.** $S := \{1 - 1/n : n \in \mathbb{Z}_{>0}\}$ has supremum 1 (Example (A Supremum Computed by Archimedes)), and $1 \notin S$, since every element is strictly below 1.

(d) **True.** Complete the square: $x^2 + x = (x + 1/2)^2 - 1/4$. So put $x := \sqrt{a + 1/4} - 1/2$, which exists because $a + 1/4 > 0$. Then $x^2 + x = (a + 1/4) - 1/4 = a$. And $x > 0$: $a + 1/4 > 1/4 = (1/2)^2$, and both $\sqrt{a + 1/4}$ and $1/2$ are non-negative, so $\sqrt{a + 1/4} > 1/2$ by Proposition 2.4.3 (g).

(e) **False.** Take S as in (c) and $c := 1$: every element of S is below 1, and $\sup S = 1$. The true statement is $\sup S \le c$, by (S2), since c is an upper bound of S.

6.1.5.3 C. Going deeper

Exercise. C1: An Archimedean field is no larger than the subsets of the rationals

Let F be an Archimedean ordered field. Prove that the map $F \to \mathcal{P}(\mathbb{Q})$, $x \mapsto \mathbb{Q}_F(x)$, is injective. What does this say about the size of F?

Solution. Let $x, y \in F$ with $\mathbb{Q}_F(x) = \mathbb{Q}_F(y)$. By Lemma 6.1.3, $x = y$. So the map is injective (Sets and Functions, Definition 2.2.1).

An injection $F \to \mathcal{P}(\mathbb{Q})$ says that F has at most as many elements as $\mathcal{P}(\mathbb{Q})$, in the sense of *Sets and Functions*, Chapter 4. In particular every Archimedean ordered field, and so (by Theorem 6.1.2) every complete ordered field, is no larger than the set of all sets of rationals.

The non-Archimedean field $\mathbb{Q}(t)$ shows that the hypothesis is needed for injectivity: t and $t+1$ have the same image.

Exercise. C2: Find the error

Here is a "proof" that a complete ordered field F is Archimedean.

> Suppose that $\iota_F(\mathbb{N})$ is bounded above, and let s be its supremum. Since s is the least upper bound of $\iota_F(\mathbb{N})$, it is the largest element of $\iota_F(\mathbb{N})$, so $s = \iota_F(m)$ for some $m \in \mathbb{N}$. But $\iota_F(m+1) > s$, a contradiction.

Is the conclusion true? Find the line that fails, give a counterexample to that line, and repair the proof.

Solution. The conclusion is true: it is Theorem 6.1.2.

The illegal step is "the least upper bound of a set is its largest element". A supremum is an upper bound that lies below every other upper bound (Sets and Functions, Definition 5.3.5); nothing says it belongs to the set. Counterexample to the step: in the constructed $\mathbb{R}$, $A \coloneqq \{1 - 1/n : n \in \mathbb{Z}_{>0}\}$ has supremum 1 by Example (A Supremum Computed by Archimedes), with $F = \mathbb{R}$ (a complete ordered field, Theorem 5.5.2) and $1 - 1/n$ read as $\iota_{\mathbb{R}}(1 - 1/n)$, and $1 \notin A$, since every element is < 1. So the step fails for this set, though the theorem's conclusion is not affected.

Repair: replace the second and third sentences by the argument of the proof of Theorem 6.1.2. Since $s - 1 < s$, $s - 1$ is not an upper bound, so $\iota_F(n) > s - 1$ for some n; then $\iota_F(n+1) = \iota_F(n) + 1 > s$, against s being an upper bound. The repair uses that s is **least** only through "$s-1$ is not an upper bound", which hands us an element of the set just below s, not s itself.

Marking scheme. (10 points) - 3: says the conclusion is true, with the reason (the theorem). - 3: names the line "the supremum is the largest element" and says which property of suprema it wrongly assumes. - 2: a counterexample **to that step**: a set whose supremum is not an element of it, with the supremum verified. - 2: a correct repair that uses that s is **least**, without assuming s lies in the set: for example, $s - 1$ is not an upper bound, so some $\iota_F(n) > s - 1$ and then $\iota_F(n+1) > s$; or, equivalently, if $\iota_F(n+1) \leq s$ for all n then $s - 1$ is a smaller upper bound (the case $n = 0$ uses $1 \leq s$). Fatal errors (0 for the step): declaring the conclusion false; a counterexample to the theorem rather than to the step.

6.2 Building the Map from the Rationals Below

Section 6.1 gave every element x of an ordered field F a record, the set $\mathbb{Q}_F(x)$ of rationals below it, and showed that in an Archimedean field the record determines the element (Lemma 6.1.3). This section turns the record into a map. Given an Archimedean field F and a complete field L, we send $x \in F$ to the element of L with the same record. The result is the main theorem of the chapter in its one-sided form: there is exactly one order-preserving field map $F \to L$. Section 6.3 applies it twice, in both directions, to compare two complete ordered fields.

Two fields are in play throughout, so every proof here keeps ι_F and ι_L visible. Only the two lemmas, each about one field, write q for $\iota_F(q)$.

6.2.1 What the map must be

Suppose $\varphi\colon F \to L$ is an order-preserving field map (Definition 6.1.1). By Proposition 6.1.1 it fixes the rationals, $\varphi(\iota_F(q)) = \iota_L(q)$, and it preserves and reflects order. So for $q \in \mathbb{Q}$ and $x \in F$,

$$\iota_F(q) < x \iff \varphi(\iota_F(q)) < \varphi(x) \iff \iota_L(q) < \varphi(x).$$

In the notation of Section 6.1: $\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x)$. *The image of x has exactly the rationals below it that x has.*

That already shows there can be at most one such map when L is Archimedean: two candidates for $\varphi(x)$ would have the same rationals below them, and rationals separate. For existence, we need an element of L with a prescribed set of rationals below it. When L is complete there is a natural candidate: the least element of L that lies above all the copies $\iota_L(q)$, $q \in \mathbb{Q}_F(x)$, that is, their supremum.

The check that this candidate respects sums and products goes through the record, not through suprema. For that we need to know how the rationals below $a+b$ and below ab are made from those below a and b.

Quick check. Let $\varphi\colon \mathbb{Q} \to L$ be an order-preserving field map into an ordered field L. Which rationals lie below $\varphi(1/2)$?

Solution. Those $q < 1/2$: $\varphi(1/2) = \iota_L(1/2)$ (Proposition 6.1.1 with $\iota_{\mathbb{Q}} = \mathrm{id}_{\mathbb{Q}}$), and ι_L preserves and reflects order.

6.2.2 Sums and products, measured by rationals

A rational q lies below $a+b$ exactly when it splits into a rational below a and a rational below b.

Lemma 6.2.1. Rationals below a Sum

Let F be an Archimedean ordered field, $a, b \in F$ and $q \in \mathbb{Q}$. Then $\iota_F(q) < a + b$ if and only if there is $r \in \mathbb{Q}$ with

$$\iota_F(r) < a \qquad \text{and} \qquad \iota_F(q - r) < b.$$

In other words, $q \in \mathbb{Q}_F(a+b) \iff \exists r \in \mathbb{Q}\ (r \in \mathbb{Q}_F(a) \wedge q - r \in \mathbb{Q}_F(b))$.

Idea. One direction is addition of inequalities. For the other, split q as $r + (q - r)$ with r a rational just below a: we need r in the gap between $q - b$ and a, which is non-empty exactly because $q < a + b$, and density supplies it.

Proof. We write q for $\iota_F(q)$; since ι_F preserves sums, it preserves differences, so $\iota_F(q-r) = q-r$ in F.

($\Leftarrow$) If $r < a$ and $q - r < b$, then $q = r + (q-r) < a + b$ by Proposition 2.4.3 (b).

($\Rightarrow$) Let $q < a + b$. Then $q - b < a$ (Proposition 2.4.3 (a)). By Theorem 4.2.5 there is $r \in \mathbb{Q}$ with $q - b < r < a$. Then $r < a$, and $q - r < b$ by the same rule.

■

For products the same idea works for positive elements and positive rationals, with q/r in place of $q - r$.

Lemma 6.2.2. Rationals below a Product

Let F be an Archimedean ordered field, $a, b \in F$ with $a > 0$ and $b > 0$, and $q \in \mathbb{Q}$ with $q > 0$. Then $\iota_F(q) < ab$ if and only if there is $r \in \mathbb{Q}$ with

$$r > 0, \qquad \iota_F(r) < a \qquad \text{and} \qquad \iota_F(q/r) < b.$$

For $q \leq 0$, $\iota_F(q) < ab$ holds always.

In words: below a product of positives, a positive rational splits as a product of a rational below each factor. Exercise B1 asks for the proof; the fold below has it.

↻ **Second reading** · *The proof of the product lemma (try Exercise B1 first)*

Idea. As for sums, with division in place of subtraction: split q as $r \cdot (q/r)$ with r a rational in the gap between q/b and a, a gap of positive elements.

Proof. We write q for $\iota_F(q)$; since ι_F preserves products and $\iota_F(r) \neq 0$ for $r \neq 0$, it preserves quotients, so $\iota_F(q/r) = q/r$ in F.

If $q \leq 0$, then $q \leq 0 < ab$ by (O4). Let $q > 0$.

($\Leftarrow$) Let $r > 0$ with $r < a$ and $q/r < b$. Since $q/r > 0$ (Proposition 3.2.2 (a), (O4)), Proposition 2.4.3 (c) gives $q = r \cdot (q/r) < a \cdot (q/r)$; since $a > 0$, it gives $a \cdot (q/r) < ab$. So $q < ab$.

($\Rightarrow$) Let $q < ab$. Multiplying by $b^{-1} > 0$ (by the same two rules) gives $q/b < a$, and $q/b > 0$ by (O4). By Theorem 4.2.5 there is $r \in \mathbb{Q}$ with $q/b < r < a$. Then $r > q/b > 0$, so $r > 0$ in $\mathbb{Q}$, since ι_F reflects order. Multiplying $q/b < r$ by $b > 0$ gives $q < rb$, and multiplying by $r^{-1} > 0$ gives $q/r < b$.

■

Quick check. In $F = \mathbb{Q}$, with $a := 3/4$, $b := 1/2$ and $q := 1$, find a rational r as in Lemma 6.2.1. Which rationals r work?

Solution. We need $r < 3/4$ and $1 - r < 1/2$, that is, $1/2 < r < 3/4$. For example $r = 2/3$: then $1 - 2/3 = 1/3 < 1/2$. Exactly the rationals of the open gap between $q - b = 1/2$ and $a = 3/4$ work, as the proof predicts.

6.2.3 The map into a complete field

Here is the theorem. Its proof is the first key proof of the chapter.

Theorem 6.2.3. Embedding an Archimedean Field in a Complete One

Let F be an Archimedean ordered field and L a complete ordered field. Then there is **exactly one** order-preserving field map $\varphi\colon F \to L$. It is given by

$$\varphi(x) := \sup_L \{\iota_L(q) : q \in \mathbb{Q}_F(x)\} \qquad (x \in F),$$

the supremum taken in L, and it satisfies

$$\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x) \qquad \text{for every } x \in F. \tag{$\star$}$$

Try it first (ten minutes): if φ exists and fixes $\mathbb{Q}$, which element of L must $\varphi(x)$ be?

Idea. The natural first attempt defines φ on the copy of $\mathbb{Q}$, by $\varphi(\iota_F(q)) := \iota_L(q)$, and then tries to "extend". It stalls at once: an element $x \in F$ that is not a rational has no formula, and nothing in F names it. What we do know, from the opening of this section ("What the map must be"), is its record: $\varphi(x)$ must have exactly the rationals $\mathbb{Q}_F(x)$ below it. So we **construct a point as a supremum** (Example (Method: construct a point as a supremum)): $\varphi(x)$ is the least upper bound in L of the copies of those rationals, and $(\star)$ says the construction has the right record. After that, sums, products and uniqueness are checked the same way: compute the record of both sides (for sums and products by the two lemmas) and let rationals separate in L. Monotonicity needs only one rational between x and y.

① $\varphi(x)$ exists; ② $(\star)$; ③ φ fixes the rationals; ④ φ is strictly increasing; ⑤ sums; ⑥ products and the unit; ⑦ uniqueness. *Before reading on: where is completeness of L used, and where the Archimedean property of F?*

Proof. By Theorem 6.1.2, L is Archimedean. The rules of an ordered field are used without comment. So Theorem 4.2.5, Lemma 6.1.3, Lemma 6.2.1 and Lemma 6.2.2 apply in L as well as in F. For $x \in F$ write $A_x := \{\iota_L(q) : q \in \mathbb{Q}_F(x)\} \subseteq L$.

Step 1: $\varphi(x)$ is defined. Let $x \in F$. Since F is Archimedean, there is $n \in \mathbb{N}$ with $-x < n \cdot 1_F$, so $\iota_F(-n) = -(n \cdot 1_F) < x$, and $-n \in \mathbb{Q}_F(x)$: the set A_x is non-empty. Applied to x, the Archimedean property gives $m \in \mathbb{N}$ with $x < m \cdot 1_F = \iota_F(m)$. For $q \in \mathbb{Q}_F(x)$,

$\iota_F(q) < x < \iota_F(m)$, so $q < m$ because ι_F reflects order, and $\iota_L(q) < \iota_L(m)$. So $\iota_L(m)$ is an upper bound of A_x. Since L is complete, A_x has a supremum in L, and it is unique; $\varphi(x)$ is that supremum. So $\varphi\colon F \to L$ is a function.

Step 2: $(\star)$**.** Let $x \in F$ and $q \in \mathbb{Q}$.

$(\subseteq)$ Let $q \in \mathbb{Q}_F(x)$. By density in F there is $r \in \mathbb{Q}$ with $\iota_F(q) < \iota_F(r) < x$. Then $r \in \mathbb{Q}_F(x)$, so $\iota_L(r) \in A_x$ and $\iota_L(r) \le \varphi(x)$, since $\varphi(x)$ is an upper bound. Also $q < r$ (reflection in F), so $\iota_L(q) < \iota_L(r) \le \varphi(x)$, and $q \in \mathbb{Q}_L(\varphi(x))$.

$(\supseteq)$ Let $\iota_L(q) < \varphi(x)$. Since $\varphi(x)$ is the **least** upper bound of A_x, $\iota_L(q)$ is not an upper bound, so there is $r \in \mathbb{Q}_F(x)$ with $\iota_L(q) < \iota_L(r)$ (totality). Then $q < r$ (reflection in L), so $\iota_F(q) < \iota_F(r) < x$, and $q \in \mathbb{Q}_F(x)$.

Step 3: φ **fixes the rationals.** Let $r \in \mathbb{Q}$. By $(\star)$ and the first example after Definition 6.1.2, used in F and in L,

$$\mathbb{Q}_L(\varphi(\iota_F(r))) = \mathbb{Q}_F(\iota_F(r)) = \{q \in \mathbb{Q} : q < r\} = \mathbb{Q}_L(\iota_L(r)).$$

By Lemma 6.1.3 in L, $\varphi(\iota_F(r)) = \iota_L(r)$. In particular $\varphi(0_F) = 0_L$ and $\varphi(1_F) = 1_L$.

Step 4: φ **is strictly increasing.** Let $x < y$ in F. By density in F there is $r \in \mathbb{Q}$ with $x < \iota_F(r) < y$. Then $r \notin \mathbb{Q}_F(x)$ (by (O1)), so $r \notin \mathbb{Q}_L(\varphi(x))$ by $(\star)$, and $\varphi(x) \le \iota_L(r)$ by totality. And $r \in \mathbb{Q}_F(y) = \mathbb{Q}_L(\varphi(y))$, so $\iota_L(r) < \varphi(y)$. Hence $\varphi(x) \le \iota_L(r) < \varphi(y)$.

Step 5: sums. Let $x, y \in F$ and $q \in \mathbb{Q}$. Then

$$\begin{aligned} q \in \mathbb{Q}_L(\varphi(x+y)) &\iff q \in \mathbb{Q}_F(x+y) \\ &\iff \exists r \in \mathbb{Q}\ (r \in \mathbb{Q}_F(x) \wedge q - r \in \mathbb{Q}_F(y)) \\ &\iff \exists r \in \mathbb{Q}\ (r \in \mathbb{Q}_L(\varphi(x)) \wedge q - r \in \mathbb{Q}_L(\varphi(y))) \\ &\iff q \in \mathbb{Q}_L(\varphi(x) + \varphi(y)), \end{aligned}$$

by $(\star)$, Lemma 6.2.1 in F, $(\star)$ twice, and Lemma 6.2.1 in L. By Lemma 6.1.3 in L, $\varphi(x+y) = \varphi(x) + \varphi(y)$.

Step 6: products and the unit. First let $x > 0$ and $y > 0$. By Steps 3 and 4, $\varphi(x) > \varphi(0_F) = 0_L$ and $\varphi(y) > 0_L$; and $xy > 0$, $\varphi(x)\varphi(y) > 0$ by (O4). Let $q \in \mathbb{Q}$. If $q \le 0$, then $q \in \mathbb{Q}_F(xy) = \mathbb{Q}_L(\varphi(xy))$, and $q \in \mathbb{Q}_L(\varphi(x)\varphi(y))$, by the last clause of Lemma 6.2.2 in F and in L. If $q > 0$, then

$$\begin{aligned} q \in \mathbb{Q}_L(\varphi(xy)) &\iff q \in \mathbb{Q}_F(xy) \\ &\iff \exists r \in \mathbb{Q}_{>0}\ (r \in \mathbb{Q}_F(x) \wedge q/r \in \mathbb{Q}_F(y)) \\ &\iff \exists r \in \mathbb{Q}_{>0}\ (r \in \mathbb{Q}_L(\varphi(x)) \wedge q/r \in \mathbb{Q}_L(\varphi(y))) \\ &\iff q \in \mathbb{Q}_L(\varphi(x)\varphi(y)), \end{aligned}$$

by $(\star)$, Lemma 6.2.2 in F, $(\star)$ twice, and Lemma 6.2.2 in L (with $\varphi(x), \varphi(y) > 0$). By Lemma 6.1.3 in L, $\varphi(xy) = \varphi(x)\varphi(y)$ for $x, y > 0$.

Now the signs. By Steps 5 and 3, $\varphi(x) + \varphi(-x) = \varphi(0_F) = 0_L$, so $\varphi(-x) = -\varphi(x)$. If $x = 0$ or $y = 0$, both sides of $\varphi(xy) = \varphi(x)\varphi(y)$ are 0_L (Step 3). If $x < 0 < y$, then $-x > 0$ and $xy = -((-x)y)$, so

$$\varphi(xy) = -\varphi((-x)y) = -(\varphi(-x)\varphi(y)) = -(-\varphi(x))\varphi(y) = \varphi(x)\varphi(y).$$

The case $x > 0 > y$ follows by commutativity, and if $x, y < 0$, then $xy = (-x)(-y)$ and $\varphi(xy) = (-\varphi(x))(-\varphi(y)) = \varphi(x)\varphi(y)$. Finally $\varphi(1_F) = 1_L$ by Step 3. With Steps 4 and 5, φ is an order-preserving field map.

Step 7: uniqueness. Let $\chi\colon F \to L$ be any order-preserving field map, and let $x \in F$, $q \in \mathbb{Q}$. By Proposition 6.1.1, $\chi(\iota_F(q)) = \iota_L(q)$. If $\iota_F(q) < x$, then $\iota_L(q) = \chi(\iota_F(q)) < \chi(x)$. If $\iota_L(q) < \chi(x)$, that is, $\chi(\iota_F(q)) < \chi(x)$, then $\iota_F(q) < x$, because χ reflects order (the remark after Definition 6.1.1). So

$$\mathbb{Q}_L(\chi(x)) = \mathbb{Q}_F(x) = \mathbb{Q}_L(\varphi(x)),$$

the second equality by $(\star)$. By Lemma 6.1.3 in L, $\chi(x) = \varphi(x)$. Since x was arbitrary, $\chi = \varphi$. This proves the theorem.

■

The answer to the prompt. Completeness of L is used in Step 1, where the supremum must exist, and in Step 2 ($\supseteq$), where its leastness turns "below $\varphi(x)$" into "below some $\iota_L(r)$". The Archimedean property of F is used in Step 1 for the bounds $-n$ and m, in Steps 2 and 4 through density in F, and in Steps 5 and 6 through the two lemmas applied in F. Completeness of L is used a third time, indirectly: it makes L Archimedean (Theorem 6.1.2), which Steps 3, 5, 6 and 7 need for rationals to separate in L and for the lemmas in L.

Remark. Step 7 is the whole of "uniqueness". Its idea, comparing the rationals below two candidate values, needs only that L is Archimedean, not that it is complete: Exercise C1 runs it for any two order-preserving field maps $F \to L$, with no appeal to $(\star)$. Section 6.3 leans on the uniqueness clause throughout: to show that the maps between two complete fields are inverse to each other, that the isomorphism they give is the only one, and that a complete ordered field has no order-preserving field automorphism other than the identity.

Remark. Neither hypothesis can be dropped from the existence half. *Without completeness of L:* take $F = \mathbb{R}$, which is Archimedean (Theorem 5.4.2), and $L = \mathbb{Q}$. An order-preserving field map $\varphi\colon \mathbb{R} \to \mathbb{Q}$ would fix 2 (Proposition 6.1.1), so for $\sqrt{2} \in \mathbb{R}$ (Section 6.1), $\varphi(\sqrt{2})^2 = \varphi((\sqrt{2})^2) = \varphi(2) = 2$ in $\mathbb{Q}$, against Theorem 3.3.1. *Without the Archimedean property of F:* take $F = \mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field) and any complete L. An order-preserving field map φ would fix each $n \cdot 1$ and be strictly increasing, so $n \cdot 1_L = \varphi(n \cdot 1) < \varphi(t)$ for every $n \in \mathbb{N}$, against the Archimedean property of L (Theorem 6.1.2).

Warning. The supremum in the definition of φ is taken **in** L, of the copies $\iota_L(q)$. In $\mathbb{Q}$ it may not exist: $\mathbb{Q}_{\mathbb{R}}(\sqrt{2})$ has the same upper bounds in $\mathbb{Q}$ as $\{p \in \mathbb{Q} : p^2 < 2\}$, which has no least one (Theorem 3.3.2). In F it always exists and is x itself (Exercise B2 of Section 6.1): an element of the wrong field.

Quick check. Take $L := F$, a complete ordered field. Which map $F \to F$ does Theorem 6.2.3 produce, and why, without computing a single supremum?

Solution. The identity. The theorem says there is exactly one order-preserving field map $F \to F$ (F is Archimedean by Theorem 6.1.2), and id_F is one (the first example after Definition 6.1.1). So $\varphi = \mathrm{id}_F$. This is the argument Section 6.3 builds on.

6.2.4 Exercises

6.2.4.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Theorem 6.2.3.

(a) Which steps use that L is complete, and which that F is Archimedean? Which steps use that L is Archimedean, and where does that come from?

(b) Run the definition of φ with $F = \mathbb{Q}$ and $L = \mathbb{R}$, the constructed reals. What is φ?

(c) State the idea of the proof in two sentences.

(d) Why may Step 1 not define $\varphi(x)$ as a supremum taken in F?

Solution.

(a) L complete: Step 1 (the supremum exists) and Step 2 ($\supseteq$) (its leastness). F Archimedean: Step 1 (the bounds $-n$, m), Steps 2 and 4 (density in F), and Steps 5 and 6 (Lemma 6.2.1 and Lemma 6.2.2 applied in F). L Archimedean: Steps 3, 5, 6 and 7, through Lemma 6.1.3 in L and the two lemmas in L. It comes from the completeness of L, by Theorem 6.1.2.

(b) $\varphi = \iota_{\mathbb{R}} = \kappa$. By Step 3, $\varphi(\iota_{\mathbb{Q}}(r)) = \iota_{\mathbb{R}}(r)$ for every $r \in \mathbb{Q}$, and $\iota_{\mathbb{Q}}$ is the identity (Chapter 4, after Theorem 4.2.2). So $\varphi(r) = \iota_{\mathbb{R}}(r)$, which is $\kappa(r)$ by Theorem 5.4.1 (d). Concretely, $\varphi(r)$ is the supremum in $\mathbb{R}$ of $\{\kappa(q) : q < r\}$, and that supremum is $\kappa(r)$.

(c) An order-preserving map that fixes the rationals must send x to an element with the same rationals below it, and in a complete L the supremum of their copies is such an element. Every field identity for φ is then checked by comparing the rationals below the two sides, which the sum and product lemmas compute, and rationals separate in L.

(d) In F the supremum of $\iota_F(\mathbb{Q}_F(x))$ always exists, since F is Archimedean, and it is x itself (Exercise B2 of Section 6.1): an element of F, not of L. So it does not give a map into L. The supremum must be of the copies $\iota_L(q)$, taken in L.

6.2.4.2 B. Practice

Exercise. B1

Prove Lemma 6.2.2: let F be an Archimedean ordered field, $a, b \in F$ with $a, b > 0$, and $q \in \mathbb{Q}$ with $q > 0$; show that $\iota_F(q) < ab$ if and only if some rational $r > 0$ has $\iota_F(r) < a$ and $\iota_F(q/r) < b$. Show also that $\iota_F(q) < ab$ for every rational $q \leq 0$.

Hint: imitate the proof of Lemma 6.2.1, with division in place of subtraction.

Solution. Write q for $\iota_F(q)$. Since ι_F preserves products, $\iota_F(r)\,\iota_F(r^{-1}) = \iota_F(1) = 1$ for $r \neq 0$, so $\iota_F(r^{-1}) = \iota_F(r)^{-1}$ (Lemma 4.1.1 (a)) and $\iota_F(q/r) = q/r$ in F.

If $q \leq 0$, then $q \leq 0 < ab$ by (O4).

Let $q > 0$. ($\Leftarrow$) Let $r > 0$, $r < a$, $q/r < b$. Then $q/r > 0$ (Proposition 3.2.2 (a), (O4)), so Proposition 2.4.3 (c) gives $q = r(q/r) < a(q/r)$, and, as $a > 0$, $a(q/r) < ab$. Hence $q < ab$.

($\Rightarrow$) Let $q < ab$. Since $b^{-1} > 0$, Proposition 2.4.3 (c) gives $q/b < a$, and $q/b > 0$. By Theorem 4.2.5 there is $r \in \mathbb{Q}$ with $q/b < \iota_F(r) < a$. Then $\iota_F(r) > 0 = \iota_F(0)$, so $r > 0$ since ι_F reflects order. From $q/b < r$ and $b > 0$, $q < rb$; multiplying by $r^{-1} > 0$, $q/r < b$. This r is as required.

Exercise. B2: Cube roots

Let F be a complete ordered field and $a \in F$ with $a > 0$. Let $S := \{y \in F : y \geq 0 \text{ and } y^3 < a\}$. The following steps prove that some $s \in F$ has $s^3 = a$.

(1) S is non-empty ($0 \in S$) and bounded above by $1 + a$: if $y > 1 + a$, then $y > 1$, so $y^3 > y > a$. Let $s := \sup S$. Then $s > 0$, since $m := \min\{1, a/2\}$ has $0 < m \leq 1$, so $m^3 \leq m \leq a/2 < a$, and $m \in S$.

(2) For $y, z \in F$, $z^3 - y^3 = (z - y)(z^2 + zy + y^2)$ and $z^2 + zy + y^2 = (z + y/2)^2 + (3/4)y^2$, which is > 0 unless $y = z = 0$ (if $y \neq 0$, then $(3/4)y^2 > 0$ by Proposition 2.4.3 (e), (c) and the square is ≥ 0, so the sum is positive by (b); if $y = 0$ and $z \neq 0$, the expression is $z^2 > 0$ by (e)). Hence $y < z \iff y^3 < z^3$, for all $y, z \in F$.

(3) The case $s^3 < a$ is impossible. Let $h := \min\{1, (a - s^3)/(2(3s^2 + 3s + 1))\} > 0$. Since $0 < h \leq 1$ and $s \geq 0$, $(s + h)^3 = s^3 + 3s^2h + 3sh^2 + h^3 \leq s^3 + h(3s^2 + 3s + 1) \leq s^3 + (a - s^3)/2 < a$, so $s + h \in S$, against s being an upper bound.

(4) The case $s^3 > a$ is impossible. *Name the move, then complete it.*

(5) Hence $s^3 = a$.

Then explain how to get a cube root of $a = 0$ and of $a < 0$, and why every $a \in F$ has **exactly one** cube root.

Solution. The move is **construct a point as a supremum**: s is built as $\sup S$, and each of the two wrong cases of trichotomy is refuted by exhibiting a point that contradicts one clause of the supremum. In step (3) it was an element of S above s, against (S1). In step (4) it is an upper bound below s, against (S2).

(4) Suppose $s^3 > a$. Let $h := (s^3 - a)/(3s^2)$; it is positive, since $s > 0$ by (1). Also $h < s$: this says $s^3 - a < 3s^3$, which holds because $a > 0$ and $2s^3 > 0$. So $s - h > 0$, and

$$(s - h)^3 - a = (s^3 - a) - 3s^2h + 3sh^2 - h^3 = h^2(3s - h) > 0,$$

since $3s^2h = s^3 - a$, $h^2 > 0$ and $3s - h > 2s > 0$. Let $y > s - h$. By (2), $y^3 > (s - h)^3 > a$, so $y \notin S$. Hence every element of S is $\leq s - h$, and $s - h$ is an upper bound of S smaller than s, against s being the least upper bound.

(5) Steps (3) and (4) leave only $s^3 = a$, by trichotomy (O1).

For $a = 0$, $0^3 = 0$. For $a < 0$, $-a > 0$ has a cube root c by the above, and $(-c)^3 = -c^3 = a$

(Proposition 2.3.2 (c)). Uniqueness: if $b^3 = c^3$, then neither $b < c$ nor $c < b$ can hold, since by (2) either would make the cubes differ; so $b = c$. So every element of a complete ordered field has exactly one cube root.

6.2.4.3 C. Going deeper

Exercise. C1: Exit check: at most one map into an Archimedean field

Let F and L be Archimedean ordered fields, and let $\chi_1, \chi_2 \colon F \to L$ be order-preserving field maps. Prove that $\chi_1 = \chi_2$.

Solution (Hint 1). Compare the rationals below $\chi_1(x)$ and below $\chi_2(x)$.

Solution (Hint 2). By Proposition 6.1.1, $\chi_i(\iota_F(q)) = \iota_L(q)$ for $i = 1, 2$ and every $q \in \mathbb{Q}$.

Solution. Let $x \in F$ and $i \in \{1, 2\}$. By Proposition 6.1.1 (which needs no order hypothesis), $\chi_i(\iota_F(q)) = \iota_L(q)$ for every $q \in \mathbb{Q}$. Since χ_i is strictly increasing and reflects order (the remark after Definition 6.1.1),

$$\iota_F(q) < x \iff \chi_i(\iota_F(q)) < \chi_i(x) \iff \iota_L(q) < \chi_i(x).$$

So $\mathbb{Q}_L(\chi_1(x)) = \mathbb{Q}_F(x) = \mathbb{Q}_L(\chi_2(x))$. Since L is Archimedean, Lemma 6.1.3 gives $\chi_1(x) = \chi_2(x)$. As x was arbitrary, $\chi_1 = \chi_2$. (This is Step 7 of Theorem 6.2.3 with χ_2 in place of φ; completeness of L is not needed, and neither is the Archimedean property of F.)

Why L must be Archimedean (remark, not proved here). The last step separates $\chi_1(x)$ from $\chi_2(x)$ by a rational strictly between them, which is density of $\iota_L(\mathbb{Q})$ in L, and density needs Archimedes (Theorem 4.2.5). In the non-Archimedean field $\mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field), the elements $1/t$ and $2/t$ are different and have the same rationals below them, namely all $q \leq 0$: a positive rational q has $q > 1/n$ in $\mathbb{Q}$ for some $n \in \mathbb{Z}_{>0}$ (Theorem 3.2.3), so $\iota_{\mathbb{Q}(t)}(q) > \iota_{\mathbb{Q}(t)}(1/n) = 1/(n \cdot 1) > 2/t$, because $t > 2n \cdot 1$ (Example (A Non-Archimedean Ordered Field)). So the separating step fails in such a field.

Marking scheme. (10 points) - 2: each χ_i fixes $\mathbb{Q}$, with the reason (Proposition 6.1.1, or directly: induction for $\mathbb{N}$, then negatives and quotients). - 3: links the rationals below x to those below $\chi_i(x)$, using that χ_i fixes $\mathbb{Q}$ and reflects order. Either gives the full equivalence $\iota_F(q) < x \iff \iota_L(q) < \chi_i(x)$, or, for the squeeze route of the next item, applies reflection by χ_1 and χ_2 to a rational strictly between $\chi_1(x)$ and $\chi_2(x)$. - 3: concludes $\chi_1(x) = \chi_2(x)$ by rationals separating, or by a rational strictly between the two, invoking the Archimedean property **of** L. - 2: states the conclusion for every $x \in F$. Fatal errors (0 for the step): using density in L without a hypothesis that gives it; assuming χ_i surjective.

Exercise. C2: Dense image, and when the map is onto

Let F be an Archimedean ordered field, L a complete ordered field, and $\varphi \colon F \to L$ the map of Theorem 6.2.3.

(a) Prove that $\varphi(F)$ is dense in L: for all $u < v$ in L there is $x \in F$ with $u < \varphi(x) < v$.
(b) Prove that if F is complete, then φ is surjective.
(c) Prove that if φ is surjective, then F is complete.

Solution (Hint 1). For (b), given $y \in L$, look at the set $S \coloneqq \mathbb{Q}_L(y)$ and its copy $\iota_F(S)$ in F.

Solution (Hint 2). For (b): $\iota_F(S)$ is non-empty and bounded above in F; let $x \coloneqq \sup_F \iota_F(S)$, and show $\mathbb{Q}_F(x) = S$.

Solution. By Theorem 6.1.2, L is Archimedean.

(a) Let $u < v$ in L. By Theorem 4.2.5 in L there is $q \in \mathbb{Q}$ with $u < \iota_L(q) < v$. By Step 3 of the proof of Theorem 6.2.3, $\iota_L(q) = \varphi(\iota_F(q))$, so $x \coloneqq \iota_F(q)$ works.
(b) Let F be complete and $y \in L$. Let $S \coloneqq \mathbb{Q}_L(y)$. Since L is Archimedean, there are $n, m \in \mathbb{N}$ with $-y < \iota_L(n)$ and $y < \iota_L(m)$. So $-n \in S$, and every $q \in S$ has $\iota_L(q) < y < \iota_L(m)$, hence $q < m$ and $\iota_F(q) < \iota_F(m)$. So $\iota_F(S)$ is non-empty and bounded above in F, and $x \coloneqq \sup_F \iota_F(S)$ exists.
We show $\mathbb{Q}_F(x) = S$. ($\supseteq$) Let $q \in S$. By density in L there is $r \in \mathbb{Q}$ with $\iota_L(q) < \iota_L(r) < y$; then $r \in S$ and $q < r$, so $\iota_F(q) < \iota_F(r) \leq x$. ($\subseteq$) Let $\iota_F(q) < x$. Then $\iota_F(q)$ is not an upper bound of $\iota_F(S)$, so $\iota_F(q) < \iota_F(r)$ for some $r \in S$; hence $q < r$ and $\iota_L(q) < \iota_L(r) < y$, so $q \in S$.
By $(\star)$, $\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x) = S = \mathbb{Q}_L(y)$, so $\varphi(x) = y$ by Lemma 6.1.3 in L.
(c) Let φ be surjective. It is injective (strictly increasing), so bijective, with an inverse $\varphi^{-1}\colon L \to F$ (Sets and Functions, Theorem 2.4.4). Let $A \subseteq F$ be non-empty with an upper bound b. Then $\varphi(A)$ is non-empty and bounded above by $\varphi(b)$, since $a \leq b$ gives $\varphi(a) \leq \varphi(b)$. As L is complete, $s \coloneqq \sup \varphi(A)$ exists; let $c \coloneqq \varphi^{-1}(s)$. *Upper bound:* for $a \in A$, $\varphi(a) \leq s = \varphi(c)$, so $a \leq c$, because φ reflects order (and is injective, for the case of equality). *Least:* if u is an upper bound of A, then $\varphi(u)$ is an upper bound of $\varphi(A)$, so $\varphi(c) = s \leq \varphi(u)$, and $c \leq u$ as before. So $c = \sup A$ (Sets and Functions, Definition 5.3.5), and F is complete.

So φ is an order-preserving field isomorphism exactly when F is complete. Section 6.3 reaches the "if" direction by a different route, which computes no supremum at all.

Marking scheme. (10 points) - 2: (a) L is Archimedean (Theorem 6.1.2), so some $\iota_L(q)$ lies strictly between u and v, and $\iota_L(q) = \varphi(\iota_F(q))$ (Step 3). - 2: (b) $\iota_F(S)$ is non-empty and bounded above in F (Archimedes in L, transferred through ι_F), so $x = \sup_F \iota_F(S)$ exists by completeness of F. - 2: (b) $\mathbb{Q}_F(x) = S$, both inclusions; ($\supseteq$) uses density in L. - 1: (b) $\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x) = \mathbb{Q}_L(y)$ by $(\star)$, hence $\varphi(x) = y$ by Lemma 6.1.3 in L. - 3: (c) for non-empty $A \subseteq F$ bounded above by b, $\varphi(A)$ is bounded above by $\varphi(b)$, and $\varphi^{-1}(\sup \varphi(A))$ is an upper bound of A and the least one, both using that φ reflects order and is bijective. Fatal errors (0 for the step): using density of $\iota_L(\mathbb{Q})$ in L without the Archimedean property of L; in (b), taking a supremum in F before checking that the set is non-empty and bounded above; in (c), assuming φ^{-1} preserves order without justification.

6.3 Two Complete Ordered Fields, One Isomorphism

Section 6.2 built, from an Archimedean ordered field F into a complete one L, **exactly one** order-preserving field map φ, and it keeps the rationals below each element: $\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x)$ (property $(\star)$ of Theorem 6.2.3). This section takes both fields complete. Then maps run both ways, and the word "exactly" makes them inverse to each other. So any two complete ordered fields are copies of each other, by one isomorphism; the only symmetry of a complete ordered field is the identity; and every first-order statement about sums, products and order has the same truth value in all of them. The section ends with what this means for the two ways to begin analysis.

6.3.1 The uniqueness theorem

Section 6.2 gave us, for complete F and L, a map $\varphi\colon F \to L$. It did not say that φ is onto. A map can preserve everything in sight and still miss most of its target: the map $\iota_{\mathbb{R}}\colon \mathbb{Q} \to \mathbb{R}$ preserves sums, products and order (Theorem 4.2.2), and the real number r of Example (The Class of the Newton Sequence), whose square is 2, is not in its image. What rules this out when the source is complete as well?

Theorem 6.3.1. Uniqueness of the Complete Ordered Field

Let F and L be complete ordered fields (Definition 5.5.1). Then there is **exactly one** order-preserving field isomorphism $F \to L$ (Definition 6.1.1). It is the map φ of Theorem 6.2.3.

Idea. Section 6.2 runs in both directions. Since F is complete, it is Archimedean (Theorem 6.1.2), so it is a legal source; since L is complete, it is a legal target. That gives $\varphi\colon F \to L$. Swapping the roles gives $\psi\colon L \to F$.

To show that ψ undoes φ, we could try to compute $\psi(\varphi(x))$ from the two suprema. We do not have to. The composite $\psi \circ \varphi\colon F \to F$ is again an order-preserving field map, and so is the identity of F. The uniqueness clause of Theorem 6.2.3, applied with F as source **and** as target, says there is only one such map. So the two are equal. **Uniqueness gives inverses.** (This is the pattern of Sets and Functions, Example (Method: existence and uniqueness): two objects with the same defining property are equal.)

Before reading on: self-explain why $\psi \circ \varphi$ is order-preserving. Which of the four conditions of Definition 6.1.1 needs more than one line?

Proof. Let F and L be complete ordered fields. By Theorem 6.1.2, each of them is Archimedean.

The two maps. Since F is Archimedean and L is complete, Theorem 6.2.3 gives an order-preserving field map $\varphi\colon F \to L$. Since L is Archimedean and F is complete, the same theorem, with the roles of the two fields exchanged, gives an order-preserving field map $\psi\colon L \to F$.

The composite $\psi \circ \varphi$ is an order-preserving field map $F \to F$. For $a, b \in F$, additivity of φ

and then of ψ gives

$$\psi(\varphi(a+b)) = \psi(\varphi(a)+\varphi(b)) = \psi(\varphi(a)) + \psi(\varphi(b)),$$

and the same two steps with products give $\psi(\varphi(ab)) = \psi(\varphi(a))\,\psi(\varphi(b))$. Also $\psi(\varphi(1)) = \psi(1) = 1$. If $a < b$, then $\varphi(a) < \varphi(b)$, since φ is strictly increasing, and then $\psi(\varphi(a)) < \psi(\varphi(b))$, since ψ is.

The composite is the identity. The identity map id_F is an order-preserving field map $F \to F$: each of the four conditions reads $u = u$ or $a < b \Rightarrow a < b$. Now apply Theorem 6.2.3 with source F, which is Archimedean, and target F, which is complete: there is **exactly one** order-preserving field map $F \to F$. Both $\psi \circ \varphi$ and id_F are such maps, so $\psi \circ \varphi = \mathrm{id}_F$.

Exchanging the roles of F and L, the same argument gives $\varphi \circ \psi = \mathrm{id}_L$.

Existence. So ψ is a two-sided inverse of φ, and φ is bijective by Sets and Functions, Theorem 2.4.4. Hence φ is an order-preserving field isomorphism $F \to L$.

Uniqueness. Let $\chi\colon F \to L$ be any order-preserving field isomorphism. Forgetting that it is bijective, χ is an order-preserving field map $F \to L$, and by the uniqueness clause of Theorem 6.2.3 there is only one such map. Hence $\chi = \varphi$. This proves the theorem.

■

The answer to the prompt: each of the four conditions passes through a composite in one line, the order condition in two ($a < b$ gives $\varphi(a) < \varphi(b)$, which ψ accepts). The weight of the proof sits in the uniqueness clause, used three times: for $\psi \circ \varphi$, for $\varphi \circ \psi$, and for χ.

A direct proof that φ is onto, computing a supremum in F, is Exercise C2 of Section 6.2.

Quick check. Why could the proof not simply **define** $\psi := \varphi^{-1}$, instead of building ψ from Theorem 6.2.3?

Solution. The inverse $\varphi^{-1}\colon L \to F$ exists only if φ is bijective (Sets and Functions, Theorem 2.4.4), and bijectivity is what the theorem is trying to prove. Section 6.2 gives φ as an injective map (it is strictly increasing), with no claim that it is onto. The map ψ is built independently of φ, and only afterwards shown to be its inverse.

Each hypothesis has a job. Completeness of L makes it a target, so φ exists; completeness of F makes it a target, so ψ exists; and completeness of each gives the Archimedean property that makes it a source. Drop the completeness of F, and the conclusion fails: for $F = \mathbb{Q}$, $L = \mathbb{R}$, the only order-preserving field map is $\iota_{\mathbb{R}}$, which is not onto.

6.3.2 One real line

The theorem says that two complete ordered fields have the same shape, with a unique dictionary between them. We want a short way to write the first half of that sentence.

Two ordered fields are the same up to relabeling when an order-preserving bijection respects the arithmetic.

For ordered fields F and L, write

$$F \cong L$$

when there is **an** order-preserving field isomorphism $F \to L$ (Definition 6.1.1). Read it "F is isomorphic to L", with "as ordered fields" understood.

Examples.

- $\mathbb{R} \cong \mathbb{R}$, by $\mathrm{id}_{\mathbb{R}}$. Any ordered field is isomorphic to itself in the same way.
- $F \cong \mathbb{R}$ for every complete ordered field F, where $\mathbb{R}$ is the field of Chapter 5. This is the corollary below.
- $\mathbb{Q} \ncong \mathbb{R}$. An isomorphism $\chi\colon \mathbb{Q} \to \mathbb{R}$ would fix the rationals (Proposition 6.1.1, with $\iota_{\mathbb{Q}} = \mathrm{id}_{\mathbb{Q}}$ by the uniqueness clause of Theorem 4.2.2), so $\chi = \iota_{\mathbb{R}}$, and $\iota_{\mathbb{R}}$ is not onto (Example (The Class of the Newton Sequence)).

Corollary 6.3.2. Every Complete Ordered Field Is a Copy of the Reals

Let F be a complete ordered field, and let $\mathbb{R}$ be the field of Chapter 5. Then $F \cong \mathbb{R}$: there is **exactly one** order-preserving field isomorphism $F \to \mathbb{R}$.

Proof. By Theorem 5.5.2, $\mathbb{R}$ is a complete ordered field. Apply Theorem 6.3.1 with $L = \mathbb{R}$. ∎

So "the real numbers" names one ordered field up to a unique relabeling. A statement about sums, products and order is then true in F exactly when it is true in $\mathbb{R}$; the subsection *What the isomorphism carries* below makes this precise for first-order statements and shows it at work on limits.

↻ **Second reading** · *Why isomorphism is an equivalence relation*

On any set $\mathcal{S}$ of ordered fields, $\cong$ is an equivalence relation (Sets and Functions, Definition 5.2.1): it is reflexive, symmetric and transitive. No proof in this subsection uses its symmetry or transitivity; this paragraph records that the word "isomorphic" behaves as the word "equal" suggests. The symmetric clause is used below: the inverse of the isomorphism of Theorem 6.3.1 is again one.

Reflexive. id_F is an order-preserving field isomorphism $F \to F$.

Symmetric. Let $\chi\colon F \to L$ be an order-preserving field isomorphism, and let $\chi^{-1}\colon L \to F$ be its two-sided inverse (Sets and Functions, Theorem 2.4.4); it is bijective by the same theorem. For $u, v \in L$, additivity of χ gives

$$\chi(\chi^{-1}(u) + \chi^{-1}(v)) = \chi(\chi^{-1}(u)) + \chi(\chi^{-1}(v)) = u + v = \chi(\chi^{-1}(u + v)),$$

and since χ is injective, $\chi^{-1}(u + v) = \chi^{-1}(u) + \chi^{-1}(v)$. The same lines with products give $\chi^{-1}(uv) = \chi^{-1}(u)\chi^{-1}(v)$, and $\chi(1) = 1$ gives $\chi^{-1}(1) = 1$. Finally, let $u < v$. Then $\chi(\chi^{-1}(u)) < \chi(\chi^{-1}(v))$, and since χ reflects order, $\chi^{-1}(u) < \chi^{-1}(v)$.

Transitive. If $\chi\colon F \to L$ and $\chi'\colon L \to M$ are order-preserving field isomorphisms, then $\chi' \circ \chi$ is bijective (Sets and Functions, Theorem 2.2.1), and it is an order-preserving field map by the

computation in the proof of Theorem 6.3.1 for $\psi \circ \varphi$, which used nothing but the four conditions for each factor.

Example. Two Models, One Square Root of Two

Let M be any complete ordered field (for instance the one *Axiomatic Set Theory* builds, whose elements are sets of rationals). Let $\varphi\colon \mathbb{R} \to M$ be the isomorphism of Theorem 6.3.1 and r the real number of Example (The Class of the Newton Sequence) ($r > 0$, $r^2 = 2$). Find $\varphi(r)$ in terms of M alone, and describe $\mathbb{Q}_M(\varphi(r))$.

Solution. The image of r. Since φ is multiplicative and fixes the rationals (Proposition 6.1.1), $\varphi(r)^2 = \varphi(r^2) = \varphi(\iota_{\mathbb{R}}(2)) = \iota_M(2)$. Since φ is strictly increasing and $r > 0$, $\varphi(r) > \varphi(0) = 0$ (φ fixes the rational 0, Proposition 6.1.1). By Proposition 6.1.4 in the complete field M, there is exactly one $b \geq 0$ in M with $b^2 = \iota_M(2)$. So $\varphi(r)$ is that b: the square root of 2 in M, however M was built.

The rationals below it. By $(\star)$ of Theorem 6.2.3, $\mathbb{Q}_M(\varphi(r)) = \mathbb{Q}_{\mathbb{R}}(r)$. By Example (The Rationals below the Square Root of Two) this is the set of rationals q with $q < 0$, or with $q \geq 0$ and $q^2 < 2$. The same set of rationals sits below $\sqrt{2}$ in both models.

The set $\mathbb{Q}_F(x)$ of an element x of an Archimedean ordered field F always has four properties. It is non-empty: some $n \in \mathbb{N}$ has $-x < n \cdot 1_F$, so $-n \in \mathbb{Q}_F(x)$. It is not all of $\mathbb{Q}$: some m has $x < m \cdot 1_F$, so $m \notin \mathbb{Q}_F(x)$. It is closed downward (Section 6.1). It has no largest element: for $q \in \mathbb{Q}_F(x)$, density (Theorem 4.2.5) gives $p \in \mathbb{Q}$ with $\iota_F(q) < \iota_F(p) < x$. A set of rationals with these four properties is called a **Dedekind cut**. *Axiomatic Set Theory*, Chapter 1 takes the cuts themselves as the real numbers, proves that they form a complete ordered field, and shows that there the rationals below a cut c are exactly the elements of c. By Theorem 6.3.1 that field is one more copy of $\mathbb{R}$; by $(\star)$ the copy of a real number y has the same rationals below it as y, so the copy of y is the cut $\mathbb{Q}_{\mathbb{R}}(y)$.

6.3.3 Automorphisms

An isomorphism from a field to itself is a symmetry: a relabeling that leaves the arithmetic unchanged. The uniqueness theorem rules out every symmetry of a complete ordered field that respects the order; the theorem after it rules out every symmetry.

Corollary 6.3.3. A Complete Ordered Field Has Only the Trivial Order Automorphism

Let F be a complete ordered field. The only order-preserving field automorphism of F is id_F. In particular, this holds for the field $\mathbb{R}$ of Chapter 5.

Proof. F is Archimedean by Theorem 6.1.2 and complete by hypothesis, so by Theorem 6.2.3 with source and target F there is exactly one order-preserving field map $F \to F$. The identity is one, and every order-preserving field automorphism is one. So every such automorphism is id_F. For $\mathbb{R}$, use Theorem 5.5.2.

■

The corollary assumes that the map respects the order. Can a map that only respects the arithmetic move anything?

Theorem 6.3.4. Field Maps of a Complete Ordered Field to Itself

Let F be a complete ordered field, and let $\sigma\colon F \to F$ satisfy, for all $a, b \in F$,

$$\sigma(a+b) = \sigma(a) + \sigma(b), \qquad \sigma(ab) = \sigma(a)\sigma(b), \qquad \sigma(1) = 1.$$

Then $\sigma = \mathrm{id}_F$. No order condition and no bijectivity is assumed.

Try it first (ten minutes): why must such a σ send positive elements to positive elements? Look for a description of "positive" that uses only multiplication.

Idea. The order condition has to be **recovered** from the arithmetic. In a complete ordered field, positivity has a description in terms of multiplication alone: an element is positive exactly when it is a non-zero square (Proposition 6.1.4 gives the square root). A map that respects products sends squares to squares, and squares in an ordered field are never negative. Then additivity turns "positives go to positives" into "σ is strictly increasing", and the uniqueness clause of Theorem 6.2.3 finishes, as in the corollary's proof (which never uses bijectivity).

Plan: ① positive to positive, through a square root; ② strictly increasing, through differences; ③ the uniqueness clause.

Before reading on: which line of the proof below needs completeness, and which would work in any ordered field?

Proof. **Step 1: σ sends positive elements to positive elements.** Let $a \in F$ with $a > 0$. By Proposition 6.1.4 there is $b \in F$ with $b \geq 0$ and $b^2 = a$. Then $b \neq 0$, since $0^2 = 0 \neq a$ (Proposition 2.3.2 (b)). So b has an inverse b^{-1}, and

$$\sigma(b)\,\sigma(b^{-1}) = \sigma(bb^{-1}) = \sigma(1) = 1.$$

If $\sigma(b) = 0$, the left side would be 0 (Proposition 2.3.2 (b)), and $0 \neq 1$ in the field F (Definition 2.3.1). Hence $\sigma(b) \neq 0$, and

$$\sigma(a) = \sigma(bb) = \sigma(b)^2 > 0$$

by Proposition 2.4.3 (e): a non-zero square is positive.

Step 2: σ is strictly increasing. First, $\sigma(-c) = -\sigma(c)$ for every $c \in F$: additivity gives $\sigma(c) + \sigma(-c) = \sigma(c + (-c)) = \sigma(0)$, and $\sigma(0) = \sigma(0+0) = \sigma(0) + \sigma(0)$ gives $\sigma(0) = 0$; so $\sigma(-c)$ is the negative of $\sigma(c)$ (Proposition 2.3.2 (a)). Now let $x < y$ in F. Then $y - x > 0$ (Proposition 2.4.3 (a)), so by Step 1

$$\sigma(y) - \sigma(x) = \sigma(y) + \sigma(-x) = \sigma(y - x) > 0,$$

and $\sigma(x) < \sigma(y)$ by Proposition 2.4.3 (a).

Step 3: $\sigma = \mathrm{id}_F$. By Step 2 and the hypotheses, σ is an order-preserving field map $F \to F$ (Definition 6.1.1). The field F is Archimedean by Theorem 6.1.2, and complete by hypothesis.

By the uniqueness clause of Theorem 6.2.3, with source and target F, there is exactly one order-preserving field map $F \to F$; the identity is one. Hence $\sigma = \mathrm{id}_F$.

■

The answer to the prompt: only the first line of Step 1, "there is b with $b^2 = a$", and Step 3 need completeness; every other line of Steps 1 and 2 works in any ordered field.

Completeness entered twice: it supplied square roots in Step 1, and, through Theorem 6.1.2, the Archimedean property that the uniqueness clause in Step 3 needs (by the remark after its proof in Section 6.2, an Archimedean target suffices there).

Remark. **What each hypothesis buys.** Without the order condition, a bijection that respects sums, products and 1 can reverse signs: *Algebraic Preliminaries*, Chapter 2 meets $\mathbb{Q}(\sqrt{2})$, an Archimedean field that is not complete, which has one exchanging $\sqrt{2}$ and $-\sqrt{2}$. The theorem above shows that this cannot happen in a complete ordered field. Without the Archimedean property, an order-preserving automorphism can move elements: in $\mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field), $t \mapsto t+1$ keeps degrees and leading coefficients (school polynomial algebra, as flagged there). And $\mathbb{Q}$ has only the identity (Exercise B1), although it is not complete: there is nothing besides rationals to move.

Second reading · *Maps that ignore the order*

Steps 1 and 2 used two things only: square roots of positive elements in the **source**, and the rule "non-zero squares are positive" in the **target**. If F is a complete ordered field, L is any ordered field, and $\chi\colon F \to L$ satisfies $\chi(a+b) = \chi(a) + \chi(b)$, $\chi(ab) = \chi(a)\chi(b)$ and $\chi(1) = 1$, then χ is strictly increasing, that is, an order-preserving field map. In particular, for complete ordered fields F and L, every bijection $F \to L$ that respects sums, products and 1 is the isomorphism of Theorem 6.3.1.

6.3.4 What the isomorphism carries

Throughout this subsection F and L are complete ordered fields and $\varphi\colon F \to L$ is the isomorphism of Theorem 6.3.1. The definition gives sums, products, 1 and order; five more facts follow in a line each. Within this subsection, (T1)–(T5) name them.

(T1) $\varphi(0) = 0$, $\varphi(-x) = -\varphi(x)$, $\varphi(x-y) = \varphi(x) - \varphi(y)$, and $\varphi(x^{-1}) = \varphi(x)^{-1}$ for $x \neq 0$. Indeed $\varphi(0) = \varphi(0+0) = \varphi(0) + \varphi(0)$, and adding $-\varphi(0)$ gives $\varphi(0) = 0$; then $\varphi(x) + \varphi(-x) = \varphi(0) = 0$; and $\varphi(x)\varphi(x^{-1}) = \varphi(1) = 1$.

(T2) $x < y$ **if and only if** $\varphi(x) < \varphi(y)$: the forward direction is the definition, and the backward one is the reflection of order shown just after Definition 6.1.1. With (T1), $x > 0$ if and only if $\varphi(x) > 0$.

(T3) $\varphi(|x|) = |\varphi(x)|$. If $x \geq 0$, then $\varphi(x) \geq 0$ by (T2), and both sides are $\varphi(x)$. If $x < 0$, then $\varphi(x) < 0$, and both sides are $\varphi(-x) = -\varphi(x)$ (Definition 4.1.1, (T1)).

(T4) $\varphi(\iota_F(q)) = \iota_L(q)$ for every $q \in \mathbb{Q}$ (Proposition 6.1.1): each rational goes to the same rational.

(T5) The inverse $\varphi^{-1}\colon L \to F$ is again an order-preserving field isomorphism (the fold after Corollary 6.3.2). So everything said about φ holds for φ^{-1} too.

The map acts on larger objects through their elements: a subset A goes to $\varphi(A)$, a sequence (a_n) to $(\varphi(a_n))$, a function f to $\varphi \circ f \circ \varphi^{-1}$, and every such object of L comes from exactly one of F.

An isomorphism is a renaming: it changes what the elements are called and keeps every relation between them.

Here is the renaming at work on a statement with three quantifiers.

Example. Limits Are Carried Across

Let (a_n) be a sequence in F and $a \in F$. Show that $a_n \to a$ in F (Definition 4.3.2) if and only if $\varphi(a_n) \to \varphi(a)$ in L.

Solution. For ($\Rightarrow$), let $\varepsilon \in L_{>0}$, and put $\varepsilon' := \varphi^{-1}(\varepsilon)$. Then $\varepsilon' > 0$ by (T2), since $\varphi(\varepsilon') = \varepsilon > 0$. Since $a_n \to a$, there is $N \in \mathbb{N}$, depending on ε' and so on ε, with $|a_n - a| < \varepsilon'$ for every $n \geq N$. For such n, by (T1), (T3) and (T2),

$$|\varphi(a_n) - \varphi(a)| = \varphi(|a_n - a|) < \varphi(\varepsilon') = \varepsilon.$$

So $\varphi(a_n) \to \varphi(a)$. For ($\Leftarrow$), apply ($\Rightarrow$) to φ^{-1}, which is allowed by (T5).

The step that needs φ to be **onto** is the first: the tolerance ε lives in L, and the hypothesis speaks of tolerances in F. For a map that is not onto the transfer can fail: the Newton sequence has no limit in $\mathbb{Q}$ (Theorem 4.6.1), while its copy under $\iota_{\mathbb{R}}$ converges in $\mathbb{R}$ (Exercise B3 of Section 5.4).

The example is one case of a general fact. The ordered field F, with $+$, $\cdot$, 0, 1 and $<$, is a structure in the sense of *Logic and Proof*, Chapter 7, and φ is an isomorphism of structures: a bijection that respects every symbol. By Logic and Proof, Theorem 7.5.1, every first-order sentence in these symbols (Logic and Proof, Definition 5.2.2) is true in F exactly when it is true in L. Statements that also quantify over subsets, sequences or functions are not first-order sentences of this kind; for those, the renaming above is checked case by case, as in the example: one renaming per object, and onto for each quantifier. We do not state a general theorem for them.

Quick check. Let F, L and φ be as above, and suppose some $x \in F$ satisfies $x^3 = x + 1$. Show that some $y \in L$ satisfies $y^3 = y + 1$.

Solution. Take $y := \varphi(x)$. By Definition 4.2.1 (b), $x^3 = ((1 \cdot x)x)x$, and φ respects 1 and products, so $\varphi(x^3) = \varphi(x)^3$. Hence $y^3 = \varphi(x^3) = \varphi(x + 1) = \varphi(x) + \varphi(1) = y + 1$.

The renaming keeps relations between elements. It does not keep what the elements are made of.

Example. A Complete Ordered Field Whose Zero Is the Empty Set

Find a complete ordered field whose zero element is $\emptyset$.

Solution. In the Cauchy model the zero is $[(0)_n]$, a class that contains the constant sequence, so it is not $\emptyset$. Let $X \coloneqq (\mathbb{R} \setminus \{0_{\mathbb{R}}\}) \cup \{\emptyset\}$ and $\beta\colon \mathbb{R} \to X$, $\beta(0_{\mathbb{R}}) \coloneqq \emptyset$, $\beta(x) \coloneqq x$ otherwise. Every element of $\mathbb{R}$ is a class $[a]$, which contains a and so is not empty; hence $\emptyset \notin \mathbb{R}$, and β is a bijection. Copy the structure of $\mathbb{R}$ to X along β:

$$u \oplus v \coloneqq \beta(\beta^{-1}(u) + \beta^{-1}(v)),$$
$$u \odot v \coloneqq \beta(\beta^{-1}(u)\,\beta^{-1}(v)),$$

and $u \prec v$ if and only if $\beta^{-1}(u) < \beta^{-1}(v)$. Then X is a complete ordered field, β is an isomorphism, and the zero of X is $\beta(0_{\mathbb{R}}) = \emptyset$. So the statement "the zero element is the empty set" is false in $\mathbb{R}$ and true in X, although the two are isomorphic. The statement is about what an element is made of, and no renaming can be expected to keep that.

Second reading · *Why the copied structure is a complete ordered field*

With $u = \beta(a)$ and $v = \beta(b)$, the definitions read $\beta(a + b) = \beta(a) \oplus \beta(b)$, $\beta(ab) = \beta(a) \odot \beta(b)$, and $a < b$ if and only if $\beta(a) \prec \beta(b)$. Each axiom for X, rewritten in this way, becomes the same axiom for $\mathbb{R}$: for (F1), $u \oplus v = \beta(a + b) = \beta(b + a) = v \oplus u$; the zero and one of X are $\beta(0)$ and $\beta(1)$, and $\beta(1) \neq \beta(0)$ because β is injective and $1 \neq 0$ in $\mathbb{R}$; (F2)–(F9) and (O1)–(O4) go the same way. For completeness, let $B \subseteq X$ be non-empty and bounded above by w. Then $\beta^{-1}(B)$ is non-empty and bounded above by $\beta^{-1}(w)$, so it has a supremum s in $\mathbb{R}$, and $\beta(s)$ is a supremum of B, because β and β^{-1} both preserve the order by the definition of $\prec$.

Warning. **Isomorphic is not equal.** The isomorphism links two models; it does not make them one set, and it does not make $\mathbb{Q}$ a subset of $\mathbb{R}$. In the Cauchy model the rational number 2 is a class of pairs of integers (Definition 3.1.1), while the real number 2 is $\kappa(2) = [(2)_n]$, a class of sequences (Definition 5.4.1); they are different sets, and, as Section 5.2 noted after Definition 5.2.1, the rational 2 is not an element of $\mathbb{R}$. "$\mathbb{Q} \subseteq \mathbb{R}$" is shorthand for the embedding $\iota_{\mathbb{R}}$, the one copy of $\mathbb{Q}$ that Theorem 4.2.2 provides, and every isomorphism respects that copy by (T4).

So "which construction is $\mathbb{R}$?" has no mathematical content: every construction gives a complete ordered field, any two are linked by exactly one isomorphism, and that isomorphism decides every first-order question in $+$, $\cdot$, $<$ (and, case by case, statements about subsets, sequences and functions) in the same way in both. Questions such as "is $\sqrt{2}$ a set of rationals?" depend on the model, and analysis never asks them.

6.3.5 Assume the axioms or build a model

(A) Assume the axioms. Declare that $\mathbb{R}$ is a set with an addition, a multiplication and a relation $<$ satisfying the field axioms (F1)–(F9) and $1 \neq 0$ of Definition 2.3.1, the order axioms (O1)–(O4) of Definition 2.4.1, and completeness, Definition 5.5.1: every non-empty subset that is bounded above has a supremum. Then prove theorems from these axioms and nothing else. *Analysis on the Real Line*, Chapter 1 begins this way. It assumes an ordered field containing $\mathbb{Q}$ in which the least upper bound axiom holds, and every theorem of that volume about real numbers is proved from this assumption, with the logic and set theory of the earlier volumes.

(B) Build a model. Start from sets and construct the objects one by one. This volume did that from a Peano system $\mathbb{N}$, whose existence Section 1.5 explains and *Axiomatic Set Theory* proves from the axioms of set theory: the integers and the rationals as classes of pairs (Chapters 2 and 3), and the real numbers as classes of Cauchy sequences of rationals (Chapter 5). The result is a complete ordered field (Theorem 5.5.2); once $\mathbb{N}$ is given, no property of the result was assumed.

Route (A) is shorter, and it is the route of most analysis courses. On its own, though, it leaves two questions open.

(Q1) *Is there a complete ordered field at all?* A list of axioms can be contradictory: a list of demands that nothing meets. Theorems proved from a contradictory list would all be worthless.

(Q2) *If there are several, which one is* $\mathbb{R}$*?* Route (A) says "let $\mathbb{R}$ be **a** complete ordered field". The article matters.

Route (B) answers (Q1). The Cauchy model of Chapter 5 is a complete ordered field, built inside set theory, from a Peano system. So if a contradiction could be derived from the axioms of a complete ordered field, the same derivation, run in the Cauchy model, would be a contradiction about one particular set built in that set theory. In short, the axioms are consistent if the axioms of set theory are, with a Peano system among the sets (*Axiomatic Set Theory* states the axioms as ZF and builds one). That is the most a construction can give: it moves the question of consistency from the axioms of $\mathbb{R}$ to the axioms of set theory.

Route (B) alone does not answer (Q2): a builder who uses Dedekind cuts gets a different set. The answer to (Q2) is Theorem 6.3.1, and the previous subsection explained why it is the right answer: one structure, not a family that agrees only on what the axioms prove.

Theorems need no uniqueness; truth does. A statement **proved** from the ordered-field axioms holds in every ordered field, but the axioms alone leave statements open: "some x has $x \cdot x = 1 + 1$" is false in $\mathbb{Q}$ (Theorem 3.3.1) and true in $\mathbb{R}$ (Proposition 6.1.4), and "some y lies above every $n \cdot 1$" is false in $\mathbb{R}$ (Theorem 6.1.2) and true in $\mathbb{Q}(t)$ (Example (A Non-Archimedean Ordered Field)). For complete ordered fields no such disagreement can occur, whether or not anybody has a proof.

The analysis volume uses route (A). The table matches each of its assumptions with the place where this series supplies or justifies it.

Assumed in *Analysis on the Real Line*, Chapter 1	Supplied or justified here
$\mathbb{R}$ is an ordered field, axioms (F1)–(F9), (O1)–(O4)	the axioms: Definition 2.3.1, Definition 2.4.1, in the same numbering (in *Analysis on the Real Line*, (F7) includes $1 \neq 0$; here it is a separate clause of the field definition); a model: Theorem 5.3.3
$\mathbb{R}$ contains $\mathbb{Q}$	every ordered field contains exactly one copy of $\mathbb{Q}$: Theorem 4.2.2; isomorphisms respect it: Proposition 6.1.1
the least upper bound axiom	the property: Definition 5.5.1; a model: Theorem 5.5.2
"the" real numbers	any two models are isomorphic, by exactly one isomorphism: Theorem 6.3.1, Corollary 6.3.2
logic and set theory	*Logic and Proof*, *Sets and Functions*; $\mathbb{N}$ in Chapter 1 of this volume (Section 1.5 for where it comes from)

Some results are proved in both volumes, from different starting points. *Analysis on the Real Line*, Chapter 1 proves from its axioms that $\mathbb{N}$ is not bounded above, that $\mathbb{Q}$ is dense in $\mathbb{R}$, and that positive numbers have square roots. Here the same three facts were proved for an arbitrary complete ordered field: Theorem 6.1.2, Theorem 4.2.5 (which needs Archimedes, supplied by the first) and Proposition 6.1.4. The two volumes start from different places, and neither volume's proofs cite the other's.

This ends the volume. *Analysis on the Real Line* begins from route (A), and this chapter is why that beginning names one structure.

6.3.6 Exercises

6.3.6.1 A. Check your understanding

Exercise. A1

Parts (a)–(c) are about the proof of Theorem 6.3.1, parts (d)–(g) about Theorem 6.3.4.

(a) Where is the completeness of F used? Name every place.

(b) Why is $\psi \circ \varphi$ order-preserving?

(c) Suppose F is only assumed to be an Archimedean ordered field, and L is complete. Which step of the proof fails first?

(d) Where is completeness of F used in the proof of Theorem 6.3.4?

(e) Why is $\sigma(b) \neq 0$ in Step 1?

(f) Why does σ fix every rational, although no order hypothesis is made?

(g) Let $\sigma\colon \mathbb{R} \to \mathbb{R}$ respect sums and products, with $\sigma(1) = 1$, and let r be the real number of Example (The Class of the Newton Sequence) ($r > 0$, $r^2 = 2$). Without quoting Theorem 6.3.4, show that $\sigma(r) = r$.

Solution.

(a) In two ways. First, through Theorem 6.1.2, completeness makes F Archimedean, so F is a legal **source** for Theorem 6.2.3: for $\varphi\colon F \to L$, and for the uniqueness clause applied to maps $F \to F$ such as $\psi \circ \varphi$. Second, it makes F a legal **target**, so that $\psi\colon L \to F$ exists. The uniqueness clause for $\psi \circ \varphi$ is cited with target F, so as stated it asks for F complete; but by the remark after its proof in Section 6.2 only the Archimedean property of F is used there, and the first use already supplies it (as in (d)).

(b) If $a < b$, then $\varphi(a) < \varphi(b)$ because φ is strictly increasing, and this strict inequality in L is a hypothesis ψ accepts, so $\psi(\varphi(a)) < \psi(\varphi(b))$. A composite of strictly increasing maps is strictly increasing.

(c) The construction of $\psi\colon L \to F$: Theorem 6.2.3 needs a complete target, and F is not one. (The map φ still exists.) The conclusion fails too: for $F = \mathbb{Q}$ and $L = \mathbb{R}$, the only order-preserving field map is $\iota_{\mathbb{R}}$, which is not onto.

(d) Twice in substance. In Step 1, Proposition 6.1.4 needs F complete to give the square root b of $a > 0$, and that is how $\sigma(a) > 0$ is proved. In Step 3, Theorem 6.1.2 makes F Archimedean, which the uniqueness clause of Theorem 6.2.3 needs of its target (the clause separates elements by rationals in the target, here F, Lemma 6.1.3). Step 3 also cites the completeness of F, because that theorem is stated for a complete target; by the remark after its proof in Section 6.2, the uniqueness clause needs only an Archimedean target, so this is the same use again.

(e) $b \neq 0$ has an inverse, and $\sigma(b)\sigma(b^{-1}) = \sigma(1) = 1$. If $\sigma(b)$ were 0, the left side would be 0, and $0 \neq 1$ in a field.

(f) By Proposition 6.1.1, whose hypotheses are additivity, multiplicativity and $\sigma(1) = 1$ only: the composite $\sigma \circ \iota_F\colon \mathbb{Q} \to F$ respects sums and products and fixes 1, so it is ι_F by the uniqueness clause of Theorem 4.2.2. Order plays no part.

(g) $\sigma(r)^2 = \sigma(r^2) = \sigma(2) = 2$, since σ fixes the rationals (Proposition 6.1.1). By Proposition 6.1.4 there is $s \geq 0$ with $s^2 = r$, and $s \neq 0$ since $r \neq 0$. So s has an inverse, and $\sigma(s)\sigma(s^{-1}) = \sigma(ss^{-1}) = \sigma(1) = 1$, which forces $\sigma(s) \neq 0$. Hence $\sigma(r) = \sigma(s)^2 > 0$ by Proposition 2.4.3 (e). So $\sigma(r)$ and r are both non-negative square roots of 2, and the uniqueness part of Proposition 6.1.4 gives $\sigma(r) = r$.

Exercise. A2

Throughout, F and L are complete ordered fields and $\varphi\colon F \to L$ is the order-preserving field isomorphism.

(a) For each statement, decide whether its truth could depend on which complete ordered field is taken as $\mathbb{R}$, and justify your answer: (i) $\sqrt{2} + \sqrt{3} > 3$; (ii) the zero element is the empty set; (iii) $\mathbb{R}$ is uncountable.

(b) Which step of Example (Limits Are Carried Across) uses that φ is onto?

(c) Why does a model of the axioms, built inside set theory, show that the axioms of a complete ordered field cannot contradict each other unless that set theory does?

Solution.

(a) (i) No. Write b and c for the non-negative square roots of 2 and 3 in F, which exist and are unique by Proposition 6.1.4. Then $\varphi(b) \geq 0$ by (T2) and $\varphi(b)^2 = \varphi(b^2) =$

$\varphi(2) = 2$ by (T4), so $\varphi(b)$ is the non-negative square root of 2 in L, by the uniqueness in Proposition 6.1.4; likewise for c. By (T2) and (T4), $b + c > 3$ if and only if $\varphi(b) + \varphi(c) > 3$. (The statement is in fact true: both sides are positive, $(b + c)^2 = 5 + 2bc$, and bc is the non-negative square root of 6, so $5 + 2bc > 9$ if and only if $bc > 2$, if and only if $6 > 4$, by Proposition 2.4.3 (g).)

(ii) Yes. It is false in the Cauchy model and true in the field X of Example (A Complete Ordered Field Whose Zero Is the Empty Set), and the two are isomorphic. The statement is about what an element is made of, which no renaming keeps.

(iii) No. A set is countable exactly when it injects into $\mathbb{N}$ (Sets and Functions, Theorem 4.2.2). If L were countable, an injection $g\colon L \to \mathbb{N}$ composed with the bijection φ would give an injection $g \circ \varphi\colon F \to \mathbb{N}$; and symmetrically with φ^{-1}. So F is countable exactly when L is.

(b) The first: the tolerance ε is given in L, and the hypothesis $a_n \to a$ speaks only about tolerances in F. Surjectivity provides $\varepsilon' = \varphi^{-1}(\varepsilon) \in F$.

(c) A contradiction derived from the axioms could be run in the model, where it would become a contradiction about one set built in the set theory; so the set theory would be inconsistent. In this volume the model is the Cauchy model (Theorem 5.5.2) and the set theory is the one *Sets and Functions* uses informally and *Axiomatic Set Theory* states as the ZF axioms, in which it builds the Peano system $\mathbb{N}$ that Chapter 1 assumes.

6.3.6.2 B. Practice

Exercise. B1

Prove that the only order-preserving field map $\mathbb{Q} \to \mathbb{Q}$ is the identity, using Proposition 6.1.1 and Theorem 4.2.2 but not Section 6.2. Does your proof use the order condition?

Solution. Let $\chi\colon \mathbb{Q} \to \mathbb{Q}$ be an order-preserving field map. It respects sums and products and fixes 1, so Proposition 6.1.1 with $F = L = \mathbb{Q}$ gives $\chi \circ \iota_{\mathbb{Q}} = \iota_{\mathbb{Q}}$. The identity $\mathrm{id}_{\mathbb{Q}}$ respects sums and products and fixes 1, so by the uniqueness clause of Theorem 4.2.2, $\iota_{\mathbb{Q}} = \mathrm{id}_{\mathbb{Q}}$. Hence $\chi = \chi \circ \mathrm{id}_{\mathbb{Q}} = \mathrm{id}_{\mathbb{Q}}$.

The order condition was never used: every map $\mathbb{Q} \to \mathbb{Q}$ that respects sums and products and fixes 1 is the identity. In $\mathbb{Q}$ there is nothing besides rationals to move.

Exercise. B2

Let F and L be complete ordered fields, and $\varphi\colon F \to L$ the isomorphism of Theorem 6.3.1. Let $A \subseteq F$ be non-empty and bounded above. Prove that $\varphi(A) \coloneqq \{\varphi(a) : a \in A\}$ has a supremum in L and that $\varphi(\sup A) = \sup \varphi(A)$ (Sets and Functions, Definition 5.3.5).

Solution. Since F is complete, $s \coloneqq \sup A$ exists. We show that $\varphi(s)$ satisfies (S1) and (S2) for $\varphi(A)$ in L; then it is the supremum.

(S1). Let $a \in A$. Then $a \le s$, so $\varphi(a) \le \varphi(s)$: equal if $a = s$, and smaller if $a < s$, since φ is strictly increasing.

(S2). Let $u \in L$ be an upper bound of $\varphi(A)$. Since φ is bijective, $u = \varphi(v)$ for some $v \in F$. For $a \in A$, $\varphi(a) \le \varphi(v)$, so $a \le v$: if $v < a$, then $\varphi(v) < \varphi(a)$, against (O1). So v is an upper bound of A, hence $s \le v$ by (S2) for A, and $\varphi(s) \le \varphi(v) = u$.

So $\varphi(s) = \sup \varphi(A)$. The proof used that φ is onto (in (S2)) and that φ reflects order; for an order-preserving field map that is not onto, the image of a supremum can fail to be the supremum of the image, because the image may have new upper bounds that are not images. For example, take $\iota_{\mathbb{Q}(t)} \colon \mathbb{Q} \to \mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field) and $A = \{1 - 1/n : n \in \mathbb{Z}_{>0}\}$. Then $\sup A = 1$ in $\mathbb{Q}$, but $1 - 1/t$ is an upper bound of the image that lies below 1, since $0 < 1/t < 1/(n \cdot 1)$ for every $n \ge 1$. (For $\iota_{\mathbb{R}}$ this cannot happen: $\mathbb{Q}$ is dense in the Archimedean field $\mathbb{R}$, so $\iota_{\mathbb{R}}$ keeps every supremum that exists in $\mathbb{Q}$.)

Exercise. B3

Let F and L be complete ordered fields, $\varphi \colon F \to L$ the isomorphism of Theorem 6.3.1, and (a_n) a sequence in F. Prove that (a_n) is a Cauchy sequence in F (Definition 4.5.1) if and only if $(\varphi(a_n))$ is a Cauchy sequence in L.

Solution. ($\Rightarrow$) Let $\varepsilon \in L_{>0}$, and put $\varepsilon' := \varphi^{-1}(\varepsilon)$. Since $\varphi(\varepsilon') = \varepsilon > 0 = \varphi(0)$, and φ reflects order, $\varepsilon' > 0$. Since (a_n) is Cauchy, there is $N \in \mathbb{N}$, depending on ε, with $|a_m - a_n| < \varepsilon'$ for all $m, n \ge N$. For such m, n,

$$|\varphi(a_m) - \varphi(a_n)| = \varphi(|a_m - a_n|) < \varphi(\varepsilon') = \varepsilon,$$

by (T1), (T3) and (T2). So $(\varphi(a_n))$ is Cauchy in L.

($\Leftarrow$) The inverse φ^{-1} is an order-preserving field isomorphism $L \to F$ (T5), so the same argument with the roles of F and L exchanged shows that $(\varphi^{-1}(\varphi(a_n))) = (a_n)$ is Cauchy in F.

6.3.6.3 C. Going deeper

Exercise. C1: Exit check: field maps of a complete ordered field

Let F be a complete ordered field and $\sigma \colon F \to F$ a map that respects sums and products and has $\sigma(1) = 1$. Using only Proposition 6.1.4, Proposition 6.1.1, Lemma 6.1.3, Theorem 6.1.2 and Theorem 6.2.3 (and the rules of an ordered field), prove that $\sigma = \mathrm{id}_F$. Write the proof without looking at the one in this section.

Solution. σ fixes 0 and negatives. $\sigma(0) = \sigma(0 + 0) = \sigma(0) + \sigma(0)$, so $\sigma(0) = 0$; then $\sigma(c) + \sigma(-c) = \sigma(0) = 0$, so $\sigma(-c) = -\sigma(c)$.

Positive to positive. Let $a > 0$. By Proposition 6.1.4, $a = b^2$ for some $b \ge 0$, and $b \ne 0$ since $a \ne 0$. Then $\sigma(b)\sigma(b^{-1}) = \sigma(1) = 1 \ne 0$, so $\sigma(b) \ne 0$, and $\sigma(a) = \sigma(b)^2 > 0$ because non-zero squares are positive.

Strictly increasing. If $x < y$, then $y - x > 0$, so $\sigma(y) - \sigma(x) = \sigma(y - x) > 0$.

Conclusion. σ is an order-preserving field map $F \to F$. By Theorem 6.1.2, F is Archimedean; it is complete. Theorem 6.2.3 with source and target F gives exactly one order-preserving field map $F \to F$, and id_F is one. So $\sigma = \mathrm{id}_F$.

(Alternatively, the last paragraph can be done by hand: σ fixes the rationals by Proposition 6.1.1, and a strictly increasing map that fixes the rationals keeps $\mathbb{Q}_F(x)$; since F is Archimedean, $\mathbb{Q}_F(\sigma(x)) = \mathbb{Q}_F(x)$ gives $\sigma(x) = x$ by Lemma 6.1.3.)

Marking scheme. (10 points) - 2: $\sigma(0) = 0$ and $\sigma(-c) = -\sigma(c)$ from additivity (on the by-hand route, also that σ fixes the rationals, with a reason that uses no order hypothesis). - 3: $a > 0 \Rightarrow \sigma(a) > 0$ through a square root of a, **including** the proof that $\sigma(b) \neq 0$. - 2: strict increase from additivity, through $\sigma(y) - \sigma(x) = \sigma(y - x)$. - 3: concludes $\sigma = \mathrm{id}_F$ by the uniqueness clause, or directly by comparing the rationals below x and below $\sigma(x)$, and in either case says that F is Archimedean by Theorem 6.1.2. Fatal errors (0 for the step): assuming σ order-preserving; using the uniqueness clause or the rationals-below comparison without saying why F is Archimedean; taking a square root of a negative element; dividing by $\sigma(b)$ without showing $\sigma(b) \neq 0$.

Exercise. C2: Find the error

Here is an argument that the map $\varphi\colon F \to L$ of Theorem 6.2.3 is additive. For subsets $A, B \subseteq \mathbb{Q}$, write $A + B \coloneqq \{a + b : a \in A, b \in B\}$.

(1) Let $x, y \in F$ and $q \in \mathbb{Q}_F(x + y)$.
(2) Then $q/2 < x$ and $q/2 < y$.
(3) So $q = q/2 + q/2 \in \mathbb{Q}_F(x) + \mathbb{Q}_F(y)$.
(4) Conversely, if $r < x$ and $s < y$, then $r + s < x + y$; so $\mathbb{Q}_F(x) + \mathbb{Q}_F(y) \subseteq \mathbb{Q}_F(x + y)$.
(5) Hence $\mathbb{Q}_F(x + y) = \mathbb{Q}_F(x) + \mathbb{Q}_F(y)$.
(6) Taking suprema in L, $\varphi(x + y) = \varphi(x) + \varphi(y)$.

Is the conclusion true? Name the first line that fails, give a counterexample **to that line**, and name a second gap. Then repair the argument with results of Section 6.2.

Solution. The conclusion is true: φ is additive by Theorem 6.2.3. The argument does not prove it.

The first illegal line is (2). From $q < x + y$ it does not follow that $q/2 < x$. Counterexample to the step, in $F = \mathbb{Q}$: $x = 0$, $y = 2$, $q = 1$. Then $q = 1 < 2 = x + y$, but $q/2 = 1/2$ is not below $x = 0$. The line would be right if $x = y$; it splits q evenly while $x + y$ may be split unevenly.

The second gap is (6). It uses, without saying so, that the supremum of a sum set is the sum of the suprema: $\sup_L \iota_L(A + B) = \sup_L \iota_L(A) + \sup_L \iota_L(B)$. That is true in a complete ordered field, but this book has not proved it, and the route of Section 6.2 was chosen to avoid it.

Repair. Line (5) is in fact true, for another reason: by Lemma 6.2.1 in the Archimedean field F, $q < x + y$ if and only if there is $r \in \mathbb{Q}$ with $r < x$ and $q - r < y$; then $q = r + (q - r)$ with

$r \in \mathbb{Q}_F(x)$ and $q - r \in \mathbb{Q}_F(y)$. So replace (2) and (3) by Lemma 6.2.1, which chooses the split r to fit x and y. Replace (6) by the rationals-below argument: by $(\star)$, $\mathbb{Q}_L(\varphi(x+y)) = \mathbb{Q}_F(x+y)$ and $\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x)$, $\mathbb{Q}_L(\varphi(y)) = \mathbb{Q}_F(y)$; by Lemma 6.2.1 in L (Archimedean by Theorem 6.1.2), $\mathbb{Q}_L(\varphi(x)+\varphi(y)) = \mathbb{Q}_L(\varphi(x)) + \mathbb{Q}_L(\varphi(y))$. So $\varphi(x+y)$ and $\varphi(x)+\varphi(y)$ have the same rationals below them, and they are equal by Lemma 6.1.3 in L.

Marking scheme. (10 points) - 3: names line (2) as the first failure and says what it would need (an even split of q, which $x+y$ does not guarantee). - 1: names line (6) as a second gap, because it assumes the sum-of-suprema rule. - 3: a counterexample to the step itself (values of x, y, q with $q < x+y$ and $q/2 \not< x$), not to the conclusion. - 3: a correct repair through Lemma 6.2.1 and the rationals below, with $(\star)$ and Lemma 6.1.3 in L; or a proof of the sum-of-suprema rule, if the reader supplies one. Fatal errors (0 for the item): declaring the conclusion false.

Exercise. C3: Bounded increasing sequences

(a) Let F be a complete ordered field and (a_n) a sequence in F with $a_n \le a_{n+1}$ for every n, whose set of terms $S := \{a_n : n \in \mathbb{N}\}$ is bounded above. The following steps prove that $a_n \to \sup S$ (Definition 4.3.2).
  (1) S is non-empty and bounded above, so $s := \sup S$ exists.
  (2) Let $\varepsilon \in F_{>0}$. Find $N \in \mathbb{N}$ with $a_N > s - \varepsilon$.
  (3) *Name the move, then complete it:* show $|a_n - s| < \varepsilon$ for every $n \ge N$, and conclude.

(b) Show that in $\mathbb{Q}$ the conclusion fails: the sequence $(-x_n)$, with (x_n) the Newton sequence, is increasing and bounded above and has no limit in $\mathbb{Q}$.

Solution. The move is **construct a point as a supremum** (Example (Method: construct a point as a supremum)): the limit is built as $\sup S$; (S1) puts every term below it, and leastness, in (2), puts a term above $s - \varepsilon$.

(a) (a) Since $s - \varepsilon < s$ and s is the least upper bound, $s - \varepsilon$ is not an upper bound of S, so $a_N > s - \varepsilon$ for some $N \in \mathbb{N}$.

(3) Let $n \ge N$. By induction on i (Theorem 1.1.2), $a_N \le a_{N+i}$, and $n = N + i$ for some i (Proposition 1.3.1 (a)); so $a_N \le a_n$. Also $a_n \le s$ by (S1). Hence $s - \varepsilon < a_N \le a_n \le s < s + \varepsilon$, and $|a_n - s| < \varepsilon$ (Proposition 4.1.3 (d)). Since ε was arbitrary and N depends only on ε, $a_n \to s$.

(b) By Theorem 4.6.1, $x_m \le x_n$ for $m \ge n$ and $x_n > 1$; so $-x_n \le -x_m$ for $m \ge n$ (increasing) and $-x_n < -1$ (bounded above). If $-x_n \to L$ in $\mathbb{Q}$, then $x_n \to -L$ (Theorem 4.4.3 (b), with $c = -1$), against the same theorem.

Marking scheme. (10 points) - 2: names the move (construct a point as a supremum) and says that every term lies below s by (S1) while step (2) uses leastness (S2). - 2: step (2): $s - \varepsilon$ is not an upper bound, so some a_N exceeds it. - 2: proves $a_N \le a_n$ for $n \ge N$ (induction, or monotonicity stated with a reason). - 2: the squeeze $s - \varepsilon < a_n \le s$ and the absolute value. - 2: (b) increasing, bounded above, and no limit, through the limit law. Fatal errors (0 for the step): assuming $s \in S$; choosing N after n.

6.3.6.4 R. Review

Exercise. R1

Let $q_n := (n+1)/(n+2) \in \mathbb{Q}$.

(a) Show that (q_n) is a Cauchy sequence of rationals and that $[(q_n)] = 1$ in $\mathbb{R}$.

(b) Every q_n is less than 1. What inequality between $[(q_n)]$ and 1 does a termwise comparison give, and why can it not give a strict one?

Solution.

(a) $q_n \to 1$ in $\mathbb{Q}$ (Example (First Limits) (c)); a convergent sequence is Cauchy (Proposition 4.5.1), so $(q_n) \in \mathcal{C}$ (Definition 5.1.1). Since $|(q_n - 1) - 0| = |q_n - 1|$, the statement $q_n \to 1$ says exactly that $(q_n - 1)_n$ is null. So $(q_n) \sim (1)_n$ (Definition 5.1.2), and $[(q_n)] = [(1)_n] = 1$ (Definition 5.4.1).

(b) Lemma 5.3.4 gives $[(q_n)] \leq [(1)_n] = 1$. A strict inequality is impossible, since (a) gives equality: strict termwise inequalities survive to classes only as $\leq$, as they survive to limits (Proposition 4.4.4).

Exercise. R2

Which of the following rules define a function $\mathbb{Q} \to \mathbb{Q}$, where $\mathbb{Q} = (\mathbb{Z} \times \mathbb{Z}_{>0})/\!\sim$?

(a) $[(p, q)] \mapsto [(p^2, q^2)]$;

(b) $[(p, q)] \mapsto [(p + q, q)]$;

(c) $[(p, q)] \mapsto [(p + 1, q)]$.

For each that does, name the function in familiar terms.

Solution. Recall $(p, q) \sim (r, s)$ iff $ps = rq$ (Proposition 3.1.1; the method is Sets and Functions, Example (Method: checking that a definition on classes is well defined)).

(a) Yes, it is $x \mapsto x^2$: $q^2 > 0$, and $ps = rq$ gives $p^2 s^2 = r^2 q^2$.

(b) Yes, $x \mapsto x + 1$: $(p + q)s = ps + qs = rq + qs = (r + s)q$.

(c) No: $(1, 2) \sim (2, 4)$, but $(2, 2) \not\sim (3, 4)$, since $2 \cdot 4 = 8 \neq 6 = 3 \cdot 2$.

Exercise. R3

Let F be a complete ordered field.

(a) Show that $\iota_F(\mathbb{Q})$ is countable.

(b) Let $\sqrt{2} \in F$ be as in Proposition 6.1.4. Show that $\sqrt{2} \notin \iota_F(\mathbb{Q})$.

(c) Let $x < y$ in F. Show that some element of $F \setminus \iota_F(\mathbb{Q})$ lies strictly between x and y.

Solution.

(a) The set $\mathbb{Q}$ built in Chapter 3 is countable; that is review exercise R2 of Section 3.3 (the countability results of *Sets and Functions*, Chapter 4, applied to $\mathbb{Z} = (\mathbb{N}\times\mathbb{N})/\sim$ and $\mathbb{Q}$; Sets and Functions, Theorem 4.2.8 is about the school rationals and is not used). $\iota_F(\mathbb{Q})$ is the image of $\mathbb{Q}$ under ι_F, so it is countable (Sets and Functions, Proposition 4.2.4).

(b) If $\sqrt{2} = \iota_F(p)$, then $\iota_F(p^2) = \iota_F(p)^2 = 2 \cdot 1_F = \iota_F(2)$ (Theorem 4.2.2), and ι_F is injective, so $p^2 = 2$, against Theorem 3.3.1.

(c) F is Archimedean (Theorem 6.1.2). By density (Theorem 4.2.5) take $q \in \mathbb{Q}$ with $x < \iota_F(q) < y$, and by Proposition 4.2.3 take $N \in \mathbb{Z}_{>0}$ with $1/(N\cdot 1_F) < (y-\iota_F(q))/\sqrt{2}$; here $\sqrt{2} > 0$, since $\sqrt{2} \geq 0$ and $\sqrt{2} \neq 0$. Put $z := \iota_F(q) + \sqrt{2}/(N \cdot 1_F)$; then $\iota_F(q) < z < y$. If $z = \iota_F(p)$ with $p \in \mathbb{Q}$, then

$$\sqrt{2} = (N \cdot 1_F)(\iota_F(p) - \iota_F(q)) = \iota_F(N(p-q)),$$

against (b). So the rational elements of F are countably many, and between any two elements of F lies one that is not rational.